A Living Machine-Checked Companion to Goertzel and Fable's Reductive–Compositional Route to the Four-Color Theorem:

Proven Structure, Refuted Shortcuts, and the Closure Frontier

The MeTTapedia formalization effort

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Abstract

This document is intentionally cumulative. It preserves earlier audits of Goertzel's v23 proposal, including negative results and abandoned repairs, while the controlling status is now the current Goertzel/Fable reductive–compositional programme described in *reductive_vs_covering (2)*, *compositional_proof_note*, *4cp_new_theory*, and *4cp-lean_playbook*. Where that package supersedes v23, the newer text controls. Section 7 records the current kernel-checked boundary. A conditional Cell–3 one-cell factorization and a lossless finite replay interface are present through the manuscript-exact two-pair profile (with the conservative three-pair carrier retained internally). Its current Lean interface requires a globally coloured closed-web-at-good-word instance, which an arbitrary direct zero-accepting-Count obstruction does not supply, and assumes global pairwise uniqueness of opened face intersections, which the dodecahedral annulus refutes. Even when the opened annulus is colourable, the transition cannot be defined by a whole-annulus colouring without making its letters depend on the unprocessed suffix. The factorization must therefore be generalized to a colouring-independent annular Cell context plus partial prefix/Cell witnesses, and refactored through the proved local retained-face transport, before it is applicable to its original Cell–3 target. The direct spine no longer needs that specialization: the physical port-tangle gluing bijection, bounded-interface Count composition law, Boolean support update, block deletion, and port relabelling invariance are machine-checked for every actual ordered noose cut. Thus a width-k cut chain has the conservative finite cumulative carrier $\mathcal{P}(\{r, b, p\}^k)$ without an executable Cell alphabet or a global colouring witness. The ordinary noose surgery is also proved to preserve connectedness, cubicity, sphericity, bridgelessness, strict size decrease, and zero accepting support. What remains is the executable minimized closure needed to make the eventual base practical, the rotation-system seam constructor and simple-map adapter, the target-forced compositional cut-chain theorem, and the route-native finite base. The former good-word closed-web base (GWCO) is refuted and is not a premise of the controlling proof. For the Lift branch, the rolling profile correctly retains the open strand relation π_{str}, but the former Lean terminal predicate counted components before the pentagonal cap was restored and was therefore vacuous on a good word; the source instead consumes the finite closure of $\pi_{\mathrm{str}} \cup \pi_{\mathrm{cap}}$. The direct Count branch has a different accepting predicate: literal terminal closure to a proper Tait colouring. Those accepting sets and their Nerode quotients are now kept separate. Direct counterexamples with triangular and square outer boundaries also show that the Seed Lemma is false for arbitrary connected annular tangles; the live statement is restricted to frontiers cut from the established cyclically-five-edge-connected, face-length-at-least-five normal form. Accordingly this companion does not claim a new proof of the Four-Color Theorem.

This is a living formalization companion to Ben Goertzel's manuscript *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem* (v23), which organizes a candidate human-readable proof of the Four-Color Theorem (4CT) around three interacting pillars: (A) an annular difference-*spanning* statement for a zero-boundary space $W_0(H)$ (Theorem 4.9);

(B) a CAP5-free normal form with radius-1 collar locality; and (C) an *executability* layer that, refusing a false additivity of Kempe switches, reduces between-region Kempe connectivity to a single *Local Kempe Reachability with fixed input* ($\mathrm{LKR}_{\mathrm{in}}$) property for one canonical three-cell collar gadget τ (Lemma 8.14), composes it along the collar chain (Lemmas 8.15, 8.18, Cor. 8.19), and assembles the wide annulus collar-by-collar (§9.2). We record what the MeTTapedia formalization has machine-checked of this route, with an honest ledger separating what holds outright, what holds modulo one named remainder, and what is open, and we foreground the places where the formalization *refutes* a step rather than discharging it.

Machine-checked outright (no `sorry`; axioms $\subseteq$ {`propext`, `Classical.choice`, `Quot.sound`}): finite encoded-model audits of the keystone $\mathrm{LKR}_{\mathrm{in}}$ property (Lemma 8.14) and two-color *transparency* (Lemma 8.15), together with completeness of the 192-state incidence enumeration — graph-level Kempe-step soundness remains open; a proper-subspace strictness theorem against a larger unrestricted boundary-zero chain space, with four uniform peel oracles refuted and conditional need-based span infrastructure — v23's actual cycle-space inclusion (Theorem 4.9) remains open; and a length-2/3 radial-closing-edge fiber census for §9.2 Step 3 — the manuscript's all-outer-stub, arbitrary-length statement remains open. The Gate-2 composition (Lemmas 8.18/8.19) is *reduced* — the manuscript's own pointwise Step-1 preparation is refuted and replaced by a frontier-automaton / fibration architecture — to one live pin combining representative real-fiber connectivity audits with DFA-mode sufficiency.

The formalization also produced a decisive negative in the encoded model. §9.2 Step 4 — the manuscript's reinterpretation of collar Kempe words as words of the ambient annulus — fails in the encoded two-layer tau-chain model: a certified, collar-input-avoiding switch whose ambient component escapes the inner boundary, re-enters, and changes the collar restriction. Its realization as a simple planar, cyclically-five-edge-connected actual collar is a named open obligation, so this is a regression countermodel against the composition *seam* as written, not a graph-level refutation — and not a refutation of 4CT; its salvage (a normal-form confinement argument, or a face-collar redesign) is open. We make no claim about the Four-Color Theorem — which is of course a theorem, via the Appel–Haken and Robertson–Sanders–Seymour–Thomas computer-assisted proofs. The human-readable v23 route remains *open*, with its foundational layer audited in an encoded model, and its most dangerous assembly seam pinned down as broken in that model.

A third development round turned to the v24 working material (*Deserts, Pivots, and the Necklace*), which replaces v23's §6–7. The v24 validation toolchain and its numerical claims were reproduced independently: the 5412 and 31704 annular censuses with their Kempe fibers, the fullerene bases, the pentagon seals, the desert and strand specimens — including the 688-edge strand refuting the package's own Short Strand Conjecture — and the historical GWCO exact search through twelve (now superseded by the counterexamples recorded below). On the structural side the development now carries a rotation-system Tait-coloring layer with exact facial linear algebra; the all-face discrete Gauss–Bonnet identity (`totalFaceCurvature_eq_twelve`); a proved *weighted* pentagon budget correcting the package's at-most-twelve pentagon count; the locked-pentagon necklace assembly with singleton-strand forcing; a hexagonal corridor theory (rung classification, clean width-four slabs, oriented interfaces, Tait transfer, a sharp chirality barrier); congruence-correct corridor and tube-ring pumping; and a framed-trail layer with square/digon/rotor fiber calculi. The exact two-defect boundary-flux identity now retires the deleted-edge Trail-Completability branch: a coloured bare deletion always extends immediately, while a frozen frame is completable exactly in its static zero-flux sector. The live endgame is instead the direct cumulative-Count descent: a locally valid exact finite transition, literal terminal acceptance, a target-forced global cut chain, the rotation-system realization of the proved support-state splice, and the route-native finite base.

Contents

1 The current proof spine: what is a theorem and what remains

This section is the controlling mathematical account of the route. It is grounded in the working manuscript's Addenda XXV–XXVIII, the accompanying formalization playbook, and the compositional proof note. The chronological material later in this companion records how the present formulation was reached; it must not be read as overriding the statuses stated here.

The intended proof is not a finite catalogue of unavoidable configurations. Its central object is an open planar cubic instance with a finite boundary interface. A corridor is cut into open cells, the behavior visible at a cut is compressed to a finite profile, and equal profiles permit a middle segment to be removed. Thus the desired contradiction has the following form:

$$\text{large minimal obstruction} \Longrightarrow \text{long composable corridor} \Longrightarrow \text{repeated interface profile} \Longrightarrow \text{smaller obstruction.}$$

The first and last arrows are geometry. The middle arrow is finite-state pigeonhole. Keeping those three arrows separate is essential: finiteness of the profile carrier does not itself construct a corridor, and equality of two codes does not itself construct the spliced planar map.

1.1 A status-local dependency chain

The following table uses "proved" only for an argument actually supplied in the cited layer. A source assertion, a finite audit still to be run, and a conditional theorem are different statuses.

There are now two logically distinct spines. The direct Count spine is the short route advertised by the compositional proof note and controls the headline. The older Kauffman–WC* development is an alternative source branch; progress there does not discharge a direct-Count premise.

Node	Direct Count spine	Present status
D_0	Tait reduction and least-counterexample normal form	the genuine spherical headline, least-counterexample selection, semantic two-sidedness, simple-graph backing, reverse Tait construction, cyclic five-edge-connectivity, and minimum face length five are machine-checked. Only the standard external bridge from abstract planarity to a spherical rotation presentation remains named conditional
D_1	Every sufficiently large least obstruction supplies a target-forced composable decomposition	open and the highest structural risk. The manuscript's linear-corridor assertion remains unsupported; in the exact tree reformulation the missing global statement is a uniform branch-width bound, equivalently exclusion of one fixed wall from least obstructions. The generic finite-tree descent, conversion of connected edge shores to connected vertex shores without width loss, the structural physical splice, and the mesh obstruction are machine-checked. A boundary-separated three-interface source rail segment now supplies a canonically oriented saturated cut, connected edge shores, and a nonempty proper majority core. When that source trail is obtained by deleting its distinguished edge from a closed cubic map, the same vertex partition now supplies connected complementary edge shores and a nonempty proper majority core in the closed ambient map as well. The standard rooted branch-decomposition interface and its conversion to the literal checked consumer are machine-checked. The Fomin–Fraigniaud–Thilikos width-preserving connectedization theorem is now machine-checked in that interface. Wall exclusion and the arbitrary-length nested exact finite-state supply are not proved
D_2	Count across every actual ordered noose cut factors through a finite boundary-word relation, and cumulative support is deletable on repetition	the physical port-tangle gluing bijection, matrix law, support determinization, exact 2^{3^k} support bound, repeated-block deletion, and relabelling invariance are machine-checked. The semantic closed composite is also proved equivalent to Tait-colourability of the literal sewn rotation system. The specialized Cell–3 closure is now an optional state-minimization and finite-base instrument, not a correctness premise of the direct descent
D_3	Equal cumulative rooted states admit a strictly	the ordinary noose construction

Node	Alternative WC*/Lift spine	Present status
W_1	Kauffman closed-cubic parity and the proposed deleted-edge prime/factored trail start	parity is proved, but the trail start is refuted by the two-defect boundary-flux identity; this branch cannot presently start
W_2	Meyniel rotor, digon, two-cut, and square transport	substantial ordinary and Lean components are proved; integration is incomplete
W_3	Pentagon Lift for the realized good-state move graph	the four-block-edge criterion is proved below; one seed in every fibre is a strictly stronger condition, false for arbitrary annuli and open in normal form
W_4	Route-specific square transport for WC*	the route-specific proof is written; arbitrary-target transport, $A \neq \varnothing$, and universal SQ3 connectivity have refutation fixtures

Consequently the route is a conditional proof, not yet a proof of the Four-Color Theorem. The point of the remainder of this section is that the conditions are now explicit enough to attack independently and cannot be blurred by prose.

The formal M0 boundary now has one exact input: the compositional core must exclude graph-backed vertex-minimal non-Tait-colourable spherical maps. From that proposition, least-counterexample selection, two-sidedness, graph backing, Tait colourability of every bridgeless spherical cubic map, and the spherical Four-Colour headline are all checked. VERIFIED VERIFIED The refuted legacy `IsPlanar` predicate and the refuted closed-web headline are not imported anywhere in this chain.

1.2 Entering the normal form without the v23 degree-four slip

The original v23 proof of minimum dual degree contains a small but real error: from "a deleted vertex has at most four neighbours" it concludes that four colours leave an unused colour. This is false when four neighbours use all four colours. The current route does not consume that sentence.

Start instead with a vertex-minimal uncolourable connected spherical cubic map, as supplied by the Tait bridge below. A cyclic cut of size two, three, or four splits the map into two smaller open cubic pieces. Cap both pieces, colour the smaller closed maps by minimality, and compare the colours on the cut. For two and three ports the boundary pattern is forced up to a colour permutation. At four ports the Klein-group boundary sum is zero, leaving the all-equal class and the two non-crossing adjacent-pair classes; if one side is all-equal, a planar Kempe escape moves it into an adjacent class. Thus the two sides always admit compatible boundary classes and glue to a Tait colouring, a contradiction. The least counterexample is therefore cyclically five-edge-connected.

This complete cut-and-glue argument is machine-checked. VERIFIED A facial cycle of length below five would expose such a small cyclic cut (the finitely small exceptional carriers are directly colourable), so every face has length at least five; this consequence is checked separately. VERIFIED Hence the curvature calculation below is entered through a sound normal form, not through the erroneous four-neighbour extension.

1.3 The abstract reductive pumping theorem

We isolate the exact ordinary-mathematics theorem used by either a direct zero-Count target or a Lift/Seed target once its physical cut chain is given.

Theorem 1 (Finite-profile reductive pumping)**.** VERIFIED *Let $\mathcal{T}$ be a class of finite open instances with a size $|\cdot|\colon \mathcal{T} \to \mathbb{N}$ and a predicate P. Assume:*

1. *every $\neg P$ instance having a ladder-reducible local feature has a strictly smaller $\neg P$ reduction in $\mathcal{T}$;*
2. *there are constants V_0, N such that every irreducible $X \in \mathcal{T}$ with $|X| > V_0$ has a corridor with more than N* pairwise suitably separated *admissible cut positions;*
3. *every cut has a profile in a set Q with $|Q| \leq N$;*
4. *whenever two suitably separated cuts have the same profile, deleting the material strictly between them constructs $X' \in \mathcal{T}$ with $|X'| < |X|$ and*

$$\neg P(X) \Longrightarrow \neg P(X').$$

Then every $\neg P$ instance descends to one of size at most V_0. If the finite base $\{X \in \mathcal{T} : |X| \leq V_0\}$ contains no $\neg P$ instance, then P holds throughout $\mathcal{T}$.

Proof. Suppose $\neg P$ occurs and choose such an X of least size. Hypothesis 1 makes X irreducible. If $|X| > V_0$, Hypothesis 2 supplies more than N pairwise suitably separated cut positions. Their profiles lie in Q, so two are equal by the pigeonhole principle. This pair satisfies the separation premise of Hypothesis 4, which then constructs a strictly smaller $\neg P$ instance, contrary to minimality. Hence $|X| \leq V_0$. The final assertion is immediate from emptiness of the base. □

This theorem is elementary; that is a virtue. It shows exactly where the difficult mathematics lives. The finite Count functor supplies Hypothesis 3 and helps formulate Hypothesis 4. It does not supply Hypothesis 2, and a quotient-code equality supplies Hypothesis 4 only after a physical splice theorem proves that the decoded pieces really glue.

1.4 What self-loops do and do not buy

The source's L2 flag is a self-loop property on relevant profiles. It gives a useful monotonicity theorem in a genuinely periodic corridor, but it does not erase the need to locate that periodic geometry.

Lemma 1 (One-rung padding)**.** VERIFIED *Let S be one fixed corridor rung with nonnegative transfer matrix M on a finite profile set. Let A, B be fixed end pieces with boundary vectors v, u. Suppose every profile occurring on an accepted path through one copy of S has a positive diagonal entry in M. Then*

$$v^{\mathsf{T}} M u > 0 \quad \Longrightarrow \quad v^{\mathsf{T}} M^2 u > 0.$$

Proof. Choose i, j with $v_i M_{ij} u_j > 0$. The profile i is alive on this accepted path, so $M_{ii} > 0$. The term $v_i M_{ii} M_{ij} u_j$ is a positive summand of $v^{\mathsf{T}} M^2 u$. □

Thus a minimal zero-count obstruction cannot contain two consecutive copies of the same physically removable rung, once the exact factorization and re-insertion theorem are available. For a corridor word $M_L \cdots M_1$ with varying letters, however, a self-loop for one local matrix does not show that deleting an arbitrary letter reconstructs the original instance. One must either find repeated physical rung data or use the equal-profile splice of Theorem 1. Accordingly profile cardinality has not been removed from the general source argument.

1.5 The curvature calculation and the missing global implication

For a connected cubic plane map whose face lengths are at least five, write p_k for the number of k-faces and

$$W_- := \sum_{k \geq 7} (k-6) p_k.$$

Euler's formula and double counting give

$$\sum_f (6 - |f|) = 12, \qquad \text{hence} \qquad p_5 = 12 + W_-. \tag{1}$$

This equality is machine-checked in the rotation-system development. VERIFIED

Equation (1) is a *signed* balance. It does not bound p_5, W_-, or the total number of nonhexagonal faces. For every m the formal face multiset

$$p_5 = m + 12, \qquad p_7 = m, \qquad p_k = 0 \ (k \neq 5, 7)$$

satisfies the same identity and has $2m + 12$ nonhexagonal faces and no hexagons. Lean records this precise arithmetic obstruction. VERIFIED

The same failure is realized by actual normal-form maps, rather than only by formal face multisets.

Proposition 1 (dodecahedral chains realize unbounded signed-defect cancellation)**.** PROSE *For every $s \geq 0$ there is a connected simple cellular spherical cubic map D_s with girth five and edge-connectivity three such that*

$$|V(D_s)| = 18s + 20, \quad |E(D_s)| = 27s + 30,$$
$$p_5(D_s) = 6s + 12, \qquad p_8(D_s) = 3s,$$

and D_s has no faces of any other length. Moreover, two disjoint untouched pentagonal faces may be designated as boundary faces. The resulting annular map has $6s + 10$ internal pentagons, $3s$ internal octagons, no internal hexagon, and

$$\sum_{f \text{ internal}} (6 - |f|) = 10$$

independently of s.

Proof. Let D be the dodecahedral map and choose opposite vertices a, b. Their incident face sets are disjoint: vertices on one pentagonal face have graph distance at most two, whereas $d_D(a, b) = 5$. Arrange $s + 1$ disjoint copies $D_0, \ldots, D_s$ in a row. At the ith join delete b from D_i and a from D_{i+1}, and match the three exposed neighbours in reverse cyclic order. Topologically this removes one open vertex-disc from each sphere and glues the two boundary circles orientation-reversingly, so the result is again a cellular map on the sphere. Every exposed neighbour loses one edge and gains one seam edge, hence the result remains cubic and simple.

Each join removes two vertices and six old edges and inserts three seam edges. It also pairs the six pentagonal face wedges at the deleted vertices into three faces. A pentagonal wedge left after deleting its vertex is a three-edge path; two such paths and the two intervening seam edges form an octagon. Since the a- and b-incident faces in an internal block are disjoint, no face is altered at two joins. Starting from $s+1$ copies, this gives

$$\begin{aligned} V &= 20(s+1) - 2s = 18s + 20, \\ E &= 30(s+1) - 3s = 27s + 30, \\ p_5 &= 12(s+1) - 6s = 6s + 12, \qquad p_8 = 3s. \end{aligned}$$

Thus $F = 9s + 12$, and both $3V = 2E$ and $V - E + F = 2$ hold.

For completeness, the local normal-form properties survive the join. The dodecahedral graph is 3-edge-connected. Restoring a deleted connector and placing it on the side containing a majority of its three neighbours turns any edge cut of size at most two in a three-edge sum into a cut of size at most two in one factor. That factor cut is trivial; iterating along the row shows that the original cut was trivial. Hence the sum is 3-edge-connected. A cycle lying in one block has length at least five. A cycle using a seam uses at least two seam edges; between two distinct neighbours of a deleted dodecahedral vertex every path has length at least three, since a shorter path together with that vertex would give a triangle or quadrilateral in D. Such a cycle therefore has length at least eight. This proves the girth assertion.

The dodecahedron has a Tait colouring. At a deleted cubic vertex its three incident colours are distinct. Before each join, globally permute the three colour names on the new block so that corresponding exposed half-edges have the same colour, and give the new seam edge that common colour. Induction therefore also gives a Tait colouring of every D_s.

At the two free ends choose a pentagon incident with the unused connector vertex. These two faces are disjoint and were not altered by a join. Declaring their interiors to be the two holes removes two pentagons from the spherical census. The internal curvature is consequently

$$(6s + 10)(6 - 5) + (3s)(6 - 8) = 10,$$

while all $9s + 10$ internal faces are nonhexagonal. This completes the construction. □

The census arithmetic is independently checked in Lean: VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED

Proposition 1 refutes precisely the geometry-only implication used in the source L1 and L6: fixed signed excess, girth five, and absence of 1- or 2-edge cuts do not force even one hexagonal face, much less a long hexagonal ladder. The family is Tait-colourable, so it does *not* refute a target-aware M1 theorem for a least zero-Count obstruction. It shows why that theorem must use zero Count and minimality essentially rather than repeating the signed-defect argument.

The annular and framed Excess Identities have the same sign issue, with boundary surplus terms added. They bound a difference of positive and negative curvature resources, not their sum. Therefore the sentence in Addendum XXV saying that the Excess Identity makes the nonhexagonal defect budget independent of V is not a consequence of that identity.

The tempting repair “add a uniform negative-curvature bound” is not a second proof obligation. It is only one sufficient route to the actual target-forced cut-chain theorem stated below. No such bound follows from the signed identity, and the proof should not spend effort manufacturing a stronger hypothesis than it consumes. The existing weighted corridor theorems correctly retain the negative-curvature weight or a caller-supplied pentagon bound; they do not manufacture a uniform constant. The live global question is therefore stated once, as Proposition 5. This is the highest mathematical risk in the route.

The strongest existing all-face extraction theorem is exact but graph-relative. If $W = W_-(M)$ and a positive corridor length L is fixed, it proves

$$|F(M)| \leq (W+7)^{(((12+2W)(W+7)+1)L)-1} \quad \text{or} \quad M \text{ contains a radius-one clean hexagonal geodesic block of length } L$$

VERIFIED The first alternative is not a finite base: its right-hand side contains the unbounded parameter $W(M)$ of the very graph being bounded. For example, allowing $W \geq |F|$ makes the inequality compatible with arbitrarily large $|F|$. Thus this theorem is a correct conditional corridor extractor, not a proof of a graph-independent threshold V_0.

The uniform weight bound is sufficient, but it is stronger than the compositional architecture actually needs. The following elementary dichotomy removes curvature from the global existence question.

Theorem 2 (bounded-face geodesic or a long face)**.** *VERIFIED* *Let M be a connected cellular spherical map with two-sided simple face boundaries and connected facial dual. Fix integers $B \geq 2$ and $L \geq 2$. If the number of faces is greater than*

$$1 + B\sum_{i=0}^{L-2}(B-1)^i,$$

then either M has a face of length greater than B, or its facial dual has a geodesic of at least L edges. In the latter case the geodesic is an induced face path, and every central face on it has length at most B.

Proof. Assume every face has length at most B. Assign to each neighbouring face a shared boundary edge. Since a primal edge is incident with at most two quotient faces, this assignment is injective, so the facial dual has maximum degree at most B. If its diameter were at most $L-1$, breadth-first layers about any root would contain at most 1 vertex at level zero, at most B at level one, and at most $B(B-1)^{i-1}$ at level $i \geq 1$: after the first step a shortest path cannot immediately return along the edge just used. Summing through level $L-1$ gives the displayed upper bound, contrary to the face count. Thus two dual vertices have distance at least L; a shortest path between them is the required geodesic. Any adjacency between two nonconsecutive vertices of that path would shorten it, so the path is induced. □

For a connected cubic spherical map, $3V = 2E$ and Euler's equality give $F = V/2 + 2$. Thus the face-count hypotheses in the theorem and corollary are effective vertex-count thresholds. In particular, the dichotomy applies to every sufficiently large member of the cubic normal form; it is not a statement about maps separately assumed to have many faces.

Corollary 1 (avoiding finitely many marked faces)**.** *VERIFIED* *Under the bounded-face branch of Theorem 2, let S be a set of h forbidden faces and put*

$$C = 1 + h(B-1), \qquad M = 1 + B\sum_{i=0}^{L-2}(B-1)^i.$$

If the facial dual has more than $h + CM$ vertices, the induced facial dual on the complement of S contains a geodesic of at least L edges. We call this an S-avoiding geodesic; no claim is made that it is geodesic in the undeleted facial dual.

Proof. Deleting one vertex of degree at most B from a connected graph increases the number of components by at most $B-1$. Deleting all h forbidden vertices therefore leaves at most C components. More than CM vertices remain, so one component has more than M vertices. Its maximum degree is still at most B; the breadth-first estimate in the preceding proof gives a geodesic of length at least L in that component, hence an S-avoiding geodesic in the stated sense. □

Both the underlying bounded-degree graph statement and its quotient-face map adapter—including the sharp component count $1 + h(B-1)$, the Moore threshold, and geodesicity in the deleted graph—are machine-checked. VERIFIED The verified theorem does not silently assert that the resulting path is geodesic before deletion.

Thus, within the bounded-face branch, the cap faces and any fixed collection of target or collar faces can be put into S at an explicit multiplicative cost. Bounded-degree dual connectivity gives this finite avoidance directly; it does not require an unsigned defect bound.

Consequently the bounded-face branch is a genuine chain of cells drawn from a finite alphabet depending only on B. A dual step records its chosen primal edge incidence; each central face has at most B such choices, and the geodesic has no nonconsecutive face adjacency. This conclusion does *not* require the stronger, presently unused assertion that two faces share at most one edge. The existing hexagonal Cell is the $B = 6$, all-central-faces-hexagonal subalphabet, not the only possible finite alphabet.

There is a useful local bound here, but it must not be confused with a bound on the complete transfer interface.

Lemma 2 (S-avoiding geodesic locality)**.** *VERIFIED* *Let S contain at most h faces and let $F_0, F_1, \ldots, F_L$ be a geodesic in the facial dual with S deleted. If a face $H \notin S$ is adjacent in the original dual to both F_i and F_j, then $|i-j| \leq 2$.*

More precisely, call an unmarked face a crossing side face at the cut after i if it is adjacent to some F_a with $a \leq i$ and to some F_b with $i+1 \leq b \leq L$. Retain the complete boundaries of (i) every unmarked neighbour of the four-face window $F_{i-1}, F_i, F_{i+1}, F_{i+2}$ (omitting nonexistent end terms), (ii) the window faces themselves, and (iii) all marked faces. Every crossing side face is retained. If all faces have length at most B, this radius-one carrier has cost at most

$$4B^2 + 4B + hB$$

darts. In particular the bound is independent of L.

Proof. Because $H \notin S$, the two dual edges F_iH and HF_j survive the deletion and form a walk of length two there. Geodesicity in the deleted dual gives $|i-j| = d_{D-S}(F_i, F_j) \leq 2$. Therefore any unmarked side face crossing the cut is adjacent to one of $F_{i-1}, F_i, F_{i+1}, F_{i+2}$ (omitting nonexistent end terms). Those four axis faces have together at most $4B$ boundary incidences, hence at most $4B$ distinct unmarked neighbours. Storing their complete boundaries and the axis boundaries costs at most $4B^2 + 4B$. A marked face need not obey the locality conclusion, but there are at most h of them and retaining each whole boundary costs at most B further darts. Adding these contributions gives $4B^2 + 4B + hB$. □

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Lemma 2 bounds the faces directly incident with the axis near one cut. It does *not* yet bound the complete interface to the unprocessed graph. Components outside the radius-one slab may join side faces met at many different axis positions; a transfer state that remembers those pending connections can therefore grow with the prefix. Proving that this cannot happen in a least Seed obstruction, or turning a deep family of such connections into a second pumpable direction, is a separate theorem. Consequently the lemma does not by itself terminate the receipt-enlargement process.

This distinction is visible even in defect-free geometry. For the Goldberg family obtained by subdividing every icosahedral triangle with frequency k and dualizing, one has twelve pentagons and only hexagons, so $W_- = 0$ while the number of vertices tends to infinity. A direct combinatorial audit of the generated maps gives the following cap-distance cut sizes. Here B_d is the set of primal vertices at graph distance at most d from the five vertices of one pentagonal cap, and δB_d is its edge boundary:

k	1	2	3	4	5	6	7	8	9	10	11	12
$\lvert V\rvert$	20	80	180	320	500	720	980	1280	1620	2000	2420	2880
$\max \lvert\delta B_d\rvert$	10	20	30	40	50	60	70	80	90	100	110	120.

This is numerical evidence, not a theorem about every possible cut. Its logical force is narrower and important: an arbitrarily large clean hexagonal region is not automatically a one-dimensional bounded-interface ladder. The deterministic generator and all per-radius arrays are in `tools/fourcolor/v24_goldberg_frontier_width_audit.py` and its JSON certificate.

1.6 The cap-composed Menu predicate

The terminal predicate must be fixed before any finite closure is measured. The source does not count bichromatic components in the open tangle alone. It restores the pentagonal cap and then counts the components obtained by composing the open strand pairing with the cap pairing. These are different objects.

Definition 1 (cap-composed pair relation)**.** VERIFIED Let w be a good inner word, let x be a proper colouring of the five cap cycle edges extending w, and fix a majority pair (p, q). Let $P_{p,q} \subseteq \mathbb{Z}/5\mathbb{Z}$ be the four inner positions whose spokes have colour p or q. On $P_{p,q}$ put two relations:

$$i \, \pi_{\text{str}} \, j \iff \text{the two spokes lie in the same } (p, q)\text{-component of the open tangle,}$$
$$i \, \pi_{\text{cap}} \, j \iff i \text{ and } j \text{ are joined in the cap 5-cycle by edges of colours } p, q.$$

The *cap-composed pair relation* is the equivalence relation generated by $\pi_{\text{str}} \cup \pi_{\text{cap}}$. The state has menu B for (p, q) exactly when this relation has at least two classes on $P_{p,q}$.

Lemma 3 (finite pentagonal cap table)**.** *VERIFIED* *Normalize the majority colour to* 0*, the singleton colours to* $1, 2$*, and let* x_i *colour the cap edge from* i *to* $i + 1$*, with indices modulo five. For the ten good words, the unique cap extension and the two cap matchings are:*

w	x	$\pi_{\text{cap}}(0,1)$	$\pi_{\text{cap}}(0,2)$
00012	21201	03 | 12	01 | 24
00021	12102	01 | 24	03 | 12
00120	12012	01 | 24	04 | 13
00210	21021	04 | 13	01 | 24
01200	20121	04 | 13	02 | 34
02100	10212	02 | 34	04 | 13
10002	21210	03 | 12	14 | 23
12000	01212	02 | 34	14 | 23
20001	12120	14 | 23	03 | 12
21000	02121	14 | 23	02 | 34

Here $ab \mid cd$ denotes the perfect matching with blocks $\{a, b\}$ and $\{c, d\}$. Global colour permutation gives every unnormalized case.

Proof. Properness at cap vertex i says $\{w_i, x_{i-1}, x_i\} = \{0, 1, 2\}$. Thus any one cap-edge colour forces the next, and cyclic consistency leaves the single extension displayed in each row. For a majority pair, each of its four active cap vertices has exactly one cap edge in the pair. Tracing those selected edges gives the two displayed blocks. The ten rows exhaust the cyclic placements of a consecutive triple together with the order of the two singleton colours. □

Lemma 4 (literal cap restoration computes the composed relation)**.** VERIFIED *The classes in Definition 1 are exactly the (p, q)-components of the graph obtained by restoring the five cap edges, restricted to the four active cap vertices.*

Proof. Every (p, q)-walk in the restored graph alternates between maximal portions inside the open tangle and maximal portions inside the restored cap. Its successive active cap vertices are therefore related alternately by π_{str} and π_{cap}. Hence its endpoints lie in the equivalence closure of their union. Conversely, every generator of π_{str} is witnessed by a walk in the open tangle and every generator of π_{cap} by a walk in the cap. Concatenating the witnesses for a chain in the generated equivalence relation gives a (p, q)-walk in the restored graph. □

Lemma 5 (transverse-chord criterion)**.** VERIFIED *For either majority pair, let the two blocks of π_{cap} be $A \mid B$. Then the cap-restored state has menu B for that pair if and only if no pair joined by π_{str} has one endpoint in A and the other in B.*

Proof. The cap relation is a perfect matching on the four active positions, while the open-strand relation is a partial matching: a bichromatic component of the open tangle has at most two inner degree-one ends. If every strand pair stays within A or within B, each generator of $\langle \pi_{\text{cap}} \cup \pi_{\text{str}} \rangle$ preserves the partition $A \mid B$, so the generated relation has at least two classes. Conversely, one transverse strand joins a vertex of A to a vertex of B; the two cap pairs then connect all four active positions into one class. By Lemma 4, these are exactly the two menu cases. □

1.6.1 The actual Lift target is weaker than one seed per fibre

Write a normalized good word uniquely as

$$w(s; m, x, y)_{s,s+1,s+2,s+3,s+4} = (m, m, m, x, y), \qquad s \in \mathbb{Z}/5\mathbb{Z},$$

where (m, x, y) is an ordering of the three colours. The coordinate s is its *block*. Thus the thirty good words form five blocks of six. The singleton-pair switch and a majority-pair menu-A switch change only the ordering (m, x, y) and not s. A menu-B switch for (m, x) joins block s to $s + 1$, while a menu-B switch for (m, y) joins block s to $s - 1$. Equivalently, if v is the unique inactive singleton position for the chosen majority pair, the installed edge of the five-cycle of blocks is

$$e = v + 1 \pmod 5,$$

where edge e joins blocks e and $e + 1$.

Theorem 3 (four-block-edge Lift criterion)**.** VERIFIED *Let $\mathcal{C}$ be the good-word colouring states of one fixed annular tangle. Suppose:*

1. *every nonempty fixed-word fibre is connected by inner-avoiding Kempe moves;*

2. *in each of the five blocks, the nonempty realized word vertices form a connected subgraph under within-block moves; and*

3. *cap-composed menu-B moves realize at least four of the five adjacent edges of the block cycle.*

Then all states in $\mathcal{C}$ lie in one Kempe class. In particular the pentagon Lift step holds for this tangle.

Proof. Four edges of a five-cycle form a spanning path, so Hypotheses 2 and 3 make the realized word graph connected. Let two colouring states be given and choose a path between their words. At each word edge, Hypothesis 1 first moves within the current fibre to the state witnessing that edge; perform the corresponding cap-composed move, and continue in the next fibre. Hypothesis 1 finally reaches the prescribed target state. Thus the full state graph is connected. Restoring the uniquely coloured pentagonal cap identifies these states and moves with the colourings and Kempe moves in the frontier before the cap was opened, which is precisely the Lift conclusion. □

The later "one seed in every nonempty good fibre" formulation is therefore a sufficient strengthening, not the target consumed by the word-graph argument. A source-faithful finite audit should retain the majority-pair label of every seed and ask which of the five block edges it installs. The reproducible pair-resolved audit is `tools/fourcolor/v24_plantri_lift_bridge_audit.py`. This weaker criterion does not erase the normal-form issue below: the triangular witness realizes no block edge and the square witness realizes only edge 3.

On the normal-form generator class, the pair-resolved audit has so far checked all 568 connected annuli at $V = 24, 26, 28, 30$. Every one realizes at least four block-cycle edges; no failure occurs. The exact per-size counts, solver version, and scope warning are recorded in `results/fourcolor/v24_plantri_lift_bridge_audit_v24_30.json`. This is evidence for Hypothesis 3 of Theorem 3, not a proof of it or of the two within-fibre connectivity hypotheses.

Proposition 2 (the open-tangle implementation predicate is vacuous)**.** PROSE *For well-formed annular boundary data, a proper Tait colouring whose inner word has shape* $(3,1,1)$ *has at least two* open-tangle *bichromatic components meeting the inner boundary, for each of its two majority pairs. Consequently the present Lean predicates* `AnnularFrontierMenuBForPair` *and* `AnnularFrontierMenuBForPairOfProfile`*, which use only the open component relation (respectively its* `connectionTable` *encoding), are true on every realizable colouring/profile in every nonempty good fibre and are not the source's menu-B predicate.*

Proof. Fix one majority pair. Exactly four inner spokes carry one of its two colours. Their four boundary vertices are distinct and have degree one in the corresponding bichromatic support graph. Every vertex of that support graph has degree at most two: an interior cubic Tait vertex has exactly the two selected colours, while a boundary vertex has ambient degree one. A finite connected graph of maximum degree two has at most two degree-one vertices. Thus one component cannot contain all four active inner stubs; at least two components meet the inner boundary. The same argument applies to the other majority pair. □

This is a semantic correction, not a refutation of the Seed Lemma. It does mean that the current Lean seed count and its profile-factorization theorems count the whole good fibre rather than the source's cap-composed seeds, so they cannot discharge the Seed Lemma as presently named. The rolling engine still computes useful data: it already retains the open strand relation π_{str}, while π_{cap} is a five-port finite function of the fixed boundary word and its cap extension. The repair is therefore to add the finite equivalence closure of Definition 1 to terminal acceptance. Lemma 4 now verifies that this finite closure agrees pointwise, on active ports, with reachability in the literally restored graph; the repair is not a reason to discard the one-cell factorization.

Proposition 3 (cap-correct terminal acceptance is profile-local)**.** VERIFIED VERIFIED *For an actual Tait colouring of the open annular tangle, let the finite uniform profile record its inner boundary word and, for each selected colour pair, the Boolean relation saying which active ports lie in the same open component. The source's Menu-B test after literally restoring the pentagonal cap is a predicate of this profile alone. More precisely, it is the cap-composed predicate of Definition 1; consequently two colourings with equal uniform profiles either both pass the restored-cap terminal test or both fail it.*

Proof. For a majority pair the active support has four of the five cap positions, so its complement is the unique inactive position. A true entry of the profile's component table joins two active positions. Apply Lemma 4 to identify restored reachability with the equivalence closure of the open component relation and the cap pairing. This description uses only the two stored profile fields, and hence is invariant under profile equality. ☐

Corollary 2 (cap-correct Seed count factors through finite profiles)**.** VERIFIED VERIFIED *Define the cap-correct Seed count at a fixed word to be the number of proper Tait colourings in that fibre which pass the restored-cap predicate of Proposition 3. This count is positive if and only if some realized finite profile at the word has positive profile count and satisfies the cap-composed terminal predicate. Equivalently, it is zero if and only if every cap-composed accepting profile at that word has literal profile count zero.*

Proof. Partition the finite Tait-colouring fibre by its uniform profile. Terminal acceptance is constant on each part by Proposition 3; taking cardinalities gives the positive form, and negating it gives the zero form. ☐

Remark 1 (cap-correct Seed audit)**.** PROSE *The Seed conclusion has now been tested with the inner cap restored before Kempe components are counted, as required by the Reformulation Theorem. For every pair of vertex-disjoint pentagonal faces in the three source laboratories, exhaustive enumeration of the literal five-by-five boundary data gives*

	pairs	*disconnected*	*good fibres*	*good states*	*menu-B states*	*seedless fibres*
C_{24}	42	24	1260	10080	9036	0
C_{30}	46	15	1380	23400	17340	0
C_{40}	46	10	1380	118320	89580	0
total	134	49	4020	151800	115956	0.

Here "disconnected" means that opening the two cap cycles isolates one edge joining an inner stub to an outer stub. These rows are retained as a boundary-data regression test because connectivity is absent from `AnnularBoundaryData.WellFormed`*; they are not controlling source instances, since the full* `ClosedWebAnnularEmbedding` *contains the hypothesis that the graph is connected. On the* 85 *connected source instances the audit has* 2550 *good fibres,* 114840 *good states,* 84066 *menu-B states, and no seedless fibre. The polar cases reproduce the manuscript's published menu-B counts* 228/240*,* 390/600*, and* 1680/3120*, which is a fidelity check on the operational definition. The deterministic audit is* `tools/fourcolor/v24_seed_lemma_census.py`*.*

This is genuine evidence for the Seed Lemma, but it is not its proof: these are three finite fullerene families and all have five outer stubs. It shows that a valid global corridor theorem may profitably use the seedless hypothesis; the manuscript's present derivation still does not construct the required bounded-interface chain from it.

Wide Goldberg audit. The same cap-restored predicate has also been encoded directly in SAT on the icosahedral Goldberg family, with a second literal component traversal used to replay every satisfying witness. On $\mathrm{GP}(1,0)$ (20 vertices), all 72 ordered vertex-disjoint cap annuli and all ten good-word representatives modulo global colour permutation are nonempty and seeded: 720/720. On $\mathrm{GP}(2,0)$ (80 vertices), the corresponding result is 1320/1320 over 132 ordered annuli. A maximally separated cap pair of $\mathrm{GP}(3,0)$ (180 vertices, cap-vertex distance 15) is seeded for all ten representatives. The exact constraints, graph hashes, and results are generated by `tools/fourcolor/v24_goldberg_seed_sat.py`.

Proposition 4 (the unrestricted Seed statement is false)**.** REFUTED *Nonemptiness of a good fibre does not imply a cap-composed menu-B seed for an arbitrary connected annular cubic tangle.*

Proof. There is a connected 14-vertex plane cubic map with a pentagonal face disjoint from a triangular face such that opening those two faces gives nonempty fibres at the normalized good words 00012 and 00021, while both fibres contain zero cap-composed seeds. Its rotation-system hash is

`96668268249e8043817aa0c8c1fff8adf23f9445fa9849a85c37de6654caa449`.

The phenomenon is not removed merely by forbidding triangles: a connected 18-vertex example of minimum face length four, opened at a pentagon and a square, has six nonempty normalized good fibres and is seedless at 10002 and 20001. Its hash is

`6fb5266b12bbf4e9c395e6ab6ea1a8ebf6dde039b48de5e0f97a67443865e039`.

In both cases SAT supplies proper colourings and an independent literal traversal restores the cap and recomputes the two majority-pair component relations. The complete embeddings, face cycles, generator hash, and fibre lists are recorded in `results/fourcolor/v24_plantri_seed_scope_triangle.json` and `results/fourcolor/v24_plantri_seed_scope_square.json`. □

Thus the hypothesis silently supplied by Step 1 of Addendum XXV is load-bearing. The route needs the Seed conclusion only for a frontier obtained from the cyclically five-edge-connected, face-length-at-least-five normal form of a least counterexample. That is the theorem which Proposition 5 now targets. Alternatively, one could retain the manuscript's unrestricted statement only after proving that triangle/digon/two-cut/square reduction transports the *cap-composed* zero count. The current rolling square identities use the open-tangle predicate of Proposition 2, so they do not provide that proof.

Exhaustive small normal-form audit. An independent generator audit uses `plantri -a -d -m5 -c5` to enumerate every generated simple cubic spherical map of face length at least five and cyclic edge-connectivity at least five through 34 primal vertices. Across 45 maps it opens every ordered pair consisting of a pentagonal inner face and an arbitrary vertex-disjoint outer face of length 5 through 8. The totals are

V	maps	boundary openings	connected annuli	seeded/nonempty good fibres
20	1	72	12	720/720
22	0	0	0	0/0
24	1	96	36	960/960
26	1	108	48	1080/1080
28	3	380	190	3800/3800
30	4	539	294	5390/5390
32	12	1872	1092	18720/18720
34	23	3978	2448	39780/39780
total	45	7045	4120	70450/70450.

On the connected source-annulus subset the corresponding total is 41200/41200. Every SAT seed witness is replayed by an independent literal cap-restored component traversal. The generator hash, solver version, aggregate counts, and explicit warning that this finite absence is not a universal proof are recorded in `results/fourcolor/v24_plantri_seed_audit_v2_20_32.json` and `results/fourcolor/v24_plantri_seed_audit_v2_34.json`; the reproducible driver is `tools/fourcolor/v24_plantri_seed_audit.py`. This substantially strengthens falsification coverage while remaining a test, not a replacement for Proposition 5.

At $V = 36$, a connected-only run over all 71 generated maps adds 8721 connected annuli and 87210/87210 seeded/nonempty normalized good fibres, with no seedless witness. Thus the cumulative connected audit through 36 vertices is 128410/128410 fibres over 12841 annuli. This longer run is recorded separately in `results/fourcolor/v24_plantri_seed_audit_v36_connected.json` because disconnected openings were connectivity-filtered before SAT rather than included in the displayed all-opening table.

These data are deliberately not promoted to a theorem. They do, however, separate the two phenomena which L1 had conflated: the same Goldberg family has growing radial frontier width, while its tested good fibres have seeds. Thus a successful global argument should use the assumption that the target count is zero; a geometry-only claim that every large hexagonal patch is a fixed-width tube is both stronger than needed and contradicted by the radial audit.

Proposition 5 (target-forced compositional cut chain)**.** OPEN *For each fixed boundary and acceptance type of the chosen direct-Count or Lift system, there are computable constants b, m, V_0 with the following property. Every ladder-irreducible target counterexample of size greater than V_0 either has a strictly smaller target counterexample, or contains more than m pairwise nested simple ordered transversals, each meeting at most b graph edges, such that:*

1. *consecutive transversals bound composable slabs containing at least one graph vertex;*
2. *the designated holes and frozen boundary lie outside the slabs;*
3. *every transversal has an exact saturated deletion side and a cycle on each complementary side; and*
4. *all transversals carry one common finite ordered geometric seam type, and cumulative target Count factors exactly through the associated boundary state.*

Here m is at least the cardinality of the finite cumulative profile carrier at width b.

This proposition is the precise L1–splice junction. It deliberately states the geometric outputs which the existing Lean interfaces consume: a simple transversal, its crossed-edge set, an exact deletion component boundary, two cyclic sides, and the correspondence used for surgery. A chord wall, an induced dual path, or a bounded local collar supplies only part of this data.

The following checked lemma supplies one of those outputs for a literal finite segment of the source corridor. It is useful precisely because its hypotheses expose, rather than suppress, the remaining global geometry.

Lemma 6 (a boundary-separated source rail segment supplies connected shores)**.** *VERIFIED* *Let three consecutive transverse source interfaces be joined on their two sides by the literal facial-dual rails of the intervening corridor cells. Assume that each composed rail is a simple path, that the two rail supports are disjoint, that every face on either rail is an internal annular face, and that neither rail visits the centre face of either endpoint interface. Close the two rails by the transverse two-edge paths at the endpoint interfaces, and let W be the resulting closed facial-dual walk. If $X(W)$ is the set of primal edges crossed by W, then W is a simple dual cycle and some connected component of $G - X(W)$ has graph boundary exactly $X(W)$. Both this component side and its complementary vertex side induce connected subgraphs. Moreover, if every edge incident with the component side is assigned to one edge shore, then that edge shore and its complementary edge shore are connected in the edge-walk sense used by the majority-shore descent.*

Proof. Split W into the path formed by the first endpoint transversal followed by one rail, and the path formed by the reverse last transversal followed by the reverse other rail. Simplicity of the rails, avoidance of the two centre faces, and mutual rail disjointness imply that the two path tails are disjoint. Appending them therefore gives a simple cycle. This first step is checked as VERIFIED.

For completeness, the saturation argument is not a Jordan-curve picture. The framed dual-cycle separator first proves that deleting $X(W)$ disconnects the primal graph and supplies a component with nonempty computed boundary. Every computed boundary edge lies in $X(W)$. Conversely, two-sided face parity around the simple dual cycle says that a nonempty cut contained in $X(W)$ must contain every crossed edge. Hence the inclusion is an equality. The generic form, independent of the corridor realization, is checked as VERIFIED.

The same saturation argument applies to every component of $G - X(W)$. Because $X(W)$ is nonempty, a component distinct from the chosen one also has nonempty boundary; parity therefore forces its boundary to be all of $X(W)$. Two distinct such components cannot coexist, since one crossed edge has only two endpoints. Thus the complement of the chosen component is connected. This generic strengthening is checked as VERIFIED.

Finally put in the chosen edge shore every ambient edge with at least one endpoint in the chosen component. A path in the component connects the component endpoints of any two such edges, and the two edges themselves attach their possible outside endpoints. Hence this edge shore is connected. Its complement consists exactly of the edges having both endpoints in the complementary vertex side, so connectivity of that induced side proves connectivity of the complementary edge shore. This construction is canonical and monotone under inclusion of vertex sides. Its generic form is checked as VERIFIED. □

Corollary 3 (the source rail segment has an oriented majority-shore certificate)**.** *VERIFIED* *Assume in addition that the source trail is well formed. Let C_{out} be the component of $G - X(W)$ containing the named outer-container root, let U_{in} be its vertex complement, and put in S_{in} every edge incident with U_{in}. Then*

$$\partial C_{\text{out}} = X(W), \qquad S_{\text{in}} \textit{ and } E(G) \setminus S_{\text{in}} \textit{ are connected edge shores,}$$

and the majority vertex side of S_{in} is nonempty and proper.

Proof. The completed boundary visits only internal faces: this is an explicit hypothesis on the two rails, while each endpoint transversal is internal by the boundary-clean source realization. Consequently $X(W)$ is disjoint from the outer-hole boundary. The named outer-container cycle therefore survives the deletion and lies in the component rooted at its own basepoint. This fixes the annular orientation without choosing a component after the fact. These two steps are checked as VERIFIED and VERIFIED.

The prescribed-component form of the dual-cycle separator now gives the exact boundary and connectedness of both oriented vertex sides: VERIFIED and VERIFIED. The incident-edge construction from the preceding lemma then gives the two connected edge shores.

It remains to check that "majority" is not vacuous. A simple dual cycle has length at least three and, by unique shared-edge incidence, crosses that many distinct primal edges. Choose two. If their endpoints on U_{in} coincide, that vertex already has two incident edges; otherwise a path in the connected inner side starts with an internal edge which, together with one crossing edge, gives the same conclusion. Hence some inner vertex has at least two incident edges, all in S_{in}, and is a majority vertex.

For the other direction, every well-formed source-trail vertex has degree at most three: ordinary internal vertices have degree three, the two defects degree two, and frozen stubs degree one. At a vertex of the surviving outer cycle, two distinct incident cycle edges lie outside S_{in}; at most one incident edge can lie inside it, so that vertex is not a majority vertex. The local degree and subcubic steps are checked as VERIFIED, VERIFIED, and VERIFIED. □

Corollary 4 (the oriented source shore survives closure of the missing edge)**.** *VERIFIED* *Let the source graph in Corollary 3 be $H = G - e$, where $e = uv$ is an edge of a finite cubic ambient graph G. Use the same outer-root component C_{out}, the same vertex partition*

$$U_{\mathrm{in}} = V(G) \setminus C_{\mathrm{out}}, \qquad U_{\mathrm{out}} = C_{\mathrm{out}},$$

but recompute the edge shore in the ambient graph by

$$S_{\mathrm{in}}^{G} = \{f \in E(G) : f \text{ has an endpoint in } U_{\mathrm{in}}\}.$$

Then $G[U_{\mathrm{in}}]$ and $G[U_{\mathrm{out}}]$ are connected, S_{in}^{G} and its edge complement are connected, and the majority vertex side of S_{in}^{G} is nonempty and proper. The exact source equality $\partial_H C_{\mathrm{out}} = X(W)$ remains available as the provenance of the partition.

Proof. Restoring e only adds an adjacency, so connectedness of both induced vertex sides in H implies their connectedness in G. The generic incident-edge-shore construction therefore gives connectedness of S_{in}^{G} and its complement. The inner witness from the source certificate remains an inner vertex, and cubicity of G makes it a majority vertex for S_{in}^{G}. Likewise the surviving outer-container cycle in H is still a cycle in G; two cycle edges at one of its vertices witness failure of majority on the outer side.

It is important that the edge shore is recomputed in G. The restored edge e is allowed to cross the vertex partition, so the ambient cut-edge set need not equal the source set $X(W)$. No noncrossing premise is being silently added. □

These results are not yet Proposition 5. They now construct, for one three-interface segment, a saturated boundary with a canonical outer-root orientation, complementary connected edge shores, and a nonempty proper majority core, and transport that oriented partition to the closed ambient cubic map. The internal-support premise is load-bearing: rail disjointness alone would not prevent the completed boundary from running through a cap.

It would be incorrect to obtain the required chain by fixing the left interface and composing these exterior rails to successively later right interfaces. The resulting closed dual walk crosses the two endpoint transversals and both longitudinal rails, so its number of crossed primal edges grows linearly with the distance between the endpoints. Such walks do not have the uniform width (b) required in Proposition 5.

The already constructed sliding local windows do not repair this defect by themselves. Their exact failure mode is checked rather than inferred from a picture.

Lemma 7 (the local-window seriality dichotomy)**.** *For two consecutive aligned two-tile source windows, the output transverse edge at each of the two positions is the input transverse edge of the next window. At each such edge, either the two enclosed-side boundary darts are α-opposite, as required for a literal serial seam, or the enclosed vertex carriers of the two windows share the base vertex of that dart. In the α-opposite branch, the common edge is internal after the two enclosed vertex sides are united, hence it is absent from the true crossing frontier of their union. Moreover, the two closed facial-dual window walks have intersecting support.*

Proof. Equality of the two transverse edge coordinates and the fact that every edge fibre consists of a dart and its α-mate give the dichotomy. If the displayed darts are equal, their common base vertex belongs to both enclosed carriers. These statements are checked as VERIFIED. For opposite darts the two dart bases are the two endpoints of the common edge, one in each enclosed side, so adjoining the sides absorbs that edge. The resulting exact alternative—absorption or a concrete shared vertex—is checked as VERIFIED. The closed windows also share the centre of their common transverse interface, so their supports are not disjoint; this is checked as VERIFIED. ☐

Lemma 8 (separated local rail supports grow linearly)**.** *Let S be a finite set of aligned two-tile window starts such that any two distinct starts differ by at least four positions. For each $i \in S$, let R_i be the two primal edges displayed by the two rail coordinates of the local six-port profile. Then the sets R_i are pairwise disjoint and*

$$\left|\bigcup_{i\in S} R_i\right| = 2|S|.$$

This is a statement about the displayed local rail carriers, not yet a claim that every such edge survives in the true crossing frontier after the enclosed vertex sides are united.

Proof. At starts separated by at least four positions, all four possible pairs of the two short facial-dual rail tracks have disjoint vertex support. In a two-sided rotation system, dual walks with disjoint support cross disjoint primal edge sets. Each local rail carrier has exactly two edges, so finite additivity gives the formula. The pairwise-disjointness and cardinality statement are checked as VERIFIED. ☐

Thus coordinate agreement of the rolling local profiles is not yet physical serial composition, and choosing one saturated component independently at each window does not supply a nested shore chain. The transverse portal is absorbed in the oriented branch, but Lemma 8 shows that the displayed lateral carriers themselves do not telescope: a separate theorem would have to absorb all but uniformly many of them from the true frontier of the united side. The remaining theorem is therefore target-forced, not a geometry-only source adapter: it must either prove precisely that additional absorption or construct, at each selected corridor position, a different individually bounded saturated ambient cut. It must prove that these cuts are simple and pairwise ordered, that their outer-root shores are strictly nested with nonempty intervening slabs, and that both holes and the frozen interface remain outside those slabs. It must also decide the α-seam branch of Lemma 7, transport the cuts to the ambient graph, prove connected complementary edge shores, nonempty majority sides, and a uniform middle-set bound, and identify the recomputed ambient shore boundaries with ordered seams. Lemma 10 then supplies the roots and ports, while the chain consumer computes the ordered Count states used by Theorem 14.

There is a second, equally compositional way to state the same structural risk. It replaces a linear corridor by a bounded-width decomposition tree. This is not asserted by the working manuscript, but it is a useful repair normal form because its generic half is elementary and its target-specific half has one unmistakable statement.

Theorem 4 (finite-interface pumping on a decomposition tree)**.** VERIFIED *Let X be an instance with a reduced rooted binary interface decomposition T. Suppose:*

1. *each node introduces at most c vertices, every vertex is introduced at exactly one node, and every proper ancestor–descendant segment contains some introduced material;*
2. *every node carries one of finitely many ordered interface types and an exact context state, with q typed states in total; and*
3. *whenever a proper ancestor u and descendant v have the same typed state, replacing the piece below u by the piece below v is a valid composition, strictly decreases size, and preserves the target truth value in the exterior context.*

If X is minimal among target counterexamples, then

$$|V(X)| \le c(2^q - 1).$$

Proof. No root-to-leaf path in T contains more than q nodes. Otherwise two nodes on that path have the same typed state. They are comparable, and Hypotheses 1 and 3 replace the higher piece by the lower one to produce a strictly smaller target counterexample, contrary to minimality.

A rooted binary tree whose longest root-to-leaf path contains at most q nodes has at most $1+2+\cdots+2^{q-1} = 2^q - 1$ nodes. Since every graph vertex is introduced once and a node introduces at most c vertices, the displayed bound follows. □

The tree theorem deliberately leaves its target-specific input visible. For the direct Tait target, the exact semantic part of that input is already a theorem.

Theorem 5 (closed Count depends only on the inner seam support)**.** VERIFIED *Let O be an outside port tangle and let I, I' be inside port tangles with the same ordered seam. If I and I' realize exactly the same seam-colour words, then*

$$O \circ I \text{ is not Tait-colourable} \quad \Longrightarrow \quad O \circ I' \text{ is not Tait-colourable.}$$

Proof. A colouring of $O \circ I$ is uniquely a seam word together with a colouring of O and a colouring of I realizing that word. Thus the closed composite is colourable exactly when the outer and inner supports intersect. Replacing the inner support by an equal set does not change this intersection. The Lean theorem proves the displayed implication; the same argument gives the converse as well. □

Theorem 6 (the semantic closed composite is the literal sewn map)**.** VERIFIED *Let L and R be finite literal open rotation tangles and let $m : B_L \simeq B_R$ match their complete boundary-dart interfaces. Regard L as a port tangle with empty input and output interface B_L, and regard R as a port tangle with input interface B_L, relabelled through m, and empty output. Then the closed serial composite used by* `ClosedColorable` *is colourable if and only if the literal rotation system* `composeRotationSystem` *is Tait-colourable.*

Proof. Write D_L, D_R for the intact dart carriers. After deleting the two empty external interfaces, the serial Count construction has interior dart carrier

$$(D_L \sqcup D_R) \sqcup (B_L \sqcup B_L),$$

whereas the literal splice has carrier

$$(D_L \sqcup D_R) \sqcup (B_L \sqcup B_R).$$

The required equivalence is the identity on intact darts and on the left seam half, and sends the right seam half b to $m(b)$. It preserves the base vertex of every dart. It also conjugates the serial edge flip to the literal matched-seam flip: old edges retain their old flips, while the serial pair (b_L, b_R) is sent to the literal pair $(b, m(b))$.

Consequently a proper nonzero serial dart-colouring pulls across this equivalence to a proper nonzero colouring invariant under the literal flip. Such a colouring descends to the two-dart edge orbits of the literal splice, giving a Tait edge-colouring. Conversely, a Tait edge-colouring of the literal splice colours each serial dart by the colour of the literal edge orbit containing its image. Flip-invariance is automatic. If two distinct serial darts based at one vertex received the same colour, their image edges would be distinct adjacent edges of the literal splice with the same colour, contradicting properness. This proves the biconditional. Notice that the vertex rotation is not used: this is exactly the representation-invariance needed to join Count semantics to the verified structural splice. □

Corollary 5 (support replacement preserves the physical zero Count)**.** VERIFIED *Fix a literal outside tangle L. Let R, R' be two literal inside tangles whose boundaries are both matched to the same ordered seam of L. If R and R' realize exactly the same genuine nonzero Tait cut words, then*

$$L \# R \text{ not Tait-colourable} \quad \Longrightarrow \quad L \# R' \text{ not Tait-colourable},$$

where both occurrences of # are the literal matched-seam rotation-system splice.

Proof. Use Theorem 6 to move the first literal splice to `ClosedColorable`. Every boundary word realized by a proper port colouring is pointwise nonzero, so equality of the finite sets of genuine Tait cut words is already equality of the raw seam supports. Apply Theorem 5, and use the same biconditional in the reverse direction for the second literal splice. □

The following is the precise finite-tree statement currently proved. It is a theorem about a supplied bounded-width decomposition, not yet a theorem constructing that supply from a plane map.

Theorem 7 (typed bounded-width descent)**.** VERIFIED *Fix a literal seam bound k. Suppose an instance X is supplied with a rooted binary decomposition whose nodes have states*

$$(j, s, S), \qquad 0 \le j \le k, \quad s \in \mathrm{SeamType}_j, \quad S \subseteq \{r, b, p\}^j,$$

and suppose:

1. $|V(X)|$ *is at most twice the number of labelled tree nodes; and*

2. *equal states at a proper ancestor and descendant produce a strictly smaller target instance.*

If X is size-minimal among target instances, then

$$|V(X)| \le 2(2^q - 1), \qquad q = \sum_{j=0}^{k} |\mathrm{SeamType}_j|\, 2^{3^j}.$$

Lemma 9 (cubic spacing forces material between two shores). VERIFIED *Let $A \subseteq B \subseteq E(G)$ be two edge shores of a finite cubic graph. For an edge set C, let*

$$\partial C = \{v : v \text{ is incident with an edge of } C \text{ and an edge outside } C\}.$$

If $|\partial A|, |\partial B| \le w$ and

$$|B \setminus A| > 6w,$$

then there is a vertex v incident with an edge of $B \setminus A$ such that every edge incident with v lies in $B \setminus A$. Thus v lies strictly in the material between the two shores, rather than on either middle set.

Proof. For every edge of $B \setminus A$, choose one endpoint. Suppose every chosen endpoint belonged to $\partial A \cup \partial B$. That union has at most $2w$ vertices, and cubicity allows at most three chosen edges at each vertex. Hence $|B \setminus A| \le 6w$, a contradiction. We may therefore choose an edge $e \in B \setminus A$ whose chosen endpoint v belongs to neither middle set.

Because $e \notin A$, if any edge of A met v, then $v \in \partial A$. Thus no edge at v lies in A. Because $e \in B$, if any edge outside B met v, then $v \in \partial B$. Thus every edge at v lies in B. Together these statements put every incident edge in $B \setminus A$, as required. □

Theorem 8 (majority shores from a connected edge decomposition). PROSE *Let G be a finite cubic graph and let $A \subseteq E(G)$. Write*

$$d_A(v) = |\{e \in A : v \in e\}|, \qquad K_A = \{v : d_A(v) \ge 2\}, \qquad M_A = \{v : 0 < d_A(v) < 3\}.$$

Suppose that the edge-induced graphs on A and on $\bar{A} = E(G) \setminus A$, with isolated vertices discarded, are connected. Then:

1. $K_{\bar{A}} = V(G) \setminus K_A$;

2. *if $K_A \neq \varnothing$, then $G[K_A]$ is connected, and likewise for $K_{\bar{A}}$;*

3. *the edge boundary of K_A has at most $|M_A|$ edges; and*

4. *if $A \subseteq B$, then $K_A \subseteq K_B$. Moreover, if every edge incident with v lies in $B \setminus A$, then $v \in K_B \setminus K_A$.*

Consequently, a connected branch decomposition of width w, restricted to non-leaf cuts for which both shores contain at least two edges, canonically supplies nested connected complementary vertex shores with at most w crossing edges. The shore is a function of the edge set alone; no independently chosen noose or perturbation is involved.

Proof. At every vertex cubicity partitions its three incident edges between A and $\bar{A}$, so

$$d_A(v) + d_{\bar{A}}(v) = 3.$$

Thus $d_{\bar{A}}(v) \geq 2$ exactly when $d_A(v) \leq 1$, proving the first claim. For the second, take $x, y \in K_A$. Connectedness of the edge-induced A-graph gives an A-edge walk from x to y. Erase repeated vertices to obtain a path. Its endpoints already lie in K_A, and every internal vertex is incident with the two distinct path edges adjacent to it. Both edges lie in A, so the whole path lies in K_A. This proves that $G[K_A]$ is connected. Apply the same argument to $\bar{A}$.

For the boundary bound, orient every edge of $\delta(K_A)$ from K_A to its complement. If the edge lies in A, send it to its outer endpoint. That endpoint has one incident A-edge but not two, hence belongs to M_A; and two different such edges cannot have the same endpoint, since that would give two incident A-edges there. If the edge lies in $\bar{A}$, send it to its inner endpoint. The inner endpoint has at least two incident A-edges and therefore, by cubicity, exactly one incident $\bar{A}$-edge; so this assignment is again injective. The images of the two cases are disjoint because the first case lands outside K_A and the second inside it. We have constructed an injection $\delta(K_A) \hookrightarrow M_A$.

Finally, $A \subseteq B$ implies $d_A(v) \leq d_B(v)$, proving monotonicity. If all three edges at v lie in $B \setminus A$, then $d_B(v) = 3$ and $d_A(v) = 0$, which proves the strict statement.

For a connected branch decomposition, M_A is its usual middle set and therefore has size at most w. An internal shore containing at least two edges has a vertex incident with two of them: otherwise its edge-induced graph would be disconnected. Hence both majority shores are nonempty on the stated cuts, and the preceding claims apply. Descendant edge shores are nested, so their majority shores are nested as well. □

The graph-local content of this theorem is machine-checked: VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED The remaining conditional in the final sentence is exactly the existence of the connected branch decomposition itself and its finite-tree packaging; it is not part of any of the six declarations just cited.

The conversion also feeds the physical splice without another geometric choice. Put $D_A = V(G) \setminus K_A$, open the ambient rotation system on K_A and D_A, and match the two boundary copies by the ambient edge flip. Suppose the ambient map is connected, bridgeless, spherical, cubic, has cyclic vertex rotations and two-sided faces; suppose both edge shores are connected and both majority shores are nonempty; and suppose the cut has two distinct boundary darts. Then the canonical first-return hub rotations on the two shores have opposite orientation, both capped shores satisfy the spherical Euler equality, and their literal composite is a connected, bridgeless, spherical cubic map with cyclic vertex rotations.

Indeed, Theorem 8 supplies the two connected vertex shores. The planar-bond rank theorem closes each shore by the inverse first-return hub rotation, the complementary-shore theorem proves that the two hub rotations are oppositely oriented under the ambient edge flip, and the seam face-count theorem proves sphericity of the composite. All other fields are transported directly through the literal splice. This entire implication is machine-checked as VERIFIED

The roots and the two distinct boundary darts in this construction are not additional decomposition data. They follow from connectedness, bridge-freeness, and cubicity.

Lemma 10 (connected shores supply a literal descent node)**.** *VERIFIED Let M be a graph-backed vertex-minimal non-Tait-colourable connected, bridgeless, spherical cubic rotation map. Let $A \subseteq E(M)$ have connected edge shore and connected edge complement, with both majority vertex sides nonempty. If the middle set of A has cardinality at most k and at most w, then A supplies a literal shore node of width at most k and middle bound w, including retained-dart roots on both sides and two distinct exterior boundary darts.*

Proof. Write K for either nonempty majority vertex side. Since its complement is also nonempty and the primal graph is connected, some edge crosses K. There cannot be exactly one such edge e: after deleting e, any path starting in K remains in K, because the first edge by which it left would be another boundary edge. This contradicts bridge-freeness, which says that the endpoints of e remain connected after e is deleted. Hence K has at least two boundary edges, equivalently at least two literal boundary darts; choose two distinct ones. This generic bridge argument is checked as VERIFIED and its boundary-dart consequence as VERIFIED.

Cubicity gives three darts at every vertex, so a chosen vertex on each nonempty side supplies its retained-dart root. For graph-backed rotation data, literal boundary darts are exactly the graph darts crossing the same vertex predicate. The injective charge from crossing darts to the edge-shore middle set in Theorem 8 therefore transfers the middle-set bound to the literal width. Packaging these choices gives the asserted node. □

Thus the remaining supplier problem is no longer "turn a branch cut into a spherical splice", choose roots, or prove that a proper shore has two ports. It is to construct and package the connected branch-decomposition cuts with strict laminar nesting and vertex accounting; the exact target-specific Count state is then computed by the literal node consumer.

Lemma 11 (equal phases are separated)**.** VERIFIED *If $i < j$ and $i \equiv j \pmod{s}$, then $s \le j - i$.*

Proof. The congruence says $s \mid (j - i)$. Since $j - i > 0$, a positive multiple of s is at least s. (When $s = 0$, the conclusion is immediate.) □

The strict-material clause in a sphere-cut supply can consequently be made finite-state rather than pictorial. Root the branch-decomposition tree and augment every typed state by the node depth modulo $6w + 1$. Equal augmented states at distinct comparable nodes are separated by at least $6w + 1$ tree edges by Lemma 11. At each step the discarded sibling subtree contains a leaf, and the sibling leaf sets at different steps are disjoint; hence the two nested edge shores differ by at least $6w + 1$ graph edges. Lemma 9 then supplies a strict slab vertex. The price is only the explicit finite factor

$$q_{\text{spaced}} = (6w + 1)q$$

VERIFIED

in the typed-state count. This settles material accounting once the two shore interfaces have been constructed; it does not itself construct their ordered open-tangle carriers or prove that the splice retains exactly those vertex shores.

There is an equivalent phase choice which is better adapted to the literal edge-shore carrier. If $A \subsetneq B$ are the descendant and ancestor edge shores, attach to the interface state of a shore S the residue

$$|S| \pmod{6w + 1}.$$

Theorem 9 (cardinality-phased shore descent)**.** VERIFIED *Let a reduced binary decomposition tree be labelled by finite edge shores, strictly decreasing along every descent, and suppose every middle set has size at most w. Let Q be any finite exact interface-state type and let $q(S) \in Q$ be the state of shore S. If equal Q-states admit a target-preserving local splice whenever the slab contains a vertex all of whose incident edges lie in the slab, then a minimal target instance has*

$$|V| \le 2\left(2^{(6w+1)|Q|} - 1\right).$$

Proof. Use the augmented state

$$\widetilde{q}(S) = \big(|S| \bmod (6w+1), q(S)\big).$$

Suppose comparable shores $A \subsetneq B$ have equal augmented states. Strict inclusion gives $|A| < |B|$, while equality of residues says that $6w+1$ divides $|B| - |A|$. Hence

$$6w + 1 \le |B| - |A| = |B \setminus A|.$$

Lemma 9 supplies a vertex lying strictly inside the slab. Equality of the second components gives $q(A) = q(B)$, so the assumed local splice produces a smaller target instance, contradicting minimality. Thus no descent repeats an augmented state. There are exactly $(6w+1)|Q|$ augmented states, and finite-tree pumping gives the displayed bound. ☐

This formulation has the same finite cost as depth phasing, but the phase is a function of the edge shore itself. Consequently the connected branch decomposition supplier need not carry or synchronize a separate node-depth annotation.

The colouring part of the exact state can now be stated without identifying the literal boundary-dart types of two different shores. Let T be an inner open tangle with boundary carrier B, let $|B| = k$, and choose a coordinate equivalence $c : B \simeq [k]$. Define

$$\mathrm{Supp}_k(T, c) = \{x \in \{1,2,3\}^{[k]} : x \text{ is realized by a proper Tait colouring of } T\},$$

where the port at coordinate i is $c^{-1}(i)$. This is an actual finite set, not a predicate retained in an ambient carrier.

Theorem 10 (normalized exact Count replacement)**.** VERIFIED *Let L be an outside open tangle whose seam is indexed by $[k]$, and let (T, c), (T', c') be inner open tangles with k-port coordinates. If*

$$\mathrm{Supp}_k(T, c) = \mathrm{Supp}_k(T', c'),$$

then replacing T by T' preserves zero Count, equivalently preserves non-Tait-colourability of the literal sewn rotation system.

Proof. After transport through c and c', both inner pieces have the same literal seam carrier $[k]$. Equality of the displayed finite sets is therefore equality of the genuine-word supports consumed by the closed Count replacement theorem. That equality is also equality of the raw colour-word supports, because every boundary letter of a proper colouring is nonzero. The gluing equivalence says that a closed composite is colourable exactly when the outside and inside supports intersect. Hence the two composites are colourable simultaneously. Finally, the physical/semantic bridge identifies this closed Count statement with Tait colourability of the literal sewn rotation systems. ☐

There are exactly

$$2^{3^k}$$

possible normalized supports VERIFIED. This is the raw logical state count; reachable-state minimization is only an optimization. The cyclic order is intentionally not forgotten. If ρ_T is the canonical hub rotation of T, put

$$\Sigma_k(T, c) = (c\rho_T c^{-1}, \mathrm{Supp}_k(T, c)).$$

This state is finite, of raw size

$$k!\,2^{3^k},$$

as machine-checked by VERIFIED.

Theorem 11 (the normalized state is a complete replacement receipt)**.** VERIFIED *Suppose an outside boundary with hub rotation ρ_L is matched orientation-reversingly to (T, c), and $\Sigma_k(T, c) = \Sigma_k(T', c')$. Compose the old matching with $c'^{-1} \circ c$, from the old inner boundary to the new one. Then the resulting matching to T' is still orientation-reversing, and the two normalized Tait supports are equal.*

Proof. Equality of the first fields says exactly that the coordinate change conjugates ρ_T to $\rho_{T'}$. Substituting this identity into the old orientation-reversal equation gives the new equation pointwise. Equality of the second fields is the support equality of Theorem 10. □

Thus equality of the full finite receipt supplies both semantic facts required by replacement: zero Count and seam orientation. For a majority shore whose literal boundary has cardinality k, choose the canonical finite-cardinality equivalence with $\mathrm{ULift}(\mathrm{Fin}\,k)$. Applying this coordinate to the literal vertex-side open tangle and its canonical first-return hub rotation gives an actual state Σ_k. Equality of two such states produces, without any extra receipt field, both the transported orientation-reversing matching and equality of the exact Tait supports: VERIFIED. Thus the coordinate-equivalence packaging is closed. The supplier must still supply literal connected shores and a strict material witness; the two disc-side hypotheses are then consequences of the planar-bond theorems, not extra receipt fields.

Theorem 12 (constructive physical replacement at equal normalized states)**.** VERIFIED *Let M be a graph-backed non-Tait-colourable connected, bridgeless, spherical cubic rotation map with cyclic vertex rotations and two-sided face orbits. Let $S_{\mathrm{new}} \subseteq S_{\mathrm{old}}$ be finite edge shores such that each shore and its edge complement are connected and each corresponding majority vertex shore and complement are nonempty. Choose the retained-dart roots and two distinct darts on the old exterior boundary. If the two interior boundaries have common cardinality k, their normalized states are equal, and some vertex has its complete cubic star in $S_{\mathrm{old}} \setminus S_{\mathrm{new}}$, then these data construct a literal rotation system M' such that*

1. *M' is connected, bridgeless, spherical and cubic, with cyclic vertex rotations;*
2. *M' is not Tait-colourable; and*
3. *$|V(M')| < |V(M)|$.*

Proof. Take M' to be the literal matched-seam composite of the exterior of the old majority shore and the interior of the new majority shore. The matching first crosses the old ambient cut by inverse edge flip and then transports through the two common finite coordinates. Equality of normalized states contains both facts needed by the splice: equality of the exact Tait-word supports and conjugacy of the canonical hub rotations. The first, together with exact boundary reindexing and complementary-shore reassembly, proves that M' is not Tait-colourable. The second transports the ambient orientation-reversal equation to the new interior.

Connected complementary edge shores give connected majority vertex shores. The planar-bond rank theorem therefore gives spherical face-simple canonical closures on the retained sides; the transported opposite hub orders and the seam face-count theorem make the composite spherical. The literal gluing theorems give connectedness, bridge-freeness, cubicity and cyclic vertex rotations. Finally majority is monotone under $S_{\mathrm{new}} \subseteq S_{\mathrm{old}}$, while the displayed cubic star belongs to the old majority shore but not the new one. Thus the retained new interior is strictly smaller than the removed old interior, proving the vertex inequality. These three conclusions are the fields of the returned replacement certificate; minimality is not used. □

Theorem 13 (equal normalized majority-shore states contradict minimality)**.** *VERIFIED* *Let M be a graph-backed vertex-minimal non-Tait-colourable member of the connected bridgeless spherical cubic rotation-map class, with cyclic vertex rotations and two-sided face orbits. Let $S_{\mathrm{new}} \subseteq S_{\mathrm{old}}$ be finite edge shores such that each shore and its edge complement are connected and each corresponding majority vertex shore and complement are nonempty. Root the old exterior, old interior and new interior literal vertex-side tangles by retained darts. Suppose the two interior boundaries have the same cardinality k, their normalized states are equal, and the old exterior boundary contains two distinct darts. Finally suppose some vertex x has every incident edge in $S_{\mathrm{old}} \setminus S_{\mathrm{new}}$. These data are impossible.*

Proof. Apply Theorem 12. It returns a strictly smaller map in the complete cap-stable structural class and proves that map non-Tait-colourable. Vertex minimality says the same returned map is Tait-colourable, a contradiction. □

Theorem 14 (direct literal corridor bound)**.** *VERIFIED* *Let M be a graph-backed vertex-minimal non-Tait-colourable member of the connected bridgeless spherical cubic rotation-map class. Fix $k, w \geq 0$. Suppose*

$$S_0 \supsetneq S_1 \supsetneq \cdots \supsetneq S_{m-1}$$

is a chain of actual edge shores. At every S_i, suppose the shore and its edge complement are connected, both majority vertex sides are nonempty, the literal interior and exterior have retained-dart roots, the literal boundary width is at most k, the edge-shore middle set has at most w vertices, and the exterior boundary contains two distinct darts. Then

$$m \leq (6w+1) \sum_{0 \leq j \leq k} j!\, 2^{3^j}.$$

Proof. Label S_i by its exact normalized seam state in the dependent sum $\coprod_{0 \leq j \leq k} \Sigma_j$, together with $|S_i| \bmod (6w+1)$. There are exactly

$$(6w+1) \sum_{0 \leq j \leq k} j!\, 2^{3^j}$$

such labels. If two positions $i < j$ had the same label, the strict inclusion $S_j \subsetneq S_i$ and equality of phases would put a cubic vertex wholly inside $S_i \setminus S_j$, by Lemma 9. Equality of the dependent seam states gives equality of the two normalized literal states at their common width. Theorem 13 would then splice out the nonempty slab and produce a smaller non-Tait-colourable map in the same structural class, contradicting vertex minimality. Hence the label map is injective, and the displayed cardinality is the claimed bound. □

This is the direct consumer for an L1 corridor. It uses cumulative states of the actual prefixes, so it needs neither the older L2 self-loop premise nor a caller-supplied replacement operation. The still-open geometric adapter is not hidden by the statement: from the source corridor positions one must construct the full ambient edge shores and verify connectedness of both edge shores, nonemptiness of both majority sides, roots, the width and middle-set bounds, two distinct exterior boundary darts, and strict nesting. The theorem above proves only that any chain carrying those literal certificates cannot be longer than the finite state carrier.

Theorem 15 (bounded-width connected-shore descent). VERIFIED *Let M be a graph-backed vertex-minimal non-Tait-colourable member of the connected bridgeless spherical cubic rotation-map class. Fix $k, w \geq 0$, and let $\mathcal{T}$ be a finite rooted binary tree. At every node t, suppose we are given an actual finite edge shore S_t such that:*

1. *S_t and its edge complement are connected, and the corresponding majority vertex shore and its complement are nonempty;*
2. *the edge-shore middle set has cardinality at most w;*
3. *the same middle set has cardinality at most k;*
4. *along every root-to-leaf path, a later shore is a proper subset of every earlier shore.*

If $|V(M)| \leq 2|V(\mathcal{T})|$, then, with

$$q_k = \sum_{0 \leq j \leq k} j!\, 2^{3^j},$$

one has

$$|V(M)| \leq 2\left(2^{(6w+1)q_k} - 1\right).$$

Proof. Complete every supplied node by Lemma 10. Thus its roots and two distinct boundary darts are derived from the minimal map, and its literal boundary width is at most k. For a node of literal boundary width $j \leq k$, compute its normalized seam state directly from its displayed shore, interior root, canonical boundary coordinate and first-return hub rotation. Embed it in the single dependent sum

$$Q_k = \coprod_{0 \leq j \leq k} \Sigma_j, \qquad |Q_k| = q_k,$$

and augment it by $|S_t| \bmod (6w+1)$. Thus the tree is labelled by exactly $(6w+1)q_k$ possible states.

Suppose two comparable nodes repeated an augmented state. Proper nesting and equality of the cardinality phases give, by Lemma 9, a vertex whose three incident edges all lie in the strict edge slab between the old and new shores. Equality in the dependent sum first identifies the two literal widths and then transports the two normalized states to an ordinary equality at that common width. This dependent transport is checked explicitly in VERIFIED; it is not a caller-supplied cast. All remaining hypotheses of Theorem 13 are fields of the two literal tree nodes. That theorem therefore makes the repeated state impossible. No root-to-leaf descent repeats a state, so finite binary-tree pumping and $|V(M)| \leq 2|V(\mathcal{T})|$ give the displayed bound. □

This theorem removes both the abstract seam-type parameter and the former application-specific replacement premise from the bounded-width consumer. The finite state is computed from literal shores, and repeated states are fed directly to the physical splice. Its vertex-accounting premise is discharged for a genuine edge-leaf branch decomposition by the following theorem.

Theorem 16 (bounded-width rooted connected branch decomposition). VERIFIED *Let M be as in Theorem 15. Root a full binary edge-leaf decomposition at one distinguished edge e_0. After removing that root leaf, write its first fork as two edge-leaf trees T_L, T_R. Suppose*

1. *e_0, followed by the leaves of T_L and T_R, is a duplicate-free enumeration of $E(M)$;*

2. *at every internal fork of T_L or T_R, the set S of leaves below that fork and $E(M) \setminus S$ are connected edge shores, their middle set has size at most both k and w.*

Set

$$q_k = \sum_{0 \le j \le k} j!\, 2^{3^j}, \qquad B = 2^{(6w+1)q_k} - 1.$$

Then

$$|V(M)| \le 4B + 6.$$

Proof. An internal fork is labelled by the finite set of edge leaves below it. Distinct leaves make the shores of its two nontrivial children proper subsets of the parent shore. Transitivity therefore gives proper nesting between every two comparable internal forks. Erasing the edge leaves produces two trees of the kind consumed by Theorem 15.

It remains to check that each displayed edge cut has vertices on both majority sides; this is a consequence, not an additional hypothesis. The fork shore contains at least two edges, one below each child. Its complement also contains at least two distinct edges: the distinguished root edge e_0, and one edge in the opposite top-level tree. These same two outside edges remain outside every descendant shore. If a connected edge shore contains two distinct edges, take a shore-edge walk between them. Either the first step is a different edge from the first chosen edge, or delete that step and continue; induction produces a vertex incident with two distinct shore edges. Thus the shore has a majority vertex. Applying the same argument to the complementary edge shore gives a vertex incident with two complement edges. Since every vertex of M has exactly three incident edges, that vertex is not a majority vertex for the original shore. Hence both majority sides are nonempty at every internal fork. The recursive construction of these witnesses is checked by VERIFIED.

The node-count half of that theorem, checked separately in VERIFIED, bounds each tree by B internal nodes.

A full binary tree has one more leaf than internal forks. Consequently the complete leaf enumeration gives

$$|E(M)| = i(T_L) + i(T_R) + 3,$$

where $i(T)$ denotes the number of internal forks of T. Cubicity gives $3|V(M)| = 2|E(M)|$. Hence

$$|V(M)| \le 2\big(i(T_L) + i(T_R) + 3\big) \le 4B + 6.$$

The one-edge cut above the first fork is deliberately absent: its complementary majority shore is empty, so inserting it as a root node would make the prettier one-tree constant depend on a false interface. □

Thus proper nesting and vertex accounting are no longer supply fields. What remains on the supply side is precise: construct the complete edge-leaf tree from the source corridor or a source-faithful connected decomposition and provide its nodewise connected edge shores and middle-set bounds. For every retained internal fork, both edge sides contain at least two edges: the fork side contains two leaf branches, while its complement contains the distinguished root edge and at least one leaf outside the fork. In a connected edge shore containing two edges, some vertex is incident with two shore edges (otherwise distinct edges would lie in different components), so the two majority sides are nonempty. The graph-theoretic implication and its complementary cubic form are checked in VERIFIED; the two-edge propagation itself is checked in VERIFIED and VERIFIED.

Accordingly, the theorem above receives no majority-nonemptiness fields. Lemma 10 then supplies the roots, literal width, and two-port fields.

For clarity, the standard branch-decomposition input is now separated from this consumer. Choose one leaf of the usual unrooted ternary tree, delete that leaf and its incident internal vertex, and call the two remaining binary edge-leaf trees T_L, T_R. The resulting rooted encoding $D = (e_0, T_L, T_R)$ remembers only the duplicate-free complete leaf labelling. Write $\mathrm{Conn}(D)$ when both edge-induced shores are connected at every tree cut, including the distinguished root-leaf cut and every remaining leaf cut, and $\mathrm{Width}_w(D)$ when every such middle set has size at most w. These are independent predicates. The checked adapter first forgets only the root and leaf cuts that the descent does not consume, then converts the remaining internal cuts to the exact literal consumer via VERIFIED and VERIFIED. Thus "a bounded decomposition exists" and "it can be connected without increasing its width" can no longer be conflated in one supply field.

For an ordered boundary of size at most k, the manuscript's data—edge colours, the two noncrossing connectivity relations, capped face progress, and the target-specific accepting bit—form a finite typed context state. Theorem 4 therefore shows that *bounded interface width is already enough*; a long literal tube is one convenient decomposition, not the logical essence of the descent.

The planar grid theorem of Robertson and Seymour ("Graph minors III: planar tree-width", 1984, doi:10.1016/0095-8956(84)90013-3) says that excluding one fixed planar wall as a minor bounds treewidth. The Seymour–Thomas sphere-cut theory supplies noose-respecting branch decompositions of plane graphs (Call routing and the ratcatcher, 1994, doi:10.1007/BF01215352). A more direct supplier is available here, and it is now checked. Fomin, Fraigniaud and Thilikos (The price of connectedness in expansions, Technical Report LSI–04–28–R, 2004, `https://hdl.handle.net/2117/98590`, Theorem 1; full proof in Barrière, Flocchini, Fomin, Fraigniaud, Nisse, Santoro and Thilikos, Connected graph searching, Information and Computation 219 (2012), doi:10.1016/j.ic.2012.08.004, Lemma 1) prove that, given a width-w branch decomposition of a 2-edge-connected graph, their algorithm *Make-it-Connected* returns a connected branch decomposition of width at most w. Here "connected" means that both sets of edge leaves at every tree cut induce connected subgraphs; there is no hidden factor in the width.

Theorem 17 (width-preserving connectedization). *VERIFIED* *Let G be a finite simple 2-edge-connected graph and let T be a complete rooted branch decomposition of G of width at most w. Then G has a complete rooted branch decomposition of width at most w in which both edge shores of every tree cut, including the leaf cuts, induce connected subgraphs.*

Proof. Root the decomposition at its root leaf and carry, with every subtree, the set of edges outside it (its *up-set*). Every internal vertex of the ternary tree is then a fork whose three directions are the two child shores and the up-set. Every tree edge between two internal vertices is a fork child of a fork. Such a tree edge is a *quartet* when one pair of directions is vertex-disjoint while the shores can be re-paired across the edge with nonempty boundaries.

Re-pairing is a rotation of the fork-of-fork. Two finite boundary facts drive the argument. First, the two new cut sizes sum to at most the sum of the two old cut sizes. If only one crossed matching is a quartet, its new boundary is contained in one old boundary. Thus some quartet replacement never increases the width. Second, at a fork with directions S_1, S_2, S_3 put

$$\psi(S_1, S_2, S_3) = \sum_{\{i,j,\ell\}=\{1,2,3\}} [\partial(S_i, S_j) = \varnothing]\,(|S_\ell| - 1),$$

and sum ψ over all forks. Direct boundary calculation shows that every allowed quartet replacement strictly decreases this natural-number potential. Strong induction therefore produces a quartet-free tree on the same edge leaves without increasing its width.

It remains to prove this tree connected. In a quartet-free tree of a 2-edge-connected graph, every fork has pairwise-touching directions. A vertex-disjoint pair can be pushed across the third direction: quartet freeness makes the merged pair vertex-disjoint from one of the two directions beyond it. Pairs involving the up-set are pushed down the rooted tree and sibling pairs are pushed up; structural induction reduces either case to a single-edge shore. Deleting that edge cannot disconnect a 2-edge-connected graph, a contradiction. Connected directions that touch at a vertex have connected union, so both shores of every cut, including leaf cuts, are connected. □

The graph-side fact that a connected graph without bridges is 2-edge-connected is checked as VERIFIED. Theorem 8 converts its edge shores canonically into nested connected complementary *vertex* shores. In a cubic graph their literal edge boundary has size at most the original middle-set width, not twice that width. The planar-bond cap and orientation theorems then construct the ordered open tangles directly from the ambient rotation system. Thus a sphere-cut noose and its local perturbation are an optional geometric realization, not part of the load-bearing D3 supply. Lemma 9 and the depth phase above settle strict-material accounting. Theorem 16 now roots the complete edge-leaf decomposition, discards its invalid one-edge top cut, derives proper nesting in the two surviving subtrees, and performs the full edge/vertex accounting. The majority shores, literal nodes, and exact ordered Count states are then computed by Theorem 8, Lemma 10, and Theorem 16.

Proposition 6 (wall-exclusion reduction, relative form)**.** *CONDITIONAL* *Fix the normal-form class and one direct-Count or Lift target. Suppose there are computable r, w such that:*

1. *no vertex-minimal zero-target instance contains a subdivision of the r-wall;*
2. *every plane instance in the class with no such wall has a complete rooted branch decomposition of width at most w;*

Then every vertex-minimal zero-target instance has an effective vertex bound. If the route-native finite base below that bound is empty, there is no zero-target instance.

Proof. Choose a vertex-minimal zero-target instance X. Hypothesis 1 excludes the fixed wall. Since a wall is subcubic, a minor model can be pruned inside each branch set to a subdivision: the at most three incident model edges are joined by a minimal tree, whose three terminal paths have one common branch point and are otherwise internally disjoint. Thus the minor and topological forms agree for this wall in a cubic host. Hypothesis 2 supplies the raw width-w tree, and Theorem 17 connectedizes it without increasing the width. The checked standard-to-consumer adapter supplies the complete connected edge-leaf decomposition. Theorem 16 derives its two strictly nested shore trees and gives the effective vertex bound; internally, a repeated normalized state is ruled out by the exact strict replacement of Theorem 13. An empty finite base rules out X. □

The high-width receipt can be split one step more finely, so that the route-specific assertion is not conflated with the conventional planar wall theorem.

Proposition 7 (exact fixed-mesh factorization of the width input)**.** *VERIFIED* *Fix positive integers a, b, w. Suppose:*

1. *no graph-backed vertex-minimal Tait counterexample contains an injective a-by-b mesh; and*

2. *every such counterexample which contains no injective a-by-b mesh has a complete rooted branch decomposition of width at most w.*

Then every graph-backed vertex-minimal Tait counterexample has a complete rooted branch decomposition of width at most w. Conversely, if that raw width-w conclusion holds and

$$B(w, w) < ab,$$

then the fixed injective a-by-b mesh is absent from every such counterexample. VERIFIED *With the route-native finite base through $B(w, w)$, the two displayed hypotheses therefore imply the combinatorial spherical Four-Colour statement.* VERIFIED

Proof. For a fixed least counterexample, the first hypothesis supplies absence of the stated injective mesh. Feed that fact to the second hypothesis; the result is exactly the raw branch-decomposition supplier consumed by the checked width-preserving connectedization and reductive assembly.

For the converse, a raw width-w supplier gives the checked vertex bound

$$|V(G)| \leq B(w, w).$$

An injective a-by-b mesh embeds its ab branch positions into $V(G)$, so it gives $ab \leq |V(G)|$, contrary to $B(w, w) < ab$. Finally, compose the raw supplier from the first paragraph with the checked finite-base assembly. □

The cut-counting mesh in the preceding proposition is intentionally more permissive than a topological wall: it does not remember the order in which its designated branch vertices occur on a path, and a path need not be simple. Those omissions do not affect mesh isoperimetry, but they do affect the two-dimensional overlap transport below. The route-specific target can therefore be weakened to the following wall-like carrier.

Proposition 8 (ordered fixed-mesh factorization)**.** *An* ordered injective a-by-b mesh *is an injective mesh together with simple row and column paths and a selected position of every branch vertex on its row and column, strictly increasing in the other coordinate. Its underlying permissive mesh is vertex-injective.* VERIFIED

Suppose that no graph-backed vertex-minimal Tait counterexample contains an ordered injective a-by-b mesh, and that absence of this carrier supplies a complete rooted branch decomposition of width at most w. Then the raw width-w supplier follows. VERIFIED *Conversely, a raw width-w supplier excludes the ordered mesh whenever $B(w, w) < ab$.* VERIFIED *Together with the route-native finite base through $B(w, w)$, the two hypotheses imply the combinatorial spherical Four-Colour statement.* VERIFIED *The stronger exclusion from Proposition 7 implies the ordered exclusion.* VERIFIED

Proof. Forgetting the positions and path-simplicity witnesses gives the old mesh; the branch-injectivity field is exactly the existing vertex-injectivity predicate. The first implication feeds ordered-mesh absence to the supplied wall-free decomposition theorem. The converse forgets the ordering and uses the same injection of the ab branch positions into $V(G)$ as in Proposition 7. The finite-base conclusion then uses the unchanged checked reductive assembly. □

The first premise of Proposition 8 is the sole route-specific high-width assertion used below. Its second premise is the conventional wall/grid-to-branchwidth implication, now expressed on a carrier which records the rectangular order needed by the overlap connection. Neither premise entails the other without the quantitative size condition in the converse, and neither is concealed in a Lean structure field. The permissive fixed-mesh proposition remains a valid stronger alternative, but it is not imposed on the wall-facing proof.

One sufficient way to establish Hypothesis 1 is the stronger statement that every zero-target instance containing the wall has either positive target or a strict zero-target reduction. The route has not proved that statement. The hard step is therefore isolated as

$$\boxed{\text{a fixed embedded wall in a zero-target normal-form instance forces a strict zero-target reduction.}}$$

The Goldberg audits show why curvature alone cannot establish the boxed assertion: they have only pentagons and hexagons while their widths grow. Their tested good fibres are seeded, which is evidence that the target predicate may exclude the wall even though the geometry does not. It is not a proof. Nor is the tree reformulation permission to count generic planar configurations or revive a classical unavoidable catalogue.

The source's formation dynamics has a strong local entrance to this boxed problem which is worth recording with its quantifiers in the correct order.

Lemma 12 (one adjacent-pair Kempe orbit contains all constant words)**.** VERIFIED *Let uv be any ordered adjacent pair in a graph-backed vertex-minimal Tait counterexample. Delete u and v, leaving the four incident defect ports in their cyclic order. There is one rotation ordering and one proper Tait colouring C of this deleted graph such that, for each nonzero Tait colour a, the Kempe orbit of C contains a proper colouring C_a whose four missing boundary colours are all a.*

Proof. The adjacent-pair planar reduction first supplies, on the literal cyclic ordering, a proper colouring C and a two-colour profile in which the first two ports belong to one Kempe side and the last two ports belong to the other. If a selected two-colour component joined the two sides, then reinserting (u), (v), and their common edge would extend the colouring to the ambient graph, contrary to the counterexample assumption. Thus the two side requests may be changed independently inside the Kempe closure.

Fix a nonzero target colour a. Recolour the first side to a while leaving the second side fixed, and then recolour the second side to a while leaving the first fixed. Both operations are Kempe switches in the deleted graph and preserve properness, so their composite gives C_a. The cyclic ordering and the initial colouring C were selected before a; hence the three target colourings belong to one common orbit rather than to three unrelated existentially chosen colourings. □

This lemma is nonlocal in precisely the way the original formation argument needs: the Kempe paths witnessing the two independent side moves may run through the whole retained graph. It is therefore not a bounded-radius reducibility certificate. Its remaining wall-facing gap is also exact. For overlapping adjacent pairs the lemma supplies a common orbit separately at each deletion, but it does not transport a colouring, a Kempe word, or an exact Count state from one deleted graph to the next. A source-faithful Programme–G proof of fixed-wall exclusion must construct that overlap transport (or a simultaneous multi-deletion orbit) and then extract a strict physical replacement. Merely pigeonholing the three constant words would discard the graph-dependent Kempe paths and would not prove the boxed assertion.

The first overlap object can now be stated without choosing a privileged representative of either Kempe orbit.

Lemma 13 (canonical finite state of two adjacent-pair orbits)**.** VERIFIED VERIFIED VERIFIED VERIFIED *Let $p = (u, v)$ and $q = (x, y)$ be two ordered adjacent-pair data in the same finite graph. Their exact common carrier is the induced graph obtained by deleting u, v, x, y. It embeds canonically in each* *of the two adjacent-pair deletions, and restriction along either embedding sends a proper nonzero Tait colouring to a proper nonzero Tait colouring of the common carrier.*

For proper base colourings (C_p, C_q) *of the two deletions, attach to a pair of proper colourings in their respective Kempe closures the following finite state:*

1. *the* 4*-by-*4 *Boolean incidence table recording which labelled port vertices of* p *and* q *coincide;*
2. *the exact four-port colour word at each deletion;*
3. *for every ordered pair of colours and every ordered pair of ports, the Boolean value saying whether those ports meet the same selected two-colour component; and*
4. *the Boolean value saying whether the two restrictions to the common carrier agree literally.*

Let $\mathcal{R}(C_p, C_q)$ *be the set of all such states realised by proper colourings in the two closures. Then* $\mathcal{R}(C_p, C_q)$ *is nonempty. Moreover, if* C'_p *and* C'_q *are reachable from* C_p *and* C_q*, respectively, then*

$$\mathcal{R}(C_p, C_q) = \mathcal{R}(C'_p, C'_q).$$

Thus this overlap relation belongs to the two Kempe orbits, not to the chosen representatives.

Proof. Deleting all four named vertices factors through either two-vertex deletion by the inclusion of retained vertices. Pulling an edge colouring back along either induced-graph embedding merely evaluates it on the corresponding ambient edge, so nonzeroness is preserved edge by edge.

Every coordinate listed above has finite domain and finite codomain; their product is therefore finite. The two base colourings themselves witness nonemptiness. Finally, a member of a Kempe closure has the same Kempe closure as the base: reachability is reflexive, symmetric, and transitive. Replacing either base by a reachable representative consequently changes neither quantified colouring set and hence changes no realised overlap state. ☐

Applied to Lemma 12, the two marginals of this relation each contain the three nonzero constant boundary words. The lemma deliberately does *not* assert that a state with literal common-carrier agreement is realised, nor that two neighbouring relations have equal Count support. Exact agreement is false in some of the checked intrinsic-short reentry geometries. The remaining wall-facing task is therefore a genuinely two-dimensional statement about composing these finite orbit relations—either a coherent flat region yields the support equality needed by the physical splice, or a non-coboundary holonomy yields a structural target. A chosen-word transport, and any additive discrepancy of endpoint potentials, is too weak for that task.

The first exact composition object and its connection to the wall carrier are now available. They remove a quantifier ambiguity which is easy to miss: pairwise relations may be witnessed by different colourings at their common corner, whereas a compositional cell must reuse one actual colouring there.

Lemma 14 (coherent orbit cell on an ordered mesh)**.** *VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED*

Let four adjacent-pair deletions be arranged clockwise around a cell, with proper base colourings at the north-west, north-east, south-east and south-west corners. Evaluate the overlap state on each side using the same concrete corner colouring for the two sides incident with that corner. The set of all cell states obtained by independently ranging the four corner colourings through their Kempe orbits is finite and nonempty, is independent of the four chosen orbit representatives, and projects *into the cycle of the four pairwise overlap supports. At every corner its two projected boundary observations agree literally.*

Moreover, every consecutive coordinate rectangle of an ordered injective mesh in a graph-backed vertex-minimal Tait counterexample canonically supplies such four deletion sites. They are anchored at the four mesh branch vertices, and the common-constant-orbit lemma supplies proper base colourings, hence a nonempty coherent cell support.

Proof. For the abstract statement, choose one proper colouring at each corner and use it in both incident pairwise overlap evaluations. The two boundary observations at that corner are then definitionally the same. Taking all four colourings in their respective Kempe closures gives a finite relation; the base colourings witness nonemptiness. Replacing a base colouring by a reachable representative leaves its whole Kempe closure unchanged, so it leaves the cell relation unchanged. Forgetting the shared witnesses gives the asserted inclusion in the naive pairwise-compatible cycle.

For the mesh statement, every step of a row or column path is an edge of the rotation-system multigraph and therefore an edge of its graph backing. At each cell corner select the first outgoing path step in clockwise order (equivalently the reverse of the last incoming step on the south and west sides). Strict monotonicity of the branch positions guarantees that these steps exist. Cubicity and the absence of a common neighbour at an adjacent pair reconstruct the canonical four-port deletion data, and the selected step begins at the corresponding branch vertex. Finally apply Lemma 12 independently at the four sites and use the nonemptiness just proved. The ordered-mesh carrier permits additional row–column intersections and shared row–column path segments, so this construction does not assert that the four intervening path segments form the boundary of an embedded disc. □

Lemma 15 (complete boundary relation for an ordered mesh cell). VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED *For one coordinate rectangle of an ordered injective mesh, orient every subdivision step of its top, right, bottom and left paths clockwise. These steps form a finite cyclic carrier: the displayed boundary list contains every step exactly once, and clockwise successor is a permutation. This includes all path steps between branch vertices, not only the four steps selected at the corners in Lemma 14.*

Each step canonically supplies an adjacent-pair deletion. The terminal vertex of a step is literally the initial vertex of its successor, including at all four turns. Therefore the two consecutive deletions compare on the canonical induced common carrier obtained by deleting their three named endpoints (with repetitions allowed). Minimality supplies one proper base colouring, and hence one selected Kempe orbit, at every site. The finite overlap-state relation from each selected site to its successor is nonempty.

Choose one realised finite overlap state on every successor edge. Exactly one of the following holds:

1. *there is one concrete deletion colouring at every boundary site which simultaneously witnesses the chosen incoming and outgoing states; or*

2. *no such boundary-wide family exists.*

In the first case the state on every edge is the overlap state of the two displayed concrete colourings. Thus a site colouring is reused, rather than chosen afresh, in its two incident relations.

Proof. The four side lengths are the positive differences of successive branch positions. List the corresponding path steps in clockwise order, reversing the bottom and left sides. Successor increments within a side and moves to the first step of the next side at a turn. A case split over the four sides, using strict monotonicity of the branch positions at the turns, proves both injectivity of successor and equality of consecutive endpoints. Finiteness then makes successor a permutation. The four lists are disjoint by their side tags and each finite index occurs once, proving completeness of the boundary list.

The common-carrier statement follows by substituting the shared endpoint in the four endpoint inequalities defining the generic two-pair deletion. Choose the adjacent-pair data and a proper base colouring once at every site. The two bases at a step and its successor realise an overlap state, so each one-step support is inhabited. Choice over the finite carrier gives a state assignment. A boundary section is, by definition, a dependent family of site colourings satisfying every selected successor relation. Excluded middle applied to the existence of that single family gives the stated dichotomy, and unfolding the successor relation gives the final state-realisation equality. □

Lemma 16 (one physical Kempe site per ordered-mesh step)**.** *VERIFIED VERIFIED* *Fix one ordered injective mesh in a graph-backed vertex-minimal Tait counterexample. There is one globally indexed forward site for every step of every row path and every column path, consisting of ordered adjacent-pair data and a proper base colouring of its two-vertex deletion. Every oriented step on the complete boundary of a coordinate rectangle injects into this global carrier after its cell-side tag and orientation are forgotten. Its unordered deleted endpoint pair is unchanged.*

Use the global site directly on the north and east sides of a clockwise cell boundary. On the south and west sides use its canonical reversal: exchange the two deleted endpoints, exchange the two two-port blocks, and transport the base colouring through the canonical isomorphism between the two orders of the same vertex deletion. This transport preserves the nonzero Tait condition.

If two coordinate rectangles share a side, reversal of the finite position interval identifies their side-step carriers exactly. Under this identification the clockwise site of one rectangle is the canonical reversal of the clockwise site of the other, and its base colouring is exactly the transported base colouring. In particular, the two rectangles do not make independent Kempe-orbit choices on their common physical side.

This identification preserves the complete finite observation at the site. More precisely, reversal precomposes the four-port Kirchhoff word by $0, 1, 2, 3 \mapsto 2, 3, 0, 1$*, and it preserves every labelled assertion that two ports lie in the same component of a specified two-colour subgraph under the same port permutation. Hence the full boundary Kempe state on one reading of a shared side is exactly the involutive reversal of the state on the other reading. This holds for east–west and north–south neighbours.*

The same transport acts on the complete two-site overlap observation. It transposes the source–target port-coincidence matrix after the four-port reversal, exchanges and reverses the two boundary states, and leaves unchanged the Boolean assertion that the two colourings agree on their common four-vertex deletion. This action is an involution. Consequently, reading an ordered pair of physical sites backwards gives exactly the reversed overlap state, rather than merely an overlap state with the same boundary words. The retained-vertex isomorphism also transports each selected bicoloured component, commutes with switching that component, and therefore identifies the two complete Kempe closures. It follows that the entire realised overlap support in one direction is the image, under this same involution, of the realised support in the other direction.

For each rectangle, the successor overlap supports built from these globally chosen sites are nonempty. Hence every finite state assignment on its complete boundary has the same exact alternative as in Lemma 15: either one concrete colouring at every globally based site simultaneously witnesses all successor states, or no such boundary section exists.

Proof. Index a physical path step by the disjoint sum of a row index and a row-step index, or a column index and a column-step index. The four cell-side formulas map into this carrier by the corresponding branch-position interval. Within one rectangle, equality of two images either equates two positions on the same side or would equate its distinct top and bottom row indices, its distinct left and right column indices, or the two summands. This proves injectivity. Minimality supplies one rotation-ordered adjacent-pair Kempe site and proper base colouring at every global index; make this choice once.

The north and east formulas traverse their path steps forward. The south and west formulas traverse them backward, so exchange the deleted endpoints and the two port blocks. Swapping the two inequalities in the retained-vertex subtype is a graph isomorphism between the two deletion orders. Pulling the base colouring back along this isomorphism preserves properness and preserves the assertion that every edge has nonzero colour.

For neighbouring rectangles, the shared row or column interval has the same length. The elementary identity

$$l + p = u - 1 - ((u - l) - (p + 1))$$

identifies a forward position on one boundary with the reversed position on the other. Substitution in the four branch-position formulas proves literal equality of their global step indices. The statements about adjacent-pair data and base colourings now follow by unfolding the globally selected site once, not by comparing two choices.

The retained-vertex swap is a graph isomorphism. It maps the incident-edge set at every retained vertex bijectively to the incident-edge set at its image, so the corresponding Kirchhoff sums agree. On each pair of colours it also restricts to an isomorphism of the two bicoloured line subgraphs; reachability therefore agrees for every pair of ports. These two facts give the boundary word and connectivity-bit formulas, and applying the port permutation twice is the identity. Substitution of the already proved shared-side data and base-colouring equalities gives the east–west and north–south state identities.

For a pair of sites, exchange source and target while applying the same retained-vertex swap to both colourings. The common four-vertex deletion is carried to the reversed common deletion by the identity on ambient vertices with the four exclusion proofs permuted. Thus literal agreement of the two restricted colourings is invariant. Together with the one-site state formula and the transposed port-coincidence matrix, this proves the complete overlap state reversal formula and its involutivity.

For the orbit statement, map a selected bicoloured component through the line-graph isomorphism induced by the retained-vertex swap. Membership in the component is preserved in both directions, so switching before transport is the same colouring as transporting before switching. Induction on a finite Kempe-move sequence gives one inclusion of closures; applying the same result after a second reversal gives the converse. Apply the concrete overlap-state formula pointwise to the two closure sets. This proves equality of the realised supports under the overlap-state involution.

Finally, two proper globally based colourings realise an overlap state on every successor edge. Choice of one such finite state on each edge gives an assignment. Excluded middle applied to a single dependent family of concrete site colourings gives the section alternative, exactly as in the preceding lemma. □

This lemma closes the carrier- and orientation-coherence problem, not the wall theorem. What remains is now intrinsically two-dimensional: prove that the relational returns of neighbouring cells cancel after the displayed support involution, or show that their failure produces a bounded cut, a strict compositional replacement, or another global invariant forbidden in a least counterexample.

No equality of opposite-shore supports, flatness, or fixed-wall exclusion is inferred merely from sharing the physical sites.

This lemma is deliberately an incidence and quantifier theorem. The second alternative is not yet a geometric obstruction, and the first does not yet identify the supports of two physical shores. In particular, an ordered injective mesh may have extra row–column intersections or shared path segments, so the cyclic carrier is not asserted to bound an embedded disc. The next wall-facing obligation is now exact: either construct a boundary section with the support equality required by the physical splice, or turn failure of a section into a bounded cut, a strict replacement, or a global invariant contradicting a least counterexample.

Corollary 6 (complete boundary relational residue)**.** *VERIFIED VERIFIED VERIFIED VERIFIED* *Cut the complete boundary immediately before its north-west first step. Let (H) be the relation on proper colourings at that root obtained by imposing every successor relation except the cut relation on one boundary-wide family, and then using the cut relation from the last west-side colouring to a second root colouring. A boundary section exists if and only if (H) has a fixed point.*

Consequently, if a chosen boundary-state assignment has no section, then all of its one-step relations are nevertheless nonempty and exactly one of the following global residues occurs:

1. *(H) is empty: the one-step witnesses cannot be concatenated into even one complete clockwise return;*
2. *the domain and range of (H) differ;*
3. *(H) branches forward or backward; or*
4. *(H) is the graph of a fixed-point-free permutation on its nonempty active support.*

Proof. The final west-side step has the last west index, so its successor is the north-west root. A section supplies a fixed point of (H) by cutting its family there. Conversely, a fixed-point witness for (H) glues the two root colourings and satisfies every successor relation, hence is a section.

If no fixed point exists, either (H) is empty or every realised return moves its starting colouring. In the latter case compare the active domain and range. If they differ, the second residue holds. If they agree but either forward or backward uniqueness fails, the third holds. Otherwise the unique successor map is a bijection of the common active support, and the absence of a fixed point gives the fourth residue. □

Only one implication is claimed:

$$\{\text{states with shared corner witnesses}\} \subseteq \{\text{pairwise-compatible cycles}\}.$$

The converse is an open lifting, or sheaf, condition: equal finite boundary observations can be represented by different intermediate colourings. Thus Lemma 14 does not prove flatness, trivial holonomy, support equality across a rectangle, or fixed-wall exclusion. It fixes the exact input of the next wall-facing theorem: either lift compatible local data to a coherent rectangular section and obtain the physical support replacement, or extract a non-coboundary holonomy with a separate structural consequence. The finite-state dichotomy itself is checked: shared-witness lifting holds, or one compatible cycle state explicitly fails to lift. This dichotomy is a specification of the two possible proof obligations, not a proof that either alternative has the required geometric consequence.

There is an exact positive criterion. Say that the boundary observation *separates an orbit* when two proper colourings in that Kempe orbit with the same boundary word and all the same labelled two-colour port-connectivity bits are literally the same colouring. If the observation separates each of the four corner orbits, then pairwise compatibility does lift: choose witnesses on the four sides; corner coherence makes the two observations at a corner equal, separation identifies the two concrete colourings, and the four identified colourings are shared witnesses for the cell. Contrapositively, failure of lifting forces a nontrivial fibre at one or more corners. Thus the missing datum is not an unspecified topological defect: it is exactly a pair of distinct proper orbit representatives with the same recorded boundary state.

The obstruction has an exact relational normal form. For a fixed side state s_i, let R_i be the relation consisting of the pairs of concrete proper orbit representatives which realise s_i. A clockwise cell state has shared corner witnesses precisely when

$$(R_{mW} \circ R_{mS} \circ R_{mE} \circ R_{mN}) \cap \Delta \neq \varnothing;$$

that is, the four-step relation on the north-west fibre has a fixed point. Consequently a lifting obstruction is equivalent to a corner-coherent finite state for which every R_i is nonempty but the composite is fixed-point-free. This is the precise sense in which the remaining datum is a relational holonomy. It does not yet give a geometric consequence for an embedded wall.

There are two distinct residues hidden in the phrase "fixed-point-free." The fourfold composite may be empty, even though each of the four side relations is nonempty; call this a *relational break*. Otherwise the composite is nonempty and every completed clockwise path returns to a different north-west witness; call this *proper relational holonomy*. These alternatives are exhaustive: if the composite is empty there is a break, while if it is inhabited then absence of a fixed point says exactly that every one of its pairs has distinct endpoints. This split must precede any signed-permutation argument, because an empty relation is not a permutation.

The nonempty horn must be refined once more before it has a sign. For an endorelation (R), let its domain and range be the witnesses which occur as first and second coordinates. Exactly one of the following descriptions applies: the domain and range differ (support drift); (R) is not unique in one of its two coordinates (relational branching); or the two supports agree and (R) is unique in both coordinates. In the last case, choice of the unique successor constructs a bijection of the common active support, and a fixed-point-free (R) induces a genuine fixed-point-free permutation there. Only this last residue can be passed to a permutation-sign invariant. If its active support has two elements, fixed-point-freeness makes the permutation's support the whole two-element carrier; it is therefore a transposition and has sign -1.

An exhaustive finite diagnostic gives an important warning about the scope of this criterion. For the four edges of a square face in the pentagonal prism, each selected proper deletion orbit is the proper part of its full edge-Kempe closure, yet a pairwise-compatible cycle fails to lift; two opposite corner observations have fibres of size two. More strongly, a failing cycle remains when all four literal common-restriction agreement bits are true. In that cycle the incident sides choose opposite representatives at exactly the two non-singleton corners, and every non-singleton fibre in the example is joined by one direct Kempe-component switch. This computation has not yet been reflected into the kernel, so it is evidence rather than a formal counterexample. In the displayed all-agreement obstruction each side relation has one witness pair, but the target chosen by one side is not the source chosen by the next at two opposite corners; its fourfold composite is therefore empty. Thus this example lies in the relational-break horn, not in the proper-holonomy horn. It nevertheless rules out treating planarity, bridgeless cubicity, bare ordered-mesh geometry, or literal pairwise agreement as a reason to assume a global section. A wall argument must either control these relational breaks using the minimal-counterexample hypothesis or derive a target-specific consequence from a proper nonempty holonomy.

Exhausting all 2 792 976 compatible cycles in the same prism separates the two residues: 6 912 have empty composite and 6 144 have a nonempty fixed-point-free composite; the remaining 2 779 920 lift. Thus both obstruction horns already occur in one planar bridgeless cubic graph. Moreover, each of the (6,144) proper composites is a singleton partial transport whose one-element domain differs from its one-element range. Hence every proper prism example lies in the support-drift residue, not the permutation residue. The calculation is again external evidence, not a kernel theorem.

There is nevertheless no colour-extension gap at a compatible checkpoint. The following statement isolates the exact planar cap which may be attached there; it will be the closed carrier on which Kauffman's formation parity can legitimately be evaluated.

Lemma 17 (compatible four-port colourings extend across the paired cap)**.** VERIFIED *Fix an adjacent pair (uv), delete (u,v), and retain the four defect ports in their cyclic order. Choose either of the two planar square pairings of those ports. The corresponding paired square reduction is Tait-colourable if and only if the four-defect graph has a proper Tait colouring whose missing-colour word is constant on each of the two chosen pairs.*

Proof. In one direction, restrict a colouring of the paired reduction to its old edges. At either endpoint of a new seam, the two retained incident colours are distinct and nonzero; their missing Klein-four colour is therefore the colour of that seam. The two endpoints of a seam consequently have equal missing colours, giving the required compatible word.

Conversely, start with a compatible deletion colouring. Restore (u,v) temporarily by assigning the four old cut edges their four missing colours and assigning (uv) the colour zero. This one-zero chain is proper at every retained vertex and has no zero on any retained or cut edge. Compatibility says exactly that the two old cut edges identified at each selected seam have the same colour. Hence their dart colouring descends through the two identifications. The zero edge (uv) lies in the discarded region, so the descended colouring is nowhere zero and is a Tait colouring of the paired reduction. □

The four-state square used by the source is narrower than an arbitrary open Kempe path, and all of its intermediate words stay compatible with one fixed cap.

Lemma 18 (the same-side Kempe square extends through one paired cap)**.** VERIFIED *Let C be a deletion colouring compatible with the pairing* $(01 \mid 23)$*. Suppose an (a,b)-component K_{01} has exact boundary support $\{0,1\}$, while another (a,b)-component K_{23} has exact boundary support $\{2,3\}$. Switch neither, either, or both components. Each of the resulting four proper deletion colourings extends to a proper Tait colouring of the same* $(01 \mid 23)$ *paired reduction.*

Proof. The four boundary words are the original word, the result of interchanging (a,b) at ports (0,1), the result of interchanging them at ports (2,3), and the result of interchanging them at all four ports. Equality within each of the pairs $\{0,1\}$ and $\{2,3\}$ is preserved in all four cases. Every corner therefore satisfies the compatibility hypothesis of Lemma 17. Applying its explicit extension construction at each corner gives the four capped colourings. □

The source needs more than endpoint extension, but less than a universal path-lifting theorem. Its two moves have exact same-side support, and such a move closes to one ordinary Kempe circuit when its matching seam is inserted.

Lemma 19 (a same-side Kempe switch closes through the paired cap)**.** VERIFIED *In the setting of Lemma 18, let $\widehat{C}$ be the explicit extension of C across the* $(01 \mid 23)$ *cap. If C' is obtained from C by switching K_{01}, then some closed (a,b)-Kempe component of $\widehat{C}$ contains the* (01)*-seam, and switching that component restricts on every old edge exactly to C'. The same statement holds for K_{23}.*

Proof. Consider K_{01}. At each of ports $0, 1$, the missing colour belongs to $\{a, b\}$. Of the two old edges incident with that degree-two boundary vertex, exactly one therefore has colour a or b; exact boundary support says that this edge lies in K_{01}. The new (01)-seam has the missing colour, so in the capped graph it joins the two open ends of K_{01}. No other selected-colour edge is available at either endpoint. Consequently K_{01} together with the seam is a connected 2-regular (a, b)-subgraph, hence one whole closed Kempe circuit.

Switching this circuit interchanges a, b on every old edge of K_{01} and on the new seam, and fixes every other old edge. Consequently its restriction to the old graph is exactly C'. This is the required closed realization of the open switch; no separately chosen extension of C' is needed. The proof for K_{23} is identical. □

This does not assert that every open Kempe path lifts through a fixed cap: cross-side switches can pass through incompatible boundary words, and an explicit planar example shows that closed formation parity can then change after recapping. The lemma instead covers exactly the two same-side switches used in Lemma 12. Its remaining checked content is the graph-level identification of the capped component: old edges are labelled by their original open component, that label is invariant across every selected cap adjacency, and the matching seam closes the intended component without fusing it to another one.

Corollary 7 (spherical parity consumer for the same-side lift)**.** VERIFIED *Let the ambient map be a graph-backed least Tait counterexample. Suppose the retained deletion is connected, its four ports occur in their physical cyclic order, and the canonical* 01 | 23 *paired reduction is bridge-free. Then the capped circuit in Lemma 19 may be chosen so that its closed switch both restricts to the required open switch and preserves Kauffman's formation parity. Equivalently, for the third colour $c = a + b$, the parity of the sum of the three bicoloured-circuit counts is unchanged.*

Proof. The two new seams join ports 0 to 1 and ports 2 to 3. Neither joined pair can already be the endpoints of a retained edge: together with the corresponding deleted vertex, such an edge would form a triangle, whereas a least counterexample is triangle-free. The two seam endpoint pairs are distinct from one another, and the old endpoint pairs remain distinct because the ambient graph is simple. Thus the capped rotation system has an endpoint-injective, canonical simple-graph presentation.

The retained connectivity and physical boundary order make the reduction spherical, connected, and cyclic at every vertex; bridge-freeness then implies that its two dart-sides lie on distinct face orbits. Transport the cap colouring, its selected bicoloured component, and the component switch through the canonical graph presentation. Kauffman's checked closed-spherical parity theorem gives equality of formation parity before and after that switch. Finally, the checked same-side component lift identifies the switched colour of every retained edge with the prescribed open Kempe switch. □

No abstract cross-curve receipt is used here. The parity theorem is applied only after the open switch has been closed inside an actual bridge-free spherical cubic map. This is the precise, source-faithful role of Kauffman's closed-curve argument; no parity assertion is being extended to an open tangle.

There is a second, checked constraint on any such overlap transport. It shows why the transfer state may not remember only an affine action on the four colours.

Lemma 20 (singleton-collar parity is not affine colour transport)**.** *Along a nonempty closed chain of admissible adjacent-pair reentries, choose at each state the canonical colour frame determined by its two ordered boundary colours. Allow every reentry an arbitrary additive translation of the four Tait colours. The composite affine action around the closed chain is even. By contrast, the rooted*

support monodromy of a singleton intrinsic collar is odd. Consequently no change of four-state coordinates conjugates the singleton geometric monodromy to that closed affine colour action.
VERIFIED VERIFIED

Proof. The linear part of a reentry is the change from the source canonical colour frame to the target canonical colour frame. These changes telescope along a path. On a closed path the initial and terminal frames agree, so the net linear part is the identity. Translation of the four-element colour group is even: every nontrivial translation is the product of two disjoint transpositions. Hence the sign of an affine word is the sign of its net linear part, namely (+1).

For a singleton collar the two sheets of the rooted support order are exchanged at exactly one endpoint, so its geometric permutation has sign (-1). Sign is invariant under conjugacy. An even affine colour action therefore cannot be conjugate to this odd geometric action. □

Lemma 20 is not, by itself, a wall-exclusion theorem. It is a state-design theorem. A valid simultaneous reentry invariant must retain the geometric orientation parity in addition to the affine colour gauge, or else separately prove that singleton collars cannot occur. Treating the colour action as the whole support transport would make the overlap step false exactly in the dense-defect geometry now under investigation.

There is a further orientation gauge that must be fixed before the odd local transport can be regarded as a charge on a particular ambient edge.

Lemma 21 (support reversal moves the negative singleton crossing)**.** *Let a singleton intrinsic collar be represented by an oriented scalar support cycle. Reverse only that support-cycle orientation, leaving the remote facial-dual cycle fixed. The reversed walk is again a valid scalar support state. After both crossing orders are rooted at the distinguished target edge, corresponding positions denote the same two physical crossing edges, but the chirality at each position is toggled. Consequently the unique negative crossing for the reversed orientation is a different ambient edge from the unique negative crossing for the original orientation.*
VERIFIED

Proof. Reversing a simple support cycle preserves its edge set, simplicity, target edge, second certified crossing, and the nonvanishing support condition. It reverses the support crossing word and leaves the remote crossing word unchanged. In the singleton branch these words contain exactly two entries. Rotating either support word to start at the target therefore gives the same target-rooted list.

At a fixed physical crossing the facial-dual step, and hence its canonical remote dart, is unchanged. The dart contributed by the reversed support walk is the reverse of the original support dart. Thus agreement of the two darts becomes disagreement and disagreement becomes agreement: both chiralities are toggled. Each orientation has exactly one negative position, so those two negative positions cannot denote the same physical edge. □

The sign (-1) of singleton monodromy is therefore orientation-gauge invariant, but the ambient edge carrying its displayed transposition is not. A global noncancellation argument must either choose support orientations coherently across reentries or formulate its charge after quotienting this gauge. Merely multiplying edge-labelled local transpositions is not a well-defined global observable.

Even after a fixed carrier has been supplied, local oddness retains only the parity of the number of reentries.

Lemma 22 (parity boundary for a closed intrinsic reentry word)**.** *Let a closed all-intrinsic reentry chain have $n > 0$ steps. Suppose that one fixed finite carrier has been chosen and that the geometric* *transport of each step is represented on that carrier by an odd permutation. Then the sign of the ordered product is*

$$(-1)^n.$$

If this product is identified, up to a change of coordinates, with the canonical affine colour transport around the same closed chain, then n is even. This conclusion is sharp at the purely algebraic level: two odd transpositions can have identity product.

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Proof. Induct on the explicitly counted reentry word. The empty product has sign $(+1) = (-1)^0$. Appending a step multiplies the geometric product by one odd permutation, so the sign is multiplied by -1. After n steps it is therefore $(-1)^n$.

The canonical affine colour word of a closed reentry chain has identity linear part, and its translations are even; hence its total sign is $+1$. Sign is invariant under a change of coordinates. Identification of the two products thus gives $(-1)^n = +1$, which is equivalent to n being even. For sharpness, take the transposition of the two Boolean states twice: each factor is odd and their product is the identity. □

Thus a coherent fixed-carrier construction is necessary but not sufficient. To turn the all-intrinsic residue into a contradiction one must additionally prove odd length, or construct a stronger nonabelian or geometric charge that cannot cancel on an even orbit. Neither property is part of the present closed-residue data. This is the exact interface the high-width argument must now supply; no parity premise is silently added.

The other terminal of the all-face machine already has the literal geometry required by the replacement layer; it is not merely a descriptive cut profile.

Lemma 23 (a large all-face cut is a literal facial-dual noose)**.** VERIFIED *Suppose the all-face machine reaches its large-remote-primal-cut outcome at a directed edge. For every anchor vertex a, there is a simple closed walk W in the actual facial dual, of length at least five, with crossed primal edge set X, such that the component C_a of $G - X$ containing a satisfies*

$$\partial C_a = X.$$

Both $G[C_a]$ and $G[V(G) \setminus C_a]$ are connected. If

$$S_a := \{e \in E(G) : e \text{ has an endpoint in } C_a\},$$

then its middle-set width obeys

$$|\operatorname{mid}(S_a)| \leq |W|.$$

Thus this outcome supplies the literal noose-side geometry consumed by the physical replacement theorem. It does not, by itself, bound $|W|$, produce a second nested noose, or prove equality of their phased states.

Proof. Unpack the large-cut certificate. Its witness is already a simple closed walk W in the literal facial dual, and its long-profile field gives $|W| \geq 5$. Let X be the primal edges crossed by successive dual edges of W, and choose the connected component of $G - X$ containing the specified anchor a. These data are exactly an oriented facial-dual noose side; no new curve or separator is chosen.

Apply spherical facial-parity saturation to this simple dual cycle. Every edge of X crosses the cycle once and therefore has endpoints on opposite sides. Conversely, a primal edge not in X has zero intersection with the cycle and cannot leave the chosen component. Hence X is the exact edge boundary of C_a. The same separation argument, applied to the two regions of the sphere cut by W, shows that the chosen component and its vertex complement are both connected.

For the width estimate, a middle vertex of S_a lies outside C_a: all edges incident with a vertex of C_a belong to S_a. Choose an S_a-edge incident with that middle vertex. Exactness of the boundary orients it uniquely from C_a to the middle vertex and therefore assigns a crossing dart of W. Distinct middle vertices have distinct terminal endpoints, so this assignment is injective. A simple dual cycle crosses one distinct primal edge at each step, giving $|\operatorname{mid}(S_a)| \le |X| = |W|$. □

The preceding local facts can be assembled without adding a geometric supply hypothesis. The result is useful chiefly because it states the remaining global obligation without hiding either of its two branches.

Theorem 18 (exact residue of the all-face formation machine). *VERIFIED* *Start the exact all-face escape-state machine at any directed edge of a graph-backed vertex-minimal Tait counterexample. The machine reaches one of the following:*

1. *a certified structural target;*
2. *a certified large remote primal cut;*
3. *a nonempty closed orbit containing a literal closure-recovery face transfer; or*
4. *a nonempty closed orbit all of whose reentries are intrinsic singleton collars with odd rooted signed-support monodromy.*

In the fourth case the sum of the exact ambient colour discrepancies is zero. Moreover, for every assignment of additive colour translations to the reentry steps, the resulting closed affine colour word has identity linear holonomy and even sign.

Proof. The finite exact-state theorem supplies an initial state at the chosen dart and either one of the first two stopping outcomes or a reachable nonempty closed reentry orbit. In the latter case, classify each reentry by the exact intrinsic-singleton/recovery alternative. If a recovery step occurs, split the closed orbit immediately before and after that step; this retains its source, target, localized loss, restoring gain, and the two ordinary reentry paths around it, giving item 3. Otherwise the same orbit is a chain entirely of intrinsic odd signed steps, giving item 4.

For the two assertions in item 4, extend each escape colouring by zero over the six omitted adjacent-pair edges. A step discrepancy is the sum of its source and target extensions. In characteristic two every intermediate extension appears twice, so a closed chain telescopes to zero. Independently, the linear colour transport of a step is the change between its endpoint canonical colour frames. These changes also telescope on a closed chain. The net linear map is therefore the identity; arbitrary additive translations of the four-element colour group are even, so the entire affine word is even. □

Theorem 18 is not wall exclusion. The recovery alternative already restarts the complete adjacent-pair fusion machine at a different ambient edge, but no well-founded global quantity is yet known to decrease under every such restart. In the all-intrinsic alternative, local oddness is also insufficient: an even number of odd local transports may cancel while both telescoping laws remain true. The source-faithful missing statement is consequently one of the following equivalent kinds of global control: a fixed ambient signed-support observable whose intrinsic charges do not cancel, or a well-founded measure that strictly decreases on recovery restarts and converts the residual intrinsic cycle into a structural target.

Remark 2 (a local Stone–Wales neutralization fails)**.** REFUTED *The most direct attempt to remove a pentagon–heptagon defect pair is not a valid shortcut. At the common twelve-port interface, the pure hexagonal flower and the radius-one Stone–Wales flower have exact supports of sizes* 16533 *and* 16854*, with intersection* 9222*; neither support contains the other. After exhaustive boundary-Kempe closure,* 4047 *hexagonal words remain outside the Stone–Wales closure and* 3924 *Stone–Wales words remain outside the hexagonal closure. In particular the all-constant word is realized by the hexagonal flower but is not in the Stone–Wales closure. Thus neither patch is a monotone replacement for the other at this interface, even after all noncrossing boundary Kempe switches. This refutes the local reverse-Stone–Wales move, not every possible larger-scale formation transport.*

Theorem 19 (mesh isoperimetry and the bounded-cut obstruction)**.** VERIFIED *Let G contain a edge-disjoint row paths and b edge-disjoint column paths, with a designated branch vertex on every row–column pair; arbitrary additional edges and vertices of G are allowed. If a vertex set S has at most $k < \min(a, b)$ boundary edges, then at most k^2 designated branch positions lie on one side of the cut.*

If

$$S_0 \subset S_1 \subset \cdots \subset S_m$$

is a nested chain of such cuts and the set of designated branch positions inside S_i strictly grows at every step, then

$$m \leq 2k^2 + 1.$$

VERIFIED

The hypothesis that the branch-position set strictly grows at every step is load-bearing. The theorem does not say that a graph containing a wall has no long bounded-cut chain. It says that at most $2k^2 + 1$ slabs of such a chain can acquire a new wall branch position. A longer chain must contain slabs which acquire none; they may run through subdivided wall edges or through material attached to the wall. Consequently a large wall does not itself furnish the manuscript's long sequence of saturated cuts. Conversely, the mesh theorem does not prove wall exclusion. It explains why the missing implication must use zero Count and minimality rather than wall size alone.

For the converse direction used in the reduction, one must distinguish mesh positions from actual vertices. The abstract mesh above permits two row–column positions to designate the same vertex, because injectivity is irrelevant to the isoperimetric inequality. A genuine subdivided wall has an injective branch map. On that exact strengthening the bounded-width descent does exclude every sufficiently large wall.

Proposition 9 (a raw width bound excludes a sufficiently large injective mesh)**.** VERIFIED *Fix w, and suppose every graph-backed vertex-minimal zero-Tait-count map has a complete rooted branch decomposition of width at most w. Put*

$$B(w) = 4\left(2^{(6w+1)\sum_{j=0}^{w} j!\,2^{3^j}} - 1\right) + 6.$$

Then no such map contains an a-by-b mesh whose ab designated branch positions are pairwise distinct vertices whenever $B(w) < ab$.

Proof. The checked width-preserving connectedization turns the supplied raw tree into the connected edge-leaf decomposition consumed by the literal-shore descent. Its two rooted subtrees and the exact Count state give

$$|V(M)| \le B(w).$$

On the other hand, injectivity of the branch map sending (i, j) to v_{ij} embeds the ab-element set of mesh positions into $V(M)$, and hence $ab \le |V(M)|$. These inequalities contradict $B(w) < ab$. $\square$

Thus the checked implication is

$$\text{uniform raw width bound} \quad \Longrightarrow \quad \text{exclusion of one sufficiently large injective wall mesh.}$$

The reverse implication still uses the planar wall/grid theorem: exclusion of one fixed wall bounds width and thereby supplies the raw branch decomposition. The proposition does not claim that theorem and does not exclude a wall directly from zero Count. It makes precise that the sole route-specific high-width task is fixed-wall exclusion for least counterexamples; all mathematics after a raw width bound is already checked.

The numerical mismatch with the direct Count state is also exact. A literal width-k cut has 3^k Tait boundary words, hence its unminimized cumulative support carrier has 2^{3^k} states. For every $k \ge 1$,

$$2k^2 + 1 < 2^{3^k}.$$

Therefore any nested width-k sweep which acquires a new mesh branch position at every step is strictly shorter than the raw support carrier and cannot meet the cardinality premise of the raw support pigeonhole theorem. Both the comparison and its direct negated-pigeonhole form are checked: VERIFIED VERIFIED This does not rule out an early accidental state collision or a much smaller proved reachable quotient. It rules out deducing the required repetition from wall size and raw-state finiteness alone. In particular, the source's L1 conclusion that an internal hexagonal ladder exists is not yet the ambient saturated serial decomposition consumed by pumping.

There is also a version which does not require a strict gain at every cut. Suppose a nested width-k sweep begins with at most k^2 mesh branch positions below it and ends with at most k^2 above it. If every consecutive slab acquires at most d branch positions, then the whole mesh has at most $2k^2 + d$ branch positions. Consequently, when $ab > 2k^2 + d$, some single slab must acquire more than d branch positions: VERIFIED VERIFIED Thus a uniformly local one-dimensional cell decomposition cannot sweep from one side of a sufficiently large two-dimensional mesh to the other. This is an obstruction to the literal L1-to-pumping bridge, not a wall-exclusion theorem: a target-aware two-dimensional replacement principle could still use zero Count and minimality in a way that the raw linear receipt does not.

The most immediate two-dimensional candidate is to iterate the positive planar resolution which makes the square rung work. That candidate already fails on one hexagonal face, at the boundary-colour level.

Proposition 10 (hexagonal planar-pairing obstruction)**.** REFUTED *Delete the six vertices of a facial hexagon, leaving six exterior ports in cyclic order. There is no fixed choice of nonnegative integer weights on the five noncrossing perfect matchings of those ports for which, for every nonzero boundary word, the number of proper colourings of the hexagon equals the weighted number of colour-compatible matching caps.*

Proof. Write the three Tait colours as (r,b,p), and take the cyclic boundary word

$$(r, b, p, r, b, p).$$

It has the proper internal hexagon colouring

$$(p, r, b, p, r, b),$$

where entry (i) colours the edge from vertex (i) to vertex (i+1). Indeed the three colours at every vertex are (r,b,p) in some order. This extension is unique. The five noncrossing matchings, written as pairs of port indices, are

$$\begin{array}{lll} (01)(23)(45), & (01)(25)(34), & (03)(12)(45), \\ (05)(12)(34), & (05)(14)(23). & \end{array}$$

Every one contains a pair whose two entries in the displayed boundary word have different colours. Hence no planar matching cap is compatible: every weighted cap count is zero, while the hexagon extension count is one. Lean checks the extension, its uniqueness, all five incompatibilities, and the result for an arbitrary natural-valued weight function. It also checks the opposite sharpness witness ((r,r,b,r,r,b)): one planar cap accepts that word although the hexagon itself does not. ☐

Thus the exact positive square identity does not extend by replacing each hexagon with a sum of planar pairing caps. This is the local shadow of the same high-width issue: after digon and square normalization, a hexagonal web retains boundary information which pairings alone do not express. The proposition does not rule out a richer planar web basis, a target-aware Kempe operation, or a replacement involving a genuinely two-dimensional region. It rules out treating a wall as a collection of independent square resolutions.

Nor can the pairing-cap proposal be rescued merely by allowing the chosen pairing to depend on an unrestricted exterior.

Proposition 11 (parity-restricted target-aware pairing reduction fails)**.** REFUTED *Let U be the set of genuine nonzero Tait words on the six cyclic ports, let $P \subseteq U$ be the support of the hexagonal cell, and let $R_0, \ldots, R_4 \subseteq U$ be the supports of its five noncrossing pairing caps. Let*

$$A = \{w \in U : \textstyle\sum_{j=0}^{5} w_j = 0\}$$

be the parity-admissible language obeyed by every physical six-edge cut. It is false that

$$\forall E \subseteq A, \quad E \cap P = \varnothing \Longrightarrow \exists i \in \{0, \ldots, 4\}, \quad E \cap R_i = \varnothing.$$

Thus no rule which chooses one of the five caps after seeing an arbitrary parity-admissible zero exterior support can replace the hexagon while preserving zero Count.

Proof. For each of the five pairings, there is a genuine nonzero boundary word which is constant on every paired pair but has no proper extension across the hexagon. The five finite witnesses are checked simultaneously by VERIFIED. Consequently $R_i \nsubseteq P$ for every i. Every pairing word has zero total Klein colour because each of its three colours occurs in an equal pair, so $R_i \subseteq A$. The cut-parity assertion itself is not a planar-picture assumption: summing the three incident colours at every vertex of any finite open cubic tangle cancels every interior edge twice and leaves exactly the boundary sum. VERIFIED VERIFIED VERIFIED

Now take the parity-admissible zero exterior $E = A \setminus P$. It is contained in A and disjoint from P, while $R_i \subseteq A$ and $R_i \nsubseteq P$ say exactly that E meets every R_i. Hence no cap works against this one exterior. Lean records both the explicit parity adversary and the displayed quantified negation. VERIFIED ☐

The obstruction now survives the first universal physical constraint, but "parity-admissible" is still weaker than "realized by a planar cubic exterior". We have not shown that $A \setminus P$, or any equally strong hitting support, is physically realized. For comparison, the unrestricted complement form and its physical-class specialization are also checked: VERIFIED VERIFIED. Therefore the remaining loophole is now exact: a route-native target-aware reduction must prove and exploit a structural restriction on the support languages of actual planar exteriors. Moving the existential quantifier or retaining an unrestricted profile does not provide that restriction.

The first natural strengthening is to remember the boundary trace of Kempe components. Fix a pair of colours $\{a, b\}$. For a boundary word w, call a port active when it has colour a or b. A *boundary Kempe witness* is a fixed-point-free matching of the active ports, noncrossing in their cyclic order, such that interchanging a and b on any union of matched pairs produces another word in the same support. A support is *boundary-Kempe closed* when every one of its words has such a witness for each of the three colour pairs.

Proposition 12 (parity plus boundary Kempe closure is insufficient)**.** *Let $E = A \setminus P$ be the parity adversary from Proposition 11. Then E is boundary-Kempe closed. Nevertheless $E \cap P = \varnothing$ and $E \cap R_i \neq \varnothing$ for every one of the five noncrossing pairing caps. Hence parity together with this abstract noncrossing Kempe-closure condition still does not force a target-aware pairing-cap replacement.* *VERIFIED*

Proof. For each genuine six-port word in E and each colour pair, the certificate records one of the five Catalan noncrossing matchings of the active ports. A verified Boolean decoder checks fixed-point-free matching on the active set, noncrossing, and closure under all 2^6 selected port sets; the last test is conditional on the selected set being a union of complete matched pairs. The flat certificate is checked independently for the red–blue, red–purple, and blue–purple pairs. The decoder and these three checks give one witness uniformly for every word and colour pair. VERIFIED The remaining assertions about intersections are exactly the preceding parity-adversary calculation. □

This proposition is deliberately one-sided. The combinatorial part of the physical bridge can be stated without a picture and is now verified. Fix one proper colouring of one literal port tangle. For one colour pair, declare two active darts adjacent when they are the two halves of an interior edge or lie at the same vertex, and take the equivalence relation generated by these local wires. All three such relations are therefore derived from the same tangle and the same colouring; they are not three independently selected boundary matchings.

Theorem 20 (common-web component switching)**.** *Let T be a literal port tangle, let C be a proper nonzero Tait colouring of T, and fix a pair of colours. Every connected class of the bichromatic dart relation is closed across interior edges and across the two active darts at a vertex. More generally, if D is any union of such classes, then swapping the two colours on precisely the darts of D gives another proper colouring of the same tangle. Its boundary word consequently lies in the same exact Tait support as the word of C.* *VERIFIED VERIFIED VERIFIED*

Proof. The generated relation has two kinds of wire. Across an interior-edge wire, the original edge-colouring equation says that both darts have the same colour; hence they are active together. Across a vertex wire, both endpoints are active by definition. Equivalence closure therefore preserves activity, and one equivalence class, or any locally saturated union of classes, is closed under both wire types.

After the switch the two halves of every interior edge still agree. At a vertex, either all active darts in the selected component union are switched or none are. Every inactive dart is fixed, and its colour lies outside the swapped pair. Thus the incident nonzero colours remain distinct (in the cubic case these are the familiar two active darts and one third-colour dart). Restriction to the ports is exactly the corresponding boundary switch, so the switched colouring itself witnesses membership of the new word in the same exact support. □

Theorem 21 (two-ended physical Kempe components)**.** *Let T be a finite cubic port tangle with a proper nonzero Tait colouring, and fix a colour pair. Every bichromatic component which contains a boundary port contains exactly two boundary ports.* VERIFIED VERIFIED VERIFIED

Proof. Replace every active dart by a vertex of an auxiliary finite graph. Join the two active darts based at each cubic vertex, and join the two active darts which are the halves of each interior edge. Properness and cubicity give exactly two active darts at every tangle vertex. Hence an active boundary dart has degree one in the auxiliary graph, while an active interior dart has degree two: it has one neighbour across its tangle vertex and one across its interior edge. These two neighbours are distinct because an interior edge is not a loop.

Now restrict to the connected component containing the chosen boundary dart. Restriction does not change degrees, since every neighbour of a vertex in a connected component lies in the same component. The component is finite and connected, all its degrees are one or two, and it has at least one degree-one vertex. The handshaking lemma makes the number of degree-one vertices even; connectedness gives at most two of them. Thus there are exactly two, and the degree characterization identifies them exactly with the boundary ports. □

Theorem 22 (the physical common web induces the exact boundary matching)**.** *Let T be a finite cubic port tangle with a proper nonzero Tait colouring, and fix a colour pair. Pair each active boundary port with the unique other active boundary port in its bichromatic component, and fix each inactive port. This defines an involution which preserves the active set and has no active fixed point. Moreover, for every union of its active pairs, switching the selected boundary coordinates produces another word in the exact Tait support of T.*

For a six-port tangle, if this physical matching is noncrossing for every proper colouring and every colour pair, then the exact Tait support of T is boundary-Kempe-closed. Noncrossing is the sole hypothesis in this last implication; the matching and every required switch witness are supplied by the one physical common web. VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED

Proof. Reachability in the auxiliary active-dart graph is exactly the equivalence relation generated by the two local wire types used in the common-web construction. The preceding theorem therefore says that a component meeting the boundary has exactly two boundary ports. The other endpoint is unique; using it on active ports and the identity on inactive ports gives the claimed involution. Equality of components is symmetric, so applying the operation twice returns to the original port, and uniqueness excludes an active fixed point.

If a set of boundary ports is a union of complete pairs, take the union of the corresponding physical components. This is a Kempe region in the sense of Theorem 20. Switching that region therefore gives a proper colouring of the same tangle. On an active selected port its boundary colour is swapped; on an unselected port it is unchanged; and on a selected but inactive port the colour is outside the chosen pair, so the formal swap is also the identity. Hence the new boundary word is exactly the requested coordinate switch and belongs to the exact support.

At six ports, adding noncrossing supplies the one remaining field of the finite boundary-Kempe witness. Quantifying this construction over every proper colouring and colour pair proves boundary Kempe closure of the whole exact support. □

The missing geometric bridge has the following precise form. We state it at the level of the spherical combinatorial map, rather than adding an uninterpreted "planar matching" field to the port-tangle carrier.

Theorem 23 (noncrossing of physical Kempe components)**.** *Let M be a finite connected spherical cubic combinatorial map with two-sided faces and connected face dual. Let S and its complement be the two connected shores of an edge bond, and suppose the boundary darts based in S are enumerated in the cyclic order given by retained facial first return. Give the literal open tangle on S a proper nonzero Tait colouring, and fix a pair of colours. Let one bichromatic component join boundary ports a and c, and let a second join b and d. Suppose b lies on the forward cyclic arc from the successor of a to c, while the forward cyclic arc from d returns to a, neither open arc meeting a or c. If b and d are distinct boundary ports, the two components cannot be distinct. Thus physical components cannot join alternating pairs of boundary ports, even when the second component consists only of the local vertex wire between two ports based at one retained cubic vertex.*

In particular, when the bond has exactly six boundary darts, the physical mate involution in cyclic coordinates is a noncrossing matching. Hence the exact ordered Tait support of the literal shore tangle is boundary-Kempe closed: its matching and all switch witnesses are supplied by the physical bichromatic components of the realizing colouring, not by an independently chosen boundary certificate. VERIFIED VERIFIED VERIFIED

Proof. It is enough to rule out two component trails with endpoints occurring in cyclic order a, b, c, d, the first joining a to c and the second joining b to d. Because the complementary shore is connected, join the outside end of the cut edge at c to the outside end of the cut edge at a by a trail in the complementary shore. Appending the two cut edges and the a–c trail in S gives a closed trail Q. The four pieces have pairwise disjoint edge sets: the two shore trails use no cut edge, the shores are disjoint, and the two named cut edges are distinct. Thus Q is a closed trail. Among boundary edges it uses exactly those at a and c.

Label every face by the parity of the number of edges of Q crossed by a dual walk from a fixed face. Sphericity, two-sidedness, and connectedness of the face dual make this label well defined; an edge has differently labelled incident faces exactly when it belongs to Q. At the boundary edge a the label therefore changes. Consecutive boundary darts in retained facial first-return order expose the two sides of one ambient face. Since all boundary edges strictly between a and b avoid Q, the label propagates unchanged from the face immediately after a to the face at b. Likewise, along the other boundary arc from d back to a, it propagates unchanged from the face at d to the face immediately before a. The faces incident with the boundary endpoints b and d consequently have opposite labels.

If the retained endpoints of b and d are distinct, their component trail is nonempty and edge-disjoint from Q. At either endpoint, cubic cyclic rotation identifies the boundary-side face label with the label beside the first or last trail dart; along the trail the label cannot change, because none of its edges lies in Q. Hence its endpoint faces have equal labels.

If the two retained endpoints instead coincide, the two boundary darts are still distinct. Their two boundary edges avoid Q. At a cubic vertex any two distinct incident darts are consecutive in one of the two rotation directions. Exact face-cut parity across the two avoided edges therefore gives the same label to the two boundary-side faces. In either case this contradicts the opposite labels forced by the two boundary arcs. Distinct bichromatic components are vertex-disjoint and therefore edge-disjoint, while Theorem 21 gives exactly two boundary endpoints for every component meeting the boundary. This proves the stated unrestricted noncrossing conclusion for each physical colour pair.

For six boundary positions, assume two chords of the physical mate cross in the standard cyclic order. Orient each chord from its smaller to its larger coordinate. The four endpoints then interleave. Since there are only six coordinates, the two complementary open arcs are finite successor lists avoiding the endpoints of the first chord. The theorem just proved applies to those two physical components and says that they cannot be distinct. But if they were the same component, uniqueness of the other endpoint would make the two chords equal, contrary to interleaving. Thus the physical mate is noncrossing. The common-web switching construction of Theorem 22, transported through the cyclic enumeration, now supplies every field of boundary Kempe closure for the exact ordered support. □

The original v23 CAP5 move-realizability proof tries to pass from this per-pair fact to a joint planar picture by contracting every edge of one colour. The contraction is planar, but the asserted cross-family disjointness does not follow.

Proposition 13 (zero-contraction does not separate different colour-pair systems)**.** REFUTED *There is a spherical properly Tait-coloured cubic combinatorial map and a red colour class whose complete contraction produces a degree-four vertex at which the surviving blue and purple through-pairs have alternating endpoints in the vertex link. Consequently the two post-contraction bichromatic systems are edge-disjoint but are not mutually disjoint as embedded arc systems away from their terminals.*

Proof. Take the tetrahedron on vertices A, B, C, D, with rotations

$$(AB, AC, AD), \quad (BA, BD, BC), \quad (CA, CB, CD), \quad (DA, DC, DB).$$

Give AB, CD colour red, AC, BD colour blue, and AD, BC colour purple. Each vertex sees the three colours, and the displayed rotation system has four facial orbits, so $4 - 6 + 4 = 2$.

Contract both red edges AB and CD. Deleting their four darts and taking first return in the vertex rotation joins the two punctured rotations at the image of AB into

$$(AC, AD, BD, BC).$$

The corresponding colour word in the link of the contracted vertex is

$$(\text{blue}, \text{purple}, \text{blue}, \text{purple}).$$

The red–blue component through AB therefore joins link positions 0 and 2, whereas the red–purple component joins positions 1 and 3. Those two chords alternate. They meet at the contracted interior point even though the surviving blue and purple edge sets are disjoint. Lean checks the source Tait colouring and spherical Euler count, identifies the contracted dart carrier with the source darts minus the four red darts, verifies the first-return rotation law at both contracted vertices and the spherical count $2 - 4 + 4 = 2$, and checks the alternating chords. Thus the failure is caused by contraction itself, not by a nonplanar model. □

This proposition does not refute per-pair noncrossing, nor by itself refute the exceptional-CAP5 conclusion. It refutes the particular v23 step which treats the two different post-contraction systems as mutually disjoint and then forms a Jordan curve from arcs belonging to both systems. A repair must retain their common contracted vertices (equivalently, the common trivalent web before contraction) and prove a genuinely joint statement.

The exact local information lost by the disjoint-arc picture is one chirality bit per contracted edge.

Theorem 24 (local zero-contraction chirality law). VERIFIED VERIFIED *Let M be a cubic combinatorial map with cyclic vertex rotations, let C be a proper Tait edge-colouring, and orient an edge by a dart d. After contracting that edge, write the four surviving darts in the cyclic link order*

$$\rho d, \quad \rho^2 d, \quad \rho\alpha d, \quad \rho^2\alpha d.$$

The two same-colour pairs have exactly one of the following forms:

$$(0,2),(1,3) \qquad \textit{or} \qquad (0,3),(1,2).$$

The first, alternating, pairing occurs if and only if

$$C(\rho d) = C(\rho\alpha d),$$

that is, if and only if the two endpoint rotations present the surviving colours in the same order. The second, adjacent, pairing occurs when the orders are reversed. This agreement predicate is unchanged when d is replaced by αd.

Proof. At the endpoint of d, cubicity and cyclicity say that $d, \rho d, \rho^2 d$ are the three distinct incident darts. Properness and the nonzero Tait condition therefore put the three nonzero colours on their three underlying edges exactly once. The same is true of $\alpha d, \rho\alpha d, \rho^2\alpha d$ at the other endpoint, and the first edges in those two triples have the same colour because d and αd are the two darts of one edge.

Consequently the ordered pair of surviving colours at the second endpoint is either the same as the ordered pair at the first endpoint or its reverse; there is no third possibility. In the same-order case the four-link colour word is (x, y, x, y), so the equal-colour chords are $(0, 2)$ and $(1, 3)$, whose endpoints alternate. In the reversed case it is (x, y, y, x), so the chords are $(0, 3)$ and $(1, 2)$, both adjacent. Changing the orientation of the contracted edge exchanges the two endpoint blocks, so it does not change whether their orders agree. Lean checks the three-dart rotation facts, proper-colouring inequalities, the exhaustive three-colour dichotomy, the same-colour mate at all four positions, and the four-point chord crossing test. ☐

Thus the v23 contraction can be repaired locally, but not by deleting the contracted vertices from the topology: a joint state must retain this chirality at every contraction site and its compatibility along the common web. The theorem does not yet supply that global compatibility law.

The reduct ‘PortTangle‘ deliberately forgets the rotation and disk, so this conclusion cannot follow at that carrier level alone. More importantly, even the full noncrossing bridge does not rescue the reduction, because the adversary already passes the resulting abstract test. We also have not realized the whole adversary simultaneously as the exact support of one planar cubic exterior. The surviving loophole is therefore stronger and genuinely joint: the three colour-pair matchings of an actual exterior arise from one and the same embedded tangle. A successful replacement theorem must exploit that simultaneous physical compatibility (or some still stronger feature of the exact exterior language), rather than parity or three separately existential boundary matchings.

The first joint law is now checked at its proper, edge-set level.

Theorem 25 (Kempe idemposition identity). *Let $C : E \to \{0, r, b, p\}$ be any edge colouring and let $D \subseteq E$ contain only red or blue edges. Switch red and blue exactly on D. If $M_{xy}(C)$ denotes the set of edges whose colours lie in $\{x, y\}$, then*

$$\begin{aligned} M_{rb}(C') &= M_{rb}(C), \\ M_{rp}(C') &= M_{rp}(C) \triangle D, \\ M_{bp}(C') &= M_{bp}(C) \triangle D. \end{aligned}$$

VERIFIED *The same statement is checked for arbitrary three pairwise distinct Klein colours.* VERIFIED VERIFIED

Proof. Outside D no colour changes. On D, every red edge becomes blue and every blue edge becomes red. Thus membership in M_{rb} is unchanged, whereas membership in each of M_{rp} and M_{bp} is toggled exactly once. Pointwise membership is therefore precisely symmetric difference with D. □

This theorem identifies why three separately chosen noncrossing boundary matchings are not yet a physical formation. A switch in one family changes the other two through the *same interior edge set* D. Its endpoint pair on the boundary does not in general determine the connected components of the two symmetric differences: that depends on where the three manifolds share interior segments. Accordingly the next realizability test must carry one common trivalent web (or equivalent edge-incidence data) and check all three equations simultaneously. Treating the three matchings as independent would discard exactly the source's idemposition coupling and cannot discharge the surviving physical-exterior loophole.

The part of this dynamics which belongs to one fixed colour pair is completely rigid, not merely existential.

Theorem 26 (same-family physical Kempe persistence)**.** *Let T be a finite cubic port tangle with a proper Tait colouring C, fix a colour pair q, and let R be a union of connected components of the q-bichromatic web. If C' is obtained by switching the two colours on R, then the q-web has exactly the same local edges and the same connected components in C' as in C. Consequently its physical boundary mate is unchanged.* VERIFIED VERIFIED VERIFIED

Proof. At every dart, switching q exchanges the two members of q and fixes every other colour. Membership in the unordered pair q is therefore unchanged. The two kinds of local web edge—the two active darts at a cubic vertex and the two halves of an active interior edge—are consequently unchanged pointwise. Their equivalence closure, hence each connected component, is the same relation before and after the switch. The physical mate is defined as the unique other boundary port in that component (and fixes inactive ports), so it too is unchanged. □

This persistence is still not a boundary characterization of a literal hexagonal patch. The failure persists even if one remembers a deterministic mate at every state rather than choosing a fresh witness after each switch.

Proposition 14 (persistent pairwise mates do not determine the hexagon)**.** *There is a sixty-word set A of six-port Tait words with the following properties.*

1. *A is nonempty and is disjoint from the exact boundary support of the literal hexagonal patch.*
2. *For every $w \in A$ and every one of the three colour pairs q, there is a deterministic noncrossing active mate $m_q(w)$.*
3. *Whenever a union of complete $m_q(w)$-pairs is switched, the resulting word w' belongs to A and $m_q(w') = m_q(w)$.*

The set is colour-blind: it is the union of the ten equality-pattern classes

$$\begin{array}{ccccc} 001001, & 001010, & 010001, & 010010, & 010212, \\ 010221, & 011202, & 011220, & 012102, & 012120, \end{array}$$

where equal digits require equal colours and distinct digits require distinct colours. VERIFIED VERIFIED VERIFIED VERIFIED

Proof. Define A by the ten displayed equality patterns. Because the definition uses only equality and inequality, it is invariant under renaming the three Tait colours. Direct finite reduction over all 3^6 boundary words gives $|A| = 60$. For each of the three colour pairs and each supported word, the certificate supplies one of the five noncrossing matchings on the active ports. The checker verifies both the ordinary active-matching conditions and all 2^6 selected port sets; for every selected set which is a union of complete pairs, it verifies membership of the switched word in A and equality of the selected mate before and after the switch. The same complete enumeration verifies that no word in A extends across the literal hexagon. Finally the equality pattern 001001, instantiated by two distinct Tait colours, supplies an explicit member of A. The flat tables used to suggest the five-way matching choices are not trusted: the Boolean witness condition is proved equivalent to the mathematical proposition, and the kernel reduces every row of the three separate certificates. □

Proposition 15 (the persistent adversary does not obstruct the cap menu)**.** *Let A be the sixty-word language of Proposition 14. In the zero-based enumeration of the five noncrossing planar pairing caps, A is disjoint from the supports of caps $0 = (01)(23)(45)$ and $2 = (03)(12)(45)$. In particular A does not meet every planar cap.* VERIFIED VERIFIED VERIFIED

Proof. The finite checker tests every word in A against the two displayed pairing relations and finds no compatible word. Either disjointness statement alone negates the assertion that A meets all five cap supports. □

This corrects an important scope distinction. The sixty-word language is a witness against compressing the common physical web to three persistent boundary mates; it is not an obstruction to replacing a hexagon by a member of the five-cap menu. The larger parity adversary in Proposition 11 is the language which meets every cap.

Nor should the universal physical cap-menu implication be treated as a cheap local search. In the route's intended use, the exterior is obtained by deleting a facial hexagon from a hypothetical bridgeless planar cubic counterexample. Exhibiting such an exterior already exhibits the counterexample, while proving that one cap always works is precisely a uniform reducibility theorem for the dual degree-six vertex. Even that strong result would not by itself bound width: it supplies no exclusion of large pentagon–heptagon wall patches. Thus the central width problem remains global.

Thus even the exact boundary shadow of Theorem 26, imposed separately for all three pairs and all of their successor states, admits a language which no literal hexagon realizes. What is absent is precisely the cross-family clause in Theorem 25: all three updates must be induced by one common selected interior edge set. The next state therefore has to retain a common trivalent web, or equivalent shared edge-incidence data; adding further independent pairwise closure fields cannot close this gap.

The corresponding unquotiented joint carrier is exact. This separates two questions which must not be conflated: whether the source's simultaneous idemposition has been represented faithfully, and whether that representation has already been compressed to a bounded interface receipt.

Theorem 27 (joint physical idemposition on one carrier)**.** *Let T be a cubic port tangle and let C be a proper Tait colouring of its common dart carrier D. Write*

$$\mathcal{M}(C) = (M_{rb}(C), M_{rp}(C), M_{bp}(C)).$$

If $R \subseteq D$ is a physical Kempe region for a colour pair q, and C' is obtained by switching q on R, then the q-coordinate of $\mathcal{M}(C)$ is fixed and the other two coordinates are both replaced by their

symmetric difference with the same *set R. Moreover the map $C \mapsto \mathcal{M}(C)$ is injective on proper Tait colourings.*
VERIFIED VERIFIED

Proof. The first assertion is pointwise. On R, the two colours of q are interchanged. Membership in M_q is unchanged, while membership in each companion bichromatic support is toggled once. Outside R, nothing changes. Because all three supports are subsets of the one dart carrier, the two companion updates use literally the same R, rather than two boundary sets merely having the same endpoints.

For injectivity, a nonzero colour is recovered from its membership pattern in the three supports:

	M_{rb}	M_{rp}	M_{bp}
r	1	1	0
b	1	0	1
p	0	1	1.

Thus equal triples give equal colours on every dart and hence equal colourings. □

This theorem is the first carrier in this section which satisfies the exact cross-family law demanded by the source. It is intentionally not yet a finite-profile theorem: $\mathcal{M}(C)$ remembers subsets of every interior dart of T. The remaining interface problem is therefore precise. One must quotient this shared carrier to bounded boundary data while proving that the quotient still determines simultaneous physical updates and the replacement support. Projecting the three coordinates separately is ruled out by Proposition 14.

Even retaining the three projected coordinates in one boundary state does not repair that loss.

Proposition 16 (projected joint idemposition does not determine the hexagon)**.** *Let*

$$\partial\mathcal{M}(w) = (A_{rb}(w), A_{rp}(w), A_{bp}(w)),$$

where $A_q(w)$ is the set of ports whose colours belong to q. There is a nonempty sixty-word language A, disjoint from the literal hexagon support, with a deterministic noncrossing mate $m_q(w)$ for every $w \in A$ and every colour pair q, such that the following holds. Whenever S is a union of active $m_q(w)$-pairs and w' is obtained by switching q on S, then

$$w' \in A, \qquad m_q(w') = m_q(w), \qquad \partial\mathcal{M}(w') = \partial\mathcal{M}(w)\,\mathrm{toggle}_q\,S.$$

In the final equation the q-coordinate is fixed and both companion coordinates are symmetrically differenced with the same set S. VERIFIED VERIFIED

Proof. Use the language and deterministic mates from Proposition 14. That certificate already proves $w' \in A$ and persistence of the operated mate. For the displayed joint equation, argue at one port. If the port is outside S, its colour and all three membership bits are unchanged. If it lies in S, its colour is one of the two colours of q; exchanging them fixes membership in q and toggles membership in each of the other two pairs. This is exactly the three-coordinate toggle by the one set S. □

The proposition does not contradict Theorem 27. It identifies the information lost by restricting that theorem to the ports. At the boundary, the three support coordinates are already determined by w, so writing them as one triple adds no shared-interior information. The next receipt must retain a quotient of how the three manifolds overlap and reconnect inside the tangle (including the contraction chirality above); a word, three mates, and their entire boundary transition graph are still insufficient.

There is also a sharp warning against treating an arbitrary *signed* pairing identity as if it were already a planar counting argument. Write a perfect matching of the six ports as, for example, 01|23|45, and let $\delta_M(w) \in \{0,1\}$ be one precisely when the boundary word w is constant on every pair of M. Enumerate the fifteen matchings by

$$\begin{array}{r|ccccc} 0 & 01|23|45 & 01|24|35 & 01|25|34 & 02|13|45 & 02|14|35 \\ 5 & 02|15|34 & 03|12|45 & 03|14|25 & 03|15|24 & 04|12|35 \\ 10 & 04|13|25 & 04|15|23 & 05|12|34 & 05|13|24 & 05|14|23. \end{array}$$

Proposition 17 (a signed pairing tensor has exactly the adversary support)**.** *Define*

$$\begin{aligned} k_0 &= \delta_8 - \delta_4, \\ k_1 &= \delta_{10} - \delta_4, \\ k_2 &= -\delta_0 + \delta_1 + \delta_3 - 2\delta_4 + \delta_{11}, \\ k_3 &= -2\delta_4 + \delta_5 + \delta_9 - \delta_{12} + \delta_{13}, \qquad K = k_0 + 2k_1 + 4k_2 + 8k_3. \end{aligned}$$

For every genuine nonzero six-port Tait word w,

$$K(w) \neq 0 \quad \Longleftrightarrow \quad w \in A \setminus P.$$

In particular every k_i, and hence K, vanishes on the literal hexagon support P, while the support of K is exactly the whole parity adversary. VERIFIED VERIFIED VERIFIED

Proof. There are three choices at each of the six ports. For each of the 3^6 words, evaluate the fifteen Kronecker-delta products and then the four displayed integer combinations. Independently test the two defining conditions for $A \setminus P$: zero total Klein colour and failure of the six-step hexagon extension recurrence. The resulting two Boolean columns agree in all 729 rows. The same exhaustive evaluation checks that each k_i is zero on every row accepted by the recurrence. Lean evaluates both closed tables from the displayed mate involutions and proves the biconditional and the four coordinate equalities. □

The coefficients above are signed. Consequently K is a formal integer linear combination of pairing contractions, not the nonnegative extension count of one planar exterior, and several of its matchings are crossing in the cyclic boundary order. The proposition therefore does not realize the adversary physically. It says something more diagnostic: ordinary linear pairing identities alone can carry the entire abstract obstruction. To rule it out one must use the positive cone and the common embedded web from which all three two-colour pairings arise. This is exactly the extra content suggested by the source's Penrose–Kauffman evaluation and idemposition calculus.

Proposition 18 (pairing-linear positivity cannot separate the adversary)**.** *Let*

$$L(w) = \sum_M c_M \delta_M(w), \qquad c_M \in \mathbb{Z},$$

be any integer linear combination of the fifteen pairing contractions. If L vanishes on the literal hexagon support P, then L is neither strictly positive nor strictly negative on all of $A \setminus P$. VERIFIED VERIFIED

Proof. Take the following five adversary words:

$$q_0 = (r, r, b, r, r, b), \quad q_1 = (r, r, b, r, b, r), \quad q_2 = (r, r, b, p, b, p),$$
$$q_3 = (r, b, r, r, b, r), \quad q_4 = (r, b, r, p, p, b).$$

Lean checks $q_i \in A \setminus P$ for all five. VERIFIED Now list eleven words accepted by the hexagon:

$$\begin{array}{lll} 0\ (r,r,r,r,r,r) & 1\ (r,r,r,r,b,b) & 2\ (r,r,r,b,b,r) \\ 3\ (r,r,b,b,r,r) & 4\ (r,r,b,b,b,b) & 5\ (r,r,b,p,p,b) \\ 6\ (r,b,b,r,r,r) & 7\ (r,b,b,r,p,p) & 8\ (r,b,b,b,b,r) \\ 9\ (r,b,p,r,b,p) & 10\ (r,b,p,p,b,r). & \end{array}$$

VERIFIED For each one of the fifteen perfect matchings M, direct evaluation gives

$$\sum_{i=0}^{4} \delta_M(q_i) = \delta_M(p_0) - \delta_M(p_1) - \delta_M(p_3) + \delta_M(p_4) - \delta_M(p_5)$$
$$- \delta_M(p_6) + \delta_M(p_7) + \delta_M(p_9) + \delta_M(p_{10}).$$

This finite row identity is checked for every M. VERIFIED Multiply its M-th instance by c_M and sum over M. Linearity turns the left side into $\sum_i L(q_i)$, while every term on the right is zero because each p_j lies in P. Thus $\sum_i L(q_i) = 0$. The five summands cannot all be strictly positive, and they cannot all be strictly negative. VERIFIED □

Thus the most direct Penrose repair is now ruled out precisely. Positivity of the *same fifteen pairing coordinates* cannot supply the missing physical-exterior restriction: their entire linear span has no strict separator. A successful argument must enlarge the observable to a common trivalent web (retaining shared-edge incidence and the three coupled idemposition equations), or exploit a nonlinear positivity statement about actual extension counts. This is a narrowing of the high-width problem, not evidence against the source's broader operator programme.

There is one sign convention hidden in that conclusion which must be audited before moving on. The source's colouring operator F_X is the *unsigned* nonnegative extension-count operator. By contrast, an open Penrose contraction written in the fifteen pairing coordinates carries a boundary sign. For a six-port word with even colour multiplicities, list the occurrences of each colour in boundary order and pair consecutive occurrences. Let

$$\sigma(w) = (-1)^{\mathrm{cr}(M(w))}$$

be the parity of the crossings among those three chords. If a colour has odd multiplicity, all fifteen pairing contractions vanish and the value assigned to $\sigma(w)$ is immaterial.

Proposition 19 (the gauge-positive pairing cone still contains the persistent adversary)**.** *Put*

$$L = \delta_4 - \delta_5 - \delta_9 + \delta_{10} + \delta_{12} - \delta_{13}.$$

Then, on every genuine nonzero six-port Tait word,

$$\sigma(w)L(w) \geq 0, \qquad \sigma(w)L(w) > 0 \quad \Longleftrightarrow \quad w \in E,$$

where E is the sixty-word persistent Kempe adversary. Moreover L vanishes on the literal hexagon support P. For each of the three four-vertex plane cubic tree shapes there is a word in its support on which σL is positive. VERIFIED VERIFIED VERIFIED VERIFIED

Proof. In the displayed enumeration of matchings, crossing parity alternates with the index. For an even-multiplicity word, the first compatible matching in that enumeration is exactly the matching obtained by pairing consecutive occurrences of each colour. Thus its index parity computes $\sigma(w)$. Evaluate this sign and the six displayed pairing coordinates on the $3^6 = 729$ boundary words. The corrected value is nonnegative in every row, and its positive rows are precisely the ten colour-blind equality patterns defining E, hence precisely its sixty words.

The vanishing on P also has a short algebraic check. In the notation of Proposition 17,

$$L = k_1 - k_3,$$

and both kernel coordinates vanish on P. Finally the three positive rows

$$\texttt{rrbrrb}, \qquad \texttt{rbrrrb}, \qquad \texttt{rbrpbp}$$

(spaces only separate the six boundary positions) are respectively accepted by the separated-path, adjacent-path, and tripod tree supports. Lean checks the complete table, the kernel identity, and these three explicit extension witnesses. □

This proposition is stronger than the ungauged linear-separator failure, but its scope is equally important. It does *not* say that L, or its positive support E, is realized by one planar cubic exterior. It says that sign-corrected linear invariant theory plus coordinatewise positivity still permits exactly the already known persistent obstruction. The next condition must therefore use the nonlinear, integral semigroup of tensors realized by actual common trivalent webs (or an equivalent shared-interior receipt). That distinction keeps this negative result aligned with the source's unsigned operator F_X, rather than mistaking a formal pairing expansion for an extension count.

The remaining gap in that last paragraph can be seen in one small graph. The positive tensor is an honest unsigned extension count, but only after the cyclic order of the six ports is allowed to cross.

Proposition 20 (the gauge adversary is a crossed pentagon–leaf tensor)**.** *Let Q be the abstract trivalent network with internal vertices $v_0, \ldots, v_5$, internal edges*

$$v_0v_1,\ v_0v_4,\ v_1v_2,\ v_1v_5,\ v_2v_3,\ v_3v_4,$$

and six boundary half-edges attached, in the prescribed cyclic port order, at

$$v_3, v_0, v_2, v_4, v_5, v_5.$$

For every nonzero Tait boundary word w, the number $N_Q(w)$ of proper internal edge-colourings is

$$N_Q(w) = \sigma(w)L(w) = \begin{cases} 1, & w \in E, \\ 0, & w \notin E. \end{cases}$$

In particular each supported word has exactly one internal colouring.

Now adjoin boundary vertices $b_0, \ldots, b_5$ at the six half-edges and the boundary cycle $b_0b_1 \cdots b_5b_0$. The resulting graph contains $K_{3,3}$ as a minor. Consequently Q has no embedding in a closed disc whose boundary ports occur in the displayed cyclic order. VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED

Proof. Name the internal edge colours, in the order displayed above, by $x_{01}, x_{04}, x_{12}, x_{15}, x_{23}, x_{34}$. At the pendant vertex v_5, properness forces

$$x_{15} = w_4 + w_5.$$

At v_1 the other two colours are therefore the unordered pair $\{w_4, w_5\}$: they are distinct and nonzero, and in $\mathbb{F}_2^2$ two pairs of distinct nonzero colours having the same sum are the same unordered pair. There are consequently only two candidate orientations,

$$(x_{01}, x_{12}) = (w_4, w_5) \quad \text{or} \quad (x_{01}, x_{12}) = (w_5, w_4).$$

Once one is chosen, the remaining vertices force, successively,

$$x_{23} = x_{12} + w_2, \qquad x_{34} = x_{23} + w_0, \qquad x_{04} = x_{34} + w_3.$$

Thus every internal colouring is one of two explicitly displayed candidates; no enumeration over arbitrary internal colourings is involved. Evaluating local properness for those two candidates on the 3^6 boundary words gives one valid candidate precisely on the ten equality-pattern classes defining E, and none elsewhere. The same finite table agrees row by row with σL. Lean proves the forcing argument, the equivalence between the two-candidate fibre and the full colouring fibre, and the 729-row boundary check.

For the topological assertion, use the following six disjoint connected branch sets in the boundary-cycle graph:

$$\begin{aligned} &A_0 = \{v_1, v_2, v_5\}, \quad &&A_1 = \{b_0, b_1\}, \quad &&A_2 = \{v_4, b_3, b_4\}, \\ &B_0 = \{v_3\}, &&B_1 = \{b_2\}, &&B_2 = \{b_5\}. \end{aligned}$$

Each A_i is adjacent to each B_j: the nine witnessing edges are, row by row,

$$\begin{aligned} &v_2v_3, \quad v_2b_2, \quad v_5b_5, \\ &b_0v_3, \quad b_1b_2, \quad b_0b_5, \\ &v_4v_3, \quad b_3b_2, \quad b_4b_5. \end{aligned}$$

Contracting each branch set therefore gives $K_{3,3}$. If Q had the claimed disc embedding, the six boundary-cycle edges could be drawn along the rim, producing a plane embedding of this augmented graph. This is impossible because planarity is minor-closed and $K_{3,3}$ is nonplanar. □

This identifies the failure more sharply than positivity alone. The tensor σL is the unsigned, integral, uniquely-colourable tensor of a tiny common trivalent network; what prevents it from being a physical exterior is the *joint cofacial order* of that network. The proposition does not claim that every possible realization of the same tensor has this particular $K_{3,3}$ model. It says that any successful local receipt must be strong enough to distinguish a disc-realizable common web from this crossed one.

There is no hidden second pairing-linear realization with different multiplicities on the sixty supported words. The following calculation shows that the preceding Penrose tensor spans the whole pairing-linear ray with that support.

Proposition 21 (rigidity of the Penrose ray on the persistent adversary)**.** VERIFIED *Let E be the sixty-word persistent adversary. Using the zero-based enumeration of the fifteen pairings displayed above, let*

$$T_c(w) = \sum_{i=0}^{14} c_i \delta_i(w), \qquad c_i \in \mathbb{Z}.$$

If $T_c(w) = 0$ for every nonzero six-port Tait word outside E, then

$$c_i = c_4 d_i \quad (0 \le i < 15), \qquad (d_0, \ldots, d_{14}) = e_4 - e_5 - e_9 + e_{10} + e_{12} - e_{13}.$$

Consequently $T_c = c_4 L$, and for every $w \in E$,

$$\sigma(w) T_c(w) = c_4.$$

Proof. The following fourteen words all lie outside E:

`rrrrrr`, `rrrrbb`, `rrrbrb`, `rrrbbr`,
`rrbbrr`, `rrbbbb`, `rrbbpp`, `rrbpbp`,
`rbrbrr`, `rbrbbb`, `rbrbpp`, `rbbbrb`,
`rbprbp`, `rbpprb`.

Evaluate the fifteen indicators $(\delta_0, \ldots, \delta_{14})$ on these rows. The resulting fourteen homogeneous equations are independent. Successive integer elimination, with c_4 left free, gives

$$c_5 = -c_4, \quad c_9 = -c_4, \quad c_{10} = c_4, \quad c_{12} = c_4, \quad c_{13} = -c_4,$$

and every other c_i is zero. The checked proof evaluates the fourteen rows directly from the displayed mate involutions and performs this elimination over $\mathbb{Z}$; hence no division or rank-over-a-field assumption is hidden. This proves $T_c = c_4 L$. On E, the already checked gauge-corrected Penrose tensor is the extension count of the crossed pentagon–leaf network and equals 1. Therefore $\sigma(w) T_c(w) = c_4$ on all of E. □

This is an algebraic rigidity result. Within the pairing-linear ansatz it closes the unequal-multiplicity loophole: support exactly E forces one constant gauge-corrected multiplicity. It does *not* prove that a positive multiple of this ray is realized by a plane cubic tangle. That remaining condition is topological, and the crossed realization above shows why cyclic order cannot be discarded.

Theorem 28 (the exact width criterion). PROSE *Let $\mathcal{M}$ be the class of vertex-minimal zero-target normal-form instances. Suppose that for every fixed width k, ordered width-k tree decompositions carry a finite exact typed context semantics, and equal comparable states admit a strict zero-target replacement as in Theorem 4. Then the following are equivalent:*

1. *the vertex counts of the members of $\mathcal{M}$ have a uniform finite bound;*
2. *the treewidths of the members of $\mathcal{M}$ have a uniform finite bound.*

If the width bound and the finite context semantics are effective, then the vertex bound is effective.

Proof. A graph on at most N vertices has treewidth at most $N - 1$, proving $1 \Rightarrow 2$. Conversely, assume every member of $\mathcal{M}$ has treewidth at most k. Convert a width-k decomposition to a reduced nice rooted binary decomposition. Its introduce nodes account for at most one graph vertex each; join, forget, and leaf nodes introduce none. By hypothesis its ordered interfaces and exact exterior behaviours form one finite typed state set, say of cardinality $q(k)$. Minimality and exact replacement forbid a repeated state on an ancestor–descendant chain. Theorem 4, with $c = 1$, gives

$$|V(G)| \leq 2^{q(k)} - 1$$

for every $G \in \mathcal{M}$. This proves $2 \Rightarrow 1$, and the displayed formula is effective when the stated inputs are. $\square$

Thus, *relative to exact finite tree semantics and D3 replacement*, the tree-form repair of D1 is precisely a width theorem for minimal zero-target instances. This is not a literal proof of the manuscript's linear hexagonal-corridor L1. It is a compositional reformulation of that failed global step. The claimed equivalence below always has these semantic and physical hypotheses in scope.

Neither the planar wall theorem nor a branchwidth dichotomy supplies the replacement by itself. Model paths of a wall subdivision may contain arbitrarily many degree-two vertices before they are viewed inside the cubic host, and ambient material may attach along them. Cutting only abstract rails therefore need not give a saturated cut of the original map; saturating it can make the interface grow with the model paths. The mesh theorem above sharpens this warning: only the number of slabs that acquire new designated branch positions is bounded. A wall becomes useful to the proof only after zero Count and minimality turn it into a strict exact-state replacement or exclude it altogether.

Proposition 22 (two exhaustive geometric branches for the global corridor)**.** CONDITIONAL *Fix constants B, L above the relevant profile thresholds. Assume both of the following source-preserving constructions, including all the transversal data in Proposition 5.*

1. *A bounded-face geodesic of length L factors through a finite manuscript profile and equal profiles admit a physical deletion splice.*

2. *A face of length greater than B admits an analogous perimeter factorization and a strictly size-decreasing, target-preserving splice.*

Then every sufficiently large irreducible obstruction is pumpable, with no bound on W_- and no claim that most faces are hexagons.

Proof. Apply Theorem 2. In its bounded-face branch, the first assumed construction supplies the required composable cut chain and splice. In its long-face branch, the second does. The vertex/face identity $F = V/2 + 2$ makes the dichotomy's face threshold an effective vertex threshold. Thus every obstruction above the maximum of the two construction thresholds is pumpable. $\square$

Theorem 2 proves that the two *geometric inputs*—a bounded-face geodesic or a long face—are exhaustive. It does not prove either transfer construction. This is nevertheless a sharper research front than a uniform-curvature hypothesis: the high-negative- curvature branch is retained explicitly instead of being discarded by an invalid unsigned-defect inference.

The original v23 manuscript proposes a different genuinely two-dimensional move: its Theorem 4.9 claims that annular Kempe-closure face generators span the whole boundary-zero cycle space. That would be highly relevant here, so we rechecked the actual proof rather than relying on its later summaries. One of its two concluding inferences is not valid as stated.

Proposition 23 (the v23 annihilator inference needs a containment lemma)**.** REFUTED *For subspaces U, W of a finite-dimensional space with a nondegenerate bilinear form, the condition*

$$U^{\perp} \cap W = \{0\}$$

does not by itself imply $W \subseteq U$.

Proof. Over $\mathbb{F}_2$, take the ambient space $\mathbb{F}_2^2$ with its standard dot product, let $W = \langle(1,0)\rangle$, and let $U = \langle(1,1)\rangle$. A vector $(x,0) \in W$ orthogonal to $(1,1)$ has $x = 0$, so $U^\perp \cap W = \{0\}$. But $(1,0) \notin U$, hence $W \not\subseteq U$. The Lean theorem proves both assertions on literal two-coordinate functions over $\mathbb{Z}/2\mathbb{Z}$. $\square$

In v23, W is $W_0(H)$ and U is the span of the purified face generators. The proof establishes the displayed annihilator condition and immediately concludes $W_0(H) \subseteq U$. It does not first establish $U \subseteq W_0(H)$; indeed the definition of a purified generator explicitly permits support on boundary edges, while $W_0(H)$ vanishes there. The dual-forest peeling also treats the unique *tree* edge incident with a leaf face as though it were the unique remaining dual-graph edge, which need not hold in a spanning forest. Proposition 23 by itself only refutes that inference. The literal spanning statement is in fact false for an additional, concrete reason.

Proposition 24 (triangular-prism counterexample to the literal v23 spanning theorem)**.** REFUTED *Let H be the triangular prism embedded as an annulus, with its two triangular faces as the two boundary cycles and its three rectangular faces as the internal faces. Define the v23 generator $X_f^{ab}(C)$ exactly as in the final Kempe-closure generator-family definition: it has coefficient equal to the third colour on those edges of ∂f whose colours lie in $\{a,b\}$, and coefficient zero elsewhere. Even if one permits generators from* every *proper Tait colouring of H, rather than only the Kempe closure of a fixed colouring, their span does not contain $W_0(H)$.*

Proof. Encode the three nonzero colours by $\mathbb{F}_2^2$. Number the top triangle edges t_0, t_1, t_2, the bottom triangle edges b_0, b_1, b_2, and the spokes s_0, s_1, s_2, with rectangular face f_i incident with t_i, b_i, s_i, s_{i+1} (indices modulo three). Define a linear functional λ on $\mathbb{F}_2^2$-valued edge chains by summing the first coordinate on

$$t_0, t_1, t_2, s_0, s_1, s_2.$$

For every proper colouring, the three colours on t_i, s_i, s_{i+1} are pairwise distinct. The first two inequalities are properness at the two endpoints of t_i; the third follows because the three top edges have pairwise distinct colours and s_i, s_{i+1} are the respective missing colours at consecutive top vertices. Hence any chosen colour pair contains exactly two of these three edges. On $X_{f_i}^{ab}(C)$ the two selected edges carry the same third-colour coefficient, so their first coordinates cancel in $\mathbb{F}_2$. The bottom edge does not occur in λ. Thus λ annihilates every generator arising from every proper colouring, and therefore annihilates their span.

Now let w have first coordinate one on s_0 and be zero everywhere else. Every vertex of this annulus lies on a boundary component, so the v23 Kirchhoff conditions, which are imposed only at interior vertices, are vacuous; moreover w vanishes on both boundary cycles. Thus $w \in W_0(H)$. But $\lambda(w) = 1$, so w is not in the generator span. The Lean model uses all 3^9 edge assignments satisfying the six literal cubic properness conditions and proves this separating-functional argument without enumerating a Kempe closure. $\square$

The triangular prism is planar, cubic, and bridgeless, and its two triangular cycles are disjoint boundary components, so it satisfies the hypotheses as printed. It is boundary-degenerate in the precise sense that it has no interior vertices. That observation does not repair the theorem: the same failure occurs with two pentagonal boundaries and ten genuine interior vertices.

Proposition 25 (positive-depth dodecahedral counterexample)**.** REFUTED *Let G be the dodecahedral graph in its standard spherical embedding, and let H be obtained by declaring the disjoint pentagonal faces numbered* 0 *and* 6 *to be its two boundary components. Use the remaining ten faces as the internal faces. For the explicit Tait colouring C_0 recorded in the certificate, the literal Definition–4.8 generators from the exact three-colour Kempe closure of C_0 do not span $W_0(H)$.*

Proof. The two selected face boundaries each contain five edges and are disjoint; the ten vertices not on them are

$$6, 7, 8, 9, 10, 11, 12, 13, 14, 18.$$

In C_0, for each of the three unordered colour pairs the corresponding bicoloured subgraph is connected. A Kempe component for two distinct colours is therefore the whole bicoloured subgraph, so switching it is just a global transposition of those two colours. Switching a colour with itself changes nothing. Induction on the Kempe reachability relation shows that every colouring in the exact source closure has the form σC_0 for a permutation σ of the three colours.

Write e_{ij} for the dodecahedral edge joining vertices i, j. Define the linear functional λ by summing the first Klein coordinate on

$$e_{01}, e_{3,19}, e_{9,10}, e_{10,11}, e_{11,18}, e_{12,13}, e_{12,16}, e_{15,16}, e_{16,17}, e_{17,18}, e_{18,19}.$$

The finite certificate checks, for all six permutations σ, all ten internal faces, and all three colour pairs, that

$$\lambda\big(X_f^{ab}(\sigma C_0)\big) = 0.$$

Hence λ annihilates the span of the literal source generators.

Let w have first coordinate one on the four-edge path

$$e_{0,10}, e_{10,11}, e_{11,18}, e_{18,19}$$

and vanish elsewhere. Its endpoints lie on a boundary pentagon, its three internal path vertices have even incidence, and all other internal vertices have incidence zero. It also vanishes on both boundary cycles. Thus $w \in W_0(H)$. Exactly three of its four edges occur in the support of λ, so $\lambda(w) = 1$ over $\mathbb{F}_2$. Therefore w is not in the generator span. Lean checks the graph, the two five-edge boundaries, their disjointness, all ten Kirchhoff equations, the exact Kempe-closure reduction, and the separating calculation. □

Thus neither boundary degeneracy nor a mere positive-depth hypothesis can repair the literal v23 theorem. A route-realizability restriction would be a new statement needing a new proof. The v23 spanning theorem cannot serve as the missing high-width M1 theorem. A corrected need-based generator theorem would have to prove that the actual route discrepancies lie in its generated subspace and then connect those algebraic moves to a strict physical replacement. This refutation does not affect weak L2 or the later profile certificates, which make different claims.

One elementary combinatorial ingredient of the long-boundary branch is the following sweep lemma.

First isolate its finite part from the geometric assertion that a particular embedded graph supplies such stacks.

Lemma 24 (bounded-stack sweep pigeonhole)**.** VERIFIED *Let A and K be finite sets. At each of n positions record one element of A and, for each of r families, a K-labelled stack of depth at most D. If*

$$n > |A| \left(\sum_{d=0}^{D} |K|^d \right)^r,$$

then two distinct positions carry exactly the same record.

Proof. A stack is encoded by its exact depth $d \in \{0, \ldots, D\}$ and a word in K^d. Thus the stack carrier has exactly $\sum_{d=0}^{D} |K|^d$ elements, and the full record carrier has exactly $|A|(\sum_{d=0}^{D} |K|^d)^r$ elements. The conclusion is the ordinary pigeonhole principle. The Lean statement uses the literal dependent sum $\Sigma_{d:\mathrm{Fin}(D+1)} \mathrm{Vector}(K, d)$, proves this cardinality exactly, and returns the two unequal indices together with their equal states. ☐

Lemma 25 (noncrossing length–depth alternative)**.** VERIFIED *Let n ordered boundary positions carry a letter from a finite alphabet A and, for each of r families, a labelled noncrossing partial matching whose arc labels lie in a finite set K. Fix $D \geq 0$ and put*

$$S = |A| \left(\sum_{d=0}^{D} |K|^d \right)^r .$$

If $n > S$, then either some matching has nesting depth greater than D, or two cut positions have identical sweep states (the local letter together with the ordered stacks of currently open labelled arcs in every family).

Proof. Sweep the boundary in its given order. Noncrossing means that the open arcs of one family close in last-in–first-out order, so its complete unresolved connectivity at a cut is an ordered stack. If every nesting depth is at most D, one stack has at most $\sum_{d=0}^{D} |K|^d$ possible values. Together with the local letter and the r independent stacks, Lemma 24 gives two equal sweep states as soon as there are more than S positions. If the depth bound fails, the first alternative holds.

In the formal statement, each arc has two ordered endpoints and a label; distinct arcs are endpoint-disjoint and arcs are listed by increasing left endpoint. For two arcs open at the same cut, a later left endpoint followed by a later right endpoint would be a crossing. Equality of the right endpoints is excluded by endpoint-disjointness. Hence their right endpoints strictly decrease, which is the claimed LIFO order. The resulting raw lists are encoded by the exact stack carrier of Lemma 24; equality of codes is then transported back to literal equality of the local letter and every stack. ☐

For the route, the two Kempe families suggest the matching stacks and capped face progress supplies additional finite labels. However those matchings belong to a colouring state, whereas the physical graph surgery must be defined before a global colouring exists. In the shallow case an equal sweep state is therefore only a candidate length interface. In the deep case nested arcs are only candidate layer boundaries. One still has to construct simple ordered transversals from the graph embedding and prove that the cumulative zero-count profile factors through them. The lemma proves a finite classification of abstract noncrossing matchings; it proves neither the graph-level dichotomy nor the splice.

1.7 The exact finite interface

The colouring part of the finite-state claim does not depend on the current Cell–3 carrier. It is the following elementary fibre-product theorem.

Theorem 29 (bounded-interface Count is a strict composition law)**.** PROSE *Let $\mathcal{C} = \{r, b, p\}$. A finite tangle T has disjoint ordered input and output port sets I, O, and $\mathrm{Col}_T(x, y)$ denotes its proper nonzero Tait colourings with input word $x \in \mathcal{C}^I$ and output word $y \in \mathcal{C}^O$. Put*

$$M_T(x, y) = |\,\mathrm{Col}_T(x, y)|.$$

Suppose $T_2 \circ T_1$ is obtained by identifying the output ports J of T_1 with the equally ordered input ports of T_2, deleting the temporary stub vertices, and making no other identification. Then

$$\mathrm{Col}_{T_2\circ T_1}(x,z) \;\cong\; \coprod_{y\in\mathcal{C}^J} \mathrm{Col}_{T_1}(x,y)\times \mathrm{Col}_{T_2}(y,z), \tag{C1}$$

naturally in x, z. Consequently

$$M_{T_2\circ T_1} = M_{T_1}M_{T_2}. \tag{C2}$$

After Booleanizing positive entries, the support relation

$$R_T = \{(x,y) : M_T(x,y) > 0\}$$

satisfies $R_{T_2\circ T_1} = R_{T_1};R_{T_2}$. For port width at most b, there are at most $2^{3^{2b}}$ such relations.

Proof. Restrict a colouring of the composite to the two tangles. The restrictions have one common colour on every identified port, hence determine a unique middle word y and an element of the corresponding summand on the right-hand side of (C1).

Conversely, take two proper colourings in one summand. Their common middle word says exactly that the two colours on every pair of identified stubs agree. Delete the temporary stubs and give the resulting seam edge that common colour. Every old internal vertex retains its three distinct nonzero incident colours, and no new vertex is created, so the glued colouring is proper. Restriction and gluing are inverse operations. Taking cardinalities of the disjoint union gives (C2), and replacing positive natural numbers by truth values gives relational composition. There are 3^b possible words on each side, hence 3^{2b} possible ordered word pairs and at most $2^{3^{2b}}$ relations. ☐

Corollary 8 (exact cumulative support update)**.** PROSE*For a serial word $T_1, \ldots, T_n$, let $S_i \subseteq \mathcal{C}^J$ be the boundary words extendable across the first i pieces. Then*

$$S_{i+1} = \{y : \exists x \in S_i,\ (x,y) \in R_{T_{i+1}}\}. \tag{C3}$$

If $S_i = S_j$ for $i < j$, deleting the block $T_{i+1}, \ldots, T_j$ leaves the final accepting support unchanged. Port reordering merely conjugates every relation by the induced permutation, so canonical relabelling is representation-invariant.

Proof. Equation (C3) is the Boolean form of (C2). Equal supports followed by the same suffix remain equal by induction over that suffix. For a port permutation σ, restriction reads the relabelled word as $x \circ \sigma^{-1}$; applying this bijection at both ends conjugates the relation and preserves composition and terminal nonemptiness. ☐

Remark 3 (machine-checked algebra after the physical gluing bijection)**.** *The graph-level restriction/gluing bijection (C1) is now machine-checked on the in-tree port-tangle model.* VERIFIED *Its cardinal convolution is machine-checked as well.* VERIFIED *Boolean support of the convolution is relational composition.* VERIFIED *Equal cumulative supports remain equal after deleting the intervening block and appending a common suffix.* VERIFIED *Port relabelling commutes with matrix composition.* VERIFIED *Finally, the Boolean support carrier has cardinality $2^{|X||Z|}$.* VERIFIED

Thus the direct zero-Count automaton never needs a global colouring of the unprocessed graph. Its state is the *set of boundary words currently extendable through the processed prefix*; its letters are literal tangle relations. Connectivity, bridgelessness, cyclic side data, and capped face progress are additional finite *geometric* interface fields needed by the physical replacement theorem. They are not licences to ask for a whole-annulus Tait colouring. At fixed width each field has finitely many values (partitions of a finite port set, bounded incidence bits, and capped lengths), and its update must be proved from the literal composition just as (C1) was.

The manuscript's profile has three kinds of observable data:

1. the colours of edges crossing the cut;
2. connectivity for the $\alpha\beta$ and $\alpha\gamma$ Kempe families;
3. capped progress of faces crossing the cut.

Proposition 26 (global opened-annulus face uniqueness is false)**.** REFUTED *Pairwise uniqueness of shared interior face edges does not hold on the literal annular carrier obtained by opening two faces of a normal-form map. In particular it cannot be supplied globally to the current rooted factorization interface.*

Proof. Open two opposite pentagonal faces of the dodecahedral map, deleting the ten cap-cycle edges but retaining their vertices as degree-one stubs. The opened map is connected, with $V = 20$, $E = 20$, and hence $F = 2$. Its ten stub edges are bridges, while each of the other ten edges is incident to both face orbits. Thus the two distinct opened faces share ten interior edges, not at most one.

For exact replay, the two deleted facial cycles are

$$(0, 1, 5, 6, 2), \qquad (14, 18, 19, 16, 15),$$

and the ten shared edges are

$$\begin{gathered}(3, 8), (3, 9), (4, 9), (4, 10), (7, 8),\\ (7, 13), (10, 11), (11, 17), (12, 13), (12, 17).\end{gathered}$$

The opened rotation has SHA-256 fingerprint `f109f292c33df0ae4eff0403a23efb424979ddba8ecc8ef2d14d16c54` The independent rotation traversal also checked every connected normal-form opening generated by `plantri -a -d -m5 -c5` through 34 vertices: all 4120 violate the global premise. The reproducible falsification audit and its compact certificate are `tools/fourcolor/v24_open_annulus_face_uniqueness_audit.py` and `results/fourcolor/v24_open_annulus_face_uniqueness_20_34.json`. The single dodecahedral witness proves the proposition; the larger sweep only shows that the problem is structural rather than exceptional. ☐

Theorem 30 (conditional Cell–3 rooted factorization of one positive Cell)**.** CONDITIONAL *Assume the positioned Cell geometry satisfies*

$$F \neq F' \quad \Longrightarrow \quad |\,\mathrm{sharedInteriorEdges}(F, F')| \leq 1. \tag{U}$$

Fix a Cell–3 closed-web-at-good-word instance with global proper Tait colouring p, and an executable corridor position together with the next two positions needed by the rolling root. Let c be a positive literal Cell colouring which agrees with the restriction of p on their actual overlap. Write $s(p)$ for the canonical rooted state at the incoming cut, $a(p, c)$ for the finite Cell–rebase letter extracted from the two colourings, and $p \diamond c$ for their right-biased splice. Then:

1. *$a(p, c)$ accepts $s(p)$ and its successor is exactly the canonical rooted state $s(p \diamond c)$ at the following cut;*

2. *every positive entry of the literal five-field Cell count supplies such a Cell colouring c, so every source-positive Cell behaviour occurs in the finite transition;*

3. *the induced transition is independent of the ambient representative: two witnesses with the same rooted source, literal Cell code, successor source-profile, and four rebase-role colours induce the same source and target rooted states, hence the same transition relation; and*

4. *the realized initial set and the realized directed-edge set have exact witness semantics. Membership is equivalent, respectively, to the existence of a compatible first-position witness and to the existence of a compatible positioned prefix/Cell witness.*

After projection to the manuscript's two Kempe families, terminal-aware compression preserves precisely the equality fibres of both endpoints. Consequently it changes neither the realized edge count nor the realizable initial count.

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Proof. For the first assertion, form the literal Cell colour on the ambient edge carrier, splice it with p on the Cell region, and then perform the boundary rebase. Positivity makes every one of the four rebase-role colours nonzero. The finite letter first checks the physical Cell profile and the equality of the shared intermediate profile. Its tracked and facial rebase maps then compute the two successor exterior relations, while the rolling maps compute the next interaction roots. Projection along the fixed carrier coordinates recovers the successor cut colours, the two connectivity relations, and the capped face-progress field. Equality in each field therefore identifies the computed state with $s(p \diamond c)$.

For the second assertion, a positive count is a nonempty finite set of literal open Tait colourings with the prescribed left and right five-field profiles. Choose one member. The overlap hypothesis is exactly the condition required to splice it with p, so the first assertion applies. Conversely, letters in the realized alphabet are defined as images of these compatible literal witnesses; hence no accepted behaviour is introduced by the extraction.

For representation invariance, retain only the rooted source, literal Cell code, output profile, and rebase-role colours. Equality of these data gives equality on the active part of the fixed rebase collar. The tracked local graphs therefore agree. The facial output is already determined by the common output profile, and the four role colours determine the transported cut colouring. Extensionality of the rooted state gives equal targets, so the two singleton transition relations coincide. Finally, both realized sets are finite images. Membership in an image is equivalent to existence of a preimage, giving the two witness characterizations. On terminal graph profiles the compression map is injective, endpoint by endpoint, which proves the final cardinality statement. □

Proposition 27 (the Cell–3 carrier is not supplied by a direct Count obstruction)**.** PROSE *An arbitrary zero-accepting-Count instance does not supply an object of type* `ClosedWebAtGoodWord.Instance` *on its opened annulus. If the annulus itself has zero Tait count then such an object is impossible; even when the annulus is colourable, the direct hypothesis supplies neither a totally closed colouring nor a good inner word.*

Proof. The type contains a proper edge colouring of the whole opened graph; properness is bundled into `EdgeColoring`, while its `tait` field supplies nonzeroness. Hence an inhabitant already proves that the open graph has positive Tait count. In the colourable case the additional `totallyClosed` and `goodWord` fields are still extra existential witnesses, absent from the direct obstruction premise. Neither follows from zero terminal acceptance of the corresponding closed map. □

The six cited Lean declarations are machine-checked with (U) as an explicit hypothesis; they do not prove (U). Proposition 26 shows that no cap-deletion adapter can establish it: opening the caps splices ambient faces and destroys global uniqueness on the intended carrier. Therefore this conditional theorem is not presently an on-route factorization result, and no closure number derived from it is authoritative.

There are therefore two independent repairs. First, factorization must be generalized from the special globally coloured `ClosedWebAtGoodWord.Instance` to a colouring-independent annular Cell context together with the partial positive prefix and literal Cell witnesses used by cumulative Count. A total colouring of the opened annulus may be used to prove that one particular run is sound, but it cannot define the autonomous letter relation: doing so would make the available transitions depend on extendability through the unprocessed suffix. In particular no theorem on the direct path may obtain prefix properness from the type of the ambient `EdgeColoring` or prefix nonzeroness from `web.tait`. Second, face lookup must be local, not a stronger global hypothesis. The closed minimal map does satisfy ambient pairwise uniqueness. VERIFIED If two corridor faces are fully retained by the opening, their open images still share at most one edge; the tree proves this both for arbitrary retained ambient faces and for every pair of retained corridor faces. VERIFIED VERIFIED The selected common-neighbour incidence calculation likewise identifies the two local flanks without reinstating global uniqueness. VERIFIED The remaining Lean task is consequently to inventory both dependencies: replace every use of the global colouring by its rooted prefix/Cell witness, replace every globally chosen shared edge by the appropriate retained-face or pointwise-incidence theorem, and keep boundary-spliced faces out of the simple interior-dual lookup. This is a carrier-generalizing, hypothesis-minimizing refactor of the factorization, not a new geometric assumption.

The tracked half of this refactor is now complete on the corrected pointwise-selected source geometry. At each corridor position the selected literal Cell has two vertices and a unique internal bond. A crossing of a strictly earlier Cell can neither equal nor meet an outgoing crossing of the successor Cell: the corresponding corridor faces are separated along the clean dual geodesic. If an edge were internal to both Cells, equality of the two endpoint sets would reduce to the same contradiction. Consequently any historical Cell edge which reappears in the successor rolling carrier is one of the four rebase-role edges, and every active successor edge is either a rebase-role edge or belongs to the current literal Cell. VERIFIED VERIFIED

The resulting finite tracked factor uses the actual interaction carrier (at most forty-nine edges), a four-role deletion mask, the literal local adjacency table, and a partial coordinate map into the next rolling carrier (at most twenty-one edges). Its update is rowwise equal to the literal successor graph on the interaction carrier. An unmapped target coordinate is isolated by the historical-overlap theorem above, so generic partial contraction is exact for vertex equality, direct adjacency, and exterior connectivity. Thus the factor's executable target state is exactly the canonical tracked exterior code on the next literal rolling carrier. VERIFIED VERIFIED

The combined finite projection now closes both rolling roots on this pointwise-selected carrier. For every compatible positive prefix witness and literal Cell colouring, the tracked target root computed by the finite rebase factor is exactly the canonical tracked prefix state at the next cut. VERIFIED

The facial target root is exact as well. Its fixed twenty-four interface slots and four persistent-port slots recover the next-cut facial connectivity, presence bits, and cap-at-five component sizes from the cap-at-six target code. The proof uses only the pointwise-selected face carrier and the local prefix/Cell witnesses; it does not reinstate global opened-annulus face uniqueness. VERIFIED

Proposition 28 (the selected finite Cell–rebase machine is formation-local). VERIFIED *Fix only the colouring-independent annular formation, a pointwise-selected literal Cell, and the following boundary rebase. Every positive prefix colour function and compatible positive literal Cell colouring extracts a supported finite letter whose source and target are its exact terminal-aware input and rebased output profiles. In particular the finite carrier, colour tables, rooted interaction receipt, supported-letter predicate, and complete Cell–rebase transition consume no global Tait colouring, total-closure witness, or good inner word. Specializing a former* `ClosedWebAtGoodWord.Instance` *factors through its colouring-independent* `toFormation` *field and gives the same transition.* VERIFIED

Proof. The finite Cell letter is computed from the selected formation, the positive prefix values, and the literal open Tait colouring of the Cell. Compatibility on the terminal overlap proves physical Cell support. The right-biased splice is positive on each of the four rebase roles, so the normalized rebase letter is supported; its input is definitionally the Cell letter's output. Their pair therefore witnesses the finite transition between the displayed source and target. The compatibility theorem applies this construction to `web.toFormation`; none of the three additional fields of `web` occurs in the resulting statement or proof. □

Proposition 29 (selected Cell–rebase realizations are one-step local). VERIFIED *Fix a selected Cell and its following boundary rebase. Let the observed prefix support be the union of the current terminal-aware cumulative region and the rebase switch. The latter is the image of the four fixed rebase roles, and therefore has cardinality at most four.* VERIFIED *Replace the former positive prefix function on the whole ambient edge set by a positive function on this observed support, retaining the same compatible literal Cell colouring. Then the realized source–target transition image is unchanged. Moreover every transition in this local realized image is a transition of the canonical finite Cell–rebase machine.* VERIFIED

Proof. Restriction is immediate. In the other direction extend a local function by one fixed nonzero colour outside the observed support; this extension is only an adapter to the former API. The source profile reads only the current terminal region. For the target, adjoin the common literal Cell. On the next terminal region, an edge in the rebase switch is already observed. An edge outside the switch belongs to the old pre-rebase region if and only if it belongs to the next terminal region. Since the pre-rebase region is exactly the union of the current terminal region and the Cell, its colour is therefore either an observed prefix colour or the common Cell colour. Hence the two spliced colourings agree throughout the next terminal region. Regional profile congruence gives equality of both exact endpoints, so restriction and extension prove the two inclusions between realized transition images. The finite-letter soundness theorem applied to the extended local witness gives the final assertion. No converse from an arbitrary abstract supported letter to a physical local witness is asserted here. □

Thus the finite one-Cell transition has now been proved exact on both of the strong rolling roots from which the manuscript's cumulative profile is projected, its finite letter semantics has been separated from the global closed-web colouring, and its physical realization witness carries no colour coordinate on the unprocessed suffix. This does not yet prove the full theorem above. The realizable initial image still sits under the surrounding closed-web parameter. The source collar or three-Cell word must therefore be made into a formation directly, and the resulting local realized transitions and initials must be connected soundly and completely to the cumulative Count and Seed semantics. In particular, completeness of the abstract supported finite-letter relation with respect to physical local witnesses remains a separate obligation. Global opened-annulus face uniqueness is no longer an obligation of either rolling transition.

The Lean transition uses a conservative three-pair refinement internally and then projects to exactly the two families named by the manuscript. It proves that every compatible positive literal Cell witness followed by the boundary rebase produces one exact source and target profile edge, and that terminal- aware compression is lossless relative to that two-pair source profile: compressed edges agree exactly when their projected two-pair edges agree. VERIFIED The realized edge image and realizable initial image both preserve cardinality under this compression. VERIFIED VERIFIED

The conservative third family has not been smuggled into the source count. At the live $(2,1,4)$ shape its carrier is exactly sixteen times larger:

$$12{,}556{,}911{,}218{,}688 = 16 \cdot 784{,}806{,}951{,}168.$$

This equality is machine-checked. VERIFIED

The local letter and the cumulative terminal receipt have different shapes, and this distinction is load-bearing. The rolling Cell letter has shape $(2,1,4)$ because it remembers one local seam terminal. Terminal acceptance is read only after passing to the already-constructed cumulative source crosscut carrier of shape $(2,5,8)$, whose five terminals are the five fixed cap-foot edges. The following proposition verifies that no additional terminal datum is needed there.

Proposition 30 (cap acceptance is a projection of the five-terminal source receipt)**.** *VERIFIED In the standard colour-name gauge, project a graph-semantic state of the existing* `BoundedCorridorCutProfile 2 5 8` *to the two Kempe families named in the manuscript and apply the lossless terminal-aware compression. The compressed state determines:*

1. *the colours of all five cap feet; and*
2. *the complete terminal connectivity relations for the $\alpha\beta$ and $\alpha\gamma$ families.*

Consequently it determines exactly whether either source pair satisfies the cap-composed Menu-B predicate. No graph lookup and no enlargement of the one-Cell letter is required.

Proof. For one terminal, the two diagonal bits record membership in $\{\text{red}, \text{blue}\}$ and $\{\text{red}, \text{purple}\}$. On a nonzero Tait colour the possible bit pairs are therefore

$$(1,1) \mapsto \text{red}, \qquad (1,0) \mapsto \text{blue}, \qquad (0,1) \mapsto \text{purple};$$

the remaining pair $(0,0)$ is impossible. This finite decoding is checked as VERIFIED, and on a literal graph-derived regional profile it gives exactly the colours of the five named terminal edges: VERIFIED.

Terminal diagonal entries are stored explicitly by the compressor. Every off-diagonal entry is stored once on the corresponding unordered pair, and symmetry reconstructs its opposite orientation. Hence compression preserves the whole cap-facing view: VERIFIED. The cap-composed predicate is a function only of this five-word and these two relations, so rewriting by that equality proves the proposition. □

The preceding statement used literal colour names only to choose coordinates. The manuscript instead names the two connectivity roles relative to the ordered majority/singleton triple of the current good word and then says “by global colour relabeling” to use the standard names. That normalization is also exact.

Proposition 31 (global colour normalization preserves cap acceptance)**.** *VERIFIED Let (α, β, γ) be the ordered majority, first-singleton and second-singleton colours of a five-terminal cap view. There is a canonical zero-fixing permutation of the four Klein-group colours sending (α, β, γ) to* (red, blue, purple). *Relabel the boundary word by this permutation while retaining the two abstract strand roles $\alpha\beta$ and $\alpha\gamma$. The arbitrary-gauge cap-composed Menu-B predicate holds before relabeling if and only if the standard-gauge predicate holds afterward.*

Thus the two-pair source receipt is sufficient in every colour gauge. This uses the fact that its pair labels are roles relative to the ordered triple; it does not *assert that two fixed literal colour pairs suffice without normalization.*

Proof. The inverse of the canonical Tait-triple equivalence sends α, β, γ to the three standard nonzero colours and fixes zero. Relabeling preserves the three boundary multiplicities, so the normalized word has the standard majority/singleton triple: VERIFIED. For either abstract source role, relabeling carries its unique inactive position to the same cap index and leaves its Boolean strand relation unchanged. Therefore its cap-composed predicate is equivalent before and after normalization: VERIFIED. Taking the disjunction over the two roles proves the claim. The independent cap-extension table is equivariant under the same zero-fixing permutation: VERIFIED. □

The corrected all-colour Seed count and the compressed cumulative receipt now meet at exactly this normalized terminal view. It is useful to state the two directions separately, because the remaining realization problem lies between them rather than inside either statement.

Proposition 32 (corrected Seed acceptance and the compressed accepting set)**.** *VERIFIED For a fixed cap word, the corrected cap-composed Seed count is positive if and only if there is a realized all-colour uniform profile of positive count and an ordered majority/singleton triple whose normalized two-role terminal view is accepted.*

For every graph-semantic bounded cumulative source profile, define its compressed code to be accepting precisely when the standard-gauge terminal view decoded from the code satisfies cap-composed Menu B. Lossless compression preserves this predicate exactly: VERIFIED. In particular this holds on the existing Cell–3 cumulative carrier of shape $(2, 5, 8)$*: VERIFIED.*

Proof. Project an all-colour profile to the two source roles $\alpha\beta$ and $\alpha\gamma$. For either role, the resulting cap relation is definitionally the corresponding component of the all-colour connection table. The corrected Seed predicate therefore says exactly that one of these two projected roles is cap-accepted. Proposition 31 then transports this disjunction to the standard colour gauge. Rewriting the already verified positive-count factorization by this equivalence proves the first assertion.

For the second assertion, the accepting set is the finite subset obtained by filtering the compressed carrier by the decoded standard-gauge cap predicate. Proposition 30 says that decoding after compression gives the same terminal word and the same two strand relations as decoding before compression, so membership rewrites to the uncompressed predicate. Specializing the bounds to $(2, 5, 8)$ gives the last statement. □

The two profile presentations in the preceding proposition are not merely abstract finite carriers. On a literal annulus their terminal coordinates coincide.

Proposition 33 (the counted terminal profile is cumulative)**.** *VERIFIED Let an annular boundary datum be backed by a rotation system on its underlying simple graph. Give the whole annulus the regional edge set $E(G)$, no crossing ports, its five cap-foot edges as terminals, and no face-fragment coordinates. Then every Tait colouring counted by the uniform Menu–B profile determines a graph-derived cumulative source profile whose five-terminal cap view is exactly the counted view. After the manuscript's global colour normalization, the equality remains literal. Consequently the corrected Seed count is positive if and only if an accepted normalized cap view is realized by such a cumulative source profile.*

Proof. Fix two selected colours. The counted profile puts two cap stubs in the same component precisely when they are connected in the subgraph consisting of edges of those colours. Each cap stub is incident with its named cap-foot edge. In any graph, two edges with chosen incident endpoints are connected in the line graph if and only if those endpoints are connected in the original graph: VERIFIED. Applying this to the selected-colour subgraph identifies the source component label with tracked-edge reachability: VERIFIED and VERIFIED.

Because the regional edge set is all of $E(G)$, the regional guards add no restriction. The diagonal entries reconstruct the five nonzero terminal colours, while the off-diagonal entries are the two source connectivity relations. Thus the entire terminal views agree. Relabel the actual edge colouring by the canonical zero-fixing colour equivalence, rather than merely renaming the view. The selected two-colour subgraphs before and after relabeling are isomorphic, so the complete connection table is preserved. This proves the normalized equality VERIFIED. Finally, positivity of a uniform-profile count supplies an actual Tait colouring, and conversely that colouring lies in its own profile fibre; substitution gives the displayed equivalence. □

The initial object of the literal serial calculation can likewise be made genuinely local. This matters because the source does not claim that a wide annulus is one bounded-width serial corridor: its finite calculation is made on a selected depth-one collar (and, at the local audit, on three consecutive cells). A witness for the first serial position must therefore not carry a second arbitrary positive-colour function on the unprocessed part of the annulus.

Proposition 34 (realizable initials are prefix-local). *VERIFIED Fix a literal source corridor and its first terminal-aware serial position. Replace the positive colour function on all edges of the opened graph by a positive colour function whose domain is exactly the active prefix region, retaining the same compatible literal Cell colouring. The resulting finite image of initial cumulative source profiles is exactly the former realizable initial image.*

Proof. Restrict an old witness to the active prefix. Conversely, extend a local witness by one fixed nonzero colour outside that prefix; this is only an API adapter and consults no suffix datum. Restriction after extension is literal. For extension after restriction, every field of the rooted source receipt observes the colour function only on its stated terminal input region: VERIFIED. Thus the complete rooted source state, and hence its bounded cumulative projection, is unchanged. The two inclusions between the finite images follow by extension and restriction. □

Together with Proposition 33, this closes the narrower representation-level excess in the initial witness: its additional positive-colour coordinate is now prefix-local. It does *not* remove the global Tait colouring, total closure, or good word already contained in the surrounding `ClosedWebAtGoodWord.Instance`. Proposition 28 completes the corresponding refactor for the finite carrier, state, supported letter, and one-step transition: those objects now require only the colouring-independent formation. Proposition 29 also removes the transition witness's ambient suffix-colour coordinate. What remains is narrower than whole-annulus coverage but still substantive: construct that formation directly from the radius-1 collar or three-Cell word actually selected by the source, identify both local realized images with the cumulative Count and Seed semantics consumed at the local target, and prove physical completeness for the abstract finite letters retained by the closure. No further equivalence between vertex-component and edge-component semantics is owed. The filtered finite accepting set has also not yet been emitted as an optimized executable certificate.

Theorem 30 checks the Cell–3 algebra under (U), but Propositions 26 and 27 prevent applying it to the direct source target. Even after both refactors it will not by itself justify a numerical closure count: the exact realized letter image is still specified as a finite image of literal witnesses, and the initial image, although local in its additional positive-colour coordinate, is still parameterized by the closed-web instance. Neither is yet a practical enumerator.

Proposition 35 (Executable optimized closure certificate)**.** *OPEN* *For each relevant corridor type and target semantics, produce a finite raw-profile array Q, a finite array A of* physical *Cell types, an initial array $I \subseteq Q$, an accepting array $F \subseteq Q$, and a letter-labelled edge array $E \subseteq Q \times A \times Q$ such that*

$$q \in I \quad \Longleftrightarrow \quad q \text{ is produced by a compatible literal first-position witness},$$

and

$$(q, a, q') \in E \quad \Longleftrightarrow \quad (q, q') \text{ is produced by a compatible positive prefix/Cell witness of physical type } a.$$

Also prove

$$q \in F \quad \Longleftrightarrow \quad q \text{ has a literal terminal realization satisfying the chosen target.}$$

The letter index must forget the colouring witness while retaining every geometric choice on which the literal Cell count depends; an unlabelled union of edges is insufficient for cumulative Count.

For $a \in A$ let $R_a(S) = \{q' \mid \exists q \in S,\ (q, a, q') \in E\}$. Prove the displayed biconditionals in the kernel, replay raw reachability, then replay the determinized closure from I under the maps R_a, and finally minimize the reachable deterministic system by equality of accepted right languages. Compare all three fingerprints and counts with an independently implemented generator:

$$|Q_{\text{raw}}^{\text{reach}}|, \qquad |\operatorname{Reach}(\mathcal{P}(Q))|, \qquad |\operatorname{Reach}(\mathcal{P}(Q))/{\sim_{\text{Nerode}}}|.$$

Then, and only then, these array lengths are the exact raw, cumulative-support, and observational state counts for that corridor type. This certificate is needed when its minimized count is used to sharpen an effective threshold or to replay a finite base. It is not needed for the existence of a finite direct-Count state space once actual ordered noose cuts have been supplied: Theorem 38 below gives the conservative state space directly.

There are two different accepting arrays. For the Lift/Seed system, F is the cap-composed predicate of Definition 1. Replaying that closure against the open-tangle predicate of Proposition 2 would certify only that a nonempty good fibre is nonempty; it would not test the Seed conclusion. For the direct Count system, F instead consists of terminal profiles whose literal boundary closure is a proper Tait colouring of the closed instance. The direct descent consumes

$$S_n \cap F = \varnothing \quad \Longleftrightarrow \quad \text{the restored closed map has zero Tait count},$$

not menu B. The transition carrier may be shared, but these accepting predicates and their Nerode quotients must not be conflated.

The measured $(5, 0)$ certificates remain valuable regression tests and base-bound optimizations, but they are not substitutes for this equality theorem when their numerical sizes are quoted: their own headers correctly say that identifying the decoded transition with every source-legal corridor is a separate fidelity obligation.

1.8 Direct cumulative-Count descent

The finite-state object needed to pump a zero count is not one raw profile. An uncolourable closed instance has no accepting colouring from which one could select a profile path. Its prefix invariant is instead the *set* of all raw profiles realized by positive prefix colourings. This determinization costs a powerset in the crude bound and is why the three counts in Proposition 35 must not be conflated.

Definition 2 (support action of a positive Count letter)**.** PROSE Let Q be finite and let $M_a \in \mathrm{Mat}_Q(\mathbb{N})$ be the Count matrix of a physical Cell a. Its Boolean support relation is

$$q \, R_a \, q' \quad \Longleftrightarrow \quad M_a(q, q') > 0.$$

For $S \subseteq Q$ put

$$R_a[S] = \{q' \in Q : \exists q \in S, \; q \, R_a \, q'\}.$$

If $x \in \mathbb{N}^Q$ is a cumulative Count row vector and $S = \{q : x_q > 0\}$, then the support of xM_a is exactly $R_a[S]$.

Proof. The q' entry of xM_a is $\sum_q x_q M_a(q, q')$. Every summand is a natural number, so the sum is positive exactly when at least one summand is positive. A product of natural numbers is positive exactly when both factors are positive. These two equivalences give precisely $q' \in R_a[S]$. □

Theorem 31 (determinized support-state deletion)**.** *PROSE Let $a_1, \ldots, a_n$ be physical Cell letters, let $I \subseteq Q$ be the initial support, and define*

$$S_0 = I, \qquad S_{k+1} = R_{a_{k+1}}[S_k].$$

If $S_i = S_j$ for some $0 \le i < j \le n$, delete the block $a_{i+1}, \ldots, a_j$. The shortened word has exactly the same final support S_n. In particular, for any accepting set $F \subseteq Q$, it has an accepting colouring if and only if the original word does:

$$S_n \cap F \neq \varnothing.$$

Consequently, if more prefix positions occur than there are reachable determinized support states, some nonempty block is deletable without changing zero versus positive total Count.

Proof. After the deletion, the support at the new position i is $S_i = S_j$. Apply the same remaining functions $R_{a_{j+1}}, \ldots, R_{a_n}$ to equal sets. Induction over this common suffix gives equality of every later support, hence equality of the final support and of its intersection with F. The last assertion is the pigeonhole principle applied to the finite reachable subset of $\mathcal{P}(Q)$. □

The repeated-block equality and terminal nonemptiness equivalence are machine-checked for arbitrary finite cut-word relations. VERIFIED The exact support cardinality and its pigeonhole consequence are checked separately. VERIFIED VERIFIED

This theorem uses no self-loop and no accepting witness. It is the direct zero-count pumping law advertised by the source's monoidal Count semantics. It is also exact about its price: raw-profile reachability alone does not justify it, and the worst-case state count is $2^{|Q|}$ before reachable-state minimization.

Theorem 32 (unrestricted support right languages separate states)**.** *VERIFIED Let W be a finite set of boundary words. Suppose a deterministic encoding $e \colon \mathcal{P}(W) \to X$ and an observer $A \colon X \times \mathcal{P}(W) \to \{\mathsf{false}, \mathsf{true}\}$ answer every abstract suffix experiment exactly:*

$$A(e(S), T) \quad \Longleftrightarrow \quad S \cap T \neq \varnothing \qquad (S, T \subseteq W).$$

Then e is injective. Hence, for $W = \mathcal{C}^k$ with $|\mathcal{C}| = 3$, every such encoding has at least 2^{3^k} states.

Proof. If $S \neq S'$, choose a word x in their symmetric difference. The singleton suffix $T = \{x\}$ accepts exactly one of S, S', because $S \cap \{x\} \neq \varnothing$ is equivalent to $x \in S$. Thus equal codes would give contradictory answers to this suffix experiment, so e is injective. The powerset of $\mathcal{C}^k$ has cardinality $2^{|\mathcal{C}^k|} = 2^{3^k}$. $\square$

The cardinal specialization is checked separately. VERIFIED This lower bound says that the raw powerset is not merely a wasteful encoding of the *unrestricted* support algebra. It does not say that every singleton suffix, or every raw support, is realized by a planar cubic tangle. A smaller automaton remains possible after restricting both prefix and suffix experiments to the physically reachable language. Consequently this theorem does not discharge the high-width supply: that would require a proved bound on the reachable Myhill–Nerode quotient, or a genuinely two-dimensional replacement theorem.

There is a second boundary on that proposed minimization. Let $\{L_a : a \in A\}$ and $\{R_b : b \in B\}$ be indexed families of literal open tangles with one common ordered seam and fixed orientation-reversing matching, and define the physical right language

$$\mathcal{L}(R_b) = \{a \in A : L_a \# R_b \text{ is Tait-colourable}\}.$$

The physical/semantic bridge gives, exactly,

$$a \in \mathcal{L}(R_b) \quad \Longleftrightarrow \quad \mathrm{Supp}_{\mathrm{out}}(L_a) \cap \mathrm{Supp}_{\mathrm{in}}(R_b) \neq \varnothing.$$

VERIFIED Consequently

$$(\forall b,\ \mathcal{L}(R_b) = A) \quad \Longleftrightarrow \quad (\forall a, b,\ L_a \# R_b \text{ is Tait-colourable}).$$

VERIFIED Thus physical suffix restriction can genuinely identify raw support states, but universality of the right languages on a target-complete family is the colourability target itself, not an independent compression lemma. Any reachable quotient used in the descent must therefore be certified from its literal transition table and terminal predicate; it cannot be justified just by saying that all testers are physical.

The smallest genuinely two-dimensional repair one might try is to replace one hexagonal cell at a time. Its exact boundary behaviour is already large enough to rule out that local move.

Let H_6 be the open tangle consisting of a six-cycle with one dangling port at each cycle vertex, the ports ordered cyclically. If b_i is the colour on port i, and x_i is the colour on the cycle edge leaving vertex i, properness at that vertex is equivalent, in $\mathbb{F}_2^2$, to

$$x_{i-1} + b_i + x_i = 0.$$

Thus, after choosing the nonzero colour x_5, the other cycle-edge colours are forced successively; the word extends precisely when none of them is zero and the last equation closes the cycle. The literal port-tangle semantics and this recurrence are machine-checked to be equivalent. VERIFIED Exactly 63 of the $3^6 = 729$ ordered boundary words extend across H_6. VERIFIED

Proposition 36 (a facial cycle has no smaller exact connected replacement)**.** *PROSE* *For $n \geq 4$, let H_n be the open tangle consisting of a labelled n-cycle with one cyclically ordered port at each cycle vertex. No connected cubic n-port tangle with fewer than n internal vertices has the same ordered boundary support as H_n.*

Proof. First count the support of H_n. A proper colouring of its cycle edges is a cyclic word $x = (x_0, \dots, x_{n-1})$ in the three nonzero Klein colours with $x_{i-1} \neq x_i$. It induces the port word

$$b_i = x_{i-1} + x_i.$$

Conversely this equation is exactly the local Tait equation at every cycle vertex, so the image of this derivative map is $\operatorname{Supp}(H_n)$.

The number of proper colourings of a labelled n-cycle with three colours is

$$2^n + 2(-1)^n.$$

For completeness, fix the colour at one vertex. If a_j and b_j count length-j proper paths ending respectively at that colour and at either one specified other colour, then

$$(a_{j+1}, b_{j+1}) = (2b_j, a_j + b_j), \qquad (a_0, b_0) = (1, 0).$$

Induction gives $a_j = (2^j + 2(-1)^j)/3$; multiplying by the three choices of the fixed colour gives the displayed cycle count.

If two cycle colourings x, y induce the same boundary word, then

$$x_{i-1} + x_i = y_{i-1} + y_i, \qquad \text{hence} \qquad x_i + y_i = x_{i-1} + y_{i-1}.$$

Thus $y_i = x_i + c$ for one constant Klein colour c. When x uses all three nonzero colours, nonzeroness of y forces $c = 0$, so the derivative fibre is a singleton. A proper cycle colouring using only two colours exists exactly when n is even. There are then six such alternating colourings; translation by the omitted third colour pairs them into three derivative fibres. Therefore VERIFIED VERIFIED

$$|\operatorname{Supp}(H_n)| = \begin{cases} 2^n - 1, & n \text{ even}, \\ 2^n - 2, & n \text{ odd}. \end{cases}$$

For $n = 6$ this recovers the checked value 63.

Now let a proposed connected cubic replacement have $r < n$ internal vertices and m internal edges. Counting internal incidences and the n ports gives

$$3r = 2m + n.$$

Connectivity gives $m \geq r - 1$, hence $r \geq n - 2$; the incidence equation gives $r \equiv n \pmod 2$. Since $r < n$, necessarily $r = n - 2$ and $m = r - 1$. Its internal multigraph is therefore a tree.

Root that tree. A proper colouring has at most $3! = 6$ choices for the three labelled incidences at the root. At each of the other $r - 1$ vertices the parent-edge colour is fixed, and the two remaining labelled incidences have at most two possible orders. Hence the replacement has at most

$$6 \cdot 2^{r-1} = 6 \cdot 2^{n-3} = 3 \cdot 2^{n-2}$$

proper colourings, and no more boundary words than colourings. For $n \geq 4$,

$$3 \cdot 2^{n-2} < 2^n - 2 \leq |\operatorname{Supp}(H_n)|.$$

VERIFIED The two ordered supports cannot be equal. □

This is a generic perimeter obstruction, not wall exclusion. It rules out an exact support-preserving reduction which replaces a single facial cycle by a smaller connected cubic tangle on the same ports; increasing the face length does not repair the deficit. It does not rule out replacing a larger two-dimensional patch, using a target-aware quotient weaker than full support equality, or performing a move whose size decrease occurs elsewhere. Those are the remaining source-faithful possibilities.

For a hexagonal cell, the stronger monotone replacement question can also be settled completely.

Theorem 33 (no smaller connected cubic monotone replacement for a hexagon)**.** PROSE *Let R be a connected plane cubic tangle with six boundary ports in a specified cyclic order, all on the outer face. If R has fewer than six internal vertices, then*

$$\mathrm{Supp}(R) \nsubseteq \mathrm{Supp}(H_6).$$

Thus no smaller connected cubic six-port tangle can replace one hexagonal cell by the monotone support-inclusion criterion.

Proof. Let r and m be the numbers of internal vertices and internal edges of R. Counting incidences at the cubic internal vertices gives

$$3r = 2m + 6.$$

Connectivity gives $m \geq r-1$, hence $r \geq 4$, while the incidence equation gives $r \equiv 6 \pmod 2$. Since $r < 6$, necessarily $r = 4$ and $m = 3$. The internal graph is therefore a tree.

There are only two abstract trees on four vertices: the path and the tripod. At an endpoint of the path two ports occur consecutively on the outer face; at either internal path vertex one port occurs. Up to a dihedral change of the six cyclic port positions, the two singleton ports are either separated by the two endpoint blocks or adjacent to one another. These are the two path types. For the tripod, the central vertex has no port and each leaf has two consecutive ports, giving one further type. Thus every R, after renaming internal vertices and applying a dihedral boundary relabelling, has one of the following port-block orders:

$$S, P, S, P, \qquad P, S, S, P, \qquad P, P, P,$$

where S is a singleton and P is the pair at a tree leaf.

For these three types respectively, the following boundary words are properly realised by the tree but do not extend across the hexagon:

$$(r, r, b, r, r, b), \qquad (r, b, r, r, r, b), \qquad (r, b, r, p, b, p).$$

For the two path types, order the internal edges as $(01, 02, 13)$ and colour them (b, p, p). For the tripod, colour the three centre–leaf edges (p, b, r). Inspection at the four internal vertices gives the three distinct colours in every incident triple. The six-step hexagon recurrence rejects each displayed boundary word. These three finite checks, stated as exact support relations rather than drawings, are machine-checked by VERIFIED and VERIFIED. Finally, hexagon extension is invariant under cyclic rotation and reflection of the boundary, so the three canonical failures cover every plane embedding of the tree. □

This closes the one-cell monotone loophole left by Proposition 36: equality was not the issue. The source-faithful high-width repair must replace a larger two-dimensional patch, exploit the restricted support language of the *actual* exterior, or make its strict size decrease elsewhere.

Theorem 34 (monotone and target-aware support replacement)**.** *Let E be a fixed exterior tangle, P the piece removed from it, and R a replacement with the same ordered interface. If*

$$\mathrm{Supp}(R) \subseteq \mathrm{Supp}(P),$$

then

$$E \# P \text{ not Tait-colourable} \quad \Longrightarrow \quad E \# R \text{ not Tait-colourable.}$$

VERIFIED *For this one fixed exterior, the sharp condition is*

$$E \# R \text{ not Tait-colourable} \quad \Longleftrightarrow \quad \mathrm{Supp}(E) \cap \mathrm{Supp}(R) = \varnothing.$$

VERIFIED *Both statements hold for the literal sewn rotation systems after restricting to genuine nonzero Tait words.* VERIFIED

Proof. By the physical gluing bijection, a colouring of a sewn tangle is exactly a boundary word realized on both sides. If a colouring of $E \# R$ existed under the displayed inclusion, its seam word would also lie in $\mathrm{Supp}(P)$, and hence would colour $E \# P$, a contradiction. Without requiring inclusion, the same gluing statement says directly that a colouring exists exactly when the two fixed support sets meet. □

The first nontrivial physical restriction on the exterior support can now be made exact. Say that a support language is *single-orbit* if all of its words differ only by one global permutation of the three nonzero Tait colour names (equivalently, by a colour equivalence fixing 0).

Theorem 35 (a single-orbit exterior admits a strict hexagon replacement)**.** *Let $E \subseteq \mathcal{C}^6$ be the ordered boundary support of the fixed exterior of one facial hexagon, and suppose that E is single-orbit. If*

$$E \cap \mathrm{Supp}(H_6) = \varnothing,$$

then one of the three canonical plane cubic trees on four internal vertices has support disjoint from E. Replacing the six-vertex hexagonal cell by that tree therefore produces a strictly smaller zero-Count splice. VERIFIED

Proof. Write T_s, T_a, T_t for the supports of the separated-path, adjacent-path, and tripod replacements classified in Theorem 33. Their common intersection is empty. There is a short structural proof, avoiding a boundary-word census.

Let $w = (w_0, \ldots, w_5)$ extend through both path trees. Write s_0, s_1, s_2 for the three internal colours of the separated path and a_0, a_1, a_2 for those of the adjacent path, in the edge orders used in the preceding theorem. The shared leaf with port colours w_4, w_5 gives $s_1 = a_2$, because two entries of a proper three-colour triple determine the third. Comparing the two proper triples at w_3 gives equality of their remaining unordered colour pairs. The exchanged alternative would give $s_0 = a_2 = s_1$, impossible at the vertex carrying (w_0, s_0, s_1). Hence

$$s_0 = a_0, \qquad s_2 = a_2 = s_1.$$

Next compare the two triples at w_0. Again the remaining pairs agree up to exchange. The exchanged alternative would give $s_1 = w_1$, impossible at the separated-path vertex carrying (w_1, w_2, s_2), since $s_2 = s_1$. Thus $s_1 = a_1$. The adjacent-path triples at w_2, w_3 are consequently (w_2, s_0, s_1) and (w_3, s_0, s_1), whereas the separated path has (w_0, s_0, s_1). Uniqueness of the remaining colour gives

$$w_2 = w_0 = w_3.$$

But any tripod extension has a proper triple (w_2, w_3, t_1) at one leaf, and therefore requires $w_2 \neq w_3$. No word lies in all three supports. The two path implication, tripod implication, and empty intersection are checked respectively by VERIFIED, VERIFIED, and VERIFIED.

Each T_i is invariant under global colour renaming. If E met all three T_i, choose one word from each intersection and rename each back to the common orbit representative of E. The representative would then belong to $T_s \cap T_a \cap T_t$, a contradiction. Hence some T_i is disjoint from E. The physical support-gluing equivalence in Theorem 34 says exactly that the corresponding replacement remains zero-Count, and it removes two internal vertices. □

Thus a locally irreducible zero-target wall cannot present a single global colour orbit to every hexagonal cell. At each cell where all three strict tree reductions fail, its actual exterior must realize at least two distinct global colour orbits. This is a genuine restriction supplied by the target, not by wall geometry. It does not yet exclude a wall: the remaining M1 task is to show that a large common web cannot maintain this multi-orbit boundary diversity everywhere, or else to use that diversity to construct a larger two-dimensional replacement.

Theorem 36 (unrestricted target-awareness collapses to support inclusion)**.** *Let U be the boundary-word universe, let $P \subseteq U$ be the support of the removed piece, and let $(R_i)_{i\in I}$ be the supports of all candidate replacements. Then*

$$\left[\forall E \subseteq U,\ E \cap P = \varnothing \Longrightarrow \exists i \in I,\ E \cap R_i = \varnothing\right] \quad \Longleftrightarrow \quad \left[\exists i \in I,\ R_i \subseteq P\right].$$

VERIFIED

Proof. For the forward implication, test the maximal abstract zero exterior $E = U \setminus P$. A candidate disjoint from E is contained in P. Conversely, if $R_i \subseteq P$, then every set disjoint from P is also disjoint from R_i. The same equivalence holds for a restricted class of physical exterior supports whenever that class itself contains $U \setminus P$. VERIFIED □

Retaining the full corridor profile does not by itself evade this quantifier boundary. This point is easy to blur because the richer state really is essential for advancing through the next Cell. It is nevertheless invisible to the final colouring splice.

Theorem 37 (rich-profile gluing erases to the boundary word)**.** *Let X and Y be arbitrary profile types with boundary-word projections $\pi_X : X \to U$ and $\pi_Y : Y \to U$. For profile languages $A \subseteq X$ and $B \subseteq Y$, define them to be compatible when some $a \in A$ and $b \in B$ have $\pi_X(a) = \pi_Y(b)$. Then*

$$A \text{ and } B \text{ are compatible} \quad \Longleftrightarrow \quad \pi_X[A] \cap \pi_Y[B] \neq \varnothing.$$

VERIFIED In particular, profile languages with equal projected word support are indistinguishable by every exterior profile language. VERIFIED

If $(B_i)_{i\in I}$ is a family of rich replacement-profile languages and the exterior ranges over arbitrary word supports, target-aware safety again holds exactly when

$$\pi_Y[B_i] \subseteq \pi_X[A]$$

for some i. VERIFIED

Proof. A compatible pair of profiles supplies their common projected word, and a word in both projected images supplies a compatible pair of profiles. This proves the first equivalence and the indistinguishability statement. Apply Theorem 36 to the two projected word supports for the last assertion. Equivalently, the complement of $\pi_X[A]$ is still the maximal abstract exterior adversary. VERIFIED □

This changes the correct formulation of the two-dimensional crux. Exact support equality is useful for pumping because it works against every suffix, but a minimal-counterexample descent only needs a replacement whose support misses the *actual* exterior. Conversely, if the original splice is uncolourable while a proposed smaller admissible splice is colourable, then the replacement support cannot be contained in the original support: VERIFIED it must introduce some boundary word that the removed piece did not realize. Thus Proposition 36 does not kill target-aware one-cell reduction; it shows precisely why such a reduction cannot be justified by exact support preservation. Theorem 36 adds an essential quantifier warning: allowing the replacement to depend on an *arbitrary* exterior does not help either. The gain can come only from a structural theorem restricting which support sets are realized by actual planar exteriors. Theorem 37 adds that merely enriching each realization by its Kempe-component pairing or face-progress coordinates does not supply such a restriction: those coordinates matter only insofar as their *globally realizable family* constrains the projected word support. The remaining constructive problem is to exploit that physical restriction while preserving the topological class and decreasing size.

Theorem 38 (raw noose Count pumping)**.** PROSE *Let a closed cubic map be cut by a chain of pairwise nested ordered nooses, each crossing exactly k edges, into a left end, serial slabs, and a right end. Identify every ordered cut word with an element of $\mathcal{C}^k$, where $|\mathcal{C}| = 3$. For the prefix ending at cut i, let*

$$S_i = \{x \in \mathcal{C}^k : \textit{the prefix has a proper nonzero Tait colouring with word } x\}.$$

Then:

1. *each slab acts on supports by the Boolean relation of its physical port-tangle Count;*
2. *there are at most 2^{3^k} possible cumulative supports S_i; and*
3. *if $S_i = S_j$ for $i < j$, deleting the intervening slabs and gluing cut i to cut j preserves whether the complete closed map has a Tait colouring.*

Thus more than 2^{3^k} cut positions force a Count-preserving repeated support. If the original map has zero accepting Count and the cuts also satisfy the geometric hypotheses of Theorem 43, that repetition produces a strictly smaller connected bridgeless cubic spherical zero-Count instance.

Proof. Cutting a slab open gives a port tangle with k ordered input and k ordered output incidences. The physical restriction/gluing bijection (C1) identifies its Count support with a relation

$$R_i \subseteq \mathcal{C}^k \times \mathcal{C}^k.$$

Consequently $S_{i+1} = R_i[S_i]$. Since $|\mathcal{C}^k| = 3^k$, its powerset has cardinality 2^{3^k}. Pigeonhole gives equal supports when there are more cut positions. Theorem 31 then applies the unchanged right-end suffix to those equal supports and proves equality of terminal acceptance. Finally Theorem 43 supplies all the asserted graph-class and strict-size conclusions. □

All algebraic assertions in this theorem are now machine-checked: physical serial gluing becomes relational composition, its matrix support is the same relation, and the cut alphabet has exactly three letters per crossed edge. VERIFIED VERIFIED VERIFIED The final spherical-map sentence remains an application of the physical noose theorem, whose representation-specific Lean constructor is D3 rather than part of this algebra module.

This theorem is deliberately insensitive to the number or internal size of the slabs. It needs no finite alphabet of physical Cell graphs and no globally chosen colouring. The only geometric information used by the physical replacement is carried by D1: the actual ordered nooses, connected saturated retained sides, a nonempty deleted region, and the common finite seam type. The richer rooted Cell carrier remains useful only to reduce the astronomical conservative bound 2^{3^k}, thereby making D4's finite floor computationally realistic.

Lemma 26 (two boundary-essential sides glue bridgelessly)**.** *VERIFIED* *Let H and K be finite connected graphs, each with $k \geq 2$ ordered boundary half-edges. Call a side* boundary-essential *if, for every bridge of the side, both components left after deleting that bridge contain a boundary half-edge. Glue the i-th half-edge of H to the i-th half-edge of K for every i. Then the glued graph is connected and bridgeless.*

Proof. Connectivity follows because each side is connected and at least one seam edge joins them. Consider an old edge e of H. If e is not a bridge of H, an H-path already bypasses it. If it is a bridge, its two components contain boundary ports i and j. Travel from one end of e inside its component to port i, cross to K, use connectedness of K to reach port j, and cross back to the other component. This is a path between the ends of e avoiding e. The same argument handles old edges of K.

Finally take a seam edge i. Choose $j \neq i$. Connectedness of H and K, together with seam edge j, gives a path between the ends of seam edge i which avoids it. Thus no edge of the glued graph is a bridge. □

Lemma 27 (a saturated side of a bridgeless graph is boundary-essential)**.** *VERIFIED* *Let G be bridgeless and let H be a connected vertex side of an edge cut, with every edge from H to $G-H$ represented by a boundary half-edge. Then H is boundary-essential.*

Proof. If a bridge e of H left a component C containing no boundary half-edge, saturation says that no edge of G leaves C except e. Then e would be a bridge of G, a contradiction. □

Hub closures and the exact face count. Let T be an open rotation tangle with boundary set B. Its *hub closure* $\widehat{T}_\rho$ has one new hub dart $\bar{b}$ for every $b \in B$, pairs b with $\bar{b}$, and rotates the hub darts by a permutation ρ giving their boundary order. The old vertex rotations and old internal pairings are unchanged. Write

$$\widehat{\alpha}_T(b) = \bar{b}, \qquad \widehat{\alpha}_T(\bar{b}) = b, \qquad \widehat{\phi}_{T,\rho} = \widehat{\rho}_T \widehat{\alpha}_T.$$

We call the cap *face-simple* when distinct hub darts lie in distinct $\widehat{\phi}_{T,\rho}$-orbits. This is the exact combinatorial condition called `HubFacesDistinct` in the implementation; it must not be inferred merely from the hub being one vertex.

Lemma 28 (orbit exchange)**.** *VERIFIED* *Let a finite family of permutations be obtained by processing ports one at a time. At each port first transpose two points in different current orbits, and then transpose two distinct points in the resulting common orbit. Suppose the unprocessed merge orbits remain pairwise distinct. Then processing all ports does not change the number of orbits.*

Proof. A transposition of points in different cycles merges precisely those two cycles, decreasing the orbit count by one. A transposition of distinct points in the same cycle cuts that cycle into two, increasing the count by one. Thus one exchange has net change zero. The stated invariant ensures that the same hypotheses hold after each processed port, so induction on the finite port set proves the result. □

Theorem 39 (the hub-capped seam face formula)**.** VERIFIED *Let L, R be open rotation tangles with equally large boundary sets, let $m : B_L \simeq B_R$ match their ports, and give their hubs rotations ρ_L, ρ_R. Assume both hub closures are face-simple and*

$$\rho_R(mb) = m(\rho_L^{-1}b) \qquad (b \in B_L). \tag{25}$$

If $k = |B_L|$, and $L\#_m R$ is the closed rotation system obtained by pairing matched real boundary darts, then

$$F(L\#_m R) + k = F(\widehat{L}_{\rho_L}) + F(\widehat{R}_{\rho_R}). \tag{26}$$

Here every F is the number of orbits of the face permutation $\phi = \rho\alpha$.

Proof. Begin with the disjoint union of the two hub closures. For each left port b, replace the two cap pairings

$$(b, \bar{b}), \qquad (mb, \overline{mb})$$

by the real seam pairing (b, mb) and the hub pairing $(\bar{b}, \overline{mb})$. On the face permutation this replacement is one merging transposition followed by one splitting transposition. The merge joins the left hub face at b to the right hub face at mb. Face-simplicity says that merge faces belonging to distinct unprocessed ports are distinct. The split belonging to b lies on the face created by the merge belonging to $\rho_L^{-1}b$; equation (25) is exactly the identity needed on the right hub to identify those two points. Therefore Lemma 28 applies and the fully exchanged face permutation has

$$F(\widehat{L}_{\rho_L}) + F(\widehat{R}_{\rho_R})$$

orbits. This intermediate equality is also checked directly by `GoertzelV24SeamExchange.orbitCount_phiS_univ`.

After all ports have been exchanged, real darts never map to hub darts and hub darts never map to real darts. On the real summand the permutation is, up to the carrier reassociation, the face permutation of $L\#_m R$. On the $2k$ hub darts it is the face permutation of the two-hub k-edge dipole:

$$\bar{b}_L \longmapsto \rho_R(\overline{mb})_R, \qquad \bar{c}_R \longmapsto \rho_L(\overline{m^{-1}c})_L.$$

Equation (25) makes this permutation an involution. It is fixed-point-free because it exchanges the left and right summands, so its $2k$ elements form exactly k two-cycles. Orbit counts add over disjoint sums and are invariant under the carrier reassociation. Subtracting these k dipole orbits gives (26). The dipole count and the reassociation are machine-checked in `GoertzelV24CompositeSphericity.orbitCount_dipolePhi` and in the theorem named by the verification badge. □

Corollary 9 (hub-capped seam sphericity)**.** VERIFIED *Under the hypotheses of Theorem 39, suppose both hub closures are spherical. Then the composite satisfies*

$$V(L\#_m R) + F(L\#_m R) = E(L\#_m R) + 2.$$

Proof. Let e_L, e_R be the numbers of intact edges of the two open sides and write F_L, F_R for the face counts of their hub closures. The two cap equalities are

$$(V_L + 1) + F_L = (e_L + k) + 2, \qquad (V_R + 1) + F_R = (e_R + k) + 2.$$

The composite has $V_L + V_R$ vertices and $e_L + e_R + k$ edges, while Theorem 39 gives $F = F_L + F_R - k$. Adding the two cap equalities and substituting these three counts yields

$$(V_L + V_R) + F = (e_L + e_R + k) + 2,$$

as required. □

Lemma 29 (opposite first-return orders give the seam orientation)**.** *VERIFIED* *For an open shore T, let s_T be the first-return permutation on its boundary darts and choose the canonical hub rotation $\rho_T = s_T^{-1}$. Let $m : B_L \to B_R$ match two boundary interfaces. If*

$$s_R(m(s_L(b))) = m(b) \qquad (b \in B_L),$$

then m is orientation-reversing for the two canonical hub rotations:

$$\rho_R(m(b)) = m(\rho_L^{-1}(b)).$$

Proof. Apply s_R^{-1} to the displayed opposite-return equation. Its left-hand side becomes $m(s_L(b))$, while $\rho_L^{-1} = s_L$ and $\rho_R = s_R^{-1}$. This is exactly the required identity. □

Lemma 30 (inverse first return makes the cap face-simple)**.** *VERIFIED* *Let T be a literal vertex shore, let s_T be first return of its retained facial walk to the exposed boundary darts, and put $\rho_T = s_T^{-1}$. In the hub closure $\widehat{T}_{\rho_T}$, distinct hub darts lie on distinct faces.*

Proof. Transport the hub closure to the carrier consisting of the old retained darts and one copy of the boundary. Eliminate the inserted hub darts by taking first return to the old retained carrier. An internal dart makes the same one-step facial move as before. At a boundary dart b, the walk visits the hub dart $\rho_T(b)$ and returns to the old carrier after two steps. Consequently the transported return permutation is

$$\phi_T^{\text{cap}}\, \widetilde{\rho_T},$$

where $\widetilde{\rho_T}$ is ρ_T on the boundary darts and the identity elsewhere. Taking first return once more, now to the boundary, gives

$$s_T \rho_T = s_T s_T^{-1} = 1. \tag{27}$$

Every hub dart is facially adjacent to an old boundary dart. Thus two hub darts on the same capped face would give two boundary darts in the same cycle of the return permutation in (27). The identity permutation has only singleton cycles, so those boundary darts, and hence the original hub darts, are equal. □

This lemma is deliberately narrower than saying that the cap is one new vertex. Equation (27) proves the facial statement `HubFacesDistinct` for every finite boundary set. Inverse first return alone does not make ρ_T one cycle. For the connected shores used by the splice, however, the exact side rank forces that conclusion by Euler's inequality; this is Theorem 41 below.

Theorem 40 (exact canonical-hub face count)**.** *VERIFIED* *Let T be a literal vertex shore with boundary darts B, and let $F_{\text{pure}}(T)$ be the number of ambient faces all of whose darts lie in that shore. Closing T with the canonical hub rotation $\rho_T = s_T^{-1}$ gives*

$$F(\widehat{T}_{\rho_T}) = |B| + F_{\text{pure}}(T). \tag{28}$$

Proof. First return of a capped face to the old retained darts partitions the faces without changing their number: every inserted hub dart returns to the old carrier after one step. Split those return cycles according to whether they meet a boundary dart. By equation (27), the boundary-meeting part has one cycle for each $b \in B$. On a cycle avoiding the boundary, the inserted hub order is never consulted; every step is literally the old ambient facial step. Forgetting the retainedness tag therefore gives a bijection from boundary-avoiding capped faces to ambient faces wholly contained in the shore. Adding the two disjoint classes proves (28). □

Corollary 10 (canonical hub closure from the exact side rank)**.** VERIFIED *Let V_T and E_T be the retained vertices and intact edges of the shore. If*

$$F_{\mathrm{pure}}(T) + |V_T| = |E_T| + 1, \tag{29}$$

then its canonical hub closure satisfies the spherical Euler equality. For a connected vertex shore and connected complement in a connected graph-backed spherical map, equation (29), and hence the conclusion, follows from the planar-bond rank theorem. VERIFIED

Proof. The hub closure has $|V_T| + 1$ displayed vertices. Its pairing orbits are the E_T intact edges and one hub spoke for every boundary dart, so it has $|E_T| + |B|$ edges. By Theorem 40 it has $|B| + F_{\mathrm{pure}}(T)$ faces. Substitution of (29) gives

$$(|V_T| + 1) + (|B| + F_{\mathrm{pure}}(T)) = (|E_T| + |B|) + 2.$$

The final assertion is exactly the connected-side equality already obtained by splitting spherical Euler across a planar bond. □

Theorem 41 (connected exact rank forces one hub)**.** VERIFIED *Let T be a connected cubic open rotation tangle whose old darts at each vertex form one rotation cycle. Let ρ be any permutation of its boundary darts, and suppose the boundary contains two distinct darts. If the cap is assigned the exact one-hub Euler rank*

$$|V_T| + 1 + F(\widehat{T}_\rho) = E(\widehat{T}_\rho) + 2,$$

then ρ is one nontrivial cycle on the entire boundary carrier.

The inequality used here is itself available directly at map level. If R is a connected finite rotation system and the rotation darts at every stored vertex form one cycle, then

$$\mathrm{orb}(R.\rho) + \mathrm{orb}(R.\phi) \leq |E(R)| + 2. \tag{29a}$$

VERIFIED Indeed, the edge involution has an explicit transposition-list presentation; connectedness makes the associated word action transitive, so the general permutation Euler inequality has exactly one component. Thus (29a) carries no cellular-embedding or genus hypothesis.

Proof. Present the fixed-point-free hub edge involution as one transposition per edge. The old rotation, these transpositions, and the hub spokes act transitively on the capped dart carrier: cyclic old rotations move within one old vertex, connectedness moves across the intact side edges, and each hub dart crosses its spoke to its old boundary dart. Hence the generic permutation Euler inequality has one component and gives

$$\mathrm{orb}(\rho_T) + \mathrm{orb}(\rho) + F(\widehat{T}_\rho) \leq E(\widehat{T}_\rho) + 2.$$

The old rotation has exactly one orbit per old vertex, so $\mathrm{orb}(\rho_T) = |V_T|$. Comparing with the displayed exact rank yields $\mathrm{orb}(\rho) \leq 1$. The nonempty boundary gives the reverse inequality, so $\mathrm{orb}(\rho) = 1$. With two distinct boundary darts this unique orbit is nontrivial; therefore ρ is a single cycle with full support. □

Corollary 11 (a connected planar bond supplies the canonical cap). VERIFIED *Let a connected spherical graph-backed cubic map have cyclic vertex rotations and two-sided faces. If a vertex bipartition has connected induced graphs on both sides and at least two boundary darts, then closing either side by the inverse of its facial boundary first-return permutation gives a face-simple spherical cap with one cyclic hub vertex.*

Proof. The planar-bond rank theorem supplies equation (29), so Corollary 10 supplies the exact one-hub rank. Lemma 30 supplies face-simplicity without any topological hypothesis. Theorem 41 then proves that the canonical hub rotation is a single cycle. □

Theorem 42 (complementary planar-bond orders are opposite). VERIFIED *In the setting of Corollary 11, let s_S and $s_{\bar{S}}$ be the actual facial first-return permutations on the two literal shores. Match their boundary darts by the ambient edge flip*

$$m(d) = \alpha d.$$

Then

$$s_{\bar{S}}(m(s_S(d))) = m(d). \tag{29b}$$

Consequently the two canonical inverse-return hub rotations satisfy the orientation-reversing seam equation. VERIFIED *Together with the two canonical cap equalities, the natural seam composite therefore has the complete connected, bridge-free, cubic, rotation-cyclic spherical-map package.* VERIFIED

Proof. Transport the deleted shore's capped facial first return to the retained shore's boundary carrier by reversing every cut edge. On that common carrier the planar-bond boundary theorem says that the retained return s_S is the inverse of the transported deleted return. Transporting this identity back through m gives exactly (29b).

The canonical hub rotations are s_S^{-1} and $s_{\bar{S}}^{-1}$, so (29b) is precisely the orientation equation used by the seam face-count theorem. Corollary 11 supplies the two face-simple spherical caps, while the literal-shore structure lemmas supply connectedness, boundary-essentiality, cubicity, and cyclic old-vertex rotations. The structural seam theorem packages these facts on the literal composite. □

Corollary 12 (canonical literal-shore splice). VERIFIED *Let two connected vertex shores of one bridgeless cubic rotation system have at least two matched boundary darts. Suppose their matching reverses the two boundary first-return orders, both shores satisfy the exact side-rank equation (29), and the ambient vertex rotations are cyclic. Then gluing the literal shores along the matching produces a connected, bridgeless, cubic, rotation-cyclic spherical map.*

Proof. Lemma 27 supplies boundary essentiality. Lemma 30 supplies the two face-simple caps, Lemma 29 supplies their orientation reversal, and Corollary 10 supplies their spherical Euler equalities. The structural seam theorem then gives connectedness, bridge-freeness, cubicity, cyclic old-vertex rotations, and sphericity simultaneously. □

Definition 3 (cyclic bond-boundary datum). VERIFIED Let T be a literal vertex shore, let B_T be its boundary darts, and let s_T be first return of the retained facial walk to B_T. A *cyclic bond-boundary datum* for T consists of an integer $k \geq 2$ and a bijection

$$\eta : \mathbb{Z}/k\mathbb{Z} \longrightarrow B_T$$

such that

$$s_T(\eta(i)) = \eta(i+1) \tag{30}$$

for every $i \in \mathbb{Z}/k\mathbb{Z}$. Thus the datum records both that the displayed ports exhaust the boundary and that their facial first-return order is one cyclic order, rather than a disjoint union of several orders.

Lemma 31 (a cyclic bond boundary is one canonical hub)**.** VERIFIED *If T has a cyclic bond-boundary datum, then both s_T and its inverse $\rho_T = s_T^{-1}$ are single cycles on B_T. Consequently the canonical closure used above really adds one vertex whose rotation is ρ_T, not one formal "hub" split into several vertices. This gives an explicit coordinate certificate for the same conclusion as Theorem 41; it is no longer a hypothesis of the physical splice.*

Suppose moreover that $m : B_L \to B_R$ obeys the opposite-return equation

$$s_R(m(s_L(b))) = m(b). \tag{31}$$

If the left shore has a cyclic bond-boundary datum, then the right first-return order is the transported inverse of the left order. In particular both canonical hub rotations are single cycles on their entire boundary carriers and the matching is orientation-reversing.

Proof. Equation (30) says exactly that

$$s_T = \eta \circ (i \mapsto i + 1) \circ \eta^{-1}.$$

The shift on $\mathbb{Z}/k\mathbb{Z}$ is one cycle when $k \geq 2$; conjugacy preserves being a cycle, and the inverse of a cycle is a cycle. This proves the first assertion.

For the second assertion, equation (31) is equivalent to

$$s_R = m \circ s_L^{-1} \circ m^{-1}.$$

Hence s_R is conjugate to the inverse of s_L. Transporting the enumeration by m and reversing its indices gives equation (30) on the right. The final orientation statement is Lemma 29. ☐

Lemma 32 (a tight transverse noose supplies the cap data)**.** PROSE *Let M be a cellular spherical map and let γ be a simple closed curve which avoids vertices, crosses k edges transversely, and is* tight*: its intersection with the interior of every face is empty or one connected arc. Cut the crossed edges at γ, retain either closed-disc side, and close its k boundary darts by a hub in their cyclic order. Each hub closure is spherical and face-simple. Under the natural matching of the two halves of every crossed edge, the two hub rotations satisfy (25).*

Proof. By the Jordan curve theorem, γ cuts the sphere into two open discs with common boundary γ. Collapse the boundary circle of either closed disc to a point. The quotient is a sphere; the cut endpoints become the darts at the new hub, in their cyclic boundary order. This is precisely the hub closure, so its cellular Euler equality is the spherical one.

Enumerate the crossings once around γ. Tightness makes the facial arc inside the retained disc carry each boundary dart to the next crossing, so this enumeration is a cyclic bond-boundary datum in the sense of Definition 3. In particular Lemma 31 proves at permutation level that the cap adds one cyclic hub vertex.

The sector at the hub between two consecutive cut darts is the image of the unique boundary arc of γ between the corresponding crossings. It lies in one old face. Tightness says that no old face contains two such arcs. Hence no face of the capped rotation system contains two hub darts, which is face-simplicity. Equivalently, the boundary first-return order is the cyclic order of crossings around the one boundary circle, so its inverse is a single cycle and Lemma 30 supplies the same face-simplicity directly at permutation level. Finally, the two boundary orientations induced on the common boundary of the two discs are opposite. Thus their computed first-return maps obey $s_R \circ m \circ s_L = m$. Lemma 29 converts this equation into equation (25). ☐

This tight-noose lemma is now an optional geometric presentation, not the load-bearing D3 adapter. For a supplied vertex bipartition with both induced sides connected, Corollary 11 obtains the spherical face-simple one-hub cap directly from the planar-bond rank and Euler's inequality; no Jordan theorem and no per-face tightness are needed. What the sphere-cut supplier must still provide is the combinatorial data actually used by the splice: nested connected edge shores and an equality of ordered seam states which transports the retained first-return order. Lemma 10 derives the roots and at least two boundary darts. Theorem 42 then makes the natural edge-flip to the complementary shore orientation-reversing automatically. A tight noose is one sufficient way to manufacture the ordered shore data, but it is not part of the algebraic seam theorem.

Lemma 33 (one cubic sphere cut can be moved to edges)**.** PROSE *Let N be a tight sphere-cut noose of a cubic plane graph. Thus N meets the graph only in a middle set Z of vertices, and at every $z \in Z$ each of the two edge sides has at least one incident edge. After choosing one of the two disc sides, N can be moved off the vertices to a tight transverse noose γ crossing at most $2|Z|$ edges. The vertices strictly in the chosen side remain there. If a laminar family of such nooses has pairwise ordered collars, these moves can be made inside the collars and preserve laminarity.*

Proof. Choose disjoint small vertex discs around the points of Z. At a cubic middle vertex, the chosen side contains either one or two consecutive incident edge germs. Replace the portion of N through that vertex by an arc in its vertex disc pushed toward the chosen side. It crosses exactly those one or two germs and no others. The replacements are disjoint, so the result is still a simple closed curve, avoids every vertex, preserves every strictly separated vertex, and has at most $2|Z|$ transverse crossings.

Two harmless simplifications may now be needed. If the curve crosses the same edge twice with no vertex on the intervening edge segment, a closed regular neighbourhood of that edge segment is a rectangle disjoint from every vertex and from every other edge. Replace the curve arc on one side of the rectangle by the parallel arc on the other side. This is a local isotopy followed by cancellation of the two transverse crossings; it does not move the curve across a graph vertex. If the curve visits one face in more than one arc, choose two consecutive arcs in that open disc. The subdisc between them contains no graph; replacing the appropriate curve subarc along that subdisc removes a repeated visit without changing which graph vertices lie on either side and without introducing an edge crossing. Each operation strictly lowers the finite number of crossings or face visits, so they terminate. The resulting curve is transverse and tight, with the same bound.

For an already laminar finite family, choose the vertex discs and the subsequent face surgeries inside mutually ordered collar neighbourhoods, starting with the innermost curve. An isotopy supported in one such collar cannot cross a curve outside that collar, so the nesting order is preserved. □

This lemma settles the local factor 2 in the state count. It does *not* assert that the nooses attached independently to all edges of an arbitrary sphere-cut decomposition have already been chosen as one laminar family, nor that every proper tree segment contains a retained vertex. Those two global compatibility clauses remain explicit fields of the sphere-cut supply consumed by Theorem 7.

Theorem 43 (physical deletion between equal nested noose states)**.** PROSE *Let G be a connected bridgeless cubic spherical map. Let γ_i, γ_j, with $i < j$, be disjoint nested simple nooses which avoid vertices and each cross exactly $k \geq 2$ edges. Suppose the two vertex shores of each cut induce connected graphs, the two shores retained for the splice are saturated, their ordered geometric seam types identify the retained-side boundary first-return orders, and the annular slab between them* *contains a graph vertex. Delete that slab and glue the paired boundary half-edges, using the ambient edge flip on the outer cut to pass to its complementary shore. Then the result G' is a strictly smaller connected bridgeless cubic spherical map.*

If, in addition, the two cuts have the same exact cumulative Count-support state and the same finite geometric seam type, then G' has an accepting Tait colouring exactly when G does. Hence zero accepting Count is preserved.

Proof. Retain the vertex shore inside the inner cut and the complementary vertex shore outside the outer cut. For each cut, the retained shore and its complement are connected; Corollary 11 therefore supplies a spherical, face-simple canonical closure with one hub. The equality of ordered seam states transports the retained-shore return at the inner cut to the retained-shore return at the outer cut. Theorem 42 says that ambient edge flip at the outer cut transports that order to the inverse order on the complementary shore. Their composite is therefore the orientation-reversing seam matching. Hence Corollary 9 proves, by face-orbit counting rather than by an implicit drawing, that the glued rotation system is spherical. Every retained cubic vertex loses exactly the cut incidences recorded at its boundary and gains the corresponding seam incidences, so its degree remains three. The two retained sides are connected. By Lemma 27 they are boundary-essential, and Lemma 26 makes the composite bridgeless. The deleted slab contains a vertex and none is added, so G' is strictly smaller.

The geometric seam type records the ordered endpoint identifications and any local equality data needed by the chosen graph representation. The literal composite is loopless. A simple-graph implementation may reject a duplicate edge in its seam type, or the loopless-multigraph output may instead be normalized by repeated digon suppression as explained below. Equality of cumulative support states is exactly the hypothesis of Theorem 31. Corollary 5 identifies that support replacement with the literal matched-seam rotation system used in the preceding structural argument. Thus the unchanged exterior and the replacement interior have the same accepting support in G' and G, proving the last assertion without a second, merely semantic composite. □

The complete replacement kernel of this argument is now checked on the literal majority-shore data in Theorem 13. The present theorem remains marked PROSE because its stronger geometric input begins with independently supplied nooses. Its formal residue is exactly the noose/decomposition-to-edge-shore adapter; no colouring, cap, Euler or minimality step remains hidden behind that adapter.

Theorem 44 (the implemented seam composite has the non-topological map fields)**.** *Let two finite open rotation tangles be glued along a nonempty matched seam. If both side multigraphs are connected, the composed rotation system is primal-connected.* VERIFIED *If, in addition, both sides are boundary-essential and the seam has at least two ports, it is edge-bridge-free.* VERIFIED *If each open side has exactly three incident darts at every retained vertex, counting its boundary half-edges, then the composite is cubic.* VERIFIED *If the darts at each side vertex form one rotation cycle, the same is true in the composite.* VERIFIED

Lemma 34 (literal vertex shores inherit the local map structure)**.** *Let $M = (D, \alpha, \rho)$ be a finite cubic rotation system whose darts at each vertex form one ρ-cycle, and let S be any set of vertices. Cut every edge with exactly one endpoint in S, retaining its dart based in S as an unpaired boundary dart. The resulting literal open tangle is cubic when its boundary half-edges are counted.* VERIFIED *Its darts at each retained vertex still form one rotation cycle.* VERIFIED *If the graph induced by S is connected, then the interior-edge multigraph of the open tangle is connected.* VERIFIED *Moreover its full boundary-dart carrier is saturated: every ambient edge leaving S contributes its unique retained* *orientation as a boundary port.* VERIFIED *If the ambient edge multigraph is bridgeless, then this literal interior-edge multigraph is boundary-essential with respect to those boundary darts.* VERIFIED

Proof. The retained-dart partition

$$\{d \in D : \mathrm{vert}(d) \in S\} \;\cong\; D_{\mathrm{int}}(S) \sqcup D_{\partial}(S)$$

only records whether the opposite endpoint is also in S; it neither adds nor removes a dart based at a retained vertex. Hence the open darts based at $v \in S$ are in bijection with the ambient darts based at v, and there are exactly three of them.

The ambient rotation preserves the base vertex, so it restricts to the retained darts. The open rotation is exactly this restriction conjugated by the displayed partition equivalence. Two open darts based at the same retained vertex therefore correspond to two ambient darts based at the same vertex; ambient rotation-cyclicity puts them in one ρ-orbit, and restriction followed by conjugation puts the original open darts in one open-rotation orbit. Finally, every adjacency of the graph induced by S is represented by an ambient dart whose two endpoints lie in S, hence by one internal dart and one interior edge of the open tangle. An induced path therefore maps edge by edge to a path in its interior-edge multigraph. If an ambient edge has exactly one endpoint in S, its dart based at that endpoint is, by definition, a boundary dart; conversely every boundary dart arises in this way. This is precisely saturation. Finally, identify an internal ambient edge with the involution-orbit of either of its two internal darts. This is a bijection preserving the unordered endpoint pair. The preceding saturated-side lemma makes the induced ambient side boundary-essential, and transport across this multigraph isomorphism gives boundary-essentiality on the literal open-tangle carrier used by the splice. ☐

Lemma 35 (complementary vertex shores reassemble the ambient Tait problem)**.** VERIFIED *Let $M = (D, \alpha, \rho)$ be a finite loopless rotation system and let $S \subseteq V(M)$. Form the two literal vertex-side open tangles on S and $V(M) \setminus S$, and match their boundary darts by the ambient edge flip $d \mapsto \alpha d$. Their closed seam composite is Tait-colourable if and only if M is Tait-colourable.*

Proof. Partition each shore's retained darts into internal and boundary darts. The composite dart carrier is

$$(D_{\mathrm{int}}(S) \sqcup D_{\mathrm{int}}(S^c)) \sqcup (D_{\partial}(S) \sqcup D_{\partial}(S^c)).$$

Undoing these two partitions and then the predicate partition $S \sqcup S^c$ gives a canonical bijection from this carrier to D. On an internal dart the composite edge flip is the restricted old flip. On a boundary dart it is the seam matching, which was defined to be the old flip. Hence the bijection conjugates the two edge involutions. The analogous predicate partition of the vertex carrier identifies the composite base vertex of a dart with its old base vertex.

Read any Tait edge-colouring on either rotation system at its darts and transport that dart-colouring across the displayed bijection. Conjugacy of the flips makes it constant on the opposite pair of every edge. Preservation of base vertices makes it locally proper, and nonzeroness is unchanged. The generic descent from an invariant locally proper dart-colouring to edge orbits therefore gives a Tait edge-colouring on the other rotation system. Applying the construction in both directions proves the equivalence. Notice that this argument uses no face or embedding claim: those are supplied separately by the planar-bond structural theorem. ☐

Corollary 13 (two literal shores feed the structural splice)**.** VERIFIED *Let S_L, S_R be connected induced vertex shores of one bridgeless cubic rotation system with cyclic vertex rotations. Give their complete* *cut-dart interfaces a matching with at least two ports. If their hub closures are spherical and face-simple and the matching reverses the two hub orders, then the literal seam composite is a connected, bridge-free, cubic spherical map with cyclic vertex rotations.*

Proof. Lemma 34 supplies connectedness, boundary-essentiality, open cubicity, and open rotation-cyclicity on the exact two side carriers. The hub hypotheses supply the face-count input. Applying the implemented structural seam theorem therefore gives all four fields on the literal composite rotation system. □

Corollary 14 (the implemented seam composite packages the full structural class). VERIFIED *Let the two finite open rotation tangles satisfy the connectedness, boundary-essentiality, open-cubicity, and vertex-rotation hypotheses of the preceding theorem. Suppose the seam contains two distinct ports. If the two hub closures are spherical and face-simple and their hub orders obey the orientation-reversing equation (25), then the literal composed rotation system is a connected, bridge-free, cubic spherical map with one cyclic rotation at every vertex.*

Proof. Theorem 44 supplies connectedness, bridge-freeness, cubicity, and cyclic vertex rotations. Corollary 9 supplies the spherical Euler equality on the same composite carrier. These are exactly the four fields of the structural record, so no transport to a second representation is required. □

The implementation obtains the first two conclusions by an explicit edge equivalence between the multigraph glue and `composeRotationSystem`; the latter two follow by transporting the side rotations through the composite carrier. Thus connectedness, bridgelessness, cubicity, rotation-cyclicity, and strict vertex accounting are no longer D3 gaps. Theorem 39 and Corollary 9 also close the face-orbit calculation once face-simple spherical caps and the orientation equation are supplied. The bare matching type cannot manufacture those witnesses. On literal graph-backed shores they now come from connected planar-bond data by Corollary 11; Theorem 42 derives the opposite first-return equation from the natural ambient edge-flip matching.

Lemma 36 (the colouring half of digon suppression). VERIFIED *Let a cubic rotation system be separated by an exact non-colliding two-edge cut, and suppose that the chosen side contains exactly two vertices. If the ambient rotation system is not Tait-colourable, then the complementary cap is not Tait-colourable.*

Proof. The cap of the two-vertex side is cubic and still has two vertices. Counting darts in two ways gives

$$2|E| = |D| = 3|V| = 6,$$

so it has exactly three edges. Choose a bijection from these edges to the three nonzero elements of $\mathbb{F}_2^2$. Distinct adjacent edges then have distinct colours, and every edge has nonzero colour, so this cap is Tait-colourable.

If the complementary cap also had a Tait colouring, globally permute its three nonzero colours so that the two cap edges receive the same colour. The exact two-edge-cut gluing theorem then joins the two cap colourings to a Tait colouring of the ambient rotation system, a contradiction. Thus this is the formal version of "colour the replacement edge, copy its colour to the two exterior edges, and use the other two colours on the digon." □

Corollary 15 (parallel seams can be normalized after gluing). VERIFIED *Let M be a finite connected bridgeless cubic spherical loopless multigraph which is not Tait-colourable, and suppose that M contains a parallel pair. Repeatedly suppress a parallel pair by the digon splice of Lemma 41. The process terminates in a strictly smaller connected bridgeless cubic spherical rotation map with* *injective endpoint map which is still not Tait-colourable. Equivalently, the terminal map has the canonical simple-graph presentation used in the graph-backed layer.*

Proof. If three parallel edges join the same two vertices, connectedness and cubicity make M the theta graph, which is Tait-colourable. Hence at every non-simple, non-theta stage some repeated endpoint pair has multiplicity exactly two. Lemma 41 deletes its two cubic endpoints, preserves connectedness, bridgelessness, cubicity and sphericity, and introduces no loop. The two exterior edges are an exact two-edge cut whose digon side has two vertices, and its complementary cap is precisely the suppressed map. Lemma 36 therefore proves, without an additional colour calculation, that uncolourability is preserved. Each suppression removes two vertices, so well-founded induction on the vertex count terminates. The terminal graph has no loop and no parallel pair, hence has injective endpoint map, and the first suppression already made it strictly smaller than M. The canonical simple-graph backing then preserves its structural class and Tait semantics. ☐

Lean carries out this well-founded normalization uniformly, returning an endpoint-injective rotation system in the same cap-stable class with strictly fewer vertices. Endpoint injectivity is exactly the condition consumed by the canonical simple-graph backing. The checked minimal form of the same argument is also available: VERIFIED a vertex-minimal zero-Tait-count member of the bridgeless spherical cubic class, with two-sided face orbits, admits no well-formed digon patch. The arbitrary-pair extraction and the terminating iteration are therefore no longer representational gaps.

Therefore a duplicate seam edge need not enlarge the finite receipt. The replacement is performed in the loopless rotation-multigraph category, which is exactly the class quantified over by vertex minimality. In that class no post-splice simplification is needed: the sewn map itself is the strictly smaller zero-Count competitor. If a later presentation insists on a simple graph, Corollary 15 removes a parallel seam, but this is an optional representation change rather than a premise of the minimality contradiction.

Thus M2 is closed in the literal-shore representation. The full finite state is the canonical hub order together with the set of genuine Tait boundary words, and Theorem 13 consumes its equality directly. What remains belongs to M1: a connected corridor or branch-decomposition supplier must construct the complete rooted edge-leaf decomposition and its nodewise connected cuts and middle bounds. The two nested shore trees, edge/vertex accounting, roots, literal widths, two boundary ports, exact states, and strict cubic star between repeated states are now derived. No ambient graph data are reread after the state has been formed. Moreover every constructive equal-state splice has a strictly smaller endpoint-simple zero-Count descendant: VERIFIED. The cap face count, cap sphericity, one-hub property, orientation reversal, connectedness, bridge-freeness, cubicity, and cyclic vertex rotations are no longer open.

Corollary 16 (phased literal-shore replacement)**.** *VERIFIED Let $A \subsetneq B$ be two certified literal connected shores in a finite bridgeless spherical cubic map of zero Tait Count. Suppose their middle sets have size at most w, their literal boundary widths have size at most k, and their phased canonical states agree:*

$$\big(|B| \bmod (6w+1),\, q(B)\big) = \big(|A| \bmod (6w+1),\, q(A)\big).$$

Then there is a strictly smaller endpoint-simple bridgeless spherical cubic map of zero Tait Count.

Proof. Proper containment and equality of the cardinality phases imply $|B \setminus A| \geq 6w+1$. The cubic spacing lemma therefore finds a vertex whose full three-edge star lies in $B \setminus A$; this is the strict material deleted by the splice. Equality of the second coordinates of the phased states unpacks to one common literal width and equality of the ordinary normalized states. The constructive majority-shore replacement then gives a strictly smaller zero-Count rotation map in the complete bridgeless spherical cubic class. Finally the terminating digon normalization removes any parallel seam while preserving that class, zero Count, and strict decrease. □

Corollary 17 (nested facial-dual nooses physically shorten the map)**.** VERIFIED *Let M be a graph-backed vertex-minimal zero-Tait-count member of the bridgeless spherical cubic class. For $r \in \{0, 1\}$, let W_r be a simple closed walk of length at most w in the actual facial dual of M, let X_r be the primal edges crossed by W_r, and choose a connected component C_r of $M - X_r$. Put*

$$S_r := \{e \in E(M) : e \text{ has an endpoint in } C_r\}.$$

Assume that $E(M) \setminus S_r$ contains at least two edges for both values of r, that $S_1 \subsetneq S_0$, and that the two computed phased canonical states agree. Then there is a strictly smaller endpoint-simple bridgeless spherical cubic map of zero Tait Count.

Proof. A simple facial-dual cycle crosses pairwise distinct primal edges. Facial parity saturation proves that X_r is exactly the graph boundary of C_r, not merely a set containing that boundary, and proves that both C_r and its vertex complement induce connected graphs. Consequently S_r and $E(M) \setminus S_r$ are connected edge shores.

It remains to compare the geometric length with the middle-set width used by the finite state. A middle vertex of S_r cannot lie in C_r: every edge incident with a vertex of C_r belongs to S_r. Choose one S_r-edge incident with the middle vertex. Its other endpoint lies in C_r, so orienting it outward gives a crossing dart of the noose. Distinct middle vertices give distinct darts because they are the distinct terminal endpoints. Hence

$$|\operatorname{mid}(S_r)| \le |X_r| = |W_r| \le w.$$

The chosen component is nonempty and cubic, so S_r contains at least three edges. Together with the assumed two exterior edges, connectedness therefore supplies both nonempty majority sides. Thus each noose constructs the complete literal connected-shore node of width w, including roots and two distinct boundary darts. The hypotheses $S_1 \subsetneq S_0$ and equality of their computed phased states now meet Corollary 16, which performs the physical splice and terminating digon normalization. □

Theorem 45 (direct Count-reductive assembly)**.** CONDITIONAL *Assume that every sufficiently large least Tait counterexample can be put, by the source's target-preserving local reductions, into a composable corridor whose cut roots satisfy Proposition 5. Use the raw finite boundary-word semantics of Theorem 38; an optimized closure from Proposition 35 may replace its conservative bound but is not a logical premise. Work in the loopless plane-multigraph category through the splice and apply Corollary 15 if a parallel seam appears. Finally, verify by the same route-native semantics that no counterexample exists at or below the resulting effective threshold. Then every bridgeless cubic plane graph is Tait colourable, and hence every planar graph in the finite combinatorial-map convention is four-colourable.*

Proof. Suppose otherwise and choose a least counterexample. The finite base puts it above the threshold. Apply the target-preserving ladder reductions and the global cut-chain theorem. Exact Count factorization and Definition 2 identify the successive cumulative supports with a path in the finite powerset system. The cut chain is longer than 2^{3^k} (or than the separately verified optimized reachable-state count), so two cumulative support states repeat. Theorem 31 preserves the empty accepting support. The supplied literal cut data meet the hypotheses of Theorem 13, which realizes the deletion as a strictly smaller noncolourable member of the same bridgeless cubic spherical rotation-system class. This contradicts minimality. The Tait bridge gives the Four-Colour Theorem. □

Theorem 46 (route-native finite-base reflection). VERIFIED *Fix a vertex bound B, a finite state set Q, a predicate* $\mathrm{Necessary} \subseteq Q$, *and a computed reachable set $R \subseteq Q$. Suppose every graph-backed bridgeless spherical cubic map M with $|V(M)| \leq B$ has a trace $\tau(M) \in Q$ such that:*

1. *$\tau(M) \in R$; and*
2. *if M is not Tait-colourable, then* $\mathrm{Necessary}(\tau(M))$.

If

$$\{q \in R : \mathrm{Necessary}(q)\} = \varnothing,$$

then every such M is Tait-colourable. Consequently the exact finite-base premise at bound B used by the spherical reductive assembly holds.

Proof. If a bounded map M were not Tait-colourable, adequacy would make $\mathrm{Necessary}(\tau(M))$ true, while coverage would put $\tau(M)$ in R. Thus $\tau(M)$ would belong to the displayed filtered set, contradicting its emptiness. This is exactly the proposition $\mathrm{TaitBaseVerifiedAt}(B)$ consumed by the checked spherical assembly. □

This theorem fixes the certificate boundary without pretending to have run the certificate. The trace must come from the compositional profile system, its adequacy theorem must be proved from the actual bounded map, and the reachable set must cover all such traces. It may safely overapproximate the realizable states. The final obligation is then one explicit finite emptiness check, suitable for the flat-payload verified-reflection discipline; it is not a classical catalogue of unavoidable configurations.

Theorem 47 (checked spherical reductive assembly). VERIFIED *Fix interface bounds k, w, and put*

$$B(k,w) = 4\left(2^{(6w+1)\sum_{0\leq j\leq k} j!\,2^{3^j}} - 1\right) + 6.$$

Suppose that every graph-backed vertex-minimal Tait counterexample carries the complete rooted connected edge-leaf decomposition of bounds k, w required by Theorem 16. Suppose also that every graph-backed bridgeless spherical cubic map with at most $B(k,w)$ vertices is Tait colourable. Then every finite combinatorially planar simple graph is vertex-four-colourable, component by component.

Proof. If a Tait counterexample exists, choose one with the least number of vertices. Bridgefreeness and the spherical Euler equation make every edge two-sided; minimality excludes parallel edges, so this least rotation map has a canonical simple-graph backing. The supplied connected decomposition and Theorem 16 give

$$|V| \leq B(k,w).$$

The bounded-base hypothesis therefore Tait-colours the same map, a contradiction. Thus no graph-backed vertex-minimal Tait counterexample exists. The checked minimal-counterexample selection and stellar Tait reduction then give the connected spherical Four-Colour statement, and the componentwise wrapper gives the displayed conclusion. Lean checks this entire implication; its only open inputs are the decomposition-producing function and the bounded-base proposition stated above. □

Corollary 18 (factored bounded-width assembly). VERIFIED *Fix w and put $B = B(w, w)$. Suppose every graph-backed vertex-minimal Tait counterexample has a complete rooted branch decomposition of width at most w, and suppose every graph-backed bridgeless spherical cubic map with at most B vertices is Tait colourable. Then every finite combinatorially planar simple graph is vertex-four-colourable.*

Proof. A least counterexample is connected and bridge-free, hence 2-edge-connected. Apply Theorem 17 to the raw width-w tree, then apply the checked rooted adapter to obtain the exact decomposition supply of Theorem 47 with $k = w$. That theorem and the bounded base give the conclusion. Lean checks this composition; the only remaining mathematical inputs at this boundary are the raw width bound and the route-native finite base. □

Corollary 19 (factored assembly from a route-native base audit). VERIFIED *Fix w, put $B = B(w, w)$, and suppose every graph-backed vertex-minimal Tait counterexample has a complete rooted branch decomposition of width at most w. Suppose also that a finite-state reflection at bound B satisfies the coverage and adequacy hypotheses of Theorem 46, and that its filtered reachable set is empty. Then every finite combinatorially planar simple graph is vertex-four-colourable.*

Proof. Theorem 46 converts the finite audit into the exact bounded-base proposition. The checked width-preserving connectedization converts the raw decomposition into the connected decomposition consumed by Theorem 47. The Lean declaration cited above checks this composition without any additional premise. □

Corollary 20 (wall-exclusion assembly). CONDITIONAL *Assume the conventional representation theorem from finite abstract planar graphs to spherical rotation presentations. Assume there is a fixed wall excluded from every least Tait counterexample, and that wall-free members of the normal-form class carry a complete rooted branch decomposition of width at most w. Assume that the resulting route-native finite base contains no counterexample. Then every planar graph is four-colourable.*

Proof. Proposition 6 bounds the size of a least Tait counterexample and constructs the raw width supply consumed by Corollary 18. Theorem 17 converts it to the literal decomposition supply, and the finite-base hypothesis is the remaining premise. That corollary gives the spherical Four-Colour statement; the separate classical representation theorem from an abstract planar graph to a spherical rotation presentation gives the customary abstract-planar formulation. □

The word "bridgeless" in the splice conclusion is load-bearing. A cubic plane graph with a bridge has zero Tait count for a trivial parity reason, so a splice producing a bridge would not contradict minimality inside the class to which Tait's equivalence applies. Planarity, cubicity, connectedness, strict size decrease, and zero Count are proved together with bridgelessness in the ordinary argument of Theorem 43. Lean now checks the literal rotation-system composite's connectedness, bridge-freeness, cubicity, and rotation-cyclicity in Theorem 44. The orientation-aware face calculation and spherical Euler equality are now verified in Theorem 39 and Corollaries 9 and 14 package these facts on the literal composite. Corollary 11 derives the cap hypotheses from connected graph-backed shores and the exact planar-bond rank, without a Jordan/noose premise. The literal composite is loopless, and Corollary 15 removes any parallel seam if a later simple-graph presentation requires it. Thus the remaining cut-side work is M1 rather than M2: produce the complete rooted edge-leaf decomposition and its nodewise connected cuts and middle bounds from the chosen corridor or connected branch decomposition. The two strictly nested trees, vertex accounting, roots, ports, strict material, ordered first-return data, and exact supports are computed by the checked consumer; the supplier does not assume in advance that two such fields are equal.

Theorem 45 is the shortest literal implementation of the companion note's instruction to open the instances, count through the finite interface, compose, and pump. It does not use Trail Completability or Kauffman factor gluing. Those remain a source-faithful alternative assembly and a valuable explanation of the formation calculus, but their synchronization seam is not on the direct reductive critical path. No classical unavoidable set, discharging rule, or configuration catalogue enters the replacement.

1.9 GWCO: the refuted shortcut and the surviving lemma

The Good-Word Closed-web Obstruction is false. REFUTED Good-word totally closed webs occur already at ten internal vertices and in a growing family. Consequently the old False-valued GWCO base, its numerical threshold, and any reductive supplier targeting that base are retired. The dodecahedral opening is the smallest witness but its two cap interfaces share one boundary edge. A non-polar C_{30} opening at cap faces (4) and (12) is the clean annular witness: the cap cycles are vertex- and edge-disjoint, and exact enumeration finds six good-word totally closed webs. This does not contradict the manuscript's polar C_{30} and C_{40} censuses, which are reproduced exactly and contain only bad-word closed webs; it refutes the universal extrapolation from those laboratories.

The purpose of GWCO was narrower than its statement: it was meant to exclude a fibre frozen in an adversarial menu-A state. At the five-outer-stub boundary used by the closed-web system, the following replacement is valid.

Lemma 37 (Totally closed at five outer stubs implies menu B)**.** *VERIFIED* *Let an annular frontier tangle have five inner and five outer stubs and a good inner word. If a colouring has a totally closed web, then it is menu B for each majority pair.*

Proof. The Cubic Closure Theorem makes every two-colour component inner-touching. At the five–five boundary, the exact endpoint census then rules out two distinct inner stubs in one component; equivalently, every such component is radial, with one inner and one outer end. Thus, for either majority pair, the open strand relation on the four active inner positions is diagonal: no strand joins two distinct positions. Restoring the pentagonal cap contributes exactly its two disjoint active blocks. Since no open strand can join those blocks, their generated equivalence relation still has at least two classes. The transverse-chord criterion identifies this condition precisely with cap-composed menu B. Lean checks the entire chain on the literal graph-derived profile relation, including both majority pairs. □

For six or more outer stubs the source's Cubic Closure counting already makes total closure impossible. Boundaries with fewer than five outer stubs are not covered by Lemma 37; they must be discharged by their own finite boundary analysis if they occur in the final assembly. This qualification is load-bearing and prevents the five-stub argument from being silently promoted to an all-boundary theorem.

1.10 The exact conditional conclusion

Remark 4 (retired Trail-Completability assembly)**.** *REFUTED* *The earlier conditional route theorem passed from a deleted edge to a coloured uncompletable trail and later applied the closed-cubic Kauffman parity theorem to defect-path moves. Both steps fail, independently, as shown in Section 6. Merely listing Trail Completability and a relative parity theorem as assumptions would conceal rather than repair the missing mathematics, so that assembly is withdrawn.*

The surviving conditional conclusion is Theorem 45, with the tree-form alternative isolated in Corollary 20. The linear dependency chain is: the Tait/minimal-counterexample bridge; source-preserving local ladder transport; Proposition 5; the raw noose Count theorem and physical support-state splice; and a route-native finite base. The tree-form chain replaces the global corridor proposition by fixed-wall exclusion plus a raw bounded-width decomposition. The checked width-preserving connectedization theorem supplies the connected tree. Both chains use the same checked majority-shore D3 splice and the same finite base.

At present the decisive open premises are therefore: either the target-forced global cut chain or fixed-wall exclusion together with the raw bounded-width decomposition supplier; and the route-native finite base. Exact Count replacement on supplied normalized majority shores is now checked. A compatible tight vertex-noose to edge-noose perturbation is optional rather than load-bearing. D3's face count, Euler arithmetic, and majority-shore structural packaging are no longer open. The mesh isoperimetry theorem proves that a large wall does not automatically supply the linear chain, but it does not exclude the wall. An executable minimized Cell closure remains important for making the base small enough to replay, while the raw finite semantics no longer depends on the refuted closed-web carrier. Neither Trail Completability nor Kauffman factor gluing lies on this direct reductive critical path.

2 Foundations: transposition surgery and two-sided faces

The compositional route needs one classical fact at its classical boundary: in a bridgeless spherical cubic map, the two sides of every edge lie on *different* faces. In the informal literature this is treated as a consequence of the Jordan curve theorem and therefore as visually obvious. It is obvious. It is also, in a permutation model of maps, a corollary of Euler's inequality, and that route is entirely finite and combinatorial.

This section gives the finite argument in full. The mathematics is standard — the surgery lemma is folklore and Euler's inequality for hypermaps goes back to Jacques — but the path is unusually short when the genus inequality is proved *first*, and it needs remarkably little: no disjointness of the transpositions, no orientability argument, and no Jordan curve theorem.

2.1 Setting

Throughout, D is a finite set of *darts* and $\mathrm{Sym}(D)$ its symmetric group. For $\pi \in \mathrm{Sym}(D)$ write $c(\pi)$ for the number of orbits of π on D, *counting fixed points as orbits.* For a family $G \subseteq \mathrm{Sym}(D)$ write $c\langle G\rangle$ for the number of orbits of the subgroup it generates.

Definition 4 (combinatorial map)**.** A *map* is a triple (D, σ, α) with $\sigma \in \mathrm{Sym}(D)$ and $\alpha \in \mathrm{Sym}(D)$ a fixed-point-free involution. We call σ the *vertex rotation*, α the *edge flip*, and $\varphi := \sigma\alpha$ the *face permutation*, whose orbits are the *faces.* Set

$$V := c(\sigma), \qquad E := c(\alpha) = |D|/2, \qquad F := c(\varphi), \qquad C := c\langle \sigma, \alpha\rangle.$$

The map is *connected* if $C = 1$, and *spherical* if it is connected and $V - E + F = 2$.

Two conventions deserve comment. Faces are orbits of $\sigma\alpha$ rather than $\alpha\sigma$; the two are conjugate, so no count below is affected. And an *edge* is a pair $\{d, \alpha d\}$ whose two elements are its two *sides* — whether they lie on one face or two is exactly the question at issue.

Definition 5 (bridge)**.** An edge $e = \{d, \alpha d\}$ is a *bridge* if deleting it increases the number of connected components. Formally, put $\alpha_0 := \alpha \circ (d\ \alpha d)$, an involution fixing d and αd and agreeing with α elsewhere, and let $C_0 := c\langle \sigma, \alpha_0\rangle$. Then e is a bridge exactly when $C_0 > C$.

Note that α_0 is *not* the edge flip of a map: it has fixed points. This is intentional and harmless, since none of the counting below assumes fixed-point-freeness. Retaining the darts $d, \alpha d$ rather than removing them is what leaves σ, and hence V, literally unchanged by the deletion, which is what keeps the bookkeeping honest.

2.2 Transposition surgery

Everything rests on one lemma about what a transposition does to an orbit count.

Lemma 38 (split/merge). VERIFIED *Let $\pi \in \mathrm{Sym}(D)$, let $a \neq b$ in D, and let $\tau = (a\ b)$. If a and b lie on the same π-orbit then $c(\tau\pi) = c(\pi) + 1$; otherwise $c(\tau\pi) = c(\pi) - 1$.*

Proof. Since $(\tau\pi)(x) = \tau(\pi x)$, the permutations $\tau\pi$ and π agree at every x with $\pi x \notin \{a, b\}$. Only two points are affected: the π-preimages of a and of b.

Suppose first that a, b share a π-orbit, written

$$(a,\ \pi a,\ \pi^2 a,\ \ldots,\ \pi^{m-1} a), \qquad \pi^m a = a, \qquad b = \pi^k a, \quad 0 < k < m.$$

Then π sends $\pi^{k-1} a \mapsto b$ and $\pi^{m-1} a \mapsto a$, so $\tau\pi$ sends $\pi^{k-1} a \mapsto a$ and $\pi^{m-1} a \mapsto b$. The single orbit therefore breaks into $(a, \pi a, \ldots, \pi^{k-1} a)$ and $(b, \pi^{k+1} a, \ldots, \pi^{m-1} a)$: two orbits where there was one, all others untouched.

For the second case note $\tau(\tau\pi) = \pi$, and that a and b do share a $\tau\pi$-orbit — indeed $\tau\pi$ carries the π-preimage of b to a and the π-preimage of a to b, welding the two orbits into one. The first case applied to $\tau\pi$ then gives $c(\pi) = c(\tau\pi) + 1$. □

The second case repays attention, because it is the only place a reader might reach for a picture. The identity $\tau(\tau\pi) = \pi$ says the two cases are one computation read in opposite directions: a transposition is its own inverse, so splitting and merging are a single operation seen from two sides.

Corollary 21 (right-handed form). VERIFIED *With π, a, b, τ as above, $c(\pi\tau)$ obeys the same dichotomy.*

Proof. $\tau(\pi\tau)\tau^{-1} = \tau\pi$, so $\pi\tau$ and $\tau\pi$ are conjugate and have equal orbit counts; apply Lemma 38. □

We use the right-handed form below, since deleting an edge multiplies the face permutation on the right.

Remark 5. *Lemma 38 is the exact orbit-counting refinement of the familiar parity statement* $\mathrm{sgn}(\tau\pi) = -\,\mathrm{sgn}(\pi)$*. Parity records that the count changes by an odd number; the lemma records that it changes by exactly one, and says which way.*

2.3 Euler's inequality

Theorem 48 (Euler bound). VERIFIED *Let $\sigma \in \mathrm{Sym}(D)$ and let $\tau_1, \ldots, \tau_n$ be transpositions,* not assumed disjoint*, with product $\beta = \tau_1\tau_2\cdots\tau_n$. Then*

$$c(\sigma) + c(\sigma\beta) \ \leq\ n + 2\, c\langle \sigma, \tau_1, \ldots, \tau_n \rangle.$$

Proof. Induction on n.

For $n = 0$ we have $\beta = 1$, the generated group is $\langle\sigma\rangle$ whose orbits are the σ-orbits, and both sides equal $2c(\sigma)$. This base case is an *equality*; all the slack in the theorem is created by the inductive step.

For the step write $\beta = \tau\beta'$ with $\tau = (a\ b)$ and $\beta' = \tau_2 \cdots \tau_n$. Put $G' := \langle\sigma, \tau_2, \ldots, \tau_n\rangle$ and $G := \langle\sigma, \tau, \tau_2, \ldots, \tau_n\rangle$, with $C' = c\langle G'\rangle$ and $C = c\langle G\rangle$. The inductive hypothesis reads $c(\sigma) + c(\sigma\beta') \le (n-1) + 2C'$.

The key algebraic step is that inserting τ on the right of σ is the same as inserting a conjugate transposition on the left of $\sigma\beta'$:

$$\sigma\beta = \sigma\tau\beta' = (\sigma\tau\sigma^{-1})(\sigma\beta') = \tau'(\sigma\beta'), \qquad \tau' = (\sigma a\ \ \sigma b).$$

Since $a \ne b$ and σ is injective, $\sigma a \ne \sigma b$, so Lemma 38 applies to τ' and $\sigma\beta'$.

Case (i): a and b lie in the same G'-orbit. Adjoining τ joins nothing new, so $C = C'$. The lemma gives $c(\sigma\beta) \le c(\sigma\beta') + 1$ in either branch, whence

$$c(\sigma) + c(\sigma\beta) \le c(\sigma) + c(\sigma\beta') + 1 \le (n-1) + 2C' + 1 = n + 2C.$$

Case (ii): a and b lie in different G'-orbits. Adjoining τ fuses exactly those two, so $C = C' - 1$. We claim the lemma must take its merging branch. Indeed $\sigma \in G'$, so σa and σb again lie in different G'-orbits; and $\sigma\beta' \in G'$, so $\langle\sigma\beta'\rangle \le G'$ and every $\sigma\beta'$-cycle lies inside a single G'-orbit. Hence σa and σb lie on different $\sigma\beta'$-cycles and $c(\sigma\beta) = c(\sigma\beta') - 1$. Therefore

$$c(\sigma) + c(\sigma\beta) = c(\sigma) + c(\sigma\beta') - 1 \le (n-1) + 2C' - 1 = n + 2C. \qquad \square$$

The two cases repay contrast. In case (i) the component count is frozen while the face count may rise, so the inequality loses ground — by exactly one unit per independent cycle, which is the genus. In case (ii) the component count falls, but the face count is *forced* to fall with it and the ledger balances exactly. The bound is tight because a new edge across two components can never create a face.

Corollary 22 (Euler's inequality for maps)**.** VERIFIED *For a map (D, σ, α) we have $V - E + F \le 2C$, with the edges taken to be the two-cycles of α.*

Proof. Present α as an explicit list of pairwise disjoint transpositions by repeatedly stripping a two-cycle: from a moved point a, replace α by $\mathrm{swap}(a, \alpha a)\,\alpha$, which fixes both a and αa and agrees with α elsewhere, so the moved set strictly shrinks and the recursion terminates. The product of the emitted list is α. Theorem 48 applied to that list gives $V + F \le E + 2C$, since disjointness makes a single α-step coincide with a step of one of the transpositions. $\square$

Remark 6 (the bookkeeping, discharged)**.** VERIFIED *That each emitted pair is a genuine two-cycle holds by construction, since a pair is produced only from a moved point; no involutivity is even needed for it. With this the map form of the bound is unconditional.*

Remark 7. *For a connected map this reads $V - E + F \le 2$, and the* genus $g := (2 - V + E - F)/2$ *is precisely the accumulated slack of case (i). Corollary 22 is the statement $g \ge 0$; spherical maps are those achieving equality.*

2.4 Bridges and two-sided faces

Theorem 49. VERIFIED *Let (D, σ, α) be a spherical map and let $e = \{d, \alpha d\}$ be an edge whose two sides lie on the* same *face. Then e is a bridge.*

Proof. Let $\tau := (d\ \ \alpha d)$ and $\alpha_0 := \alpha\tau$ as in Definition 5, and let C_0 be the number of components after deleting e. Three observations.

First, σ is untouched, so the vertex count of the deleted structure is still V.

Second, α_0 is a product of the $E - 1$ disjoint transpositions of α other than τ, together with two fixed points, and $\sigma\alpha_0 = \sigma\alpha\tau = \varphi\tau$. Since d and αd lie on a common φ-orbit, Corollary 21 gives $c(\sigma\alpha_0) = F + 1$.

Third, Theorem 48 applied to σ and those $E-1$ transpositions gives $V + c(\sigma\alpha_0) \leq (E-1) + 2C_0$, the generated group having exactly the components of the deleted structure as in Corollary 22.

Combining, $V + F + 1 \leq E - 1 + 2C_0$. Sphericity gives $V + F = E + 2$, so $E + 3 \leq E - 1 + 2C_0$, that is $C_0 \geq 2$. The original map was connected, so deleting e increased the component count and e is a bridge. ☐

Corollary 23 (two-sidedness)**.** VERIFIED *In a connected spherical cubic rotation map with cyclic vertex rotations, if deleting any edge leaves the primal graph connected, then every edge has two distinct faces on its two sides.*

Proof. Fix a dart d of the edge to be deleted and remove its transposition from the edge involution. Bridge-freeness makes the edge-deleted primal graph connected. A primal walk then lifts, edge by edge, to a word in the vertex rotation and the remaining edge transpositions, so those generators have one dart component. If d and αd lay on one facial orbit, Theorem 49 in its permutation form would force at least two such components. This contradiction proves two-sidedness.

Equivalently, this is the contrapositive of the same-face bridge theorem, but the checked consumer above also proves that the route's concrete `EdgeBridgeFree` field has exactly the connectivity content needed by that contrapositive. ☐

Remark 8 (the converse, and what it actually costs)**.** PROSE *The converse — a bridge has one face on both sides — is also true, but it is* not *obtainable from Theorem 48. Running the same computation with $C_0 = 2$ yields only $c(\varphi\tau) \leq F + 1$, which admits both branches. It was tempting to conclude that the converse therefore needs the equality case of Euler; it does not. It has an elementary proof of a different kind, given next, which uses no counting at all.*

Theorem 50 (the walk lemma)**.** VERIFIED *Let φ be a permutation of D, let* side: $D \to \{0, 1\}$*, and let $d, o \in D$. Suppose the first step off d changes side, and that from any point on the far side other than o the next step stays on the far side. Then d and o lie on one φ-orbit.*

Proof. The orbit of d is finite and returns to d, so among $k \geq 1$ there is a least one with $\mathrm{side}(\varphi^k d) = \mathrm{side}(d)$. It is not $k = 1$, since the first step crosses. So $\varphi^{k-1}d$ lies on the far side by minimality, while $\varphi^k d$ does not. The only far-side point whose next step leaves the far side is o; hence $\varphi^{k-1}d = o$. ☐

Applied to a bridge, with side recording which component of the edge-deleted map a dart's vertex lies in, this is precisely the converse: a facial walk crossing a bridge has exactly one door back, namely the bridge's other dart. Note what the hypotheses do *not* include — that o itself lies on the far side is not needed, and the formal statement omits it.

Theorem 51 (the converse, packaged)**.** VERIFIED *If deleting an edge separates its two endpoints — that is, if the edge is a bridge — then its two sides lie on one face orbit.*

Proof. Take side(x) to record whether x is reachable from a in the edge-deleted map. Then side(a) holds by reflexivity; the first step off a lands on σb, which is reachable from b and hence not from a, so the step crosses; and from any far-side point $x \neq b$ the edge product cannot carry x onto a or b — both are fixed by it, and it is injective — so the step is a rotation composed with a remaining flip, neither of which changes the component. Theorem 50 applies. □

Corollary 24 (bridges are exactly one-sided edges)**.** PROSE *In a spherical map, an edge is a bridge if and only if its two sides lie on the same face.*

Proof. Theorem 49 and Theorem 51. □

Both halves are now finite and combinatorial. The forward direction counts; the converse walks. Neither looks at a picture.

2.5 Kempe components and boundary stubs

One further counting fact belongs here, because the compositional route's terminal analysis uses it and because it is easy to state in the wrong place.

Lemma 39 (a Kempe component holds at most two stubs)**.** VERIFIED *Let H be a finite graph in which every vertex has degree at most two, and let K be a connected component of H. Then K has at most two vertices of degree one.*

Proof. Let K have n vertices, m edges, and d vertices of degree one. Connectedness gives $m \geq n-1$. Every vertex has degree at most two, and the d special ones have degree exactly one, so

$$2m \;=\; \sum_{v\in K} \deg v \;\leq\; 2n - d.$$

Combining, $2(n-1) \leq 2m \leq 2n - d$, whence $d \leq 2$. □

In a properly 3-edge-coloured cubic graph every vertex meets exactly one edge of each colour, so a two-coloured subgraph has degree exactly two at every internal vertex and degree one at a boundary stub. Lemma 39 therefore says a Kempe component of a colour pair contains at most two stubs — it is a path between them, or a cycle carrying none.

The degree hypothesis is already available in this development rather than needing to be re-established: `colorPairGraph_degree_eq_two_of_cubic_tait` gives degree exactly two at any cubic vertex of a Tait colouring, and `cubicInterior_colorPairGraph_degree_eq_two` specialises it to the framed model's interior. Wiring those to Lemma 39 requires only a bound at the boundary vertices and a connectivity hypothesis at the use site; the adapter is deliberately not written here, since no consumer yet exists and a speculative one would be a guess at the interface its caller wants.

Remark 9 (where this must and must not be applied)**.** *The lemma is about the* open *tangle, where stubs are genuinely of degree one. In the closed map the same two-coloured subgraph has degree two everywhere and its components are cycles, which may pass through arbitrarily many former stub positions. Counting components in the open tangle and counting them in the closed map are therefore different measurements, and a bound established for one says nothing about the other. The distinction is easy to lose and expensive to lose: an argument that silently moves between the two proves nothing.*

Remark 10 (what was avoided). *The usual proof of Corollary 23 runs: a non-bridge lies on a cycle; the cycle is a Jordan curve; the two sides of the edge are separated by it; hence they lie in different faces. That argument is correct and is what a reader should picture. It is also, in a formal development, expensive: it requires a combinatorial Jordan curve theorem for embedded graphs, which no current library provides and which is a substantial development in its own right. The route above replaces it with Lemma 38 and one induction. The picture is the motivation; the counting is the proof.*

Remark 11 (on the order of work). *It is worth recording why the order mattered. The target, Corollary 23, is a statement about faces, and the natural first instinct is to attack it directly: take an edge with one face on both sides, walk around that face, and try to extract a separation. That is the Jordan route, and in a formal setting it does not terminate in reasonable time. Proving the* more general *Theorem 48 first — a statement about arbitrary permutations and arbitrary transpositions, with no map, no sphere, and no edges in sight — turns the target into three lines of arithmetic. The generalisation is easier than the special case precisely because it removes the structure that made the special case tempting to visualise.*

3 Graph-backed presentation, and why minimality is not optional

The compositional machinery is stated over rotation systems; the classical bridge of Section 4 is stated over simple graphs. To join them one needs a bridgeless spherical cubic map to be presentable as the rotation system of a *simple* graph.

The tempting statement is that every such map is so presentable. That statement is false, and the counterexample is small enough to be worth stating first, because it tells us exactly which hypothesis is doing the work.

3.1 The obstruction

Example 1 (the spherical theta multigraph). PROSE Let $D = \{u_0, u_1, u_2, v_0, v_1, v_2\}$, let the edge flip exchange u_i with v_i, and let the vertex rotations be

$$\sigma_u = (u_0\, u_1\, u_2), \qquad \sigma_v = (v_0\, v_2\, v_1).$$

The face permutation is then $(u_0\, v_2)(u_1\, v_0)(u_2\, v_1)$, so

$$V = 2, \qquad E = 3, \qquad F = 3, \qquad V - E + F = 2.$$

This map is spherical, cubic, connected, and bridgeless. But no simple graph on two vertices is cubic — a simple graph on two vertices has maximum degree one — so there is no equal-vertex-count presentation of it by a simple graph.

Two vertices joined by three parallel edges: the multiplicity is exactly what a simple graph cannot express. The unrestricted presentation claim is therefore false, and any development asserting it is asserting something untrue.

The repair is not to weaken the conclusion but to remember the hypothesis that was available all along. We do not need to present *every* map. We need to present a *least counterexample*, and the theta map is not one:

Remark 12. *PROSE The theta map is Tait colourable. Its three edges are incident with both vertices, so assigning them three distinct colours properly colours both stars at once. Hence it is not a counterexample to anything, and no statement restricted to counterexamples is threatened by it.*

That is the shape of the whole section: multiplicity is exactly what minimality removes.

3.2 Setting

Fix a map M that is connected, spherical, cubic, and bridgeless, and that is a *least counterexample*: M admits no Tait colouring, while every map in the same class with strictly fewer vertices does.

Lemma 40 (no loops)**.** PROSE *M has no loop. This uses bridgelessness only; minimality is not needed.*

Proof. Suppose v carries a loop ℓ. The loop occupies two of the three darts at v, so exactly one further edge e is incident with v, and its other end is some vertex w. If $w = v$ then e is a second loop at v, giving v four darts and contradicting cubicity; so $w \neq v$ and, in particular, M has at least two vertices.

Delete e. The only remaining edge at v is the loop ℓ, so $\{v\}$ is a connected component of $M - e$, while $w \neq v$ lies in another. Thus deleting e increased the number of components: e is a bridge, contradicting bridgelessness. □

3.3 Removing parallel edges

This is where minimality is spent.

Lemma 41 (the digon splice)**.** VERIFIED *Suppose distinct vertices u, v of M are joined by exactly two parallel edges, and M is not the theta map. Assume the two dart-sides of every edge lie on distinct face orbits, as in the class predicate `OrbitFacesTwoSided`. Let $a = uu'$ and $b = vv'$ be the remaining edges at u and at v. Then $u' \neq v'$, and the map M' obtained by deleting u, v, the two parallel edges, a and b, and inserting a single new edge $c = u'v'$, is again connected, spherical, cubic, and bridgeless, with exactly two fewer vertices.*

Proof. Each of u and v has three darts, two consumed by the parallel pair, so a and b are well defined and unique. By Lemma 40, $u' \neq u$ and $v' \neq v$. Moreover $u' \neq v$ and $v' \neq u$: either equality would make the remaining edge a third uv edge; cubicity and connectedness would then make M the theta map, which has been excluded. Thus the splice really removes u, v and joins two surviving vertices.

First, $u' \neq v'$. Suppose $u' = v' = w$. Then w is incident with a and with b, and being cubic it has exactly one further edge f. Now $\{u, v, w\}$ meets the rest of M only along f: the parallel pair, a and b are internal to it, and u, v have no other darts. Deleting f therefore separates $\{u, v, w\}$ from the remainder, so f is a bridge unless there is no remainder — and if there is none, w's third dart has nowhere to go without violating cubicity. Either way we contradict a standing hypothesis, so $u' \neq v'$.

Cubicity. Every vertex other than u', v' is untouched. Each of u' and v' loses one incident edge (a, resp. b) and gains one (c), so retains degree three.

Connectedness. Any path of M through the deleted region entered and left along a and b, and now traverses c instead.

Bridgelessness. Suppose c were a bridge of M', separating $M' - c$ into a part $A \ni u'$ and a part $B \ni v'$. Then in M the only connections between A and B run through u and v, hence through a and the parallel pair and b. Deleting a from M would then separate A from the rest, making a a bridge of M and contradicting bridgelessness. For an edge $e \neq c$ of M', expand every traversal of c to the path $u'au, e_1, vbv'$ in M, and contract every traversal of the deleted two-vertex network in M back to c. These operations preserve walks after deleting e. Hence, if $M' - e$ were disconnected, then $M - e$ would be disconnected as well, contrary to bridgelessness of M.

Sphericity. The count is consistent: $V' = V - 2$, $E' = E - 3$, and one face is destroyed — the face bounded by the parallel pair — so $F' = F - 1$ and

$$V' - E' + F' = (V - 2) - (E - 3) + (F - 1) = V - E + F = 2.$$

Here the face assertion is literal, not inferred from Euler. Orient the rotations so the digon orbit is (q_1, p_2) as in the proof of Theorem 52. Replacing the two external chains a, e_1, b and a, e_2, b by the two darts of c shortens each of the two non-digon face orbits without joining or splitting it; every orbit disjoint from the six local darts is unchanged, and the orbit (q_1, p_2) disappears. Thus the surviving face orbits are in bijection with all old face orbits but the digon, proving $F' = F - 1$.

The step that must be discharged is the claim that the parallel pair bounds a face — it does not follow from the counts. Under the explicitly stated two-sidedness predicate, Theorem 52 supplies it by a finite dart-orbit computation. The checked splice then identifies the cap face permutation with first return on the retained darts, proves that precisely one old face orbit disappears, and performs the Euler arithmetic above. Thus the splice uses no separation theorem. □

3.4 What the digon face really needs

The splice above rests on the parallel pair bounding a face. That claim decomposes further than it first appears, and the decomposition is worth recording because half of it is free.

Write the digon's darts as p_1, p_2 at u (on the two parallel edges e_1, e_2) and q_1, q_2 at v, with $\alpha p_i = q_i$, and let p_a, q_b be the third darts at u and at v.

Lemma 42 (a cubic digon always shares a face)**.** VERIFIED *In a cubic map, some face orbit contains a dart of e_1 and a dart of e_2. No sphericity and no counting are needed.*

Proof. The rotation σ_u is a three-cycle on $\{p_1, p_2, p_a\}$, so either $\sigma p_1 = p_2$ or $\sigma p_1 = p_a$. In the first case $\varphi(q_1) = \sigma(\alpha q_1) = \sigma p_1 = p_2$, so q_1 and p_2 lie on one face. In the second case $\sigma_u = (p_1\, p_a\, p_2)$, whence $\sigma p_2 = p_1$ and $\varphi(q_2) = p_1$, so q_2 and p_1 lie on one face. With only three darts at a cubic vertex, two of them are cyclically adjacent one way or the other, and there is no third possibility. The formal statement takes that dichotomy as its hypothesis, which is precisely what a three-dart rotation supplies. □

Remark 13 (where sphericity actually enters)**.** *What the splice needs is stronger: that the shared face is* exactly *the digon, with boundary $\{e_1, e_2\}$ and nothing more, since only then does deleting the pair destroy precisely one face and keep the Euler count of Lemma 41 honest. Continuing the computation above in the first case, $\varphi(p_2) = \sigma(q_2)$, and this returns to q_1 – closing the two-cycle – exactly when $\sigma q_2 = q_1$, that is, when the rotation at v is oriented compatibly with the one at u. Both orientations give legitimate rotation systems; they differ in genus. So sphericity is precisely the hypothesis that rules the incompatible one out, and this is the point at which the digon lemma stops being free.*

A small exhaustive check corroborates this before any proof is attempted. Take the cubic multigraph on $\{u, v, w, x\}$ with a parallel pair joining u to v, a parallel pair joining w to x, and single edges uw and vx; enumerate all $2^4 = 16$ rotation systems on it. Of these, four are spherical and twelve have genus one. In all four spherical systems the pair $\{e_1, e_2\}$ bounds a face of length exactly two; among the genus-one systems only four do, and eight do not. So sphericity is sufficient here and visibly not necessary — which is the asymmetry the remark predicts, and the direction the splice actually consumes. This is corroboration on one family rather than a proof. The next subsection supplies the conditional ordinary proof; this preliminary computation carries no independent status marker.

3.5 The digon face, from bridgelessness

The exhaustive check suggests the claim; here is the argument. It is worth setting down because it consumes no new hypothesis: the third edge at u would be a bridge, and bridgelessness is already standing.

Lemma 43 (a parallel pair bounds a face)**.** CONDITIONAL *Let M be connected, spherical, cubic and bridgeless, let distinct vertices u, v be joined by parallel edges e_1, e_2, and suppose M is not the theta map of Example 1. Then $e_1 \cup e_2$ bounds a face of M.*

Proof. Since $u \neq v$ and e_1, e_2 meet only at u and v, the union $\gamma = e_1 \cup e_2$ is a simple closed curve on the sphere. By separation, its complement has exactly two components D_1, D_2, each an open disc with boundary γ, and γ carries no vertex other than u and v.

Cubicity leaves u exactly one dart p_a off γ, and v exactly one dart q_b. Each points into D_1 or into D_2.

They cannot point into different discs. Say p_a points into D_1 and q_b into D_2, and let a be the edge carrying p_a, with other end u'. By Lemma 40, $u' \neq u$. If $u' = v$ then a is a third uv edge, saturating both vertices, and connectedness makes M the theta map — excluded. So $u' \in D_1$.

Now every walk from u' to v must leave D_1, and it can only leave through γ; but γ carries no vertices besides u and v, and the only dart at either of them pointing into D_1 is p_a. So every such walk traverses a, and $M - a$ separates u' from v: the edge a is a bridge, contradicting bridgelessness.

Hence both point into the same disc, say D_1. Then no dart at u or at v points into D_2. A vertex in the interior of D_2 would reach the rest of M only through γ, hence through such a dart, and there is none; connectedness then forces the interior of D_2 to be free of vertices, so also of edges. Thus D_2 is a face, bounded by e_1 and e_2 and nothing else. □

3.6 The digon face, without separation

The proof above spends a separation statement. It need not. The same conclusion follows from two facts already machine-checked here — the bridge characterisation and Euler's *inequality* — and from the observation that a digon in a cubic map is attached to the rest of the map by exactly two edges.

The only global-looking part of the dart trace is the assertion that an exterior face which leaves the digon must return through one of its two attachments. The following finite-orbit lemma isolates that assertion.

Lemma 44 (two-door return)**.** VERIFIED *Let φ be a permutation of a finite set, let $s\colon D \to \{0, 1\}$, and let $d, a', b' \in D$. Suppose the first step from d changes side, and every point on the far side stays there under φ except possibly a' and b'. Then the φ-orbit of d contains a' or b'.*

Proof. The orbit is finite and returns to d. Choose the least positive k for which $s(\varphi^k d) = s(d)$. The first step crosses, so $k \geq 2$, and by minimality $\varphi^{k-1} d$ is on the far side. Its next step changes side; the side-invariance hypothesis therefore says that $\varphi^{k-1} d$ is one of the two doors. Hence that door lies in the orbit of d. □

Lemma 45 (the local digon orbit closes)**.** VERIFIED *Use local darts p_1, p_2, q_1, q_b and exterior darts a', b'. Assume that $\varphi(q_1) = p_2$, while the next step from p_2 is either q_1 or q_b. Assume also that the exterior has only the two doors a', b', that $\varphi(a') = p_1$, and that the edges with dart pairs $\{q_b, b'\}$ and $\{q_1, p_1\}$ are two-sided. Then*

$$\varphi(q_1) = p_2, \qquad \varphi(p_2) = q_1,$$

and the face orbit of q_1 is exactly $\{q_1, p_2\}$.

Proof. If the second step is q_b, Lemma 44 says that its exterior orbit contains a' or b'. Two-sidedness of the b-attachment excludes b', so the orbit contains a', hence also $\varphi(a') = p_1$. But $q_1 \to p_2 \to q_b$, so this puts q_1 and p_1 on one face, contradicting two-sidedness of e_1. Thus the second step is q_1. Iterating the resulting two-cycle shows that its orbit has precisely the two displayed darts. $\square$

Lemma 46 (the map-level digon orbit closes)**.** VERIFIED *Let (D, σ, α) be a cubic rotation map whose rotations are cyclic on each vertex, and suppose every edge is two-sided. Let e_1, e_2 be parallel edges between distinct vertices u, v. Write b' for the far dart of the third edge at v. If the vertex of b' is not u (and hence, by looplessness, is not v either), then one of*

$$(q_1, p_2), \qquad (q_2, p_1)$$

is an exact two-cycle of the face permutation. In particular the corresponding quotient face has precisely the two edges e_1, e_2.

Proof. Cubicity identifies the stars at u and v with $\{p_1, p_2, p_a\}$ and $\{q_1, q_2, q_b\}$, and cyclicity leaves exactly two orientations at each star. Fix the orientation at u. The first face step across one of the parallel edges then has the form $q_1 \mapsto p_2$ or $q_2 \mapsto p_1$. At v the next step either closes that two-cycle or exits through q_b.

Distinguish the two digon vertices from the exterior by a Boolean side. Away from a' and b', applying α cannot carry an exterior dart to u or v: the complete cubic stars show that the only exterior darts paired to local darts are exactly a' and b'. Rotation preserves the vertex, so the face permutation preserves this side away from those two doors. Since b' is exterior, the step from q_b crosses the side. Lemma 44 therefore says that an exiting facial walk returns through a' or b'. Two-sidedness excludes return through b'; return through a' would put both darts of the chosen parallel edge on one face, again contradicting two-sidedness. Thus the walk cannot exit, and Lemma 45 gives the asserted exact two-cycle. $\square$

Theorem 52 (the digon face, combinatorially)**.** PROSE *Let M be a connected bridgeless cubic map carrying a parallel pair e_1, e_2 between distinct vertices u, v, and suppose M is not the theta map. If M is spherical then $\{e_1, e_2\}$ is a face of M.*

Proof. Write p_1, p_2, p_a for the darts at u and q_1, q_2, q_b for those at v, with $\alpha p_i = q_i$; let a be the edge carrying p_a, with second dart a', and b the edge carrying q_b, with second dart b'. Since u and v have no further darts, every other dart lies in the exterior, and M meets the exterior only along a and b.

Corollary 23 makes every edge of M two-sided. The dart b' cannot be based at v: then b would be a loop, contributing two darts at v in addition to the darts of e_1, e_2, contrary to cubicity. If b' were based at u, then b would be a third uv edge. Both cubic stars would be saturated by the three parallel edges, and connectedness would make M the theta map, contrary to hypothesis. Thus b' is based at neither u nor v.

All hypotheses of Lemma 46 now hold. One of the two parallel-edge side pairs is an exact face-permutation two-cycle, so its quotient face has boundary $\{e_1, e_2\}$ and no other edge. $\square$

Remark 14 (the last embedding obligation, discharged)**.** PROSE *Theorem 52 uses no curve and no separation. Its inputs are the bridge characterisation (through two-sidedness), cubicity, and a finite permutation-orbit argument. The exterior is not opened: it is summarized by a Boolean side and the fact that the patch has exactly two doors. This is the same compositional principle used later by the corridor profiles, in its smallest nontrivial form.*

An exhaustive check accompanies the theorem: over three cubic multigraphs carrying a digon, all $16 + 64 + 256$ *rotation systems were enumerated, and no spherical bridgeless one failed to have the pair as a face.*

Lemma 43 is retained for the picture it gives, but nothing depends on it, and with it goes the last obligation in this development that mentioned a plane.

Remark 15 (what this costs)**.** PROSE *Nothing beyond what the route already owes. The load-bearing proof is Theorem 52: the bridge characterisation's two-sidedness corollary, cubicity, and a finite dart-orbit calculation. Lemma 43 remains only as the geometric picture and is not consumed. The exclusion of the theta map is harmless, since by Remark 12 it is Tait colourable and so is never a least counterexample.*

Lemma 47 (no parallel edges)**.** VERIFIED *M has no two vertices joined by more than one edge.*

Proof. Suppose it does. By cubicity there are either exactly two or exactly three edges between the repeated pair of vertices. In the latter case those two vertices have no other incident darts, so connectedness makes M the theta map, which is Tait colourable by Remark 12; this contradicts that M is a counterexample. Hence there are exactly two, M is not theta, and we may form M' as in Lemma 41. Then M' lies in the same class and has two fewer vertices, so by minimality of M it admits a Tait colouring χ'.

Lift χ' to M. Give a and b the colour $\chi'(c)$, and give every other edge of M its χ'-colour. It remains to colour the parallel pair. At u the edge a already carries $\chi'(c)$, so assigning the two parallel edges the two remaining colours makes the star at u proper; the same two colours make the star at v proper, since b also carries $\chi'(c)$. Every other star is unchanged from χ' — at u' the edge c has been replaced by a of the same colour, and likewise at v'.

So M admits a Tait colouring, contradicting the assumption that it is a counterexample. □

3.7 The presentation

Theorem 53 (graph-backed presentation of a least counterexample)**.** VERIFIED *A least counterexample M is the rotation system of a simple graph on its own vertex set, and Tait colourings transfer between the two descriptions.*

Proof. The rotation-system record is loopless by construction, and Lemma 47 says that its endpoint map is injective. Let G be the computed primal simple graph on the same vertex set. Every edge of M determines its unordered endpoint pair in G; surjectivity is immediate from the definition of the primal graph, and injectivity is exactly Lemma 47. Hence the literal edge carrier is canonically equivalent to $E(G)$.

The same construction on oriented edges gives a dart equivalence. It commutes with dart reversal, and transporting the vertex rotation through it commutes with σ and therefore with the facial permutation $\varphi = \sigma\alpha$. Consequently the vertex stars, facial orbits, and their cyclic orders are carried bijectively to the graph-backed rotation. In particular, cubicity and two-sidedness are preserved. The recomputed primal graph of the graph-backed rotation is literally G, so connectedness is preserved. Deleting an edge identifier before the transport is the same as deleting its corresponding edge of G, so bridgelessness is preserved. Finally the vertex carrier is unchanged while the edge and face carriers are bijective, and hence the Euler equation is preserved.

The edge equivalence is an isomorphism from the rotation-system edge adjacency graph to the line graph of G. Pullback and pushforward along this isomorphism preserve proper edge colourings and the condition that every edge has nonzero colour. Thus Tait colourability transfers in both directions. The vertex count is unchanged, so the same leastness hypothesis supplies the `smallerColorable` field of the resulting graph-backed minimal counterexample. ☐

Remark 16 (what the restriction bought). PROSE *The corresponding statement over the whole class is false by Example 1. What minimality supplies is precisely Lemma 47, and it supplies it in the only way available: by producing a smaller member of the class and colouring it. This is worth emphasising because the formal statement must carry the minimality data as a hypothesis — a version quantified over all bridgeless spherical cubic maps is not merely unproved but refuted, and a development asserting it would be unsound rather than incomplete.*

Remark 17 (status summary for the classical boundary). PROSE *With this section the ordinary mathematics of the classical bridge is closed. Two-sidedness is now machine-checked directly from the route's `EdgeBridgeFree` class field, and both directions of the bridge characterisation are machine-checked at the underlying permutation level. Theorem 52 removes the only geometric premise from the digon splice, so minimality eliminates loops and parallel edges and yields Theorem 53 without a separation axiom.*

The graph-backing transport and its least-counterexample constructor are now machine-checked; there is no remaining representation hypothesis at this junction. The companion reduction in Section 4 likewise uses no maximal-plane-graph argument: stellar subdivision is an explicit dart substitution, duality is the replacement of σ by $\sigma\alpha$, and looplessness of the subdivision makes its dual bridgeless. The classical maximal triangulation lemma and the geometric digon lemma remain in the text as recognizable comparison routes; their OPEN/CONDITIONAL markers describe those statements themselves, not dependencies of the controlling proof. The only remaining classical front-end boundary is the named theorem converting an abstract planar graph into a spherical rotation presentation; it lies outside graph-backed presentation and outside the connected-spherical headline.

4 The classical bridge: Tait's reduction

The compositional argument produces a statement about *cubic maps*: no bridgeless spherical cubic map fails to be 3-edge-colourable. The Four-Colour Theorem is a statement about *planar simple graphs*. Tait's 1880 reduction connects them, and this section gives it in full, with the status of each ingredient marked, because the ingredients are not uniformly available.

The reduction is classical and no part of it is in doubt. What is worth setting down precisely is which pieces the present development already supplies — the algebraic heart is one of them — and which remain to be built, so that the remaining labour is visible rather than assumed.

4.1 Statement

Definition 6 (Tait colouring)**.** A *Tait colouring* of a cubic map is a proper 3-edge-colouring: an assignment of one of three colours to each edge such that the three edges at every vertex receive three distinct colours.

Throughout we identify the three colours with the three nonzero elements of the Klein four-group

$$\mathbb{F}_2^2 \;=\; \{\,0,\; \mathsf{a},\; \mathsf{b},\; \mathsf{a}+\mathsf{b}\,\},$$

which is legitimate because any bijection between colour sets will do, and which buys the identity we shall use twice: *the three nonzero elements of* $\mathbb{F}_2^2$ *sum to zero.*

Definition 7 (combinatorial planarity). VERIFIED A finite simple graph is *combinatorially planar* when each connected component with at least three vertices carries a connected cellular spherical rotation-system presentation. Thus the darts over every vertex have one cyclic order, the primal graph is the given component, and the face orbits satisfy Euler's spherical equation.

Theorem 54 (Tait's reduction for finite combinatorial maps). *VERIFIED* *Suppose every bridgeless spherical cubic map admits a Tait colouring. Then every finite combinatorially planar simple graph admits a proper vertex colouring with four colours.*

Corollary 25 (topological-drawing formulation). *CONDITIONAL* *Suppose every bridgeless spherical cubic map admits a Tait colouring. Then every finite planar simple graph, with planarity defined by a crossing-free topological drawing in the plane, admits a proper vertex colouring with four colours.*

Proof. Apply Lemma 48 componentwise and then Theorem 54. □

Proposition 37 (audit of the legacy Lean planarity predicate). *VERIFIED* *The predicate currently named `Mettapedia.GraphTheory.IsPlanar` is not a formalization of planarity. Every finite simple graph satisfies it, while K_5 satisfies it and is not 4-colourable.*

Proof. The current structure records only a finite family of edge sets in which every edge occurs twice. For an arbitrary graph take two formal faces and declare the boundary of each to be the entire edge set. This satisfies every field. On the other hand, a colouring of the complete graph on five vertices is injective, so no map from its vertices to a four-element palette can be a proper colouring. Both statements, and their conjunction, are checked by the declaration named above. □

This proposition concerns only the repository's legacy vocabulary, not Theorem 54. The controlling formal statement uses the genuine combinatorial notion already used throughout the route: a graph-backed rotation system with cyclic vertex rotations, connected primal graph, and spherical Euler equation. The stellar construction below then acts on that presentation. No theorem may use the legacy `IsPlanar` premise as a substitute, since that would make the formal headline false before the compositional argument begins.

Lemma 48 (the abstract-planar presentation bridge). *CONDITIONAL* *Every finite connected planar simple graph with at least three vertices admits a connected spherical graph presentation: a dart rotation system whose primal graph is the given graph, whose rotation is cyclic at every vertex, and whose face orbits satisfy*

$$|V| - |E| + |F| = 2.$$

Proof. Choose a crossing-free embedding of the abstract planar graph in the sphere. Take the two oriented incidences of every edge as darts, reverse a dart to obtain the fixed-point-free edge involution, and read the cyclic order of the incident darts around each vertex from the orientation of the sphere. The resulting primal graph is the original graph. Its face permutation traces the complementary regions of the embedding, and Euler's formula gives the displayed identity. This is the standard equivalence between a cellular sphere embedding and a spherical rotation system.

The marker is CONDITIONAL, rather than PROSE, because the repository currently has no machine-checked constructor implementing this topological representation theorem. The ordinary argument above fixes the intended bridge exactly; it must not be replaced by the refuted legacy predicate of Proposition 37. This lemma is the sole extra input for Corollary 25; it is not a premise of the combinatorial-map Four-Colour statement in Theorem 54. □

The proof occupies the rest of the section. It is assembled from five ingredients, stated separately so that each carries its own status.

4.2 The ingredients

Lemma 49 (componentwise spherical reduction)**.** VERIFIED *Suppose every connected graph on at least three vertices which carries a connected spherical graph presentation is four-colourable. Then every finite graph whose nontrivial connected components carry such presentations is four-colourable.*

Proof. A component on at most two vertices is coloured injectively into the four-element palette. Apply the connected hypothesis to every other component. The connected components partition the vertex set, and every edge has both endpoints in one component, so the component colourings combine to one proper colouring of the whole graph. The abstract-planar input is kept separate: Lemma 48 supplies the required presentation to each connected planar component. □

Lemma 50 (triangulation)**.** OPEN *Every connected planar simple graph G on at least three vertices is a spanning subgraph of a plane simple graph T on the same vertex set in which every face is bounded by a triangle.*

This is the standard maximality argument: fix a plane embedding and add non-crossing edges between non-adjacent vertices as long as any can be added; the process terminates because the graph is simple and finite, and a maximal plane simple graph on at least three vertices is a triangulation. It is classical, and it is *not* formalized here. Of the ingredients it is the one carrying real formalization cost, since it needs a plane embedding to be manipulated rather than merely assumed. It is also avoidable: Theorem 58 builds a triangulation outright, on darts, and the reduction can be routed through that instead — at the price of letting the vertex set grow, which the reduction does not mind. This lemma is retained in its classical form, and keeps its OPEN marker, because that is the statement it makes; the reroute does not prove it.

Because a proper colouring of T restricts to a proper colouring of any spanning subgraph, it suffices to colour T.

4.3 Duality, at the level the reduction uses it

The dual is usually introduced by drawing a vertex inside each face and an edge across each primal edge, and that picture makes duality look like a statement about embeddings. At the level this reduction needs, it is not. It is a one-line change of generators.

Definition 8 (the dual map)**.** PROSE Let $M = (D, \sigma, \alpha)$ be a map, with $\varphi = \sigma\alpha$ its face permutation. Its *dual* is

$$M^* \;=\; (D,\ \varphi,\ \alpha).$$

The darts and the edge flip are unchanged; only the vertex rotation is replaced by the face permutation.

Lemma 51 (what the dual inherits, for free)**.** VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED *Let $M = (D, \sigma, \alpha)$ be a map. Then:*

1. *M^* is a map: α is unchanged, hence still a fixed-point-free involution, and φ is a permutation of D.*

2. *The face permutation of M^* is $\varphi\alpha = \sigma\alpha\alpha = \sigma$. So the faces of the dual are the vertices of the primal: $V^* = F$, $E^* = E$, $F^* = V$.*

3. *$\langle\varphi, \alpha\rangle = \langle\sigma, \alpha\rangle$, since $\varphi = \sigma\alpha$ and $\sigma = \varphi\alpha$. Hence M and M^* have the same orbits on darts and the same number of components; one is connected exactly when the other is.*

4. $V^* - E^* + F^* = F - E + V$*, so* M^* *is spherical exactly when* M *is.*

5. $M^{**} = M$ *on the nose: the dual of* (D, φ, α) *is* $(D, \varphi\alpha, \alpha) = (D, \sigma, \alpha)$*.*

Proof. Every item is the displayed computation. Item (v) is worth remarking on: duality here is an involution up to *equality*, not up to isomorphism, which is what makes it usable without transport lemmas. □

Corollary 26 (loops and bridges exchange)**.** VERIFIED VERIFIED VERIFIED *Let* M *be a connected spherical map. An edge* e *is a loop of* M *if and only if it is a bridge of* M^**. Consequently the dual of a connected spherical loopless map is bridgeless.*

Proof. Write $d, \alpha d$ for the darts of e. That e is a loop of M means its two darts sit at one vertex, that is, in one σ-orbit. By Lemma 51(ii) the σ-orbits are exactly the face orbits of M^*. So e is a loop of M iff its two darts lie on a common face of M^*, and by the two directions of the bridge characterisation — Theorems 49 and 51, both applied to the map M^* — that holds iff e is a bridge of M^*. □

Remark 18 (what this changes in the accounting)**.** PROSE *Nothing in the last two results mentions a plane, a curve, or a face interior. Duality, the exchange of loops with bridges, and the preservation of sphericity and connectedness are all consequences of replacing* σ *by* $\sigma\alpha$*. So of the embedding obligations this section owes, the* duality *one is not an embedding obligation: it is elementary permutation algebra sitting on top of a bridge characterisation that is already machine-checked. Stellar subdivision below supplies the only triangulation the reduction consumes, and the graph-backed argument of Section 3 has a combinatorial digon proof. Thus no separation principle remains in the controlling ordinary-mathematics route.*

Lemma 52 (the dual of a triangulation is cubic and bridgeless)**.** PROSE *Let* T *be a connected spherical loopless map whose face orbits have length three and meet three distinct vertices. Then its dual map* T^* *is connected, spherical, cubic, and bridgeless.*

Proof. By Lemma 51, connectedness and sphericity pass to the dual. Each face orbit of T has three darts, so each vertex rotation of T^* has three darts: T^* is cubic. The distinct-vertex hypothesis records that these are genuine triangular faces; it will be supplied by stellar subdivision.

Bridgelessness is subtler than it first appears, and it is worth stating exactly which implication is needed. Since T is loopless, what must be shown is

$$\text{no primal loop} \implies \text{no dual bridge}, \qquad \text{equivalently} \qquad \text{dual bridge} \implies \text{primal loop}.$$

Now observe that the dual of the map (σ, α) is the map (φ, α) with $\varphi = \sigma\alpha$: its faces are the orbits of $\varphi\alpha = \sigma$, that is, the primal vertices. Consequently a *primal loop* is exactly an edge of the dual whose two sides lie on one dual face, and $V' + F' = F + V = E + 2$, so the dual of a spherical map is spherical with the same edge count.

Corollary 26 now supplies exactly this: an edge is a loop of T precisely when it is a bridge of T^*, so the loopless primal has a bridgeless dual, with no appeal to an embedding. □

Corollary 27 (the explicit stellar dual has the complete Tait class)**.** VERIFIED *Let* M *be a connected spherical graph presentation with at least three old vertices, let* T *be its stellar subdivision, and let* T^* *be the facial dual. Then* T^* *is a connected, bridgeless, spherical cubic rotation map with one cyclic rotation at every vertex.*

Proof. The stellar map is connected and spherical, every stellar face orbit has three darts, and every stellar edge has two distinct face sides. Replacing the stellar vertex rotation by its face permutation therefore makes the dual cubic and preserves connectedness and Euler characteristic. If a dual edge were a bridge, the two darts of the corresponding stellar edge would lie in one dual face orbit; dual faces are stellar vertex orbits, so that stellar edge would be a loop, contrary to the looplessness of the stellar construction. The declaration above packages these facts as exactly the map class consumed by the Tait-colouring hypothesis. ☐

Remark 19 (a correction worth recording). PROSE *It is tempting, having listed the residual obligations of this section and of Section 3, to conclude that they are all facets of a single plane-duality interface. That is very nearly right and is wrong in one place. Adding a non-crossing edge, maximality forcing triangles, and a digon's parallel pair bounding a face are indeed three statements about plane embeddings and their duals. The bridgelessness of T^* is not: by the duality just described it is a statement about permutations, namely the converse of Theorem 49, and it requires no embedding machinery at all.*

The distinction is worth keeping because the two kinds of debt have very different costs. The converse is an isolated combinatorial lemma — now proved, by a walk argument that never counts anything; the embedding interface is a development. Merging them in the ledger would have hidden a cheap item inside an expensive one.

We record the fact that makes the algebra work.

Lemma 53 (the Klein identity). VERIFIED VERIFIED *The three nonzero elements of $\mathbb{F}_2^2$ sum to zero, and any three pairwise distinct nonzero elements of $\mathbb{F}_2^2$ are all three of them.*

Proof. $\mathsf{a} + \mathsf{b} + (\mathsf{a} + \mathsf{b}) = 0$ since the group has exponent two. There are exactly three nonzero elements, so three pairwise distinct ones exhaust them. ☐

Theorem 55 (facial boundaries span the cycle space). VERIFIED VERIFIED *For a connected spherical map with two-sided faces and connected dual, the image of the facial-boundary map is exactly the $\mathbb{F}_2$ cycle space of the literal edge carrier of the map.*

This is the algebraic heart of the reduction and it is already available. Note its hypothesis of two-sided faces: that is supplied by Corollary 23 of the previous section, so the two halves of this development meet here. The second checked declaration is the form consumed below. It uses the map's edge type as the columns of the incidence matrix, so two parallel edges remain two different coordinates. This is load-bearing for stellar subdivision: if an old face revisits an old vertex, two different spokes may have the same pair of endpoints, and projecting to a simple-graph edge set would identify them.

4.4 Proof of the reduction

Proof of Theorem 54. By Lemma 49 we may assume G is connected with at least three vertices. Let M be the spherical rotation-system presentation supplied by Definition 7; it is connected and loopless because its primal graph is the connected simple graph G. Apply Definition 9 to obtain $T = M'$. By Theorem 58, T is connected, spherical and loopless, it contains every old vertex and edge of G, and every face is a 3-cycle meeting three distinct vertices; the same theorem also gives two distinct face sides at every edge. Hence a proper colouring of T restricts to one of G, and the two-sided hypothesis used in Step 2 below is available.

By Lemma 52 the dual T^* is a bridgeless spherical cubic map, so the hypothesis of the theorem furnishes a Tait colouring of T^*. Primal and dual share an edge set, so this is a function

$$\omega\colon E(T) \longrightarrow \mathbb{F}_2^2 \setminus \{0\}$$

with the property that the three edges bounding any face of T — equally, the three edges at the corresponding vertex of T^* — receive three pairwise distinct nonzero values.

Step 1: ω vanishes on facial boundaries. Fix a face of T. Its three edges receive three pairwise distinct nonzero values, which by Lemma 53 are all three nonzero elements, and these sum to zero. So ω sums to zero over the boundary of every face.

Step 2: ω vanishes on the whole cycle space. By Theorem 55 the facial boundaries span the $\mathbb{F}_2$ cycle space of T. Summation of ω over an edge set is $\mathbb{F}_2$-linear in that set, applied coordinatewise in $\mathbb{F}_2^2$; a linear functional vanishing on a spanning set vanishes identically. So ω sums to zero over every element of the cycle space.

Step 3: potentials exist. Fix a root $r \in V(T)$. For $v \in V(T)$ choose a path $P_{r,v}$ from r to v — one exists by connectedness — and set

$$c(v) \;:=\; \sum_{e \in P_{r,v}} \omega(e) \;\in\; \mathbb{F}_2^2.$$

This is well defined: if P and P' are two r–v paths, their symmetric difference is an element of the cycle space, so by Step 2 the two sums agree. More generally, the same value is obtained from any r–v walk: retain the edges traversed an odd number of times (even repetitions cancel in $\mathbb{F}_2^2$); its symmetric difference with $P_{r,v}$ has even degree at every vertex and therefore lies in the cycle space.

Step 4: the potential is a proper colouring. Let $uv \in E(T)$. Appending the edge uv to a path from r to u gives a walk from r to v, so the preceding walk-independence gives $c(v) = c(u) + \omega(uv)$, that is

$$c(u) + c(v) \;=\; \omega(uv) \;\neq\; 0,$$

so $c(u) \neq c(v)$. Thus $c\colon V(T) \to \mathbb{F}_2^2$ is a proper colouring with four colours, and its restriction to G is one too. □

The algebraic implication used in Steps 1–4—from a facial Tait colouring, two-sidedness, connected primal and dual, and spherical cubic map data to a proper four-colouring—is now machine-checked both on the earlier graph-backed carrier and on the literal multigraph carrier used by stellar subdivision. VERIFIED VERIFIED VERIFIED

The final consumer is checked as well. A Tait colouring of the explicit stellar facial dual is integrated on literal stellar edges; the resulting four-colouring of the stellar primal map is then restricted along the old vertex inclusion. Thus the entire reverse Tait implication is verified for the genuine connected spherical presentation used in this section. VERIFIED VERIFIED VERIFIED VERIFIED

Theorem 56 (the sound M0 least-counterexample wrapper)**.** *VERIFIED VERIFIED* *Suppose there is no graph-backed vertex-minimal non-Tait-colourable member of the connected, bridgeless, spherical cubic rotation-map class. Then every bridgeless spherical cubic rotation map is Tait-colourable, and every finite graph carrying spherical presentations on its nontrivial connected components is four-colourable.*

Proof. If a non-Tait-colourable map existed, choose one with the least number of vertices. Corollary 23 constructs its two-sided face property from bridge-freeness; the digon normalization of Section 3 excludes parallel edges by minimality. The canonical primal simple graph therefore supplies the graph-backed vertex-minimal counterexample forbidden by the hypothesis. Hence every map in the class is Tait-colourable. The checked stellar construction and Klein-group integration above give the connected spherical Four-Colour statement, and Lemma 49 combines the component colourings.

This well-ordering wrapper is important logically: two-sidedness and graph backing are conclusions inside the least-counterexample argument, not extra premises exported to the compositional core. □

4.5 What the reduction costs

It is worth being explicit about where the remaining labour sits. The reverse Tait implication is now complete both as ordinary mathematics and as a formal theorem over connected spherical graph presentations.

Steps 1 through 4 are machine-checked algebra over $\mathbb{F}_2^2$ resting on Theorem 55. The other ingredients are now equally explicit: Definition 9 and Theorem 58 replace maximality by a finite permutation construction; Definition 8 is a change of generator; and Corollary 26 supplies dual bridgelessness. The componentwise wrapper is checked by Lemma 49, so the finite combinatorial-map form of Tait's reduction has no remaining premise besides the Tait-colourability hypothesis it is supposed to reduce from. The named representation bridge of Lemma 48 is needed only to translate the separate topological-drawing convention of Corollary 25. No legacy `IsPlanar` premise is used.

4.6 Audit of the unused maximality route

The next discussion records why the textbook maximal-plane-graph route was not chosen. It is retained as an audit of an attractive detour, not as a dependency of Theorem 54; the controlling proof uses stellar subdivision instead.

Adding a non-crossing edge is a construction, not a theorem. Let d, d' be darts on one face orbit. Adjoin two darts x, y; set $\alpha' = \alpha \cup (x\,y)$; and splice x into the rotation at d's vertex and y into the rotation at d''s vertex, so $\sigma' d = x$, $\sigma' x = \sigma d$, and symmetrically at d'. No embedding is consulted: the rotation system *is* the embedding. The Euler bookkeeping is immediate, since $V' = V$, $E' = E + 1$, and the face through d, d' splits, so $F' = F + 1$ and $V' - E' + F' = V - E + F$. Sphericity is preserved, and the face count is governed by the split law of Lemma 38 rather than by a separation argument.

Maximality is where the content actually sits. The step that resists is not the construction but the claim that a maximal simple plane graph on at least three vertices has only triangular faces. Given a face of length at least four, one wants two of its vertices that are *not already adjacent elsewhere in the graph*, since joining adjacent ones would destroy simplicity. That non-adjacent pair must be shown to exist, and its existence is a genuine combinatorial fact about plane simple graphs, not a corollary of the construction above.

Why counting does not finish it. It is tempting to hope Euler settles maximality directly, as it settled two-sidedness. It does not, and the gap is instructive. If a simple plane graph with $V \geq 3$ has one face of length at least four and the rest at least three, then $2E \geq 3F + 1$, and substituting $F = 2 - V + E$ gives $E \leq 3V - 7$. So the graph falls short of the triangulation bound $E = 3V - 6$. But that shows only that it is not edge-*maximum*; being *maximal*, in the sense that no further edge can be added, is a weaker condition. Counting establishes a deficit without exhibiting an edge to add, and producing that edge is what needs an argument about which vertices the face separates.

Remark 20 (where the last classical debt lands)**.** PROSE *Within the unused maximality route, combinatorial separation for rotation systems is available in dual form below. An earlier draft then claimed that, on a face of length at least four, a chord through the first and third vertices separates the second vertex from the fourth. That inference is wrong: the second vertex lies on the resulting three-cycle. Recovering the maximality route would require the additional rotation-at-the-second-vertex argument discussed below; the controlling stellar route needs none of it.*

This is retained only as an audit of a discarded proof route. The generic dual separation result is useful elsewhere, but it does not by itself prove the vertex non-adjacency asserted by the former maximality argument.

Separation may already be available. Before treating combinatorial separation as a development to be built, it is worth noticing that the pieces for it are in hand. Let C be a cycle. It lies in the $\mathbb{F}_2$ cycle space, so by Theorem 55 it is a sum of facial boundaries, say $C = \sum_{f \in S} \partial f$ for some set S of faces. By Corollary 23 each edge has two *distinct* faces, so $e \in \partial f$ exactly when f is one of them, and therefore

$$e \in C \quad \Longleftrightarrow \quad \bigl|\{\text{the two faces of } e\} \cap S\bigr| \text{ is odd, that is, exactly one.}$$

Writing $\mathrm{side}(f) := (f \in S)$, an edge outside C has its two faces on the same side and an edge of C has them on opposite sides. That is separation: C is precisely the edge cut between the two classes, and nothing outside C crosses.

Theorem 57 (the boundary map is the two-sided sum)**.** VERIFIED *With two-sided faces, the boundary map at an edge returns the sum of the coefficients of that edge's two faces.*

Corollary 28 (separation, dual form)**.** VERIFIED *An edge lies in the image of a face-set indicator exactly when its two sides fall on opposite sides of the set. Hence the image is the edge cut between that set and its complement, and no edge outside the cut crosses it.*

With Theorem 55, every cycle is such a cut: the combinatorial separation that planar arguments normally import from topology is available here without importing anything.

Remark 21 (what remains of the maximality step, and two routes to it)**.** PROSE *Corollary 28 separates* faces. *The maximality argument consumes a statement about* vertices*, so a transfer is still needed. Since that transfer is now the only unwritten step between this section and a complete classical bridge, it is worth setting down the two candidate routes precisely, neither of which is established here or consumed by the controlling proof.*

Suppose, for contradiction, that every two vertices on a face of length $k \geq 4$ are adjacent.

Route A, through separation. *The cycle C formed by two face edges v_1v_2, v_2v_3 and the chord v_3v_1 is a cycle, so by Corollary 28 it is a face cut. But v_2 lies* on C, *while the conclusion wanted concerns v_2 and v_4 being non-adjacent rather than lying on opposite sides. The step from a face two-colouring to a vertex non-adjacency is not immediate, and no transfer supplying it has been identified.*

Route B, through forbidden subgraphs. *If $k \geq 5$ the face's vertices induce K_5, which is not planar. If $k = 4$ they induce K_4 with all four vertices on one face; but every face of G lies inside a face of the subgraph K_4, and each of those is bounded by only three vertices, so no face of G can meet all four. This route needs the non-planarity of K_5 and the containment of faces under passage to a subgraph — both classical, neither present in the current libraries.*

Route B does not use Corollary 28 at all, which is worth noticing on its own: the separation result may be valuable elsewhere in this development without being the tool this particular step wants.

It is tempting to call Route B the shorter, and that judgement does not survive checking. No current library supplies planarity in any form, so the non-planarity of K_5 would have to be proved here, and it does not follow from the Euler bound of Theorem 48: the classical derivation $2 = V - E + F$ together with $F \leq 2E/3$ yields $E \leq 3V - 6$, but what is available here is the inequality $V - E + F \leq 2C$*, and two upper bounds in the same direction do not combine to bound E. Route B therefore needs sphericity as a theorem about embeddings rather than as the hypothesis it presently is.*

Both routes accordingly require real work, and no preference between them is recorded here. The estimate that one was shorter is withdrawn.

Remark 22 (a third route, which avoids maximality entirely)**.** PROSE *Both routes above exist only to serve Lemma 50, and that lemma was stated in the classical way — extend G to a* maximal *plane graph — because that is how the textbooks do it. Maximality is not what the reduction needs. What it needs is* some *plane triangulation containing G, and one can be built outright.*

Place a new vertex u_F inside each face F and join it to every vertex on F's boundary. A face of length k becomes k triangles. Nothing is removed, so G survives as a subgraph; a proper colouring of the result restricts to G, since restricting a proper colouring to a subset of vertices is again proper; every face is a triangle, so the dual is cubic; and planarity is immediate because u_F sits inside F.

Crucially, no edge is added between existing vertices*. The entire difficulty of maximality was finding two non-adjacent vertices on a long face, and this construction never needs one.*

It is tempting to attach a caveat here — that the construction yields a simple *graph only when each face boundary has distinct vertices, hence only for 2-connected G — and to trade maximality for a reduction to blocks. Tracing where simplicity is actually used shows the caveat is unnecessary.*

Simplicity enters the reduction exactly once, in Lemma 52, and only to conclude that the dual is bridgeless: a dual bridge corresponds to a primal loop*, and simple graphs have none. What is needed is therefore looplessness, not simplicity. Multiple edges are harmless, since a multiple edge is not a loop and creates no dual bridge.*

Re-reading the construction with that in hand: if a face walk repeats a vertex, joining u_F to each occurrence *gives u_F several edges to that vertex — multiple edges, not a loop. Each new face is still a triangle (u_F, v_i, v_{i+1}), and $v_i \neq v_{i+1}$ because consecutive equality would be a loop in G, which a simple G does not have. So stellar subdivision of a loopless plane graph yields a loopless plane triangulation, with no connectivity hypothesis at all.*

Two identities corroborate the construction before any of it is built. The subdivision adds one vertex per face and one edge per dart-corner, so $V' = V + F$, $E' = E + 2E = 3E$, and each old face of length k becomes k triangles, giving $F' = 2E$. Then

$$V' - E' + F' = (V + F) - 3E + 2E = V - E + F = 2,$$

so Euler is preserved exactly; and a map all of whose faces are triangles satisfies $3F' = 2E'$, which here reads $6E = 6E$. Both hold with no slack, which is what a correct triangulating construction must give and what an incorrect one generally does not.

What this route costs, then, is the construction itself and the verification that its faces are triangles — and the next subsection carries both out on darts, so that even the phrase "u_F sits inside F" is discharged rather than appealed to. What it does not cost is any argument about non-adjacent pairs, hence neither separation nor forbidden subgraphs. That is a different kind of obligation from Routes A and B, not merely a smaller one — though, given how often the size *of this residual has been misjudged here, the comparison is offered for checking rather than as a conclusion.*

4.7 Stellar subdivision on darts

The construction above was described by placing a vertex inside a face. That phrase is the only geometry in it, and it can be removed: the whole subdivision is a substitution on darts, and the triangles come out of a three-line computation.

Definition 9 (stellar subdivision)**.** PROSE Let $M = (D, \sigma, \alpha)$ be a map with face permutation $\varphi = \sigma\alpha$. For each face orbit $F = (d_0, d_1, \ldots, d_{k-1})$, so that $\varphi(d_i) = d_{i+1}$ with indices mod k, adjoin $2k$ fresh darts $a_0, \ldots, a_{k-1}$ and $b_0, \ldots, b_{k-1}$, and set

$$\alpha'(a_i) = b_i, \qquad \sigma'(\alpha d_i) = a_i, \qquad \sigma'(a_i) = d_{i+1}, \qquad \sigma'(b_i) = b_{i-1},$$

with $\alpha' = \alpha$ and $\sigma' = \sigma$ on all darts not named. Write $M' = (D', \sigma', \alpha')$.

That σ' is again a permutation is exactly the identity $\sigma(\alpha d_i) = \varphi(d_i) = d_{i+1}$: the new dart a_i is spliced into the rotation between αd_i and d_{i+1}, where the old rotation went directly. Distinct faces name distinct darts αd_i, since the face orbits partition D, so the clauses do not collide.

Theorem 58 (the subdivision triangulates)**.** *VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED* *The faces of M' are exactly the 3-cycles*

$$(\, d_i,\ a_i,\ b_{i-1}\,),$$

one for each face F of M and each index i. Consequently $V' = V + F$, $E' = 3E$, $F' = 2E$, and M' has the same Euler characteristic as M. The underlying graph of M is a subgraph of that of M', and connectedness of M implies connectedness of M'. If M is loopless then so is M', and each face of M' meets three distinct vertices; moreover the two darts of every edge of M' belong to distinct face orbits.

Proof. Compute $\varphi' = \sigma'\alpha'$ on the three darts in question:

$$\varphi'(d_i) = \sigma'(\alpha d_i) = a_i, \qquad \varphi'(a_i) = \sigma'(b_i) = b_{i-1}, \qquad \varphi'(b_{i-1}) = \sigma'(a_{i-1}) = d_i.$$

So each such triple is a φ'-orbit of length three. Every dart of D' occurs in exactly one triple — each d_i in its own, each a_i in the i-th, each b_{i-1} in the i-th — so these are all the faces.

For the counts: the darts $b_0, \ldots, b_{k-1}$ form a single σ'-cycle, one new vertex per face, and no old vertex is split, so $V' = V + F$; the new edges are the pairs $\{a_i, b_i\}$, one per dart of M, so $E' = E + 2E = 3E$; and the triples number $\sum_F k_F = 2E$, so $F' = 2E$. Then $V' - E' + F' = (V + F) - 3E + 2E = V - E + F$.

No old edge is removed. Each new vertex is the b-cycle belonging to an old face and is joined by its new edges to vertices incident with that nonempty face orbit. Hence the old underlying graph is a subgraph of the new one, and every new vertex attaches to it; connectedness is therefore preserved.

Each new edge joins the vertex carrying αd_i to the new vertex of F, which is not an old vertex, so no new loop appears; old edges are unchanged. Finally the face (d_i, a_i, b_{i-1}) meets the tail of d_i, the head of d_i (since a_i sits at the vertex of αd_i), and the new vertex of F; these are three distinct vertices precisely because d_i is not a loop.

For two-sidedness, an old edge has distinct old darts d and αd, and the displayed partition puts exactly one old dart in each new face, so its two darts lie in distinct faces. A new edge $\{a_i, b_i\}$

meets the faces indexed by d_i and d_{i+1}. These are distinct because a face orbit of length one would give $\sigma(\alpha d_i) = d_i$, placing both darts of the old edge at one vertex and making it a loop. Thus every edge of M' has two distinct sides. □

Remark 23 (what is left of the embedding)**.** PROSE *Definition 9 names four values of two permutations; the proof evaluates $\sigma'\alpha'$ three times. No face interior, no curve, and no plane occurs in either. Since the Euler characteristic is preserved, a spherical M yields a spherical M', which is what Lemma 50 was wanted for — with containment of M in place of the spanning condition, which is all the reduction uses, a proper colouring restricting to a subgraph regardless of whether the vertex set grew.*

Remark 24 (why the maximality lemma remains open)**.** PROSE *Lemma 50 keeps its* OPEN *marker because stellar subdivision does not prove that stronger, same-vertex-set statement. But it is no longer a premise of Theorem 54. The reduction needs a loopless triangulating supermap containing G, not a maximal simple supergraph on exactly the old vertex set, and Theorem 58 supplies precisely the weaker object. This is a change of lemma, not a proof of the old lemma, and the status labels preserve that distinction.*

Remark 25 (the shape of the debt)**.** PROSE *The combinatorial-map reduction needs neither new algebra nor a Jordan curve theorem. At the ordinary-mathematics level it consists of explicit finite permutation constructions plus the already-stated cycle-space theorem. The stellar construction, dual transport, literal-edge Klein-group potential, and restriction to old vertices are now checked. Translating a topological plane drawing into this finite representation is a separate representation theorem, recorded explicitly in Lemma 48; it does not weaken the kernel statement about finite combinatorial maps.*

Remark 26 (how this closes the classical debts)**.** *With Corollary 23 established, the route's internal graph-backed presentation and Theorem 54 are checked, including the least-counterexample wrapper of Theorem 56. The standard finite plane-embedding representation theorem remains named only in the separate topological Corollary 25; keeping that boundary explicit prevents the refuted legacy planarity predicate from re-entering the kernel headline.*

5 The Meyniel ladder and the repaired square seam

The reduction ladder removes a facial triangle, a digon, a two-edge cut, or a facial square before the compositional argument reaches its pentagonal frontier. Two different invariants occur in the working manuscript: nonnegative target counts, used by the Seed Lemma, and connectivity of the Kempe reconfiguration graph, used by the stronger WC* statement. They agree on the first three rungs but diverge at a square. The distinction is decisive: an additive identity transports a zero count without any path between its two summands, whereas connectivity requires an actual migration path.

For later reference, fix the source's quantifiers explicitly. If $\mathrm{Col}(X)$ is the set of proper edge colourings of a tangle X, K_X its Kempe reconfiguration graph, and W a set of boundary words, write

$$\mathrm{WC}^*(X, W) \quad :\Longleftrightarrow \quad (\exists D \in \mathrm{Col}(X),\ \mathrm{word}(D) \in W) \Longrightarrow (\forall C \in \mathrm{Col}(X),\ \exists D, C\ K_X^*\ D \wedge \mathrm{word}(D) \in W).$$

Thus WC* is deliberately true when the target fibre is empty. Every use below either retains that proviso or separately assumes the global target set is nonempty. This is the loaded definition in Addendum II of the working manuscript, not an abbreviation for connectivity of K_X.

5.1 The three unbranched rungs

For a facial triangle, properness forces its three internal edges to have the three different colours. At each corner the outer edge therefore has the colour of the opposite triangle edge. Contraction to one cubic vertex is consequently canonical, and the three possible two-colour paths through the contracted vertex expand to the three corresponding paths through the triangle. Both the colour law and the three path expansions are checked in the rotation-system development. VERIFIED VERIFIED The resulting finite portal profile is equal to the contracted-vertex profile, including composition with an arbitrary exterior profile. VERIFIED

For a digon, its two parallel edges form one maximal two-colour component. Switching that component exchanges the two local extensions, so the local fibre is connected. Suppression preserves the boundary connectivity profile, with multiplicity two. VERIFIED VERIFIED

Across a two-edge cut, the two crossing edges have equal colours. This is the Klein-group boundary-sum law on either side of the cut. Capping the two sides is exact in both directions: an ambient colouring restricts to cap colourings, and cap colourings may be aligned by a global colour permutation and glued. VERIFIED VERIFIED The remaining work on these rungs is integration: package these local equivalences as the exact target-count and reconfiguration transports consumed by the common ladder interface. No unproved geometric principle is hidden in that packaging.

5.2 The exact square fibre

Let $Q = wxyz$ be a facial square, and let (c_w, c_x, c_y, c_z) be the colours on its four outer edges in cyclic order. There are two planar reductions. The first joins the outer edges at w, x and at y, z; the second joins those at x, y and at z, w. Write them as G_0, G_1.

Theorem 59 (square fibre table)**.** *VERIFIED* *For a nonzero boundary word, the square has two internal colourings when all four boundary colours are equal, one internal colouring for either adjacent pair pattern*

$$(p, p, q, q) \quad \textit{or} \quad (q, p, p, q), \qquad p \neq q,$$

and no internal colouring otherwise. In particular, the opposite-pair pattern (p, q, p, q) is impossible.

The two states in an all-equal fibre differ by switching the square's own two-colour cycle. Thus every nonempty local fibre is internally connected. VERIFIED

For one exterior colouring profile η, the two internal states over an all-equal word have different through-square pairings: one belongs to the G_0 reduction and the other to G_1. An adjacent-pair word belongs to exactly one reduction. Keeping multiplicity therefore gives, for every target interface profile π,

$$\mathrm{Count}_G(\eta, \pi) = \mathrm{Count}_{G_0}(\eta, \pi) + \mathrm{Count}_{G_1}(\eta, \pi). \tag{2}$$

This is proved for arbitrary multisets of exterior profiles, not merely for one census graph. VERIFIED

Lemma 54 (zero targets descend additively)**.** *PROSE* *Let S be any finite set of target profiles and put $N_H(S) = \sum_{\pi \in S} \mathrm{Count}_H(\pi)$. If Equation (2) holds for every π, then*

$$N_G(S) = N_{G_0}(S) + N_{G_1}(S), \qquad N_G(S) = 0 \iff N_{G_0}(S) = N_{G_1}(S) = 0.$$

Proof. Sum Equation (2) over S and distribute the finite sum. All summands are natural numbers, so a sum of the two totals is zero exactly when both totals are zero. ☐

Consequently the square rung for the Seed target is additive rather than dynamical: if a square instance has no seed, neither reduction has a seed. Here the counted state is pointed by a choice of majority pair: it is a pair (C, P) of a colouring and one of its two majority pairs. Thus the two pair sectors form a disjoint state space even when one colouring is a seed for both pairs. Positivity of its total count is exactly the existence of a colouring and a majority pair witnessing the Seed conclusion; no inclusion–exclusion assertion about overlapping sets of colourings is being used.

Theorem 60 (graph-level square count transport)**.** PROSE *For a facial square whose two planar reductions are valid tangles, the profile-resolved count of pointed colourings satisfies Equation* (2)*. Consequently the pointed Seed count is additive and a least zero-Seed instance descends through both smaller reductions.*

Proof. Delete the four internal square edges but retain its four ordered outer half-edges. A global colouring is then uniquely a colouring of this common exterior together with one member of the local square fibre over its boundary word. A colouring of either reduction is uniquely the same exterior colouring together with the corresponding compatible planar pairing. For a fixed exterior colouring, fixed majority pair, and fixed target profile, the equality of these two multisets is exactly the checked local profile identity after composition with that exterior profile. Summing over the finite multiset of exterior colourings proves the graph-level identity.

Now sum over the finite set of Seed profiles in the pointed state space and apply Lemma 54. If the total upstairs is zero, both reduction totals are zero; since a facial-square reduction removes four vertices, either valid reduction is a strictly smaller zero-Seed instance. No migration statement occurs in the argument. ☐

5.3 The universal $A \neq \varnothing$ shortcut is false

Addendum XXVI asserts that, for any tangle in the ladder's scope and any two disjoint cofacial edges E_1, E_2 arising as the joined edges of a square reduction, some Tait colouring has $\chi(E_1) = \chi(E_2)$. Equivalently, the inverse square has a colouring in its all-equal boundary fibre A. The following specimen refutes that universal statement.

Proposition 38 (forced-unequal square reduction)**.** REFUTED *There is a connected planar cubic tangle with two stubs on one face and two vertex-disjoint cofacial edges E_1, E_2 which are the joined edges of a planar square reduction, such that the tangle has six Tait colourings and $\chi(E_1) \neq \chi(E_2)$ in all six.*

Proof. Take internal vertices $0, \dots, 5$, stub vertices $6, 7$, and edges

$$01, 02, 03, 14, 15, 23, 25, 34, 46, 57.$$

At the internal vertices use cyclic rotations

v	0	1	2	3	4	5
ρ_v	$(1,2,3)$	$(0,4,5)$	$(3,0,5)$	$(4,0,2)$	$(1,3,6)$	$(2,1,7)$.

The stub rotations are (4) at 6 and (5) at 7. Directly following the face permutation gives

$$(0,1,5,2), \quad (0,2,3), \quad (0,3,4,1), \quad (1,4,6,4,3,2,5,7,5).$$

Hence $V - E + F = 8 - 10 + 4 = 2$, both stubs lie on the last face, and $E_1 = 01, E_2 = 34$ are disjoint edges of the face $(0,3,4,1)$. Every non-boundary edge lies on one of the displayed ordinary cycles, so there is no internal bridge.

In any Tait colouring, write $x = \chi(02), y = \chi(03), z = \chi(23)$. The triangle 023 forces x, y, z to be the three distinct colours. Properness at vertex 0 then forces $\chi(01) = z$, while properness at vertex 3 forces $\chi(34) = x$. Thus the marked colours are always different. Conversely, once the ordered triple (x, y, z) is chosen, properness forces

$$\chi(14) = y, \quad \chi(15) = x, \quad \chi(25) = y, \quad \chi(46) = \chi(57) = z,$$

so each of the six permutations of the three colours gives one colouring.

Finally delete $01, 34$, insert the facial square $8, 9, 10, 11$, and attach its vertices in cyclic order to $0, 1, 4, 3$. One planar reduction restores $01, 34$. The other adds a second copy of each of $14, 03$, which is a legitimate multigraph reduction. The displayed rotation extends to a spherical rotation with the new square as a face; the complete rotations and face audit are recorded by the cited executable fixture. Thus the marked pair is exactly a square reduction pair, as required. □

The failure is not caused by disconnected Kempe dynamics. Exhaustion of the inverse square gives six colourings in one Kempe class and no all-equal boundary state. The first reduction has six colourings in one class; the parallel-edge reduction has none. Its square identity is therefore $6 = 6 + 0$. The specimen disproves the shortcut $A \neq \varnothing$, but it does not disprove the square conclusion.

The converse warning is equally important: the full universal SQ3 connectivity statement is also false, although a slightly larger specimen is needed.

Proposition 39 (universal SQ3 migration is false). REFUTED *There is a connected planar cubic tangle with two stubs on one face and a facial square such that each of its two planar reductions has a connected Kempe reconfiguration graph, but the original tangle's reconfiguration graph has two components.*

Proof. Use the planar Frucht graph with edge set

$$01, 06, 07, 12, 17, 23, 28, 34, 39, 45, 49, 56, 5, 10, 6, 10,$$
$$7, 11, 8, 11, 89, 10, 11.$$

The disjoint edges $06, 34$ lie on the face $(0, 6, 5, 4, 3, 2, 1)$. Delete them, insert a square with vertices $12, 13, 14, 15$, and attach those vertices in cyclic order to $0, 6, 4, 3$. Finally open the unrelated edge 01 into stub edges $0, 16, 1, 17$. The resulting tangle has sixteen cubic vertices and two stubs. The rotation cycles in the cited fixture give nine face orbits, including the square $(12, 13, 14, 15)$ and one face containing both stubs; thus $V - E + F = 18 - 25 + 9 = 2$. Deleting any non-boundary edge leaves the tangle connected.

The finite audit assigns the three colours by backtracking, rejecting a partial assignment as soon as two assigned edges at one cubic vertex agree. It obtains exactly eighteen Tait colourings. For every colouring and each of the three colour pairs, it computes every maximal two-colour component, switches that whole component, and identifies the resulting colouring in the same eighteen-state table. The resulting Kempe graph has components of sizes six and twelve. Every state in the six-state component has square word of type (p, p, q, q), and every state in the twelve-state component has the other adjacent-pair type; no all-equal state exists.

The first planar reduction restores $06, 34$ and has six colourings in one Kempe class. The other joins $64, 30$ and has twelve colourings in one Kempe class. Its two lifted sides are exactly the two components just described. This proves the claimed failure. All rotations, face walks, colourings, and component switches are emitted and asserted by the cited deterministic fixture. □

This second counterexample still does not refute the route's loaded WC* target. Its two stub colours agree in every one of the eighteen states, so every state already has the extendable two-stub word. The target reachability statement is true even though the stronger single-class statement is false. The repair must therefore retain the actual target predicate instead of trying to recover universal reconfiguration connectivity.

5.4 The correct reachability obligation

Let $\mathcal{K}(G)$ be the graph whose vertices are the Tait colourings of G and whose edges are legal maximal-component Kempe switches. Let L_0 consist of the colourings whose square boundary is all-equal or of the adjacent-pair type represented by G_0, and define L_1 symmetrically. The fibre table gives $\mathrm{Col}(G) = L_0 \cup L_1$; their intersection is exactly the all-equal set A.

Lemma 55 (connected-union criterion)**.** PROSE *Suppose each nonempty induced reconfiguration subgraph on L_0, L_1 is connected. Then $\mathcal{K}(G)$ is connected precisely when one of the sets is empty or some Kempe path has one endpoint in L_0 and the other in L_1.*

Proof. Necessity is immediate. For sufficiency, join any colouring first inside its own connected side to the appropriate endpoint of the cross-side path, traverse that path if the target lies on the other side, and then join to the target inside that side. If one side is empty, the other is the whole colouring set. ☐

Full connectivity is stronger than the route needs. For a target set W of stub words, put $W_s = \{C \in L_s : \mathrm{word}(C) \in W\}$.

Lemma 56 (target-aware square criterion)**.** PROSE *Assume that, whenever $W_s \neq \varnothing$, every state of L_s can reach a state of W_s while staying in L_s. If the global target set is nonempty, then* $\mathrm{WC}^*(G, W)$ *holds if and only if every state in a target-free side L_s can reach some side L_t with $W_t \neq \varnothing$.*

Proof. For necessity, a path from a target-free side to a global target has a final state in some target-bearing side. Conversely, a state already in a target-bearing side reaches W by the assumed sidewise statement. A state in a target-free side first takes the supplied path to a target-bearing side and then uses its sidewise path to W. ☐

The target cannot be left abstract in the remaining square theorem. Even colour-permutation invariance, planarity, bridgelessness, and one-face placement of the stubs do not make generic WC* transport true.

Proposition 40 (generic WC* square transport is false)**.** REFUTED *There are a bridgeless planar cubic tangle G with four stubs on one face, a facial square with two bridgeless planar reductions G_0, G_1, and a colour-permutation-invariant set W of stub words such that*

$$\mathrm{WC}^*(G_0, W) \quad \textit{and} \quad \mathrm{WC}^*(G_1, W) \qquad \textit{but} \qquad \neg\mathrm{WC}^*(G, W).$$

Proof. Start with the same planar Frucht rotation used in Proposition 39. Replace its cofacial edges $06, 34$ by the square $12\,13\,14\,15$, attached in cyclic order to $0, 6, 4, 3$. Instead of opening 01, open the two exterior edges $5\,10$ and $8\,11$. Denote the resulting stub edges, in that order, by

$$5\,16, \quad 10\,17, \quad 8\,18, \quad 11\,19.$$

Let W consist of the words

$$(a, a, b, b), \qquad a, b \in \{0, 1, 2\}, \quad a \neq b.$$

This set is invariant under every permutation of the three colours.

The rotation is obtained from the closed Frucht rotation by replacing the neighbour 10 at 5 by 16, the neighbour 5 at 10 by 17, the neighbour 11 at 8 by 18, and the neighbour 8 at 11 by 19. Its face permutation has eight orbits, including $(12, 13, 14, 15)$ and a single orbit containing $16, 17, 18, 19$. Thus $V - E + F = 20 - 26 + 8 = 2$, the square is facial, and the four stubs lie on one face. The two reductions respectively restore $06, 34$ and insert $64, 30$. Each has 16 vertices, 20 edges, six face orbits, and one face containing all four stubs. Deleting each internal edge in turn leaves all vertices connected in all three maps, so all three tangles are bridgeless. The complete rotations and face walks are asserted in the cited fixture.

It remains only a finite colouring calculation. Assign colours to indexed edges by backtracking, rejecting an assignment as soon as two assigned edges at a cubic vertex agree. For each complete colouring and each of the three colour pairs, find every maximal two-colour component and switch the whole component. Looking the result up in the complete colouring table gives the exact Kempe graph. This calculation yields

	$\lvert \mathrm{Col} \rvert$	Kempe-class sizes	W-states per class
G	48	18, 30	6, 0
G_0	30	30	0
G_1	18	18	6.

Every search is exhaustive; there is no random or heuristic step.

By the source definition, $\mathrm{WC}^*(G_0, W)$ is true because no colouring of G_0 has a word in W, so its proviso is false. The graph $\mathcal{K}(G_1)$ is connected and contains six target states, so $\mathrm{WC}^*(G_1, W)$ is true. Upstairs the target is nonempty but the thirty-state Kempe class contains none of it. Hence $\mathrm{WC}^*(G, W)$ is false. $\square$

This proposition does not refute the route's surviving square claim. Its target W is not the pentagonal cap's extendable-word set, and the specimen is not asserted to be a least obstruction. It proves that both of those qualifiers carry mathematical content: no theorem quantified over an arbitrary colour-invariant target can replace them.

The local square transparency theorem supplies the expected sidewise lifting mechanism. A colouring of G_s expands to L_s; its local fibre is connected by the square-cycle switch; and a Kempe component through a joined edge expands to the corresponding boundary-to-boundary path through the square. Establishing this simultaneously for every ambient maximal component is an exact graph-level path-lifting theorem, distinct from the finite local fibre computation.

Theorem 61 (square-side path lifting)**.** PROSE *For each reduction side s, projection $L_s \to \mathrm{Col}(G_s)$ is surjective, has connected fibres, and lifts every legal Kempe edge of G_s to a path in the induced reconfiguration graph on L_s. Hence connectedness of $\mathcal{K}(G_s)$ implies connectedness of L_s.*

Proof. Fix a reduction colouring. Give each of the two outer half-edges replacing a joined edge the colour of that joined edge. The resulting square boundary word is compatible with side s. The fibre table therefore supplies a lift. An adjacent-pair word has one lift; an all-equal word has two, and the two are joined by the square-cycle Kempe move. Projection is consequently surjective with connected fibres.

Consider one downstairs Kempe move, switching colours a, b on a maximal component K. For the present boundary word, the checked local profile identity says that the multiset of (a, b)-portal connectivity relations of the square extensions is the multiset of the compatible reduction pairings. If necessary, perform the square-cycle fibre move first; one of the lifts now has exactly the portal relation of side s for (a, b).

Glue the common exterior to the local tracked-colour graph. Components of the glued graph which meet the exterior are obtained by taking exterior components and identifying their incident portals according to the local portal relation. Equal portal relations therefore give a bijection between the downstairs maximal components and the upstairs maximal components which meet the exterior, preserving every exterior edge. In particular K lifts to one maximal component $\widetilde{K}$. Switch a, b on $\widetilde{K}$. At the four square vertices this is an ordinary Kempe switch on a maximal component, so properness is preserved; outside the square its support is exactly the support of the downstairs switch. The resulting colouring therefore projects to the prescribed successor of the reduction colouring and remains compatible with side s.

Thus one edge of $\mathcal{K}(G_s)$ lifts to at most two upstairs edges: an optional internal square-cycle move followed by the lifted component move. Lift a path downstairs one edge at a time, and use connectedness of the final fibre to reach any chosen lift of its endpoint. This proves the last assertion. ☐

The source target supplies one further fact which an arbitrary set of words does not: a target word is exactly the restriction of a colouring after the fixed completion cap is restored. Minimality can therefore prove that each reduction side contains a target state.

Theorem 62 (route-specific square transport from cap nonemptiness). PROSE *Let G be a tangle with a facial square disjoint from its ordered boundary, let G_0, G_1 be its two valid planar reductions, and let C be the fixed completion cap at that boundary. Let W_C be the set of boundary words which extend across C. Assume*

1. *each capped reduction $G_s \cup C$ is Tait-colourable;*
2. $\mathrm{WC}^*(G_s, W_C)$ *holds for $s = 0, 1$; and*
3. *square-side path lifting (Theorem 61) holds.*

Then $\mathrm{WC}^*(G, W_C)$ *holds.*

In particular, a facial square cannot occur in a least obstruction of the source's strong induction whenever attaching the same cap to either reduction gives a strictly smaller closed instance: closed-instance minimality supplies the first assumption and the tangle induction supplies the second.

Proof. Restrict a Tait colouring of $G_s \cup C$ to G_s. Its boundary word extends across C by the colouring just removed, so

$$W_s = \{D \in \mathrm{Col}(G_s) : \mathrm{word}(D) \in W_C\}$$

is nonempty for both $s = 0, 1$.

Now take any colouring D of G. The square fibre table puts D in at least one lifted side L_s. Project it to a colouring $\overline{D}$ of G_s. Since $W_s \neq \varnothing$, the assumption $\mathrm{WC}^*(G_s, W_C)$ supplies a Kempe path from $\overline{D}$ to some $\overline{E} \in W_s$. Lift that path one edge at a time by Theorem 61; its proof permits an optional square-cycle move inside a fibre and then the corresponding maximal component move. The lifted endpoint is a lift E of $\overline{E}$. Projection changes only the square and preserves every ordered boundary edge, hence $\mathrm{word}(E) = \mathrm{word}(\overline{E}) \in W_C$. Thus every colouring of G reaches the target.

The target of G is nonempty as well: choose either $\overline{E} \in W_s$ and use the surjectivity part of square-side lifting to lift it. Therefore the proviso in the definition of WC^* is respected, and the claimed statement follows. ☐

This proof identifies precisely why the generic counterexample is harmless. There G_0 has no target colouring, so its capped-colourability premise fails; the vacuous truth of $\mathrm{WC}^*(G_0, W)$ cannot create a target. In the source induction, target nonemptiness comes from the smaller capped instance, not from an all-equal square state. Hence neither universal $A \neq \varnothing$, universal SQ3 migration, nor arbitrary-target transport may be restored.

Thus the square ledger is now exact:

$$\begin{aligned}
\text{local fibre table and fibre connectivity} &\;:\; \text{machine-checked,}\\
\text{profile-resolved additive identity} &\;:\; \text{machine-checked,}\\
\text{Seed zero-count descent} &\;:\; \text{ordinary proof; Lean formalization pending,}\\
\text{universal } A \neq \varnothing &\;:\; \text{refuted,}\\
\text{universal SQ3 migration} &\;:\; \text{refuted,}\\
\text{generic target-aware WC}^* \text{ transport} &\;:\; \text{refuted,}\\
\text{route-specific WC}^* \text{ square transport} &\;:\; \text{ordinary proof; Lean formalization pending.}
\end{aligned}$$

6 The Kauffman trail spine and its exact composition seam

The final step of one source branch is expressed in the language of Spencer–Brown formations and Kauffman trails. An audit of the graph-side translation separates three different claims: Kauffman's parity theorem for closed cubic formations; a deleted-edge completion predicate; and a new factor-to-ambient composition claim. The first survives. The second collapses under the global Klein-group conservation law, and the third is therefore not presently connected to the Four-Colour conclusion. This section records the exact boundary so that the independent direct-Count descent is not made to inherit a false trail bridge.

Our primary published reference for the first part is L. H. Kauffman, "Reformulating the map color theorem", *Discrete Mathematics* 302 (2005), 145–172, especially Sections 3–4. This is the expanded and clarified version of "On the map theorem", *Discrete Mathematics* 229 (2001), 171–184. The statements below are also the ones reproduced in Section 3 of Goertzel's v23 manuscript. The present section makes explicit where the v24 assembly asks for more.

6.1 Coloured trails and completion

Definition 10 (coloured trail)**.** A *coloured trail* consists of a plane subcubic graph $G(T)$, two distinguished degree-two vertices u, v, a proper 3-edge-colouring C of $G(T)$, and a designated missing edge $e = uv$. The extended graph $G^*(T)$ is obtained by inserting e. In formation language, the two distinguished boundary curves are the containers and the remaining formation lies in the annular between-region.

At a degree-two vertex exactly two colours occur, so there is a unique missing colour. Write $\mu_C(u)$ and $\mu_C(v)$ for those colours.

Lemma 57 (one-edge completion criterion)**.** VERIFIED *The colouring C extends across e precisely when $\mu_C(u) = \mu_C(v)$. More generally, if an allowed sequence of between-region Kempe moves takes C to a colouring C' with equal missing colours, then $G^*(T)$ has a Tait colouring.*

Proof. The new edge must receive a colour absent at both endpoints, which proves necessity. Conversely, colour it by the common missing colour. Properness is unchanged away from u, v, and at each endpoint the three incident colours are then distinct. Kempe moves preserve properness, so the same argument applies after an allowed move sequence. ☐

Theorem 63 (two-defect boundary-flux identity)**.** VERIFIED *Let every vertex of a finite properly Tait-coloured graph be cubic except for two degree-two defects u, v and a finite set B of degree-one frozen boundary stubs. If e_b is the unique edge at $b \in B$, then*

$$\mu_C(u) + \mu_C(v) + \sum_{b \in B} C(e_b) = 0 \qquad \text{in } \mathbb{F}_2^2. \tag{T}$$

Proof. Sum the incident edge colours over all vertices. Every edge is counted at both endpoints, so the total is zero. A cubic vertex contributes the sum of the three distinct nonzero Klein-group elements, hence zero. A degree-two defect contributes its missing colour, and a degree-one boundary stub contributes the colour of its unique edge. These are all the remaining terms, which proves (T). □

Definition 11 (source completability)**.** The coloured trail (T, C) is *completable* if a finite sequence of the source's allowed between-region moves takes C to a colouring C' with $\mu_{C'}(u) = \mu_{C'}(v)$.

Corollary 29 (the bare deleted-edge predicate collapses)**.** VERIFIED *If $B = \varnothing$, every proper Tait colouring of the edge-deleted graph already extends across the missing edge; no move is required. Consequently, if a cubic graph is not Tait-colourable, none of its one-edge deletions is Tait-colourable.*

Proof. With $B = \varnothing$, (T) says $\mu_C(u) + \mu_C(v) = 0$, which over $\mathbb{F}_2^2$ is precisely $\mu_C(u) = \mu_C(v)$. Lemma 57 extends the colouring. The contrapositive gives the last assertion. □

Thus deleting an edge from an uncolourable cubic graph does *not* produce the coloured trail required by Definition 10. The class described in the earlier draft as "coloured but not extension-colourable" is empty.

The framed boundary does not turn this static constraint into a reachability problem. Put

$$\Phi(C) := \sum_{b \in B} C(e_b).$$

Every legal framed move is disjoint from the frozen interface, and therefore leaves Φ unchanged. VERIFIED

Theorem 64 (framed completability is the zero-flux sector)**.** VERIFIED *For a well-formed properly Tait-coloured framed trail whose legal moves fix the boundary edges,*

$$C \text{ is completable by framed moves} \iff \Phi(C) = 0.$$

Proof. If $\Phi(C) = 0$, identity (T) gives equal missing colours in the initial state, so the empty move sequence completes it. Conversely, suppose a legal sequence reaches C' with equal missing colours. Legal moves preserve properness and every frozen-edge colour, so $\Phi(C') = \Phi(C)$. Applying (T) to C' and using $\mu_{C'}(u) = \mu_{C'}(v)$ gives $\Phi(C') = 0$, hence $\Phi(C) = 0$. □

Defect-ending paths are indeed outside the *closed-circuit parity theorem* below, and their exact graph/source alignment is machine-checked. VERIFIED But they cannot change the boundary flux. Therefore the v24 assertion that every colourable framed trail becomes completable is either immediate on a proved zero-flux boundary sector or false on a nonzero-flux sector; internal Kempe dynamics cannot repair the latter. A genuinely relative Kauffman theory would need a different accepting object, not merely the present frozen frame.

6.2 The published parity theorem

For a proper Tait colouring C of a cubic graph G, let $N_{ab}(C)$ be the number of components of the two-colour subgraph on colours a, b, and define

$$\Delta(C) = N_{rb}(C) + N_{rp}(C) + N_{bp}(C), \qquad \pi(C) = \Delta(C) \pmod 2.$$

Each two-colour subgraph is 2-regular, so its components are circuits.

Theorem 65 (Spencer–Brown–Kauffman parity lemma). PROSE *If G is a plane cubic graph and C' is obtained from C by switching two colours on one two-colour circuit, then $\pi(C') = \pi(C)$.*

Proof. It is enough to work in the connected component containing the operated circuit. Such a component has no bridge: each colour class is a perfect matching, whereas deleting a bridge in a cubic graph leaves an odd number of vertices on either side, forcing every perfect matching to contain that bridge. Three disjoint colour classes cannot all contain it. In particular the two sides of every edge are distinct, and the face-cell accounting below has no repeated bridge incidence.

Identify the three edge colours with the three nonzero elements a, b, c of the Klein group $\mathbb{K} = \mathbb{F}_2^2$, so $a+b = c$. We first integrate the edge colouring on the dual map. Fix one face f_0 and put $q(f_0) = 0$. For any other face f, choose a dual walk from f_0 to f and define $q(f)$ to be the sum in $\mathbb{K}$ of the colours of the crossed primal edges.

This is independent of the chosen walk. Indeed, over $\mathbb{F}_2$ the cycle space of a spherical dual map is generated by its facial boundaries, and a dual face is the star of a primal vertex. The sum around such a face is $a + b + c = 0$, because the three edges incident with a cubic vertex have the three distinct colours. Thus every closed dual walk has colour sum zero. Across a primal edge e with incident faces f, f' we consequently have

$$q(f') - q(f) = C(e).$$

In particular q is a proper four-colouring of the faces. The three faces incident with any primal vertex have three distinct potentials.

The integration step and its edge-incidence identities are machine-checked on the graph-backed spherical rotation carrier: VERIFIED VERIFIED VERIFIED

For $x \in \{a, b, c\}$ let X_x be the union of the closed faces whose potentials lie in the dyad $\{0, x\}$. Locally at a cubic vertex, one or two of the three incident faces belong to X_x; hence X_x is a compact subsurface of the sphere with boundary. If $\{x, y, z\} = \{a, b, c\}$, an edge lies on ∂X_x exactly when its two face potentials lie on opposite sides of

$$\{0, x\} \mid \{y, z\},$$

which is exactly when its edge colour is y or z. Therefore ∂X_x is the (y, z) two-colour subgraph and

$$b(X_x) = N_{yz}(C).$$

Every component of a subsurface of the sphere has genus zero, so $\chi(X_x) = 2k(X_x) - b(X_x)$. Reducing modulo two gives

$$N_{bc}(C) \equiv \chi(X_a), \qquad N_{ac}(C) \equiv \chi(X_b), \qquad N_{ab}(C) \equiv \chi(X_c). \tag{1}$$

This boundary/component identification is not supplied by the picture. On the literal dart carrier, the selected face discs are glued across the internal (x)-edges by a finite list of transpositions.

The resulting ribbon permutation returns to its boundary after one or two turns; its first-return permutation is transported to the ordinary complementary two-colour line subgraph. Successor and predecessor are its two graph neighbours, so its permutation cycles are exactly graph connected components. Consequently the parity form of (1), including the absence of a closed selected component hidden from the boundary, is machine-checked: VERIFIED VERIFIED VERIFIED VERIFIED

We next add the three Euler characteristics cell by cell. Every primal vertex belongs to all three X_x, and therefore contributes $3 \equiv 1$. A face of potential 0 belongs to all three, while a face of nonzero potential x belongs only to X_x; hence the total face contribution is $F(G)$ modulo two. Finally, an edge incident with a zero-potential face belongs to all three X_x, whereas an edge whose two incident face potentials are distinct nonzero elements belongs to exactly two. If $d_0(C)$ denotes the sum of the boundary lengths of the zero-potential faces, no edge is counted twice in $d_0(C)$ because q is proper. Thus (1) yields the concrete parity formula

$$\Delta(C) \equiv V(G) + F(G) + d_0(C) \pmod 2. \tag{2}$$

For comparison with the literal ribbon carrier used in the formalization, the cubic handshake identity makes $V(G)$ even. The verified form of (2) therefore reads $\Delta(C) \equiv F(G) + d_0(C) \pmod 2$. Both the evenness reduction and the complete three-dyad summation are machine-checked: VERIFIED VERIFIED

Now let K be the (a, b)-circuit on which the switch is performed. Combinatorial separation for spherical maps says that the edge set of the cycle K is the boundary of a set S of faces: an edge belongs to K if and only if exactly one of its incident faces lies in S. This is the dual form of the fact that facial boundaries span the cycle space; it is precisely Corollary 28, so no unstated Jordan-curve step is being used here.

More precisely, the binary cycle-space vector is checked to equal the boundary of a finite quotient-face set, and an edge has value one exactly when its two dart-side faces have opposite membership: VERIFIED VERIFIED For the operated component itself, even incidence at every cubic vertex and the resulting exact separator are also machine-checked: VERIFIED VERIFIED

Define

$$q'(f) = \begin{cases} q(f) + c, & f \in S, \\ q(f), & f \notin S. \end{cases}$$

Edges off K have either both incident faces in S or neither, so their differences are unchanged. Along K, adding c at exactly one end sends a to b and b to a. Hence q' is the face potential of the switched colouring C'.

The underlying translation algebra, including the separating-edge and same-side cases, is machine-checked. Moreover, the resulting boundary is proved to be $a + b$ precisely on the operated component, and the translated edge chain is proved equal to the actual Kempe-switched colouring: VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED VERIFIED

Only the faces in S can change their contribution to d_0. Inside S, translation by c exchanges potentials 0 and c and exchanges a and b. Consequently

$$d_0(C') - d_0(C) \equiv \sum_{\substack{f \in S \\ q(f) \in \{0, c\}}} |\partial f| \pmod 2. \tag{3}$$

In the sum on the right, an edge joining two selected faces is counted twice. The edges counted once form the boundary of the selected face set $\{f \in S : q(f) \in \{0, c\}\}$. Every such edge separates the dyad $\{0, c\}$ from $\{a, b\}$, so it has colour a or b. At each cubic vertex this boundary has degree zero or two; it is therefore a disjoint union of alternating (a, b) circuits. Every such circuit has even length. The right-hand side of (3) is thus zero, and $d_0(C') \equiv d_0(C) \pmod 2$.

The relative-boundary cycle-space calculation and the resulting preservation of zero-potential boundary-length parity are machine-checked: VERIFIED VERIFIED

The map itself has the same numbers of vertices and faces before and after the switch. Formula (2) now gives $\Delta(C') \equiv \Delta(C) \pmod 2$, as required. This is the Tutte-style form of the Spencer–Brown–Kauffman parity argument: planarity enters exactly through face potentials and spherical cycle separation. On the exact graph-backed carrier, the translated potential with preserved zero-fibre parity, the full formation-parity theorem, and its two-cross-pair corollary are machine-checked: VERIFIED VERIFIED VERIFIED VERIFIED □

Planarity is essential. Kauffman's one-edge-deleted Petersen formation changes the curve count from five to four under a simple operation. A nonplanar counterexample is therefore a regression fixture for any eventual formal statement: a theorem that omits the spherical-map hypothesis is false.

6.3 Prime trails and Kauffman's equivalence

In Kauffman's terminology, a same-context trail is *factored* if, after the two contextual curves are removed by idemposition, the remaining trail structure has more than one component. In the different-context case it is factored if an extra formation curve misses both endpoints of the empty edge. A trail is *prime* if no recolouring in its orbit has a factored representative.

Lemma 58 (Kauffman's curve-count dichotomy)**.** *SOURCE* *Let T be a nonempty prime uncolourable planar trail.*

1. *In an unfactored representative whose contextual curves have the same colour, the total two-colour circuit count is five.*
2. *In an unfactored representative whose contextual curves have different colours, the total two-colour circuit count is four.*

The source's informal count uses *context curve*, *trail curve*, *alternator*, and *idemposition* as formation-level objects. Those objects and the exhaustive endpoint case split have not been defined in this companion. The former prose proof therefore did not prove the displayed claim; this row deliberately inherits source status instead.

Proposition 41 (the deleted-edge primality bridge cannot start)**.** *VERIFIED* *Under Definition 10, deleting an edge from a non-Tait-colourable cubic graph does not produce a coloured trail. Hence the displayed graph-side trail model cannot establish the claimed Four-Colour/Primality equivalence.*

Proof. Corollary 29 gives the stronger statement that any proper Tait colouring of the deletion extends immediately. Taking the contrapositive, an uncolourable cubic graph has no properly Tait-coloured one-edge deletion. But such a colouring is part of Definition 10; the proposed minimal trail is therefore not constructed. □

There is a second, independent mismatch. Theorem 65 is a theorem about closed cubic graphs and switches on two-colour circuits. At a degree-two trail vertex the putative dual face potential accumulates the two incident colours, whose sum is the nonzero missing colour. Thus walk-independence fails exactly at the defects, and the theorem cannot be applied to defect-path moves. Recovering Kauffman's original formation argument would require a boundary-relative potential or torsor together with formation definitions supporting Lemma 58; neither is presently supplied by the source or this companion.

6.4 The extra theorem required by the compositional route

The source now proposes to prove its Trail Completability theorem by strong induction. In a factored minimal obstruction, each factor has a smaller extended graph and is therefore colourable. Applying Trail Completability to each factor produces factor-local completion sequences. One further theorem is needed before those sequences may be composed in the original formation.

Lemma 59 (finite product reachability). PROSE *Let I be finite. For each $i \in I$, let R_i be a relation on a state space X_i, let $x_i \in X_i$, and suppose $x_i \, R_i^* \, y_i$. On $\prod_i X_i$, let R_Π be the asynchronous product relation: one step changes exactly one coordinate by an R_i-step and fixes every other coordinate. Then*

$$(x_i)_{i \in I} \; R_\Pi^* \; (y_i)_{i \in I}.$$

Consequently, if y_i belongs to a local accepting set A_i for every i, the initial product state reaches $\prod_i A_i$.

Proof. Choose an order on the finite set I. Lift the first coordinate's finite R_i-path to product states by holding all other coordinates fixed, then lift the second coordinate's path, and so on. Concatenating these lifted paths gives the required R_Π^* path. Previously completed coordinates are never changed by a later lifted step. The accepting-set claim is immediate from the endpoint tuple. □

The preceding lemma is deliberately only a scheduling theorem. It does not say that an isolated factor move remains a Kempe move after the contextual curves are identified. That missing assertion cannot be repaired by freezing all factor seams: doing so can change the reachability problem itself. The correct finite replacement is the following elementary closure criterion.

Lemma 60 (finite synchronization criterion). PROSE *Let $(P, 1, \star)$ be a finite monoid of exact factor-interface behaviours, let $G \subseteq P$ be the behaviours realized by strict locally completable factors, and let $A \subseteq P$ be the behaviours whose contextual composite is completable. Write $\langle G \rangle_+$ for the set of nonempty finite products of elements of G. Then every finite composite of locally completable factors is completable if and only if*

$$\langle G \rangle_+ \subseteq A. \tag{4}$$

Since P is finite, (4) is decided by breadth-first closure from G under right multiplication by G. If the inclusion fails, the predecessor tree returns a shortest word of locally completable factor behaviours whose composite is not completable.

Proof. Every nonempty factor list $(g_1, \ldots, g_k)$ evaluates to the monoid product $g_1 \star \cdots \star g_k$, and these are precisely the elements of $\langle G \rangle_+$. Thus (4) is exactly the asserted universal gluing property, in different notation. For the decision procedure, initialize a queue with G. Whenever p has been reached, insert every $p \star g$, $g \in G$, not seen before. Finiteness of P makes the process terminate, induction on word length proves that it reaches all and only $\langle G \rangle_+$, and stored predecessors reconstruct a word for any reached element outside A. □

Proposition 42 (source factorization has product semantics). OPEN *Construct a finite monoid P of exact Kauffman factor-interface behaviours, a set $G \subseteq P$ of behaviours realized by strict factors to which Trail Completability applies, and an ambient-completion set $A \subseteq P$, such that:*

1. *every reachable source factorization has a nonempty factor word $g_1 \cdots g_k$ with every $g_i \in G$;*
2. *the profile of the contextual composite is exactly $g_1 \star \cdots \star g_k$, not an over-approximation;*

3. *a contextual composite is completable exactly when its profile lies in A;*

4. *every represented factor is strict in the induction measure; and*

5. *the kernel-replayed closure satisfies $\langle G\rangle_+ \subseteq A$.*

This is the precise missing bridge between the topological word "factored" and the algebraic word "independent". It is a synchronization seal, not a claim that the factors move asynchronously. The strong monoidal Count semantics in the source supplies the intended construction: retain the boundary colours, the two relevant Kempe connectivities, and the capped face progress, Booleanize positive counts, and quotient only after proving that composition and acceptance factor through the quotient. Lemma 60 then reduces the universal number of factors to one finite closure.

Proposition 43 (factor completion lifts and glues)**.** CONDITIONAL *Let a reachable representative of a coloured source trail T be factored into strict factors $T_1, \ldots, T_k$. Suppose that, for every i, the specific colouring induced on T_i is completable by the source's framed between-region moves. Then there is a sequence of legal moves of T whose restriction realizes the required factor completions and after which T itself is completable.*

Proof. Apply Proposition 42. The factors determine a nonempty word in G. Its product lies in $\langle G\rangle_+$, hence in A by item 5 and Lemma 60. Exactness in items 2–3 turns this finite accepting profile into an ambient completion. □

This is not a formal nicety. A two-colour component that is closed in an isolated factor can continue through a contextual curve when the factor is placed back in the ambient formation. Switching only its factor portion is then not an ambient Kempe move. Moreover, distinct factors share contextual curves, so arbitrary completion sequences need not have disjoint supports. The proposition must therefore be proved by one of the following exact mechanisms, not by the sentence "the factors are disjoint":

1. the finite synchronization closure of Proposition 42;

2. a stronger ambient-component lifting theorem which proves that the closure is automatically accepting; or

3. an interface-relative completion theorem for the particular seams produced by a Kauffman factorization, together with a proof that those seams are one of its valid boundary classes.

The third item cannot be replaced by an unrestricted "freeze every seam" principle. A deterministic regression audit on the source C30 polar annulus deletes one interior edge, enumerates all 15,192 proper colourings, and reproduces the 5,412 restored-edge colourings exactly. With either one five-edge boundary frozen, every Kempe class reaches a restored-edge colouring. With both boundaries frozen there are 11,067 classes, of which 7,524 contain no such colouring. This refutes only the generic twice-frozen strengthening—the specimen is not asserted to be a Kauffman factor—but it is enough to forbid that strengthening as a proof rule. The replayable audit is generated by `tools/fourcolor/v24_factor_interface_audit.py`.

The current framed-trail API supplies frozen-interface vocabulary and exact source/graph move alignment, but no theorem presently establishes any of these three alternatives for a Kauffman factorization. Proposition 42 is consequently ordinary mathematics still to be done. It must not be delegated to a finite census unless the source first proves a uniform bound on the factor interface.

Proposition 44 (requirements for a repaired Kauffman assembly). *OPEN To recover a Kauffman-style final assembly from the present development, one must supply all of the following in one compatible formation model:*

1. *a boundary-relative replacement for Theorem 65 which applies to the allowed defect-path operations;*
2. *formation-level definitions and a proof of the curve-count dichotomy in Lemma 58;*
3. *an accepting condition not reduced by Theorem 64 to frozen boundary flux; and*
4. *exact factor product semantics and ambient lifting as in Proposition 43.*

Only after these are proved may a primality implication be assembled.

The Kauffman layer is therefore not a presently valid terminal bridge. Its closed-cubic parity theorem and the generic finite synchronization lemma remain useful mathematics, while the deleted-edge bridge is retired and the relative formation theorem is open. No Lean development should package the four requirements above as hypotheses merely to recreate the old conclusion. The direct cumulative-Count descent in Theorem 45 is independent of this retired branch.

7 Current controlling status: the reductive–compositional programme

Source precedence and route identity. The present development formalizes the reductive route: a sufficiently large counterexample contains a finite-state corridor, two equal interface states permit a physical splice, and the splice produces a strictly smaller counterexample. This is not the classical unavoidable-set, reducibility, and discharging architecture. No configuration catalogue, discharging system, hubcap family, or per-configuration ring-coloring table is admissible as a substitute. The current Fable manuscripts listed in the abstract govern this section; v23 remains a historical and mathematical source where it is consistent with them.

Trail-branch retirement. The graph-side deleted-edge translation in v23 and Addendum XXVIII does not supply the terminal implication. Summing the Klein-group edge colours over all vertices shows that a proper colouring of a cubic graph with one edge deleted always extends across that edge. With frozen boundary stubs, the two missing colours differ by the frozen boundary flux, and legal framed moves leave that flux unchanged. Thus defect-path moves do not create a dynamic completion mechanism. Kauffman's closed-cubic parity theorem remains valid, but it cannot be applied to the noncubic defect paths. This retirement does not touch the direct reductive Count theorem, which is the controlling assembly below.

7.1 The idea in plain language

Imagine walking along a long annular corridor and recording, at every transversal, only what the rest of the graph can observe through that cut. A full profile records boundary colors, tracked-color connectivity, face continuation, fragment-to-port incidence, and capped face progress. There are only finitely many such records. A sufficiently long walk therefore repeats a record. If the two equal records occur on actual separated simple transversals, the portion between them should be removable: the two exposed ends carry the same information, so the exterior should not notice the splice.

The word "actual" is the geometric heart of the proof. Finite-state algebra can prove what follows *from* suitable simple transversals or layer boundaries; it cannot construct them merely by naming them. The current Fable splice checklist requires such geometric representatives, but does not name the particular Lean type used below to package a pair of facial-dual crosscuts. The source's named global formation task is instead L1: a finite local classification showing that a sufficiently large all-hexagonal patch contains a ladder corridor longer than the profile count. Any eventual crosscut construction must be proved as part of a faithful realization of that source geometry, not promoted to an independent source flag by the Lean API.

Opening a cap is not yet opening an annulus. The source's Cell–3 setting starts with a frontier tangle: opening its interior pentagon leaves five ordered inner stubs while its already-free outer interface remains available for the annular argument. The formalization now checks the local half exactly. For a cubic closed graph, deleting only the five cap-cycle edges makes precisely the five cap vertices degree one; every other vertex remains cubic. Consequently a well-formed boundary datum that uses those five vertices as its inner interface has *zero* outer stubs. This is a useful negative fact, not a substitute for the missing formation: an outer interface cannot be invented by the cap-opening operation. A source-faithful construction must instead retain the outer frontier tangle and then identify its annular embedding with the five ordered cap stubs.

7.2 What is presently machine-checked

The following items are green Lean developments. Their scope matters: several are consumers of a crosscut pair rather than constructors of one.

Layer	Kernel-checked content and boundary
Finite profiles and L7	A generic finite carrier, decidable equality, exact cardinality formulae, and finite-profile pigeonhole machinery.
Relational transfer	Open-tangle relations, heterogeneous transfer words, composition, reachable-set pumping, and colorability-reflection consumers. The primitive transfer is relational; determinism is not assumed.
Literal profile semantics	The five profile coordinates are related to actual Tait colorings. Tracked connectivity and occurrence-sensitive facial progress are updated under one regional seam condition.
Portal completeness	Both tracked components and face occurrences escaping a literal prefix are routed to named interface portals.
Splice consumers	A literal Tait-coloring gluing equivalence, full-profile diagonal audits, and strict output-size decrease. These consume the missing physical realization and reflection theorem.
Source structure	Sector Alternation, two-rail separation, local Cell–3 formation, consecutive side-interface matching and orientation, and both local rail-edge identifications, with their exact successor cyclic positions, packaged as a canonical one-step rail cell. Consecutive surviving side slots now also give a literal exterior facial-dual rail edge using only cubicity at the displayed interior corner, and the finite cyclic calculation classifies every nonadjacent cell as the exact rail shape (0+2), (1+1), or (2+0). Each case now constructs two literal simple facial-dual walks, each of length at most two and with combined length exactly two; the two local supports are disjoint. Every face in either walk is additionally certified adjacent to the Cell–3 center that generated it. Combining this provenance with the boundary-clean geodesic exclusion now proves all four track combinations disjoint whenever their centers are at least three positions apart. The two bounded neighboring gaps remain a finite local obligation. For two consecutive cells, that obligation is now isolated as the proposition `CommonNeighborsExact`: the only common full-dual neighbours of the two centres are the two named flank faces of their shared rung. Assuming exactly this proposition, the literal successor rails rebase at the preceding exits, append to two simple facial-dual paths, and retain mutual support disjointness. This is a complete conditional two-cell append theorem, not yet a derivation of the condition from the source annulus. A literal successor theorem carries each outgoing flank to an incoming flank of the next cell on the same ambient edge and exterior face, then returns one dependent successor package containing the next outgoing flanks and the exhaustive local shape certificate. The literal walks are generated definitionally from that retained certificate, rather than returned by an independent existential choice. Thus a later induction cannot pair a case certificate with unrelated paths or forget which of the three literal cases generated them. Lengthwise simplicity and end-cap attachment remain open. These are local formation facts, not yet a global annular crosscut pair.

The local/global distinction is now also explicit in the kernel. The four theorems `firstFace_ne_innerHole`, `firstFace_ne_outerHole`, `secondFace_ne_innerHole`, and `secondFace_ne_outerHole` derive from the pair's `interiorFaces` certificates that neither local endpoint is either named annular hole. Thus this pair cannot be silently re-presented as the hole-to-hole crosscut needed by the splice. This is deliberately a scoped negative fact: it does not say that a longer, source-realized extension to the holes cannot be constructed.

This layer also contains a useful discipline theorem: equality of boundary color words alone cannot replace equality of full profiles. A two-state machine-checked counterexample admits an off-diagonal transition while having matching projected color support and no full-profile diagonal. Connectivity and face data are therefore mathematical content, not bookkeeping.

7.3 A Lean-side crosscut interface, not a source flag

The type `SeparatedAlignedSimpleDualCrosscuts` is a useful Lean-side interface: it packages two aligned simple dual crosscuts, disjoint transverse supports, and nondegeneracy, and much splice machinery is parametrized by it. It is nevertheless an interface chosen by this formalization, not a theorem or named obligation in the current Fable manuscript. No theorem currently produces a long/end-capped instance from the Cell–3 annular carrier. That is an open proposed realization of the interface, not evidence that the source reduces 4CT to this one existence statement.

Source-alignment correction (13 August 2026). Earlier local-rail modules used "L1" as a convenient internal tag. That was misleading. In the current Fable manuscript and playbook, L1 is the bulk statement that a sufficiently large frontier instance contains a long hexagonal ladder corridor. The bounded common-neighbour premise used to append two locally constructed rails is not stated as Fable's L1, and no current-source theorem yet identifies it as the construction of L1's corridor. The original v23 *boundary locality* argument is historical evidence about radius–one collars, but it assumes confinement and bounded-interaction claims that the corrected Fable work records as refuted. It cannot license the present rail assembly.

Accordingly, the local rail developments below are retained as elementary, kernel-checked geometry and as a conditional assembly interface; they do not count as a source realization or as progress through a Fable flag. The source-faithful priority is the source's actual L1 classification and then the other named flags. Separately, if the formalization needs this dual-crosscut interface to implement the manuscript's simple-transversal splice condition, it must construct and justify that connection explicitly. Until then, neither the L8 radial witness nor the closed-map opening transport is silently treated as a crosscut construction.

There is, however, now a direct handoff for the current Lean clean-block extraction which narrows this question further. A `ClosedWebAtGoodWord.Instance` already contains the actual `ClosedWebAnnularEmbedding` and its `AnnularFrontierGeometry`. The theorem `exists_sourceRealizedBoundaryCleanOrbitHexCorridor_of_weightedCleanBlock` therefore turns the manuscript's explicit weighted size premise into a boundary-clean corridor together with the literal source geodesic and selected block. It does not reopen the source through a cap-deletion model. This is a real conclusion of the current Lean extraction, conditional only on the stated quantitative bound. It is not yet Fable's full L1 ladder corridor. The separate local rung work below is an exploratory Cell–3 transversal candidate, not a source-identified construction of that L1 corridor. One initially apparent obstacle to such local work has now been removed without imposing a closed-minimal hypothesis. The new `SelectedCorridorRungs` object chooses one genuine shared primal edge for each consecutive dual step, rather than requiring all pairs of faces to share at most one edge. Lean then proves that the selected incoming and outgoing rungs of every internal Cell–3 hexagon occupy distinct, non-adjacent positions on its six-dart boundary. The proof uses only the source carrier's interior-face simplicity, local cubicity, and geodesic separation. This is a local rung-placement result, not a proof of the remaining finite adjacency classification, a long separated pair of crosscuts, or a splice.

The next finite condition is now stated at exactly the scale on which it is used. If one displayed pair of consecutive Cell–3 corridor faces has at most one shared interior primal edge, then each selected side slot adjacent to the outgoing rung is provably emphdifferent from the next corridor centre and hence has a genuine one-edge facial-dual rail step to it (`selectedPlacementSideFace_adjacent_next_of_*Outgoing_of_successorSharedInteriorEdges_card_le_one`). The proof is the contrapositive of a literal collision theorem: equality would exhibit two distinct shared edges, namely the selected rung and the side edge. This replaces neither the source formation map nor the global closed-map uniqueness theorem. It reduces the relevant transport obligation to one successor-pair cardinality fact and keeps the alternative explicit until that fact is actually supplied.

Three nearby results delimit the gap.

- Generic dual connectedness constructs one hole-to-hole crosscut. It does not produce a separated pair tied to the selected corridor.
- The local Cell–3 layer produces a separated local interface package, but its endpoints are not the required distant annular endpoints.
- Consecutive six-edge aligned slab-boundary walks are proved non-disjoint: they share their intervening dual interface. They are useful local windows, not the two radial cuts of the splice.
- The source-facing `LongRadialSectorWitness` already supplies two vertex-disjoint *primal* radial paths, distinct anchors on both boundary components, sector alternation, and a quantitative length witness. It does not itself supply facial-dual crosscuts, and no module presently relates this witness type to `SeparatedAlignedSimpleDualCrosscuts`.
- The coverage clause is now explicit in the ambient graph: `interior_vertex_mem_ambientRadialPath_union` transports `RadialPathPair.cover_support` to prove that every interior vertex lies on one of those two radial paths. This is precisely the "jointly covering every internal vertex" part of Addendum XXII; it still does not construct the two complementary sector sides. It does mean that any later sector side proved disjoint from both paths is immediately free of interior vertices.

The last item is a scoped refutation of a tempting Lean construction, not a refutation of Goertzel's proof. It says that the separated cuts must be assembled at the correct global scale rather than identified with adjacent local windows. Nor is the live problem a general Menger-type existence theorem.

Annular-separation implementation decision (14 August 2026). An audit of the pinned Mathlib found no usable development of planar graph separation, combinatorial maps, rotation systems, or Menger's theorem. Adding an external planarity library would not remove the work: it would introduce a second embedding language and require a compatibility proof with the rotation system and face-orbit infrastructure already used here. The intended next mathematical object is therefore a narrow annular separation lemma stated in this development's own vocabulary. Its input cannot be merely an arbitrary `ClosedWebAtGoodWord.Instance` or one dual-connectedness slit: the source requires the finite L1 local ladder classification to supply the relevant non-identification and endpoint data. This is an implementation decision and a boundary on the theorem still to prove, not a claim that the separation construction has been completed.

A framed-Euler separator specialization (14 August 2026). One narrow part of that implementation boundary is now closed. The existing `GoertzelV24FramedDualCycleSeparator` proves that the primal edges crossed by a simple facial-dual cycle disconnect a framed carrier from its internal-dual connectedness, primal connectedness, exact Euler identity, and unique-interior-shared-edge hypothesis. In particular it does not require the whole open carrier to be cubic or globally two-sided. The crosscut separator, deletion-component, and splice-boundary layers now expose `_of_euler` specializations, and a local source-layer pair can obtain its deletion component and exact ordered boundary through `localLayerPair_sourceCrosscutBoundaryData_nonempty_of_euler`.

This is a repair of an invalid wrapper premise, not a completion claim. A well-formed framed source trail has defect stubs and is not globally cubic, but the new theorem still takes an actually constructed pair of aligned simple crosscuts, two-sidedness for the exact-boundary equality, and unique shared interior edges. Fable L1 must still supply the distant, end-capped ladder pair. The local pair is not silently promoted to that pair, and no source-splice or profile reflection has yet been instantiated.

There is an important source-level separation here. In Addendum XXVII and the formalization playbook, the radial-path witness is the L8 input to the L6 length/depth dichotomy, whereas the boundary-clean hex corridor is the independent L1 output. The manuscripts do not instruct us to dualize the L8 witness into splice crosscuts. Indeed, v23's meridional construction chooses a primal boundary-to-boundary path and then chooses a dual radial arc separately. A primal path is not itself a facial-dual path, so converting the former into the latter requires an additional transversality construction, not just the shared words "radial" and "disjoint." The absence of a Lean module relating the two types is therefore not by itself a missing source theorem, and we do not silently identify them.

The literal Cell–3 rail construction is a Lean-side exploratory candidate for a physical corridor realization; the current sources do not yet identify it with Fable's L1 construction. Its remote noncollision is proved, and its adjacent append has been reduced to `CommonNeighborsExact`. Lean now proves that this face-level statement is equivalent to an edge-level statement: for every common neighbour, the canonical primal edge from the left centre to that face is one of the two literal flank edges of the shared rung. The proof uses the fact that an edge of a planar cellular map is incident with at most two face orbits, together with a new exact identification of each literal flank edge with the canonical shared edge of its centre and side face. Thus the bounded local gap is not a second, independent classification of faces; it is the transport of one closed-map primal-edge classification into the literal open Cell–3 carrier. This equivalence is a reduction of the obligation, not its discharge.

There is also a type-level boundary on what the current rail package can mean. A `SourceCornerAlignedRailPair` contains one path from the first incoming flank to the first outgoing flank and another from the second incoming flank to the second outgoing flank. Its proved numerical invariant is only that their lengths add to two: the three literal cyclic shapes are $(0,2)$, $(1,1)$, and $(2,0)$. In contrast, `SeparatedAlignedSimpleDualCrosscuts` asks for two paths with the same start and finish and with equal lengths. Thus the two raw rails cannot be retyped as that crosscut pair merely because they bound the same local strip; the unbalanced cases rule out even the equal-width part. An eventual annular construction must attach and verify genuine end caps (or introduce a different, source-justified boundary object), rather than hiding this work in a coercion. This is a check on the Lean representation, not a new claim about Fable's L1 classification.

An extra common neighbour would form precisely the small separating/nonfacial dual triangle excluded by v23's historical minimal-counterexample radius–one collar normal form. The present open-annular `Instance`, however, records well-formed boundary stubs, coloring, closure, and the good word, but not the closed minimal-counterexample provenance from which that bounded-interaction fact is derived. That old normal form is not a current-source justification for adding this classification to the Cell–3 instance. Any future use needs a separate, current-source formation theorem; until then the result is only a conditional Lean interface.

The closed half of that argument is now machine-checked. Given three pairwise adjacent faces in a graph-backed vertex-minimal Tait counterexample,

`exists_singleton_primalCut_of_dual_triangle`

constructs their literal length-three facial-dual cycle and proves that its three crossed primal edges separate a component containing exactly one vertex; moreover, the crossing-edge set of that singleton component is exactly the three edges read from the dual cycle. Lean now also extracts the local consequence used by the radius–one collar description:

`exists_vertex_mem_all_crossed_edges_of_dual_triangle`.

The consumer-facing theorem

`exists_vertex_mem_three_shared_edges_of_dual_triangle`

then identifies the three canonical pairwise shared primal edges themselves and proves that one ambient vertex belongs to all three. Thus the dual triangle is the link of one primal vertex, rather than an arbitrary separating triangle. This is a closed-map bounded-interaction classification under stronger historical minimal-counterexample hypotheses; it is not itself the current Fable L1 flag.

The closed classification is now complete in `GoertzelV24MinimalCorridorCommonNeighbors`. Its theorem

`commonNeighborEdge_eq_outgoing_flank`

starts with a literal hex-corridor placement in the closed graph-backed minimal counterexample. For any face adjacent to both a corridor face and its successor, the theorem proves that the canonical shared primal edge from the first centre is exactly one of the two face-cycle edges immediately flanking the outgoing rung. The proof combines the common vertex extracted from the dual triangle with cubicity and two-sided face incidence: at either endpoint of the outgoing rung, the face has exactly two incident boundary edges, namely the rung and its immediate flank. Since the target edge is not the rung, it is the flank. Thus the bounded local classification is no longer an assumption on the closed map.

The possible transport problem is now precise. The current Fable theory begins in the v23 normal form: a bridgeless cubic plane graph with no cyclic edge cut of size at most four (equivalently, no separating triangles or four-cycles in the dual). Its expanded Addendum XXV Seed-Lemma step spells out only the girth and two-cut reductions, so that step must cite the already stated normal form rather than be read as deriving cyclic five-edge connectivity by itself. Lean independently derives the same closed property from graph-backed vertex-minimality by capping the two sides of each small cut and gluing their colorings.

Thus the obstruction is not a new separating-three-edge-cut classification. It is the one source-formation morphism still absent from the formalization: carry the relevant closed normal-form corridor and its retained local faces through the actual cap opening, and identify the computed opening with the literal Cell–3 annulus. Until that construction exists, the closed theorem cannot discharge `CommonNeighborsExact` on the open carrier. The degree-one boundary stubs make adding closed minimality directly to that open instance inconsistent, not a shortcut.

The source convention for that opening is now stated literally, rather than as a vague "deleted cap." The manuscript says to delete the five cap vertices while retaining their spokes as stubs. The accompanying Fable toolchain realizes the intended tangle by removing the five *cycle edges* of the cap and keeping its five original vertices; each retained cap vertex is then a degree-one endpoint of its outward spoke. The new Lean module `GoertzelV24PentagonCapOpening` computes this concrete toolchain operation from a finite pentagon-cap presentation. It constructs the edge-deleted graph, transports the ordered five spokes into its edge type, and proves that each named cap vertex has precisely its one surviving spoke. It consequently computes the five-stub inner interface of `AnnularBoundaryData`, with no invented outer boundary. The manuscript-side carrier is now present separately as `GoertzelV24DeletedRegionSlitGraph`: it deletes a chosen vertex region, keeps precisely the exterior vertices, and creates one fresh degree-one stub for every directed edge from that exterior into the deleted region. Thus the two presentations have been retained rather than silently choosing one. The comparison is now constructed in `GoertzelV24PentagonCapOpeningComparison`. Its `openingGraphIso` maps the fresh slit port at ordered position i to the toolchain's retained cap vertex at that same position i. Exactness is not a cardinality argument: the cubic incident-edge list at every deleted cap vertex proves that the five outward spokes exhaust all manuscript-side ports. Thus the graph isomorphism also preserves the ordered five-stub boundary data, which a bare graph isomorphism allowed to permute the stubs would not do. The next local formation fact is now explicit in `GoertzelV24PentagonCapOpeningWellFormed`. Under closed cubicity, every vertex outside the cap retains degree three in the toolchain opening, while each ordered cap vertex has degree one on its named outward spoke. Lean therefore constructs the corresponding well-formed five-stub inner boundary datum, with an intentionally empty outer interface. This is useful local source plumbing, not an annulus in disguise: it neither chooses the source cap nor constructs the outer boundary, annular embedding, or crosscut pair. This settles a presentation comparison, not the source formation theorem: it neither selects a pentagon in the closed normal-form map nor proves that a Cell–3 corridor is retained, constructs the outer hole, or identifies the dependent face-fragment carriers of the two openings.

The first literal two-boundary opening. The source's annular laboratory removes two separated pentagon caps, and this has now been made into a separate graph construction rather than being silently inferred from a one-cap opening. In `GoertzelV24TwoPentagonCapOpening`, a pair of cap presentations carries three explicit separation conditions: disjoint cap-vertex supports and no spoke from either cap into the other cap support. The latter two conditions are necessary: vertex disjointness alone still permits one edge to be both an inner and an outer boundary edge. From a cubic closed graph, simultaneous deletion of the two cap cycles produces a well-formed `AnnularBoundaryData` with five ordered inner and five ordered outer stubs. The proof checks the degree-one and edge-disjoint boundary facts and that every other vertex stays cubic. This closes the graph/interface half of the two-cap model. It still does not select such a separated pair in a minimal counterexample, give the opened graph its annular rotation embedding, or construct the Cell–3 corridor and its crosscuts; those are the remaining formation geometry.

This distinction is source-important. Addendum V begins its mathematical setting with an already supplied frontier tangle X and one interior pentagon; the two-cap C_{30} construction is an annular laboratory, not a theorem that every closed minimal counterexample canonically supplies two such caps. The Lean two-cap construction is therefore reusable opening plumbing, not a claimed closed-to-annular formation result.

No hidden boundary ports. The manuscript-side version deletes the union of the two cap-vertex supports and creates one fresh stub for every outward dart. The new `GoertzelV24TwoPentagonCapOpeningBoundary` proves this boundary is not merely of cardinality ten: its ordered port carrier is explicitly equivalent to `Fin 5` $\oplus$ `Fin 5`. Each port is uniquely one of the five inner spokes or the five outer spokes; the cubic incidence list rules out the two cycle-edge alternatives. Thus the laboratory's two interfaces are exact, ordered data before any transfer profile is placed on them. This still compares graph boundaries only: it does not establish the ambient annulus, its rotation/face carrier, or any profile/count semantics.

The subsequent carrier comparison is now a graph isomorphism, not merely a matching cardinality. It fixes old vertices, sends each fresh outward stub to the cap vertex at its deleted endpoint, and checks adjacency in all four cases: old–old, old–stub, stub–old, and stub–stub. The inner and outer ordered-stub compatibility theorems both hold by `rfl`: coordinate i cannot silently be transported to a permuted cap vertex. This is a deliberate definitional guarantee, recorded so that a later refactor does not weaken it to a merely propositional correspondence. Dependent face-fragment carriers and their cyclic orders remain separate obligations; this is not yet an annular rotation embedding.

There is no parallel open-region graph tower behind this comparison. The generic theorem `slitGraphIsoOpenPrimalGraph` in `GoertzelV24DeletedRegionSlitOpenRegionComparison` identifies the manuscript slit graph for any deleted vertex region with the simple primal graph computed by the existing literal open-region rotation. It preserves retained–retained edges, each fresh stub and its unique retained neighbour, and the absence of stub–stub edges. Thus subsequent formation work may use the rotation and face-transport theorems through this one isomorphism rather than re-proving them for a second graph presentation. It does not turn a graph isomorphism into equality of dependent face-fragment data or construct the source-specific annular region.

The two-cap specialization now composes the two graph isomorphisms in `GoertzelV24TwoPentagonCapOpeningOpenRegionComparison`. For either chosen outer dart of the literal open-region rotation, it identifies the toolchain's simultaneous cycle-deletion graph with that rotation's computed primal graph. This eliminates a duplicate two-cap graph tower while keeping the necessary outer-dart choice explicit. It still proves neither an annular rotation embedding nor the cyclic face-order comparison that a source profile construction will require.

The carrier comparison now transports the complete graph-backed rotation as well. The generic module `GoertzelV24SimpleGraphRotationIsoTransport` carries oriented darts, their vertex rotation, the complete face permutation, and the chosen outer dart across any graph isomorphism; `transportData_phi_eq_permCongr` proves the permutation conjugacy and `transportData_phi_apply` gives its one-face-step form. `openingGraphData` applies this to the literal two-cap graph. Thus the toolchain graph is no longer missing a rotation or its unlabelled facial cycles merely because the computed open-region graph has a different vertex type. This is deliberately below the annular claim: a graph isomorphism still selects no distinguished hole faces, their cyclic order, or Euler proof.

The comparison now carries the emphordered annular boundary datum as well, not just its graph and unlabelled face permutation. The new generic module

`GoertzelV24AnnularBoundaryIsoTransport` maps both ordered stub embeddings and both ordered boundary-edge embeddings through a graph isomorphism, and proves preservation of every field of `AnnularBoundaryData.WellFormed`. Its two-cap specialization, `openingBoundaryData`, places the literal five-plus-five interface on the very open-region primal graph used by the face-transport layer. This removes a carrier mismatch: future Cell–3 formation need not rebuild its boundary data merely to use open-region faces. It remains exactly a graph/interface result—it neither chooses the annular hole faces nor proves their order or embedding.

There is a sharp division inside that remaining face work. The existing generic `GoertzelV24OpenRegionFaceTransport` already transports every ambient face whose complete dart orbit stays on the retained side. In a cubic two-sided rotation, `GoertzelV24CubicFaceVertexSeparation` discharges that condition for a face dual-nonadjacent to a specified deleted face. The present `PentagonCapPair`, however, is graph-side data: it is not yet identified with two ambient facial orbits, nor is the needed dual separation proved. Hence those results carry the genuine interior facial geometry once the source formation supplies its hypotheses; the faces incident to either cap remain the new collar geometry. This is a useful reduction of the formation theorem, not a proof of it.

The retention component is now also available for two deleted faces. The generic theorem `faceFullyRetained_compl_union_orbitFaceVertices_of_not_adj` turns explicit distinctness and dual-nonadjacency of an interior face from each cap face into full retention after deleting the union of their vertex supports. It does not select the two cap faces or establish those separation facts; it removes only the avoidable duplication once the source formation has supplied them.

The collar calculation has a finite, literal target. The cap-adjacent faces are not covered by that retained-face transport, but they no longer require an informal assertion that "the five stubs make a hole." The opened rotation has a finite first-return permutation on its exposed boundary darts. `GoertzelV24OpenRegionBoundaryOrbit` proves that one step of this return permutation is followed by an actual orbit of the opened face permutation from one fresh stub to the next. Consequently, if the five explicitly enumerated outward darts of a facial cap form one return-permutation cycle, then all five lie on one literal opened face orbit. This is a reduction of the collar task to a finite source-specific cyclic order calculation; it is not that calculation itself. The still-open fact is to derive that five-dart cycle from the closed cap face and the retained-side geometry. The separate one-edge facial-surgery theorem supplies useful local splicing plumbing, but does not by itself make the five splices compose.

The same distinction now reaches the whole clean corridor rather than only one selected face. The theorem `openCleanOrbitHexCorridorSkeleton_compl_union_twoCaps` transports a closed clean hex corridor through the literal opening that deletes two displayed cap-face vertex supports, provided the corridor and every one of its facial-dual neighbours avoid *each* cap. This one-ring premise is essential: retaining the centres alone would still allow an open neighbour to be a boundary-cut face fragment. The theorem therefore gives the future formation map exactly the two-cap corridor transport it needs, but deliberately does not select caps, prove the avoidance facts, or construct the opened annulus and its transverse crosscuts.

There is already a precise serial open-tangle adapter, but its quantifiers are important. Given an actual equivalence from the literal exposed boundary darts to a displayed left/right port sum, it transports Tait colourings in both directions between the serial presentation and the literal opened rotation. Thus a later construction of the two boundary components will not need a second colouring-semantics proof. The adapter takes that boundary equivalence as input: it does not divide an arbitrary opened region into the two annular interfaces, prove their order, or construct the source crosscuts. It is reusable downstream plumbing, not the missing source formation in a different name.

The strictly local open-carrier implication is now machine-checked as well. The predicate `CommonNeighborVertexIncidence` asks only that, after the opening, the outgoing rung and the canonical edge from the centre to each common neighbour still share one primal endpoint. The theorem

`commonNeighborEdgesExact_of_commonNeighborVertexIncidence`

uses cubicity at that retained endpoint, edge-simplicity and local two-sidedness of the displayed interior face, and parity of face-boundary incidence to force the second edge to be one of the two immediate flanks. A second theorem then derives `CommonNeighborsExact`. Thus the open classification no longer needs the entire closed minimality package: it needs one concrete incidence fact preserved by the source opening. This is a reduction of the formation gap, not its solution. The incidence predicate is not assumed by the source instance and the new module does not construct the Cell–3 opening that must establish it.

The generic opening calculation needed by that construction is now available in `GoertzelV24OpenRegionEndpointIncidence`. If two ambient edges meet at a vertex and both edge occurrences lie on fully retained faces, their computed literal-open edges meet at the corresponding retained vertex. The vertex-retention fact is derived from full face retention rather than supplied as an extra premise, and both open edges live in the same fixed opened rotation system. Consequently the remaining incidence transport is no longer an abstract claim that opening "preserves local geometry": it is the concrete source formation, the full-retention proof for the displayed Cell–3 faces, and their already named edge identifications.

The closed classification and the generic opening calculation have now also been joined in one constructive theorem. In `GoertzelV24OpenRegionMinimalDualTriangleIncidence`, the theorem

`exists_common_open_endpoint_of_retained_dual_triangle`

starts from a facial-dual triangle in the closed graph-backed vertex-minimal counterexample and assumes that its first ambient face is wholly retained by a literal opening. It does not accept two arbitrary open edge coordinates. Instead, it extracts the two canonical closed shared edges, constructs their actual positions on the first face cycle, computes the corresponding open edges, and returns a retained open vertex incident with both. Hence the generic cap-deletion question at this corner is settled: under full-face retention, the common primal corner survives as literal open incidence. The remaining burden is source-specific and deliberately visible: construct the Cell–3 opening from the closed map, prove the selected source face is fully retained, and identify its named rung/flank coordinates with these constructed positions. The theorem neither assumes nor constructs that formation, and it does not by itself prove `CommonNeighborsExact` or the separated crosscut pair.

The full-retention premise in that theorem is no longer a purely qualitative condition. The generic cubic-map lemma in `GoertzelV24CubicFaceVertexSeparation` proves that two distinct facial cycles meeting at any primal vertex must be adjacent in the full facial dual. Therefore two distinct dual-nonadjacent faces have disjoint primal vertex supports. Taking the literal opening to retain precisely the complement of the vertices visited by a chosen cap face gives the constructive corollary

`faceFullyRetained_compl_orbitFaceVertices_of_not_adj`.

Thus any corridor face proved boundary-clean—distinct and dual-nonadjacent from the pentagonal cap—is automatically wholly retained by cap deletion; no separate vertex-by-vertex retention assumption is needed. This settles the generic closed-map side of that premise. It does not select the source pentagon, prove that the displayed Cell–3 faces are nonadjacent to it, build the two-hole annulus, or realize the long separated crosscuts. Those are remaining source-specific formation statements. They are useful generic preparation for the source's bulk Cell–1 ladder theorem, but they must not be relabelled as a construction of the separate Cell–3 geometry.

The next bounded local collision has now been classified on the closed map as well. Geodesicity rules out any dual edge between corridor faces two steps apart, but by itself it does not say that their displayed middle face is their only common neighbour. If an additional common neighbour exists, the two length–two routes form a simple dual four-cycle. In `GoertzelV24MinimalDualFourCycleClassification`, cyclic five-edge connectivity converts this cycle into the already established adjacent-pair primal collar and supplies a chord joining opposite cycle positions. The endpoint chord is impossible by geodesicity, so Lean proves

`middle_adj_other_of_two_common_neighbors`.

Its corridor specialization says that every such extra common neighbour must also be adjacent to the displayed middle face. Thus the residual two-step collision is not an arbitrary global interaction: it is forced into the finite adjacent-pair collar geometry named by L1. This is a genuine classification theorem, not a conditional rail-assembly wrapper. It does not yet prove that the extra face is impossible, discharge `CommonNeighborsExact` in the cut-open Cell–3 carrier, or construct the source opening; those are the next finite local and formation steps.

The theorem is already consumed by the closed corridor API rather than left as an isolated candidate. The derived result `twoStepCommonNeighbor_eq_middle_or_outgoing_flanks` combines it with the two adjacent dual-triangle classifications: a common neighbour at index distance two is either the middle corridor face itself, or its shared edge from each adjacent centre belongs to that centre's named flank pair. Consequently all rail-collision distances now have a distinct proved status: remote collisions are impossible by geodesicity, consecutive collisions are classified by one flank pair, and distance-two collisions are trapped between the two adjacent flank pairs. The remaining closed local task is the finite compatibility of those two flank classifications, not an unbounded path existence theorem.

The corresponding generic opening step is now proved in `GoertzelV24OpenRegionMinimalDualFourCycleClassification`. Its theorem

`open_middle_adj_open_other_of_retained_two_common_neighbors`

starts with the closed graph-backed minimal-counterexample four-cycle and a literal vertex-region opening. If the middle and extra facial orbits are fully retained, the forced middle–extra dual edge remains an edge of the opened facial dual. The proof uses the exact retained-face adjacency equivalence, so it cannot manufacture an adjacency at a boundary stub or lose one by comparing only cardinalities. This is the four-cycle analogue of the retained dual-triangle corner theorem. It is still a generic transport theorem: the source-specific Cell–3 formation must identify its literal faces with these open images and establish their full retention.

One tempting strengthening is false and is excluded from the assembly design. A middle Cell may have a zero-length rail on one track (the `forwardTwo` or `forwardFour` shape), so an incoming flank can equal the same-track outgoing flank. Such a repetition is the legitimate connector removed by the tail convention in walk append, not a self- intersection. The remaining finite theorem must therefore show that every distance-two overlap is this same-track connector; it must not claim that all distance-two overlaps are impossible.

This carrier distinction also rules out a superficially convenient adapter. The rotation system in a well-formed `SourceTrail.AnnularEmbedding` is already the cut-open framed graph and is proved not globally cubic, whereas the minimal-counterexample theorem concerns the original closed cubic map. One may not assume both structures on that same carrier and call the resulting conditional theorem a formation result. This does not form a Cell–3 source transversal: its source instance already stores a `ClosedWebAnnularEmbedding` and the required annular geometry, and the weighted clean-block theorem now returns a source-realized dual geodesic from those fields. The missing bridge includes a genuine simple-transversal or layer-boundary construction at the selected radial positions; the local rail calculations are exploratory Cell–3 geometry, not the source's Cell–1 ladder construction. The generic cap-deletion construction remains relevant only if a later source theorem identifies that particular opening with the Cell–3 carrier; it is not silently inserted as the current L1 construction.

There is a further, easy-to-miss boundary on the current two-cap library. Its `PentagonCap` record is deliberately graph-theoretic: it names a five-cycle, its five outward spokes, and their incidence. It does *not* say that the five-cycle is the boundary of one facial orbit of the closed spherical rotation. That omission is mathematically material. Deleting an arbitrary five-cycle has no reason to create a named hole face, to give the two required cap collars, or to preserve the Euler presentation needed by the annular cellulation. The new `GoertzelV24FacialPentagonCap` records the necessary strengthened input: each cap carries a closed-rotation face dart whose face-edge support is exactly its five cycle edges, and a separated pair carries this datum on both sides. It proves the facial support has five edges, contains exactly the named cycle edges, excludes every outward spoke, and distinguishes the two cap faces. Its first oriented refinement records only the source coordinate order `cycleEdge i = {vertex i, vertex (i+1)}`; that alone does not say which direction the face permutation takes around the cycle. The stronger `FacialPentagonCapBoundaryWalk` therefore now records the literal five-dart face walk: its root is the designated facial dart, its successive darts carry the five named edges, and the rotation-system face permutation advances the sequence modulo five. This completes the *facial-cap datum*, not the surgery. The next theorem must construct the two opened collar orbits and prove that all five boundary-spoke darts on each side share the appropriate orbit. That is weaker, and more accurate, than saying the face contains only five edges: a degree-one-stub graph may have further darts around the same boundary face. Treating the graph-level opening as an annulus before this surface calculation would still be precisely the unproved picture argument this formalisation is avoiding. This is a constraint on that laboratory, not a claim that the Cell–3 source corridor is obtained by selecting two such caps: Addendum V supplies the laboratory tangle, whereas Addendum XXVII requires its own simple transversal or layer-boundary construction.

That non-vacuity boundary is now a kernel theorem rather than only a warning: `graphBackedVertexMinimalTaitCounterexample_elim_of_wellFormed` derives `False` from a well-formed source trail, its annular embedding, and a closed graph-backed minimal-counterexample package on that same rotation. The closed-to-open transport layer remains usable; what is ruled out is the invalid shortcut of applying it to the already-open source carrier without an actual formation map.

The first generic piece of this formation is now constructed in `GoertzelV24OpenRegionFaceTransport`. Let a closed ambient face orbit be wholly based at vertices retained by the literal open-region construction. Lean proves that following the ambient face permutation never enters a fresh boundary stub, that the old-dart embedding commutes with every face step, and that the complete ambient and open dart cycles are equivalent. All transported faces now live in one fixed literal-open rotation system, rather than in separately rooted copies. In that common map Lean proves that distinct retained ambient faces remain distinct, that equal ambient edge occurrences compute the same open edge even when they belong to different faces, and hence that a shared ambient boundary edge remains a genuine facial-dual adjacency after opening. Conversely, every open dual adjacency between two such transported faces is pulled back to a shared ambient edge occurrence. Consequently Lean now proves the graph-level equivalence

`openFaceOrbit_adj_iff_ambientFaceOrbit_adj`,

so the literal opening induces exactly the same adjacency relation on this family of retained face orbits: it neither loses an ambient dual edge nor creates a spurious one. Thus an untouched face is transported as an actual finite orbit without splitting or identifying its old boundary edges, not merely identified by a matching cardinality or assumed through a wrapper. This is generic carrier plumbing for any future cap-deletion formation. It neither supersedes the direct Cell–3 local handoff nor constructs the separated facial-dual crosscut pair from its source-realized corridor.

The next formation step is now explicit in `GoertzelV24OpenRegionHexCorridorTransport`. First, Lean proves that a fully retained two-sided face has exactly the same boundary-support cardinality after opening. This is not inferred from two matching numbers: the retained face-cycle equivalence is combined with an edge-level injectivity proof that reflects equality of opened edges back to the ambient face. It then constructs `openOrbitHexCorridorSkeleton`: any closed hex corridor all of whose selected faces are fully retained becomes an actual hex corridor in the one computed open rotation system, with the same length, injective face indexing, hexagonal boundaries, consecutive adjacencies, and separated non-adjacencies. Thus the generic transport of an L1 corridor is complete. It remains conditional on a cap/collar opening being the relevant source carrier; the current Cell–3 clean-block path already has its own direct annular realization. Neither route supplies the source's full ladder incidence or the long, end-capped crosscut pair, which remain the live geometric obligations.

There is also a cap-specific constructor, `openOrbitHexCorridorSkeleton_compl_cap`. In a cubic, cyclic, two-sided closed map, suppose every selected corridor face is distinct and dual-nonadjacent from a chosen cap face. The previously proved cubic face-separation theorem then shows that deleting precisely the cap-face vertices fully retains every corridor face, and the generic transport constructor produces the open corridor. This reduces the cap-opening part of L1 to a source-visible premise—the displayed dual non-adjacencies—rather than a vague claim that opening should preserve the picture. It still does not select the source cap or prove those non-adjacencies.

Retaining only the selected corridor faces is not sufficient for the clean L1 interface. A neighbouring open face could otherwise be a cut fragment whose facial cycle was closed through a fresh boundary stub. The module `GoertzelV24OpenRegionCleanHexCorridorTransport` closes this loophole. Its one-sided neighbour-pullback theorem starts from an actual edge shared by a retained face and an arbitrary open neighbour, recovers the opposite ambient dart, and uses the two-face incidence bound to prove that the open neighbour is exactly the image of that ambient face. Consequently `openCleanOrbitHexCorridorSkeleton` transports an entire clean corridor whenever its radius-one ambient neighbourhood is fully retained. The cap specialization packages the geometric premise as `CorridorOneRingAvoidsFace`: every centre and every adjacent face is distinct and dual-nonadjacent from the cap. Cubic face separation then constructs all retention proofs automatically. Thus boundary cleanliness is preserved as a theorem about actual face orbits, rather than inferred from a picture or from center-face retention alone.

The adjacent common-neighbour classification has now been upgraded from its edge form to its literal face form. The same closed module defines the face across a side-slot dart using only the ambient rotation system and proves

`commonNeighbor_eq_outgoing_sideFace`.

Thus a face adjacent to both consecutive corridor centres is not merely known to use one of two flank edges: two-sided facial incidence proves that it is exactly the opposite-dart face at one of those two flank positions. This generic definition is deliberately independent of the later closed-web namespace; the proof uses the closed minimal-counterexample hypotheses only where they supply the edge classification.

The new module `GoertzelV24OpenRegionMinimalCorridorCommonNeighbors` then joins that closed theorem to the boundary-clean opening. An arbitrary open common neighbour is pulled back, from each of the two corridor centres, to a fully retained ambient neighbour. Injectivity of retained face-orbit transport shows that the two pullbacks represent the same ambient face, after which the closed theorem classifies it as one of the two literal side faces. The result `exists_ambient_side_rep_of_open_commonNeighbor` therefore proves the face-level bounded interaction through the computed opening without assuming cubic minimality of the noncubic open carrier, invoking a general disjoint-path theorem, or matching carriers by cardinality. It still leaves the source specialization visible: the actual Cell–3 cap/corridor selection must establish radius-one retention and identify the generic transported side faces with its named local rail data.

The representative choice in that statement has now also been removed. Fully retained roots on one ambient facial orbit have equal images in the fixed opening, by

`openFaceOrbit_eq_of_dartOrbitFace_eq`.

Consequently `open_commonNeighbor_eq_sideFaceImage` states the consumer-facing result directly: every open common neighbour is one of the two explicit open facial orbits rooted at the opposite darts of the outgoing flanks. This is stronger than retaining an unspecified ambient representative and then identifying its quotient face. It remains a generic literal-opening theorem, however; it does not assert that the computed opening and the source-local closed-web annulus are definitionally the same carrier. The next source formation adapter must prove that identification or an isomorphism preserving these named roots.

The opened rotation is now also backed by an actual computed simple graph in `GoertzelV24OpenRegionGraphBacking`. Its primal adjacency is defined from the opened dart endpoints. Lean proves that mapping an opened dart to its ordered pair of endpoints is bijective: internal darts are recovered from the two retained endpoints, while a fresh boundary vertex remembers the exact exposed ambient half-dart. Transporting the opened vertex rotation across this equivalence produces a `SimpleGraphDartRotation.Data` package on the computed graph. The equivalence is proved to commute with edge reversal, vertex rotation, and hence the facial step. Thus later Cell–3 formation may move between the literal-open rotation carrier and an ordinary graph-backed carrier without silently changing face or edge semantics. Moreover, if the ambient rotation is genuinely cyclic at every vertex, the opened rotation and its graph-backed presentation inherit that property: retained vertices keep their complete cyclic dart fibre, while every fresh boundary vertex has a singleton fibre. The same module now constructs the transparent graph homomorphism from the ambient graph induced on the retained vertices into the opened graph. Every fresh boundary vertex is proved adjacent to the retained endpoint of the exposed ambient half-dart that created it. It follows that a connected retained induced region remains connected after opening: the new vertices are leaves, not unaccounted-for components. These results discharge the cyclic-rotation and connectedness fields of a future annular cellulation, once source-specific formation supplies the corresponding ambient cyclicity and retained-region connectedness proofs. This remains generic carrier plumbing: it neither selects the Cell–3 retained region nor constructs its two hole faces, planar cellulation, a Cell–3 source transversal, or separated crosscuts.

There is a second, deliberately separate formation result in the tree. The first carrier-changing part of the *single-edge deletion* model is now constructed, rather than postulated. This is the formation geometry used by the missing-edge framed trail of Addendum XXVIII (Cell–4); it is not the annular Cell–3 opening of Addendum XXVII. The generic permutation lemma

`eraseTwoPointsSubtype`

removes the two oriented darts of the distinguished edge from the ambient vertex rotations: each removed dart becomes fixed in the auxiliary permutation, and its old predecessor is spliced directly to its old successor. Restricting to the complement and transporting along the exact dart equivalence gives

$$\texttt{deletedEdgeData} : \texttt{Data G} \longrightarrow \texttt{Data (DeletedEdgeGraph G u v)}.$$

Every surviving dart remains based at the same vertex. More importantly for face geometry, Lean proves

`deletedEdgeData_phi_ambient_eq_of_ne`.

If an ambient face step does not land on either orientation of the removed edge, the face step computed in the deleted graph maps back to exactly the old ambient face step. Thus deletion changes face tracing only where the missing edge was actually encountered; remote face steps are not reconstructed by an unproved picture. This is still only a bare rotation-system construction and a local transport theorem. It does not transport planar-geometry fields, choose the source's missing edge, or finish Cell–4 formation. More importantly for the present closure frontier, it does not construct Cell–3's cap-deleted annulus, container cycles, two holes, or long separated crosscut pair. Specializing it to Cell–3 merely because both constructions produce open graphs would confuse two different source objects.

The Cell–4 deletion result has now been lifted to complete untouched faces in `GoertzelV24DeletedEdgeFaceTransport`. The predicate `FaceAvoidsDeletedEdge` says that every dart in one ambient face orbit avoids both orientations of the missing edge. Under exactly that condition, `deletedFaceCycleEquiv` gives a bijection between the whole ambient dart cycle and the whole computed face cycle of `deletedEdgeData`. Surjectivity is derived from the commuting face step and finite cycle cardinality; it is not assumed. This proves exact orbit and face-length preservation for untouched faces. The same module now proves that two distinct untouched ambient faces remain distinct, transports equality of their shared primal edge in both directions, and packages the result as

`deletedFaceOrbit_adj_iff_ambientFaceOrbit_adj`.

Thus single-edge deletion induces exactly the old facial-dual adjacency relation on the family of untouched face orbits: it neither loses a shared edge nor invents one between their images. This is enough to transport a closed-map dual triangle through the Cell–4 missing-edge opening once its three faces are proved untouched. It is not enough to transport the triangle into Cell–3, whose annular carrier comes from a different region/cap opening. The live Cell–3 bridge must instead specialize the generic open-region transport to the actual closed-map annulus and prove that every open common neighbour under consideration is the image of an eligible ambient face.

Selected local rungs. The direct Cell–3 local handoff no longer assumes that every pair of interior faces shares at most one primal edge. That global uniqueness premise belongs to a closed minimal-counterexample carrier and is incompatible with silently reusing it for the source's open annular tangle. The new module `GoertzelV24ClosedWebSelectedRungPlacement` instead selects one actual shared edge for every consecutive pair of faces in a boundary-clean source corridor. At each internal hexagon it proves, from local face incidence and the displayed Cell–3 separation, that the selected incoming and outgoing rungs occupy distinct positions and cannot be adjacent. Consequently their cyclic type is one of the two nonadjacent cases, `oneEdgeBetween` or `opposite`; deleting the two selected rung positions leaves four side slots and exactly two directed cyclic side steps. Each selected slot now determines its literal opposite-dart face, which Lean proves is facial-dual adjacent to the corridor centre and an annular interior face. This is a genuine finite local Cell–3 classification, and it removes a false global hypothesis from the local construction. It does not establish that distinct slots give distinct side faces, construct simple rails, or produce the required long separated crosscut pair.

At the generic crosscut boundary, Lean now records the corresponding representation repair in `GoertzelV24SelectedDualPathTransversal`. A selected transversal is a simple facial-dual path together with one certified, actually crossed primal edge at each step. Its crossing support is proved injective from path simplicity and the local at-most-two-face incidence law alone; it does not require the stronger claim that every ambient adjacent face-pair shares a unique edge. The Cell–3 selected-rung record now carries the same certificate directly: interior-support membership and membership on both consecutive face boundaries. Its selected edge map is now also injective in the corridor-step index, proved from the injective face order and the same local two-face incidence bound—not from global unique intersections. This makes the literal selected-rung geometry admissible to a future transversal construction. It is deliberately not that construction: the module neither produces the long source path nor a separated pair of paths.

The opposite dart of a selected outgoing rung now lands on the next literal corridor face, by `selectedOutgoingAlphaFace_eq_nextCenter`. This uses the selected source edge and its actual two incident faces; it does not reintroduce a global face-intersection-uniqueness premise. A later rail-extension argument therefore has its correct outgoing endpoint, while any collision of side faces remains an explicit finite case rather than an implicit identification.

One further local branch is now checked exactly. If any surviving side slot of that Cell–3 hexagon has the next corridor centre as its opposite-dart face, `selectedSideFace_eq_nextCenter_implies_two_distinctSharedInteriorEdges` exhibits the selected outgoing rung and that side edge as two distinct shared interior primal edges. This does not exclude the branch or construct a crosscut; it records the branch as the finite exceptional geometry which the source's L1 classification must decide.

The selected interface is also now oriented rather than merely matched by its underlying edge: `selectedNextIncomingDart_eq_alpha_outgoingDart` proves that a successor Cell–3 placement reads the common selected rung at the opposite ambient dart. This is the local seam datum required by a future rail append. It neither asserts that appended rail segments avoid one another nor turns the local Cell–3 windows into the long transversal pair.

For each of the two side slots flanking that outgoing rung, Lean now gives the same explicit alternative at the next centre: the opposite face is either already the next centre or is facial-dual adjacent to it through the literal third edge at the cubic corner (`selectedPlacementSideFace_eq_or_adjacent_next_of_beforeOutgoing` and `..._afterOutgoing`). The equality alternative is a zero-step, not a disguised edge. It remains part of the finite local case analysis needed before segments can be appended into simple rails.

The exclusion of that global premise is now itself kernel-checked rather than merely recorded in prose. Every dart over a named inner boundary edge belongs to the inner-hole orbit, as does its α-opposite; hence `ClosedWebAnnularEmbedding.not_orbitFacesTwoSided` proves that the literal Cell–3 annulus cannot satisfy global face two-sidedness. The correct replacement is the already-developed local two-sidedness certificate on the interior faces actually traversed by a layer boundary.

The first remaining collision has now been reduced without pretending it is impossible. If two distinct selected side slots lead to the same opposite face, Lean exhibits two distinct interior primal edges shared by that face and the central hexagon. Thus the open-carrier question is a finite two-shared-edge local case. It is not a license to import the closed minimal-counterexample theorem that would rule the collision out globally.

What the closed theorem does *not* discharge. A source audit makes that last caution precise. The current Fable manuscript first derives only the Seed-minimal frontier conditions "girth at least five" and "no 2-cut," then names L1 as the finite local classification of the resulting all-hexagonal patch. By contrast, `GoertzelV24MinimalDualTriangleClassification` obtains its common-corner conclusion from a *closed*, spherical, vertex-minimal Tait counterexample; the formal proof uses the consequent cyclic-five-edge-connectivity of that closed map. The literal `ClosedWebAtGoodWord.Instance` carries neither that closed-map carrier nor cyclic-five-edge-connectivity: its annular geometry supplies simple interior face boundaries, the lower bound five, and connected internal facial dual only. Thus no type coercion or generic opening transport can turn the closed theorem into `CommonNeighborsExact` for the source annulus. Doing so would silently strengthen the source hypotheses and still fail to construct the required formation.

There is a sharper provenance boundary here. The Fable source derives "girth at least five" and "no 2-cut" for an irreducible minimal-counterexample frontier before its annular opening. The current `ClosedWebAtGoodWord.Instance`, by design, is instead a colored open annulus: its fields give boundary well-formedness, connectivity, the annular embedding, internal face-length/dual-connectivity geometry, and good-word closure. It does not contain the closed minimal carrier or a no-two-cut consequence. Thus the source formation must transport the relevant irreducible local facts into this annulus. This observation does not claim that a no-two-cut fact alone resolves the two-shared-edge case; it prevents that unproved provenance from being used implicitly.

The unformalized two-cut transport. Addendum XVIII of the current Fable manuscript asserts more than the familiar fact that two capped halves can be glued to a coloring. For a pendant two-edge-cut piece it states a positive multiplicative identity *at every routing profile*; Addendum XXV then uses that identity to make a zero seed count descend to a smaller frontier. Lean has already proved the coloring-level part exactly: `twoEdgeCut_bicolored_portals_reachable` supplies the forced matched bichromatic route through either side of an exact two-edge cut, and `TwoEdgeCutPairData.taitColorable_iff_caps` proves the cap/gluing equivalence for Tait colorings. It has not proved the stronger source claim. In particular, no theorem currently relates a two-edge cap to `AnnularBoundaryData`, preserves its five named inner boundary edges, or transports the selected-component table that defines Menu B. Thus the exact seed predicate `annularFrontierMenuBSeedCount` is available, but its zero count does not yet descend through a two-edge cut. The next construction is therefore a boundary-preserving cap transport—with the cut-off component disjoint from the named annular interfaces—followed by a literal profile/count factorization. This is an open formalization obligation, not a refutation of the manuscript's corollary and not a license to substitute the closed vertex-minimal no-two-cut theorem.

The common count direction is now isolated independently of any particular gadget. `GoertzelV24AnnularFrontierSeedDescent` defines a *seed reflection*: every Menu–B coloring in a reduced fixed-word fiber must lift to a Menu–B coloring in the original fixed-word fiber. Its kernel-checked descent theorem then carries a zero source seed count to the reduced frontier, even when the two graphs have different vertex carriers. This is deliberately weaker than the manuscript's positive multiplicative/additive count identities, but it is exactly the implication minimality consumes. It neither supplies the two-cut cap nor asserts that the required lift preserves the five named boundary positions; those are now the concrete remaining obligations for the two-cut specialization.

The honest next object is consequently smaller and sharper than a general Jordan or Menger theorem: a source-specific finite local classification that derives the required common-vertex incidence on the actual Cell–3 annular carrier, or an actual formation from the same Seed-minimal frontier to that carrier which preserves the needed local hypotheses. The existing Lean theorem proves that this incidence implies the literal rail classification; it does not supply the incidence. This records a real boundary in the current construction, not a new Fable flag and not a conditional completion.

The next finite case is now explicit as well. For two consecutive surviving side slots around a selected Cell–3 hexagon, `selectedPlacementSideFaces_eq_or_adjacent_of_forwardStep` proves exactly one of two alternatives: the opposite-dart faces coincide, or they are joined by the literal exterior facial-dual edge at their common locally cubic corner. The first alternative is deliberately retained and is fed to the two-shared-edge collision theorem above; the second is the actual one-step rail edge. This is the intended finite local classification, carried out on the source annulus, rather than an unproved assertion that every local rail step is automatically simple.

The dichotomy now has an intentionally modest walk-level form as well:

`exists_selectedPlacementSideWalk_of_forwardStep`.

For a forward side step it constructs a literal facial-dual simple walk of length at most one. The adjacent-face branch has length one; the coincident face branch has length zero. Recording the latter rather than coercing it into a one-edge rail is important: it preserves the finite collision case for the source's L1 classification. This is one local segment, not a simple crosscut, a disjoint rail family, or a construction of the long end-capped pair required by the splice.

An exhaustive selected local-layer alternative. The next source-local step is now a theorem rather than an inherited closed-map assumption. For the two selected side slots flanking an outgoing rung, `exists_selectedLocalLayerPair_or_collision` returns exactly one of two results. In the ordinary branch it constructs the existing `LocalLayerPair` directly from literal Cell–3 faces. In either exceptional branch it retains a `SelectedLocalLayerCollision`: a flank is the successor centre, or two distinct selected side slots denote the same face. The collision is informative rather than a dead end: Lean extracts two distinct shared interior primal edges between the current centre and a named neighbour. Thus the finite local case analysis has a complete, source-side entry point without assuming global pairwise uniqueness of face intersections. It neither eliminates the collision cases nor concatenates the successful local pairs into long, end-capped, hole-to-hole crosscuts; those remain the formation geometry required by the splice.

The finite connector shape itself no longer depends on the older closed-map uniqueness premise. The theorem `selectedSourceLocalRailShape_of_cell3` classifies the four flank slots of every literal selected Cell–3 placement into exactly the three source cases

$$0+2, \qquad 1+1, \qquad 2+0.$$

This is a six-cycle coordinate calculation plus the proved Cell–3 nonadjacency. The zero is meaningful: one rail may be a stationary identity connector while the other traverses two local pieces.

Treating that case as an automatic failure would exclude valid source geometry. The theorem is a finite shape certificate only; appendwise simplicity, mutual rail separation, and the two literal end caps remain to be constructed.

The same module realizes every classified shape as two literal simple facial-dual paths whose combined length is at most two. Each one-step piece comes from the selected opposite-dart adjacency theorem; if the bounded local geometry repeats a face, standard walk loop erasure removes the repetition without changing either endpoint. Thus the path constructor covers all three shapes without smuggling closed-map uniqueness into the annulus. It also retains the local provenance erased by a bare path type: every face in either support is proved adjacent to the literal Cell–3 centre that generated the connector. Combining this invariant with the boundary-clean geodesic condition gives the four remote noncollision theorems in `GoertzelV24ClosedWebSelectedLocalRailSeparation`: for centres more than two positions apart, either selected rail at the left has support disjoint from either selected rail at the right. The two rails of one local connector are still not asserted mutually disjoint. The bounded neighbouring cases, append induction, and the two end caps remain the finite L1 formation obligations before these paths become the long source crosscut pair.

The same-cell ambiguity is now fail-closed rather than assumed away. `GoertzelV24ClosedWebSelectedLocalRailCollision` checks every cross-track coordinate pairing in the three finite connector shapes. It returns either two genuinely support-disjoint selected rail paths, or an existing `SelectedLocalLayerCollision.repeatsSide` certificate naming the two distinct literal side slots whose opposite-dart faces coincide. The latter is the concrete two-shared-edge exceptional geometry already exposed by the source-local formation layer. This completes the finite same-cell alternative; it does not yet refute that exceptional branch or settle either neighbouring-cell gap.

The first neighbouring seam is now literal as well. The selected successor module packages the four flank coordinates around two consecutive Cell–3 faces together with the proved fact that the common rung is represented by opposite ambient darts. Cubicity at the two endpoints then gives two exact face identities: the exterior face immediately before the outgoing rung is the exterior face immediately after the successor's incoming rung, and the other two flanks agree analogously. This is a direct three-dart rotation calculation, not an appeal to global uniqueness of shared face edges. With those identities in hand, the next cell constructs its two mutually disjoint local rails or returns its exact finite collision certificate. What remains is the append invariant across arbitrarily many successful cells and the two hole end caps; a one-seam successor package is not yet a long crosscut.

The algebraic append invariant is now explicit as well. A `SelectedSourceLocalRailAssembly` stores two accumulated simple rails and their mutual support disjointness; a single separated Cell–3 connector is its base case. The theorem `appendSuccessor` extends the assembly by one literal successor and preserves both path simplicity and mutual rail separation. It exposes exactly four geometric premises: each old rail must avoid the tail of its own continuation and the tail of the other continuation. The new local pair's internal separation is already certified. This cleanly separates composition from geometry: remote segments discharge the relevant premises by the proved gap-separation theorems, but the bounded neighbouring interactions and end caps still require construction. Thus the new theorem is an iteration-safe carrier, not yet an existence proof for an arbitrary long source crosscut.

The first bounded neighbouring reduction is now available on this literal selected carrier. The proposition `CommonNeighborsExact` says that every full-dual face adjacent to both consecutive Cell–3 centres is one of the two named flank faces of their shared rung. From exactly this proposition Lean derives all four old/new tail-disjointness conditions above; `appendLocalSuccessor` therefore constructs the two-cell assembly without accepting four unrelated collision hypotheses from its caller. This is a reduction of the adjacent geometry to one quantified local statement, not a proof of that statement. Its source-opening transport, the distance-two interaction needed by an arbitrary prefix induction, and both annular end caps remain open.

The negation of that adjacent classification is now geometric data rather than an opaque failed premise. The fail-closed theorem `appendLocalSuccessor_or_adjacentDualTriangle` returns either the actual two-cell selected rail assembly or a literal third common facial-dual neighbour. In the latter branch Lean constructs the resulting simple closed dual walk of length three through the two consecutive Cell–3 centres. This is the exact local separator candidate that the next reduction or rerouting must consume; it is not yet an exclusion of that branch. In particular, the Addendum–V open-frontier induction has so far supplied girth at least five and absence of two-edge cuts, whereas the stronger closed-minimal three-edge-cut theorem has not been transported to the opened stub carrier. The module therefore records the dual triangle without silently borrowing closed minimality and does not report L1 closed.

The sufficient classification has now been separated from the geometry the append actually consumes. The four premises of `appendSuccessor` mention only intersections between an accumulated rail support and the tail of one of the two literal continuations. `GoertzelV24ClosedWebSelectedLocalRailAppendCollision` tests those four intersections directly. It returns either the constructed two-cell assembly or a face belonging to one of the four actual old/new support collisions; in the latter case Lean proves that this very face is the third vertex of the adjacent dual triangle. Thus failure of `CommonNeighborsExact` at an unused third common face no longer blocks a valid append. This is a strict refinement of the fail-closed fork, not its elimination: an actual support collision still produces the same separator branch, whose cyclic case remains to be discharged on the source-faithful frontier.

The append itself admits one further normalization. A revisit of a rail by its own continuation need not be excluded: loop erasure of the concatenated walk preserves its endpoints and produces a simple path. In `GoertzelV24ClosedWebSelectedLocalRailAppendBypass`, both concatenated rails are therefore normalized by `Walk.bypass`. The mutual separation proof then needs only the two crossed conditions—old first rail against new second tail, and old second rail against new first tail. The resulting exact alternative returns either two simple support-disjoint rails or a face in one of those two genuine cross-track intersections. Same-track revisits are now constructions rather than obstructions. Cross-track collisions remain positive dual-triangle data and are not claimed impossible.

The separator candidate has now been promoted to an actual graph separator without restoring global uniqueness of shared face edges. The generic module `GoertzelV24SelectedDualCycleSeparator` lets a simple facial-dual cycle carry the primal edge selected at each of its steps. Ordinary at-most-two face incidence makes these selected crossings injective, and the exact face-boundary/cycle-space calculation proves that deleting their underlying graph edges disconnects the framed carrier. Specializing this result in `GoertzelV24ClosedWebSelectedLocalRailAppendSeparator`, Lean proves that the obstruction triangle crosses exactly three distinct primal edges and that deleting them disconnects the literal closed-web annulus.

This is not yet a cyclic three-edge cut. The missing source-local statement must classify the deletion components relative to both degree-one boundary families. In particular, the previously proved remote separation of two cap boundary-face families does not imply that the triangle-enclosed component avoids every inner and outer stub. Once that avoidance is established, the cubic degree sum has a sharp dichotomy: a tree side is a singleton star and recovers the common-neighbour data needed by the append, while a cyclic side gives the three-edge cut to be pulled back to closed minimality. Until the stub classification is proved, neither conclusion is used.

The outer half of that component classification is now literal. The selected crossing support has been identified exactly with the finite edge-deletion support used by the component API, and all three faces of the obstruction triangle are proved to be annular-interior faces. Hence none of its three selected crossings lies on either named hole boundary. In `GoertzelV24ClosedWebSelectedLocalRailAppendComponent`, Lean chooses an actual component of the deleted graph opposite a distinguished outer-hole dart, follows the complete rooted outer-face walk, and proves that every one of the five named outer stubs lies outside that component. This is stronger than boundary-edge disjointness, but it is still only the outer half: no comparison walk has yet put the inner hole on the same side as the outer hole. Therefore inner-stub exclusion, boundary saturation, cyclicity, and the cubic degree count remain open and are not inferred here.

The inner half is now discharged by the source's own radial-escape count, rather than by a new Jordan assumption. The generic finite-cut form counts twenty oriented inner Tait incidences against four possible oriented color-pair slots per deleted edge. A three-edge triangle supplies only twelve slots, so one literal inner-to-outer radial path avoids the selected cut. Starting from the already excluded outer endpoint, this path puts one inner stub outside the chosen component; the complete rooted inner-face walk then puts all five inner stubs outside it. Thus the selected component avoids all ten named boundary stubs. Boundary saturation and the subsequent singleton-star-versus-cyclic degree classification are the next graph-level obligations; they are not bundled into this counting theorem.

The cubic degree classification is now proved without imposing global cubicity on the opened carrier. A new local form of the side-dart count in `GoertzelV24CubicSmallBoundaryCycle` assumes degree three only on the chosen side. Applied to the actual selected deletion component—whose ten degree-one boundary stubs have just been excluded—it shows that the induced side either contains a cycle or has exactly one vertex. This is the sharp three-edge degree sum: an acyclic connected cubic side with at most three outgoing edges can only be a singleton star. It is not yet an unconditional append or a cyclic three-edge-cut contradiction. The cyclic branch still needs a cycle on the complementary side before closed minimality can be used. The rooted outer-face walk does not supply that cycle: its construction deliberately proves a closed facial walk rather than `Walk.IsCycle`, as boundary faces may revisit underlying edges. A direct degree count measures the opened-carrier cost precisely. If the complement is acyclic, with its ten degree-one boundary stubs internal to that side and at most three crossing darts, then

$$3n - 2 \cdot 10 = 2(n-1) + c, \qquad c \leq 3,$$

so $n \leq 21$. Thus the cyclic branch requires either a genuine size input or the source formation transport from the closed cubic carrier; no outer cycle is inferred from the displayed hole walk.

The singleton branch has now closed independently. The generic selected-cycle constructor can now pin one specified primal crossing at one specified dual step while retaining choice at the other steps. Its specialization to the obstruction triangle pins the first crossing definitionally to the literal selected outgoing corridor rung. This removes the provenance ambiguity without a global unique-shared-edge premise. For a one-vertex deletion component, the three outgoing darts inject into its computed crossing boundary; that boundary is already contained in the three selected crossings. Equality of cardinalities therefore proves exact boundary saturation and yields one ambient vertex incident to every selected crossing. The existing local cubic incidence proof has been factored through a pointwise interface. Applied to the selected crossing from the third face back to the left centre, it forces that third face to be one of the two named flanks, contradicting the obstruction witness. Consequently the chosen outer-remote component of every obstruction triangle contains a cycle. The complementary-side cycle/closed-carrier transport remains open, so L1 is not reported closed.

The complete adjacent-append alternative is now packaged at exactly this strength in `GoertzelV24ClosedWebSelectedLocalRailAppendCyclicObstruction`. After loop erasure, either both selected rails extend, or an actual cross-track collision returns its facial-dual triangle together with a deletion component away from the outer root which contains a primal cycle. All ten ordered boundary stubs lie outside that component, and the selected crossing support has cardinality three. This is stronger than returning an unused common neighbour, but it is not a discharge: Addendum XVII contracts a literal primal triangle, while the component returned here may be the inside of a larger cyclic three-cut.

The collision branch now also carries its first constructive repair. In `GoertzelV24ClosedWebSelectedLocalRailAppendReroute`, Lean cuts the old rail at the exhibited cross-track collision, follows the opposite local continuation from that face, and loop-erases the result. This gives a simple route to the opposite outgoing endpoint; its support is proved to remain in the two displayed source pieces, and on the new side it lies strictly in the continuation tail rather than reusing the unmatched seam start. Boundary-clean neighbour containment also proves that every face of either reroute remains in the annular interior. It is one crossed route, not yet a new separated pair: simultaneous uncrossing, the endpoint permutation, mutual separation, and the two literal end caps remain open.

The repaired path also reaches the source-facing crossing representation. `GoertzelV24ClosedWebSelectedLocalRailAppendRerouteTransversal` chooses an actual shared primal edge at every dual step from the step's local adjacency witness, without restoring global shared-edge uniqueness. Its endpoints are still corridor side faces rather than the two hole faces, so it is a selected local transversal fragment and not the final crosscut. A single consumer theorem now returns the two possible crossed endpoint orientations directly from the actual collision witness.

There is now also a precise endpoint-swap theorem. A single cross collision does not connect the other two unmatched seam faces; Lean rejected that tempting construction at the endpoint types. If both cross-track tests exhibit collisions, however, the two reroutes switch together. Under the two remaining same-track separation facts, `GoertzelV24ClosedWebSelectedLocalRailAppendSwap` constructs two simple mutually support-disjoint rails with their outgoing endpoints exchanged. The double-cross pattern is therefore repaired; the lone-cross pattern remains.

The four local support tests have now been assembled into one exhaustive classifier. The constructors of `GoertzelV24ClosedWebSelectedLocalRailAppendClassification` return, respectively, a straight assembly, a swapped assembly, an exactly one-cross collision, or two cross collisions together with a same-track collision. In the last two cases the result retains the actual common faces rather than only the negations of disjointness. Thus the finite Cell–3 problem no longer has an implicit Boolean branch: every append either constructs the next pair or returns the precise local pattern still to be consumed. This is a classification theorem, not yet a proof that the two residual patterns are absent, an arbitrary-length rail induction, or the end-capped crosscut pair.

One of those two residual patterns is now impossible for a purely local reason. If both cross-track tests collide, all four displayed local rails are nonstationary. Each Cell–3 pair has total rail length at most two, so all four rails have length one. Any additional same-track collision would then identify one incoming face first with one outgoing face and, through the opposite cross collision, with the other outgoing face. This contradicts the right local pair's support disjointness. The theorem `not_doubleCrossSameTrack` formalizes this count and endpoint argument, and the refined classifier now has only three kinds of result: straight assembly, swapped assembly, or exactly one cross-track collision. The lone-cross case remains; neither arbitrary-length iteration nor either annular end cap follows from this local theorem.

The lone-cross case has now been reduced one step further by a literal centre bridge. At the exhibited collision Lean follows the crossed reroute to the opposite outgoing flank. It connects the two unmatched seam flanks by the two-edge walk through the displayed Cell–3 centre, appends the remaining continuation, and loop-erases that walk. Both orientations are implemented in `GoertzelV24ClosedWebSelectedLocalRailAppendCenterBridge`. If the crossed reroute avoids the unmatched old rail, the unmatched continuation tail, and the centre, this constructs a simple mutually support-disjoint pair with swapped outgoing endpoints. The endpoint exclusions at the two seam flanks are proved from path simplicity and the already established same-cell rail separation; they are not extra hypotheses.

The accompanying exhaustive classifier does not assume the three tests. After applying the centre bridge, every adjacent append either returns a straight or swapped assembly, or exhibits exactly one of four smaller residues: a same-track old/new collision on the first rail, the analogous collision on the second rail, or the first/second next-cell continuation literally revisiting the previous Cell–3 centre. Thus a cross-track collision itself is no longer a terminal branch. The four displayed residues still require the finite length-and-coordinate classification; this result is not an arbitrary-length induction, an end-cap construction, or closure of L1.

Those four residues now carry exact length equations. The provenance-preserving classifier in `GoertzelV24ClosedWebSelectedLocalRailAppendResidueLengths` keeps the original cross-collision witness together with the later same-track or centre witness. Two distinct collision faces on one path, both different from its seam endpoint, force that path to have length two. The stored bound that the two rails in one Cell–3 connector have total length at most two then forces the companion rail to be stationary; when both rails have positive length, both have length one. Consequently every surviving branch is a literal (2+0), (1+1), or (0+2) pair on each side of the seam. This is the finite coordinate input for the next rerouting audit, not yet a proof that all six oriented residue patterns construct an append.

The endpoint coordinates are now classified as well. The elementary core is that a length-two walk has only one interior support position: two distinct support faces which both avoid its finish must include its start, and dually two distinct faces in its support tail must include its finish.

`GoertzelV24ClosedWebSelectedLocalRailAppendResidueCoordinates` applies this fact to all six oriented residues. In each same-track case, either the original cross-collision face or the same-track collision face is the forced start or finish of the length-two rail. In each centre case, either the cross-collision face or the previous Cell–3 centre is the forced finish. These are kernel-checked binary coordinate alternatives with no new separation, connectivity, or embedding hypotheses. They are the final finite choices for the local reroute; the file does not yet construct the six reroutes, iterate them, or attach either annular end cap.

The first symmetric pair of those six reroutes is now constructed in `GoertzelV24ClosedWebSelectedLocalRailAppendResidueReroute`. In the old-(2+0), new-(1+1) same-first residue, the endpoint alternative has a literal interpretation. If the same-track collision is the incoming first endpoint, the first output rail is stationary and the second continuation is retained. If the cross-track collision is that endpoint, the crossed output rail is stationary and the other rail runs through the previous Cell–3 centre before taking the first continuation. Loop erasure supplies the path certificate in the latter case, and the local rail separation plus the successor separation prove the two complete supports disjoint. Thus this residue returns either a straight or a swapped assembly without any global topological hypothesis. The same module proves the rail-exchanged old-(0+2), new-(1+1) same-second case by the mirrored construction. Thus two of the six finite residues are discharged. The seam-reversed pair is now constructed too: with old lengths (1+1) and successor lengths (0+2), the dual-triangle non-endpoint fact forces the relevant old collision to be its one-edge rail's start. The same stationary-path and centre-bridge construction then applies at the outgoing endpoint. Four of the six finite residues are therefore discharged without a cyclic-cut or embedding premise; only the two cases in which a successor continuation revisits the previous centre remain open.

Those last two centre-revisit cases are now constructive as well, in `GoertzelV24ClosedWebSelectedLocalRailAppendResidueCenterReroute`. For a length-two successor rail, the cross face and the previous centre are the two tail positions. If the cross face is the outgoing endpoint, Lean truncates the old crossed rail at that face and sends the other rail through the complete two-edge centre bridge. If the centre is the outgoing endpoint, the offending middle face is omitted and the unmatched flank takes its literal one-edge adjacency directly to the centre. The rail-exchanged case is proved separately. Thus all six finite residual shapes now produce a straight or swapped separated assembly. They still await one exhaustive wrapper and a permutation-aware arbitrary-length iteration; neither end cap nor L1 is closed by the local result alone.

The six constructors are now assembled into one exhaustive local theorem. `GoertzelV24ClosedWebSelectedLocalRailAppendComplete` defines `appendLocalSuccessorComplete`: for two consecutive literal selected Cell–3 rail pairs, Lean always returns two simple support-disjoint rails. The output records whether the two outgoing labels remain straight or are exchanged. There is no residual collision, triangle, connectivity, or embedding branch. This closes the adjacent two-cell append obstruction. A long corridor still needs a finite-state fold followed by the two literal annular end caps; those are the remaining formation obligations. The state cannot consist only of the endpoint permutation and the two finished walks: some finite repairs replace part of the preceding Cell–3 window, so the next transition must retain the shape and provenance of that mutable window.

The remote half of that stronger invariant is now proved in `GoertzelV24ClosedWebSelectedLocalRailWindowSeparation`. A repaired two-cell rail is allowed to use either Cell–3 centre itself as well as faces adjacent to either centre. If two such closed-neighbourhood windows are separated by the established corridor-index gap, then any two supports they carry are disjoint. The proof checks the possible collisions directly: distinct centres cannot be equal, separated centres cannot be adjacent, and source boundary-cleanliness forbids a face adjacent to both remote centres. This proves that a provenance-tracked finite window has exactly the remote separation needed for induction. The interface is connected to every local constructor. For the two principal cases, both the ordinary loop-erased straight append and the double-cross swapped append are proved to stay inside the five-piece source carrier consisting of the old centre and the four old/new local rails. The two immediate centre-bridge exits are certified at their constructors too. Each of the six finite residue repairs now proves the same positive support-containment statement beside its private bridge construction. A uniform theorem then lifts that membership to the closed neighbourhood of the two selected centres. Hence every finite straight-or-swapped repair carries the remote-separation invariant without erasing its endpoint permutation. The public theorem `appendLocalSuccessorComplete_hasWindowProvenance` packages this certificate across all ten constructive exits of the exhaustive classifier: the two principal cases, the two immediate centre bridges, and the six residue repairs. What remains is the transition table which advances a mutable terminal two-cell window; this is not yet an arbitrary-length rail or either annular end cap.

There is an important coordinate distinction here. The previously proved pair of equal-profile, support-disjoint local layer loops belongs to the *depth* branch of the two-directional pumping argument. Each is a bounded layer boundary built from two local crosscuts. The selected rails under construction here belong to the *length* branch: after end-capping they must be genuine hole-to-hole transversals. Addendum XXVII's dichotomy may return either branch, so the depth-layer theorem cannot be used as a constructor for the missing length transversal. The formal API now also retains the straight-or-swapped result type for provenance certification, rather than forgetting the endpoint permutation while preparing the finite-state transition.

That local invariant now has a typed carrier rather than remaining a theorem about a bare result. The structure `CertifiedSelectedLocalRailTerminalWindow` stores the complete straight-or-swapped adjacent append together with its two-centre provenance, so it preserves the endpoint permutation instead of quotienting the two outcomes together. Lean proves that its two projected rail supports are mutually disjoint and that either support is disjoint from every sufficiently remote future certified window, using the already established corridor-index gap. This is the canonical seed and the remote half of a length-direction induction invariant. It is deliberately not called a transition system: a neighbouring repair may replace part of the current final two-cell window, so the next theorem must retain enough finite branch data to update that mutable suffix before freezing the remote prefix. Arbitrary length, both annular end caps, and L1 therefore remain open.

The algebraic splice needed after that finite transition is now isolated in `GoertzelV24ClosedWebSelectedLocalRailAssemblyAppend`. Given two ordered simple separated rail pairs with matching middle endpoints, `appendAssembly` returns their pairwise append once the two same-track and two cross-track prefix/suffix support conditions are proved. Its terminal-window wrapper handles both certified endpoint orders and returns a straight-or-swapped global assembly without forgetting which occurred. The four conditions remain visible by design: this theorem attaches a frozen prefix after the local geometry has certified compatibility; it does not pretend that compatibility follows from the types alone. Thus the remaining formation step is now specifically the finite mutable-suffix transition which must discharge those four premises. The companion operation `appendAssemblyCrossed` handles the exchanged middle order: it continues the first prefix rail along the second suffix rail and conversely, and records the resulting exchanged final order in the type. Thus a swapped local window is composition data, not a failed state or a reason to choose a noncanonical next connector. The helper `rebaseAssemblyStart` changes only the two named starting faces; the facial walks and their support lists remain unchanged.

The coordinate hand-off between consecutive seams is now proved separately. Although successor frames are constructed independently, `GoertzelV24ClosedWebSelectedLocalRailSuccessorCompatibility` shows that they select literally equal before/after outgoing slots on their shared Cell–3 placement. This is a six-cycle coordinate uniqueness calculation, not a global uniqueness claim about faces. Therefore the separated right-cell rail pair stored by one successor can be reused, with dependent types aligned, as the left-cell rail pair for the next successor. This opens a smaller induction design: construct disjoint two-cell windows and join them across the canonical intervening seam, instead of retaining an ever-mutating overlapping window. The global support-separation proof for that tiling and the collision alternatives in successor existence remain to be discharged. The corresponding face-level seam equalities are now public as well, so the next two-cell window can be rebased to the previous window's endpoints before the four genuine support-interaction tests are classified.

The pure pair-composition algebra has now been strengthened independently of the source geometry. Before testing mutual separation, Lean loop-erases each same-track concatenation. A same-track revisit is therefore always removed constructively and is no longer returned as an obstruction. The ordered and crossed classifiers now return either a composed pair or one of exactly two cross-track collisions, still carrying the literal common face. This reduces the finite repair table but does not discharge its two remaining collision kinds or construct the arbitrary-length chain.

The residual test is now made on the paths that actually survive loop erasure, rather than conservatively on their raw supports. In both endpoint orders, a bad result carries a face belonging to both final simple candidate paths. Lean also proves from the internal disjointness of the two input pairs that this retained face must have one of the two cross-track raw origins. A raw intersection erased by bypass is therefore no longer reported as a geometric obstruction. This origin is retained as proof-relevant data and projects canonically to the previous raw cross-collision witness, so the source-local geometric classification can be reused without reconstruction. That projection is now wired: both endpoint orders recover the five geometric cases together with the exact piece envelopes of both source windows. The retained bottleneck itself is not yet repaired.

That classification is now explicit and fail-closed. The generic four-way classifier returns either both same-track and both cross-track disjointness proofs, or an actual common face tagged by the interaction that failed. Its source specialization takes four consecutive selected Cell–3 placements and two certified non-overlapping two-cell windows, aligns them through the canonical intervening seam, and composes the straight/swapped endpoint order by parity. Lean therefore returns either a literal four-cell rail assembly or the exact ordered/crossed support collision. No collision has yet been repaired at this new scale, so this is a finite transition alternative rather than an arbitrary-length rail construction; the induction, both annular end caps, and L1 remain open. The collision witness now also exposes its common face. Its two membership proofs, combined with the certified provenance of both windows, place that face simultaneously in the closed dual neighbourhoods of the first and last centre pairs. The remaining repair is therefore a bounded four-cell intersection calculation, rather than an unlocalized global separation claim. That calculation is now kernel-checked. Boundary cleanliness leaves exactly five possibilities: either central corridor face itself, a common neighbour across the central seam, or a common neighbour across one of the two adjacent two-step spans. The classification applies to both endpoint orders and is the finite local case split prescribed by the playbook. It does not yet construct the corresponding reroutes; in particular, the two-step cases mark the exact place where closed-minimal corridor geometry must be transported to the opened source carrier or replaced by an equivalent direct argument.

The terminal window now preserves the stronger provenance needed for that direct argument. Every constructive adjacent-append branch already proved that its two output supports lie in the literal union of five pieces: the previous centre, the two incoming rails, and the two successor rails. The new exact terminal-window package retains this information and has a proved projection to the earlier coarse two-centre certificate. Thus existing remote-separation arguments remain unchanged, while the next bounded collision calculation can distinguish the actual source pieces involved. This is bookkeeping of proved geometry, not a new separation hypothesis or a reroute; four-cell repair, arbitrary-length iteration, both end caps, and L1 remain open.

The four-cell collision theorem now consumes this exact package as well. For both ordered and crossed endpoint parity, the common collision face carries two independent finite coordinates at once: one of the five four-centre geometric positions above, and membership in the literal five-piece support envelope of each terminal window. Thus the next repair is a finite product of named source pieces, not an appeal to a coarse neighbourhood or a global face-uniqueness principle. No product case is eliminated and no reroute is constructed by this classification alone.

Specializing the loop-erased pair algebra to these two exact windows removes the same-track branches from the actual four-cell outcome as well. Endpoint parity is still explicit: the result is a straight assembly, a swapped assembly, an ordered cross-track collision, or a crossed cross-track collision. Thus no collision has been assumed away, but the remaining local repair has exactly two kinds rather than four. Those two branches, the arbitrary-length fold, and both annular end caps remain open.

The source-facing transition now also uses the stronger post-bypass test. Both candidate rails are first loop-erased and mutual separation is tested on the supports which actually remain. All four straight/swapped parity cases retain their concrete prefix and suffix assemblies, the canonical seam rebase, and equations tying those assemblies to the two exact terminal windows. A negative result therefore contains a face surviving in both final simple candidates, rather than a raw intersection that the construction later erases. This is a sharper finite obstruction, not its repair: the retained face still has to be eliminated or rerouted, and arbitrary iteration, both end caps, the separated crosscut pair, and L1 remain open. For each of the four endpoint-parity combinations, a separate consumer now reconstructs the exact window-indexed cross witness while preserving that same face definitionally. The surviving face therefore carries both the five-case four-centre classification and membership in the literal source piece envelope of each window. No dependent cast or loss of endpoint parity separates the classifier from the finite repair table.

The adjacent repair's finite construction branch is now retained before the public API erases it. The existing length-resolved outcome already contains the necessary proof-relevant data: in each of its six residual constructors it stores the cross collision, the relevant same-track or centre witness, and the forced local length equations. The new `ExactSelectedLocalRailConstructionTrace` packages that value together with its equality to the canonical classifier and its exact five-piece support certificate. Its projection computes the same straight-or-swapped complete outcome as before. This is the missing work ticket for a mutable terminal window, not the rolling transition itself: the next theorem must still use the recorded branch to split a frozen remote prefix from the bounded suffix which the following Cell–3 repair may replace.

The first consumer now runs that construction trace on both sides of a four-cell transition and on the seam between them. The value `ExactSelectedLocalRailTracedFourCellTransition` is constructed from the three actual adjacent classifiers, applies the post-bypass four-cell classifier to the two outer exact windows, and stores all three proof-relevant local branches beside the result. The middle trace is the canonical repair of the very seam at which a retained collision is detected; omitting it would leave the subsequent finite table without its central work ticket. If the result is a collision, the same packet also carries its five-position four-centre classification and both literal piece envelopes. Thus the remaining finite table is restricted to canonical repairs actually produced by the source construction; arbitrary certified windows have been removed from that consumer boundary. The packet still returns a collision alternative and therefore is not the rolling induction or a construction of the final crosscut pair.

The literal source-piece product has now been pruned before the repair table is opened. In `GoertzelV24ClosedWebSelectedLocalRailTracedCollisionBands`, corridor distance rules out every retained collision at the first centre, every old-first-window-rail collision with the third centre, and every old-first-window-rail collision with either fourth-centre rail. Thus the twenty-five raw pairs of five source pieces reduce to four proof-relevant bands: old/old, successor/third-centre, successor/old, and successor/successor. The theorem is applied directly to the canonical traced transition, so the next table sees only source-produced repair branches and one of these four bands. None of the four surviving bands is claimed empty; their elimination or rerouting, the rolling induction, and both end caps remain open.

The first construction-sensitive refinement of those bands is now proved at the four stationary-residue constructors. The former five-piece envelope forgot which rail contributed a face; the new `AssemblySumSupportedByExpectedSelectedRailTracks` records, for each straight or exchanged output, its named old rail and endpoint-selected successor rail. A tempting stronger statement—that no output ever touches the opposite old track—is false at the literal connector endpoint: the two-edge centre bridge contains both seam flanks. The checked invariant therefore names those two flanks and the Cell–3 centre explicitly, while excluding every interior face of the opposite track. The export is now complete. It covers all four same-track length residues, both centre-revisit residues, both immediate centre bridges, the principal straight append, and the principal crossed repair. The invariant is propagated through every finite classifier to the public complete append and to `ExactSelectedLocalRailConstructionTrace`. Thus every canonical terminal window now carries a receipt for its actual old and successor rail tracks. This does not itself make any of the four surviving collision bands empty; it supplies the branch-sensitive input from which that elimination or rerouting must be proved.

The retained four-cell collision now consumes both receipts. Its four endpoint-parity branches each store one of two exact cross origins, giving eight proof-relevant cases in all. For example, the straight/straight first–second case can use only the first window's old-first/successor-first track and the last window's old-second/successor-second track, together with the three explicitly named connector faces of either window. The other seven cases record the analogous track exchange rather than collapsing it into an unordered source-piece union. This is the finite track table which the next distance and seam argument consumes. The packet also exposes the track receipt of its middle repair trace. The compatibility constructor which presents one successor's right rails as the next successor's left rails is now explicit rather than an opaque dependent cast; its two support equalities are proved once at the producer. Consequently the first window's two successor tracks are literally the middle repair's two old tracks, and the middle repair's successor tracks are literally the last window's two old tracks. The retained collision is therefore expressed in the same track vocabulary as the repair which must consume it. No origin has yet been declared impossible.

One of the four broad bands has now been consumed. In the successor/old case, the first-window successor support is the old support of the stored middle repair, while the last-window old support is that repair's successor support. Splitting the latter support at its initial vertex gives either a strict-tail append collision or one of the two shared seam endpoints. A generic lemma for both ordered and exchanged retained bypasses excludes those endpoints using path simplicity and mutual rail disjointness. Consequently the successor/old band is replaced by an actual collision witness for the canonical middle-seam append. The remaining alternatives are old/old, successor/third-centre, successor/successor, or this exact middle collision. This is a genuine band reduction, but not yet a reroute: the collision witness selected by the outer retained classifier must still be related to the finite branch stored in the middle construction trace before the rolling suffix can advance.

The unbounded part of that rolling construction has now been separated from the bounded repair. The predicate `SupportSeparatedFromFutureSelectedWindows` says that a support frozen at a corridor cutoff is disjoint from every sufficiently remote certified two-centre window. Each canonical terminal window satisfies this predicate by the proved gap-≥ 4 separation theorem; moving the cutoff forward preserves it; and both ordered and exchanged assembly append preserve it. Thus a completed remote prefix can be accumulated without reopening its geometry. What remains is exactly the neighbouring four-cell transition which produces the next terminal window, followed by the two literal end caps. This invariant is not that transition and is not L1 closure.

A tempting constructor-level use of the invariant has been refuted. One cannot mark the *entire* endpoint-carrying assembly as frozen and then prepend a later terminal window: each old rail ends at the very face where the corresponding new support begins, whereas the invariant requires their full supports to be disjoint. Lean proves this contradiction separately for both rails and for the paired assembly. The correct rolling state must therefore split the genuinely frozen support from a bounded live suffix, with the latter owning the named splice endpoints. This negative result prevents a green but vacuous prepend theorem from being mistaken for the induction. Constructing that split state, its neighbouring four-cell transition, and the two literal annular end caps remains open; L1 is not closed.

The corrected interface is now formalized as well. Instead of freezing a complete rail support, it freezes `support.dropLast`: the old interior is immutable, while the final face remains the live splice endpoint owned by the bounded terminal window. A certified terminal window constructs this state, and the interface-aware prepend theorem advances it across any remote prefix satisfying the established gap-≥ 4 certificate. Same-track endpoint sharing is discharged by path simplicity;

crossed endpoint contact is excluded by the terminal pair's mutual separation. Most importantly, the returned ordered-or-exchanged assembly again carries the interior-frozen certificate at the new cutoff, using the exact support identity for appended walks. Thus the unbounded prefix algebra is now a non-vacuous inductive interface. The neighbouring four-cell repair must still construct each compatible live window, and both literal annular end caps and L1 remain open.

The closed-to-open face-intersection step has now been isolated at the same boundary. For two fully retained ambient faces, every interior edge shared by their opened images is reflected to an ambient shared edge and reconstructed as the exact opened image of that edge. Therefore the closed minimal carrier's bound $|\texttt{sharedInteriorEdges}| \leq 1$ survives literal opening. The source-facing theorem applies this directly to any two distinct fully retained faces of a closed vertex-minimal hex corridor. This is the local uniqueness fact needed by the collision consumer; it is not a global uniqueness assertion for arbitrary faces of an annulus. In particular, the remaining C–4 formation theorem must still identify the selected Addendum–V carrier with the literal opening and prove that the selected corridor faces are fully retained. Neither premise is inferred from the open annulus alone.

The canonical four-cell transition now also carries a support-provenance receipt. Loop erasure in either the ordered or exchanged retained bypass can remove faces but cannot introduce them: every face of either successful output rail is proved to lie on one of the four literal input rails. This containment is lifted through all four endpoint-parity cases of the exact traced transition, while collision alternatives retain their existing exact witnesses. Importantly, the strengthened packet is constructed from the three actual adjacent source classifiers; the same receipt is not postulated for an arbitrary inhabitant of the older transition type. This closes the proof-relevance gap needed before defining the bounded live suffix of the rolling state. It does not perform that rolling induction, attach either annular end cap, construct the separated crosscut pair, or close L1.

That receipt now transports the remote-window invariant as well. The two supports of the first terminal window are separated from all sufficiently future windows at its first centre, and monotonicity advances their cutoff to the third centre; the last terminal window supplies the same certificate there directly. Since every successful four-cell output face belongs to one of those four supports, both output rails inherit full-support separation from every later remote window. The result is a constructed property of the canonical support-certified transition, not a new premise. It is the safety certificate needed before freezing a successful bounded suffix; collision alternatives, the actual rolling constructor, both end caps, the separated crosscut pair, and L1 remain open.

These two receipts now feed a genuine bounded rolling constructor. Given an older paired assembly whose rail interiors are frozen behind a remote cutoff, the canonical four-cell transition proves the four required append disjointness conditions from its literal window provenance. In either successful endpoint order it appends both simple rails, advances the endpoint to the fourth placement and the frozen cutoff to the third centre, and returns the same interior-frozen invariant. If the four-cell classifier collides, the result instead retains the unchanged exact transition and therefore its exact collision witness; no append is manufactured. This is one fail-closed rolling step, not yet its arbitrary-length iteration. Eliminating the collision branch, constructing both annular end caps and the separated crosscut pair, and closing L1 remain open.

The middle collision is now tied to its actual repair without identifying two arbitrary witness choices. The outer retained classifier may select a different collision face from the one encountered internally by the adjacent classifier, so witness equality would be an unjustified strengthening. The new `GoertzelV24ClosedWebSelectedLocalRailMiddleCollisionRepair` instead evaluates the stored canonical middle trace. Its concrete straight-or-exchanged assembly has disjoint rail supports; consequently, for the particular face named by the outer collision, at least one repaired rail avoids that face. The strengthened middle-band constructor records this face-specific avoidance together with the trace's exact source-track provenance. The same collision triangle proves that its face is neither the displayed centre nor either seam flank, so the general provenance disjunction specializes further: if the face occurs on a repaired rail, it occurs on the corresponding old rail or literal successor continuation, with no connector alternative. Thus the remaining finite consumer can choose the clear repaired track and its exact source origin without synchronizing collision-choice functions. This does not yet reroute the outer collision, eliminate the other three bounded bands, perform the arbitrary-length fold, attach either end cap, construct the separated crosscut pair, or close L1.

The parity-matched half of that finite repair table is now deterministic. In `GoertzelV24ClosedWebSelectedLocalRailMiddleCollisionClearance`, a straight repair of a first/first or second/second collision forces the opposite repaired rail to avoid the collision face; after an endpoint exchange, the analogous statement holds for first/second and second/first collisions. The proof combines connector-free track provenance with the two already-proved input rail separations. Thus four of the eight repair-parity/collision-origin combinations no longer require a clear-rail choice. The four parity-mismatched combinations still carry only the proved disjunctive avoidance certificate. That certificate is now also exposed as proof-relevant data: one of four constructors retains the actual straight-or-swapped assembly, the chosen first or second rail, and its proof of avoiding the specified face. This gives the next path-surgery theorem a literal simple rail rather than a bare existential. No outer four-cell reroute is claimed yet.

That rail now participates in a literal, fail-closed middle replacement. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacement` keeps the first-cell rails and fourth-cell continuations fixed, rebases the canonical two-cell middle repair across the proved seam-face equalities, and tests both joins after loop erasure. A successful branch therefore returns an actual separated four-cell assembly, with straight or exchanged endpoint order recorded. A failing branch does not collapse back to an unmet disjointness premise: it records whether the surviving collision occurred at the left or right join, together with the exact repair parity, concrete assemblies, and classifier equation. The packet separately retains the proved middle rail which avoids the original collision face. Thus the overlapping window has been replaced by a real path construction and the remaining failure has moved to one of two named boundary joins. Successful rails and surviving collision faces also carry exact support provenance: every such face belongs to the first-cell packet, the rebased canonical middle repair, or the fourth-cell continuations. Hence the two remaining joins can be attacked with the existing remote/local separation theorems without guessing where loop erasure obtained a face. The first such attack is now complete: a right-join collision whose prefix contribution came from the first cell would meet the fourth-cell continuation, but those literal rail pieces are generated at centres three positions apart and the proved remote-separation theorem makes that impossible. Therefore every surviving collision lies in one of exactly two bounded adjacent bands: first/repaired-middle or repaired-middle/fourth. This removes the sole nonlocal source term. Moreover, track provenance unfolds the repaired-middle side of either band into seven literal atoms: two old bridge tracks, two successor tracks, the Cell–3 centre, and its two seam flanks. The remaining finite table is therefore expressed entirely in source pieces rather than an opaque loop-erased path. Since every one of the four source rails has length at most two, Lean now expands membership in each rail into its start, optional middle, or finish. Thus the repaired side of either join has at most fifteen explicit coordinates: three for each short rail, plus the centre and two seam faces. Coordinates are allowed to coincide for stationary and one-step rails; no false distinctness is assumed. The untouched first- and fourth-cell rails are now expanded by the same length-two argument. Both sides of both residual joins are therefore finite coordinate data; this exposes the complete bounded case interface but does not yet eliminate its cases. Lean now collapses that coordinate table to five exact centre geometries: the second-cell centre itself; a common neighbour of the first and second centres; a common neighbour of the first and third centres; a common neighbour of the second and fourth centres; or a common neighbour of the third and fourth centres. The two adjacent-centre cases are literal dual-triangle geometries, and the two distance-two cases are bounded alternative routes. This is a complete classification of the surviving local geometry, not an elimination of any of its five constructors. A further equality split turns either distance-two relation into the intervening corridor centre or a genuine four-step boundary. The resulting six source-ladder shapes are: the second or third centre, a dual triangle on the first/second or third/fourth centre pair, or a dual square spanning the first/third or second/fourth centre pair. This is the exact rotor/centre/square input interface; no reduction or reroute is claimed at this point. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementCycles` turns each of the four non-centre shapes into a literal simple closed facial-dual walk of length three or four. The walk contains the collision face, and Lean proves that its complete support stays in the annular interior. Thus the bad branch now returns either one of two named centres or a proof-relevant short interior dual cycle; applying the rotor or square reduction remains the next step. The companion module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSeparator` turns each such short dual cycle into literal primal separator data. One actual shared primal edge is selected at every dual step; the selected set has cardinality exactly three or four, is disjoint from both named hole boundaries, and its deletion disconnects the annular carrier. This is an application of the generic selected-cycle parity separator to the constructed annular cellulation, not an assumption of global face-intersection uniqueness. It does not yet apply the rotor or square reduction and does not eliminate the two centre cases. The next companion constructs an actual deletion component on the side away from the distinguished outer-hole root. Propagating component membership around the two proved hole-face orbits, and using a radial path which avoids any selected cut of cardinality below five, proves that both complete hole boundaries and all ten named degree-one stubs lie outside this component. The remaining vertices on its side are therefore locally cubic. A generic selected-cycle bond theorem now propagates any nonempty exact side boundary around the literal selected dual cycle, using two-sidedness only for the faces visited by that cycle. Consequently the computed component boundary is exactly the full selected set, without global face-intersection uniqueness. Exact handshaking then sharpens the trichotomy: the component contains a primal cycle, or a three-edge selected boundary encloses exactly one vertex, or a four-edge selected boundary encloses exactly two vertices. In the singleton case every selected crossing is incident to the unique centre; in the two-vertex case the vertices are adjacent and every selected crossing is incident to one of them. This is the literal star-or-bond attachment classification. The short-cycle construction now also retains one literal selected corridor rung at a canonical dual step. The separator uses that edge at the same step, and the component calculation proves that the singleton centre, or one endpoint of the two-vertex bond, is incident to it. This records source provenance without asserting a globally unique shared edge. It does not identify the facial-dual cycle with a primal facial boundary, and it does not eliminate the cyclic branch, turn a facial-dual short cycle into a primal facial triangle or square, apply a source reduction, or eliminate the two centre cases. One atom is also directionally resolved. Both adjacent-triangle cases now reuse the earlier append obstruction rather than acquiring a second local theory. The surviving collision face is proved distinct from both named flanks of the relevant interface, so the first/second and third/fourth shapes each instantiate a literal three-edge facial-dual separator whose selected outer-remote side contains a primal cycle. This is a cyclic append obstruction, not a rotor reduction: the source's rotor is a primal facial triangle, whereas the object here is a facial-dual triangle with selected primal crossings. The combined reduced-frontier theorem therefore returns exactly a second/third corridor centre, one of the two cyclic adjacent-triangle obstructions, or a typed distance-two square packet which retains the literal equality that its facial-dual cycle has length four. That packet feeds the cyclic-side-or-two-vertex-bond classifier directly; a consumer can no longer lose the four-step provenance by first weakening it to the generic three-or-four short-cycle interface. The cyclic side, the two square cases, and the two centre cases therefore remain open. The displayed second-cell centre cannot occur on either fourth-cell continuation. The second and fourth corridor centres are nonadjacent in the skeleton, whereas every face in a fourth-cell continuation is adjacent to the fourth centre. Consequently, if the collision face is the displayed second-cell centre, Lean forces it into the first/repaired-middle band and rules out the repaired-middle/fourth band. Symmetrically, the third-cell centre cannot lie on either first-cell rail and is therefore forced into the repaired-middle/fourth band. Each centre residue is now attached to one exact adjacent join. This does not eliminate either bounded band in general. Neither join is yet proved collision-free, so this is not collision elimination, the arbitrary-length fold, either end cap, the separated crosscut pair, or L1 closure.

The retained-collision API now also has a literal assembly-level consumer. For either endpoint order, Lean cuts the appropriate accumulated rail at the stored collision face, follows the opposing new rail from that same face, and loop-erases the result. The output is a simple route with the intended outer endpoints, together with a proof that its support lies in exactly those two displayed rail pieces. This lifts the earlier one-cell reroute to arbitrary accumulated assemblies and applies directly at either localized join. It is deliberately only one route: no claim is made that a compatible second route can be chosen disjointly, so this is not yet a repaired rail pair or a crosscut constructor.

The four-step residue is now sharpened independently of any source reduction. Its selected remote component is either cyclic, or Lean constructs exactly two distinct adjacent vertices whose support is the whole component and whose endpoints meet every one of the four selected crossings. In particular the literal corridor rung retained by the short-cycle packet meets one of those two vertices. The singleton-star branch is impossible because the exact handshake equation assigns it a three-edge, not four-edge, boundary. This is the complete graph-level square residue; it is not an identification with a primal facial square and does not invoke the manuscript's square reduction. The adjacent-pair branch now carries its exact endpoint arithmetic as well. The edge joining the two component vertices is proved internal and therefore absent from the selected cut. Local cubicity, derived separately at each endpoint from exclusion of all ten boundary stubs, then proves that exactly two selected crossings meet each endpoint. Thus the four-edge residue is a literal (2+2) boundary around one internal bond. This is the input needed for a source-local rotation-order or rerouting argument; it is not itself such an argument, and it still does not turn the facial-dual four-cycle into the source's primal facial square.

The local rotation calculation now consumes this exact endpoint split. At each stub-free cubic endpoint, the first dart after the internal bond belongs to the selected crossing boundary. Consequently the two facial sides of the internal bond are distinct faces already present in the selected four-cycle, and the bond itself witnesses their adjacency in the interior dual graph. This is a literal dual adjacency, not yet a chord theorem: the same two faces could still share a selected boundary edge. Accordingly it neither identifies the four-cycle with the manuscript's primal facial square nor supplies a rail reroute. The remaining finite question is now exact: is the bond adjacency off the selected walk, or does it coincide with one of the four selected steps?

That question is now a kernel-checked dichotomy. If the bond adjacency is not one of the four walk edges, Lean packages it as an actual walk chord. If it is a walk edge, Lean instead exhibits the selected primal crossing at that step and proves that it is distinct from the internal bond while both edges lie on the same pair of cycle-support faces. Hence the non-chord branch is no longer an unspecified degeneracy: it is exactly a two-edge face-pair multiplicity. No global unique-shared-edge property of the opened carrier is assumed.

The chord branch is now split once more, without changing proof routes. A generic four-cycle calculation cuts the selected facial-dual square along the internal chord into two literal simple three-cycles. The strengthened theorem retains the provenance that every face of both triangles belongs to the original four-cycle support. Each triangle is then equipped with actual selected primal crossings, with the internal bond pinned as the crossing at the chord step. The selector now retains edge provenance as well as face provenance: every non-chord step reuses a crossing already selected on the original four-cycle, rather than choosing a fresh edge from the same pair of faces. Thus the split introduces exactly one new crossing edge, the internal bond, while its complete support remains among the annulus's interior faces. Conversely, every edge of the original four-cycle belongs to at least one of the two chord triangles, and the selector transports the crossing attached to that exact old step. In particular the literal source rung pinned at the replacement square's anchor survives in one selected triangle; it is not replaced by an arbitrary edge having the same incident faces. These triangles are therefore ready for the existing selected separator machinery without forgetting which literal source edges they cross. They remain facial-dual triangles: no claim identifies them with the manuscript's primal rotor triangle, and coverage alone does not prove that the three crossings meet at one vertex. The cyclic component and parallel face-pair branches therefore remain distinct open residues.

The two chord triangles now reach that separator machinery literally. Each is repackaged as the established selected short-cycle datum, with the original square's internal bond retained as its distinguished primal crossing. For a component chosen away from the outer root, Lean proves the exact alternative: either the component contains a cycle, or it consists of a single star centre incident to every selected crossing of the triangle. In the star alternative the centre is additionally proved to be an endpoint of the original internal bond. Thus this branch has ceased to be an untyped “small triangle” residue; its constructive case is anchored at a literal bond endpoint, while its cyclic case is an explicit remaining reduction obligation. This theorem still does not perform the rail repair or discharge the cyclic case.

A stronger local construction now removes that cyclic alternative from the task of finding the two endpoint stars. The internal-adjacency packet retains the literally oriented primal bond. Iterating the cyclic rotation at either stub-free cubic endpoint enumerates the bond and the two other incident edges; their three face corners form a selected facial-dual triangle. Lean proves for each endpoint that the triangle has length three, remains in the original four-cycle's interior support, and crosses exactly the full incident-edge finset of that endpoint. Reversing the bond supplies the second triangle by the same theorem. Thus both common primal centres are constructed directly, without closed-map minimality, global face-pair uniqueness, or a cyclic-cut premise. The earlier chord-versus-parallel classification remains true, but is no longer a prerequisite for this endpoint construction. This still does not identify the facial-dual triangles with primal rotor triangles or perform the rail repair that consumes the two centres.

The two endpoint triangles now carry an exact decomposition receipt. Their selected crossing sets meet only in the internal bond, and their union is the internal bond together with the four original selected crossings of the replacement square. Consequently the local split loses no literal source edge and introduces no crossing other than the bond. This is the edge-level input to a rail reroute, not yet the face-path construction or its separation proof. The face-side geometry is equally explicit: any two distinct faces in either endpoint triangle's support are adjacent. Therefore a collision face can be bypassed by one dual edge once the two surviving rail attachment faces are shown to lie in the same endpoint triangle. That incidence statement, and the resulting separation from the companion rail, remain to be proved. Moreover every face visited by the original selected four-cycle belongs to at least one endpoint triangle. The proof takes an incident selected crossing, uses the exact union identity to assign it to an endpoint triangle, and then uses the local two-sided face law to recover the visited face. Hence the triangle split covers the complete original face support; it does not yet say that both surviving attachment faces receive the same assignment.

The packaged existential keeps the useful special case explicitly: together with the two endpoint centres and both edge identities, it returns a proof that the collision face itself lies on one of the two endpoint triangles. This is a choice receipt for the reroute, not the reroute. For that chosen triangle the remaining two faces are distinct and joined by one literal dual edge. Thus Lean now constructs the complete local bypass of the collision face. A subsequent type audit blocks a tempting unsupported identification: the other triangle faces are drawn from the displayed corridor-centre square, whereas the retained rails carry side faces adjacent to those centres. The current interfaces prove no uniform equality with the rail predecessor and successor. A source-faithful uniform repair must instead join those rail positions through a short chain of corridor centres; the triangle faces cannot simply be renamed as the rail positions. The generic splicing operation itself is now machine-checked. Given two simple facial-dual walks meeting at an internal face and an edge from the predecessor on the first walk to the successor on the second, `crossSpliceAround` cuts away the shared face, inserts that bypass edge, and loop-erases the result. The returned walk is proved simple and its support is proved not to contain the removed face. Its strengthened companion `crossSpliceAroundWithBridge` accepts a complete walk between the two attachment positions and proves the same avoidance theorem when that bridge avoids the removed face. This separates generic path surgery from source geometry cleanly: the remaining local obligation is the explicit predecessor–centre-chain–successor bridge and its separation from the other rail. No iteration or annular end cap follows merely from the generic splice.

The explicit centre-chain bridge is now machine-checked for both alternating replacement-square branches. The predecessor on the old local path is joined to its corridor centre, then across the two consecutive centre adjacencies, and finally from the third centre to the successor on the new local path. Its support is exactly the five displayed faces. The square incidence and the internal-endpoint conditions therefore prove that the collision face is absent, after which the generic bridge splice returns a simple route avoiding that collision. This is a source-faithful local repair, conditional only on extracting the two surrounding local path pieces and their already-recorded centre-adjacency invariants from the exact collision packet. It does not yet prove separation from the companion rail, iterate repairs along an arbitrary corridor, attach the two annular end caps, or construct the final separated crosscut pair. The returned path now carries an exact support receipt as well: loop erasure uses only the old local path, the three displayed corridor centres, and the new local path. A generic corollary therefore reduces disjointness from a companion rail to disjointness from precisely those ingredients. Discharging those local separation facts from each exact collision constructor is the next source-facing step; it is not assumed here. All four literal source-track specializations are now complete for a first–third square collision lying on a first-cell rail and on a continuation into the third cell. Lean handles the 2×2 choices of first/second old rail and first/second continuation separately. In each case it reverses the old rail so that the corresponding already-proved collision-free seam flank is the splice start. The two flank-exclusion theorems certify that the collision is internal at both ends, and the centre-chain construction yields a simple route to the matching third-cell outgoing flank which avoids it. This closes the continuation-pairing table, not the square branch: middle source atoms outside the third-cell continuations, the second–fourth mirror, and separation from the companion rail remain open. Each of the four source specializations now retains that exact support receipt rather than discarding it at the source boundary. The public predicate `FaceInFirstThirdSquareSourceSpliceSupport` says that every route face comes from the selected old rail, one of the three corridor centres, or the selected new continuation. Its disjointness theorem reduces companion-rail separation to those five literal ingredients. It does not assert the three centre-avoidance facts; legal zero-step local shapes make those facts part of the remaining source geometry.

The superficially symmetric second–fourth specialization has a further endpoint issue. The available four-cell packet proves that the collision is not either third-cell seam flank, but it does not prove that the collision is different from the fourth-cell continuation's far endpoint. A route ending at that face cannot simultaneously avoid it. Thus this mirror needs either one more successor cell, making the collision internal, or a separately proved far-end exclusion; it is not obtained by renaming the first–third theorem.

The first adjacent band is now consumed exhaustively in `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementFirstBandResolution`. The theorem reopens the exact source-piece provenance instead of classifying only by an abstract common-neighbour relation. Its seven middle atoms yield exactly three outcomes: the displayed second-cell centre, a literal first–second dual triangle, or one of the four constructed first–third source splices. Each splice retains endpoint parity, simplicity, collision-avoidance, and the support receipt above. If the original collision lies in the opposite middle–last band, that entire source packet is returned unchanged. Thus the first band is constructive, while the second band and companion separation remain explicit rather than being absorbed into a conditional "resolved" wrapper.

The non-endpoint part of that opposite band is now constructive as well. The theorem `ExactSelectedLocalRailMiddleReplacementCollision.resolveSecondBand` reopens the same seven source atoms and the two literal fourth-cell tracks. Contacts carried by a third-cell continuation give the exact third–fourth dual triangle. Contacts carried by either second-cell track give one of four literal second–fourth centre-chain splices, covering the full 2×2 track table and retaining path simplicity, collision avoidance, endpoint parity, and an exact support receipt. The displayed second-cell centre is excluded from the fourth-cell packet by corridor separation. The only remaining constructors are the displayed third-cell centre and equality with one of the two fourth-cell far endpoints. Thus the apparent asymmetry is now proved to consist of exactly those endpoint values rather than an unclassified second band. The next rolling step must advance the endpoint before consuming that residue; this theorem does not pretend that a route can both avoid and end at the same face, and it does not yet prove companion separation or close L1. The far-end packet also retains the geometry already established on that branch: which old rail contains the face, adjacency to the second and fourth corridor centres, and inequality with the third centre. These are proof-relevant inputs to the later centre bridge, not facts to be reconstructed after the rolling transition. The resulting centre control is now generic. Adjacency to the second centre, together with the corridor skeleton's proved non-adjacency of separated centres, excludes the retained face from every later centre with an intervening position. Independently, exact four-cell support provenance proves that every face on a successful returned rail lies near one of the two literal terminal centre pairs. Thus the next bridge has a finite, source-derived choice of landing neighbourhood and a proved face-avoiding centre chain; collision outcomes remain explicitly outside that successful-face statement. The finite landing choice is now realized by a literal walk. For each of the four closed-neighbourhood alternatives at either terminal pair, the new centre bridge runs from the old attachment through consecutive corridor centres two through at most seven and ends at the future suffix attachment. Its support is proved to contain only those centres and the two attachments, and the advanced collision face is absent. This is the bridge consumed by the next cross-splice; it is not yet the spliced rail or a two-rail assembly.

The required endpoint movement is now quantified rather than guessed. A legal local $0 + 2$ connector may leave one track stationary, so advancing by a single Cell–3 is not a uniform repair. The theorem `SecondFourthFarEndpoint.not_mem_futureRailSupports` instead uses the source corridor's proved remote-separation law: once the future selected-cell centre lies at least three positions beyond the fourth-cell centre, either fourth-cell far endpoint is absent from both future local rail supports. This is a derived geometric fact, not a new hypothesis on the rolling assembly. It fixes the exact look-ahead cost of the residue. At that stage it did not show whether the intervening loop erasure preserved the collision-bearing prefix, so the next construction had to account for both possibilities.

That apparent prefix requirement has now been removed without assuming it. The next canonical four-cell classifier is run on the actual source rails beginning in the fourth cell. Its new endpoint-advance theorem is exhaustive: the classifier either returns its exact collision-bearing constructor, erases the old endpoint face from both successful rails, or retains the face on exactly one rail while proving that it is not that rail's remote finish. Thus loop erasure cannot lose the argument: if it deletes the face, the collision is already cleared; if it preserves the face, the proved three-centre gap has made the occurrence nonterminal for the combined rolling route. This is the canonical future-window advance, not yet the centre-chain splice for the surviving case or elimination of a newly returned collision.

The surviving case is now consumed by an actual source-supported splice. The old far-end packet retains not only which old rail contains the face but also that the face is not that rail's starting endpoint, so the old prefix can be cut at the retained occurrence without adding a hypothesis. The future four-cell packet supplies the successful rail, path proof, nonterminal occurrence, and exact terminal-centre neighbourhood. Lean joins those two literal rail pieces through the proved face-avoiding centre bridge and returns a simple walk whose support is contained in the old rail, corridor centres two through seven, and the new rail. The canonical theorem is exhaustive: a new collision constructor remains explicit, successful output either clears the old face from both rails, or one of the four old/new rail combinations carries this concrete splice. This closes the surviving-face routing obligation; it does not yet prove separation from the companion rail, rebuild the full two-rail assembly, iterate the rolling step, construct end caps, or close L1. The splice now carries the corresponding finite separation interface as a theorem. If a companion walk is disjoint from the selected old source rail, avoids each of corridor centres two through seven, and is disjoint from the selected future source rail, then it is disjoint from the complete loop-erased splice. The proof consumes the exact support receipt above; it introduces no new geometric assumption. This theorem deliberately stops at the interface: the companion walk and those source-local premises still have to be constructed from the retained cross-collision parity data. Thus it is a reusable reduction of the separation proof, not a conditional declaration that the two-rail assembly already exists.

The complementary source pieces are now retained rather than reconstructed after the parity choice. In every one of the four old-rail/future-rail splice combinations, Lean packages the opposite old rail, the opposite future rail, and the two disjointness theorems already present in the selected source pairs. Consequently the old-side and future-side premises of the preceding separation rule are supplied by construction. The remaining premise is geometric and sharply local: the two complementary pieces must still be joined through the middle by a walk avoiding corridor centres two through seven. No such complete companion walk, rebuilt two-rail assembly, rolling iteration, annular end cap, separated crosscut pair, or L1 closure is claimed by this refinement.

There is now a constructive payoff from retaining that parity. When the future splice agrees with the canonical middle classifier—first-to-first or second-to-second for a straight middle, and first-to-second or second-to-first for a swapped middle—the opposite rail of the middle assembly already joins the two complementary source pieces. Lean appends it to the future piece, loop-erases the result, and records a support receipt for the complete companion walk. The actual transition has not yet been proved to choose one of those matching parities: for either middle outcome, two crossed mismatch cases remain. Separation from the repaired primary splice, arbitrary-length iteration, end caps, and L1 therefore remain open.

That remaining parity fork is now represented explicitly by the canonical classifier. Its proof-relevant trace retains either the literal straight or literal swapped middle assembly. For each of the four possible source-splice pairings, Lean now returns either the complete companion path in the compatible case or a typed mismatch packet proving that the classifier pairing and splice pairing differ. Hence there is no longer an unnamed "wrong parity" failure, but the mismatch packet is deliberately not counted as a repair: centre avoidance, mutual separation, rolling iteration, end caps, and L1 remain open.

The classification has also been lifted into the actual canonical seven-cell transition. On a successful future outcome, the theorem either proves that the retained endpoint has disappeared or returns the literal companion-piece packet together with a complete companion path or its exact middle/splice parity mismatch. A newly produced collision remains an explicit constructor. Thus the parity analysis now consumes the source transition that the rolling argument uses; it is not merely a theorem about four hypothetical path choices. Opposite-parity repair and mutual separation are still open.

The mutual-separation interface has nevertheless become strictly sharper. The cross-splice now records that its loop-erased support uses only the old prefix before the collision, the six-centre bridge, and the future suffix after the collision. Consequently no hypothesis is imposed on the discarded old suffix or future prefix, where further source collisions may legitimately occur. In every parity-compatible branch, Lean packages the primary splice and the constructed companion route as a literal two-rail assembly once three retained-piece separation facts and the six-centre avoidance fact are given; the fourth retained-piece fact follows automatically from the already separated future source pair. This is an explicitly conditional assembly interface, not a construction from the canonical transition. The remaining retained-piece facts, centre avoidance, opposite-parity repair, arbitrary-length iteration, both end caps, the separated crosscut pair, and L1 remain open.

The simultaneous four-cell repair now carries the future-separation receipt needed by the rolling construction. Every face of a successful straight or swapped middle replacement is traced to one of three literal source packets: the first local rail pair, the repaired middle pair, or the final continuation pair. Each packet lies in an adjacent two-centre window, so the proved gap-at-least-four theorem separates it from every sufficiently later selected window. Lean therefore freezes either successful repaired pair at the third centre without adding a centre-avoidance or global-disjointness hypothesis. The exact collision branch remains explicit. This is the safety receipt for advancing a repaired bounded window.

The repaired window can now be attached to an older frozen prefix as well. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementFrozenPrepend` uses the prefix's interior-only certificate and the repair's three source packets to prove all four old/new support separations. It then appends both simple rails, allowing precisely their two common splice endpoints, and moves the frozen cutoff to the third corridor centre. Both straight and swapped repairs are constructed without a centre-avoidance premise; an exact local collision is returned unchanged. This is one bounded rolling step, not yet collision elimination, an arbitrary-length fold, either end cap, the separated crosscut pair, or L1 closure. In fact Lean now records the precise iteration warning: `immediateNextMiddleReplacement_gap_false` proves that the third-centre cutoff produced by this step cannot meet the gap required by an immediately adjacent next block beginning at the fourth centre. A genuine iteration must therefore retain several recent cells as a bounded mutable suffix and freeze only the older prefix; deleting only the final support occurrence is insufficient.

A source-and-type audit now fixes the scope of that recurrence repair. In the v23 collar decomposition the long coordinate is the serial sequence of cross-section positions, while the interfaces are radial transversals; the current Fable manuscript likewise pumps a shallow corridor through finite cross-section profiles. The Lean local Cell–3 pair nevertheless has internal endpoint faces, proved distinct from both annular holes. Its longitudinal rail fragments must therefore still be assembled and end-capped before they can supply the two hole-to-hole paths consumed by the splice. The rail route is not being discarded. What has failed is the present state representation: no current module maps a `SelectedSourceLocalRailAssembly` into the crosscut API, `forwardFour` measures flank distance inside one shared hexagon rather than a four-cell jump along the corridor, and the second-band `toSeparatedAssembly` remains conditional on retained-piece and centre-clear premises. The proof-relevant construction trace records which finite repair occurred and where its output support came from, but it does not show that a named internal seam survives loop erasure. The next valid recurrence must consequently retain a bounded raw terminal window alongside the frozen path prefix (or first prove a seam-retention theorem); splitting a flattened path at an assumed seam would silently reintroduce the missing claim.

That corrected state now exists in Lean. The `BoundedLiveMiddleReplacementState` keeps the interior-frozen old prefix and the exact proof-relevant four-cell repair as separate data while also carrying the already-verified flattened prepend result. Its equality receipts identify the stored repair with the canonical finite classifier and the stored flattened result with the proved prepend operation. Thus the next rolling step can inspect the literal terminal construction without asserting that any internal seam survived flattening. Construction traces with the same literal inputs are now proved equal, and a two-trace live suffix has a literal one-position advance: its later trace becomes the next suffix's earlier trace, while only the newly exposed successor is classified. The shifted suffix now feeds an actual fail-closed recurrence step as well. The new seam is tested by the existing append classifier; a clean seam returns its literal separated assembly, whereas a bad seam returns the concrete collision and immediately runs the canonical wider middle replacement using the shifted trace. This is a construction, not a conditional wrapper, and it adds no geometric premise. The branches deliberately retain different endpoint spans, so the next rung must reconcile those spans before advancing the frozen prefix. Arbitrary- length iteration, both annular end caps, the separated crosscut pair, and L1 remain open.

That span reconciliation is now machine-shaped as well. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixCommonSpan` constructs the support-certified canonical four-cell transition on the shifted cells 2–5 beside the raw seam classification. Two equality receipts prove that the future transition's first trace is the stored suffix's later trace and that its bridge trace is the newly classified trace of the shifted suffix. Thus the common-span future packet is the continuation of the actual stored calculation, not an unrelated inhabitant with matching endpoints. Its successful branches span cells 2–5, and its collision branches retain their exact witnesses. A direct formal attempt to append the old cell 1–cell 2 trace revealed that consecutive flattened traces overlap on a complete selected rail piece, not only at a terminal face. Appending both wholesale would therefore double-count literal source geometry.

The rolling cut is now constructed on the correct side of that overlap. `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedPrefix` extends the genuinely frozen old prefix through only the first live local pair, derives all four old/new support separations from the stored source gap, and transports its terminal faces across the proved successor-frame equalities. The resulting simple two-rail prefix ends exactly where the common-span future transition begins and retains an interior-frozen certificate at the first live centre. No separation assumption is added. The packet now also retains its source decomposition explicitly: every face on either rebased output rail is proved to come from the corresponding old frozen rail or from the literal first live pair. This receipt is essential for localizing any collision found by the following append; merely choosing an inhabitant of the weaker prefix type would not preserve that information. The matching bounded append is now fail-closed as well. `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedAppend` runs the retained-support classifier between the constructed shared-cut prefix and the constructed cells 2–5 future. A successful future yields a simple two-rail assembly in its literal ordered or exchanged far-end order. A collision already found by the four-cell classifier remains stored in that exact future packet, while a new collision against the older prefix retains the corresponding support witness separately for the ordered and exchanged far-end cases. The latter two cases use the same ordered append operation: "exchanged" changes the far endpoints, not the two faces at which the future begins. Thus no collision is erased and no endpoint permutation is inferred. This closes one bounded prefix-to-future classification, not its collision elimination or arbitrary-length iteration; both end caps, the separated crosscut pair, and L1 remain open. The same module also packages the entire calculation directly from the stored rolling state: it selects the proved shared-cut prefix, constructs the shifted common-span packet from the actual source classifiers, and records the append result with definitional equality receipts for all three computations. This rules out replacing any stage by an unrelated inhabitant with matching types, while the stronger shared-prefix field retains the source-support provenance needed by its collision consumer.

That consumer now localizes every newly retained prefix/future collision. `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollision` uses the future packet's exact terminal-window provenance on the singleton support containing the collision face. The old state's frozen `dropLast` invariant excludes that face from either arbitrarily long old rail; the shared-prefix provenance then forces it onto one of the two literal first-live rails. The live endpoint is treated correctly as part of the local rail rather than the frozen interior. Thus a collision introduced by this append is now bounded source geometry, not an unbounded-prefix debt. The theorem localizes rather than eliminates that collision, and collisions already internal to the four-cell future remain a separate exact branch.

The two bounded residues are now returned through one interface rather than left in parallel prose. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionBands` proves that every constructed shared-prefix transition either succeeds or retains one of two proof-relevant finite packets. An internal future collision is promoted to the existing exact four-cell collision band, including the stored middle repair when applicable; a collision introduced by joining the prefix is first placed on the literal first-live rail pair and in one of the shifted future's two exact terminal windows. The more remote of those windows lies at corridor centres four and five. Exact five-piece window provenance, together with the first-live rails' adjacency to centre one, makes any shared face there either a forbidden distant centre adjacency or a forbidden common neighbour. Hence that last-window alternative is empty: every newly retained collision is confined to the first live pair and the shifted future's first terminal pair, a literal three-cell window. Consequently no failed rolling step can refer to an unspecified location in the unbounded prefix, and one of its two terminal-window alternatives has been eliminated rather than merely named. The first-window collision class and the internal exact four-cell bands remain; this is not arbitrary iteration, an end cap, a crosscut constructor, or closure of L1.

The surviving three-cell residue is now normalized in the source ladder's own geometric vocabulary. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionFirstWindowGe` keeps the exact first-live and terminal-window support receipt and proves that its shared face is exactly one of three kinds: the displayed second centre, a first–second adjacent dual triangle, or a first–third distance-two square. Equality with the third centre is impossible by the corridor skeleton's proved separation. The outcome-level theorem attaches this classification to every newly retained collision while leaving internal future collisions in their exact four-cell packets. This is a finite geometric normalization, not an application of the rotor or square consumer: none of the three shapes is silently declared harmless, and arbitrary iteration, end caps, separated crosscuts, and L1 closure remain open.

The non-centre alternative is now physical rather than merely facial-dual. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionFirstWindowS` selects one primal crossing at every step of the short cycle. The resulting cut has cardinality three or four, is disjoint from both named hole boundaries, and its deletion disconnects the annular carrier. A genuine deletion component away from the outer root is retained in the packet, and the locally cubic count classifies it exactly as a cyclic side, a one-vertex star behind three crossings, or a two-vertex bond behind four. Thus every newly retained collision is now either the displayed second centre or a literal small separator with its component geometry; collisions already inside the canonical future retain their exact four-cell bands. The cyclic case is not eliminated here, and neither the star nor bond witness has yet been converted into a mutually separated two-rail repair.

The acyclic component alternatives are now constructive as well. `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionFirstWindowC` replaces the singleton count by an actual primal vertex lying in the chosen component and incident to every selected crossing, and replaces the two-vertex count by an actual `SquareBondRealization`: two distinct adjacent vertices whose pair is the whole component and which collectively carry all four crossings. The cyclic component remains an explicit first constructor. This strengthened trichotomy is attached to every newly retained append collision. The existing square-bond stack can construct an endpoint-triangle bypass, but its proved interface intentionally stops before identifying the two surviving bypass faces with the predecessor and successor of the retained rail collision. That endpoint matching, followed by companion-rail separation, is therefore the exact next local repair obligation; merely possessing some local bypass is not counted as a rail repair.

The corresponding track provenance is now retained rather than reconstructed after localization. The module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionFi` proves that each newly retained collision has one of exactly two literal origins: old first rail versus future second rail, or old second rail versus future first rail. In both cases the future occurrence remains in the tail of the named rail, so it is not merely the shared starting endpoint. The outcome-level repair packet carries this exact cross-track origin together with the centre/cyclic/star/bond component resolution. This repairs a real information loss in the earlier first-window packet; it does not yet identify the endpoint-triangle bypass faces with the predecessor and successor on the named track, nor prove the modified track disjoint from its named companion. Those two statements remain the next local consumer obligations.

The successful retained-bypass branch now keeps the same track information. In `GoertzelV24ClosedWebSelectedRailPairRetainedSupport`, ordered composition confines the two returned supports respectively to the old-first/new-first and old-second/new-second track pairs; crossed composition confines them to old-first/new-second and old-second/new-first. This is stated at the support level, before the independent mutual-disjointness test, so the provenance is not obtained by assuming that a bypass candidate is already a valid two-rail assembly. The terminal-window specialization and its use in the endpoint-matching argument were the next open steps.

That specialization is now machine-checked in `GoertzelV24ClosedWebSelectedLocalRailTracedFourCellTra` Every successful canonical four-cell transition retains either the ordered terminal-track pairing or the crossed pairing, and the classifier-built transition is packaged with this certificate. Thus the future rail named by the earlier collision-origin packet can now be traced back to one particular first-window track plus one particular last-window continuation, instead of only to their four-way union. The next step is to exclude the remote continuation at the shared collision and thereby obtain the literal future first-window track needed by the endpoint-coordinate argument. No collision has yet been repaired, and neither companion separation nor C–5/L1 closure is claimed.

That remote-continuation elimination is now complete in GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionFirstWindowE For every newly retained collision, the exact alternatives are now old-first against the shifted future's literal first-window second track, or old-second against its literal first-window first track. The future membership remains in the assembled rail's tail, so this does not collapse to the shared starting endpoint. This closes the support-provenance loss which previously prevented the endpoint-coordinate question from even being stated on the correct rail; it does not answer that coordinate question or construct a mutually separated repair.

The next positional refinement is now kernel-checked as well. The generic module GoertzelV24DualPathPointedOccurrence proves that a vertex in the tail of a simple path is either the terminal endpoint or has a strict internal index with literal predecessor and successor vertices; their two incident adjacencies and pairwise distinctions follow from the path invariant. The source-local specialization GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementBoundedSuffixSharedCollisionFirstWindowP attaches exactly this dichotomy to the future occurrence in the retained collision packet. The terminal case is deliberately not discarded. This turns anonymous support membership into the proof-relevant coordinate needed by the endpoint-triangle comparison, but it does not prove that comparison, construct a bypass, or establish companion-rail separation.

The coordinate now retains edge provenance as well as vertex provenance. For a strict internal occurrence, GoertzelV24DualPathPointedOccurrence names the incoming and outgoing edges and proves that both occur literally in the path's edge list. In GoertzelV24ClosedWebSelectedRailPairRetainedSupport, every edge of a successful loop-erased append is correspondingly traced to one of the two named input tracks, with the ordered and crossed pairings kept distinct. Thus loop erasure may remove edges but cannot manufacture the two edges used at the pointed collision. This is the prerequisite for comparing its actual rail neighbourhood with the existing square-bond endpoint triangle; that comparison, the terminal-endpoint case, and companion-rail separation remain open.

The edge receipt has now been lifted through the actual canonical four-cell classifier as well. The extension of GoertzelV24ClosedWebSelectedLocalRailTracedFourCellTrackSupport proves, in each of its four straight/swapped input cases, that every successful output edge retains the corresponding ordered or crossed terminal-track origin. A generic rebase lemma records that changing only a walk's typed starting face preserves its edge list. The remaining local question is now strictly smaller: exclude the remote terminal track for the two incident edges at the pointed collision, and only then compare them with the square-bond endpoint triangle.

That remote edge alternative is now excluded in GoertzelV24ClosedWebSelectedLocalRailMiddleReplaceme For either retained cross-track collision, source separation forces both the incoming and outgoing edge of every strict internal occurrence into the corresponding literal first terminal-window track. The terminal endpoint is kept as a separate constructor rather than silently treated as internal. Consequently the endpoint-coordinate problem is now genuinely finite and local: compare two named edges of one literal terminal track with the existing square-bond endpoint triangle. That comparison, the terminal case, the bypass itself, and separation from the companion rail remain open; no collision elimination or closure of the source's bulk-corridor flag L1 is claimed.

The same terminal-window module now exposes the next coordinate layer without guessing it. Every endpoint of an edge on either canonical output track is classified into the exact old rail, successor rail, or one of the three named connector faces, with straight-versus-swapped endpoint parity retained. In particular, the predecessor and successor attached to an internal collision now inherit a finite source-origin split from their localized incident edges. What remains is the genuinely geometric finite comparison between those source alternatives and the faces of the square-bond endpoint triangle; no such equality is inferred merely from support membership.

The first missing source-coordinate receipt in that comparison is now machine-checked in `GoertzelV24ClosedWebSelectedLocalRailStepReceipt`. A forward pair of surviving Cell–3 side slots is classified without a global face-uniqueness assumption: either the two opposite-dart faces coincide and the connector is stationary, or the connector crosses the literal third primal edge at their named locally cubic outside corner. The nonstationary branch retains the successor-dart equation, the exact third edge, and its membership in both face boundaries. The former length-at-most-one walk interface retained only the dual adjacency and therefore could not support the endpoint comparison by itself. This receipt restores that information, but has not yet been propagated through every canonical terminal track or identified with either square-bond endpoint triangle; the bypass and companion separation remain open.

The receipt is now attached to the older choice-built local walks in `GoertzelV24ClosedWebSelectedLocalRailStepEdgeReceipt`. If an edge belongs to one of the selected forward walks of length at most one, the walk cannot be stationary; hence the edge has exactly the two displayed endpoint faces and carries the named third primal crossing edge, successor-dart equation, and locally cubic corner supplied by the source-coordinate theorem. This is an edge-level provenance result, not yet its propagation through a two-step rail, a bypassed terminal track, a middle replacement, or the final long crosscut assembly.

That first propagation is now proved in `GoertzelV24ClosedWebSelectedLocalRailShapeEdgeOrigin`. For each of the canonical (0+2), (1+1), and (2+0) rail shapes, every edge surviving loop erasure, reversal, and append retains a literal selected forward-step origin and hence the same exact primal crossing receipt. The proof uses only that bypassed edges come from the raw walk, appended edge lists split between their factors, and reversal preserves undirected edge membership. It has not yet strengthened every abstract separated-rail value or transported the receipt through terminal-window middle replacement and the long crosscut assembly.

The successful separated-rail interface now retains this fact rather than erasing it. Each of its two rail-edge families carries a proof-relevant selected forward-step origin; the three canonical shape constructors supply those fields, the successor-cell rebasing transports them through copied walks, and the two continuation theorems preserve them across the literal Cell–3 seam. Rebuilding through the existing pointed first-window collision module checks that this strengthening composes with the complete append, four-cell trace, and middle-replacement tower. Connector edges created by a terminal repair still require their separate finite coordinate classification before the square-bond endpoint triangle can be identified.

That connector classification is now proved in `GoertzelV24ClosedWebSelectedLocalRailCenterEdgeReceipt`. The literal two-edge walk through a selected Cell–3 centre has no anonymous edge: each of its two centre-to-side edges is tied to the corresponding selected side slot and face-cycle dart, and the crossed primal edge is proved to lie in the shared interior boundary of the centre face and the selected side face. This closes the bounded edge-receipt gap for the explicit centre connector. The centre-bridge repair and both oriented residue repairs now use that public receipted connector in place of private duplicated walks. A rebuild through the pointed-edge-origin terminal-window module checks the entire current append, separator, component, square-bond, and middle-replacement tower with that shared definition. The same module now exports receipted one-edge side-to-centre and centre-to-side half-connectors. The centre-residue repairs use these public terms when a finite repair stops at the Cell–3 centre, so no private connector shape remains outside the crossing-receipt API. The remaining step is to propagate the resulting old-rail, successor-rail, and centre-connector alternatives through every repaired terminal track and then compare the pointed predecessor/successor crossings with the square-bond endpoint triangle; this does not yet construct the arbitrary-length crosscuts required by the splice.

The first branch-independent propagation layer is now kernel-checked in `GoertzelV24ClosedWebSelectedLocalRailConstructionTraceEdgeReceipt`. It packages the exact terminal-edge alternatives—an old selected forward step, a successor selected forward step, or a named old-centre connector—as a walk invariant and proves that copying, append, prefix/suffix truncation, reversal, loop erasure, and the two canonical cross-collision reroutes preserve it. Consequently both rails of the ordinary successor append and of the double-cross repair retain source receipts on every surviving edge. The same invariant is now discharged for the two immediate centre-bridge repairs and all six finite residual repairs, including the two cases that truncate an old rail or stop at the old centre. It is then propagated through the exhaustive classifier to the public complete append and attached to `ExactSelectedLocalRailConstructionTrace`. Thus every edge surviving any canonical adjacent terminal repair carries a literal receipt. This does not yet identify the pointed predecessor/successor collision edges with the replacement-square endpoint triangle, iterate the construction through an arbitrary corridor, attach both end caps, or close L1.

The disposable end-to-end type probe has also been extended through the logical endpoint of the GWCO descent. Once a supplier constructs a strictly smaller packed closed-web-at-good-word instance above a threshold, Lean now assembles the carrier-changing reductive system and proves that exclusion of all such instances is equivalent to its `BaseVerified` claim. This checks the reduction-to-base interface, but it is deliberately not a 4CT headline theorem: the source's subsequent trail-completability assembly from GWCO to colorability remains a separate open interface, while the supplier must still reconstruct every output invariant named above.

The ignored probe now also imports and resolves the live construction-trace receipt theorem. This is a deliberately small synchronization check: the physical geometry lane reaches a canonical adjacent outcome with literal edge receipts, while the reductive lane still exposes its single missing supplier of a strictly smaller packed closed-web instance. It makes interface drift between those ends visible early; it does not pretend that the still-open arbitrary-length and end-cap construction already supplies that reduction.

The pointed first-window consumer now retains the same provenance rather than forgetting it at the collision site. Every edge of either exact terminal track inherits the complete classifier, and in each strict internal collision alternative both the incoming and outgoing pointed edges therefore carry an old selected-step, successor selected-step, or named old-centre receipt. The terminal alternatives remain separate because they do not have a pair of internal incident edges. The end-to-end type probe imports these projections alongside the reductive endpoint. This still does not identify the two receipted edges with the replacement-square endpoint triangle, construct the rolling bypass, attach end caps, or close L1.

The next comparison layer now puts all three receipt alternatives into one common form: two distinct named faces, their dual edge, and a concrete primal crossing shared by those faces. When that same crossing occurs on one of the replacement square's endpoint triangles, the rotation system's incidence-at-most-two theorem identifies the receipted dual edge with the literal triangle edge. Consequently, if the incoming and outgoing crossings at a strict pointed collision both meet the same square endpoint, the triangle gives a one-edge predecessor-to-successor bypass. No global uniqueness of a shared edge between two faces is used. The exact bond decomposition now also gives the weaker statement justified without guessing the endpoint: provided the two literal receipt crossings belong to the selected four-edge square boundary, each is assigned to one of the two endpoint triangles. Either both assignments use one endpoint and the bypass follows, or the incoming and outgoing edges are returned on opposite endpoint triangles. This corrects an overly strong intermediate expectation that the source coordinates must put both crossings at one endpoint. The remaining source-local obligations are to prove the two square-boundary memberships and consume the explicit opposite-endpoint alternative. The older first–third and second–fourth splice packages consume an earlier four-cell collision type and do not by themselves inhabit the later bounded rolling state. The ignored end-to-end probe imports this comparison beside the reductive endpoint. Thus neither the rolling iteration, the end caps, nor L1 is claimed here.

The selected-cycle interface now supports the corresponding provenance repair directly. Given two distinct steps of a simple dual cycle and literal shared primal edges at those steps, `SelectedDualCycle.ofWalkWithCrossingsAtTwo` retains both crossings definitionally while selecting the remaining crossings from adjacency. Thus the pointed square need not recover its two rail crossings from arbitrary choices after construction. What remains is the source-coordinate theorem locating the incoming and outgoing receipts at the two collision-adjacent steps of that square.

The adapter into the established replacement-cycle tower is now explicit as well. A `CrossingSelection` chooses a shared primal edge at every step of the already-proved fixed dual walk. Its anchor-only instance recovers the existing separator presentation, while `anchorAndTwoSelection` retains three pairwise-distinct source crossings: the corridor-rung anchor and the two pointed rail edges. This changes no facial-dual walk geometry and lets the separator consumers migrate incrementally. The next theorem must still locate the two rail receipts at the collision-adjacent square steps and run the bond classification for that selection.

The incidence bridge needed by that adapter is now public and checked. A literal receipt crossing shared by its original two faces remains a shared interior edge at any supplied walk step whose unoriented facial-dual edge is equal to the receipted edge. This transports incidence only: it neither chooses the collision-adjacent steps nor asserts uniqueness of a primal edge between two faces. The remaining local task is therefore a finite coordinate identification, after which the three-crossing selector can be instantiated without replacing either receipt by an arbitrary choice.

A source-coordinate audit narrows that identification and corrects the last expectation. A terminal-track receipt can arise from an old-placement side-to-side step, a successor-placement side-to-side step, or an explicit old-centre connector. In the first two cases the retained datum displays a side-to-side dual edge; it supplies no equality with either of the replacement square's centre-to-collision steps. The formal packet now preserves this three-way provenance and, in every branch, proves that the named old or successor source corner is an actual endpoint of the literal primal crossing. Thus terminal-track membership alone does not license an instantiation of `anchorAndTwoSelection`. The correct finite remainder is the primal comparison identifying each named source corner with one of the two realized square-bond endpoints; the already proved endpoint- triangle allocation can then consume it. This correction changes no proved square or receipt theorem, and it does not yet construct the bypass, rolling iteration, end caps, or L1.

The endpoint-triangle consumer has now been aligned with that corrected source packet. The two endpoint triangles cover exactly the four selected square-boundary crossings together with their internal bond. A terminal side-to-side receipt may use that internal bond, so demanding membership in the outer four-edge boundary would be stronger than the source geometry. The new exact-packet theorem accepts either kind of crossing. Moreover, once a retained source corner is identified with either endpoint of the realized bond, endpoint-star coverage proves that its named crossing belongs to this packet. For the incoming and outgoing origins, the consumer then returns either the literal predecessor–successor bypass or the explicit opposite- endpoint allocation. Thus the remaining local atom is no longer an ambiguous square-membership claim: it is the finite coordinate theorem that each named source corner is one of the two bond endpoints, followed by the opposite- endpoint discharge. Neither statement is assumed or claimed here.

Trying to prove that last endpoint statement exposed a representation defect one layer earlier.

The replacement short-cycle packet retained one literal anchor crossing, but its remaining primal crossings were arbitrary choices. Its deletion component and realized square bond therefore had no formal dependence on the later pointed receipts. The packet now retains the complete crossing selection. All existing constructors preserve their former choice, while `CrossingSelection.reselect` and `anchorAndTwoCycle` can replace only that choice, retaining the proved facial walk, simplicity, support, and anchor. The two supplied receipt crossings are then definitional projections of the replacement cycle, so the existing separator, component, and bond tower can be reused rather than duplicated. This repairs provenance, not geometry: the finite placement of the two receipts at distinct source steps, the component-side orientation, the opposite-endpoint case, rolling iteration, end caps, and L1 remain open.

The pointed-edge face coordinates are now explicit as well. Each incoming receipt identifies the unordered pair consisting of the rail predecessor and the collision face with its two named source faces; each outgoing receipt does the same for the collision face and successor. Lean retains both possible orientations rather than choosing one from a drawing, and packages the two crossing origins together with these exact facial allocations. This removes face identification from the remaining finite comparison. What remains is to orient the named primal corners onto the chosen component side, identify them with the two square-bond endpoints, and consume the explicit opposite-endpoint alternative. No rolling bypass or L1 closure is claimed by this incidence allocation alone.

The allocation is now wired into the two strict internal alternatives of the actual retained first-window collision. The first-to-second case obtains both receipts from the literal second terminal track, and the symmetric second-to-first case obtains them from the first track; each therefore constructs the complete pointed source-face packet without a new premise. Terminal collisions remain terminal alternatives rather than being coerced into an internal occurrence. The unresolved geometric step is unchanged but now sits at the consumer boundary; the next paragraph records the first construction on that boundary.

The opposite-endpoint branch is now discharged constructively. The two endpoint triangles share the literal facial-dual edge crossing the internal bond of the two-vertex residue. One of that edge's two distinct faces is not the pointed collision face, and the two complete endpoint triangles join the rail predecessor to that face and that face to the successor. Lean packages the result as an actual replacement walk of length at most two and proves that its support omits the collision. The remaining local atom is therefore only the source-placement theorem putting the two pointed crossings into the exact endpoint-triangle packet, followed by splicing this bounded bypass into the retained rail and rechecking separation from its companion. Arbitrary-length iteration, both end caps, and L1 remain open.

That consumer boundary is now sharper. The selected square residue is a literal two-vertex component, so membership of a named source corner in its support proves that the corner is one of the two bond endpoints. With this premise for both pointed crossing origins, Lean consumes both the shared- and opposite-endpoint allocations and constructs the bounded bypass. The premise is stated explicitly: the actual source placement has not yet been proved to put both corners on that component side.

The named-corner formulation is sufficient but not intrinsic, because a crossing edge has one endpoint on either side of the deletion component. The new component-incidence theorem removes this artificial orientation: Lean proves that the exact endpoint-triangle packet consists precisely of the primal edges with *some* endpoint in the selected two-vertex component. Two such edge/component incidence receipts therefore construct the same collision-free bypass without choosing which endpoint lies inside. The remaining source-local statement is correspondingly smaller but still open: the two literal first-window crossing edges must be shown to meet that selected component. No companion-rail separation, rolling iteration, end cap, separated-crosscut constructor, or L1 closure is claimed here.

There is now no extra incidence premise for the selected square boundary itself. The exact component-boundary theorem computes an inside endpoint for every selected primal crossing, and two pointed receipts identified with those boundary crossings feed the complete one- or two-edge bypass directly. Thus the residual local coordinate statement has become a finite classification: each actual first-window pointed crossing must be the crossing of a selected square side or the internal square bond. That classification, companion-rail separation, rolling iteration, both end caps, and L1 remain open.

The canonical first–third square now also carries public source coordinates. Its four facial vertices are definitionally the first, second, and third Cell–3 centres followed by the collision face, and Lean proves that the first two centre-to-centre sides are crossed by the literal outgoing rungs of the first two interiors. A proof-relevant reselection retains both rung crossings while leaving the facial cycle unchanged. This removes an arbitrary crossing choice from the finite comparison, but does not identify either actual pointed rail edge with a collision-adjacent square side or prove that it meets the selected primal component. That incidence remains the live local atom.

The bounded bypass is now also installed in the entire selected rail. Path simplicity identifies the strict occurrence with the unique support index; this proves that the bypass endpoints are exactly the predecessor and successor exposed by splitting the rail. Generic bridge splicing and loop erasure then produce a path with the original outer endpoints, omit the collision face, and prove that every retained face came from the old rail or the two-hop bypass. This is a whole-rail repair, not yet a paired-rail one: separation from the companion, literal pointed-crossing classification, arbitrary-length iteration, both end caps, and L1 remain open.

The selected-boundary case now traverses that whole local stack in one checked statement. If the two pointed primal receipts are members of the selected short-cycle crossing set, Lean constructs the bounded bypass and returns the entire repaired simple rail with its support bound. This remains a conditional integration theorem: it does not prove the finite source classification into selected square sides versus the internal bond, nor companion separation, iteration, end caps, or L1.

The paired-rail obligation has now been cut at its exact support boundary. Every repaired-rail face comes from either the old rail or the supplied two-hop bypass, so Lean proves that disjointness of both pieces from a companion implies disjointness of the repaired whole rail from that companion. This is preservation, not a new geometric construction: the source must still prove that the concrete bypass avoids the companion, as well as the pointed crossing classification, rolling iteration, end caps, and L1.

The sibling parallel face-pair residue also remains open. The existing generic shared-face two-edge separator theorem is stated for a connected spherical cubic rotation map with globally two-sided faces. The present carrier is an opened annulus with boundary stubs, so importing that theorem here would silently strengthen the hypotheses and is not done. Its next consumer must instead be annular and source-local, either a rotation-order reroute or an annular shared-face-pair separation theorem at exactly the strength used by the rail.

The opened-carrier cost is now also kernel-checked rather than merely counted on paper. For any connected component after deleting the three selected crossings, let s be the number of its degree-one boundary stubs, $c \leq 3$ its outgoing darts, and n its vertices. If that component is acyclic, handshaking gives $3n - 2s = 2(n-1) + c$, hence $n \leq 2s + c - 2$. Since the annular carrier has ten named stubs, every such acyclic component has at most 21 vertices. This statement is componentwise and does not assume that the whole complement is connected.

The cyclic branch has a correspondingly sharp conditional consumer. A complementary-side cycle together with cyclic edge-connectivity at least four on the opened carrier contradicts the computed component boundary, whose cardinality is at most three. Boundary saturation is unnecessary: the computed boundary itself is the cut. The theorem is deliberately labelled conditional because neither closed minimality nor the missing outside cycle has yet been transported to the ten-stub opened carrier. A generic closed-walk-to-cycle extractor exists in the tree, but the proved outer-hole walk backtracks at each degree-one stub and no current theorem exhibits an interior facial walk disjoint from this selected component.

The graph-level part of that transport is now proved separately in `GoertzelV24DeleteEdgesCyclicCutPullback`. If a cyclic cut is present after deleting a set of edges, and every restored edge has both endpoints on the same cut side, the two cycle witnesses and the finite cut support pull back injectively to the ambient graph with unchanged cardinality. Thus ambient cyclic edge-connectivity bounds this particular opened cut directly; one need not assert cyclic edge-connectivity for the whole stub-bearing opened carrier. What remains source-specific is to construct the Addendum-V frontier from the closed minimal counterexample and prove that its restored cap/frontier edges are noncrossing for the selected remote component.

For the literal pentagon opening, the cap half of that last premise is now discharged in `GoertzelV24PentagonCapCyclicCutPullback`. Avoiding the five named inner stubs forces every deleted cap-cycle edge to lie entirely on the complementary side, because both endpoints of every such edge are cap vertices. The selected-component theorem already proves this stub avoidance abstractly. Accordingly a cyclic cut on the annulus closes back to a same-cardinality cyclic cut of the pre-opening frontier tangle. The remaining transport is the outer frontier's relation to the induction or minimal-counterexample carrier; it is not hidden in this one-cap theorem.

The complementary cycle created by closing a cap is now explicit rather than pictorial. In `GoertzelV24FacialPentagonCapSideCycle`, the oriented five cap coordinates construct a literal five-edge primal walk; coordinate injectivity proves that walk is a genuine cycle. Consequently, if all five cap vertices lie on one side of a cut, the restored pentagon supplies a `HasCycleOnSide` witness on that side. This removes the need to extract a cycle from the backtracking outer-hole walk when a deleted cap is restored. It does not construct a closed antecedent for an abstract Addendum–V frontier; that source-formation comparison remains separate.

The deletion transport now also has the deliberately asymmetric form needed by that observation. The theorem `pullbackDeleteEdgesOfAmbientOutsideCycle` assumes only an inside cycle in the edge-deleted graph, an outside cycle in the restored ambient graph, exact crossing support of size at most four, and that the restored edges do not cross the chosen side. It then constructs the ambient small cyclic cut with unchanged support cardinality. In particular, it does not pretend that the degree-one-stub presentation already has cycles on both sides.

That generic theorem is now specialized back to the literal facial-cap opening. The constructor `pullbackOpenGraphOneSided_of_vertices_outside` takes exact crossing support and an enclosed cycle in the opened graph; avoidance of the five cap vertices simultaneously proves that restored edges do not cross and places the restored pentagon cycle on the opposite side. The output is an ambient small cyclic cut with the same cardinality. This closes the one-cap, one-sided transport without a boundary-face cycle extraction and without a cyclic-connectivity assertion about the stub-bearing carrier.

The simultaneous two-cap laboratory now verifies the same asymmetric closure all the way to its closed ambient graph. In `GoertzelV24FacialPentagonCapPairCyclicCutPullback`, avoidance of all ten cap vertices proves that none of the ten restored cycle edges crosses the chosen side. An enclosed cycle in the opened graph and the restored inner facial pentagon on the opposite side therefore construct a literal ambient small cyclic cut, with unchanged support cardinality. This is a useful end-to-end test of the transport, not the missing Addendum–V source theorem: the source starts from a frontier tangle with primitive outer stubs and opens one interior cap, rather than deleting two caps from a supplied closed graph.

On a closed graph-backed vertex-minimal Tait counterexample, this laboratory branch is now excluded outright. The module `GoertzelV24MinimalFacialPentagonCapPairNoSmallCut` composes the two-cap pullback with the already-proved cyclic five-edge-connectivity of the closed normal form. Hence no one-sided opened cut of cardinality at most four can have an enclosed cycle while avoiding both cap vertex sets. The theorem uses no cyclic-connectivity premise on the ten-stub opened graph. Its scope remains the supplied two-cap laboratory; it does not identify Addendum V's primitive-outer-stub frontier with an opening of the closed minimal carrier.

The first generic step toward comparing those presentations is now explicit in `GoertzelV24PentagonCapSubgraph`. Whenever all five cycle edges and all five spokes of a cap survive in a subgraph, the cap datum restricts to that subgraph, including its dependent edge embeddings and the exact three-edge incidence equation at each cap vertex. Thus a sequential "open the outer cap, then open the inner cap" factorization no longer needs to rebuild cap combinatorics. The current theorem does not itself prove survival under the outer opening; that is the next separated-pair specialization.

That specialization is now proved in `GoertzelV24TwoPentagonCapFrontierFactorization`. For a separated pair, opening the outer cap first constructs a literal five-stub `FrontierBoundaryData`; if the closed carrier is cubic, this frontier is well formed. Separation proves that all five inner cycle edges and all five inner spokes survive, so the inner cap restricts to an actual interior pentagon cap of the frontier. Thus the two-cap laboratory now has the source-ordered intermediate carrier

$$\text{closed graph} \longrightarrow \text{frontier with outer stubs} \longrightarrow \text{annulus.}$$

This remains a construction for a supplied pair, not a theorem that every abstract Addendum–V frontier has a closed two-cap antecedent. Identifying the second sequential opening with the simultaneous two-cap graph is the remaining comparison at this level.

That graph-level comparison is now exact. In `GoertzelV24TwoPentagonCapSequentialOpening`, the transported inner cap is proved to delete the same five value-level edges as the original inner cap, while the simultaneous pair deletes precisely the union of the inner and outer cap cycles. The standard union law for edge deletion then gives an equality

$$\mathrm{open}_{\mathrm{inner}}(\mathrm{open}_{\mathrm{outer}}\, G) = \mathrm{open}_{\{\mathrm{inner},\mathrm{outer}\}}\, G.$$

Thus the laboratory's simultaneous carrier is literally the carrier obtained in the source order, not merely an abstractly isomorphic graph. The scope is unchanged: this is a comparison for a supplied separated cap pair, not a claim that every abstract Addendum–V frontier has such a closed antecedent. Transport of the dependent boundary and rotation data is still separate.

The ordered boundary part of that transport is now closed in `GoertzelV24TwoPentagonCapSequentialBoundary`. The graph equality is viewed as the identity-on-vertices graph isomorphism, and the generic annular boundary transport is applied to the sequential package. The result is equal to the simultaneous package: all five inner and five outer stub coordinates, and all ten named boundary edges, agree. Thus the comparison cannot hide a cyclic permutation at either hole. This remains laboratory carrier plumbing; it constructs neither a closed antecedent for an arbitrary frontier nor the two source crosscuts.

One generic part of that source-opening transport is now stated at the layer where it belongs. For fully retained ambient faces, `GoertzelV24OpenRegionFaceTransport` proves the biconditional

$$\text{opened face images are equal} \quad\Longleftrightarrow\quad \text{their ambient face orbits are equal.}$$

The forward direction says that literal opening cannot fuse two old complete faces; the reverse direction says that the opened face is independent of the chosen dart naming its ambient orbit. Both directions are derived from the computed complete face-cycle transport, not from an equal-cardinality surrogate. The literal-opening minimal-corridor consumer now uses this generic API instead of carrying its own copies, and the same theorem is available for the distance-two argument. It does not prove that the source-selected faces are fully retained, identify the selected carrier with the literal opening, or establish `CommonNeighborsExact`; those remain the formation step.

The remaining mismatch is structural rather than notational. The proved literal-opening classification starts with a closed graph-backed vertex-minimal Tait counterexample and transports its dual-triangle classification through a fully retained opening. The selected rail assembly starts with a `ClosedWebAtGoodWord.Instance`; this supplies the open annular cellulation and the source corridor, but not the closed antecedent or an identification with that computed opening. Local cubicity does not by itself fill the gap: the third face of a separating dual triangle can be a common neighbour without being one of the selected rung's endpoint flanks. Thus the next source theorem must either construct the Addendum–V closed-to-open identification or eliminate that separating alternative by a source-local reduction. Installing `CommonNeighborsExact` as another field of the annular package would merely make the desired conclusion a hypothesis and is not used.

Selected incidence bridge. The selected Cell–3 carrier now has the missing local consumer at the exact strength supplied by a literal opening. For each dual face adjacent to two consecutive selected corridor centres, it suffices to exhibit one actual centre–face boundary edge whose retained endpoint also lies on the selected outgoing rung. Local cubicity at that endpoint, local two-sidedness of the displayed interior face, and parity of its boundary incidence force this edge to be one of the two named flanks. Lean therefore derives the exact `CommonNeighborsExact` proposition used by selected rail append without assuming globally unique shared edges on the opened annulus. This is a proved incidence-to-flank theorem, not yet the source formation: the Addendum–V construction must still identify the selected `ClosedWebAtGoodWord` carrier with the literal opening and transport the required common endpoint into it.

Do not confuse L8 radial paths with splice crosscuts. Lean already proves `exists_longRadialSectorWitness_at_five`: under its closed-web hypotheses it supplies two vertex-disjoint *primal* radial paths, distinct anchors on both boundary components, sector-alternation data, and a quantitative length alternative. This is an L8 input to the L6 analysis. The splice instead requires `SeparatedAlignedSimpleDualCrosscuts`, a pair of simple *facial-dual* transversals satisfying a dual-walk disjointness condition and the crosscut alignment hypotheses. No theorem currently relates these types. Thus the radial witness is a proved neighbouring asset, not yet a constructor for a splice pair; treating it as one by renaming or duality intuition would hide precisely the missing primal–dual construction. The radial package nevertheless now supplies its exact end-cap contribution: the two distinct anchor positions select two complementary arcs on each named hole boundary, and Lean proves that the two arcs cover the complete inner face-dart boundary and likewise the complete outer one (`innerHoleArcDarts_true_union_false` and `outerHoleArcDarts_true_union_false`). Their intersections are now proved to be exactly the two anchor darts (`innerHoleArcDarts_true_inter_false` and `outerHoleArcDarts_true_inter_false`), so no unrecorded overlap of the two cap choices remains. This is useful fixed boundary data for a future sector construction. It is not an assertion that the caps have been joined by facial-dual side paths, that the resulting walls are simple, or that the two Boolean choices are the complementary annular sectors.

L8 wall audit (14 August 2026). The tempting next construction is not to promote the four-piece walk `RadialSectorWallWalk` into the splice. Its useful local facts are genuine: the two radial cores are disjoint, and, given five further cross-piece disjointness facts, the assembled walk is a trail containing a simple cycle. But the module is an isolated conditional endpoint: no other Lean module imports it, and its five disjointness facts remain inputs rather than consequences of the Cell–3 carrier. In contrast, the exact-cut, hole-free-side and drainage results on which Sector Alternation depends are already consumed by `ClosedWebSectorAlternation`. Thus the L8 sector argument is not being reopened under a new name. The live task stays the source-specific construction of the L1/L6 simple transversals or layer boundaries, whose deletion components are exactly the input required by `ChordTransversalDeletionAssignment`. This is a dependency audit, not a negative result about the manuscript's sector argument. There is a useful but narrower interaction already formalized: a counting argument finds one radial component that avoids the four primal edges of a chosen local Cell–3 layer wall. That path certifies retention of the inner hole *after* the local paired-crosscut splice data have been supplied. It neither constructs the two facial-dual side paths nor discharges the local pair's endpoint-separation premise. This is evidence that the radial and facial data cooperate, not evidence that either data type can be substituted for the other.

The actual L6 geometric handoff. The finite pigeonhole itself is already source-realized: `GoertzelV24ClosedWebLengthDepthDichotomy` proves that a sufficiently long actual radial path has either a clean shallow block or a `HasDeepChordTransversal`. The latter is deliberately not called a layer stack. Sector alternation and chord noncrossing organize its many spanning chords, but the profile layer correctly refuses to treat a facial chord wall as a primal cut. The explicit Lean interface for the deep/layer branch is `ChordTransversalDeletionAssignment`. For each selected sector chord it asks for a separate finite deleted-edge transversal, a connected component whose crossing boundary is exactly that deleted set, cycles on both sides, and a uniform width bound. Its conversion to the cyclic-cut/profile machinery is already proved; only this physical construction is absent. A direct re-reading of the current Fable manuscripts finds the same boundary: Addendum XXVII names transversal positions and layer boundaries, and uses their simplicity in the splice checklist, but supplies no deleted-edge construction or graph-side component for either coordinate. This is therefore a source gap now made explicit in Lean, not a claim that the manuscript's pumping argument has been refuted. On 14 August the interface was tightened: L6 first selects one large laminar sector, so the new `ChordSectorSideAssignment` only requests graph-derived sides for the chords in that selected sector. Its nested-equal-profile theorem is kernel-checked. This removes an accidental requirement to construct transversals at every unrelated cut, but it is not itself a transversal construction. The shallow branch remains separately honest: each chosen `radialBlockCut` still needs its own simple transversal. Thus the assignment is one explicit half of the L6 geometry, not a substitute for the whole dichotomy. In particular, the wall itself is not a shortcut: Lean proves that if a chord wall were an exact primal cut, its closed walk would have even length. This records a necessary parity condition rather than inventing the source's separate simple transversal.

The annular-separation implementation boundary. This is not a place to import a large unrelated planarity library. The pinned Lean dependencies contain neither a Jordan-curve theorem nor a rotation-system or combinatorial-map separation development, and a foreign representation would first require an equivalence with the rotation-system carrier already used by the closed-web construction. More importantly, the present development has already isolated the exact narrow interface required here: `CycleSideCertificate` assigns a vertex side to a specified chord cycle, proves alternation on the cycle and preservation off it, and retains a cycle witness on each side. From that interface the exact finite edge cut is mechanical. Thus the live construction problem is a source-specific annular lemma which produces that data (or directly produces the equivalent deleted edge set and component) for the actual radial/chord geometry. It is neither an appeal to Menger's theorem nor an assertion that the facial dual labels alone select a primal side. This boundary keeps the eventual theorem at the strength used by the corridor, while making every genuinely topological input visible to kernel checking.

What the deep chord work already discharges. The face-side part of this construction is no longer conjectural. For every eligible same-path chord, `exists_exact_chordCycle_faceCut_with_holeFreeSide` supplies binary labels on the incident faces, proves that their disagreement set is exactly the chord-cycle edge support, and selects a side away from both named holes. This is useful, but it is deliberately only facial-dual information: it must not be read as a vertex side merely because it has the right cut edges. The existing constructor `cycleSideCertificate_of_exactCutLabelSide_of_rotated_external_ports` shows the remaining bridge exactly. It needs the local wall-to-endpoint port orientation and cycle witnesses on the selected vertex side and its complement; a reusable transport can provide a selected-side cycle when its face-component reachability premise is supplied, but it does not manufacture the complementary witness. Consequently no further profile state or finite audit is being hidden here. The open work is the primal annular-separation argument turning the certified facial cut into the two cycle-bearing graph sides required by the deletion assignment.

The rotated-port interface itself has also been corrected. Its first form asked for an orientation for every pair of vertices belonging to a wall edge, which inadvertently included the same endpoint twice and was impossible for a simple edge. The repaired `ChordWallRotatedExternalPorts` asks exactly for one oriented wall dart per wall edge whose two adjacent rotational darts leave the wall. Its two endpoint memberships are then derived from that dart rather than supplied as an over-quantified premise. This is a type-level repair of the conditional bridge, not a construction of those oriented darts from the annular geometry.

What one dual cycle now yields, and what it does not. The framed separator has been tightened at exactly this representation boundary. For any supplied simple facial-dual cycle it now re-expresses the crossed primal support as a finite graph-edge carrier and extracts a deletion component with a nonempty computed boundary: `exists_componentCrossingEdges_nonempty_of_dualCycle`. This is useful because a later source construction need not guess the retained component or the finite edge carrier after it has built its actual dual cycle. It is deliberately not a transversal construction: the theorem does not supply that cycle, prove its computed boundary equals an intended wall, choose the annular side, or establish the two side-cycle witnesses. Those are precisely the source-specific annular-separation obligations still visible in `CycleSideCertificate`.

7.4 The closure chain

The remaining dependency order is:

1. construct the source's genuine simple transversals/layer boundaries in both L6 branches: at shallow radial positions and, for the deep branch, through the explicit `ChordTransversalDeletionAssignment` package; then identify the resulting physical cut data needed by the splice;

2. use it to discharge the literal source `SemanticProfileBridge` or the weaker sufficient `ColorabilityReflection`;

3. discharge the named output receipts and use the now-constructed graph-backed output instance to supply the carrier-changing `ReductiveSystem`. Strict decrease and the complete packed certificate assembly are already checked; the missing content is the physical large-input supplier, not another abstract spine;

4. derive an explicit threshold and implement `BaseVerified` from the route's own finite base. No explicit threshold or base witness exists at this revision;

5. apply the abstract induction theorem to expose the concrete source-facing Four-Color statement.

An end-to-end Lean type probe has now checked this assembly boundary rather than extrapolating it from the separate lemmas. At the bare rotation-system level, a completed splice and its spherical-output certificate package directly as a strictly smaller counterexample-preserving object, and a supplier of such certificates constructs the abstract `ReductiveSystem`. The primary manuscripts impose a stronger and more informative invariant at Cell 3, however: Addendum XXVII descends through closed-web-at-good-word instances, represented here by `GoertzelV24ClosedWebAtGoodWord.Instance`. A source-faithful packed object must therefore carry its varying simple graph, annular boundary data, coloring, annular embedding, geometry, total closure, and good word. The corresponding reductive spine also typechecks, and exposes one exact remaining contract: every sufficiently large packed instance must construct a strictly smaller packed instance. Thus the existing size and colorability-reflection theorems genuinely fit the final spine, but they are not alone a concrete Cell–3 reduction; the splice must reconstruct the full output instance. Recording that distinction now prevents a late carrier mismatch without weakening any conditional theorem into a construction.

The same type audit reached the first semantic splice constructor and found a sharper mismatch. Addendum XXVII performs its preservation checklist on an open annular tangle with boundary stubs. The existing Lean completed-step constructor, by contrast, requires global cubicity of that annular carrier. This is not merely an inconvenient premise:

```
GoertzelV24ClosedWebAtGoodWord.Instance.not_isCubic
```

is a direct theorem, because the first named inner stub has degree one.

The first repair is now a kernel-checked construction. The source boundary package retains one actual dart based at its deleted-side vertex, obtained from connectedness when that package is built. That datum now roots the literal complementary open region and its two-sided serial-tangle presentation without a global degree premise. The concrete boundary agreement and the theorem gluing two Tait-coloured open pieces back to an ambient Tait colouring likewise have source-facing versions with no cubicity hypothesis. Older signatures are preserved as compatibility wrappers, and the previous completed-step consumer still rebuilds, so this localization did not fracture the established profile/count tower.

That remaining adapter is now kernel checked. From the literal complementary open region rooted by the retained source-boundary dart, it computes the two complete five-coordinate interface profiles and the associated finite profile-pair count by filtering actual Tait colourings. Positivity of an entry is therefore equivalent to a literal colouring witness. A one-way diagonal lift supplies the reverse completion needed by Addendum XXVII, and a generated finite seam-profile audit now yields a no-global-cubicity completed splice, counterexample preservation, and strict size decrease. The axiom audit of these headline interfaces contains only `propext`, `Classical.choice`, and `Quot.sound`.

The word "audit" remains load-bearing. The theorem assumes equality of the finite seam-word supports generated by the retained and complementary full profile diagonals; it does not yet prove that equality for the source's selected corridor. Likewise it does not construct the physical separated crosscuts (Lean-flag L1), reconstruct the shortened packed closed-web-at-good-word instance, derive the threshold, or instantiate the carrier-changing reductive system. What has closed is the formerly inconsistent no-cubicity splice interface, not the formation pillar or the Four-Color Theorem.

The carrier-changing logical endpoint is now also a public Lean interface. The module `GoertzelV24ClosedWebReductiveAssembly` packages a varying finite simple graph with its ordered annular boundary, colouring, and complete closed-web-at-good-word witness. Given a named supplier that constructs a strictly smaller packed witness above a freely chosen natural threshold, it builds the corresponding `ReductiveSystem` and proves that exclusion of all packed witnesses is equivalent to `BaseVerified`. The universe of the changing carrier is pinned explicitly, and the axiom audit again contains only `propext`, `Classical.choice`, and `Quot.sound`.

This is deliberately a conditional assembly theorem. No closed term of the supplier exists yet: it must combine the physical crosscut construction, the finite full-profile audit, and reconstruction of every closed-web output invariant. Publishing the interface now fixes the final carrier and exposes that one contract without counting it as the reduction itself.

An end-to-end type probe then exposed the first reconstruction joint inside that contract. The literal splice returns a rotation system with a changed edge carrier, whereas the packed descent interface expects graph-backed data. The module `GoertzelV24RotationSystemPrimalEdgePresentation` now proves that every rotation-system edge maps onto an edge of its computed primal simple graph, and that endpoint injectivity upgrades this map to a canonical edge equivalence. For the actual ordered-cut output, the existing finite `OrderedCutEndpointSimple` test supplies exactly that injectivity. This closes the edge-carrier part of the reconstruction without assuming that an arbitrary splice is simple. The source construction must still discharge the finite endpoint test before a packed shortened witness has been constructed.

The dart-and-rotation half of the same reconstruction joint is now closed as well. Under precisely that endpoint-injectivity premise, `GoertzelV24RotationSystemPrimalGraphBacking` gives a canonical equivalence from the literal splice darts to the darts of the computed primal simple graph. It proves compatibility with dart reversal and with the vertex rotation, and therefore with facial steps, then packages the result as actual `SimpleGraphDartRotation.Data`. Its ordered-cut specialization consumes the same finite `OrderedCutEndpointSimple` receipt; it does not silently derive that receipt from arbitrary local crosscuts. Thus the remaining representation work at this joint is narrower still. The module `GoertzelV24RotationSystemPrimalColorTransport` proves that the canonical edge equivalence is an isomorphism from the rotation system's edge-adjacency graph to the line graph of its computed primal graph. It then transports the literal proper splice colouring and its nonzero Tait condition, including an ordered-cut specialization using the already-proved equal-profile splice colouring. This result consumes the same finite endpoint receipt; it does not construct it. What remains is to derive that receipt from the physical source cut pair, transport the two named hole faces, and reassemble the shortened packed closed-web witness. This is a checked representation joint, not L1/C–5 closure.

The named-face part of that remaining list is now transported too. For graph-backed source rotations, `GoertzelV24SimpleGraphRotationEndpointInjective` proves directly that unordered endpoint sets identify edges. The literal dart equivalence then descends to a facial-orbit equivalence in `GoertzelV24RotationSystemPrimalFaceTransport`. Finally, `GoertzelV24ClosedWebSpliceGraphHoleTransport` applies that equivalence to the two already-retained source holes and proves that their graph-backed output faces are distinct. The hypotheses remain exact: both hole faces must be wholly outside the pumped region, and the ordered-cut endpoint-simplicity receipt must be supplied. Thus this is not yet an output annular cellulation; the boundary data, connectivity and Euler fields still have to be assembled, and the physical corridor still has to produce the endpoint receipt.

The ordered boundary carrier is now reconstructed on the same graph-backed output. The module `GoertzelV24ClosedWebSpliceGraphBoundaryData` chooses literal internal darts over each wholly retained source boundary edge, sends those old edges through the canonical primal edge equivalence, and proves that the five inner and five outer edge coordinates remain injective. The retained source stubs likewise give ordered vertex embeddings. Together these data form an actual `AnnularBoundaryData` on the shortened simple graph. Its graph-level validity is now proved in `GoertzelV24ClosedWebSpliceGraphBoundaryWellFormed`. The two ordered stub families and their named edges remain disjoint, every named edge remains incident to its retained stub, all ten stubs retain degree one, and every other retained vertex remains cubic. The proof uses the literal retained-edge endpoint calculation and the canonical dart equivalence to the computed primal graph; it does not replace the open tangle by a globally cubic carrier.

The ignored end-to-end type probe now reaches an actual `StrictReductionCertificate`. Given the already named physical splice receipts (equal profile, both holes retained, and endpoint simplicity), the boundary-validity and output Tait fields are constructed, and strict decrease is discharged by the retained-vertex count. The remaining output-instance arguments were therefore explicit: connectedness, the graph-backed annular embedding, its frontier geometry, total closure, and preservation of the good inner word.

The good-word item is now constructed rather than passed to that probe. `GoertzelV24ClosedWebSpliceGraphGoodWordTransport` traces each ordered inner boundary edge through the retained internal splice edge and the canonical graph-edge presentation, proving that its output color is the literal equal-profile color. When that profile coloring is the source graph coloring transported to the source rotation system, the five output colors are definitionally the source inner word, so the source's proved $(3, 1, 1)$ block supplies the output good-word field. The coloring-alignment premise is essential: an arbitrary coloring with equal cut profiles need not preserve the distinguished source boundary word.

The output annular cellulation is now constructed as well, rather than passed to the type probe. The module `GoertzelV24ClosedWebSpliceGraphAnnularCellulation` proves that ordered deletion and sewing preserve cyclic vertex rotations, transports connectedness from the retained ambient primal carrier, derives the output Euler equation from the splice's local vertex–edge–face balance, and packages the two already-transported distinct hole orbits into a literal `FramedAnnularCellulation` on the computed output graph. This uses no global cubicity premise on the stub-bearing annulus.

The stronger source embedding is now constructed as well. The module `GoertzelV24ClosedWebSpliceGraphAnnularEmbedding` first proves a quotient-safe retained-dart transport theorem: every retained dart on a protected source face lies on its canonical output face, using the bijection on the complete face cycle rather than choosing a convenient orbit representative. The exact two-dart edge-fiber theorem then handles both orientations over each retained boundary edge. Hence the graph-backed output root lies on the transported outer hole, every dart over the five ordered inner edges lies on the transported inner hole, and every dart over the five ordered outer edges lies on the transported outer hole. These incidences assemble the existing cellulation into an actual `ClosedWebAnnularEmbedding`; no conditional incidence bundle remains.

The minimum-face part of the output frontier geometry is now constructed too. The module `GoertzelV24ClosedWebSpliceGraphGeometry` first observes that the two source holes each contain their five injectively named boundary edges, while the interior source faces already carry the certified lower bound. The finite seam-return predicate therefore applies the existing global splice theorem to every literal output face. A canonical equivalence between the complete dart fibers of corresponding quotient faces then transports facial orbit cardinality through the simple-graph backing. Consequently `outputGeometry` derives the minimum-five field rather than accepting it as output data.

The ignored end-to-end probe still typechecks after consuming this constructor. Its remaining output-instance receipts are now the finite seam-face return test, simplicity of every internal output facial boundary, connectedness of the internal output facial dual, and total closure. The latter two topology properties and total closure are not consequences of a cardinality transport, and remain explicit. The source side must still construct the physical splice receipts—equal profiles, coloring alignment, both retained-hole receipts, endpoint simplicity, retained-side connectedness, and the local Euler balance. Thus the graph-backed annular-embedding reconstruction joint and the minimum-face geometry receipt are closed, but this remains an assembly audit rather than a supplier for arbitrary large inputs: the physical L1 crosscuts, the two remaining output topology facts, and total closure are not yet constructed.

The remaining topology is now stated on only one representation. The module `GoertzelV24RotationSystemPrimalGeometryTransport` proves that canonical simple-graph backing sends the boundary support of every literal quotient face exactly to the corresponding graph-backed support. It also constructs a graph isomorphism between the facial duals induced on any selected literal face set and its canonical image. The splice specialization identifies the internal faces of the computed output cellulation with the image of the literal non-hole faces. Hence `outputGeometryOfLiteral` takes facial-boundary simplicity and internal-dual connectedness on the literal splice rotation and transports them to the actual graph-backed output.

This removes a representation debt, not a topological hypothesis. The literal splice still has to be proved to have simple internal facial boundaries and connected internal dual, and the resulting output still needs `TotallyClosedWeb`. The ignored end-to-end probe composes the literal topology interface through the actual graph-backed output geometry and checks the public `StrictReductionCertificate` and `ReductiveSystem` interfaces. It does not yet construct that certificate: reconstruction of a complete smaller closed-web witness remains a separate joint. This keeps the probe useful without presenting a conditional interface check as a source construction or Four-Color theorem.

The facial-boundary receipt has now been reduced to its actual finite seam content. The module `GoertzelV24ClosedWebSpliceBoundarySimplicity` splits every literal output face into the two cases already used by the minimum-face proof. If a face avoids the seam, its complete dart cycle and its retained output edges transport to one source face. For a non-hole output face that source face cannot be either protected hole, so the Cell–3 interior geometry supplies its simple boundary. If the face meets the seam, the ordered seam-position bijection chooses a canonical finite root and the new `SeamFacesBoundarySimple` predicate supplies exactly that face's receipt. Global two-sidedness is neither assumed nor true for the annular source: a degree-one hole edge has both dart sides on the same named hole.

Consequently the end-to-end type probe now reaches the actual output-geometry constructor while replacing the infinite-looking "every literal internal face is simple" premise by one finite test over the ordered seam positions. This is a real topology reduction, not yet the noncrossing proof of that finite test. Literal internal-dual connectedness and `TotallyClosedWeb` remain independent output receipts, and the physical crosscut constructor still must supply the previously listed source splice data.

That finite test now has a topological normal form. The predicate `SeamFacesTwoSided` asks only that a face reached from an ordered seam dart not also contain the opposite dart over the same edge. Lean proves this pointwise condition equivalent to `SeamFacesBoundarySimple`: either condition makes the edge map injective on every seam-root face, and boundary injectivity conversely excludes the fixed-point-free opposite dart. The ignored end-to-end probe consumes this two-sided form and reaches actual output geometry. Thus the open noncrossing step is no longer an opaque face-cardinality receipt: it is the local statement to be derived from the source's simple transversals and their finite seam-return successor. This equivalence does not construct those transversals, prove internal-dual connectedness or total closure, or close L1.

The canonical return arcs make this local statement explicitly finite. `GoertzelV24ClosedWebSpliceSeamArcNoncollision` partitions every seam-touching face into the half-open arcs between successive seam hits. Its finite receipt has exactly two clauses: the edge trace within one arc is injective, and distinct arcs joined into one completed face have disjoint edge supports. Lean proves their conjunction equivalent both to seam-boundary simplicity and to seam-local two-sidedness. The end-to-end reduction probe reaches actual output geometry from this form and separately checks the public carrier-changing reduction interface. Thus the remaining source proof separates into the simple retained ambient face segment inside each arc and the noncollision of different arcs under the simple-transversal gluing; neither clause is assumed proved here.

The first of those two clauses is now constructed from the actual source geometry. In `GoertzelV24ClosedWebSpliceSeamArcAmbientSimplicity`, positive positions on a canonical arc transport to consecutive darts of one ambient source face, and equality of their retained output edges reflects back to equality of the source edges. The position-zero seam edge is new and therefore cannot equal any positive retained edge. Finally, `HoleFaceKept` excludes both named holes from being that ambient face, so the already-proved simplicity of interior source-face boundaries supplies injectivity. The strengthened end-to-end probe therefore assumes only `SeamArcsPairwiseEdgeDisjoint`; cross-arc noncollision under the homeomorphic simple-transversal gluing is the remaining seam-boundary atom. This does not prove internal-dual connectedness, reconstruct the output closed web, construct the physical crosscut pair, or close L1.

The final bookkeeping ambiguity inside this seam receipt has now been removed. In `GoertzelV24ClosedWebSpliceSeamArcClosureCriterion`, retention of the two named hole faces supplies internal arc simplicity, and Lean proves that pairwise edge-disjointness of distinct canonical arcs is equivalent to seam-local two-sidedness. The ignored end-to-end probe now takes the latter, source-topological formulation and derives the former before constructing the actual output geometry. A separate audit rules out a tempting shortcut: the already-proved spherical splice package records cubicity and Euler characteristic, but not two-sidedness. Hence the remaining L1 atom must come from the homeomorphic simple-transversal gluing (or an equivalent proof that no seam-face edge becomes a bridge), not from Euler arithmetic and not from a global two-sidedness assumption, which is false at the degree-one holes.

The finite return pattern has now been calculated at arbitrary cut width, for both possible boundary orientations. In `GoertzelV24OrderedSeamReturnAdjacency`, the ordered cap return is proved to be an involution. A left seam root and a right seam root belong to the same nontrivial return cycle exactly when their offsets differ by one, with the direction reversed when the boundary orientation is reversed. Consequently same-side seam roots belong to one return cycle only when they are equal. Every distinct two-root return cycle therefore joins neighboring offsets on opposite sides, reducing the outstanding cross-arc receipt to a uniform neighboring-strip disjointness statement for the long, end-capped source crosscuts. This is only the finite permutation part of

Addendum XXVII's homeomorphic-splice claim; it neither constructs those crosscuts nor proves the resulting geometric arcs edge-disjoint. In particular, it does not revive the already-refuted width-two local layer as a global splice boundary.

The calculation is now connected to the literal splice rather than remaining an abstract permutation model. The module `GoertzelV24ActualSeamReturnOrientation` extracts the return equality that had previously appeared only as a local step inside the planar-bond face count: for either coherent source orientation, the actual first-return map of the constructed splice is exactly the corresponding capped boundary return. It follows in the actual splice that distinct same-side roots never share a completed seam face, while any opposite-side pair that does share one has adjacent offsets. Thus no unexamined nonlocal pair remains in the seam receipt. This transport still assumes the already-constructed planar bond; it is not a constructor for the source crosscut pair, and the adjacent-strip edge-disjointness statement remains the physical L1 obligation.

The exact reduced receipt is now public in `GoertzelV24OrderedCutAdjacentSeamArcCriterion`. Lean proves, under that actual two-orientation return calculation, that pairwise edge-disjointness of all canonical seam arcs is equivalent to checking only those actual left/right arc pairs whose offsets are adjacent. Same-side and non-neighboring pairs disappear by proof, rather than by an informal locality assertion. The new criterion is intentionally not counted as L1 closure: its right-hand side is the bounded neighboring-strip geometry which the long source crosscuts must still supply.

The carrier-changing reconstruction has now been assembled end to end in a public theorem rather than only sampled by the disposable type probe. Given the still-visible physical receipts, `GoertzelV24ClosedWebSpliceOutputInstance` constructs the complete shortened `ClosedWebAtGoodWord.Instance` on the computed primal graph: boundary validity, connectedness, annular embedding, frontier geometry, the transported Tait colouring, total closure, and the good inner word all inhabit one value. The same module packages that value as the public reductive spine's `StrictReductionCertificate`, with strict decrease supplied by the proved splice-size theorem. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

This closes an integration risk, not the physical formation theorem. The constructor still takes both protected-hole receipts, endpoint simplicity, retained connectivity, the local Euler balance, the seam-five and seam-two-sided conditions, literal internal-dual connectedness, and output `TotallyClosedWeb`. In the live pointed-rail repair, membership of the two named source corners in the selected square component is likewise only a sufficient orientation premise: a crossing edge has one endpoint on each side, so terminal-track provenance alone does not decide it. The source rotation must orient those corners, or the short cycle must be reselected with the pointed crossings at the required steps. No large-input reduction supplier, numerical threshold, base verification, or Four-Color headline is claimed here.

After the pointed bypass and companion-separation refinements, the disposable end-to-end type probe was rerun through the actual carrier-changing strict reduction certificate, the public `ReductiveSystem`, and `target_all_iff_baseVerified`. It remains green. Thus the present closure risk is not a hidden mismatch between the splice output and the abstract descent interface. It is the visible mathematical supplier: for each sufficiently large counterexample, construct the literal separated cuts and discharge every output-geometry receipt required by the shortened closed web. No such universal supplier is claimed by this probe.

The older six-centre companion construction was checked again against its window-provenance theorems. Those theorems already prove separation from all genuinely remote selected windows. They do not settle the local centre condition, because the exact closed-neighbourhood provenance intentionally allows the adjacent Cell–3 centres and the centre-bridge repairs may traverse them. The current local frontier is therefore unchanged and sharper: prove that the two actual pointed crossing edges meet the selected two-vertex square component, prove the resulting bypass disjoint from the companion rail, and then iterate and attach the two literal end caps.

The square-separator interface now retains the source coordinates needed for that comparison. The strengthened constructor `exists_sharedCollisionSeparatorReceipt_eq` returns the physical separator together with the kernel equality identifying its selected cycle with the input cycle. Its first–third specialization starts from the literal two-step corridor-square witnesses and returns the canonical four-step source cycle with the same equality. This repairs an information-loss seam in the API: later pointed-edge work no longer has to reconstruct that an arbitrary short separator came from the displayed Cell–3 square. It does not yet show that the two pointed rail edges occupy the square boundary or internal bond, so the component-incidence atom and L1 remain open.

The canonical square now also exposes its four facial-dual vertices in source order—first centre, second centre, third centre, collision face—through public kernel-reducing coordinate lemmas. The second centre-to-centre side is proved to carry the literal outgoing rung of the second corridor interior, in addition to the first outgoing rung already stored as the cycle anchor. Thus the endpoint comparison may use two physical source anchors without making the false stronger claim that either pointed rail edge is itself one of the two collision-adjacent square sides.

The acyclic square residue now consumes both of those anchors. In `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareSourceRungIncidence`, the realized two-vertex bond is proved to meet the second literal centre-to-centre rung, while the earlier anchor theorem supplies the first. This is an unconditional graph-level incidence fact obtained from the exact selected crossing set. It does not choose the endpoint met by either rung, place either pointed rail corner in the component, or yet construct the local bypass; that finite source-coordinate comparison remains the live L1 atom.

The two independent rung incidences are now retained as one finite source packet. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareSourceRungAllocation` classifies the literal first and second rungs into exactly four cases: both meet the first bond endpoint, both meet the second, or they split in either orientation. It then constructs the two exact endpoint triangles, preserves their singleton internal-edge intersection and their union with the selected square boundary, and transports the same four-way allocation to the triangles' selected crossing sets. This is the bounded case distinction required before the pointed comparison; it still does not prove that either pointed crossing meets the bond, so no bypass, rolling iteration, end cap, or L1 closure is claimed.

The complementary square-boundary packet is now equally explicit. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareSourceRungRemainder` proves that the two literal consecutive source rungs occupy distinct steps of the selected simple dual four-cycle and that deleting them leaves exactly two selected primal crossings. This is a cardinality and provenance theorem, not the pointed comparison: it does not identify the two remaining crossings with the two actual pointed rail edges. That finite source-coordinate identification, companion separation, rolling iteration, both end caps, the separated-crosscut constructor, and L1 remain open.

Those two remaining edges now have exact square coordinates. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareCollisionSideCrossings` proves that the remainder is precisely the selected crossing pair at square steps two and three—the third-centre–collision side and the collision–first-centre side. Thus no anonymous two-edge set remains in the local comparison. The still-open source theorem is to show that the actual pointed incoming and outgoing rail edges, whose proof-relevant origins have three constructors, occupy those two named sides; no companion separation, iteration, end cap, separated-crosscut constructor, or L1 closure follows from the coordinate theorem alone.

The four-edge boundary now also has a single exact normal form. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareCrossingPacket` proves that the selected crossing set is precisely the first source rung, the second source rung, and the two collision-adjacent crossings above. This eliminates an implicit set-reconstruction step from every later consumer. It does not classify either actual pointed crossing into that packet or into the endpoint triangles' additional internal bond, so the local source comparison and all subsequent rail-repair obligations remain open.

The endpoint table for that normal form is now finite and exact. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareEndpointAllocation` first classifies the two collision-side crossings by which endpoint of the realized bond they meet. Combining that four-way allocation with the earlier four-way source-rung allocation appears to create sixteen cases. Local cubicity proves that exactly two selected square crossings meet each endpoint, so ten cases are impossible and exactly six incidence patterns remain. Each case carries all four literal endpoint-incidence proofs. This is not yet the pointed comparison: it does not identify either actual pointed rail crossing as one of the four square-boundary crossings or as the endpoint triangles' internal bond. Companion separation, rolling iteration, both end caps, the separated-crosscut constructor, and L1 therefore remain open.

The six cases now have an exact geometric consequence rather than a diagrammatic one. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareCornerResolution` proves that two distinct packet edges sharing a source corner force precisely one of three alternatives: the corner is the first bond endpoint, it is the second bond endpoint, or it is exterior and the two edges are exactly the two arms joining that corner to the two endpoints, with their orientation retained. The exterior fan is not assumed impossible. This also explains why the later pointed receipts cannot be replaced by mere first-window membership: that membership does not determine which way the complete rail leaves the collision. The next finite comparison must apply this trichotomy to the literal pointed origins and either establish their bond/component incidence or retain the exterior fan as an explicit obstruction. No companion separation, iteration, end cap, separated-crosscut constructor, or L1 closure is claimed here.

The third arm at such a corner is now classified as well. Module `GoertzelV24ClosedWebSelectedLocalRailMiddleReplacementSquareThirdEdgeResolution` proves that a third distinct incident edge either meets the selected two-vertex residue at the identified endpoint, or, in either exterior-fan orientation, avoids both endpoints. The latter conclusion is forced by simple-graph uniqueness: meeting an endpoint would make the third edge equal to the already named arm. The result is specialized to both the source-rung pair and the collision-side pair. What remains is the literal coordinate attachment showing that each actual pointed origin is the corresponding third edge; no companion separation, rolling iteration, end cap, crosscut constructor, or L1 closure is inferred from this local dichotomy.

The first proof-relevant origin case is now eliminated without such an attachment. An old-centre receipt joins the second Cell–3 centre to one selected side face. At a strict collision different from that centre, its unordered orientation is forced: on entry the predecessor is the centre, and on exit the successor is the centre. A simple rail therefore cannot carry old-centre receipts on both sides of the collision, since its predecessor and successor would coincide (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeCenterCoordinate`). The public statement excludes even the simultaneous existence of both old-centre receipts, rather than depending on a particular stored origin tag. The remaining finite comparison concerns the old- and successor-forward corners. In particular, no exterior-fan case has been discarded and L1 is still open.

Each surviving forward receipt now exposes the complete primal star at its source corner. The two consecutive displayed Cell–3 boundary arms and the crossed third edge are pairwise distinct, and all three meet that corner (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeForwardCornerCoordinate`). This uses the receipt's local three-dart fibre rather than global cubicity of the opened carrier. Thus the remaining coordinate comparison has been narrowed again: it must identify the appropriate two boundary arms with the corresponding pair in the exact source-square packet. The exterior-fan alternative remains explicit; no endpoint identification, repair, end cap, crosscut constructor, or L1 closure follows yet.

A further coordinate check rules out one tempting identification rather than assuming it. The two displayed arms of a forward receipt are exactly the face-cycle edges at its two non-rung side slots. Boundary simplicity of the certified interior hexagon therefore makes each arm different from both the incoming and outgoing selected rungs of that placement (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeForwardArmRungExclusion`). Thus the next comparison cannot simply rename the two arms as the two source rungs. It must use the actual mixed/collision-side square coordinates, or prove component-boundary incidence directly. This is a checked exclusion, not a construction of the pointed bypass, rolling repair, end caps, or L1.

The complete square packet rules out the stronger arm attachment as well. For an old-forward receipt based at the second Cell–3 placement, neither boundary arm is any of the canonical first–third square's four selected crossings (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeOldForwardSquarePacketExclusion`). The two rung alternatives are the non-rung exclusions above. Each of the two remaining collision-side crossings already belongs to the two consecutive faces of the square; since a primal edge is incident with at most two facial orbits, it cannot also belong to the second-centre boundary containing either arm. Thus the square cannot attach to this forward corner through either displayed arm. The live comparison is now the receipt's crossed third edge, or another literal component-boundary incidence. This is again a negative coordinate theorem, not a pointed bypass, rolling repair, end cap, separated crosscut constructor, or L1 closure.

That exclusion also fixes the component-side orientation of an old-forward corner. If the corner were one of the selected square component's two vertices, both distinct arms would meet the component. The exact endpoint packet would then force both arms, already excluded from the four-edge boundary, to equal the single internal bond—a contradiction. Hence the corner lies outside the component, and the crossed third edge cannot be the internal bond (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeOldForwardSquareComponentExclusion`). For this origin, membership of the third edge in the endpoint-triangle packet is therefore exactly membership in the source-square boundary, equivalently the intrinsic condition that its other endpoint meets the selected two-vertex component. This refutes the older named-corner-inside route for this case and isolates the remaining finite atom as the far-end incidence of the crossed third edge. It does not yet prove that incidence or construct the bypass, rolling repair, end caps, separated crosscuts, or L1.

That remaining endpoint is now literal rather than existential. The module `GoertzelV24ClosedWebSelectedLocalRailPointedEdgeOldForwardSquareFarEndpoint` names the endpoint of the crossed third edge opposite the source corner and proves the exact unordered endpoint equation. Since the source corner is outside the selected component, membership of the third edge in the source-square boundary is equivalent to membership of this one named far endpoint in the selected two-vertex component. Thus the local question has been reduced to one coordinate proposition, but that proposition is not yet proved; neither the bypass, rolling repair, end caps, separated crosscuts, nor L1 is claimed.

The occurrence-sensitive face allocation changes how that endpoint criterion is consumed.

At an actual pointed collision on the canonical first–third square, an old-forward origin based at the second Cell–3 placement makes the collision face one of that placement's side faces. It is therefore adjacent to the second corridor centre, the opposite vertex of the four-cycle. Lean now proves that this is the literal missing diagonal of the source square (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeOldForwardSquareChord`). For each of the incoming and outgoing pointed edges, and jointly for the pair, the result is fail-closed: either this chord is exhibited or the corresponding proof-relevant origin is not old-forward. Thus the actual old-forward branch does not require an unsupported far-end-membership assertion; it enters the already named square-chord reduction lane. Consuming that chord and finishing the remaining successor-forward/old-centre origin combinations, companion separation, rolling iteration, both end caps, the separated-crosscut constructor, and L1 remain open.

The residual origin table is also now exact. In the non-chord branch neither incident edge has an old-forward origin, while simplicity of the pointed rail forbids old-centre origins on both sides of the same non-centre collision. Consequently Lean reduces the table to precisely three rows: successor/successor, successor/old-centre, and old-centre/successor (`GoertzelV24ClosedWebSelectedLocalRailPointedEdgeSourceSquareOriginReduction`). This is an exhaustive local classification, not yet a consumption of the chord or a construction of the rolling repair, end caps, separated crosscuts, or L1.

The primal crossing of that diagonal is now retained without conflating two different displayed edges. The old-forward third edge locates the collision side face; the edge that actually crosses the second-centre–collision diagonal is the centre-to-side edge at that slot. The module `GoertzelV24ClosedWebSelectedLocalRailPointedEdgeSourceSquareChordCrossing` returns the exact slot, face equality, shared-interior-edge witness, and chord proof together. Constructing the selected chord triangles and consuming the source reduction remain open.

The chord branch has now been turned into selected separator geometry. Using the generic four-cycle chord theorem, `GoertzelV24ClosedWebSelectedLocalRailPointedEdgeSourceSquareChordTriangles` constructs two simple length-three facial-dual cycles. On each cycle the new chord step crosses the exact centre-to-side primal edge, while every old step inherits the corresponding selected crossing of the canonical source square. Thus the pointed allocation now returns two concrete selected triangle separators or exactly the three residual successor/old-centre origin rows. Consuming those triangles and rows, rolling the repair, attaching both end caps, constructing separated crosscuts, and L1 remain open.

Both selected triangles now feed the generic separator classifier directly, without the older square-bond wrapper. Their crossings are inherited from the reselected source square, so one triangle retains the literal outgoing rung of the middle Cell–3 placement. For that triangle Lean constructs the component away from the outer root and proves an exact alternative. If its inner side is a one-vertex star, the common vertex is incident to both the literal rung and the retained chord edge, which constructs the pointwise common-neighbour incidence consumed by the adjacent rail bridge. Otherwise the inner side is cyclic. This cyclic alternative remains a real open obligation; no minimality property of the closed carrier is assumed on the opened annulus.

The producer and consumer now compose in one kernel-checked statement. Given the actual pointed face allocation and the two non-flank receipts supplied by the collision packet, every old-forward row produces one of those selected chord triangles with a cyclic inner side. The complete local output is thus a concrete cyclic triangle separator or exactly one of the three residual successor/old-centre rows. This is an end-to-end reduction of the source-square branch, not yet a discharge of its cyclic separator or the three rows.

Those three rows are now consumed by literal local bypasses. In the successor/successor row the predecessor and successor side faces route through the successor Cell–3 centre. In the two mixed rows the same centre joins the successor side face to the old centre across the displayed corridor interface. All three walks have length two and avoid the collision face. Consequently the complete source-square local alternative is now a concrete cyclic selected chord triangle or a concrete pointed two-hop bypass. The cyclic separator, rolling iteration, end caps, separated crosscuts, and L1 remain open.

The bypass alternative now reaches the whole selected rail rather than ending as a local walk. The module `GoertzelV24ClosedWebSelectedLocalRailPointedEdgeSourceSquareBypassSplice` feeds each of the three constructed two-hop bypasses to the generic interior cross-splice. Its output is a simple rail with the original endpoints, with the collision face removed from its support, and with every new support face accounted for by the old rail or the bypass. Thus the exact source-square alternative is now a cyclic selected chord triangle or a repaired whole rail. This does not yet eliminate the cyclic branch, prove separation from the companion rail, iterate the repair, attach the end caps, construct the separated crosscuts, or close L1.

The residual-row constructor now retains a stronger proof-relevant receipt: the bypass support is contained in its predecessor, the displayed third Cell–3 centre, and its successor. Since the two endpoints already lie on the old rail, companion separation after the splice reduces to one source-local fact, namely that the companion support does not contain that third centre. The theorem `support_disjoint_companion_of_sourceSquare` proves the complete implication from the old pair's separation and this one centre-avoidance fact to separation of the repaired whole rail. This is a preservation theorem, not yet a proof of the centre-avoidance premise; the latter is the exact live geometric obligation before this repair can enter the rolling pair assembly.

The centre-avoidance premise now has an exact source-local discharge on the non-collision branch. Module `GoertzelV24ClosedWebSelectedLocalRailTerminalRightCenterAvoidance` traces every old terminal-track support face back to a selected side slot and proves that either exact terminal track avoids the successor Cell–3 centre whenever the two displayed consecutive centres share at most one interior edge. If a side slot were that centre, the source placement layer would exhibit two distinct shared interior edges; the successor rails themselves avoid their centre by irreflexivity. The cardinality bound is not asserted globally. Its complementary double-edge case remains an explicit bounded L1 collision to reduce or bypass before the whole rolling pair is constructed.

The mirror terminal calculation and the four-cell provenance layer now carry that local exclusion to the actual companion candidates. Module `GoertzelV24ClosedWebSelectedLocalRailTerminalPreviousCenterAvoidance` proves that either exact terminal track also avoids the immediately preceding Cell–3 centre under the analogous one-shared-edge bound. Module `GoertzelV24ClosedWebSelectedLocalRailFourCellMiddleCenterAvoidance` then observes that the first terminal window avoids its right centre and the last terminal window avoids its previous centre—the same middle face—and uses the already-proved terminal-window support provenance to exclude that face from both rails of either successful four-cell outcome. Thus the third-centre premise needed by the source-square companion-separation splice is available for a successful shifted four-cell companion under two explicit local cardinality bounds. This remains the good branch of the finite L1 classification: neither cardinality bound is asserted globally, and each double-edge complement remains a named collision branch.

That premise now reaches the actual repaired pair. Module `GoertzelV24ClosedWebSelectedLocalRailFourCellSourceSquareCompanionSeparation` handles all four proof-relevant cases: straight or swapped successful future, with either its first or its second rail repaired. The old successful pair's support disjointness supplies the unchanged part; the middle-centre theorem supplies the only new face exclusion; and the source-square splice theorem then proves that the repaired whole rail is disjoint from its untouched companion. This closes the companion-preservation bookkeeping for the good local branch. It does not prove the two cardinality bounds, eliminate their double-edge complements, iterate the rolling transition, attach end caps, or construct the separated crosscut pair.

The good branch now returns an assembly rather than a detached repair certificate. The module `GoertzelV24ClosedWebSelectedLocalRailFourCellSourceSquareRepairOutcome` reconstructs the complete two-rail object after the splice: the repaired rail retains its original endpoints and is simple, the untouched rail is unchanged, and their full supports are disjoint. All four cases are explicit — straight or exchanged endpoint order, with either rail repaired. The finite source classifier therefore returns either this actual separated assembly or the already exhibited selected chord triangle with a cyclic inner side. It does not suppress that cyclic residue or the two local double-edge branches, and it does not yet iterate the repaired transition, attach the end caps, construct the separated crosscuts, or close L1.

The finite-base threshold reached its first end-to-end numerical composition at this stage of the development. Module `GoertzelV24FramedBaseThreshold` now fixes the four-port separated-repeat length to 6164740, using the then-current lossless port-incidence carrier below. It proves that this length meets the literal source-layer spacing inequality and feeds it to the existing framed boundary-clean corridor bound. Above the resulting explicit face threshold, Lean constructs a clean corridor of the required length or proves that the caller's proposed interior-pentagon bound was false. This is not yet a numerical V_0: Addendum XXVII and the playbook use at most twelve ambient pentagons, whereas the proved spherical identity is $p = 12 + W_-$ when faces longer than six occur. The new theorem therefore keeps the pentagon bound as a visible input. A uniform negative-curvature bound (or another source-faithful discharge of that alternative), plus the local-cubic replacement for the older globally-cubic source-profile adapter, remain the two quantitative interfaces before this formula becomes the base cutoff. The later fragment-reindex quotient recorded below strictly improves this initial labelled-carrier baseline.

The negative-curvature alternative now has a literal length-versus-congestion form rather than an unspecified appeal to weighted counting. Module `GoertzelV24FramedCurvatureRouteLength` uses the weighted-L9 pairing already constructed in the formalization: all but the fixed source-foot allowance of the negative-curvature tokens are paired with distinct internal pentagons, and each pair carries a shortest path in the internal facial dual. If one such path is long enough for the boundary-contamination budget, the new geodesic adapter in `GoertzelV24FramedBoundaryCleanCorridor` extracts an actual boundary-clean hexagonal block of the required length. If no path is that long, Lean proves the explicit capacity bound

$$\#\{\text{paired routes}\} \;\le\; F_{\text{int}}(W_- + 7)^R, \qquad R = (B(W_-) + 1)L - 1,$$

where F_{int} is the number of internal faces, $B(W_-)$ is the already proved contamination budget, and L is the requested block length. Thus the long-route half of the high-curvature entrance now reaches the literal L1 corridor object. A further audit also proves why the displayed global congestion inequality cannot be the missing half: paired routes have distinct pentagon endpoints, so their total number is already at most F_{int}, while $(W_- + 7)^R \ge 1$. Consequently the inequality holds for every radius even without the short-route hypothesis (`pair_count_le_global_pow_capacity`). This is a kernel-checked refutation of the naive bounded-ball strategy, not a uniform cutoff. The remaining source-facing obligation must retain geometry discarded by the arbitrary exact-cardinality pairing—cyclic order, separation, or the actual two-sector laminar depth—rather than merely sharpen this global count. No fixed twelve-pentagon bound is being assumed.

The first geometry-sensitive replacement for that arbitrary pairing is now kernel checked. Module `GoertzelV24FramedMinimumCurvaturePairing` minimizes the total internal-dual distance among the same finite weighted-L9 credit/debt bijections. A generic finite-assignment lemma then proves the exact local exchange inequality: for any two distinct credits, their assigned pair cost is no larger than the cost obtained by swapping the two debt endpoints. This is the quantitative premise needed for the next uncrossing step. It does not yet prove that the chosen shortest dual routes are noncrossing or laminar, and so it is not a uniform cutoff or a completed L6 argument.

The metric uncrossing atom needed after that minimization is now kernel checked as well. Module `GoertzelV24WalkTailExchange` proves that two graph walks meeting at a vertex may exchange their suffixes while preserving their combined length exactly. Module `GoertzelV24FramedMinimumCurvatureUncrossing` applies this to two geodesics selected by the minimum weighted-L9 pairing. Minimality supplies the reverse inequality, so the crossed endpoint distances are equal to the original distance sum and both exchanged walks are again geodesics. This is the precise metric content of uncrossing; it does not yet choose a simultaneous family of routes minimizing crossing complexity, prove that a forbidden intersection strictly decreases such a secondary measure, or construct the annular laminar depth required by L6. Those are the remaining geometric steps before the high-curvature branch yields a uniform cutoff.

The geometry-sensitive assignment now also has its own literal finite route family. Module `GoertzelV24FramedMinimumCurvatureRouteFamily` decomposes every internal negative-curvature token according to whether the minimum assignment pays it with an internal pentagon or one of the fixed source-foot credits. Hence all but the fixed source-foot allowance belong to the new pentagon-backed family; its negative-curvature tokens and pentagon endpoints are each injective, and every member carries a canonical internal-dual geodesic. The module packages the preceding tail-exchange theorem directly for two intersecting members. This retires an interface mismatch with the older route family, which was tied to an arbitrary exact-cardinality pairing. It does not yet make the canonical geodesics simultaneously noncrossing: that requires a secondary finite route-choice optimization and a rotation-sensitive strict-decrease theorem.

The simultaneous finite optimizer is now constructed. Module `GoertzelV24FramedMinimumCurvatureRoutingSelection` ranges over every weighted-L9 assignment attaining the minimum total dual distance and over a finite table of geodesics for every possible pentagon/negative-curvature endpoint pair. The full table is deliberate: an uncrossing move may swap two assigned endpoints, so selecting routes only after freezing one assignment would not be closed under the move. For any natural-valued secondary complexity the module chooses a globally least candidate and proves its universal minimality property. The next theorem must instantiate that complexity with actual rotation-sensitive crossings and prove that a forbidden crossing yields a strictly smaller candidate. The generic optimizer itself is a construction, but it is not that geometric strict-decrease theorem and does not close L6.

The subsequent dependency audit found that the source's rotation-sensitive crossing theorem should not be rebuilt in this new route vocabulary. Module `GoertzelV24ClosedWebChordRotationNoncrossing` already proves, for the actual majority chords on a source radial path, that strictly interleaving chords have opposite intrinsic rotation sectors. Its L6 corollary converts a deep transversal into a large pairwise-laminar sector, and the existing laminar-profile modules continue that object to nested equal graph-derived depth profiles. The minimum-curvature routing remains potentially useful for supplying geometry in the unbounded-defect branch, but a second generic crossing calculus would duplicate rather than connect the source theorem.

Module `GoertzelV24ClosedWebSourceLengthDepth` now closes the exact L8-to-L6 path-selection seam. The constructed L8 witness says that at least one of its two radial paths is quantitatively long. Lean now performs the case split and applies the existing L6 theorem either to the first path or to the swapped radial pair. For every two genuinely bounded contamination sets, the result is therefore a shallow clean block on the long path or an explicitly large pairwise-laminar rotation sector. Module `GoertzelV24ClosedWebRadialDefectPositions` now constructs the two contamination sets rather than accepting them from the caller. A radial cut is marked exactly when its actual path edge lies on a defect face, a dual neighbour of a defect, or a face incident to either named hole. Simplicity of the radial path injects marked cuts into the union of these forbidden facial edge supports; the source-annulus geometry then supplies the cardinality bound. The resulting source-facing theorem constructs the L8 witness, chooses the long path (swapping the pair when necessary), constructs its bad positions, and feeds them directly into the L6 length/depth dichotomy.

The arithmetic used for this construction also corrects a subtler multiplicity issue. Counting nonhexagonal faces is insufficient when one long face can meet many radial edges. For any face-length family bounded below by five, Lean proves

$$\sum_{f:\,|f|\neq 6} |f| \;\leq\; 5p + 7W_-.$$

On a well-formed framed source trail, the weighted excess identity gives the derived bound $12W_- + 5\rho$, where ρ is the boundary surplus. These are edge-incidence budgets rather than mere face counts. The framed source-trail and closed-web annular-frontier packages are kept as distinct carriers; no comparison theorem between them is being assumed here.

The initial source-annulus estimate paid for every forbidden face as though it had the largest boundary in the instance. That safe estimate obscured the actual scaling. The strengthened incidence proof instead sums the real boundaries: the closed dual neighborhood of all defects has cardinality at most twice the total defect-boundary length, an arbitrary internal face family costs six edges per face plus one copy of the global defect support, and the two hole neighborhoods are charged to their literal perimeter. The resulting bound on the actual forbidden radial-edge support is

$$14(12W_- + 5\rho) + 6P_{\mathrm{holes}}.$$

Thus the quantitative entrance is linear, rather than polynomially wasteful, in the honest curvature weight. This improves the eventual cutoff substantially but does not make it uniform: W_- itself remains unbounded by the signed excess identity.

This is still not a uniform V_0. The constructed closed-web budget grows with W_-, so a sufficiently long radial path is obtained only relative to the instance's curvature. The remaining numerical problem is now sharply localized: high negative curvature must itself produce the second pumpable geometric coordinate (or an equivalent source collar), rather than being silently replaced by the manuscript's false constant pentagon count.

There is now a source-specific way to remove that curvature dependence from the *numerical L6 split*, without claiming a clean hexagonal corridor. Module `GoertzelV24ClosedWebRadialOrderDichotomy` uses total closure itself: the constructed L8 witness places every interior vertex on one of two radial paths, so if

$$2L < |V_{\mathrm{int}}|,$$

one actual radial path has at least L cuts. Applying the already proved cut-load/rotation-sector dichotomy with the empty marked set and defect budget zero gives, independently of W_-, either L consecutive cuts of bounded actual chord load or a rotation sector containing more than the requested number of pairwise-laminar spanning chords. This is the two-coordinate conclusion stated in Addendum XXVII, derived from the closed web's literal radial order rather than the false fixed-defect premise.

This theorem is not yet a uniform V_0 or a shrinking lemma. Its two conclusions still have to reach the physical splice: the bounded-load run must be realized by source-local simple layer boundaries carrying the finite length profile, while the laminar family must be realized by deletion transversals carrying the depth profile. The correction is nevertheless structural: the unbounded-curvature branch is no longer a missing numerical estimate. It has been replaced by the two explicit geometric realization morphisms that the profile/splice layer already expects. The earlier weighted incidence and minimum-routing results remain valid auxiliary constructions, but they are no longer being treated as the only possible supplier of L6.

The initial numerical junction with L7 was made explicit in Module `GoertzelV24ClosedWebProfileCalibratedRadialOrder` substitutes the proved lossless four-port carrier into the radial-order theorem. The gap-at-least-four length is the checked value

$$L = 6164740,$$

and hence its curvature-free interior-vertex threshold is

$$2L = 12329480.$$

Above this threshold Lean constructs the actual L8 radial witness and returns either the full consecutive run or a laminar sector larger than the selected finite L7 carrier. This is the first honest uniform numeral at the L6/L7 junction; it is not yet V_0. The numeral becomes a shrinking threshold only after the two branches are realized by the source's simple layer boundaries or deletion transversals and connected to the already proved splice. The later fragment-reindex quotient reduces the corresponding serial block length without changing these geometric obligations.

The depth-side realization has also been localized to what the theorem actually returns. Module `GoertzelV24ClosedWebSelectedSectorDepthProfile` shows that L7 needs a bounded cyclic-cut realization only for the chords in the single selected laminar sector, not for every chord at every radial cut. Exact cut equality then computes the crossing width and the existing pigeonhole returns two nested equal graph-derived profiles. This adapter does not construct those cuts; it reduces the remaining depth geometry to one selected-sector family and prevents a needlessly global Jordan assignment from becoming part of the formal target.

The v23 three-cell micro-enumeration is base material, not a serial corridor generator. Conversely, a raw bound on the entire over-encoded profile carrier is a finiteness witness, not a practical base census. The eventual base must bound the reachable state space and replay compact certificates generated by the manuscript's own definitions. If the argument were instead found to require an externally searched unavoidable catalogue, that would be a route failure requiring an explicit decision, not an implementation detail.

The generic tracked-connectivity escape step no longer assumes a portal classification. Module `GoertzelV24RegionalTrackedVertexBoundary` proves that, for a regional two-colour graph induced by a finite vertex side, odd regional degree together with even ambient tracked degree forces the edge to cross the actual vertex cut. The missing ambient tracked neighbour supplies an endpoint outside the side, while membership in the regional carrier supplies an endpoint inside. Componentwise handshaking then constructs a distinct reachable edge on that true crossing carrier. This is the generic parity-to-boundary theorem; it does not identify every historical crossing of a cumulative literal-Cell prefix with one of the two named moving-cut portals. That source-specific rail-absorption and exact-frontier theorem remains the connectivity half of portal completeness.

The corresponding generic cancellation atom is now kernel checked as well. Module `GoertzelV24BoundaryDartUnionAbsorption` proves that two exposed boundary darts related by the ambient edge involution carry the two endpoints of their shared edge into the union of the retained vertex sides. The shared edge therefore disappears from the true crossing frontier of that union. This proof uses no cubicity hypothesis. The live selected-rail layer already proves the required opposite-dart equation at each successor seam, but it does not yet package its literal Cells as retained vertex sides. Thus the remaining source junction is precise: construct those sides and identify their surviving frontier; do not infer portal completeness merely from the oriented seam.

There is also a quantitative warning at the finite-base end. The current literal-Cell carrier `BoundedCorridorCutProfile 2 0 8` has exactly

$$74853595474508474029607634310238208$$

states under the proved counting formula. The heterogeneous transfer-word pumping theorem remembers a whole reachable `Finset` of such profiles, so its generic pigeonhole bound is the power set size $2^{74853595474508474029607634310238208}$. These are valid finiteness bounds but not a feasible base census. A dependency audit found that this generic heterogeneous-word fallback is currently an isolated module: no downstream file imports its source specialization, and the actual framed base threshold uses the ordinary repeated source-prefix profile. Thus this power set is not the present V_0 bottleneck. It remains a warning for any future consumer of that stronger acceptance-preserving fallback; merely introducing a smaller reachable carrier as a new hypothesis would relocate, rather than prove, its compression.

The first lossless compression of that over-encoding is now machine checked. Module `GoertzelV24WidthTwoProfileCompression` observes only laws proved for graph-derived width-two profiles: tracked connectivity is symmetric, its diagonal is forced by the two boundary colours, and face continuation is symmetric with a forced diagonal. It deletes exactly those redundant matrix entries and proves that equality of the compressed codes reconstructs equality of all five fields of the original cut profile. At the four-fragment bound used by the cumulative source-prefix word, the exact carrier drops from

$$801805237456896 \quad \text{to} \quad 1536871176,$$

a factor strictly greater than 521712. Module `GoertzelV24FramedCorridorCompressedPrefixProfileRepeat` proves that the literal source-indexed prefix profiles satisfy these semantic laws and reruns the separated L7 pigeonhole argument on the smaller carrier. The required number of prefix positions is therefore 6147484705, while equality of the complete raw splice profiles is still the conclusion.

Two further source laws reduce this carrier without changing that conclusion. Module `GoertzelV24WidthTwoBoundarySemanticCompression` proves that a tracked off-diagonal entry can be true only when both displayed boundary colours belong to the tracked colour pair, that every indexed boundary fragment contains a displayed port, and that every capped fragment length is positive. It proves a second lossless codec on profiles satisfying those laws and computes its exact four-fragment carrier as 78419184. Module `GoertzelV24FramedCorridorBoundarySemanticPrefixProfileRepeat` proves these laws for the literal source prefix and lowers the separated repeat demand to 313676737.

Finally, the source face-continuation table is equality of literal face labels, so it is transitive as well as symmetric. Module `GoertzelV24WidthTwoFaceEquivalenceCompression` encodes that equivalence relation by its six upper-triangular bits, retains only valid equivalence codes, and proves the encoding lossless on actual profiles. There are exactly fifteen such codes on four fragments. The resulting exact carrier has 18641184 states, more than 43012570 times smaller than the original raw carrier. Module `GoertzelV24FramedCorridorFaceEquivalencePrefixProfileRepeat` connects the literal cumulative source prefixes to this quotient and proves a complete raw-profile repeat at gap at least four from 74564737 positions.

There is one more graph-semantic restriction before any reachability argument. Each displayed cut edge has exactly two actual darts, so it can occur in at most two occurrence-sensitive boundary fragments. Module `GoertzelV24WidthTwoPortIncidenceCompression` maps every fragment containing a fixed edge injectively to one of those darts. The proof is valid even when a bridge contributes both occurrences to the same quotient face; it assumes no global two-sidedness. The resulting lossless carrier has 1541184 states at four fragments, about 12.1 times fewer than the preceding face-equivalence carrier and more than 520 million times fewer than the raw carrier. Module `GoertzelV24FramedCorridorPortIncidencePrefixProfileRepeat` discharges the new semantic law for the literal cumulative source prefix and proves a complete raw-profile repeat at gap at least four from 6164737 positions. This established the labelled-carrier baseline of block length 6164740; the presentation quotient recorded below later reduces the route-correct serial block to 422980 while returning the required fragment permutation.

The literal source geometry sharpens the remaining face-count coordinate. Module `GoertzelV24FramedCorridorPrefixThreeFaceBound` proves that the two displayed crossings of a Cell meet only its three named dual-path faces. It then proves that the central face contributes one retained fragment, while either outer face contributes at most one, so the cumulative prefix has at most three boundary fragments. The same bound holds for the terminal-aware input: retaining the shared rung adds a terminal coordinate but no face fragment. Module `GoertzelV24FramedCorridorThreeFacePrefixProfileRepeat` narrows only the proved dependent count, retains the lossless port-incidence codec, and computes the exact 0–3 carrier as 191184 states. Four residue classes therefore force a complete raw-profile repeat at gap at least four from 764737 positions, an 8.06-fold reduction from the four-fragment carrier. No equality of the fragment count with three is assumed. The two existing threshold connectors now consume this stronger theorem: the required clean block length is 764740, and the curvature-free radial entrance is the checked numeral 1529480 interior vertices. A five-by-five closed web has exactly ten boundary stubs, so the corresponding total-carrier threshold is 1529490 vertices; Lean proves the exact partition and the resulting total-size entrance theorem. These are explicit L6/L7 constants for Cell 3, not yet the final whole-route graph-size threshold V_0.

This is a substantial carrier reduction, but more than seven hundred thousand positions still make a direct base audit impractical. The primary manuscripts require an explicit L7 upper bound, not a separately postulated reachable-state quotient. The older twenty-mode Lemma 8.18 laboratory concerns the narrow τ/mirror-τ automaton and is not used as a supplier for the general source corridor. Further proved compression remains valuable, but the live F obligations are the ones exposed by the actual threshold: discharge the uniform negative-curvature/pentagon alternative, connect the resulting corridor repeat to the strict reduction supplier, and verify the route-native finite base. A smaller carrier introduced merely as a hypothesis would remain a conditional wrapper, not such a discharge.

The first coordinate of a faithful reachable-state transition is now machine checked. Module `GoertzelV24FramedCorridorPrefixFiniteConnectivityUpdate` keeps the shared rung as one fixed terminal of the incoming cumulative profile. It then proves that every outgoing crossing-to-crossing connectivity bit is exactly the reflexive–transitive closure of a finite relation obtained from three pieces only: that terminal-aware incoming profile, the literal six-edge Cell profile, and the two residual seam turns computed from their stored colours. This is stronger than a graph-size finiteness observation: it is the semantic update law for the actual source Cell. It also explains why simply forgetting terminals was not a sound compression—a strand may pass through the shared rung and disappear from the zero-terminal view.

The terminal refinement does not duplicate the face state. Module `GoertzelV24FramedCorridorPrefixTerminalFaceCoordinates` proves that forgetting the shared-rung terminal recovers the established cumulative-prefix profile exactly. In particular, the finite fragment count, occurrence-sensitive face equivalence, crossing-incidence rows, and capped face-progress coordinates are unchanged. Thus the existing face-factor laws can be reused on the same incoming state as the finite connectivity update. This is an interface identification, not yet the joint finite transition: the local attachment and face-factor update must still be coordinateized together with the connectivity law before a reachable-state closure is authoritative.

The graph-dependent connectivity carrier has now also been eliminated. Module `GoertzelV24FramedCorridorPrefixFiniteConnectivityCoordinates` proves a canonical equivalence between its eight literal edges and $\mathrm{Fin}\,2 \sqcup \mathrm{Fin}\,6$: the two old cut edges followed by the six cyclic Cell-boundary edges. The two blocks remain distinguished definitionally, so this is a stable semantic coordinate system, not an arbitrary enumeration of eight ambient edges. Connectivity therefore no longer obstructs an enumerable one-Cell relation. The remaining finite coding obligation is the face attachment itself: the current witness still quantifies over graph-dependent prefix factors, and must be replaced by a bounded coordinate code with a proved adequacy theorem before a closure measurement is meaningful.

The connectivity relation itself is now graph-free as well. Module `GoertzelV24FramedCorridorFiniteConnectivityCode` defines the old-prefix, six-edge Cell, and two-edge seam steps on the fixed carrier $\mathrm{Fin}\,2 \sqcup \mathrm{Fin}\,6$, constructs an explicit graph isomorphism to the literal eight-edge interface, and proves that reflexive-transitive reachability is preserved exactly. Its finite geometry receipt records one shared-rung and two outgoing Cell positions, so the raw receipt alphabet has exactly

$$6 \cdot 6^2 = 216$$

elements. Every actual graph-derived one-Cell transition is proved to satisfy this relation. The number 216 is an ambient upper bound rather than a measured reachable count, and the relation becomes a complete source letter only after it is joined, for the same literal coloring, with the at-most-125 face-update receipt below.

That join is now a single checked interface. Module `GoertzelV24FramedCorridorFiniteOneCellSupport` defines the exact support of a literal Cell as existence of one Tait coloring realizing the fixed three-face input and output profiles. It then packages, from that same coloring, the full six-edge Cell profile, the 216-way connectivity geometry receipt, the semantic first/centre/second output-face coordinates, and the at-most-125 face-update table. The receipt checks all five profile fields: crossing colors, tracked two-colour connectivity, face identity, port incidence, and capped face progress. The resulting theorem is deliberately one-way:

$$\text{exact literal support} \quad \Longrightarrow \quad \text{joint finite receipt.}$$

The converse adequacy theorem is still open. Consequently an abstract receipt closure need not equal the exact source closure. Equality is not needed for a safe upper bound, however: once the initial source states and every literal transition are proved to enter the receipt closure, spurious receipt steps can only enlarge the computed reachable set. Such an over-approximate closure may therefore lower the pigeonhole carrier soundly if it remains small; if it explodes, the measurement is merely inconclusive rather than false.

Two further interfaces now make that experiment precise. Module `GoertzelV24WidthTwoPortIncidenceCodec` upgrades the preceding injection/counting argument to an actual equivalence: every one of the 191184 finite codes decodes to a complete five-coordinate semantic profile, and encoding after decoding (or decoding after encoding) is exact. This does not assert source realizability; it makes the already proved semantic carrier executable without losing information. Module `GoertzelV24FramedCorridorReachableProfileStep` then defines a common one-Cell receipt step by retaining the shared-rung terminal on input and forgetting it at the cumulative boundary. Lean proves that every Tait-colored literal source Cell is admitted and that this step transports exactly through the codec. Thus the one-Cell inclusion required by source-to-finite soundness is complete; the finite seed set and its iterated-closure induction remain to be packaged with the successor generator.

The remaining measurement atom is computational representation, not a new profile field. A dense test of all 191184^2 pairs would repeat the original over-encoding mistake, while the present receipt contains large ambient witness carriers (notably a full six-edge Cell profile). The next module must generate sparse successors from source-admissible finite Cell and terminal receipts, or prove a smaller equivalent decision procedure. Only after that generator is kernel-related to the green receipt should its closure cardinal replace 191184 in L7. The finite base below the resulting V_0 remains a separate obligation.

That face-factor dependency has now been removed for one literal outgoing fragment. Module `GoertzelV24FramedCorridorPrefixFactorSubsingleton` proves that the old-prefix factors contributing to such a fragment have cardinality at most one. Two cut-touching factors resolve to the same incoming boundary support; two residual factors are both the singleton shared rung; and a mixed pair would put an incoming cut edge on the new Cell boundary, contradicting their proved disjointness. Module `GoertzelV24FramedCorridorFiniteFaceAttachment` therefore replaces the dependent factor family by the finite alphabet

$$\{\mathsf{fresh},\ \mathsf{continues}(i),\ \mathsf{sharedRung}\},$$

which has exactly $n+2$ letters for an input with n face coordinates. Its coverage theorem retains exact ambient-face and edge-support data, and its cap theorem proves that the same code computes the exact capped face-length update already established from the rotation geometry. At the proved input bound $n \leq 3$, one outgoing fragment thus has at most five attachment receipts. This is not yet the complete finite transition: the outgoing fragment carrier must still be bounded and padded, and the per-fragment receipts must be joined with color, connectivity, face equivalence, and port-incidence coordinates before any reachable-closure measurement is authoritative.

The outgoing side now has the same proved three-fragment bound as the input. Module `GoertzelV24FramedCorridorRightThreeFaceBound` calculates directly on the right two-edge simple dual path: every incident face is one of its three named faces, the complete central boundary contributes one retained fragment, and either outer face contributes at most one. The proof uses neither a successor Cell nor an assumption that all three fragments occur. Module `GoertzelV24FramedCorridorFiniteProfileCoordinates` therefore packages the actual one-Cell input and output in the fixed carriers

$$\texttt{BoundedCorridorCutProfile 2 1 3} \quad \longrightarrow \quad \texttt{BoundedCorridorCutProfile 2 0 3}.$$

This narrowing changes only the proof-carrying outer bound, retains all five profile coordinates definitionally, and reflects equality of the complete original profiles. It fixes the types of the finite transition but does not yet construct that transition or identify literal Cells having different local data merely because they share one of the three geometric orientations.

The bound is now sharpened to exact semantic coordinates. Module `GoertzelV24FramedCorridorRightFaceRoles` proves that the three occurrence-sensitive outgoing fragments correspond canonically to the first, centre, and second faces of the literal two-step dual path. This is not an arbitrary enumeration of a three-element type: each coordinate is tied to its named ambient face by a proved equivalence. Module `GoertzelV24FramedCorridorFiniteRightFaceUpdate` consequently computes the two purely outgoing face fields. Face continuation is equality of the three roles, while their two crossing-incidence rows are respectively

$$(1, 0), \qquad (1, 1), \qquad (0, 1).$$

These matrices are theorems about the actual source path, not freely chosen Boolean masks. The result removes two further over-encoded coordinates from the reachable-state problem. It does not yet package the attachment and capped-length receipt with the finite connectivity law, so it is not by itself the complete one-Cell support relation.

The attachment and length coordinate is now finite simultaneously across all three roles. Module `GoertzelV24FramedCorridorFiniteFaceUpdateCode` assigns to each role one of the exact receipts `fresh`, `continues` i, or `sharedRung`. Its realization theorem constructs such a three-entry table from every literal source Cell and proves that the table computes all three outgoing capped face-progress values. Because one receipt has at most $3 + 2 = 5$ values at the proved input bound, the entire face-update alphabet has at most

$$5^3 = 125$$

elements. This is a kernel-checked upper bound on the semantic face update, not a dense enumeration and not an empirical reachable count. The face table must still be joined with the finite tracked-connectivity update arising from the same literal coloring before it defines a complete source letter.

This result sharpens the remaining finite-state programme without changing the meaning of the base. The old twenty-mode Lemma 8.18 closure concerns the narrow τ/mirror-τ laboratory; no typed map currently identifies those modes with the cumulative profiles used by the general source corridor. Moreover, the current Fable manuscript and the route-identity note define the terminal base as the target predicate on every bounded instance below the derived threshold. A small reachable profile closure can reduce that threshold once a full five-coordinate update and a coverage theorem are proved, but it is not by itself `BaseVerified`. The three-cell audit from v23 remains genuine local base material; where it was read as the entire bounded base, Addendum XXVII's newer finite-base statement supersedes that reading. Thus the honest order is: complete the terminal-aware joint update, measure and kernel-check its reachable closure, use the proved smaller carrier to lower V_0, and then verify the resulting route-native bounded base.

The literal outgoing Cell admits a substantially smaller exact presentation. Modules `GoertzelV24FramedCorridorSourceOutgoingProfileCompression` and `GoertzelV24FramedCorridorSourceOutgoingProfileCodec` use the proved first/centre/second face roles to omit two tables that the source geometry has already fixed: face continuation is equality of roles, and the two displayed port rows are left-only, both, and right-only. What remains is one of 24 strand states, one of the six orders of the three roles, and one of five positive capped lengths for each role. Lean therefore computes the exact carrier cardinality

$$24 \cdot 3! \cdot 5^3 = 18000.$$

This is not merely a count. The encoder is extracted from the same joint one-Cell receipt as the literal transition; the decoder reconstructs all five coordinates of the complete outgoing profile; and equality of compact codes is proved to reflect equality of those profiles. Module `GoertzelV24FramedCorridorSourceOutgoingProfileRepeat` then samples this code at every literal Cell in the heterogeneous source corridor, using the older complete-hex geometric prefix. Four residue classes force two positions at gap at least four, so Lean obtains equality of the complete original outgoing geometric-prefix profiles from

$$4 \cdot 18000 + 1 = 72001$$

indexed Cells. This is a proved compression laboratory, not merely a proposed count.

The route-correct reachable-state and threshold claims do not yet inherit this number. The compositional prefix is the source-ordered union of literal complementary Cell regions in `GoertzelV24FramedCorridorSerialPrefixRegion`; it is not the union of complete hex boundaries used by the geometric-prefix sampler above. Moreover, one local receipt exits around the next hexagon's incoming rung, whereas the next receipt enters around that hexagon's outgoing rung; no uniform equality between those differently presented cuts is proved. The live finite-state obligation is therefore a lossless finite code for `sourceCorridorSerialInputBoundedProfileAt`, or a proved factorization of that serial-prefix state through the local code, followed by a verified reachable closure. The currently packaged graph-size threshold still uses the earlier 191184-state geometric-prefix theorem. No route-correct reachable closure, new numerical V_0, or bounded-base verification is claimed here.

The first route-correct serial compression is now machine checked in `GoertzelV24FramedCorridorSerialPrefixProfileCompression`. A literal serial input profile is an occurrence-sensitive graph-derived profile on two displayed cut edges. Each cut edge has only two dart occurrences, so it can belong to at most two boundary fragments; face continuation is the actual face-equivalence relation. The existing lossless width-two codec therefore applies directly to `sourceCorridorSerialInputBoundedProfileAt`. Lean computes the exact ambient semantic carrier size

$$1541184$$

and proves that equality of its codes recovers equality of all five original profile coordinates. Adding four residue classes yields two equal complete serial-prefix profiles, separated by at least three intervening positions, from

$$4 \cdot 1541184 + 1 = 6164737$$

literal positions. Unlike the 18000-state result above, this theorem is stated on the source-ordered union of literal complementary Cell regions. It retires the raw Boolean carrier as the route-correct repeat bound. It does not yet measure the subset reachable under the heterogeneous one-Cell support word, derive a graph-size V_0, or verify the bounded base.

The same module retains this labelled count as a conservative baseline, while the fragment-reindex quotient is now wired into the framed geometric entrance in `GoertzelV24FramedSerialBaseThreshold`. The shorter clean block long enough to supply the separated serial repeat modulo the explicit splice permutation has the kernel-checked length

$$4 \cdot 105744 + 4 = 422980.$$

Lean substitutes this numeral into the existing pentagon-bounded clean-corridor face threshold and proves the exact alternative: above that threshold, either the proposed interior-pentagon bound fails or a clean corridor of the required serial length exists. This is the route-correct quantitative entrance, not yet a uniform V_0. In particular, the high-curvature branch, the executable reachable closure, the strict reduction supplier, and the bounded base remain visible obligations.

The route-correct serial face factor has now been separated from the local geometric-prefix shortcut. Module `GoertzelV24FramedCorridorSerialFaceFactorReceipt` defines a finite receipt with the exact input and output fragment counts, the finite relation from outgoing fragments to incoming fragment coordinates, and the capped old-prefix, literal-Cell, and overlap contributions. Its source constructor is extracted from the actual heterogeneous serial prefix. Lean proves that the old and local slices cover each entire outgoing fragment and proves the overlap-corrected cap-at-five formula. A strengthened finite inclusion–exclusion law removes the earlier small-overlap premise: the capped size of a union is determined by the capped sizes of both factors and their intersection, even when the raw intersection already reaches the cap. The literal serial face cap is therefore now an unconditional arithmetic function of the finite receipt. No unproved tiling claim is needed, and the singleton-factor theorem for the older small geometric prefix is not reused on an arbitrarily long serial prefix. The remaining face-compression theorem is narrower: prove that the incoming profile coordinates and the literal Cell letter reconstruct the receipt's old-slice incidence and capped overlap. Only then may the receipt be joined to the tracked seam code and used for a reachable-closure measurement.

That remaining dependence is now fail-closed rather than implicit. The receipt's incoming-incidence bit has first been corrected to require both support inclusion and equality of the ambient face: inclusion alone could confuse the two facial sides of one boundary edge. The new module `GoertzelV24FramedCorridorSerialFaceInputResidue` then unions exactly the same-face incoming fragments contained in each outgoing fragment and defines the part of its old-prefix slice left uncovered. Lean proves a disjoint partition

$$\text{old slice} = \text{selected input union} \mathbin{\dot\cup} \text{old-input residue}$$

and the corresponding cap-at-five sum law. The residue itself is observed by one value in $\mathrm{Fin}\,6$. Empty residue is therefore an optional strengthening, not a prerequisite for finite-state compression. Module `GoertzelV24FramedCorridorSerialFaceResidualCode` keeps the residue as an explicit finite coordinate and removes the receipt's redundant old-cap field. Lean proves that the selected incoming fragments are pairwise edge-disjoint, so their capped contribution is computed directly from the incoming profile's face caps and the occurrence-sensitive incidence matrix. Together with the residue, literal-Cell, and overlap caps, this gives an exact finite decoder for every outgoing face-length cap. Thus no uncapped prefix cardinality and no global internal-interface cancellation enter this part of the serial letter. Abstract-code realizability, the joint five-coordinate support relation, the reachable-state count, and the resulting V_0 and base verification remain open and are not claimed here.

Module `GoertzelV24FramedCorridorSerialFaceTransition` now packages those facts as one finite facial transition between the route-correct bounded serial profiles. Its code retains the occurrence-sensitive use of incoming fragments, the single capped residue, the literal-Cell and overlap caps, and the outgoing continuation and port-incidence tables. The outgoing face cap is not copied into the code: Lean decodes it from the incoming profile and the four finite cap ingredients. Every actual literal source Cell produces such a transition, covering jointly the continuation, port-incidence, and capped-progress coordinates of the manuscript's face data. This is the source-image direction only. The tracked two-colour strand law, a common finite Cell witness for both tracked and facial data, converse adequacy, the executable reachable closure, and the resulting V_0 and base verification remain open.

The tracked strand law and the common finite witness are now proved. Module `GoertzelV24FramedCorridorSerialTrackedTransition` separates each outgoing two-colour component into two finite cases: it reaches a component represented at one of the two displayed input coordinates, or it remains in an old-prefix residue which reaches neither input coordinate. The visible case is reconstructed through the incoming profile's `strandConnected` matrix; one Boolean matrix retains the residual case. This distinction is necessary because the displayed two-edge cut has not been proved to be the complete vertex frontier of an arbitrary source prefix. It therefore avoids silently assuming the still-open exact portal cancellation theorem.

Module `GoertzelV24FramedCorridorSerialFullTransition` combines that tracked code with the facial transition and the two outgoing edge colours. Its witness type is finite, and Lean proves that every literal source Cell yields one joint transition governing all five profile fields: edge colour, tracked connectivity, face continuation, fragment/port incidence, and capped face progress. This closes the finite source-image half of the one-Cell letter. It does not yet prove the converse adequacy statement that accepted finite codes are realized by literal source Cells, compute the heterogeneous reachable closure, derive a numerical V_0, or verify the resulting bounded base. Those are the next distinct obligations rather than consequences claimed by the present packaging.

The source-image relation is now executable as a conservative finite letter. Module `GoertzelV24FramedCorridorSerialFullTransitionDecoder` packages the dependent input and output fragment counts with the five-coordinate receipt and defines a graph-free deterministic decoder. Lean proves that every literal source transition is the decoded output of some such finite letter. This is sufficient for sound over-approximation: adding abstract letters can add reachable states but cannot hide a literal source state. It is not yet an exact local alphabet. In particular, the tracked and facial residue fields may still depend on old-prefix information not represented at the displayed input cut. Exact portal/residue control, or an equivalent state augmentation, is therefore the remaining transition-locality obligation before a small reachable closure can be claimed.

The tracked locality debt has now been narrowed to an exact geometric statement. Module `GoertzelV24FramedCorridorSerialPrefixTrackedBoundary` applies regional handshaking to the genuine vertex-side carrier of the accumulated literal-Cell prefix. In a cubic Tait colouring, every odd endpoint of a restricted two-colour component reaches a distinct edge of the true prefix boundary. The source boundary classification then proves that this edge is an input or output crossing of one of the historical Cells already present in the prefix. Thus a residual strand cannot disappear into the interior. The theorem deliberately stops short of identifying that historical crossing with one of the two currently displayed portals: cancellation of the intermediate interfaces, or a finite coordinate retaining their effect, is still required before the executable alphabet is exact.

The scale of the state side is no longer unknown. The already proved port-incidence semantic carrier has exactly 1,541,184 states at the four-fragment bound, while the three-fragment L6/L7 calibration gives the checked closed-web entrance 1,529,490. These are conservative carrier bounds rather than measured reachable counts, and neither number verifies the route-native base below it. The finite-state explosion has thus been reduced from the raw profile encoding to a million-scale semantic carrier; the remaining major risk is exact transition admissibility followed by the separate bounded-base proof.

The inter-Cell presentation change is now explicit rather than hidden behind a false equality of cuts. Module `GoertzelV24FramedCorridorSerialBoundaryRebase` proves that, after one literal Cell has been adjoined, the next input region is exactly the same accumulated prefix together with the next two displayed crossing edges. For both tracked two-colour connectivity and occurrence-sensitive facial continuation, the enlarged graph is exactly the supremum of the old prefix graph, the graph on those two edges, and the canonical residual seam. Every residual turn touches one of the two new crossings. Cubicity therefore puts the complete residual support in their closed edge neighbourhood, of cardinality at most fourteen; two-sidedness gives the same bound for facial occurrences.

Module `GoertzelV24FramedCorridorSerialBoundaryRebaseCode` turns this support bound into an actual finite letter. One canonical carrier retains the two named, distinct crossings and jointly transports both tracked factors for all three tracked-colour pairs; adjacency and reachability are preserved exactly. The facial residual seam is likewise transported to a finite graph code of size at most fourteen. Thus the heterogeneous corridor word has two machine-localized ingredients at each step: the literal complementary Cell and the bounded change of boundary presentation. The remaining finite-state obligation is to align the facial rebase code with the four boundary-fragment coordinates and then combine both ingredients into one full five-coordinate support letter. No ambient-colouring existential, reachable-closure count, new V_0, or bounded-base verification is claimed by these modules.

The two ingredients now form one exact state-to-state geometric letter in module `GoertzelV24FramedCorridorSerialOneStepRegion`. Between consecutive serial input cuts, the added region is definitionally the current literal Cell together with the successor presentation's two displayed crossing edges. The next input region is proved equal to the old input region union this local extension. For every tracked colour pair and for every ambient face orbit, Lean factors the next graph as the old-state graph, the local one-step graph, and their canonical attachment seam. It also factors the local graph itself as the Cell graph, the two-edge rebase graph, and their internal seam. This identifies the exact support object that the finite one-Cell transition must encode and rules out treating consecutive cuts as definitionally equal. It is an exact geometric factorization, not yet a proof that the two residual seams are determined by the old profile and a finite local code; no reachable closure or new threshold is claimed.

The facial rebase code is now aligned with the actual successor profile in module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceCoordinates`. The shared carrier in `GoertzelV24BoundedFaceSeamFamilyCode` stores an exact fragment count and one bounded support-graph code for each genuine fragment, rather than padding arbitrary face roots. For the literal source rebase, the stored count is at most four and is proved equal to the successor five-field profile's own `faceFragmentCount`. Each coordinate is taken from that successor cut's canonical occurrence-sensitive fragment enumeration, and the ambient seam on the fragment's named face is proved graph-isomorphic to the stored code. This closes the coordinate-alignment obligation left by the previous rebase module. The remaining theorem is semantic adequacy: the old profile and the finite local Cell/rebase receipts must determine exactly the complete successor relation. That theorem, reachable closure, and V_0 remain open.

The same exact alignment now holds on the other side of the local step. Module `GoertzelV24FramedCorridorSerialPrefixFaceCoordinates` indexes the facial seam between the cumulative old prefix and the literal Cell by the canonical fragments of the Cell-extended prefix output. Its exact fragment count agrees with that output five-field profile, and the ambient seam on each named face is graph-isomorphic to the stored fourteen-vertex code. Hence both facial seams in the two-stage state update now speak the coordinates of the profiles they update. This remains a finite presentation of the exact local geometry, not a claim that the old state and these codes already determine the new state.

The abstract serial transition now also has an exact executable boundary. Module `GoertzelV24FramedCorridorSerialFullTransitionExecutable` proves that an output is produced by some packaged deterministic decoder letter if and only if it satisfies the joint finite transition relation on all five profile fields. It also supplies the resulting finite decision procedure. Thus a later sparse closure checker may use the decoder without introducing a second, silently different transition semantics. A certificate-shaped closure predicate and its path induction theorem reduce kernel replay to checking the listed states are closed under the executable step; they do not prescribe a dense all-pairs algorithm. This is adequacy of the executable presentation, not geometric completeness of the abstract alphabet: some finite letters may still be unrealizable by a literal source Cell, and the boundary-rebase composition, measured reachable closure, derived V_0, and bounded base remain separate obligations.

There is now also a deterministic decoder at the narrower, source-literal Cell boundary. Module `GoertzelV24FramedCorridorFiniteOneCellDecoder` packages the exact terminal-aware interface used by one Cell: a profile with two crossings and the shared-rung terminal enters, and a profile on the two outgoing crossings leaves. A finite letter retains the Cell's full six-edge profile, its literal connectivity placement, the three named outgoing face roles, and the capped face-attachment table. The decoder computes all five outgoing profile fields, and Lean proves that every complete one-Cell receipt—therefore every actual Tait-coloured literal source Cell—is accepted by some letter. The converse is now proved at the same finite boundary: every successful decoder application carries a complete receipt, so the executable test is exactly the declared receipt relation rather than a second, potentially looser semantics. This exactness is about the finite receipt language; it does not assert that every abstract receipt is geometrically realized by a source Cell. The word remains heterogeneous; no theorem identifies these semantic letters merely from the three coarse slab orientations. This closes executable source inclusion for the local Cell half, not the whole serial step: the separate boundary rebase must still be composed, abstract-code realizability is not asserted, and no reachable count, V_0, or bounded base follows yet.

The second letter now has a single finite receipt as well. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseLetter` packages the two successor crossing colours, the three tracked-colour local and residual graphs on their common sixteen-edge carrier, and the occurrence-sensitive facial seams indexed by exactly the successor profile's real fragments. The source construction supplies this finite object, and Lean proves the component and fragment-count identifications. This deliberately retains the boundary rebase as a distinct letter in the heterogeneous source word. It is not yet the semantic decoder: the next theorem must show that the previous complete profile and this receipt determine the successor complete profile. In particular, the construction does not replace the explicit residual data by the false assertion that the two displayed portals are the whole accumulated frontier.

The rebase receipt now retains both sides of that semantic question in one finite coordinate system. The original fourteen-edge seam neighbourhood named the two successor crossings but did not necessarily contain the old profile's two displayed crossings. Module `GoertzelV24FramedCorridorSerialBoundaryRebasePortCarrier` adjoins the old pair, proves that the enlarged carrier has at most sixteen edges, and assigns stable coordinates to the old and new pairs while keeping every tracked local and residual graph on the same carrier. Each pair is proved internally distinct; no unwarranted claim is made that an old port can never coincide with a new one. This removes a finite-addressing defect, not the mathematical locality obligation. An exact decoder still requires the source-specific theorem that every relevant attachment of the accumulated old prefix to the local rebase geometry is represented through the retained old-profile interface, together with its facial analogue. Until those statements are proved, the receipt is finite and source-supplied but not a complete transition semantics, and no reachable-state count is claimed.

The enlarged carrier is now also proved semantically exact for the graph data it stores. For either tracked factor—the two-edge local rebase or its residual seam—adjacency and reachability between arbitrary retained carrier edges are equivalent before and after finite-coordinate transport. In particular, the bundled rebase letter exposes exact path queries between any of its four named old-or-new ports. Thus the remaining connectivity question is no longer whether the finite code loses a local path. It is the sharper portal-completeness statement that every excursion through the arbitrarily large accumulated prefix which can affect the rebase is represented by the old profile's named interface. The facial analogue remains equally load-bearing, and neither is inferred from carrier finiteness alone.

The semantic target of that quotient is now explicit. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFullProfileRelation` packages the two graph-derived profiles on either side of a literal source rebase into one joint relation. Edge colours and tracked connectivity are recorded together with face continuation, boundary-port incidence, capped face length, and the exact three-factor facial closure, all over the same old-prefix and local rebase regions. Thus a future finite decoder has a precise adequacy statement to meet. The module does not yet prove that the old profile and the finite receipt determine the successor profile: that still requires the source-specific tracked and facial portal theorems saying that every relevant old-prefix attachment factors through the two named old ports. Until those theorems and the Cell/rebase composition land, no reachable-closure count or numerical V_0 is inferred.

The old end of a rebase seam is no longer geometrically unclassified. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseOldBoundary` proves, separately for tracked connectivity and for face-orbit occurrences, that every seam endpoint belonging to the accumulated prefix lies on the prefix's genuine vertex-side crossing frontier. The earlier source theorem then classifies it as an input or output crossing of a historical literal Cell. Thus no arbitrary ambient old edge can influence the rebase decoder. This still stops short of identifying the frontier with the two currently displayed old ports: the historical interfaces internal to the Cell union must be absorbed by the literal connector geometry before the finite receipt is a complete transition semantics.

The tracked half of that remaining absorption now has a parity normal form. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedEscape` orients every tracked rebase-seam adjacency from its old-prefix endpoint to the newly displayed edge. The old endpoint is itself classified as a crossing of a historical literal Cell. Since a cubic Tait two-colour graph has degree two there, while the displayed new neighbour is absent from the old region, its old regional degree is strictly smaller: it is either zero or one. In the degree-one case regional handshaking produces a distinct reachable crossing of a historical literal Cell. Thus a nontrivial old excursion is reduced to a pair of concrete historical Cell interfaces; no arbitrary ambient endpoint remains. The theorem does not yet prove that those historical interfaces cancel to the two surviving displayed old ports, and it has no occurrence-sensitive facial analogue yet. Those are the two explicit locality obligations before the finite rebase receipt is an exact complete-profile decoder.

The tracked historical range is now bounded by the source geodesic rather than merely named. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedLocality` first proves that every literal input crossing lies on its Cell's centre face and every literal output crossing lies on the successor centre face. Edge adjacency in the cubic two-sided rotation system then forces the two corresponding centre faces to agree or to be adjacent in the facial dual. The corridor's geodesic separation eliminates every remote possibility: an input crossing adjacent to the successor rebase belongs to the current Cell, while an output crossing belongs to the current Cell or its immediate predecessor. Thus the formerly unbounded historical search has exactly three local cases. The remaining theorem must still absorb those cases into the incoming profile; bounded locality is not yet full profile factorization.

That bounded historical range is now packaged at the literal seam, for both connectivity and face progress. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedAttachment` orients every tracked rebase adjacency and proves that its old endpoint has exactly one of three source roles: an input crossing of the current Cell, an output crossing of the current Cell, or an output crossing of the immediately preceding Cell. Its facial theorem gives the same three-role classification for every occurrence-sensitive face-seam step after projecting the face position to its ambient primal edge. The new endpoint remains explicitly in the two-edge successor presentation in both statements. Hence the rebase no longer consults an unbounded historical alphabet at its point of attachment. This is not yet the full two-port decoder: after entering the accumulated prefix, a tracked component or a face fragment may still carry information to a different true-frontier edge. The remaining quotient theorem must show that such excursions are represented by the incoming profile, or retain an equivalent finite residue, before any reachable-closure count is sound.

The three roles now have an exact finite coordinate type rather than a nested historical existential. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseAttachmentCode` retains two current input coordinates, two current output coordinates, and the two outputs of the unique immediate predecessor when that predecessor exists. Lean proves that this dependent type has cardinality at most six and that every tracked and occurrence-sensitive facial rebase adjacency is named by one of its coordinates. At the initial offset no nonexistent predecessor is represented by a padding value. This closes finite addressability of the old seam contact. It does not close semantic adequacy: after entering the accumulated prefix at that contact, a tracked component or a face fragment may still carry information farther through the prefix. That excursion must be factored through the incoming five-field profile, or through an equivalent finite invariant preserved by the literal transfer word, before measuring a reachable-state closure.

The algebraic part of the tracked excursion factorization is now proved. Module `GoertzelV24SimpleGraphPortFactorContraction` gives a generic exact criterion: reachability on a finite interface is reconstructed by attaching each active interface coordinate to a named port, consulting the finite port-to-port relation, and returning to the interface. Its completeness premise is restricted to supported ports, which is essential because the profile deliberately records a false diagonal at an inactive port even though graph reachability is reflexive there. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseProfileFactor` specializes this theorem to the at-most-six exact rebase roles and reads the two-port relation directly from the actual incoming serial profile. It adds no state depending on the arbitrarily long accumulated prefix. The remaining tracked obligation is consequently a single source-geometric statement: every supported old attachment role reaches one of the two named incoming ports. The analogous occurrence-sensitive facial factorization is still open, so this is not yet a complete five-coordinate rebase decoder or a reachable-state measurement.

That last tracked obligation has now been sharpened without assuming portal coverage. Module `GoertzelV24SimpleGraphPortResidualFactorContraction` proves the exact generic alternative: a finite-interface component either meets a named persistent port, in which case the stored port relation determines its move, or it meets no such port, in which case one finite residual-component bit retains it. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseResidualProfileFactor` specializes this statement to all three tracked colour pairs on the at-most-six old rebase roles. Combined with the actual incoming profile's two-port relation, the new code recovers literal old-prefix reachability with no attachment-soundness or portal-coverage hypothesis. This does not yet make the rebase a history-independent letter: the residual matrix is presently extracted from the accumulated prefix. The remaining tracked task is the finite recurrence which computes it from the preceding refined state and the literal heterogeneous Cell/rebase receipts. The occurrence-sensitive facial analogue is also still open, so no reachable-state count, numerical V_0, or base verification is claimed.

The switching boundary of that recurrence is now finite and exact. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFiniteSwitch` writes the successor tracked graph as its already proved three-factor union: accumulated old prefix, the two newly displayed crossing edges, and the residual seam. The six possible old attachment edges together with those two new edges form a literal interface of cardinality at most eight. Lean proves that every non-isolated seam vertex belongs to this interface and that reachability between the new cut edges is exactly the reflexive–transitive closure of the three factor-component relations restricted to it. Thus the recurrence no longer ranges over an unbounded ambient graph when factors switch. This is still a structural contraction rather than the completed finite letter: the old component relation must be read from the preceding residual-aware state, the new and seam relations must be decoded from the literal Cell receipt, and the occurrence-sensitive capped-face update must be treated alongside them. No reachable-closure count, numerical V_0, or base verification follows at this checkpoint.

The tracked switch equation is now finite on both sides. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedStepCode` transports the at-most-eight-edge switch through its carrier equivalence and stores, for each of the three tracked colour pairs, the old-prefix component relation as a Boolean matrix on those finite coordinates. The two local factors are the finite new-crossing and residual-seam graphs already supplied by the literal rebase receipt. Lean proves that reflexive–transitive closure of these three finite component moves is equivalent to literal successor tracked reachability between either pair of successor crossings. This is an exact graph-free state extraction, not yet a history-independent source letter: the old matrix is still extracted from the accumulated prefix. Its recurrence from the preceding residual-aware state and the next literal receipt remains open, as does the occurrence-sensitive capped-face update. Accordingly this checkpoint supplies neither a reachable-closure count nor a numerical V_0 or base verification.

The old-role part of that finite state is now connected to the preceding profile rather than merely extracted twice. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedStepState` assigns every one of the at-most-six literal old attachment roles its exact coordinate in the eight-edge switch. On any two such coordinates, Lean identifies the switch code's old component bit with the already proved residual-or-port relation: the incoming profile supplies the two displayed port components and the finite residual code supplies precisely the port-free alternative. Hence this six-role submatrix is not a second unconstrained history field.

The remaining switch coordinates are now identified with the successor state in `GoertzelV24FramedCorridorSerialBoundaryRebaseSuccessorRoles`. Definitionally, the two newly displayed rebase crossings are the next Cell's two input crossings, while the current Cell's two output crossings become the predecessor-output roles retained by that successor. Lean proves that the resulting four-port role map is injective and that each named role denotes the corresponding literal edge. It then reads every tracked-connectivity bit of the actual successor input profile exactly from reflexive–transitive closure of the finite rebase step code. Thus the tracked half of the alternating Cell/rebase recurrence is now a finite decoder equation rather than an ambient-graph observation. The occurrence-sensitive facial recurrence is still separate, so this checkpoint does not yet supply a reachable-closure count, a numerical V_0, or the bounded-base verification.

The facial switch now has the same exact local bound. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFiniteSwitch` orients every occurrence-sensitive seam adjacency: one endpoint is one of the six possible old attachment roles and the other lies in the two-edge rebase region. Since two-sidedness makes the face-position-to-primal-edge map injective, each participating facial seam has at most eight positions, rather than the earlier fourteen-position bound obtained from a larger ambient closure. Lean transports each seam to an eight-position finite graph and packages the exact successor-fragment family as a finite 4-by-8 interface, without padding the actual fragment count. This removes another accidental carrier inflation, but it is not yet the capped-face recurrence: the incoming face state and literal Cell/rebase receipt still have to determine successor continuation, port incidence, and capped length. No reachable-state count, numerical V_0, or base verification is claimed here.

The facial recurrence now has a common finite coordinate system rather than only a separately compressed seam at each face. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceStepCode` retains all occurrences on a chosen ambient face whose underlying edge belongs to the literal eight-edge switch. This includes a switch occurrence that is isolated in the seam but participates in the old-prefix or newly displayed regional factor. Two-sidedness makes the occurrence-to-edge projection injective, so the carrier still has at most eight positions. The two local face graphs are transported through one carrier equivalence, the accumulated old-prefix component relation is stored as a Boolean matrix on the same coordinates, and Lean proves that their three-factor closure computes the literal successor face connectivity exactly. This is the occurrence-sensitive analogue of the finite tracked step. It does not yet make the face transition history-independent: the incoming profile and its finite residue must determine the old component matrix, after which the continuation, port-incidence, and cap-at-five equations must be joined to the same step. Hence no reachable-state count, numerical V_0, or bounded-base verification follows at this checkpoint.

The face state now also has a coordinate system stable across ambient faces. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceDartCarrier` replaces an edge-only global address by the actual dart occurrences over the literal eight-edge switch. There are at most sixteen such occurrences, and the two sides of one primal edge remain distinct. For every face root, Lean proves that the preceding per-face switch-position carrier is equivalent to exactly the fiber of this global carrier on the root's orbit face. This is the occurrence ABI needed to state the predecessor-state equation without silently identifying two facial events over a bridge. It is not yet that equation: the incoming five-field profile together with a finite residual must still determine the old-component submatrix, and continuation, port incidence, and capped length must still be updated jointly. No reachable-state count, numerical V_0, or bounded-base verification follows here.

The predecessor facial component relation now factors through finite interface data without identifying the two occurrences of a bridge edge. Generic module `GoertzelV24RotationFaceRegionalDartGraph` defines regional face connectivity directly on literal darts, proves it equivalent to the earlier root-relative cyclic-position graph, and shows that the canonical dart representatives of two predecessor boundary fragments are connected exactly when their fragment coordinates are equal. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceResidualProfileFactor` then extracts, on the at-most-sixteen switch occurrences, an exact finite alternative: a component either meets one named predecessor face port, or it is retained by a port-free residual bit. Lean proves that this residual-or-port relation is equivalent to literal old-prefix facial reachability. This is the occurrence-sensitive facial analogue of the earlier tracked factorization, but it is still extracted predecessor state. The next obligation is a history-independent recurrence for these residual bits together with the literal continuation, port-incidence, and cap-at-five update. No measured reachable closure, numerical V_0, or bounded-base verification is inferred.

That predecessor state is now connected to the facial step code rather than merely extracted beside it. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceStepState` sends every coordinate of the per-face eight-position switch code to its literal dart in the stable global occurrence carrier. On every pair of such coordinates, Lean proves that the stored old-component matrix bit is exactly the preceding residual-or-boundary-port relation. Consequently the accumulated-prefix matrix is not a second independent history field. This is the facial counterpart of the earlier tracked step/state equation. The remaining history-independent obligation is to transport the resulting successor components to the residual state of the next literal rebase and to join that transition with continuation, port incidence, and capped length. No reachable closure, numerical V_0, or bounded-base verification follows yet.

A support-sensitivity audit exposed one additional finite state coordinate before that transport can be sound. Graph reachability is reflexive even at an isolated coordinate, so the diagonal of a component matrix cannot distinguish an active singleton from a dart or edge absent from graph support. The residual attachment relation, however, is deliberately support-sensitive: both an interface coordinate and its persistent port must actually occur in the graph. Generic module `GoertzelV24SimpleGraphSupportedPortResidualFactorContraction` therefore adds finite interface- and port-activity tables to the existing residual code and proves that its canonical extraction records literal support membership exactly. Modules `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceSupportedState` and `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedSupportedState` specialize the refinement to the stable facial dart carrier and to all three tracked-colour predecessor states. Forgetting activity recovers the previous residual codes definitionally, while every facial switch coordinate reads its old activity from the same stable state as its old component relation. Thus no earlier theorem is weakened, and a later decoder can no longer invent an attachment by confusing reflexivity with support. The actual successor transport across the heterogeneous Cell/rebase pair, the remaining three face coordinates, the reachable closure, the numerical V_0, and the bounded-base verification are still open.

The support bit now participates in an exact finite successor calculation, rather than merely being retained for future use. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceSupportedStep` extends the finite facial rebase letter by one Boolean activity table. Its output coordinate is active exactly when that occurrence was active in the old prefix, occurs in the newly displayed regional factor, or occurs in the residual seam. The bounded-carrier encoder is proved to preserve support membership, cyclic face positions are proved to agree with their literal dart occurrences on support, and Lean then proves that this Boolean union law is equivalent to support in the actual successor regional face graph on every switch occurrence. Hence the finite letter now computes both connectivity and physical presence; it cannot create an attachment from the reflexive diagonal of an isolated coordinate. This is the support coordinate only. Transport to the next residual carrier, the intervening literal Cell update, face continuation, port incidence, cap-at-five, the complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

Successor face identity is now aligned with the same stable occurrence coordinates. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceSuccessorState` maps every canonical boundary fragment of the successor cut injectively to a dart occurrence over the finite rebase switch. Lean proves that the successor profile's face-continuation bit is true exactly when the two mapped darts have the same ambient orbit face. This is deliberately not regional reachability: disconnected fragments of one ambient face must still be recognized as continuations. The theorem supplies the exact successor coordinates needed by the recurrence; it does not yet compute them from the predecessor state and literal Cell letter, nor update port incidence or capped face length. The complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification therefore remain open.

The successor port-incidence coordinate is now finite and exact as well. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceSuccessorPortIncidence` proves that a successor fragment contains one of the two newly displayed crossing ports precisely when its canonical switch occurrence reaches some occurrence over that named port edge in the finite facial step closure. The existential occurrence is essential rather than cosmetic: the two facial sides of a bridge must remain distinct even though they project to one primal edge. Face identity and port incidence are therefore both aligned with the same bounded occurrence carrier. Capped face length, the intervening literal Cell update, the complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The successor cap-at-five coordinate now has an exact geometric update law. Module

`GoertzelV24FramedCorridorSerialBoundaryRebaseFaceSuccessorCap` splits each successor face fragment into its old-prefix occurrence slice and the slice contributed by the two newly displayed rebase edges. Lean proves that the stored capped length is their two capped cardinalities, minus the genuine shared occurrences counted by both slices, capped again at five. Two-sidedness identifies occurrence cardinality with the edge cardinality stored in the boundary profile, and the two-edge rebase bounds the common seam strictly below the cap. Thus the picture-level instruction "join the two face pieces without counting their seam twice" is now a theorem. This is not yet a graph-free recurrence: the capped old-prefix slice still has to be decoded from the predecessor profile and its finite residual state. The intervening literal Cell update, complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The predecessor contribution to that law is now retained by a finite state rather than left as an ambient-history query. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceCappedState` assigns to every stable switch occurrence both exact old-region presence and the cap at five of its full old-prefix component. The distinction is load-bearing: a one-position regional factor is physically present but is isolated in the induced adjacency graph, so adjacency support alone would incorrectly assign it cap zero. The explicit presence bit retains that factor with cap one; an absent coordinate has cap zero, and coordinates belonging to the same stored component provably carry the same presence and cap. Forgetting these new fields is definitionally the preceding support-sensitive state. The remaining decoder must count each old component once when the local rebase joins several of them, then add the new local slice and subtract the genuine overlap before transporting the result to the successor residual carrier. Thus the predecessor weights are finite and exact even for singleton factors, but the complete Cell/rebase recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The predecessor weights can now be consumed without counting the same old face once per switch occurrence. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceComponentRepresentatives` turns the finite old-component relation into a partition, chooses the least occurrence of each component as its canonical representative, and retains exactly the representatives of physically present components. Lean proves existence and uniqueness of this representative for every present occurrence and proves that the stored cap is unchanged by the replacement. The literal source rebase supplies all required semantic laws from exact regional reachability and support. What remains is to select the representatives that contribute to one successor fragment and insert their weights once into the already-proved overlap-corrected cap equation. The complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The resulting finite old-prefix sum is now executable as well. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceComponentCapSum` retains the active canonical old components connected to one successor switch occurrence, adds each stored component cap exactly once, and caps the total at five. Lean proves that every active reachable occurrence contributes through its unique representative and with the same stored weight. The remaining adequacy statement is correspondingly sharp: this decoded finite sum must be identified with the literal old-prefix position slice of each successor face fragment. Once that equality is proved, it can be substituted directly into the existing overlap-corrected successor cap law. The complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

Coverage of that decoder by the literal source geometry is now proved, including isolated one-position old factors. The generic module `GoertzelV24SimpleGraphThreeFactorSelectedAnchor` shows that, in a union of three graph factors, every point of the first factor reachable from a selected output has a selected first-factor anchor whenever every genuine factor switch is selected. Its source specialization in `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceOldFactorAnchor` uses the old prefix, the newly displayed two-edge region, and the facial seam as those three factors. Consequently every old-prefix occurrence inside a successor face fragment reaches an old-component anchor on the finite rebase switch, and the exact finite closure connects the successor coordinate to that anchor. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceOldFactorFragment` then decodes every present coordinate to its literal old regional connected component and proves equality between the stored component-position set and the fragment's actual cyclic positions. Thus no contributing old face component can escape the finite decoder merely by being a singleton or by meeting the switch away from a preferred occurrence. The remaining adequacy step is injective accounting: identify canonical finite representatives with the literal old regional factor fragments and transport their capped weights across that correspondence. The complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

That predecessor adequacy statement is now proved. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFactorRepresentativeEquiv` constructs an exact equivalence between the contributing canonical finite representatives and the literal old-prefix factor fragments inside each successor face. Surjectivity uses the three-factor selected-anchor theorem; injectivity uses equality of literal connected components and the uniqueness of the finite canonical representative. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceOldCapAdequacy` then transports the individual capped weights and their finite sum across that equivalence. Lean proves that the decoder's predecessor sum is exactly the cap at five of the successor fragment's literal old-prefix occurrence slice, including an isolated singleton factor. This closes the predecessor half of the capped face recurrence. Substitution into the already-proved successor inclusion–exclusion law, the newly displayed Cell contribution, the complete heterogeneous recurrence, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The predecessor decoder has now been inserted into the successor equation. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFinitePredecessorCap` replaces the literal old-prefix slice in the inclusion–exclusion law by the finite canonical-component cap sum. The equality is exact, including the isolated-singleton case; it does not query the unbounded prefix after the finite predecessor state has been supplied. The only terms left at this stage are the newly displayed two-edge slice and its genuine overlap with the old region.

Those last two terms now have a finite exact decoder as well. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFiniteLocalCaps` stores the local and overlap contributions as two functions into `Fin 6`, indexed by the actual successor fragments. Every literal rebase supplies such a finite table, and Lean proves that the successor profile's capped face length is precisely the decoded predecessor sum plus the stored local cap minus the stored overlap cap, capped at five. Thus the face-length coordinate of the boundary rebase no longer contains an ambient geometric query. Packaging this decoder jointly with the already finite face-continuation, port-incidence, and tracked-connectivity laws, then composing it with the distinct literal Cell letter, remains open; no reachable closure, numerical V_0, or bounded-base verification is claimed yet.

The port-incidence law is now graph-free after extraction as well. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFinitePortLabels` attaches two finite port-label bits to each occurrence coordinate of the at-most-eight-position facial switch. Its decoder says that a successor fragment contains a displayed port exactly when the fragment's canonical coordinate reaches an occurrence carrying that label. Lean proves this is equivalent to the literal profile's `fragmentContainsPort` field, so no ambient primal-edge comparison remains at the use site. Moreover, the extracted labels obey a one-hot law on each two-sided orbit face: one named primal port cannot occupy two occurrence coordinates. This is an exact restriction on the source image rather than an arbitrary Boolean table. The joint finite rebase receipt, its composition with the distinct literal Cell letter, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The one-hot table itself is now compressed without loss in `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFinitePortPositions`. For a facial switch with $n \leq 8$ occurrence coordinates, each displayed port is stored as either one coordinate or absence. The exact code count is therefore $(n+1)^2 \leq 81$, rather than the 2^{2n} possibilities of an unrestricted Boolean table. A generic round-trip theorem proves that the optional position is present exactly at the original true label, and Lean transports the literal successor port-incidence iff through this compact representation. This reduces one field of the genuine source image; it does not yet provide the joint heterogeneous transition or its reachable closure.

The successor face-continuation field has now received the same exact finite treatment. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFaceFiniteContinuation` uses the already-proved fact that a graph-derived continuation relation is equality of ambient orbit-face labels, hence an equivalence relation. It stores the canonical upper-triangular equivalence code and proves that the graph-free decoder recovers every literal successor `faceContinues` bit exactly. At the established four-fragment bound there are at most fifteen such codes, rather than an unrestricted Boolean matrix. Together with the finite cap decoder and compact port positions this closes the finite representation of the three face-progress subcoordinates; their joint packaging with the tracked-strand recurrence and composition with the distinct literal Cell letter remain open. No reachable count, numerical V_0, or bounded-base verification is claimed.

Those three coordinates are now packaged in one finite receipt. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseFiniteFaceReceipt` stores, for each successor fragment, its finite component step, distinguished output coordinate, two compact port positions, and capped predecessor contribution; the receipt also stores the canonical face-equivalence code and the local/overlap cap table. Its three adequacy theorems recover the literal successor profile's face continuation, displayed-port incidence, and capped face progress exactly. The port and cap fields deliberately retain their respective canonical representatives of the same orbit face instead of identifying dependent finite carriers by an unchecked cast. Thus the complete face part of the literal boundary-rebase letter is one finite graph-free ABI. The exact tracked-strand join, composition with the distinct literal Cell letter, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The finite face ABI is now joined to the already-proved tracked decoder in `GoertzelV24FramedCorridorSerialBoundaryRebaseFiniteOutputReceipt`. The resulting code stores the two successor crossing colours, the finite tracked component step, and the complete finite face receipt. Five adequacy theorems recover every field of the literal successor profile: edge colour, tracked two-colour connectivity, face continuation, port incidence, and capped face progress. This closes the finite output description of the boundary rebase. It does not yet close its state recurrence: the tracked step's old-component matrix is still extracted from the literal old prefix, rather than computed from the incoming finite state. That recurrence, the distinct Cell/rebase composition, reachable-closure measurement, numerical V_0, and bounded-base verification remain open.

The remaining tracked matrix has now been given an exhaustive literal coordinate presentation rather than a padded one. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedSwitchRoles` proves that every position of the at-most-eight-edge switch is named either by one of the at-most-six old-prefix attachment roles or by one of the two newly displayed successor crossings. The resulting map onto switch coordinates is surjective and has a chosen right inverse; it is deliberately not asserted injective, because the same literal edge may have more than one valid source role. On the entire old/old block, the accumulated component bit is already the exact residual-or-port relation computed from the incoming finite state. Consequently the live tracked recurrence has been narrowed to the matrix blocks involving a newly displayed crossing. Computing those blocks through the distinct rebase and Cell receipts, composing the heterogeneous letters, measuring their reachable closure, deriving V_0, and verifying the bounded base remain open.

The new-coordinate remainder now has a sharp exact boundary. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseTrackedSwitchCoverage` proves that, if every newly displayed successor crossing which was already in the old prefix coincides with one of the recorded old attachment roles, then the complete old-component matrix is precisely identity plus the incoming residual-or-port relation transported along equal finite coordinates. The statement allows old and new roles to name the same literal edge; it does not replace that quotient by a disjoint padded carrier. The coverage premise is not asserted here. Region membership alone cannot prove it, because the vertex-side regional carrier also contains edges with both endpoints on the old side, while the available historical role theorem classifies true frontier crossings. Accordingly the remaining tracked atom is an endpoint-side/orientation theorem for the successor crossings, or a finite receipt which explicitly retains the internal-overlap case. The unconditional old-role-plus-new-crossing switch remains valid in either event. This conditional exact decoder does not yet produce the heterogeneous recurrence, reachable closure, numerical V_0, or bounded-base verification.

The module `GoertzelV24FramedCorridorSerialBoundaryRebaseFiniteOutputDecoder` now turns the complete finite rebase receipt into one graph-free successor profile. The dependent face count is carried by the letter itself, and Lean proves that every literal source rebase decodes exactly to its actual serial input profile: the two edge colours, all three tracked-connectivity matrices, face continuation, displayed-port incidence, and capped face progress agree field by field. Thus the output half of the rebase ABI is now a single finite decoder rather than five separate observations. The receipt is not yet proved computable from the preceding finite state; exact old/new overlap and composition with the distinct Cell letter remain before a route-correct reachable closure, numerical V_0, or finite-base claim.

The dependent predecessor coordinate has also been stabilized without inventing geometry. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseOldAttachmentSlots` uses six named slots: two current inputs, two current outputs, and two outputs of the immediately preceding Cell. Actual source roles embed injectively and round-trip through a partial decoder. At the first offset the two predecessor slots decode to “absent”, rather than to padded edges. The literal-edge map commutes with this round trip. This gives the tracked residual recurrence a fixed carrier while preserving the source distinction between an unavailable predecessor and an inactive but real contact.

Module `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedState` then transports the complete support-sensitive tracked predecessor code for all three colour pairs onto those six slots. Missing slots have false activity, attachment, and residual-component bits; persistent-port activity is unchanged; and every real role recovers exactly the earlier dependent state. In particular, both predecessor slots are proved wholly inactive at the first source offset. The tracked predecessor state is therefore one uniform finite type without padded geometry. Computing its successor under the literal Cell-plus-rebase letter is the remaining tracked recurrence step.

The first exact interpretation of that stable state is provided by `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedStep`. Against the complete incoming serial profile, its residual-or-port relation recovers literal old-prefix tracked reachability on every pair of the six actual predecessor roles. This closes the old/old block without retaining the accumulated graph. It also identifies a real state-design boundary: six slots do not name the two successor crossings whose old-prefix component relations are queried by the boundary-rebase decoder.

Module `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedSwitchState` therefore adds exactly those two crossings, producing a fixed eight-slot tracked switch state. The extension is lossless rather than speculative: dependent source roles round-trip through the stable names, and the incoming profile plus the support-sensitive residual tables recover old-prefix reachability for every old–old, old–new, and new–new role pair. The named roles surject onto the exact finite switch, so Lean further proves that this state computes every entry of the formerly prefix-extracted `oldComponent` matrix. The arbitrary accumulated-prefix matrix has therefore been removed from the tracked rebase letter. What remains is a genuine recurrence: compute the next eight-slot state from the present state and the literal heterogeneous Cell-plus-rebase receipt, and perform the corresponding stable transport for capped facial state. No reachable count, numerical V_0, or bounded-base verification is inferred yet.

The coordinate slide needed by that recurrence is now exact in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedSwitchSuccessorSlots`. Four current slots have literal successors: the two Cell outputs become the successor's predecessor outputs, while the two crossings exposed by the current rebase become its current inputs. Lean proves that these four pairs name definitionally equal ambient edges. The other four successor slots are exactly its two Cell outputs and its two following-rebase crossings; an explicit equivalence partitions all eight slots into these inherited and fresh halves. This is only an edge-name transport theorem. It deliberately does not preserve the tracked state values, since adjoining the next literal Cell may change reachability around an inherited edge. Computing that change from the heterogeneous Cell-plus-rebase receipt remains the next semantic recurrence obligation.

The carrier on which that update must run is fixed in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedTransitionSlots`. Two consecutive eight-slot switches are not treated as sixteen unrelated names. Their four inherited edges are shared, so the transition carrier has exactly twelve slots: all eight current names plus the four genuinely fresh successor names. Both switches embed injectively. The successor embedding identifies its predecessor-output and current-input slots with the matching current names and reserves new coordinates only for its Cell outputs and following-rebase crossings. The partial source interpretation decodes every actual current role and every inherited or fresh successor role to its exact ambient edge; only the two unavailable predecessor names at the first offset remain absent. This is the stable carrier for the next finite tracked update, not yet that update itself.

The semantic region traversed by that update is now exact as well. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedTransitionRegion` proves that the successor outgoing-prefix region is the union of the current prefix with precisely two local pieces: the current boundary rebase and the successor literal Cell. Its tracked graph consequently factors as the current accumulated graph, the local rebase-plus-Cell graph, and the residual seam between them. This fixes the temporal convention needed by a replay table—outgoing prefix to outgoing prefix—without replacing the heterogeneous source word by repetitions of one abstract letter. The result does not yet contract the local graph and seam onto the twelve finite slots; that contraction is the remaining tracked-recurrence obligation before any reachable-mode count is meaningful.

That transition is also expanded in its literal source order in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedTransitionFactorization`. The successor tracked prefix is exactly the five-factor union of the current prefix graph, the boundary-rebase graph, the rebase seam, the successor Cell graph, and the Cell-input seam. The eventual finite replay may therefore compose the already verified rebase and one-Cell receipts, rather than postulating a new homogeneous transition. Contracting both seams onto the twelve stable slots remains open.

The source names are now realized as one actual finite edge carrier in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedTransitionFiniteSwitch`. The carrier is the image of the twelve stable transition names, so names that denote the same literal edge are quotiented rather than falsely declared distinct. At the initial offset, each unavailable predecessor name is sent to an already present current crossing, introducing no fictitious edge. Lean proves that the resulting carrier has cardinality at most twelve and contains every edge in both the current and successor exact eight-edge switches. The reverse containment is also proved: this carrier is exactly the union of those two switches, including at the initial offset where the fallback is already a current crossing. Thus the local five-factor transition now has a uniform, quotient-aware edge boundary on which a finite contraction can be stated. No tracked-connectivity recurrence, reachable-state count, or numerical V_0 is claimed by this carrier theorem; contracting the five factors and their two seams onto it remains the next semantic step.

The tracked output half of one boundary rebase is now history-free in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedStepDecoder`. The decoder takes only the incoming complete profile, the stable support-sensitive eight-slot state, the literal local two-factor code, and the quotient-aware coordinate-to-slot map. It reconstructs the formerly ambient `oldComponent` matrix. Lean proves agreement with every literal old-prefix component bit, then proves that replacing the extracted matrix preserves the entire local component relation and its reflexive–transitive closure. Thus successor two-colour connectivity no longer requires rereading the unbounded prefix graph. This does not yet compute the successor eight-slot residual state, and the capped facial state is still extracted from literal geometry. Those two obligations remain before a source-faithful reachable-state count, numerical V_0, or finite base verification can be claimed.

That history-free tracked decoder now passes through the complete profile interface in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFiniteOutputDecoder`. The module replaces only the tracked subreceipt of the existing literal finite-output decoder and proves that the resulting output is still exactly the source successor profile in all five coordinates: boundary colours, two-colour connectivity, face continuation, port incidence, and capped face lengths. Thus the finite tracked reconstruction composes with the full semantic interface rather than merely proving an isolated connectivity fact. The remaining prefix-history extraction is now localized to the capped facial receipt. Computing that stable facial state and the successor residual states remains open; this theorem does not supply a reachable-state count, a numerical V_0, or a bounded-base verification.

That facial history has now been placed on a fixed carrier in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceCappedState`. The literal rebase switch contains at most sixteen face-dart occurrences. Lean embeds every real occurrence into a named `Fin 16` slot and uses a partial decoder, so an unused slot is distinguished from an inactive but present occurrence. On every occupied slot the stable state recovers exactly the source activity bit, literal-presence bit, attachment to each displayed port, residual face-component relation, and component size capped at five. The cap is measured on the whole regional dart-orbit class and therefore does not depend on choosing a representative root. This removes the dependent occurrence carrier, but it is not yet the facial transition: the root-free cap must still be identified with the per-output-face capped receipt, after which the successor sixteen-slot state must be computed from the present state and the literal heterogeneous Cell-plus-rebase letter. No reachable-state count, numerical V_0, or bounded-base verification is claimed at this checkpoint.

The first of those two obligations is now discharged in `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceCappedStep`. For every displayed face coordinate, Lean identifies the cardinality of its predecessor face-component support with the root-free dart-component cardinality stored in the stable state. Consequently the stable component cap at five is exactly the per-output-face cap used by the literal rebase receipt; it is not merely an upper bound and it is independent of the chosen displayed dart.

Modules `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceCappedStepDecoder`, `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceFixedState`, and `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceFixedStepDecoder` then turn that identification into a graph-free finite interface. The exact predecessor face receipt is decoded from a common carrier of sixteen occurrence slots and four boundary-fragment slots together with the literal local Cell–rebase face code. Unused slots are padded with false data, while every real port and occurrence has a proved round trip through its fixed coordinate. The decoder reconstructs exactly the predecessor component relation, activity, literal presence, and cap used by the source transition. The four port slots use the general bound proved for the serial rebase carrier; no sharper three-slot claim is assumed.

This is a representation closure, not yet a dynamical one. The predecessor facial state is now a genuinely fixed finite object and one literal local letter consumes it exactly, but Lean does not yet compute the successor fixed state from that pair. The next load-bearing step is that successor recurrence, followed by the heterogeneous reachable-closure measurement. Hence no reachable-state count, compressed pigeonhole threshold, numerical V_0, or route-native bounded-base verification follows yet.

The fixed facial state now reaches the complete successor profile rather than stopping at an internal component decoder. Modules `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceFixedStepExact` and `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFaceFixedReceipt` prove that the fixed occurrence and predecessor-port coordinates reconstruct each complete successor-face row: continuation, port incidence, and capped face progress all agree exactly with the literal source receipt. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseStableFixedFiniteOutputDecoder` then combines that facial reconstruction with the already stable tracked decoder. For every literal heterogeneous Cell–rebase step, the resulting finite letter decodes to exactly the actual successor profile in all five coordinates. Thus the decoded output no longer consults the accumulated prefix graph at all.

This closes the one-step output interface, not the state dynamics. The next step must still compute the successor eight-slot tracked residual and the successor Fin(16)-by-Fin(4) facial residual from the present fixed states and the literal local letter. Only after that recurrence is proved can the heterogeneous reachable closure be measured and used to derive a compressed pigeonhole threshold. No reachable-state count, numerical V_0, or bounded-base verification is claimed here.

The next transition seam is now bounded without introducing a third local letter. Module `GoertzelV24TerminalProfileSeamUnion` proves, for both tracked edge connectivity and occurrence-sensitive facial continuation, that the residual seam from a left region into a union of two right regions is contained in the supremum of the two seams obtained by adjoining those regions in order. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseStableTrackedTransitionSeam` specializes this fact to one literal source step: first the boundary rebase, then the successor Cell. Every tracked turn of the combined transition seam is supported on the union of the quotient-aware twelve-edge transition switch and the successor Cell's bounded fourteen-edge outgoing neighbourhood. Thus the carrier contains at most twenty-six actual ambient edges; repeated source names are identified rather than counted twice.

This is a finite-support theorem, not yet the recurrence itself. It proves that the combined seam contains no independent history-sensitive ingredient, but it does not yet contract the accumulated prefix factor or compute the successor fixed tracked and facial residual states. The heterogeneous reachable closure, numerical V_0, and route-native bounded base therefore remain open.

The overlap in that presentation is now handled exactly rather than silently treated as disjoint. Modules `GoertzelV24TerminalProfileExclusiveRegionFactorization` and `GoertzelV24FramedCorridorSerialBoundaryRebaseStableExclusiveTransition` assign every transition edge already present in the accumulated prefix to the old factor, retain the set difference as the genuinely new local factor, and recompute both the tracked and occurrence-sensitive facial seams. Lean proves that these three factors recover the complete successor graphs exactly. It also proves that every old endpoint of either recomputed seam is a genuine vertex-side crossing of the current prefix, which is the source-geometric boundary needed for the finite successor-state recurrence.

A natural shortcut fails: the recomputed exclusive seam is not in general a subgraph of the original seam for the overlapping factors. An edge assigned to the old side may itself lie in the original right region, so the original seam omits a turn that the exact exclusive presentation must retain. Thus the existing twenty-six-edge support bound cannot simply be inherited by monotonicity. The next theorem must explicitly classify these overlap-created frontier turns into the fixed transition roles. This is a sharper recurrence obligation, not a reachable-state count or a numerical V_0.

That classification is now exact. Module `GoertzelV24FramedCorridorSerialBoundaryRebaseStableExclusi` proves, for tracked connectivity and separately for literal face-cycle occurrences, that every exclusive residual turn has one of three forms. It either touches one of the two named rebase edges, belongs to the ordinary successor-Cell seam and therefore touches one of that Cell's two named output crossings, or has an old true-prefix frontier edge which itself belongs to the successor Cell region. The first two branches are already supported on the fixed rebase and Cell carriers. The third is retained as one explicitly oriented Cell-overlap predicate, including the ambient adjacency and every old/new membership fact; it is not weakened to a new hypothesis or declared empty. Thus the remaining geometric question is no longer "what can the recomputed seam contain?" but whether this one overlap-created Cell turn is impossible or has a fixed source-role code. The result does not identify the raw ambient prefix with a physical serial gluing, compute the successor stable states, measure a reachable closure, derive V_0, or verify the bounded base.

The four-edge Cell frontier now has a sharp closed-carrier consequence. Modules `GoertzelV24CubicFourEdgeConnectedSides` and `GoertzelV24FramedCorridorLiteralCellFiniteCarrier` prove that, when the ambient graph is cubic and cyclically five-edge-connected and both sides of the exact four-edge boundary are connected, one side consists of exactly two vertices. If the literal Cell is that side, its complete regional edge carrier contains at most six edges. The source-facing theorem does not choose an orientation: it returns either this six-edge bound or the precise residue that the retained complement has two vertices.

This is intentionally not reported as closure of the overlap branch. The well-formed opened source carrier has boundary defects and is already proved not globally cubic, whereas the new count is a theorem about a closed cubic ambient graph. Applying it to the actual source therefore requires an explicit closed-to-open transport with the Cell side oriented, or a local interior-cubic replacement of the global premise. The dichotomy records the small missing interface without assuming it; it does not compute the successor stable states, the reachable closure, V_0, or the bounded base.

The local replacement has now been proved on the actual opened carrier. Module `GoertzelV24ClosedWebLocalLayerFiniteCellCarrier` first shows that the side opposite the source-selected retained component is connected, using local two-sidedness on the four-edge Cell loop rather than the false global two-sidedness of the annulus. A radial comparison path puts the complete inner interface on the retained side, while the local-loop avoidance theorem does the same for the outer interface. Hence every vertex on the literal Cell side avoids all ten degree-one stubs and has degree three. Exact degree counting now gives the non-vacuous source dichotomy: the Cell side either contains a primal cycle, or it has exactly two vertices and its full regional edge carrier has at most six edges. The former is now the sole geometric residue of this carrier reduction. No one-sided-cycle exclusion, successor state, reachable count, numerical V_0, or bounded-base verification is claimed at this checkpoint.

The non-cyclic branch now has an exact finite tracked-connectivity code. Module `GoertzelV24ClosedWebLocalLayerFiniteTrackedLetter` retains the four literal source-crosscut ports in left-then-right order and transports the three tracked two-colour graphs onto the same at-most-six-edge Cell carrier. Adjacency and complete regional reachability are both preserved exactly by the finite coordinates. The code is attached to the actual indexed Cell, not merely to its one-of-three oriented geometric shape: the present development has not proved that the semantic transfer relation factors through orientation alone.

This is deliberately called the tracked-connectivity part of a letter, not a complete profile letter. The source profile also carries face continuation, boundary-port incidence, and face progress capped at five. Their graph-independent finite update, together with the explicit cyclic-Cell branch above, remains to be proved before measuring the heterogeneous reachable closure. Consequently no reachable-state count, numerical V_0, or bounded-base verification follows from this checkpoint.

The complete local counting semantics can now be stated on the actual opened carrier. The older complementary-side profile API threaded a global cubicity argument through its declarations even though the removed-side dart needed to root the open region was already present in the source boundary datum. That premise is false for the source annulus, whose named boundary stubs have degree one. Module `GoertzelV24AnnularCrosscutComplementProfileOfBoundary` therefore roots the complementary open region directly at the stored boundary dart and reconstructs the same five-coordinate left and right profiles. Its `Count` entry is the cardinality of the finite fiber of literal Tait colourings, and positivity is proved equivalent to the existence of one such colouring realizing the two stated profiles. No coordinate or support notion is weakened; the old declarations remain as compatibility wrappers.

Module `GoertzelV24ClosedWebLocalLayerLiteralCount` specializes this boundary-rooted semantics to each actual indexed source Cell. Its literal four-edge wall bounds the number of boundary-local face fragments by eight, so the two exact dependent profiles embed injectively into a common `BoundedCorridorCutProfile 2 0 8` carrier. The resulting decidable heterogeneous letter holds exactly when the corresponding literal `Count` entry is positive, equivalently when an actual Tait colouring of that Cell realizes both full profiles. Thus the opened source now has a non-vacuous local letter carrying crossed-edge colours, all tracked two-colour connectivity, face continuation, port incidence, and capped face progress.

This is still a local Cell observation, not the live serial recurrence. The next factorization theorem must identify its left and right observations with the predecessor and successor cumulative-prefix profiles and compose the separate boundary-rebase receipt. The cyclic-Cell branch also remains to be excluded or finitely encoded. Until those steps and a checked reachable-mode table are present, this result supplies neither a reachable-state count nor a numerical V_0 or bounded-base verification.

The two local observations are now tied together at their source rather than merely realized independently. Module `GoertzelV24AnnularCrosscutComplementCombinedCountOfBoundary` proves that a positive boundary-rooted `Count` entry is witnessed by one literal Tait colouring and one realizable combined four-port profile. Its left and right matrix indices are the restrictions of that same complete five-coordinate state. The source-Cell specialization in `GoertzelV24ClosedWebLocalLayerLiteralCount` therefore supplies one joint semantic receipt for every admitted local transition. This is the atomic morphism needed by the cumulative recurrence; it is not yet that recurrence. Identifying the two restrictions with consecutive cumulative prefixes, composing the boundary rebase, handling cyclic Cells, measuring the reachable modes, deriving V_0, and verifying the bounded base remain open.

The cumulative carrier needed for that comparison is now constructed on the actual opened source annulus. Module `GoertzelV24ClosedWebLocalLayerSerialPrefixRegion` defines the prefix before a Cell to be the union of the preceding literal complementary Cell regions and proves that adjoining the current Cell gives exactly the successor prefix. Both ordered incoming and outgoing crossings are proved to belong to the literal Cell, and the source colouring therefore yields complete five-coordinate graph-derived profiles on the two consecutive prefixes. This construction does not import the false global-cubicity or global face-two-sidedness premises of the older serial tower. It is a carrier identity, not yet transition adequacy. The seam-terminal refinement is recorded next; factorization of the joint local `Count` receipt through these cumulative profiles, the separate boundary rebase, the cyclic-Cell residue, reachable closure, the numerical V_0, and the route-native bounded base remain open.

The seam state between the cumulative prefix and one Cell is now explicit. Module `GoertzelV24ClosedWebLocalLayerSerialTerminalProfile` retains the canonical shared corridor rung as one terminal of the actual predecessor, constructing a complete `BoundedCorridorCutProfile 2 1 4`. Every crossing and the terminal are proved to be genuine regional ports. Adjoining the literal Cell is then proved to give exactly a zero-terminal pre-rebase successor region, on which the corresponding complete five-coordinate profile is constructed. The rung is retained in this intermediate region rather than silently assumed redundant. The next paragraph localizes the separate boundary rebase; its five-coordinate semantic update and the adequacy theorem tying this cumulative step to the joint literal `Count` receipt remain open. No reachable count, numerical V_0, or bounded-base verification follows yet.

The boundary rebase is now localized to a uniformly finite literal support. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebase` assigns four stable roles to the old shared rung, the two newly exposed crossings, and the next shared rung. Lean proves that their image has cardinality at most four, that the pre-rebase successor and the next terminal-aware input agree on every edge outside this image, and equivalently that their symmetric difference is contained in it. Thus the next semantic theorem no longer has to compare two unbounded regions: it must update connectivity and facial state only across this explicit four-edge switch. That five-coordinate update and its joint-`Count` adequacy are not yet proved; neither are the reachable closure, numerical V_0, or bounded base.

The tracked predecessor state on that switch is now finite and exact. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseTrackedState` stores the nonzero colour and actual predecessor-region membership of each of the four roles, together with a support-sensitive component code for each of the three tracked colour pairs. Its adequacy theorem identifies the graph-free decoder exactly with the tracked-connectivity predicate used by the complete graph-derived profile. In particular, reflexive reachability at an absent or untracked coordinate cannot create a false diagonal entry: the regional and colour guards are retained separately. This removes the unbounded predecessor graph from the tracked half of the rebase state, but it does not yet compute the successor state or encode the occurrence-sensitive facial update. The joint five-coordinate recurrence, reachable closure, numerical V_0, and bounded base remain open.

The facial predecessor state on the same switch is now finite without a two-sided-face assumption. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFaceState` retains both literal darts over each of the at most four switch edges, hence at most eight occurrence slots. For every occupied slot it records predecessor region membership, exact reachability in the regional face-dart graph, and a face-progress cap in `Fin 6`. The cap counts the image of the complete dart component under `edgeOf`, so a primal edge visited through both darts is counted once rather than twice. Adequacy theorems recover occupancy, membership, occurrence connectivity, the capped distinct-edge count, and the same-fragment predicate on every actual switch occurrence. This is still the predecessor facial interface: reconstructing the successor rows from it and the literal Cell geometry, combining that update with the tracked state, and proving joint-`Count` adequacy remain open, as do reachable closure, the numerical V_0, and the bounded base.

The additional residual datum needed by edge deletion is now explicit as well. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFaceResidualState` enumerates the pre-rebase two-edge cut's actual boundary fragments in at most four fixed slots. Each switch occurrence records which old fragment it attaches to and whether it lies in a component meeting no old fragment. Lean then proves the exact factorization: for any two present switch occurrences, predecessor facial reachability holds if and only if they are connected in a port-free residual component or both attach to one common occupied old-fragment slot. This distinction is necessary because removing a switch edge can split a predecessor component; retaining only the full predecessor connectivity matrix would lose that information. The result is still predecessor data, but is not by itself a successor-row calculation: a facial path can enter or leave the changed support through an immediate neighbour in the face permutation. That calculation, the combined tracked and facial recurrence, joint-`Count` adequacy, reachable closure, the numerical V_0, and the bounded base remain open.

The complete finite facial collar of the boundary rebase is now bounded. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFaceCollar` starts from the at most four changed primal edges, retains their at most eight literal dart occurrences, and adjoins the immediate predecessor and successor of every occurrence under the face permutation. The resulting computation support has at most twenty-four darts. Lean proves that the predecessor and successor edge regions have a common core after removing the switch, that any facial adjacency incident to a changed occurrence lies wholly in this collar, and that regional facial adjacency is exactly unchanged outside it. Thus the eight switch occurrences are the observed coordinates, while the twenty-four-dart collar is the finite support required to compute their update. This is a locality theorem, not yet the collar/core reachability recurrence or the successor face-state decoder. The combined five-coordinate recurrence, joint-`Count` adequacy, reachable closure, the numerical V_0, and the bounded base remain open.

The collar now carries an exact semantic recurrence rather than merely a locality bound. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFaceCollarRecurrence` decomposes the successor regional face graph into the unchanged core and a canonical local residual whose support lies wholly inside the at-most- twenty-four-dart collar. Between any two collar darts, successor reachability is exactly the reflexive–transitive closure of whole-core component moves and local-residual component moves on the collar. Thus an arbitrarily large prefix is consulted only through its component relation at the finite interface; paths outside the collar are not enumerated. This theorem is stated on actual darts; its bounded coordinate transport is recorded next. Recovery of the core-component matrix from the predecessor receipt remains open, as do the combined five-coordinate recurrence, joint-`Count` adequacy, reachable closure, the numerical V_0, and the bounded base.

The same facial recurrence now has a single bounded code. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFaceCollarCode` transports the inherited whole-core component relation and the local successor residual through one common carrier with at most twenty-four coordinates. A Boolean tag selects the core or local graph, and Lean proves both decoding theorems and the exact statement that successor facial reachability is the reflexive–transitive closure generated by those two coded graphs. Thus each actual rebase yields a finite graph-free facial computation value; no path in the unbounded prefix is replayed. The result does not yet prove that the predecessor finite receipt and literal Cell data determine the core-component matrix. Consequently the complete five-coordinate letter, joint-`Count` adequacy, reachable closure, the numerical V_0, and the bounded base remain open.

The actual successor boundary fragments now enter that bounded code without an ambient-face shortcut. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFaceCollarFragment` gives the at most four occurrence-sensitive fragments fixed slots and proves that each canonical fragment dart lies in the collar, because the two displayed successor crossings are literal rebase edges. Most importantly, the coded closure connects two canonical coordinates if and only if they are the same regional fragment. Ambient orbit-face equality remains a separate coordinate: one ambient face can have two disconnected regional fragments, and conflating those relations would change the source profile. The finite ambient-face relation, port incidence, capped progress, complete five-coordinate letter, joint-`Count` adequacy, reachable closure, the numerical V_0, and the bounded base remain open.

Those distinct facial observations are now packaged in a single finite receipt. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteFaceReceipt` combines the at-most-twenty-four-coordinate collar code with four padded successor-fragment rows. For every actual fragment Lean recovers its occupied slot and collar coordinate, ambient orbit-face continuation, incidence with each named port, and cap-at-five regional progress; the embedded collar closure still recognizes equality of occurrence-sensitive regional fragments exactly. This is a finite relational witness for one literal source rebase, not a claim that the predecessor profile determines a unique successor. The tracked/facial join, support-level adequacy for the heterogeneous literal letter and its positive `Count` relation, reachable closure, the numerical V_0, and the bounded base remain open.

The receipt is now tied back to the literal profile rather than certified only internally. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteFaceReceiptProfile` uses the canonical actual-fragment enumeration and proves that its embedding into the four receipt slots agrees with all three facial coordinates of the successor terminal-aware bounded profile: ambient-face continuation, incidence with every crossing or seam port, and cap-at-five regional progress. The finite facial packet is therefore an exact profile decoder for one actual rebase. Joining it to the tracked receipt, proving support-level adequacy for the positive literal `Count`, measuring the heterogeneous reachable closure, deriving the numerical V_0, and verifying the bounded base remain open.

The tracked half has now been aligned to the same side of the rebase. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseSuccessorTrackedProfile` extracts a finite tracked state from the next terminal-aware input region, rather than pairing the successor facial receipt with an older pre-rebase state. Its port-role map names the two successor crossings and seam terminal among the four stable rebase roles. Lean proves that this state recovers the literal successor profile's two crossing colours and every tracked two-colour connectivity bit on all three ports. This is exact successor-side relational evidence, not yet a deterministic update from the predecessor. The joint five-coordinate packet, positive-`Count` adequacy, reachable closure, the numerical V_0, and bounded-base verification remain open.

The two time-aligned halves are now one finite successor packet. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteProfileReceipt` pairs the successor tracked state with the successor facial receipt and constructs that packet from every literal source rebase. Its five fieldwise decoder theorems recover exactly the actual bounded profile's crossing colours, tracked-pair connectivity, ambient-face continuation, port incidence, and cap-at-five regional progress. Thus finiteness and literal source adequacy of the rebase output packet are established without postulating a deterministic update. What remains is the finite support relation identifying which predecessor/successor packet pairs have positive literal `Count`, followed by its heterogeneous reachable closure, the numerical V_0, and bounded-base verification.

The fieldwise packet now has one graph-free dependent output decoder. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteOutputDecoder` packages the actual successor fragment count with the joint receipt and decodes a complete terminal-aware bounded profile. Lean proves a single structure equality: every literal source rebase supplies such a finite letter, and its decoded value is exactly the next source profile in all five coordinates. Thus the output side of the rebase is no longer a collection of parallel observations. The theorem deliberately does not infer the letter from the predecessor state: computing that same receipt from an arbitrary compatible cumulative-prefix witness and positive literal Cell `Count` witness remains the live adequacy obligation before reachable closure or a numerical V_0 can be claimed.

The local matrix is now known to be inhabited by the actual source geometry, rather than merely defined as an abstract finite relation. Module `GoertzelV24ClosedWebLocalLayerLiteralCountWitness` restricts the ambient Tait colouring to the literal complementary Cell through the existing open-region inheritance map. Lean proves that the two exact full profiles computed from this one open colouring select a positive entry of the boundary-rooted `Count` matrix, and hence inhabit the already-defined finite support relation. This is only the validation direction for an actual Cell: it does not replace the matrix by the chosen colouring, and it does not yet identify the local two-boundary profiles with the cumulative predecessor and successor packets. That prefix-factorization theorem remains the next semantic junction before the heterogeneous closure can be measured.

The validation now reaches the ambient graph-derived semantics on both sides of the literal Cell. Module `GoertzelV24RegionalProfileColorCongruence` proves that the complete five-coordinate regional profile depends only on colours inside its stored finite edge region. Module `GoertzelV24AnnularCrosscutComplementInheritedProfile` then proves that the boundary-rooted inherited colouring agrees there with the ambient Tait colouring, and identifies both literal open profiles with the corresponding ambient graph-derived cut profiles. The source-Cell witness specializes these equalities to its two cuts. Thus the positive `Count` entry and the cumulative construction now speak the same exact profile semantics; the remaining junction is geometric factorization of the cumulative predecessor plus this Cell into the decoded successor packet. No reachable-state count, numerical V_0, or bounded-base verification is inferred yet.

That geometric factorization is now exact at the two graph semantics that carry the nontrivial recurrence. Module `GoertzelV24ClosedWebLocalLayerSerialCellFactorization` uses the proved identity "terminal-aware prefix region union literal Cell region equals the pre-rebase successor region." For each tracked colour pair, Lean identifies the successor connectivity graph with the supremum of the predecessor graph, the literal Cell graph, and their explicit residual seam; the identical three-factor theorem is proved for occurrence-sensitive face positions. Successor reachability in either graph is consequently equivalent to the reflexive–transitive closure of moves inside either factor or across the seam. These are unconditional equalities on the actual opened source carrier: they use neither global cubicity nor global two-sidedness. This closes the hidden regional-composition premise, but not yet its finite support-letter coding: the seam closure must still be transported through the predecessor, literal `Count`, and decoded successor receipts before the heterogeneous reachable set or V_0 can be measured.

That residual seam now has exact portal completeness on the same opened source carrier. Module `GoertzelV24ClosedWebLocalLayerSerialCellPortalCompleteness` proves that the literal Cell's vertex cut consists of its two incoming and two outgoing source crossings. Because the incoming pair is already retained by the terminal-aware predecessor, every genuinely residual tracked adjacency touches an outgoing crossing, and every non-isolated tracked seam component reaches one of the two named outgoing portals. The occurrence-sensitive facial seam obeys the same statement. No global cubicity of the opened graph is assumed: a consecutive facial step with distinct underlying edges is already an edge adjacency, whereas a degree-one boundary-stub bounce repeats one edge and cannot cross between the two exclusive regional factors. Thus the open carrier's stubs do not create hidden residual communication. The tracked two-portal relation is now encoded exactly in `GoertzelV24ClosedWebLocalLayerSerialCellTrackedSeamFamilyCode`. Lean first weakens the old cubic neighborhood estimate to its actual premise, at most three incident rotation edges at every vertex. The annular boundary axioms discharge it with degree one at the ten stubs and degree three elsewhere. Closing the two outgoing crossings under edge adjacency then gives one common carrier of at most fourteen edges for all three tracked colour pairs. The two portals remain distinct named coordinates, and the finite graphs preserve both adjacency and ambient seam reachability exactly. This closes the connectivity residue without pretending that the facial residue is only an edge graph. The occurrence-sensitive facial code still has to retain face identity, port incidence, and capped progress jointly before a complete support letter, reachable closure, numerical V_0, or bounded-base theorem follows.

The occurrence-sensitive half is now finite on the same opened carrier without restoring the false two-sidedness premise. Module `GoertzelV24ClosedWebLocalLayerSerialCellFacialSeamCode` counts actual face-cycle positions rather than distinct underlying edges. Each of the two outgoing portal edges contributes at most its two dart occurrences on a fixed face, and each occurrence has only a successor and predecessor. Portal completeness therefore places every non-isolated facial seam in a carrier of at most $2 \cdot 2 \cdot 3 = 12$ positions, even on a hole face where one stub edge appears twice. Lean transports the exact support-induced occurrence graph to that carrier and packages the family over precisely the real pre-rebase output fragments. Finally a finite source-Cell receipt bundles the bounded input and output profiles, the fourteen-edge tracked seam family, and the twelve-position facial family; the existing theorem proves that its two profiles satisfy the complete five-coordinate source relation. This is the sound finite-image direction. Converse geometric adequacy, equivalence with positive literal `Count`, composition with the distinct boundary-rebase receipt, a reachable closure, numerical V_0, and the bounded base remain open.

The incoming seam-terminal profile now has a separate lossless semantic compression. Module `GoertzelV24TerminalAwareProfileCompression` reconstructs every crossing diagonal from its displayed colour, retains the terminal diagonals, and stores each symmetric tracked or facial relation only once. Module `GoertzelV24ClosedWebLocalLayerSerialTerminalProfileCompression` proves that the actual opened-source predecessor profiles satisfy these laws and that equality of their compressed codes reconstructs equality of all five raw coordinates. At the occurrence-sensitive four-fragment bound the kernel-checked carrier has 12,556,911,218,688 elements, over 33 million times smaller than the arbitrary-matrix fallback. This is an exact measurement, but not yet a useful reachable bound: the corresponding gap-four pigeonhole demand is 50,227,644,874,753. The live serial repeat already uses the much smaller 105,744-state zero-terminal fragment quotient; the new result instead makes the intermediate one-Cell input finite without losing its retained rung. Further source-specific incidence and reachable-mode compression, plus literal-`Count`/rebase adequacy, remain necessary before deriving V_0.

The five coordinates are now packaged against that same factorization rather than left as parallel observations. Module `GoertzelV24ClosedWebLocalLayerSerialCellFullProfileRelation` defines the actual source-indexed relational update from the terminal-aware prefix profile to the pre-rebase profile. Its fields characterize both boundary colour words, every tracked two-colour component bit, ambient-face identity, port incidence, cap-at-five progress, and the exact facial closure. Lean constructs this relation for an arbitrary edge colouring nonzero on the two displayed cuts, and in particular for the actual source colouring on the bounded profile packages. This retires the risk that the five independently proved observations do not form one relation. It is not yet the full support letter: both profiles here are computed from one common ambient colouring. The next adequacy theorem must instead start from an arbitrary compatible cumulative-prefix witness and an arbitrary positive literal `Count` witness, glue their colourings at the seam, and show that the decoded successor receipt is exactly the resulting output.

The positive literal `Count` now reaches the source's categorical carrier without restoring the false global-cubicity premise. Module `GoertzelV24AnnularCrosscutComplementSerialColoring` gives the complementary open-region colouring a boundary-rooted two-sided serial presentation and proves its Tait condition and both ordered boundary words. Module `GoertzelV24AnnularCrosscutComplementLiteralSerialSupport` then proves that every positive five-coordinate profile entry supplies such a serial Tait colouring, whose left and right boundary words are exactly the colour projections of the displayed full profiles. Thus the generated matrix support and the literal open morphism are now linked on the actual stub-bearing annular carrier. This is a projection from full-profile support to colour-word support, not a reduction of the state to boundary colours: connectivity and all three facial observations remain part of the matrix indices. The open adequacy step is still to glue an arbitrary compatible cumulative-prefix witness to this arbitrary Cell witness and recover the complete decoded successor receipt; reachable closure, the numerical V_0, and bounded-base verification remain open.

The same support statement now composes over the actual heterogeneous source word. Module `GoertzelV24ClosedWebLocalLayerLiteralSerialTangle` coordinatizes each literal complementary Cell by its two ordered crossing positions, proves that the bounded five-coordinate transfer for that Cell projects to support of the same open-tangle morphism, and then applies the generic serial composition law to the source-ordered list of Cells. Consequently an exact path through the literal five-coordinate transfer word produces a Tait colouring of the assembled serial composite at the projected exterior colour words. No homogeneous-letter or three-orientation quotient is used. This closes the support-level categorical composition of the literal Cell word; it does not yet identify the tagged composite carrier with the cumulative ambient prefix, prove the converse full-profile adequacy, compute the reachable closure, or derive V_0.

The cyclic escape in the finite Cell carrier is now excluded on the literal two-cap source laboratory. Module `GoertzelV24MinimalFacialPentagonCapPairLocalCellCarrier` starts from the exact cycle-or-six-edge dichotomy for one source-local Cell. If the Cell side were cyclic, its four displayed crossing edges would form a one-sided small cut while both ordered hole-stub families remain on the retained side. Closing the two facial caps pulls that cut back to the closed graph-backed vertex-minimal counterexample, where the proved cyclic five-edge-connectivity contradicts its cardinality. Lean therefore proves, without a cyclic alternative, that every literal Cell on this carrier has at most six regional edges. The same module instantiates the common six-slot code for all three tracked two-colour graphs and proves that decoding preserves their complete regional reachability relation in both directions. Thus this is not merely a cardinality estimate: the connectivity coordinate of the literal Cell letter now has an exact graph-independent finite representation. An indexed corollary gives the same six-edge bound at every offset of the actual heterogeneous source Cell word; it neither replaces that word by a homogeneous power nor quotients its letters by their three geometric orientations. At each such offset Lean now also constructs the exact six-slot tracked-family code, with the two incoming crossings followed by the two outgoing crossings, and proves that decoding preserves complete regional tracked reachability in both directions. Its scope remains exact. It does not identify every abstract Addendum–V frontier with a two-cap opening.

The support-composition audit isolates the next interface more sharply. The generic serial-open-tangle theorem already glues arbitrary Tait colourings whose boundary words agree, and positivity of the literal Cell's full-profile `Count` already supplies the required Cell colouring. What is not yet proved is the comparison between the resulting tagged heterogeneous serial composite and the actual cumulative source-prefix region on which face continuation, port incidence, and capped face progress are computed. Boundary colour support alone cannot supply that comparison. The source selection/formation comparison, integration of the already-proved finite facial code with this literal carrier, that cumulative-prefix comparison and converse full-profile support adequacy, heterogeneous reachable closure, numerical V_0, and bounded base remain open.

The permissible scope of the older abstract executable transition is now machine checked rather than inferred from its finite type. Module `GoertzelV24FramedCorridorSerialFullTransitionOverapproximation` constructs, for every input and every output whose asserted tracked strands have the colours named by their tracked pair, an abstract letter decoding that input to that output. Conversely every admitted abstract transition satisfies exactly that endpoint-colour condition. Hence the relation is characterized by the output condition alone and is independent of the input. The construction uses no incoming coordinate: it stores the output tracked relation in the hidden residue and stores each output face cap in the facial residue. Thus the abstract transition remains a sound conservative envelope for source transitions, but its reachable closure is not a geometric compression and must not be measured as one. The live restriction is now exact: admissible letters must be tied to the literal Cell and boundary-rebase receipts, including their attachment to the incoming profile. No reachable count or smaller threshold follows from this negative scope theorem.

One presentation-only redundancy has now been removed without changing that semantic frontier. The occurrence-sensitive face fragments are indexed by a finite enumeration, but the splice interface already accepts an explicit permutation of those fragment coordinates. Modules `GoertzelV24WidthTwoFragmentReindexFinite`, `GoertzelV24WidthTwoFragmentReindexCompression`, and `GoertzelV24WidthTwoFragmentReindexSplice` quotient the complete fragment payload by exactly this relabelling action. They retain the tracked strand state, the full face-equivalence relation, every port-incidence row, and every capped face length. Burnside's lemma, with kernel-evaluated component fixed-point sums, proves exact quotient fibers $1, 15, 240, 1325, 2825$ for zero through four fragments. Including the unchanged twenty-four strand states gives the exact route-correct bounded carrier

$$24(1 + 15 + 240 + 1325 + 2825) = 105744,$$

and the distance-four pigeonhole therefore requires 422977 successive states. Module `GoertzelV24FramedCorridorSerialFragmentReindexRepeat` applies that theorem to the actual source-ordered heterogeneous word of literal Cells. It returns two cuts separated by at least four positions, a common fragment count, and a concrete fragment permutation under which the complete decoded five-field profiles agree in the existing splice vocabulary. Thus this is a lossless quotient of presentation, not an identification of distinct geometric states. It reduces the former 1541184-state route-correct serial carrier by a factor of about 14.6. The heterogeneous one-step support, its reachable closure, the numerical V_0, and the route-native bounded base remain separate open obligations.

A second, orthogonal presentation redundancy has now been removed at the literal Cell letter. Module `GoertzelV24FiniteHexCellProfileCompression` proves that the three raw 6×6 Boolean tracked-connectivity matrices of every actual six-edge Cell obey their graph laws: tracked support, diagonal support, symmetry, and transitivity. It replaces those matrices by the proper six-color word and one canonical component label at each of the six positions for each tracked color pair. The encoder and executable decoder satisfy an exact kernel-checked round trip on every graph-semantic Cell profile, and every literal source Cell is proved to lie in that semantic image. A flat finite code was used deliberately: storing Mathlib's general finite-partition type as the enumerated payload induces a powerset-of-powerset implementation, whereas position-bounded representative labels give the same lossless interface without that elaboration cost. Minimality of the representative puts its label at position i in the structural type $\mathrm{Fin}(i+1)$, leaving only $1 \cdot 2 \cdots 6 = 720$ ambient label maps for each tracked pair. The proved cardinal ceiling is

$$272097792000,$$

compared with the raw Cell factor $3^6 2^{108} = 236574025616993083824920739000090624$, a reduction by more than 10^{23} before reachability is used. This is still far too large for a dense closure computation and it concerns only the local Cell's color and tracked-connectivity coordinates. The terminal-aware face-update code, the distinct boundary-rebase letter, their joint five-coordinate support relation, the reachable closure, the numerical V_0, and the bounded base remain open.

The compact Cell field is now wired into the established terminal-aware decoder rather than left as an isolated counting lemma. Module `GoertzelV24FramedCorridorCompressedOneCellDecoder` replaces the raw Cell matrix in a one-Cell letter by the preceding code, expands it through the proved decoder, and reuses the existing terminal, face-role, port-incidence, and capped-progress computation unchanged. Every compact transition is therefore already a transition of the declared finite receipt over-approximation; conversely, any raw decoder witness whose Cell profile obeys the graph laws has an identical compact witness. This is the safe direction for an upper-bound closure: compact codes may remain spurious but no semantic raw transition is lost. The same module now carries that result all the way back to the source. It defines a semantic one-Cell receipt whose existential data retain the graph-law certificate for the very Cell profile used by the edge-colour, tracked-connectivity, face-equivalence, port-incidence, and capped-length equations. Every actual Tait-coloured literal source Cell constructs such a receipt, and Lean proves that it is decoded by a compact letter with exactly the same output profile. This closes the finite source-image half of the compact one-Cell alphabet. It does not yet compose the separate boundary-rebase decoder, measure the heterogeneous reachable closure, derive a numerical V_0, or verify the route-native bounded base.

The compact source-image theorem is now connected to the homogeneous geometric-prefix step interface in `GoertzelV24FramedCorridorCompressedReachableProfileStep`. A compact step existentially restores the single shared-rung terminal, applies one semantic compact Cell letter, and returns a complete three-fragment outgoing profile. Lean proves both directions needed for safe measurement: every such compact step is an earlier full receipt step, and every actual Tait-coloured source Cell supplies such a compact step. The same relation transports losslessly to the 191184-state port-incidence codec. Source re-anchoring also fixes its precise scope: these profiles use the complete-hex `corridorPrefixEdgeRegion`, whereas the route-correct v24 word uses the union of literal complementary Cells in `sourceCorridorSerialPrefixRegion`. It is therefore a proved compression laboratory, not yet a refinement of the live serial reachable relation. The latter still needs the literal Cell/rebase receipts tied to the incoming serial profile. Consequently no route-correct closure count, numerical V_0, or bounded-base verification is inferred here.

The route-correct fragment quotient can nevertheless be used immediately in the already proved curvature-free L6 arithmetic. The module `GoertzelV24ClosedWebFragmentReindexRadialThreshold` substitutes the checked serial block length 422980 into the radial-order dichotomy. It thereby lowers the uniform interior entrance to 845960 vertices and, after the exact ten boundary stubs are added, the five-by-five closed-web entrance to 845970. Above that threshold Lean constructs either the required long radial coordinate with bounded actual chord load or a rotation sector with a large pairwise-laminar chord family. This resolves the earlier astronomical raw-carrier concern at the L6/L7 counting junction, but it is still not the final V_0: the shallow radial cuts must be realized as literal serial layer boundaries, and the selected laminar chords must be realized as genuine deletion transversals before either branch can feed the strict splice.

The pumping bound can now also be stated against a verified presentation of the reachable sets themselves, rather than against the full powerset of the profile carrier. Module `GoertzelV24ReachableSetModeCompression` packages a finite mode type with an encoder, decoder, and an explicit admissibility predicate. The only required round trip is on admissible complete reachable sets. If every prefix reachable set of the heterogeneous source word is admissible, equality of two mode codes reconstructs equality of the complete reachable sets. Lean then produces a nonempty removable middle subword and proves that rejection is preserved by its deletion. The resulting pigeonhole threshold is the number of verified modes, not the cardinality of the unrestricted powerset.

This is the proof interface for a compressed reachable-state certificate, not such a certificate itself. The literal Cell and boundary-rebase relations must first be connected to the incoming serial profile, and a concrete mode table must be accompanied by a checked decoder and a proof that every source prefix reachable set is admissible. In particular, the older twenty-mode τ/mirror-τ laboratory is not silently reused for the heterogeneous v24 word. No numerical V_0 or bounded-base conclusion follows at this checkpoint.

The end-to-end reductive interface is now checked at this same boundary.

$$\left.\begin{array}{r}\text{shallow radial block} \longmapsto \text{strict-reduction certificate,}\\ \text{deep laminar sector} \longmapsto \text{strict-reduction certificate}\end{array}\right\} \Longrightarrow \texttt{StrictReductionSupplier}(845970).$$

The module `GoertzelV24ClosedWebFragmentReindexReductionSupplier` proves this implication for the actual universe-packed closed-web carrier used by the reductive spine. It consumes the proved forward-or-swapped radial alternative, then dispatches by the two geometric branches; orientation is bookkeeping, not an extra mathematical premise. Thus the towers have been type-checked together through the numerical entrance. The two displayed maps remain the honest missing constructions: a shallow block must become a literal pair of serial cuts, while a deep laminar family must yield the deletion transversals and complete reconstructed output web required by the existing strict splice.

This numerical improvement does not discharge the route's finite base. Addendum XXV, Step 5, explicitly states L3 as verification of the instances with $V \leq V_0(n_{\mathrm{out}})$, and the Lean playbook gives the same specification for L3/L5: verified reflection over the remaining irreducible instances below the effective bound. The existing twenty-mode Lemma–8.18 closure is a computation for the narrower τ/mirror-τ laboratory; there is no proved coverage map from that table to the present heterogeneous Cell corridor, and it is therefore not a base certificate. A smaller proved profile carrier can lower the eventual V_0, but no reachable-profile computation by itself proves `BaseVerified`. At the current checkpoint the honest order is thus: finish the two geometric realization maps that turn the checked 845970 entrance into a strict reduction threshold, derive the actual V_0, and only then decide how the source-prescribed irreducible finite base is to be verified. No bulk base enumeration is claimed or begun here.

The tracked half of one literal opened-source transition now also has a single finite coordinate system. Module `GoertzelV24ClosedWebLocalLayerSerialCellTrackedTransitionCarrier` takes the at-most-six-edge Cell region, its at-most-fourteen-edge residual seam neighborhood, and the retained shared rung, and places their union in a carrier of at most twenty-one literal edges. The two incoming crossings, two outgoing crossings, and rung terminal are named in the same carrier, and the Cell and seam factors preserve complete tracked reachability under the common coordinates. This removes a coordinate mismatch in the successor-state recurrence. It does not yet compute that recurrence: a prefix path may leave and re-enter the local carrier, so the finite attachment relation to the incoming prefix state must be reconstructed explicitly rather than hidden in the local letter. The facial successor state and reachable closure remain open.

The accumulated-prefix contribution has now been put on that same fixed carrier without truncating paths to the local neighbourhood. Module `GoertzelV24ClosedWebLocalLayerSerialCellTrackedPrefixAttachmentState` records, for each tracked colour pair, which of the twenty-one slots are active, which attach to the terminal-aware input's three named ports (the two crossings and the retained rung), and which lie in the same component meeting no named port. This is finite source-image state. Lean proves that the state together with the complete incoming profile recovers reachability through the literal accumulated prefix on every represented carrier edge, even when the connecting path leaves the local carrier and later returns. No successor relation is stored in the Cell letter. The next tracked obligation is now sharply finite: take the reflexive–transitive closure of this predecessor relation with the already coded Cell and seam relations, then project the result to the outgoing profile. The facial recurrence and the heterogeneous reachable closure remain separate open steps.

The tracked three-factor closure is now finite as well. Module `GoertzelV24ClosedWebLocalLayerSerialCellTrackedFiniteClosure` first proves that every nontrivial switch among the accumulated prefix, the literal Cell, and their residual seam occurs on the common twenty-one-edge carrier. It then transports the complete component closure to finite coordinates. Consequently, for every literal source Cell and tracked colour pair, Lean proves an iff between reachability in the actual enlarged prefix and reflexive–transitive closure of a graph-free relation determined by the terminal-aware incoming profile, its finite attachment state, and the coded Cell and seam graphs. The same module projects this closure to the two outgoing profile ports, including the exact tracked-colour guards, so every output `strandConnected` bit is now decoded by the finite relation. This closes the tracked-connectivity recurrence on the literal carrier; it does not identify heterogeneous Cells merely by their three geometric orientations. The finite facial attachment/cap recurrence, the combined positive-support letter, reachable closure, numerical V_0, and bounded-base verification remain open.

The facial recurrence has now acquired the corresponding exact algebraic split, before any finite encoding is imposed. Module `GoertzelV24RotationFaceRegionalDartSeam` defines, on literal darts, the new facial adjacencies created when a prefix region and one Cell region are adjoined; their union is proved to be exactly the prefix face graph, the Cell face graph, and this residual seam. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceDartFactorization` specializes that identity to every actual opened-source Cell and proves the corresponding iff for complete facial reachability. Using literal darts is important: predecessor fragments and local Cell occurrences can be named from different roots of one face orbit, while the darts themselves inhabit one fixed carrier. This theorem assumes neither global two-sidedness nor global cubicity, and it does not yet claim a finite facial alphabet. The next obligation is to encode the predecessor attachments and local seam on a bounded dart carrier, retain the capped face-progress data, and prove the finite closure adequate for the five-coordinate output profile.

That common carrier is now explicit. The generic module `GoertzelV24RotationFaceRegionalDartCarrier` proves that the closed face-permutation neighbourhood of a finite edge set has at most six literal darts per edge, counting both occurrences even on a one-sided boundary face. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceTransitionCarrier` then proves that every residual-seam adjacency of an actual source Cell touches one of its two named outgoing crossings. The darts over the at-most six Cell edges and the closed neighbourhood of those crossings therefore form a common carrier of at most twenty-four darts. The Cell and seam face graphs are transported through one finite coordinate system, with complete reachability equivalent to their ambient literal-dart reachability. The cardinality theorem takes the local six-edge Cell bound as an explicit premise; it neither assumes global cubicity nor identifies the two occurrences over a stub edge. What remains is to summarize the accumulated predecessor's attachments to these slots, compose the three finite factors, and decode face continuation, port incidence, and capped progress in the outgoing profile.

The accumulated facial predecessor has now been summarized without replacing its global components by paths induced on the local carrier. Module `GoertzelV24ClosedWebLocalLayerSerialCellFacePrefixAttachmentState` uses the actual occurrence-sensitive fragments of the terminal-aware incoming cut as persistent ports; there are at most four. It records on fixed 24×4 coordinates the exact predecessor component relation, attachment to those fragments, literal regional presence, and the cap at five of the distinct primal edges in each component. Lean proves that every represented dart recovers its exact presence and cap, and that the padded relation is iff complete reachability in the literal predecessor face graph. Thus a face may leave and re-enter the Cell neighbourhood without being lost, while the state stores no successor relation. This is the predecessor input to the finite facial recurrence, not the recurrence itself: the next obligation is its three-factor closure with the already coded Cell and seam graphs, followed by projection to face continuation, port incidence, and capped progress.

That three-factor facial closure is now finite. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteClosure` proves that every switch among predecessor, Cell, and residual seam occurs on the common twenty-four-dart carrier. It then transports the complete closure through the finite coordinates. For every represented pair of literal darts, successor facial reachability is therefore iff the reflexive–transitive closure of a graph-free relation determined by the finite predecessor attachment state and the two coded local factor graphs. This matches the already-closed tracked recurrence and removes the unbounded ambient graph from the facial component update. The output decoder is still open: face continuation, port incidence, and capped progress must be projected jointly before this becomes the facial half of a full one-Cell support letter.

The first outgoing facial decoder is now proved. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteOutputFragment` sends every actual, occurrence-sensitive regional fragment of the pre-rebase output cut to its canonical cut-dart occurrence in the common twenty-four-slot carrier. Lean proves that two such output fragments are equal exactly when the graph-free finite Cell closure connects their two coordinates. The scope is deliberately regional rather than global: the opened carrier is not assumed to be two-sided, so equality of ambient orbit faces is not inferred from a bare edge or a quotient-face identification. This closes the first projection from the finite facial recurrence while leaving three honest decoder obligations: ambient face continuation, port incidence, and the capped size of the distinct-edge union of each merged component. Only after those coordinates are updated jointly with tracked connectivity and crossed-edge colours does the source-faithful one-Cell support letter exist.

Outgoing port incidence is finite as well. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteOu` first proves an occurrence-sensitive regional fact: an edge belongs to a fragment exactly when one of its two literal dart occurrences reaches that fragment's canonical dart. It then indexes those two occurrences by Fin 2, places them in the common twenty-four-slot carrier, and transports the test through the graph-free Cell closure. The decoded predicate is proved equivalent to the actual graph-derived profile's `fragmentContainsPort` bit. This remains valid when a bridge's two occurrences lie on one quotient face and uses no global two-sidedness assumption. Thus regional-fragment identity and port incidence are now decoded. Ambient face continuation and the capped cardinality of each merged component's distinct-edge support remain open before the full facial output packet exists.

The capped-progress target is now identified at the same occurrence-sensitive level. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteOutputCap` proves that an outgoing regional fragment's distinct primal-edge support is exactly the image of the full regional dart component rooted at its canonical cut occurrence. Consequently capping the fragment cardinality at five is identical to capping that canonical component support. This matters because the source coordinate measures distinct primal edges, not a walk length and not a sum of factor sizes: predecessor, Cell, and seam contributions can overlap and must be counted only once. The theorem identifies the exact finite-decoder target without assuming two-sidedness or unfolding the full dependent profile construction. Computing that component cardinality from the finite predecessor caps and the local Cell/seam code, with exact overlap correction, remains open; so this is not yet a complete facial output packet, a reachable-state count, or a numerical V_0.

The finite state now also carries the deduplication key needed to compute that cardinality. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteEdgeState` records, on the fixed twenty-four dart slots, whether two represented occurrences are the same primal edge and whether a represented edge belongs to the newly adjoined Cell. Both tables are proved exact on every live slot. Combining the Cell-membership bit with the predecessor-presence bit already stored in the facial attachment state decides membership in the complete pre-rebase output region exactly. Thus the cap decoder need not identify edges from face adjacency or count the two darts of one edge twice. What remains is the finite representative sum: select each predecessor component once, select each genuinely new Cell edge once modulo this equality table, restrict both to the finite output component, and prove that their disjoint union is the canonical support identified above. That sum, ambient face continuation, and the combined one-Cell support letter remain open.

The first exact half of that representative sum is now machine checked. Modules `GoertzelV24ClosedWebLocalLayerSerialCellFaceFinitePredecessorRepresentatives`, `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteNewEdgeRepresentatives`, and `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteNewEdgeAdequacy` choose one canonical live slot for each represented predecessor component and one canonical live slot for each genuinely new Cell edge reaching the selected output fragment. The latter selector is proved, by an explicit equivalence, to enumerate exactly the literal distinct Cell edges in that output component which were absent from the predecessor region. Thus two dart occurrences of one edge are counted once, and an edge already present in the predecessor is not counted again. The predecessor selector also carries the stored cap of each selected old component, but the theorem that its capped sum equals the literal old portion of the output support remains open; so does the final cap-at-five combination with the new-edge cardinality. Consequently this is an exact deduplication theorem, not yet the complete facial decoder, the five-field support letter, a reachable-state count, or a numerical V_0.

The old-component sum now has its required non-overlap theorem. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFinitePredecessorDisjoint` first follows each outgoing fragment's canonical cut edge into the second literal layer and proves that the fragment's ambient face is an annular interior face. Thus its cyclic boundary is simple locally, without the false claim that every face of the opened carrier is globally two-sided. From this local injectivity Lean proves that two distinct canonical predecessor components contributing to the same output fragment have disjoint primal-edge supports. The stored predecessor caps may therefore be added without double-counting two occurrences of one bridge edge. What remains is to identify each stored cap with the corresponding literal old-component cardinality, sum those disjoint supports, and combine the result with the already exact new-Cell edge cardinality at cap five. The full five-field letter and its reachable closure are not claimed yet.

The old/new support split and capped-length computation are now exact. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteOldEdgeAdequacy` proves first that every old-prefix edge in a selected output component has an anchor on the fixed transition carrier. This includes singleton old components, whose presence cannot be recovered from graph support alone. Consequently the union selected by the finite predecessor receipt is exactly the old-region portion of the literal output component. Lean partitions the whole component into this old union and the genuinely new Cell edges, proves the two parts disjoint, and combines their already-verified cardinalities at cap five. The resulting graph-free decoder is proved equal to the actual `faceLengthCap` of the regional output fragment. Thus capped facial progress is no longer estimated by path length or by a possibly overlapping sum. This is one coordinate of the literal heterogeneous Cell letter, not the letter itself: ambient face continuation must still be integrated with the crossed-edge colours, tracked connectivity, and port incidence before a positive-support relation or reachable-state closure is claimed.

Ambient face continuation is now finite and exact on the opened source. Module `GoertzelV24ClosedWebLocalLayerSerialCellFaceFiniteAmbientContinuation` follows each outgoing fragment through its canonical cut edge into the second literal two-step layer. Its ambient orbit face is therefore one of exactly three named roles: first endpoint, central corridor face, or second endpoint. Those three faces are proved distinct, and equality of their three-valued role codes is proved equivalent to the actual output profile's `faceContinues` bit. The theorem does not identify distinct regional components merely because they lie on one ambient face, and it assumes no global face-two-sidedness of the opened annulus. The five profile coordinates now each have a separate exact finite decoder. Their joint literal-Cell support relation, its heterogeneous reachable closure, the numerical V_0, and the route-native bounded-base verification remain open.

Those five decoders are now assembled into one finite, source-indexed support letter. Module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteSupportLetter` retains the complete terminal-aware input and zero-terminal output profiles, the finite predecessor states, the coded Cell/seam factors, and only fixed interface coordinates: twenty-one tracked-edge slots, twenty-four facial-dart slots, two outgoing ports, at most four output fragments, and three ambient face roles. Lean proves that every actual literal source Cell inhabits this finite support relation and that all five outgoing profile coordinates are decoded jointly: crossed-edge colours, the three tracked two-colour connectivities, ambient face continuation, fragment–port incidence, and distinct-edge face progress capped at five. This is the sound inclusion needed for a conservative reachable-state computation. It deliberately does not collapse heterogeneous Cells to the three geometric orientation labels, and it does not yet prove the converse equivalence with positive entries of the manuscript's `Count` matrix. Although the letter carrier is a Lean `Fintype`, its present support decision uses the propositional closure interface; an executable Boolean reflection of those finite closures, followed by the heterogeneous reachable table, remains the next obligation. No numerical V_0 or bounded-base verification is claimed here.

The first computational reflection of that letter is now exact. Module `GoertzelV24BoundedCarrierBoolGraphFamilyCode` replaces each proposition-valued local graph by an explicit Boolean adjacency table on the same dependent finite carrier. Reconstructing a simple graph from the table is proved to recover the original graph, so its executable reachability query is equivalent to the proof-facing component relation. Module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteStepReflection` then reflects both complete one-step unions used by the letter: the tracked two-colour predecessor/Cell/seam step and the facial predecessor/Cell/seam step. In each case Lean proves an iff with the original proposition-valued relation. Thus this reflection neither enlarges nor shrinks the literal Cell semantics and retains all five profile coordinates. The remaining computational work is to reflect the capped face computation and assemble the heterogeneous source word. No reachable-state count, numerical V_0, or bounded-base verification is claimed yet.

The finite closure of both Boolean steps is now executable and exact. Module `GoertzelV24FiniteBoolRelationClosure` repeatedly adds directed successors on a finite carrier. A cardinality proof shows that one more round than the carrier size reaches a fixed point, and the main theorem identifies Boolean membership in that fixed point with `Relation.ReflTransGen` of the original directed step relation. No symmetry assumption is used, so the calculation does not manufacture reverse transitions on unrealizable members of an over-encoded carrier. The literal Cell reflection applies this closure to all three tracked-colour relations and to the facial relation, then packages executable fixed-slot queries on the same twenty-one- and twenty-four-slot interfaces used by the finite support letter. Reflecting a proof-facing graph code is proved equivalent to the letter's original tracked and facial connectedness predicates. The local closure interface is therefore no longer merely proposition-valued. Capped face semantics, the complete executable five-field letter, the heterogeneous reachable table, its measured state count, the derived V_0, and the bounded base remain open.

The two law certificates used by the capped-face decoder are now executable rather than hidden proof arguments. Module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteCapSemanticReflection` checks, over the fixed twenty-four-slot carrier, that predecessor connectivity is an equivalence relation on which presence and stored cap are constant. For each output component it also checks that the stored primal-edge equality table is reflexive, symmetric, and transitive on precisely the genuinely new Cell edges selected by the directed facial closure. Each Boolean check is proved equivalent to the corresponding semantic structure already supplied by every literal source Cell. Thus the eventual evaluator need not inspect a proof witness in order to validate its finite tables. Computing canonical representatives and their capped sum, integrating that query into all five fields, and closing the heterogeneous source word remain open.

The representative sum is now executable as well. Module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteCapReflection` chooses the least live slot of each predecessor component and each genuinely new primal edge, sums every old stored cap and every new distinct edge exactly once, and saturates at five. Assuming only the two Boolean law checks just described, Lean identifies this computation with the earlier source-adequate mathematical decoder. Its final Boolean cap-at-slot query is proved equivalent to the support letter's proposition-valued cap relation on every reflected graph code. Thus capped face progress has joined tracked and facial connectivity as an exact executable coordinate. The next unit is the single Boolean predicate combining all five coordinates of one literal Cell; the heterogeneous reachable table, its measured state count, the derived V_0, and the bounded base remain open.

That joint Boolean predicate is now proved exact. Module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteBoolSupportLetter` retains the complete source-indexed finite letter and evaluates all five outgoing coordinates in one decision: crossed-edge colour, tracked two-colour connectivity, ambient face continuation, fragment–port incidence, and capped face progress. Its reflection theorem says that a proof-facing letter evaluates to `true` if and only if its original complete support proposition holds. Thus the local Cell letter is executable without altering its semantics, adding reverse transitions, or collapsing the heterogeneous source word to the three geometric orientation labels. The next computational obligation is the word-level recurrence and its reachable table. Converse equivalence with positive entries of the manuscript's `Count`, the measured state count, the derived V_0, and bounded-base verification remain open.

The boundary rebase's output half is now executable too. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteBoolOutput` observes that the rebase state has no persistent ports, so its remaining proposition-valued tracked-connectivity query is exactly the stored residual Boolean bit after the explicit activity and colour guards. Lean proves this reflection and uses it with the four already-Boolean facial coordinates to decode the complete terminal-aware successor profile. For every literal source rebase, that native decoder is proved equal to the actual profile consumed by the next Cell. This closes the executable output half only: the predecessor-to-successor rebase support relation, its composition with the distinct Cell letter, the heterogeneous reachable table, the numerical V_0, and bounded-base verification remain open.

The tracked predecessor side of that rebase is now factored at the correct boundary. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseTrackedInputAttachmentState` observes that the four changing switch roles are not the two crossings shown by the old zero-terminal profile. It therefore retains those two crossings as persistent ports. An exact finite residual code proves that the complete old profile reconstructs predecessor tracked reachability among every pair of switch roles. The module then makes the deletion issue explicit: since a removed switch edge could have been internal to a predecessor path, full predecessor connectivity is not itself a safe update state. A second finite state is extracted from the common old/new core after deleting the switch; it stores the core relation between the two old ports and proves exact core reachability among all four roles. Thus the next tracked obligation is sharply local: combine this core state with the finite successor residue and take their directed closure. The complete five-field rebase letter, the alternating reachable table, the numerical V_0, and the bounded-base audit remain open.

The tracked rebase recurrence is now finite as well as exact. Modules `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseTrackedCoreRecurrence` and `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseTrackedCollarCode` prove, for each tracked colour pair, that the successor graph is precisely the union of the unchanged-core component relation and a canonical local residual. The residual cannot leak arbitrarily far: every edge it touches is in the closed edge-adjacency neighbourhood of the at-most-four changing switch edges. Since the opened carrier is subcubic, this gives an explicit bound of twenty-eight edges. Lean transports the two factors through one common finite carrier, retains all four rebase roles as named coordinates, and proves that closure of the six coded graphs (three colour pairs, each with a core and local factor) is equivalent to literal successor reachability. This closes the semantic and finite-coordinate tracked update; native Boolean reflection and its combination with the already finite facial collar remain before the complete rebase support letter. The alternating reachable table, its measured state count, the numerical V_0, and the route-native bounded-base audit are not claimed yet.

The two finite collar recurrences are now executable as well. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteStepReflection` reflects the at-most-twenty-eight-edge tracked collar and the at-most-twenty-four-dart facial collar to Boolean adjacency tables, computes the directed closure of the inherited-core and local-residual steps, and proves in both cases that the Boolean answer is equivalent to literal successor reachability. Thus no tracked or facial connection is added or lost by the executable representation. This is still one indexed literal rebase, not a quotient by the three geometric orientation labels, and it does not assert the converse from an arbitrary finite code to a positive `Count` witness. Combining these closures with the existing output decoder into the complete five-coordinate rebase letter is the next unit; the heterogeneous reachable table, measured state count, numerical V_0, and bounded-base audit remain open.

The output receipt itself is now a native finite value. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteBoolReceipt` replaces its last proof-facing field—the facial collar family—by a Boolean adjacency table, while retaining the finite fragment occupancy, coordinates, continuation, port incidence, and capped lengths. Its decoder is proved equal to the earlier five-coordinate decoder and hence, on every literal source rebase, to the terminal-aware profile consumed by the next Cell. The Boolean facial closure is also proved to recognize equality of actual successor regional fragments exactly. This is an executable output receipt, not yet a transition system: the joint predecessor-to-successor support predicate and the alternating reachable closure remain the next obligations, followed by the measured state count, numerical V_0, and bounded-base verification.

That joint rebase predicate is now executable and source-sound. Module `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseFiniteBoolSupportLetter` retains the zero-terminal predecessor and terminal-aware successor as the two endpoints of a finite letter. One Boolean decision checks that the successor is the complete five-coordinate receipt, that every guarded tracked bit agrees with the exact inherited-core-plus-local closure, and that every occupied facial row has a collar coordinate whose closure component recognizes exactly that row. Ambient face continuation remains a distinct stored coordinate; it is not identified with regional-fragment closure. Lean proves that every actual literal source rebase evaluates to `true`. Consequently both alternating letter kinds—Cell and boundary rebase—now have executable source-sound support predicates without an orientation quotient. Their endpoint-compatible heterogeneous recurrence and reachable table are the live next obligations. Converse adequacy with positive entries of the manuscript's `Count`, the measured state count, the derived V_0, and bounded-base verification remain open.

The two alternating receipts now also meet at an exact typed seam. Module `GoertzelV24ClosedWebLocalLayerSerialFiniteBoolStepLetter` pairs one complete Cell receipt with the following boundary-rebase receipt, checks exact equality of their zero-terminal middle profiles, and presents the resulting terminal-aware-to-terminal-aware witnessed transition as a Boolean relation. For every actual source offset with a successor, Lean constructs such a pair, proves its support predicate true, and identifies its endpoints with the two consecutive cumulative source profiles.

This does not yet make that one receipt the source transfer letter. A physical Cell's letter is the complete support relation of its `Count` matrix, ranging over every positive endpoint pair; one ambient colouring supplies only one member of that relation. A provisional construction which listed those single receipts as though they were the heterogeneous word was therefore discarded before being committed. The live finite-state obligation is to assemble the full positive-`Count` relation over the executable receipts and then certify its reachable closure. This distinction prevents a witnessed path from being silently substituted for the matrix semantics prescribed by the source.

The first coordinate of that comparison is now kernel checked. Module `GoertzelV24ClosedWebLocalLayerSerialCountColorAdequacy` proves, at every actual source offset, that the inherited literal-Cell colouring selects a positive entry of the complete five-coordinate `Count`, that the paired Cell–rebase receipt is supported, and that their ordered input and pre-rebase output colour words agree exactly. The comparison is proved by passing through the literal complement's two cut-data profiles and the cumulative serial profiles; it is not a definitional identification of the two carriers. This closes only boundary-colour adequacy. Tracked connectivity, facial identity and port incidence, capped face progress, the arbitrary positive-entry converse, the reachable table, and the numerical V_0 remain open.

The tracked local factor is now universal over that positive support, rather than only over the inherited ambient colouring. Module `GoertzelV24ClosedWebLocalLayerLiteralCountTrackedAdequacy` proves that every positive literal-Cell `Count` entry supplies an actual open-Cell Tait colouring and hence a six-slot finite graph-family code. For each of the three tracked colour pairs, the code preserves the complete reachability relation on the literal Cell region. Thus an arbitrary matrix witness, not merely the one selected by the ambient colouring, has a finite tracked local factor. This is intentionally not yet the cumulative one-step transition: the predecessor witness and Cell witness must still be glued across their common seam, and the facial/incidence/capped-progress coordinates must be carried through that same compatible composition before the heterogeneous reachable table or a numerical V_0 is claimed.

The tracked part of that arbitrary-witness seam is now closed at the source's combined-profile level. Module `GoertzelV24AnnularCrosscutComplementCombinedConnectivityOfBoundary` removes the global-cubicity premise from the earlier combined-connectivity calibration. Directly from a boundary-rooted literal open-Cell colouring, Lean proves that the four-port combined profile satisfies the exact finite serial-connectivity law between its left and right restrictions. Equivalently, every positive entry of the boundary-rooted five-coordinate `Count` supplies one realizable combined state with that law. The ten degree-one annular stubs therefore do not obstruct the local tracked morphism, and no ambient-prefix colouring is used. This does not yet establish the analogous combined adequacy for ambient face identity, fragment–port incidence, and capped face progress; those three jointly implement the source's facial coordinate and remain the last local semantic comparison before the complete positive-support relation and reachable table can be certified.

That local semantic package is now complete. Module `GoertzelV24AnnularCrosscutComplementCombinedFacialSemanticsOfBoundary` reads the other three fields from the same boundary-rooted literal colouring: two fragments continue precisely when they name the same ambient face, a port belongs to a fragment precisely when its literal edge occurs in that regional face fragment, and the stored progress is precisely the regional fragment length capped at five. Consequently a positive `Count` entry is equivalent to one realizable combined profile carrying both the exact tracked serial law and all three facial observations. This finishes the local five-coordinate meaning of an arbitrary matrix-support witness. It does not yet compose that witness with an arbitrary cumulative prefix: the remaining factorization must express the prefix–Cell face attachments through the existing finite receipt and then compute the heterogeneous reachable closure.

7.5 What the programme may explain—and what is not yet a theorem

The finite-state route suggests an information-theoretic explanation for why a short informal proof may be unavailable: the reduction can require preserving a large amount of interface information, and the terminal finite verification may contain irreducible case information. This does *not* imply that no human-friendly proof of 4CT can exist in every proof system.

A formal non-compressibility result must first specify a presentation or proof class, an encoding, a description-length or verification-complexity measure, and uniform quantifiers. Only then can a lower bound be meaningful. The future formal endpoint is therefore the strongest such theorem supported by the compositional transfer presentation, together with an explicit boundary between proved mathematics and philosophical interpretation. A large state count by itself is evidence about one representation, not a universal impossibility theorem.

7.6 How to read the rest of this companion

The remaining sections preserve earlier investigation strata: the v23 three-pillar audit, machine-checked obstruction and repair attempts, the composition ladder, the transition to the v24/Fable work, and the graveyard of superseded routes. They are retained because negative knowledge is expensive and because deleting it makes old mistakes attractive again. When an older status statement conflicts with this section or with the current Fable manuscripts, this section and those primary sources control.

8 The problem and Ben Goertzel's three-pillar route

8.1 Tait's reduction and the minimal-counterexample target

The Four-Color Theorem asserts that every plane triangulation G admits a proper vertex 4-coloring. Passing to the cubic dual $K = G^*$, Tait's classical equivalence (v23, Thm. 2.3) restates 4CT as: every bridgeless planar cubic graph admits a *proper 3-edge-coloring*, i.e. a map $\chi : E(K) \to \{r, b, p\}$ assigning distinct colors to the three edges at each vertex. The manuscript encodes the three colors as the three nonzero elements of $\mathbb{G} = \mathbb{F}_2^2$, with $r = (1, 0)$, $b = (0, 1)$, $p = (1, 1)$, so that "properness at a vertex" is exactly $\chi(e_1)+\chi(e_2)+\chi(e_3) = r+b+p = 0$ in $\mathbb{G}$, and "integrating" an edge-coloring across the planar dual becomes a path-sum in $\mathbb{F}_2^2$.

Following Spencer–Brown and Kauffman, the manuscript reformulates colorings as plane *formations* (red/blue Jordan curves with purple overlaps), defines Kempe switches as the standard two-color interchange, and proves the *Parity Lemma* (planar invariance of the mod-2 count of two-color circuits). The role of §3 is to reduce 4CT, via Kauffman's parity/primality argument (Theorem 3.1), to the nonexistence of a minimal counterexample containing a *between-region annulus* H whose boundary-compatible colorings are *not* connected by Kempe moves supported inside the annulus. Thus the entire analytic burden is shifted to a *between-region Kempe connectivity* statement for an annulus $H \subseteq K$ with outer and inner boundary cycles Γ_∞, Γ_0.

8.2 The three pillars

The manuscript's conceptual overview (v23, §1.1) states the argument as three interacting pillars.

Pillar A: annular difference spanning (the linear "accounting" layer). Working in a between-region annulus H, one encodes boundary colorings and their differences as vectors in an $\mathbb{F}_2$-space. Let $W(H)$ be the coordinatewise cycle space of $\mathbb{G}$-chains on $E(H)$ satisfying the Kirchhoff/even-degree constraints at interior vertices, and let

$$W_0(H) \;:=\; \{\, x \in W(H) : x(e) = 0 \text{ for all boundary edges } e \in B_1 \cup B_2 \,\}$$

be the *zero-boundary* subspace. For a fixed proper coloring C_0 of H with Kempe closure $\mathrm{Cl}(C_0)$, the manuscript defines a family $\mathrm{KC}(C_0)$ of *Kempe-closure (polarized face) generators* $X_f^{ab}(C)$ supported in depth-1 collars (v23, Def. 4.8), and proves:

Theorem 66 (Spanning theorem for $W_0(H)$, v23 Theorem 4.9)**.** *Let H be an annular subgraph of a bridgeless planar cubic graph, with boundary cycles Γ_0, Γ_∞. Fix any proper 3-edge-coloring C_0 of H and let $\mathcal{G} := \mathrm{KC}(C_0)$. Then $W_0(H) \subseteq \mathrm{span}_{\mathbb{F}_2}(\mathcal{G})$; equivalently, if $z \in W_0(H)$ satisfies $z \cdot g = 0$ for every $g \in \mathcal{G}$, then $z = 0$.*

The manuscript proves Theorem 66 in strong-dual annihilator form: it shows $U^\perp \cap W_0(H) = \{0\}$ for $U = \mathrm{span}_{\mathbb{F}_2}(\mathcal{G})$, by a planar orthogonality peeling along a dual spanning forest, after a "purification" step (Lemma 4.4) that removes the polarization of the face generators.

Pillar B: CAP5-free geometry and radius-1 locality (the geometric locality layer). Assuming a minimal counterexample, the manuscript eliminates the CAP5 separating obstruction and works in a CAP5-free normal form (Corollary 6.9). A tube-forcing argument then reduces every "escape" of an anchored Kempe chain to a finite three-cell gate enumeration, yielding radius-1 confinement of anchored Kempe cycles together with a cap-aware disjointness/commutativity package (Lemmas 7.8, 7.9, 7.10). This is what licenses peeling the wide annulus into radius-1 collars.

Pillar C: executability without false additivity (the algebra-to-geometry bridge). The decomposition $\Delta = \sum v_{f,g}$ provided by Pillar A is a statement in $W_0(H)$; it does *not* by itself imply that one may *execute* the corresponding mixed color-pair Kempe switches as if they formed an abelian sum. The manuscript is explicit that cross-pair interference is real: an (a,b)-switch can destroy the (b,c)-structure a later switch needs (v23, §8.1), and Kempe switches using different color pairs on shared support need not commute (v23, §8.3). To execute the decomposition honestly, the manuscript moves to an operator model: proper colorings form an orthonormal basis of a Penrose–Kauffman state space, and each Kempe switch is an involutive linear operator (generally noncommuting with switches of other color pairs). Using radius-1 locality, between-region Kempe connectivity is reduced to a single local property:

Definition 12 (Local Kempe reachability with fixed input, v23 Def. 8.12)**.** Let X be a planar cubic tangle with ordered input stubs S_{in} and output stubs S_{out}, and fix an input coloring c. Let $\Omega_X(c)$ be the set of proper 3-edge-colorings of X extending c on the input stubs. The *Kempe graph* $\mathsf{K}_X(c)$ has vertex set $\Omega_X(c)$, with an edge whenever two colorings differ by a single Kempe switch supported on a two-color component touching no input-stub edge. We say X satisfies $\mathrm{LKR}_{\mathrm{in}}$ if $\mathsf{K}_X(c)$ is connected for every c for which $\Omega_X(c) \neq \emptyset$.

The chain of Pillar C is then:

- **Lemma 8.14**: the canonical three-cell collar gadget τ satisfies $\mathrm{LKR}_{\mathrm{in}}$ (and so does its mirror);
- **Lemma 8.15**: two-color *transparency* of τ — every two-color component is a boundary-to-boundary path with no internal cycle;

- **Lemma 8.18 / Corollary 8.19**: $\mathrm{LKR}_{\mathrm{in}}$ *composes* across a transparent interface, so an entire cut-open radius-1 collar tangle $\widehat{A} = G_m \circ \cdots \circ G_1$ satisfies $\mathrm{LKR}_{\mathrm{in}}$;
- **§9.2 assembly**: peel the wide annulus H into nested radius-1 collars and use the collar $\mathrm{LKR}_{\mathrm{in}}$ property to rewrite the current coloring to the target collar-by-collar from outside to inside, the cap-aware locality lemma (7.9) ensuring no switched component reaches back to change a fixed boundary.

8.3 Why Lemma 8.14 is the keystone

The manuscript states that its proof is human-readable and "the only finite verification required is a small by-hand audit of a single three-cell gadget" (v23, Abstract and §1). In the three-pillar architecture, every other step is either an explicit local lemma or a structural reduction; the one place where a finite, exhaustive check is genuinely unavoidable is the $\mathrm{LKR}_{\mathrm{in}}$ property of the canonical gadget τ. Formalizing Lemma 8.14 therefore certifies exactly the finite core that the manuscript isolates as load-bearing. The remainder of this companion describes that formalization (§9), the resolution of the Pillar A spanning question into a need-based repair (§10), the Pillar C composition ladder — what is proved, what is reduced to one pin, and where a decisive counterexample was found (§11) — and a precise status ledger (§13). Superseded routes are recorded in appendices.

9 The Pillar-C keystone, machine-checked: Lemma 8.14 and transparency 8.15

We restate the manuscript's keystone in its own notation and then describe the Lean encoding.

Lemma 61 ($\mathrm{LKR}_{\mathrm{in}}$ for the canonical three-cell gadget, v23 Lemma 8.14)**.** *Let τ be the canonical three-cell collar gadget. It is a planar cubic tangle with internal vertices $F_0, \dots, F_5$ and boundary stubs $B_0, \dots, B_7$, whose internal edges form the tree*

$$F_1 - F_0 - F_2 - F_4 - F_5, \qquad \textit{with a leaf } F_3 \textit{ attached to } F_2,$$

and whose boundary stubs are attached by F_0B_0, F_1B_1, F_1B_2, F_3B_3, F_3B_4, F_5B_5, F_5B_6, F_4B_7. Take input stubs $\{B_0, B_1, B_2, B_3\}$ and output stubs $\{B_4, B_5, B_6, B_7\}$. Then τ satisfies $\mathrm{LKR}_{\mathrm{in}}$*. The same holds for the mirror gadget.*

The manuscript proves Lemma 61 by cutting τ along the internal interface edge $I = F_2F_4$ into $\tau = \sigma_{\mathrm{R}} \circ \sigma_{\mathrm{L}}$ and doing a small case analysis on the two pieces. Our formalization instead audits the *full* gadget directly, by exhibiting the proper-coloring state space and an explicit Kempe-path certificate connecting every fixed-input fiber; this is a strictly more direct realization of the same $\mathrm{LKR}_{\mathrm{in}}$ claim, and it does not rely on the $\sigma_{\mathrm{L}}, \sigma_{\mathrm{R}}$ split or on the composition Lemma 8.18.

9.1 The canonical gadget and its 192-state coloring space

The Lean module fixes the colors and the gadget incidence exactly as in Lemma 61: the three nonzero colors are an inductive type, and the gadget's edges (five internal edges plus the eight boundary stubs) and internal vertices are inductive types with an explicit incidence map.

```
inductive LColor | r | b | p
  deriving DecidableEq, BEq, Repr, Inhabited
```

```
inductive TauEdge
  | F1F0 | F0F2 | F2F3 | F2F4 | F4F5
  | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7
  deriving DecidableEq, BEq, Repr, Inhabited

def tauInputEdges : List TauEdge := [TauEdge.B0, TauEdge.B1, TauEdge.B2, TauEdge.B3]
```

A proper 3-edge-coloring of τ is represented by the finite choices left after properness is imposed at every cubic vertex: an ordered color triple at the central vertex F_2, one binary choice for the forced edge F_1F_0, and one ordered-pair (swap) bit at each remaining boundary fork — $6 \times 2^5 = 192$ states. That this enumeration is exactly the proper-coloring state space is a checked fact (192 entries, no duplicates, proper at every vertex).

9.2 The Kempe audit and the machine-checked theorems

A Kempe step switches a two-color component *disjoint from the input stubs* (Definition 12). The $\mathrm{LKR}_{\mathrm{in}}$ property is witnessed by an explicit indexed-path certificate: for each of the 192 states it records the representative of that state's fixed-input fiber and a path of at most two input-disjoint Kempe switches reaching it, plus a coherence audit that any two states with the same input edges share a representative (so fibers are well defined). The three finite audits each reduce to `true` by the kernel's decision procedure; the only axiom dependency is `propext`.

```
theorem tauStateSpaceAudit_ok : tauStateSpaceAudit = true := by decide
theorem lemma814PathCertificate_rows_ok :
    validIndexedRowsFrom 0 lemma814PathCertificate = true := by decide
theorem representativeCoherenceAudit_ok : representativeCoherenceAudit = true := by
    decide

/-- Lemma 8.14, closed by the finite state-space and indexed path certificate. -/
theorem lemma814_tau_lkrIn_audit : lemma814_tau_lkrIn_finiteCheck = true := by
  simp [lemma814_tau_lkrIn_finiteCheck, tauStateSpaceAudit_ok, tauLKRInAudit,
    lemma814PathCertificateAudit_ok]
```

The mirror clause holds definitionally (reflection changes only the planar order of an isomorphic incidence tree). The incidence enumeration is complete for the encoded tau graph: every proper incidence coloring is one of the 192 enumerated colorings. The finite certificate audit is proved, but graph-level soundness of every encoded step remains the named obligation `Lemma814EncodedMovesAreGraphKempeSwitchesTarget`. Thus the current Lemma 8.14 result is a finite encoded-model audit, not yet the manuscript graph theorem.

Remark 27 (Transparency 8.15, finite encoded-model audit)**.** *The same module formalizes Lemma 8.15 (two-color transparency): for each of the 192 colorings, every nonempty two-color component is a boundary-to-boundary tree path (`transparentComponent`), aggregated as `tauTreeTransparencyAudit`. That audit is now* proved — *`lemma815_tau_tree_transparency_audit` reduces to `true` by a range decomposition of the finite check, axiom-clean. The encoded transparency audit is proved. Its use as graph-level composition (§11) shares Lemma 8.14's named graph-fidelity obligation, so the local graph-theoretic core is not yet sealed.*

10 Pillar A: a proper-subspace obstruction, and its need-based repair

Pillar A (the spanning Theorem 66) is where the formalization first found — and then routed around — a real gap. We give both halves, because together they are a clean example of what a companion of this kind is for.

10.1 The obstruction: the spanning *equality* over-reaches

On every *closed-walk exact shell* — a concrete annular configuration carrying the boundary-source and interior-endpoint witness data that Theorem 66 is meant to cover — the boundary-zero target that the Kempe-closure detector actually pins down is a *strict* subspace of the full boundary-zero chain space.

```
/-- On every closed-walk exact shell, the Theorem-4.9 boundary-zero target is a
proper subspace of the full selected-boundary-zero chain space. -/
theorem
   theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule_of_closedWalkExactShell

    [Fintype G.edgeSet] {emb : PlaneEmbeddingWithBoundary G}
    (shell : ClosedWalkExactShell emb) :
    theorem49BoundaryZeroKirchhoffSubspace emb
        (selectedBoundaryInteriorEdgeEndpointVertices emb) <
      planarBoundaryZeroSubmodule emb
```

The strict inequality `<` is the load-bearing content: a proved *proper*-subspace inclusion, not a heuristic. The structural reason (every closed-walk shell forces two distinct interior-support edges onto one face boundary) refutes the natural one-collar repairs, and a companion audit shows the four proposed *uniform geometric oracles* that would each close Theorem 66's synthesis are all false:

```
theorem not_interiorDualPeelOracle         : ¬ InteriorDualPeelOracle
theorem not_wellFoundedPeelOracle          : ¬ WellFoundedPeelOracle
theorem not_previousBoundaryGeometryOracle : ¬ PreviousBoundaryGeometryOracle
theorem not_residualBoundaryGeometryOracle : ¬ ResidualBoundaryGeometryOracle
```

Lean proves that the formal detector is strictly smaller than a larger unrestricted selected-boundary-zero chain space. This does not refute v23's actual cycle-space/Kirchhoff-constrained claim $W_0(H) \subseteq \operatorname{span}_{\mathbb{F}_2}(\mathcal{G})$. That inclusion is now named `V23Theorem49W0ContainedInKempeClosureGeneratorSpanTarget` and remains open; the four uniform oracle routes that would have closed it uniformly are refuted.

10.2 The repair: the route never needs the full space

The resolution is that the synthesis does not *use* the full boundary-zero space; it uses only the discrepancy classes of boundary-compatible coloring pairs, and those lie in the detector subspace the manuscript's own peeling witnesses span. Formally, the requirement actually consumed downstream is *need-based*: it suffices that (1) the detector subspace is contained in the projected Kempe-closure generator span, (2) each execution step preserves that subspace, and (3) every route-used discrepancy class lies in it. All three are machine-checked in the route's natural cubic setting (module `PillarASmallerSpace`):

```
-- (3) discrepancy classes lie in the detector subspace (cubic setting):
theorem routeUsedDiscrepancyClassesLieInDetectorSubspace_of_hasCubicIncidentEdgeTriples
    ...
-- (2) a Kempe-switch execution step preserves detector membership:
theorem routeKempeSwitchExecutionStep_preserves_detectorSubspace ...
-- assembled sufficiency (inclusion form), discharging the requirement:
theorem
    needBasedPillarARequirement_of_detector_le_span_of_stepPreserves_of_routeDiscrepancies
     ...
```

Proposition (3) has a two-idea proof one can state aloud: a Tait coloring is a Klein-four flow at each cubic vertex, so two colorings agreeing on the selected boundary differ by a *boundary-zero flow*, which is exactly what the Kirchhoff detector certifies. The manuscript's Theorem 4.9 states an *inclusion*, not an equality. The existing need-based theorems are conditional on detector containment, step preservation, discrepancy membership, and embedding-specific peel data. They are conditional algebraic-span infrastructure, not a proof of Theorem 4.9. What remains a genuine per-embedding obligation is the *supply* of a peeling witness for the synthesis's actual embedding class; that is a composition-stage task, not a spanning gap.

11 Pillar C: the composition ladder

Between the local keystone (§9) and 4CT, v23's own argument climbs a ladder: single-gadget $\mathrm{LKR}_{\mathrm{in}}$ $\to$ transparent composition to a whole collar chain (Gate-2) $\to$ closed re-glued collars (§9.2 Step 3) $\to$ execution in the ambient annulus (§9.2 Step 4) $\to$ the wide-annulus peel. This companion has now walked a substantial part of that ladder, with one step proved-modulo-a-pin and one step *refuted*.

11.1 Gate-2: transparent composition, reduced to one pin

Corollary 8.19 lifts $\mathrm{LKR}_{\mathrm{in}}$ from the single gadget to a whole cut-open radius-1 collar chain across a transparent interface. The manuscript's own Step-1 proof of Lemma 8.18 (a pointwise preparation via Lemma 8.16) does *not* go through — the formalization refutes it directly:

```
theorem lemma818_pointwise_lift_preparation_obstruction : ...   -- axiom-clean
```

In its place the development builds a frontier-automaton / fibration architecture: a finite frontier state over the chain transcript, generated parent-row certificates, and a fibered-connectivity decomposition (base connected + fibers connected + lifts). The executable chain test `chainLKRInAudit` passes on all words up to length seven (real-fiber labs, all input fibers connected, zero failures). The former metadata-only classification audit has been replaced: the actual per-target test is `representativeTargetFiberKempeOrbitConnectivityAudit`, and Gate-2's single live pin is

```
Gate2RepresentativeFiberKempeOrbitConnectivityPin
```

combining fourteen representative real-fiber audits with DFA-mode sufficiency; the all-chain implication is proved conditional on this pin. (A separate refuted route, `not_concreteChainFiberAppendPrefixFibrationClosed`, records a shortcut that provably fails and is kept dead.)

11.2 S2: length-2/3 radial-closing-edge fiber census

Gate-2 certifies $\mathrm{LKR}_{\mathrm{in}}$ with the *first* gadget's input stubs fixed; §9.2 Step 3 needs the *cyclically re-glued* collar with all outer-side stubs fixed — different fibers, no bridging lemma in the manuscript. What is proved is a finite census: it ranges over twelve tau/mirror words of length 2 or 3 and fixes the four radial-closing edges (a lab found all 972 such fibers connected-or-empty — 216 nonempty, 756 empty, zero disconnected — and the census target was discharged with zero pins). It does *not* fix every outer stub and does not range over arbitrary collar length. The actual manuscript claim is named `ManuscriptSection92S2AllOuterStubsFixedArbitraryCollarTarget` and remains open.

```
def closedCollarLength23RadialClosingEdgeFiberCensus : Prop := ...   -- proved
def ManuscriptSection92S2AllOuterStubsFixedArbitraryCollarTarget : Prop  -- open
```

The route survey had rated S2 the sharpest mismatch in the manuscript — a statement never written down in v23 and not implied by Gate-2 — so proving it on the finite class is a real advance.

11.3 S4: abstract-chain ambient-escape regression

§9.2 Step 4 reinterprets a collar Kempe word as a word of the *ambient* annulus, and confines the switched components with a face-only locality lemma (7.9). The formalization proves an ambient-escape regression in the encoded tau-chain model — a collar plus one deeper radial ring sharing the interface. No theorem currently realizes that witness as a simple planar, cyclically-five-edge-connected actual collar; graph-facing salvage is conditional on `ClosedWalkActualCollarRotationFaceIncidenceCutLiftObligation`, which remains open.

```
/-- A collar-certified, collar-input-avoiding Kempe switch whose ambient
two-layer component leaves through the inner side, hits the fixed outer input,
and does NOT restrict to the certified collar move. -/
theorem section92Step4TwoLayerCounterexampleClass :
    Section92Step4TwoLayerCounterexampleClass         -- axioms [propext]
```

Concretely (outer word $\tau\tau$, inner word τ, input key r, r, b, r): a switch that is input-avoiding *in the collar alone* balloons, in the two-layer object, into a component that crosses the shared inner boundary into the deeper ring, returns, meets the fixed outer input edge, and lands the collar on a *different* coloring than the certified move. This refutes the reinterpretation as a regime-free, purely-local principle: the ambient switch is a *different* move on the collar than the one Gate-2/S2 certified.

We are precise about what this does and does not say. It is a defect in the manuscript's assembly *seam* as argued (Step 4 invokes 7.9, a lemma proved for single-face collars, as if it were a global confinement), not a refutation of 4CT.

The cheapest repair has also been tested, and *refuted*. Strengthening the certification so that switches avoid the entire outer interface — both input *and* output stubs, not only the inputs — does eliminate the escape (a two-layer companion run finds zero escapes under full-interface avoidance), but at a fatal cost: on the same finite class, two colorings with the same fixed input are connected *only* by a switch whose component touches the output stubs, so forbidding output touches disconnects the fibers (module `GoertzelLemma818TwoSidedAvoidance`: 432 open and 216 closed nonempty fibers disconnect). The escape channel is therefore *load-bearing for connectivity*: the very moves that let a component escape the collar are the moves the fibers need to remain connected. Confinement and connectivity are in direct tension, so any repair must confine the escaping components *without* severing collar connectivity — a needle the simplest fix cannot thread.

What remains is a genuinely regime-level salvage: a proof that the minimal-counterexample *regime* (CAP5-free normal form) excludes the disconnecting/escaping configurations, or a §9 redesign that only ever peels *face* collars where 7.9 applies as stated. Neither is established; both need planar-normal-form infrastructure the development does not yet have; and the abstract gadget lab cannot decide them, since the tension may or may not dissolve in the actual normal form. As of this writing the seam is open, with both the escape and the failure of the cheapest repair pinned down as checked theorems.

11.4 Sharpening the open salvage: a winding invariant and a cyclic 2-cut reduction

Since the escape was pinned down, the formalization has *sharpened* — though not closed — the open salvage, by asking a Spencer–Brown/Kauffman-native question: which two-color components can escape? On the closed annular collar a two-color component carries a well-defined mod-2 *winding class* around the annulus generator — the parity of its crossing count with a fixed radial cut — a Parity-Lemma-style invariant, well defined per component and tracked across Kempe switches. In this language the escape is exactly a *winding* move: the components that leave through the inner side and return are those of nonzero winding class. The precise form of the S4 freedom is then a homological statement, machine-checked on the witness class: *winding is not boundary-determined* — a fixed-boundary fiber can carry components of different total winding class, so the escaping (winding-1) moves are not forced by the boundary data alone.

This localizes the difficulty to one geometric question: can a winding-freedom escape be *realized on a genuinely simple planar collar*? The formalization reduces it to a classical connectivity fact. Any simple planar realization that preserves the winding profile must replace the escape's parallel endpoint bundle by a simple two-pole side; a tree-rank (handshaking) count — $2\,|\text{side edges}| = 3\,|\text{internal}| + 4$ forces $|\text{internal}| + 1 < |\text{side edges}|$, hence a cycle — turns that side into a *cyclic 2-cut* of the collar. A collar admitting no realized cyclic 2-cut therefore admits no escape realization:

```
-- every simple planar winding-escape realization forces a cyclic two-cut:
theorem ...Escape_forces_cyclicTwoCut_allSizes : ...
-- so a collar forbidding cyclic two-cuts (e.g. cyclically 5-edge-connected)
-- admits no escape realization:
theorem ...Escape_not_simplyRealizable_allSizes
    (h : ...ForbidsCyclicTwoCut G) : ¬ ...SimplePlanarEscapeRealization G
```

We are precise about what this buys. It is a *reduction of the open salvage to one named residual*, not a repair: the theorem is conditional on the collar forbidding cyclic 2-cuts, and the current realization datum carries the separating-pair structure as an obligation rather than deriving it from the collar's embedding. What it does is convert the vague "the minimal-counterexample regime should exclude the escape" into a concrete checkable target — *the escape collar carries no cyclic 2-cut*, i.e. it is cyclically sufficiently edge-connected, which is exactly the kind of property the minimal-counterexample normal form is expected to supply. And this residual is not an unformalizable abstraction: the development already proves cyclic-five-edge-connectivity for concrete graphs (and refutes it for others), so discharging or refuting it *for the escape collar* is a reachable next question — the honest current frontier of the S4 salvage.

12 The v24 route: deserts, pivots, and the necklace

Goertzel's v24 working package (July 2026) replaces v23's §6–7 with a new architecture: pentagonal *pivots*, hexagonal *deserts*, a *necklace* decomposition of the sphere relative to the twelve pentagons, winding and monodromy invariants along corridors, and a final *Trail Completability* target from which the minimal-counterexample contradiction is to follow, conditional on a dependency chain L1–L10. The package proposes a universal Short Strand Conjecture and then refutes it itself with a 688-edge, reach-40 strand, so the route requires a replacement bound. This section records what the formalization has established of this route so far; the formalization targets combinatorial rotation systems throughout, not topological planarity.

Independent reproduction of the validation toolchain. Every numerical claim consumed from the package was recomputed independently before use: the annular censuses (5412 and 31704) with their Kempe fibers, the fullerene finite bases, the pentagon finite seals, tube seed growth, the desert and strand specimens (including the Short-Strand refuter), Goldberg normal-form specimens, and the GWCO exact base through twelve. A recurrent tube-ring transfer theorem replaces the package's Pascal-triangle extrapolation (`GoertzelV24TubeRingTransfer`).

Rotation systems and facial linear algebra. A Tait-coloring layer is built over rotation systems, with each edge's two faces identified, face simplicity reduced to two-sidedness, facial dual connectedness, and the exact facial linear algebra: facial boundary rank, primal cycle-space rank, the equality of the facial span with the cycle space, and the face winding classification (`GoertzelV24FaceBoundaryLinearAlgebra`, `GoertzelV24OrbitFaceCycleSpaceEquality`, `GoertzelV24WindingClassification`).

Curvature. The all-face discrete Gauss–Bonnet identity is proved from rotation-system data alone: the total face curvature $\sum_f (6 - \ell(f))$ equals 12 (`totalFaceCurvature_eq_twelve` in `GoertzelV24CurvatureScope`). The package's *at-most-twelve pentagon* step, which is false as stated under minimum face size five, is corrected by a proved *weighted* pentagon budget charging faces by $6 - \ell(f)$; the fullerene hexagonal bulk is derived on orbit faces, and curvature-to-bulk corridor bounds connect the budget to corridor extraction (`GoertzelV24OrbitFaceCurvatureBulk`, `GoertzelV24BulkCorridor`).

Necklace and corridors. The locked-pentagon necklace is assembled and the singleton necklace strand forced (`GoertzelV24Necklace`). A hexagonal corridor theory is developed: defect-free corridor neighborhoods extracted from geodesic face paths; an internal rung classification with induced rung chains and the exclusion of adjacent induced rungs; clean width-four slabs with oriented interfaces and the matching of consecutive interfaces; Tait transfer across slabs with finite color-transition alphabets and color-state words; and a sharp chirality barrier under which transfer across a chiral reversal forces the melted state (`GoertzelV24CleanHexCorridor`, `GoertzelV24HexFaceRungType`, `GoertzelV24OrientedHexSlab`, `GoertzelV24HexCorridorColorTransfer`, `GoertzelV24HexCorridorChiralityBarrier`).

Opening a pentagon is not yet the Cell–3 formation. Addendum V starts from a supplied frontier tangle X with an interior pentagon and then forms its annular X'; it does not say that merely opening a pentagon in an arbitrary closed cubic graph supplies the free outer interface. The Lean comparison now makes this distinction exact. Given an explicitly selected `PentagonCap`, the manuscript-side deletion (five fresh ordered spoke ports) and the toolchain-side deletion of the five cap-cycle edges are isomorphic as graphs; all boundary ports are the five ordered spokes, and the isomorphism sends stub i to cap vertex i definitionally (`openingGraphIso`, `openingGraphIso_ordered_stub`). But the resulting one-cap opening has *no* outer stubs (`outerCount_eq_zero_of_innerStub_eq_capVertex`). Thus these theorems close a representation mismatch, rather than manufacturing the annulus required by the source. The still-open formation theorem must add the actual outer interface, its annular embedding, and the transport of the selected clean corridor; it cannot be replaced by a cardinality argument or a second cap chosen without source geometry.

The two-cap laboratory is now an exact graph construction, not yet the source formation. The separate modules `GoertzelV24TwoPentagonCapOpening`, `...Boundary`, `...Comparison`, and `...OpenRegionComparison` now make the C_{30}-style laboratory literal. From a supplied pair of pentagon caps with disjoint vertex supports and the necessary no-spoke-contact conditions, Lean deletes both cap cycles and constructs a well-formed 5+5 `AnnularBoundaryData`. It proves that every exposed port is uniquely one of the two ordered spoke families, and identifies the manuscript slit graph, the toolchain opening, and the computed open-region primal graph by explicit graph isomorphisms; the two ordered families are preserved by construction. This is useful because it eliminates a representation mismatch without pretending that ten boundary stubs alone make an annulus. The subsequent `FacialPentagonCapPair` strengthens the supplied laboratory datum by identifying both cap cycles with literal closed face orbits. The unbuilt source-facing part remains selecting such a facial pair from the relevant minimal frontier, calculating the two named opened hole faces and their rotation-level embedding, and transporting the chosen clean corridor and its local hypotheses. Addendum V itself treats the two-cap examples as an annular laboratory, not as a theorem that every minimal counterexample canonically supplies this formation.

The collar calculation has now begun at the right level of generality. `GoertzelV24OpenRegionBoundaryStubFaceStep` proves the exact opened face-permutation equation at one fresh degree-one stub: a ϕ-step crosses the stub edge and then applies the retained rotation at its old endpoint. The immediately following literal-boundary calculation `GoertzelV24FacialPentagonCapOpenBoundary` proves that those stubs are neither assumed nor incompletely enumerated: after deleting one cap's five vertices, the five source-indexed outward spokes are injective exposed darts and exhaust the open rotation's `BoundaryDart` carrier. Thus the next face trace has exactly the source's five named ports to account for. This is the mechanical first move of the cap-collar orbit calculation, not a complete surface surgery or an assertion that the stubs already form either named hole face. The same generic rotation calculation now also identifies the old dart and the fresh stub dart of each individual exposed spoke as one opened facial orbit. The reverse, cap-facing half-darts have now been enumerated by the same five positions and are carried back to the retained boundary by the literal edge involution. Thus neither side of the finite cap boundary is being hidden behind a choice or a cardinality count.

The deleted-side half of that inter-spoke calculation is now literal. The oriented facial walk, together with cubic cyclic rotation at each cap vertex, forces two capped deleted-side face steps from spoke (i) to spoke (i-1). The resulting finite five-position calculation proves that all five cap-side boundary darts occupy one capped face cycle, and transports that fact through the deleted-to-retained boundary equivalence to the complementary first-return order. The new module `GoertzelV24FacialPentagonCapOpenHoleOrbit` completes the one-cap transfer. The oriented pentagon cycle itself proves connectedness of the deleted induced side; the planar-bond boundary-order theorem then transfers the five-cycle to the retained first-return permutation, and the literal open-region calculation proves that all five fresh stubs lie in one actual opened face orbit. This is not a collar hypothesis: it is equality of the computed `dartOrbitFace`s. The retained complement's connectedness is no longer an extra formation premise in the intended minimal-counterexample setting. The module `GoertzelV24FacialPentagonCapRetainedConnectivity` proves it by a finite five-spoke argument. Every retained connected component must own at least three cap spokes: if it owned at most two, deleting a pair covering its spokes would disconnect the closed graph, contrary to the already proved two-edge-cut normalization from vertex minimality. Two distinct components would then need at least six disjoint indices among the five spokes. Thus the retained complement is connected, and the five-stub face-orbit equality follows directly for a graph-backed vertex-minimal Tait counterexample. This closes the one-cap C–2 bridge without a picture assumption. The theorem is nevertheless still for one cap opening. The remaining C–2 work is to transport the two selected cap calculations to the simultaneous two-cap carrier and prove that the two ordered collar orbits remain distinct there.

There is a real typing constraint on that last transport. The one-cap planar-bond theorem cannot simply be reapplied to the union of the two cap supports: the deleted side then has two components, whereas that theorem is a one-bond result with connected induced sides. Moreover the current graph-level `PentagonCapPair` excludes shared cap vertices and direct spoke-to-cap contact, but does not by itself exclude an ambient face that meets both cap neighbourhoods. Thus simultaneous opening needs the stronger remote facial separation supplied by the source corridor selection, together with connectivity of each retained complement. This is a local sequential transport obligation, not permission to postulate a two-component bond theorem or to identify the two collar faces from graph-level separation alone.

The planar-bond layer now exposes the precise local invariant needed for that assembly. For a connected deleted side, `cutFacesHaveUniqueRetainedBoundaryDart_of_planar_bond` proves that each touched ambient facial cycle has exactly one retained boundary occurrence. Accordingly the simultaneous-opening proof need not repeat the Euler/rank calculation: it must combine the two one-cap uniqueness results with the source-corridor fact that no ambient face meets both selected cap interfaces. The generic helper does not assert that last, source-specific separation. The follow-up `GoertzelV24FacialPentagonCapPairBoundaryFaces` makes the boundary fully explicit. Its predicate `RemoteBoundaryFacesSeparated` says exactly that no ambient face cycle contains both a named inner interface dart and a named outer one. Lean classifies all ten simultaneous-cut occurrences into the two five-port families and, from minimality plus this predicate, proves unique retained occurrence on every touched face. This is a genuine conditional consumer, not the missing construction: the corridor selection in C–4 must still prove remote facial separation. What remains locally is to lift each cap's five-step deleted-side cycle into the two-component deleted carrier and thereby obtain the two simultaneous opened hole orbits. The input type for that lift is now explicit: `FacialPentagonCapBoundaryWalkPair` packages the two literal five-dart facial walks together with exactly the separation fields of the underlying `PentagonCapPair`. It adds neither an annular embedding nor remote facial separation. This prevents the simultaneous calculation from silently replacing ordered collar coordinates by the weaker unordered cap datum. The local simultaneous-deletion calculation is also now complete. `GoertzelV24FacialPentagonCapPairDeletedCycle` proves that enlarging a deleted carrier preserves the cap's two-step face turn whenever the enlarged set still contains the cap and excludes its five spoke endpoints. Applying this once to each member of the pair puts each ordered five-port family in one actual cycle of the same two-cap deleted-face permutation. This is local non-interference. The same module now proves the stronger separation fact: the deleted face permutation preserves the inner cap component, and hence no inner boundary dart can share an orbit with an outer boundary dart. The next transport is now kernel-checked as well. `GoertzelV24FacialPentagonCapPairOpenHoleOrbit` combines those two deleted cycles with touched-face uniqueness under the explicitly named `RemoteBoundaryFacesSeparated` condition. The complementary boundary order is the inverse of the deleted order, and the literal open-region rotation therefore places all five inner stubs in one actual face orbit and all five outer stubs in another actual face orbit. A new generic converse, `GoertzelV24OpenRegionBoundaryOrbitClassification`, proves that two fresh stubs can share an opened face only when their retained boundary coordinates share a first-return cycle. Combining this with the invariant cap components proves `openedInnerHoleFace_ne_openedOuterHoleFace`: the two computed collars are distinct literal face orbits. Thus the local C–2 collar calculation is complete as a consumer of the explicitly named `RemoteBoundaryFacesSeparated` premise. The source corridor must still construct that premise in C–4, and C–3 must still package these two face orbits, with the Euler comparison, as the literal annular cellulation.

The carrier part of that C–3 passage is now explicit rather than implicit. `GoertzelV24OpenRegionGraphFaceTransport` proves that the canonical dart equivalence from the literal opening to its computed simple graph conjugates the full face permutation. It therefore induces an equivalence of quotient face carriers which sends each dart-named literal face to the correspondingly named graph-backed face; in particular, the two opened collars remain distinct after transport. This does not prove the annular cellulation: connectivity of the simultaneously retained complement and the Euler equation remain separate C–3 obligations. `GoertzelV24FacialPentagonCapPairGraphHoleFaces` now specializes this transport to the literal simultaneous cap opening: it names the two graph-backed hole faces, proves that each receives all five of its ordered ports, and proves them distinct. Consequently three of the C–3 cellulation fields—the two named holes and their inequality—are no longer schematic; the remaining fields are the simultaneous connectivity and Euler assertions.

The numerical carrier half of the Euler comparison is now exact as well. `GoertzelV24TwoPentagonCapOpeningCounts` proves that the two disjoint pentagon cycles comprise exactly ten distinct ambient edges. On both the toolchain opening and its graph-backed manuscript presentation, the vertex count is unchanged and the opened edge count is the ambient edge count minus ten. This is not yet an Euler theorem: the corresponding face-orbit count and simultaneous retained-complement connectivity remain the two open C–3 assertions.

The first face-count term is now equally literal. `GoertzelV24CutTransitionFaceCount` proves generically that unique retained boundary occurrence makes the boundary-dart carrier equivalent to the touched ambient-face carrier. Its facial-cap specialization identifies the simultaneous boundary with the ordered ten coordinates $\mathrm{Fin}(5) + \mathrm{Fin}(5)$ and, under the explicitly retained C–4 remote-separation premise, proves that exactly ten ambient faces are touched. The two pure deleted cap faces and the opened quotient-face count remain to be related before the Euler equation is complete.

The pure-deleted term is now exact as well. `GoertzelV24FacialPentagonCapPairPureFaces` first proves that an oriented facial pentagon has exactly one ambient face supported wholly on its five cap vertices. The named facial orbit supplies existence; connectedness of the induced pentagon and the spherical cycle-rank inequality exclude any second such orbit. It then proves that a pure face on the union of two separated cap supports cannot move between the supports: a facial-permutation step doing so would exhibit an ambient edge joining them, contrary to the literal cap-opening separation. Thus the simultaneous pure-deleted family is the disjoint union of the two one-cap families and has cardinality two. Together with the ten touched faces, this fixes both ambient-side terms of the C–3 face census. The opened quotient-face comparison and simultaneous retained-complement connectivity remain unproved; no annular cellulation is claimed at this point.

The boundary term on the opened side is also now counted. `GoertzelV24FacialPentagonCapPairBoundaryOrbitCount` proves that every retained boundary dart is literally one of the five inner or five outer source coordinates. Each family is one cycle of the computed first-return permutation, and the two cycles are distinct. Hence that boundary permutation has exactly two quotient cycles: precisely the two opened faces that meet fresh stubs.

The remaining face-count bridge is now proved generically in `GoertzelV24OpenRegionFaceCount`. A face of the literal opening which avoids every fresh stub consists entirely of retained old darts; forgetting the subtype witnesses identifies its whole cycle with one ambient face lying wholly on the retained side. Conversely every such ambient face lifts to a stub-free opened face, and the two maps are inverse. Boundary-successor cycles classify the complementary opened faces. Thus the opened face count is the number of wholly retained ambient faces plus the number of boundary-successor cycles; this is an equivalence of quotient carriers, not a diagrammatic assertion.

Specializing in `GoertzelV24FacialPentagonCapPairOpenEuler` completes the numerical C–3 Euler comparison. Under the still-explicit C–4 remote facial-separation premise, the ambient faces split as ten touched faces, the wholly retained faces, and two pure deleted cap faces, while the opening contains the same retained faces and two boundary faces. Consequently the opening has exactly ten fewer faces, matching its already-proved ten-edge loss and unchanged vertex count. The opened rotation remains a spherical map with two distinguished faces; deleting the interiors of those two named faces gives the annular Euler characteristic zero. The remaining C–3 obligation is now the connectivity of the simultaneously retained complement and the final cellulation packaging. No conditional connectivity wrapper is being counted as that construction.

For the two-cap laboratory, the packaging itself is now literal and machine-checked in `GoertzelV24FacialPentagonCapPairAnnularCellulation`. Given the two formation properties of the selected pair—remote separation of its two boundary-face families and connectivity of the simultaneously retained complement—Lean constructs a `FramedAnnularCellulation` on the actual graph-backed opening. Its two holes are the computed face orbits, their inequality is transported rather than asserted, and its Euler field follows from the exact vertex, edge, and face equations above. Thus no annular cellulation is postulated in this special case. Equally importantly, this theorem does not promote its two hypotheses into universal source geometry. Addendum V starts instead with a frontier tangle that already has its outer interface and opens one interior pentagon; the general formation pillar must construct its crosscuts in that supplied annulus, not select two pentagonal corridor ends merely to instantiate the laboratory.

One of the laboratory's two hypotheses has now been reduced to a quantitative separation condition rather than left as an opaque topological assumption. In `GoertzelV24FacialPentagonCapPairRemoteSeparation`, the ambient face on the retained side of every named spoke is proved adjacent in the full facial dual to the corresponding pentagonal cap face. Therefore two selected cap faces with no common dual neighbour—in particular, at dual distance greater than two—satisfy `RemoteBoundaryFacesSeparated`. This is an exact reusable theorem for a supplied two-cap pair. It does not assert that the source corridor ends are pentagons. The genuinely source-facing remainder is instead to assemble the L1 boundary-clean corridor's selected Cell–3 rail steps into the separated facial-dual crosscuts used by the splice. Addendum XXVII uses the primal radial paths for the L8-to-L6 length/depth dichotomy; they are not the splice's transverse cuts and must not be dualized into them merely because both geometries live in the same annulus.

The paired transversal interface has accordingly been repaired at the type level. `GoertzelV24SelectedDualPathTransversalPair` defines two equal-width, separated simple facial-dual paths carrying the literal primal edge selected at every step. Their closed composite is proved to be a dual cycle, and the union of their selected crossings has exactly twice the common width. Unlike the older interface, this carrier does not assume globally that every pair of annular faces shares at most one edge—a condition known to fail on the literal boundary. The older stronger carrier maps into the new one in one direction. This is a representation repair, not the missing construction: the selected Cell–3 rail steps still have to be assembled and end-capped to inhabit it.

That repaired carrier now performs its first literal downstream operation. Module `GoertzelV24SelectedDualPathTransversalSlitRotation` forms the simultaneous slit rotation system from the union of the two selected primal crossing supports and proves, pointwise, that each selected crossing opens to its own fresh boundary stub. This is a construction on the selected-edge representation itself, not a conversion back to global shared-edge uniqueness. It narrows the C–to–D integration boundary: the long end-capped pair remains to be built, and the mature splice-boundary/profile tower must still be transported to this selected slit rather than silently invoked through the older stronger carrier.

The successful local branch now inhabits that repaired carrier literally. For any source-derived `LocalLayerPair`, Lean selects one actual shared primal edge at each of the four dual steps and proves that the two selected crossing supports are disjoint. The proof uses only the local incidence law: if the same primal edge occurred on both two-step paths, it would lie on the boundaries of three distinct faces, whereas an embedded edge has at most two incident face orbits. No global unique-shared-edge premise is restored. This closes one Cell–3 tile, not the formation gate: the alternative `SelectedLocalLayerCollision` still has to be eliminated or resolved, and the surviving local tiles still require append-safe assembly and literal end caps before they form the source's long transverse pair. The corresponding source-facing theorem preserves that boundary as an explicit disjunction: a selected rung placement yields either such a paired selected tile or the already classified local collision witness. Thus the representation repair exposes, rather than conceals, the one remaining finite Cell–3 branch.

The source-level carrier mismatch has now been removed without promoting the two-cap laboratory into the general proof. Addendum V begins with a frontier tangle X whose ordered outer stubs already exist and opens one interior pentagon. The module `GoertzelV24FrontierPentagonCapOpening` represents exactly this input by an outer-only `FrontierBoundaryData`, together with a well-formedness predicate saying that the outer stubs have degree one and every other vertex is cubic. Its `InteriorPentagonCap` relation states that the selected cap is separated from the old outer interface. Deleting the cap cycle then computes an `AnnularBoundaryData`: the cap vertices are the five new inner stubs, while the old outer stubs and their ordered boundary edges are retained literally. Lean proves that no outer edge is deleted, both interfaces remain degree one and disjoint, and all other vertices remain cubic. Thus this is the graph-carrier part of $X' = X - \mathrm{cap}(f)$, not a diagrammatic identification. It deliberately does not yet construct X from the minimal counterexample, identify the two hole face orbits on the common opened rotation, or supply the long separated C–5 crosscuts.

The corresponding representation bridge is also now literal rather than postulated. The module `GoertzelV24FrontierPentagonCapOpenRegionComparison` composes the ordered one-cap slit/toolchain isomorphism with the generic slit/open-region isomorphism. Consequently the toolchain graph carries the dart rotation computed by the existing open-region construction, and the complete ordered inner/outer boundary datum transports to that same graph-backed carrier with well-formedness preserved. This closes a presentation mismatch only: it does not assert that either interface is already a face orbit. In particular the closed-sphere one-cap hole theorem cannot simply be reused here, because the pre-existing outer stubs of the frontier tangle make global cubicity false. The remaining geometric step is therefore precisely to retain the supplied outer boundary orbit of X while computing the new inner orbit created by the cap opening.

That first half is now explicit. The module `GoertzelV24FrontierPentagonCapOuterFace` records the source-level statement that both darts over every named old outer-interface edge lie on one ambient outer face. Provided that complete ambient face orbit avoids the deleted cap, Lean transports it through the literal one-cap opening and then through the computed simple-graph backing. Every named outer-interface dart is proved to remain on the transported orbit, while that orbit is proved distinct from the face rooted at each fresh cap stub. Thus the old rim and the newly cut boundary cannot be identified by a convenient drawing. This still does not compute the common inner orbit of the five fresh stubs on the frontier carrier, derive the old outer-face premise from source selection, or package the resulting pair as the source annulus; those are the remaining formation obligations before the C–5 crosscuts can be realized.

The complementary inner-face calculation is now valid on that same frontier carrier without restoring false closed-map hypotheses. The module `GoertzelV24FrontierPentagonCapInnerFace` first localizes cubicity to the five cap vertices; frontier well-formedness proves precisely those five degree equations even though the old outer stubs have degree one. It then isolates the finite collar condition actually used by the orientation comparison: among the five retained-facing cap spokes, no ambient facial orbit contains two distinct occurrences. That condition implies the generic cut-face uniqueness predicate, and Lean now proves the converse as well by classifying the cut boundary by the five ordered cap spokes. Consequently these are two presentations of one geometric invariant, not two independent premises. When that invariant holds, the retained first-return order is the inverse of the already computed deleted-side pentagon order. Lean therefore proves that all five fresh stubs occupy one literal opened face, carries this face to the graph-backed rotation, and proves it distinct from the retained outer face. This is a conditional bridge rather than the finished source formation: the Addendum–V selection theorem must still derive the five-spoke face-occurrence condition (equivalently, generic cut-face uniqueness), as well as the previously stated retained outer-face premise, from the actual frontier tangle. No global cubicity or global two-sidedness assumption has been smuggled back into the annulus.

The corresponding one-cap face arithmetic is now intrinsic as well. In `GoertzelV24FrontierPentagonCapFaceCounts`, the five source-ordered spokes are proved to enumerate the entire retained cut boundary, so the cut has exactly five boundary darts and five crossing edges. Under the same five-occurrence collar invariant used above, these give five distinct touched ambient faces. The complementary first-return calculation also proves that the retained boundary permutation is transitive on the whole cut carrier, so its quotient has exactly one boundary orbit. Thus two numerical terms in the literal Addendum–V face surgery are no longer inherited from the closed two-cap laboratory. The remaining term is stated separately: one must prove, on the already-open frontier and without global two-sidedness, that the deleted pentagon side supports exactly its one named pure face. Only after that local theorem may one conclude that opening the cap removes five faces and derive the one-cap Euler equation.

That local theorem is now proved. The argument in `GoertzelV24FrontierPentagonCapPureFace` uses only the literal cyclic order at the five cap vertices. A dart following the recorded pentagon orientation lies on the named cap face. A dart in the reverse orientation is forced by the degree-three rotation onto the unique spoke, after which its face successor is based outside the cap. It therefore cannot belong to a face wholly supported on the cap vertices. The named pentagon is consequently the unique pure deleted face even though the ambient frontier has degree-one outer stubs and is not globally two-sided. The module `GoertzelV24FrontierPentagonCapOpenEuler` combines this fact with the five touched faces and the single retained boundary orbit to prove the exact one-cap surgery equation

$$|F_{\mathrm{open}}| + 5 = |F_{\mathrm{frontier}}|.$$

What remains at this stage is not another cap count: the source selection must construct the ambient frontier's Euler/cellulation presentation and discharge the already isolated collar-uniqueness invariant before the graph-backed annulus can be packaged.

The carrier arithmetic has now been closed as well. In `GoertzelV24FrontierPentagonCapOpeningCounts`, complement counting proves that the toolchain opening deletes exactly the five cap-cycle edges, while the ordered opening isomorphism proves that the manuscript carrier has the same number of vertices as the frontier: the five removed cap vertices are replaced by five fresh stubs. Combining these two equalities with the face-surgery equation above, `GoertzelV24FrontierPentagonCapAnnularEuler` proves the conditional transport

$$\chi(X') = 2 \qquad \text{from the explicit premise} \qquad \chi(X) = 2.$$

This is not yet the formation theorem. Its name, `graphBacked_euler_of_ambient_euler`, records the remaining direction of dependence: the source-local construction must still produce the Addendum–V frontier X with its own Euler/cellulation datum and prove the five-spoke collar-uniqueness invariant. No closed two-cap sphere has been substituted for that task.

The corresponding downstream carrier is now explicit in `GoertzelV24FrontierPentagonCapAnnularCellulation`. Given the four source-formation outputs just isolated—connectedness of the retained side, complete retention of the old outer face, the ambient frontier Euler equation, and the five-spoke collar-uniqueness invariant—it packages the literal opening as the exact `FramedAnnularCellulation` used by the corridor theory. The inner hole is the computed cap-stub face, the outer hole is the transported old outer face, and their inequality is proved. This is intentionally a conditional C–3 constructor. It makes the C–4 debt readable in one type signature; it does not discharge that debt.

The interface from that cellulation into the pre-existing closed-web tower is now exact as well. The module `GoertzelV24FrontierPentagonCapClosedWebAnnularEmbedding` proves that both orientations of every fresh inner boundary edge lie on the computed inner hole and both orientations of every retained outer boundary edge lie on the transported outer hole. This is stronger than merely placing each unoriented edge in a hole boundary: the proof classifies the two darts over each edge and transports their face orbits through the literal opening and its graph isomorphisms. It then constructs the `ClosedWebAnnularEmbedding` consumed by the corridor, radial-path, and crosscut developments. The result is still deliberately conditional. It connects the source carrier to the downstream tower, but it does not select the frontier tangle or derive retained connectedness, outer-face retention, the ambient Euler equation, cap-collar uniqueness, or the separated crosscuts. Those are the remaining C–4/C–5 formation obligations rather than hidden fields of the bridge.

The first genuine surgery atom is now kernel-checked in `GoertzelV24DeletedEdgeFaceMerge`. Deleting one graph edge removes its two darts from the vertex rotations; the theorem computes the resulting face step exactly: the predecessor that formerly entered one deleted dart now jumps to the old successor on the opposite dart side. Its orbit-level corollary identifies those two surviving positions as one post-deletion face orbit. This is the correct local mechanism by which a removed facial cycle can join its neighbouring facial sides. It does *not* yet say that the edge-deletion rotation is the already constructed literal slit rotation; it is therefore complementary to, rather than a prerequisite for, the now-proved one-cap slit-orbit bridge.

Pumping. The corridor acceptance language is proved periodic and corridor pumping congruence-correct; with the recurrent tube-ring transfer these are the working tools of the Short-Strand replacement search (`GoertzelV24CorridorPumping`).

Framed trails and local fiber calculi. The framed-trail layer proves completion semantics and defect freedom — the unique admissible missing color and the characterization of proper trail extensions (`missingColorAt_is_unique_admissible`, `hasMatchingDefectColors_iff_exists_properTrailExtensionColor` in `GoertzelV24FramedTrail`) — supported by exact local calculi: the square fiber table with its realized Kempe switch and local and compositional profile identities, the digon Kempe rung, rotor local color routes, and the two-edge-cut color parity (`GoertzelV24Square` /`SquareKempe`/`SquareComposition`, `GoertzelV24Digon`, `GoertzelV24Rotor`, `GoertzelV24TwoEdgeCut`).

The minimality normal form (proved). The full minimality connectivity ladder is now machine-checked on rotation systems, with no `sorry` and axioms in {`propext`, `Classical.choice`, `Quot.sound`}. The two-edge, cyclic three-edge, and cyclic four-edge cut reductions are each proved by capping both sides into strictly smaller cubic rotation systems and gluing their Tait colorings across the cut; a vertex-minimal Tait counterexample admits none of them, so it is cyclically five-edge-connected (`cyclicallyFiveEdgeConnected_of_vertexMinimalTaitCounterexample`). The four-edge rung is the classically hard one: at a four-edge cut the four boundary colors sum to zero in the Klein four-group, giving three boundary classes (all-equal and the two non-crossing adjacent pairings), and the two capped sides always share a reducible class. The load-bearing step — that a side admitting an all-equal boundary escapes, by a Kempe swap, into a non-crossing adjacent class (never the crossing Birkhoff pairing) — is *proved*, not assumed, from the planarity of the interior Kempe chains (`GoertzelV24FourEdgeCutKempeEscape`), and a minimal counterexample is correspondingly forced to have minimum face size five. The corridor-existence side is lifted off the fullerene premise entirely: on any spherical cubic map of minimum face size five, the pentagon count equals 12 plus the total negative curvature weight, and either the face count lies below a weight-dependent threshold or a clean hexagonal corridor exists (`GoertzelV24WeightedOrbitFaceCorridor`) — no restriction to pentagonal and hexagonal faces is used anywhere.

In progress and open. On this normal form the endgame reduction is under active construction: a fusion-chain / rebase-pumping dynamics that, from any adjacent pair of the counterexample, produces either a fusion terminal or a strictly descending rebase pumping (`exists_rotationOrderedFusionTerminalOrRebasePumping_of_vertexMinimalTaitCounterexample`), together with an exhaustive scope classification of the remote facial-dual cycles it generates (length $\geq$ 5 exact cut, length-three single-vertex cut, length-four adjacent-pair cut with an internal dual chord). What remains *open* is the coloring transport that turns each classified case into a contradiction with minimality — the large-cut splice/interface law and the explicit consumption of the singleton-star and adjacent-pair-collar cases — and hence Trail Completability, the L1–L10 chain, and a proved Short-Strand replacement. The v23-era collar obligations are taken up only where this route consumes them. Nothing in this section claims the Four-Color Theorem or the completion of the v24 route.

Canonical data at reentry joins. The rotation ordering removes a small but potentially serious compositional ambiguity. If two adjacent-pair presentations have the same ordered central edge, then their rotation-ordered four-port data are equal, regardless of the original enumeration of the four boundary ports. We therefore refine the recursive relation so that each step retains its source fusion normal and its certified pumping launch. Given two consecutive steps, the target Kempe package of the first supplies a normal with a genuine same-base Kempe escape, and the second supplies its source normal; the two recovered rotation-ordered data agree exactly. Finite iteration consequently yields a structural terminal, a large exact remote separator, or a nonempty closed orbit whose successive normal forms are canonically aligned. This alignment is a bookkeeping theorem, not yet a coloring-transport theorem: transport between different deleted adjacent pairs remains the next local collar problem.

13 Status and scope: machine-checked, reduced, and open

Historical snapshot. This section records the companion's 21 July 2026 v23/v24 status. It is kept for provenance and should not be read as the present closure ledger; the controlling August status is Section 7.

We state the scope exactly. The full route is a ladder $S0, \dots, S9$ (Gate-2, then v23 §9 and §3); the table records each step's manuscript reference and current Lean status.

Step	Content	v23 ref	Lean status
—	single-gadget $\mathrm{LKR}_{\mathrm{in}}$	Lem. 8.14	finite encoded-model audit; graph move-soundness ope
—	two-color transparency	Lem. 8.15	finite encoded-model audit; shares graph-fidelity boun
—	Pillar A	Thm. 4.9	actual W_0 inclusion open; unrestricted-oracle strictnes
S0	Gate-2 chain $\mathrm{LKR}_{\mathrm{in}}$	Lem. 8.18/8.19	one actual fiber/orbit plus mode-sufficiency pin
S2	radial-closing-edge fiber census	§9.2 Step 3	proved only at length 2/3; manuscript target open
S4	ambient escape regression	§9.2 Step 4	abstract-chain countermodel; graph realization open
S1	collar $\Rightarrow \tau$/mirror chain	Lem. 8.5–8.6	open (structural)
S3	annulus peeling well-definedness	§9.2 Step 2	open (structural)
S5	interface to Kauffman Thm. 3.1	§9.2 / §3	open (statement-level)
S6	Kauffman parity/primality internals	§3.2–3.3	open (structural)
S7	minimal-counterexample normal form	Lem. 5.2–5.3	open (structural)
S8	CAP5 reducibility	§6	partial
S9	minimality wrap (+ Tait)	Def. 5.1 / Thm. 2.3	open (structural)

Machine-checked outright. The finite encoded-model audits for Lemmas 8.14 and 8.15, together with completeness of the 192-state incidence enumeration; unrestricted-oracle strictness and conditional need-based Pillar A infrastructure (with four uniform peel oracles refuted); and the length-2/3 radial-closing-edge S2 census. The corresponding graph move-soundness, actual v23 $W_0(H)$ inclusion, and manuscript S2 statement remain open. All with no `sorry` and axioms $\subseteq \{\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}\}$.

Reduced to one pin. Gate-2 (Lemmas 8.18/8.19): the manuscript's pointwise Step-1 preparation is refuted and replaced by the frontier-automaton / fibration architecture, whose real-fiber labs pass to length seven; the whole composition rests on a single named structural remainder.

Regression countermodel. §9.2 Step 4's collar-word-to-ambient-word reinterpretation fails in the encoded two-layer tau-chain model. A simple planar cyclically-five-connected graph realization has not been proved; the actual-collar cut-lift obligation remains open. In the encoded model this lowers the odds that the v23 route completes *as written*; its salvage is open.

v24 scope. See §12. The validation toolchain and censuses are independently reproduced; the all-face curvature identity and the corrected *weighted* pentagon budget are proved; the necklace/corridor, pumping, and framed-trail layers are in the tree; the endgame (fusion chains, chiral monodromy, Trail Completability, L1–L10, and a proved Short-Strand replacement) is open.

Open (the non-finite heart). Steps S1, S3, S5–S9: the realization of abstract chains as actual collars, the wide-annulus peeling, the interface to Kauffman's reduction, and the Kauffman/normal-form/CAP5 internals. Their common cost is one shared planar-embedding/duality/Jordan infrastructure the development does not yet have; most carry a known proof shape, so their risk is formalization cost rather than mathematical doubt — with the exception of the S4 salvage, which is genuinely open.

What is not claimed. We do *not* claim the Four-Color Theorem, and we do not claim that v23 proves it. The 4CT is a theorem via the computer-assisted proofs; what is at issue is solely the viability of the v23 human-readable route, which remains *open* — its foundational layer now machine-checked, its most dangerous assembly seam pinned down as broken.

Hardened interfaces for the remaining assembly. Four further boundaries are now machine checked. First, explicit finite fixtures show that capped subtraction requires its order premise, arbitrary letter deletion has an exponential orbit, unordered five-field boundary data does not determine a cyclic order, and the planar curvature identity permits thirteen pentagons when a heptagon is present. Thus the route must use its weighted curvature budget and may not infer order or monotonicity from a smaller payload. Second, a cubic connected bridge-free spherical theta map shows that a rotation carrier need not admit a simple-graph presentation on the same vertex type; graph backing must therefore be proved for the selected minimal-counterexample carrier rather than postulated globally.

The corrected capped law is also specialized to the two literal source updates, whose new-contribution/overlap pairs are respectively $(5, 1)$ and $(1, 0)$. Hence its side condition is discharged by kernel-checked corollaries rather than left as prose at the application boundary.

Third, `GoertzelV24PhysicalSpliceInterfaces` states and proves the generic composition contracts demanded by Addendum XXVII: a complete three-edge seam star with single-use incidence receipts and three distinct colours, a support-union rule preserving global edge, internal-vertex, and face separation of two rails, and an explicit side-reversing identification of the ordered boundary. Counterexamples in the same module show why weak shared-edge agreement and pairwise-valid local rail steps do not suffice. These are interfaces, not yet their source-specific inhabitants or the final quotient theorem.

Finally, `GoertzelV24FiniteClosureInterfaces` isolates the finite endgame without assuming a symmetric pumping equivalence. Deleting a middle word is sound under one-sided inclusion of right languages; a forward simulation supplies that inclusion, while exact reachable-set equality is retained as a corollary; a rooted literal-letter presentation factors the raw transition through a finite code; and a sparse CSR closure certificate is split into independently replayable pages and carries collision-free state indices, exhaustive rows, local witnesses, realizable initials, exact accepting and necessary bits, and frontier closure. No numerical reachable table, numerical V_0, or bounded base verification follows merely from these interfaces. The live order remains: construct the complete five-field literal Cell letter, prove its rooted factorization, replay its actual reachable closure, derive V_0, and verify the resulting route-native base.

Closure-readiness audit and route verdict (25 August 2026). The proposed next experiment—compute one reachable-closure cardinality—is the right way to measure the finite endgame, but the presently executable letters do not yet define the exact source automaton. In particular, `GoertzelV24ClosedWebLocalLayerSerialFiniteBoolStepLetter` is a finite Boolean Cell–rebase receipt extracted from one actual colouring. Its proved scope is one supported singleton transition, not the complete support of the physical Cell's `Count` matrix. The older homogeneous `LocalLayerFiniteReceiptStep` and its compressed version are explicitly sound overapproximations; the latter also omits the distinct boundary rebase and is computed on a different geometric-prefix presentation. Running BFS on either object is useful as a labelled diagnostic upper-bound experiment, but its state count is not an authoritative reachable count and cannot define V_0.

The missing gate is now one mathematical factorization rather than another interface design: every positive literal `Count` entry, composed with an arbitrary compatible cumulative-prefix witness, must produce exactly one of the finite rooted Cell–rebase transitions, and every accepted finite transition must have such a source witness. The local half is green: positive `Count` support already supplies one realizable combined profile with all five semantic coordinates. The cumulative-prefix attachment, its finite rooted code, the source initial set, and the exact converse remain to be joined. Once that theorem lands, the next checkpoint is deliberately numerical and narrow: independently enumerate and replay the first exact closure, compare two implementations, and report the raw carrier size, reachable subset count, and right-language (Nerode) quotient count.

This measurement is urgent. The proved conservative carrier formula has orders of magnitude about $10^{51}, 10^{77}, 10^{109}, 10^{189}$ at widths $4, 5, 6, 8$, respectively, so using it directly would make the bounded base physically meaningless. The earlier twenty-mode frontier table is encouraging evidence that actual dynamics can occupy a tiny subset, but it belongs to the narrow τ/mirror-τ Lemma 8.18 laboratory and no typed theorem identifies it with the present cumulative corridor profiles. It is precedent for the experiment, not its answer.

The resulting verdict has two levels. The current manuscript is not valid *as literally written*: among other things, Addendum XXVII invokes at most twelve internal pentagons because the ambient minimal counterexample allegedly has exactly twelve, while the proved curvature identity permits thirteen pentagons in the presence of a heptagon; several weaker seam and local-rail readings are also refuted below. These are refuted shortcuts, not a refutation of the proof strategy. They should be reported to the author with their in-route replacements, and no further effort should be spent extending the shortcuts themselves. The reductive–compositional programme explicitly asks for a finite factorization of `Count`; the weighted curvature law, complete physical-splice interfaces, and exact local five-field semantics are viable repairs inside that same architecture. Before the exact census below, the honest report was therefore "source repairs and one decisive closure gate remain," not "the route is dead." The new census supplies a negative verdict for the GWCO endpoint itself; the following paragraph records why this is not a negative verdict on the Cell–1 compositional strategy.

GWCO refutation and the surviving route repair (25 August 2026). A subsequent exact census invalidated the terminal predicate to which the foregoing closure machinery had been aimed. Deleting the cycles of two disjoint pentagonal faces numbered 4 and 12 in the manuscript toolchain's C_{30} fullerene gives a proper five-by-five annulus with no shared boundary vertices or boundary edges. Exact Tait-colouring enumeration gives 3696 colourings, of which 12 are totally closed webs and 6 have a good inner word. An independent replay of the same enumerator reproduced these numbers. As a fidelity control, the same code on the polar C_{30} pair reproduces the manuscript's

5412 colourings and 360 totally closed webs, all at bad words. A smaller dodecahedral opening gives 936 colourings, 36 totally closed webs, and 12 at good words, although one boundary edge is shared between its two cap interfaces; the proper non-polar C_{30} witness is therefore the clean counterexample.

Thus the Good-Word Closed-web Obstruction of Addenda XXII–XXVII is false as stated. In particular, the reductive carrier `GoertzelV24ClosedWebReductiveAssembly.PackedInstance`, whose target is definitionally `False`, must not be supplied with a purported verified base or used as an endpoint for a numerical V_0. Packaging the finite C_{30} witness as a literal Lean inhabitant remains a useful regression fixture; until that packaging is complete, the exact executable census rather than a kernel theorem is the status of this refutation.

The refutation does *not* refute the compositional strategy. It removes an unnecessarily strong intermediate claim. Addendum VII's Reformulation Theorem says that menu B for a majority pair is equivalent to at least two distinct components of that pair meeting the cap. At five outer stubs, total closure forces every strand to be radial, so the four active inner spokes of either majority pair lie on four distinct components. Direct inspection of all six good-word C_{30} witnesses confirms the count $(4, 4)$ for the two majority pairs. Hence a totally closed web is already strongly menu B and cannot be a frozen adversarial state. The source-faithful repair is therefore

$$\text{totally closed at five outer stubs} \Longrightarrow \text{all strands radial} \Longrightarrow \text{menu B},$$

not nonexistence of the web. The exact cumulative-prefix/Cell factorization remains valuable, but its first authoritative reachable closure must now be the parametric transfer used by the Cell–1 Seed Lemma of Addendum XXV, not a base audit for GWCO. This correction distinguishes a refuted proof cell from the still-viable proof strategy and prevents further work from polishing a false terminal target.

The first post-correction factorization unit is now machine checked in `GoertzelV24ClosedWebLocalLayerSerialCellStateFactorization`. It splits the former finite Cell record losslessly into (i) the complete cumulative prefix state, (ii) a candidate outgoing profile, and (iii) a bounded local Cell-plus-seam factor containing no prefix state. Reassembly is inverse in both directions, and the native Boolean reflection preserves the resulting three-argument support relation exactly. This repairs the structural error in which a prefix-dependent receipt could be mistaken for a pumpable physical letter. It is deliberately not yet the positive-`Count` converse: gluing an arbitrary positive literal Cell witness to an arbitrary compatible prefix, computing the successor attachment state through the following boundary rebase, and only then measuring the reachable closure remain the live steps.

The tracked local factor has now been made genuinely colouring-parametric as well. The common twenty-one-edge Cell–seam carrier no longer has to read the single ambient closed-web colouring: it accepts an arbitrary edge-colour function, and Lean proves that its two tagged graph families preserve complete reachability in the literal Cell and in the residual seam. The former ambient-colouring constructor is retained as an exact specialization, so no existing source witness changes meaning. This is the local construction needed by an arbitrary positive `Count` witness; the remaining theorem must splice that witness's colour function with the cumulative prefix on their actual overlap and prove that the boundary profile supplies precisely the required compatibility. No reachable-state count is claimed before that splice theorem.

That splice is now explicit in `GoertzelV24ClosedWebLocalLayerSerialCellColorSplice`. It uses the Cell witness on the literal Cell region and the prefix witness elsewhere, with compatibility stated on the actual finite intersection of those two regions. Lean proves that the splice restricts to the Cell colouring on every Cell edge, restricts to the original prefix colouring on every prefix edge, and therefore leaves the complete regional tracked graph of the prefix unchanged. The two outgoing crossing colours are definitionally supplied by the Cell witness. The same module now exposes the complete pre-rebase output profile over an arbitrary colour function and specializes it to the spliced colouring. Lean proves that its outgoing word is exactly the Cell witness's outgoing word and that splicing the ambient colouring with itself recovers the previous ambient five-field profile exactly. Thus the first and fifth coordinates are not being rebuilt in a parallel carrier; the colour-parametric adapter feeds the existing occurrence-sensitive profile.

The next geometric obligation is sharp: identify the actual overlap with the source interface data (or otherwise show that the stored boundary packet determines all colours on it). Until that theorem is proved, agreement of the two displayed colour words is not silently promoted to agreement on the whole overlap.

The tracked-connectivity coordinate of that splice is now closed in `GoertzelV24ClosedWebLocalLayerSerialCellSplicedTrackedFiniteClosure`. The underlying three-factor theorem has first been generalized: for any colour function and any cumulative finite state whose input profile and attachment record recover the realized prefix graph, pre-rebase reachability is exactly the reflexive–transitive closure of the finite prefix, Cell, and residual-seam factors. The splice then discharges this contract from its compatibility theorem and the already exact ambient-prefix attachment state. Consequently the complete `strandConnected` field of the spliced output profile is equivalent, in both directions, to that finite closure, guarded by the two displayed output colours. This is an arbitrary compatible Cell witness result, not merely an ambient-colouring specialization.

The remaining three facial coordinates are now joined to it in `GoertzelV24ClosedWebLocalLayerSerialCellSplicedFiniteSupportLetter`. They require no second transition construction: face continuation, port incidence, and capped face length are independent of the edge colouring, so the proved occurrence-sensitive twenty-four-slot facial carrier and its decoders are reused unchanged. Lean therefore packages all five pre-rebase coordinates of any compatible cumulative-prefix/literal-Cell splice into one supported finite letter. This is still deliberately conditional on agreement over the *actual* shared edge region. The next physical theorem must derive that compatibility from the source boundary packet; after it, the following boundary rebase must be composed with this letter before the cumulative transition can be enumerated.

That compatibility has now itself been factored through finite data, without assuming a smaller geometric overlap than the source construction proves. The module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteColorCompatibility` assigns each edge of the common transition carrier to one of twenty-one stable slots and stores a partial colour table: `some c` exactly when the slot represents an edge retained by the selected region, and `none` for an inactive or padding slot. Lean proves that agreement of the prefix and Cell tables wherever both are active is equivalent, in both directions, to pointwise colour agreement on the literal intersection of the two graph regions. Raw reachability is deliberately not used for this test: it is reflexive at isolated coordinates, while graph support can omit a tracked boundary edge with no tracked neighbour inside the region. Thus the next transition layer may consume an executable finite compatibility predicate, with no hidden boundary-only or non-isolation assumption.

The tracked part of that transition has now been rebuilt from the same finite data rather than retained as a prefix-specialized receipt. The module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteTrackedColorCode` separates the local letter into an uncoloured twenty-one-slot adjacency code, the prefix partial colour table, and the Cell partial colour table. Their finite splice gives the Cell colouring precedence on the literal overlap and reconstructs both the Cell tracked graph and the residual seam graph. For every tracked colour pair and for both graph tags, Lean proves equality with the corresponding source graph, and hence equivalence of complete graph reachability. This closes the specific factorization defect identified above: the seam consumed by a transition is computed from the current cumulative state and a prefix-independent local letter, rather than stored inside that letter after specializing it to one ambient prefix. The next unit must lift this reconstruction into the cumulative transition structure and show that arbitrary positive literal `Count` witnesses realize it through the following boundary rebase; no reachable-closure count is yet claimed.

The same reconstruction is now executable in the native Boolean transfer representation. The module `GoertzelV24ClosedWebLocalLayerSerialCellFiniteTrackedBoolColorCode` checks prefix–Cell colour compatibility by enumerating all twenty-one stable carrier slots, and proves that this Boolean test is equivalent to the exact proof-facing compatibility predicate. From the uncoloured local geometry and the two partial colour tables it then computes the Cell and residual-seam adjacency tables. For each tracked colour pair and both factor tags, Lean proves equality of the reconstructed simple graph with the proof-facing graph and equivalence of complete reachability; the literal-source specialization recovers the formerly stored source factors exactly. Thus the current prefix is no longer hidden inside the purported physical letter even at the native execution layer. This is still a representation theorem, not the promised closure measurement: the next theorem must feed every positive literal `Count` witness into this reconstruction, compose the resulting Cell support with the distinct boundary-rebase receipt, and prove the exact cumulative transition before any reachable-state count is authoritative.

That positive-`Count` lift is now proved for the Cell half of the alternating source transition. The module `GoertzelV24ClosedWebLocalLayerSerialCellNativeFactorization` replaces the remaining prefix-specialized tracked table by a coloured cumulative state and a physical local factor containing only uncoloured Cell geometry and the Cell partial-colour table. Applying the factor reconstructs the Cell graph and prefix–Cell seam dynamically. Lean proves that this native Boolean code is literally the reflection of the proof-facing reconstruction, including its carrier and named coordinates, and that every compatible literal source splice satisfies the resulting executable five-field predicate. The module `GoertzelV24ClosedWebLocalLayerSerialCellCountNativeFactorization` then uses the literal definition of `Count`: every positive entry produces an open-Cell Tait colouring, hence a nonzero outgoing word and a physical native factor; whenever that witness agrees with the cumulative prefix on the actual overlap, the reconstructed transition evaluates to `true`. This is the promised arbitrary-witness direction, not a receipt for one ambient colouring.

The remaining composition boundary is now narrower and explicit. The existing boundary-rebase support letter is specialized to the ambient web colouring. It may therefore be paired with the ambient Cell receipt, but not yet with an arbitrary positive-`Count` colouring. The next theorem must parameterize the rebase collar and its five-field decoder by the compatible spliced colouring, prove the ambient receipt as a specialization, and only then form the finite rooted Cell–rebase relation. No closure count or numerical threshold is claimed before that colour-parametric rebase is green.

The colour-parametric rebase and the first complete alternating factor are now green. The modules `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseTrackedColorParametric`, `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseOutputColorParametric`, and `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseSupportColorParametric` recompute the rebase collar at an arbitrary nonzero local colouring. Lean identifies its tracked closure exactly, decodes all five successor-profile fields (edge colours, tracked connectivity, face continuation, port–fragment incidence, and capped face progress), and proves the resulting finite letter supported. The earlier ambient receipt is therefore a specialization, not a hidden premise of the executable transition.

The module `GoertzelV24ClosedWebLocalLayerSerialCellRebaseNativeFactorization` then composes that rebase with the physical Cell factor. Every positive literal five-field `Count` entry supplies a Cell colouring nonzero on the complete literal Cell carrier; whenever it agrees with the cumulative prefix on their actual overlap, the physical Cell and the following rebase both satisfy their finite support predicates and meet at a literally equal pre-rebase profile. This is one exact source alternating letter, rather than one ambient-colouring receipt or an executable over-approximation. It does not yet supply the reachable closure: the remaining proof must derive the compatibility test from the finite input packet, prove source realizability and representative invariance for supported codes, enumerate the initial states, and only then compute the certified closure. Accordingly no reachable-state count, numerical V_0, base certificate, or headline theorem is claimed here.

The remaining compatibility premise is now internal to the executable letter. Module `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExecutableFactorization` defines the native acceptance bit by comparing the cumulative state's and the physical Cell factor's partial colour tables at all twenty-one stable carrier slots. For every literal source Cell, Lean proves an equivalence: the complete Cell–rebase factor is supported if and only if that Boolean bit is `true`. Consequently every positive five-field `Count` entry produces an exact finite candidate whose acceptance is decided entirely by data already present in the cumulative state and local factor. This removes the proof-facing compatibility hypothesis from the transition relation; it does not remove the distinct remaining obligations of source-realizability, representative invariance, initial-state construction, and certified reachable closure.

The target of that executable letter is now a state of the same cumulative type as its source, rather than only a decoded boundary-profile receipt. Module `GoertzelV24ClosedWebLocalLayerSerialTrackedPrefixAttachmentColorParametric` evaluates the twenty-one-slot tracked prefix-attachment state at an arbitrary colour function and proves its exact finite reachability reconstruction. The following module `GoertzelV24ClosedWebLocalLayerSerialColoredCumulativeStateForColor` packages, at the literal prefix/Cell splice colouring, the next cut's boundary profile, tracked attachments, facial attachments, and partial carrier colours. Lean proves definitionally that the boundary rebase's decoded output is the input-profile field of this next cumulative state. This fixes the codomain of the source's Phase-5 state transition. It is not yet the finite recurrence: the next tracked, facial, and colour-code fields must still be derived from the predecessor state and the local Cell–rebase factor before representative invariance, initial-state enumeration, or a certified reachable closure can be claimed.

The moving carrier at the next Cell is now connected to its predecessor by an exact finite change of coordinates. Module `GoertzelV24ClosedWebLocalLayerSerialCellRebaseCarrierTransport` decodes each of the next twenty-one slots either as one predecessor slot or as one of the four named boundary-rebase roles, with the rebase role taking precedence when an edge has both descriptions. Lean proves that this finite transport computes the next cumulative colour field exactly once one literal overlap statement holds. That statement has itself been reduced to a single pairwise geometric atom: if a target-local edge also occurs in a strictly earlier Cell, then it must already lie in the current carrier or in the four-edge rebase switch. Incoming crossing edges and the retained rung require no additional premise; their coverage follows from the carrier definitions. Thus the remaining colour-coordinate debt is geometric locality, not another finite-state interface. The older framed-trail development proves the same kind of locality under global cubicity and global two-sidedness, hypotheses which are false on the opened annulus because of its boundary stubs. The needed local face-adjacency replacement is now proved in

GoertzelV24ClosedWebLocalFaceBoundaryAdjacency. It uses only two intrinsic facts: every edge has exactly two oriented darts, and a vertex incident to two distinct edges cannot be one of the degree-one boundary stubs. Therefore adjacent primal edges on distinct named faces force adjacency of those faces in the full facial dual on the literal opened carrier, without either false global premise. The remaining step is to apply this local theorem to the prior-Cell/target-carrier intersection and then invoke the source corridor's geodesic separation. The crossing part of that application is now proved in GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseLocality. Every incoming crossing lies on its Cell's centre face and every outgoing crossing on the successor centre face; consequently a strict historical incoming crossing adjacent to the successor presentation comes from the current Cell, while an adjacent historical outgoing crossing comes from the current or immediately preceding Cell. These are the opened-carrier versions of the old framed-trail locality lemmas and use neither global cubicity nor global two-sidedness. What remains is the region-to-portal reduction: show that an edge in the intersection of a strict prior Cell and the successor transition carrier either is a named portal or forces adjacency of named portals, after which the new index-locality lemmas apply. No unconditional recurrence or reachable-state count is claimed until that final pairwise overlap atom is discharged.

The first half of that reduction is now sharp rather than cardinality-only. Module GoertzelV24ClosedWebLocalLayerFiniteCellCarrier exposes the degree count which its earlier six-edge theorem used internally: a literal Cell side either contains a cycle or has exactly two vertices. In the actual two-cap minimal-counterexample laboratory, GoertzelV24MinimalFacialPentagonCapPairLocalCellCarrier now names the one-sided-cut contradiction separately and therefore excludes the cyclic branch at every indexed Cell. Each Cell side thus has exactly two vertices; its four boundary crossings leave room for at most one non-boundary internal edge in the simple graph. The generic endpoint lemma edge_eq_of_mem_region_not_mem_crossing_of_card_eq_two and its source specialization make this uniqueness literal, and the resulting sourceLocalLayerCellRegionAt_card_le_five sharpens the previous six-edge envelope to four crossing edges plus at most one internal edge. This is the finite region-to-portal shape needed by the remaining overlap argument; the existing six-slot code remains a conservative padded presentation. The result does not by itself identify that internal edge across distinct Cell indices, discharge the overlap atom, or compute a reachable-state count.

Existence and index locality of that internal bond are now proved as well. The generic connected-two-vertex lemma in GoertzelV24RotationVertexCutProfile constructs a regional edge which does not cross the chosen side. Its source specialization shows that every literal Cell has exactly one such bond and that the bond's endpoint set is exactly the Cell's two-vertex side. The module GoertzelV24ClosedWebLocalLayerSerialCellInternalOverlap then proves that a strict historical Cell and the successor Cell cannot share an edge which is internal to both: a shared bond would identify their two vertex sides; a genuine outgoing portal of the successor would consequently occur on both separated corridor-face boundaries, contradicting the clean corridor's facial-dual distance condition. This first closes the internal–internal branch of the pairwise overlap atom without global cubicity or global two-sidedness. The subsequent module GoertzelV24ClosedWebLocalLayerSerialCellPastOverlap closes the mixed branches as well. The internal bond is adjacent to every genuine portal of its two-vertex Cell, while a strict historical portal can neither equal nor touch an outgoing portal of the successor Cell: either relation would make two corridor faces adjacent beyond the clean corridor's permitted distance. Successor incoming portals and the retained rung are precisely named members of the four-edge rebase switch; successor internal and outgoing-carrier edges cannot recur from a strict prior Cell. Consequently every old edge which can affect the successor's moving twenty-one-slot carrier is represented by the bounded Cell–rebase window, and Lean proves the resulting cumulative colour coordinate recurrence unconditionally on this literal two-cap source carrier. This is the colour-coordinate part of the source's Phase-5 factorization, not the full factorization: exact tracked and facial successor fields, representation invariance, realizable initial states, and the first certified reachable-state count remain open.

The tracked coordinate now has its exact cumulative recurrence as well. The generic module `GoertzelV24InterfaceDeletionComponentFactorUpdate` proves that adjoining a graph supported wholly on a bounded interface changes only literal interface adjacency: equality and attachment through components strictly outside the interface remain unchanged. Module `GoertzelV24ClosedWebLocalLayerSerialCellTrackedDeletionStablePreRebaseState` specializes this law to the actual twenty-one-edge Cell carrier. It Boolean-ORs the already-verified finite Cell and residual-seam adjacency tables into the cumulative deletion-stable prefix state, proves that this code presents the literal pre-rebase tracked graph, and reconstructs its reachability under *every* interface-deletion mask by closure over the surviving coordinates. Thus the path/cycle counterexample no longer blocks the tracked recurrence: the state retains exactly the exterior information that deletion cannot change. The total masked relation is now exact even when a named port is deleted. The generic module `GoertzelV24InterfaceDeletionComponentFactorTotal` proves that two arbitrary named coordinates are reachable after a mask precisely when they denote the same ambient edge (the isolated reflexive case), or both survive and the finite closure connects them. Module `GoertzelV24ClosedWebLocalLayerSerialCellTrackedDeletionStablePreRebaseTotal` specializes this theorem and retains the two old outgoing crossings as literal coordinates of the current Cell carrier.

The next tracked junction is narrower but is *not* a definitional identification. The current carrier has twenty-one coordinates, whereas the boundary switch is presented independently by four roles (old rung, two successor incoming crossings, successor rung), and the existing carrier transport deliberately uses their sum. The exact interface therefore uses that sum rather than forcing every role into an old coordinate. Deleting the role summand is now proved to select exactly the literal switch, including fresh roles and aliases, and to leave exactly the tracked core. The generic module `GoertzelV24InterfaceDeletionComponentFactorTotalGraph` turns the resulting finite total relation into a literal component graph without changing reachability. Module `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedCoreState` then decodes this graph through the two old output crossings into the existing finite tracked-core state and proves exactness on all four semantic roles. Thus the expanded deletion-safe code now reaches the input type of the proved core recurrence without an active-role coverage premise. What remains is a rolling construction of the expanded code from the cumulative state and the bounded local receipt, followed by successor-carrier reindexing.

The reindexing operation is now fixed at graph level. A plain restriction of the expanded relation to successor coordinates would be wrong: an expanded coordinate which is not retained by the successor becomes part of the successor's exterior and may still join two retained coordinates. Module `GoertzelV24InterfaceDeletionComponentFactorForget` instead masks the coordinates retained by the smaller interface and proves that the resulting graph is exactly the exterior graph of that interface. Its partial-map form also handles padded target carriers, provided an unmapped target coordinate is inactive. Module `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedSource` constructs the literal partial map from the successor carrier to the predecessor-plus-role interface. Returned coordinates are proved to denote the same ambient edge, and the clean-corridor past-overlap theorem proves that every unmapped coordinate is isolated in each successor tracked graph. Consequently the generic forgetting mask is unconditionally the literal successor exterior graph on this source carrier. The same generic module constructs the partially reindexed code and proves its equality and direct-adjacency rows exact. Literal target coordinates are injective; a mapped coordinate inherits the exact expanded adjacency, while an unmapped coordinate contributes no adjacency because its isolation is already a theorem.

The remaining exterior row is now materialized and proved exact in `GoertzelV24InterfaceDeletionComponentFactorForgetExterior`. It is not a restricted submatrix. The executable calculation first tests the old exterior row and otherwise enters a surviving forgotten coordinate, computes the verified finite reflexive–transitive closure of the three old primitive rows among all forgotten coordinates, and exits to the other retained coordinate. Survival is computed from the vertex-equality row, so two coordinate names for one ambient edge are deleted together. A traversal invariant proves both directions: every accepted finite chain expands to a literal path in the smaller exterior, and every literal smaller-exterior path decomposes at its first and last forgotten coordinates into exactly such a chain. The partial-map proof includes inactive padded coordinates rather than assuming them away.

The source specialization in `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedSource` now therefore proves that contracting the exact predecessor-plus-role code gives an exact three-row primitive-step code on the literal successor carrier. This closes the graph-theoretic associativity-of-contraction junction. The rolling construction has now advanced through the next exact seam. The generic update law in `GoertzelV24InterfaceDeletionComponentFactorUpdate` identifies the updated three rows with the exact code of the union graph field by field, and `GoertzelV24ClosedWebLocalLayerSerialCellTrackedDeletionStablePreRebaseState` specializes that equality to the cumulative predecessor and the positive literal Cell receipt. The expanded-source module then gives an executable, alias-safe partial coordinate map from the predecessor carrier plus four semantic role occurrences back to the cumulative carrier. Every mapped coordinate denotes the same ambient edge, including coincident role names.

The attempted smaller expansion also identified, and then retired, one more shortcut. A role edge absent from the predecessor carrier need not be inactive in the pre-rebase tracked graph: the clean-corridor past-overlap theorem leaves the literal switch case real. Thus “fresh role is inactive” is not used as an assumption. The replacement in `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedReceipt` carries the complete mask-independent exterior receipt on the old coordinates plus the four role occurrences. It has at most twenty-five occurrences, records aliases through its equality row, and is packaged as one closed finite type whose index is only the predecessor cardinality, never an ambient edge name. Its graph-free role mask deletes an old coordinate whenever that equality row says it aliases a removed role. Lean proves, for every compatible colour function and every pair of expanded coordinates, that the resulting finite relation is exactly reachability in the literal unchanged core. A second decoder turns the same closed receipt into the already-established finite tracked-core state and proves exactness on all four successor roles. This resolves the tracked bounded-lookahead fork without freshness, coverage, or source-named state.

The deletion part of that calculation has now been separated from this particular twenty-five-occurrence presentation. The generic module `GoertzelV24InterfaceDeletionComponentFactorMaskUpdate` proves, for any injectively named finite interface, that deleting selected interface vertices is exactly a row-wise Boolean mask of its exterior code. The equality row is unchanged; literal adjacency and connection through the unchanged exterior survive exactly when both endpoints survive. This is the representation-free algebraic half of the successor update, and it composes with the already-proved interface-supported adjacency update without rebuilding ambient component tables.

This theorem also fixed the next proof order. Rather than answer every newly visible collar edge by enlarging the bespoke receipt again, the source specialization now proves the structural bounded-interface statement in `GoertzelV24ClosedWebLocalLayerSerialCellUniformInterface`. The literal Cell transition and following rebase use a common carrier of at most forty-nine primal edges and forty-eight facial darts, independently of the corridor length and Cell offset. Every tracked or facial seam adjacency meets one of the two named outgoing portals; every local and rebase graph factor is supported on the corresponding bounded carrier; and any strict historical Cell edge that reappears in the successor moving carrier is already visible in the current carrier or the four-edge switch. These conclusions assemble the previously separate portal-completeness, past-overlap, and local-collar support theorems into a termination certificate for the local receipt width. They do not yet identify the exact cumulative five-field recurrence, and no closure count is authoritative until that recurrence and its realizable initial states are executable.

The tracked half of that fixed-width recurrence is now exact in `GoertzelV24ClosedWebLocalLayerSerialCellUniformTrackedRecurrence`. The uniform carrier is transported to injective finite coordinates, and one closed update performs precisely the source operation: it masks every coordinate naming one of the four switch edges and then Boolean-ORs the local successor-collar adjacency row. Lean proves equality, row by row, between the resulting graph-free exterior-component code and the canonical exact code of the literal successor tracked graph. Consequently its total finite closure is sound and complete for successor reachability between every pair of the at-most-forty-nine coordinates. Aliases are absent because the carrier itself is a finset; equivalence with the source's semantic-role deletion is proved before the update. The result is also packaged in the source-independent finite type `BoundedInterfaceExteriorFamilyCode 49 TrackedColorPair`. The facial half is now exact as well in `GoertzelV24ClosedWebLocalLayerSerialCellUniformFaceRecurrence`. Its coordinates are the literal darts of the fixed at-most-forty-eight-dart interaction carrier. Lean proves that the coordinate mask selects exactly all dart occurrences over the four switch edges; masking the cumulative pre-rebase facial graph therefore gives the canonical unchanged face core, not an approximation to it. Boolean-ORing the existing local facial residual then produces a code equal row by row to the exact exterior-component code of the literal successor face graph. Its total finite closure is consequently sound and complete for successor facial reachability between every pair of named dart coordinates. The same generic mask-and-add algebra now drives the forty-nine-edge tracked recurrence and the forty-eight-dart facial recurrence; only their source interpretations differ.

The coordinate names used by these fixed-width codes are now proved semantically irrelevant in `GoertzelV24InterfaceDeletionComponentFactorReindex`. Exact extraction of all three rows—ambient-vertex equality, literal interface adjacency, and connection through the strict exterior—commutes with every bijective change of finite coordinates. The proof retains the load-bearing exterior row by showing that a bijective renaming represents literally the same set of ambient vertices and hence the same induced exterior graph. Deletion masks and local adjacency additions commute with the same reindexing, and the resulting notion of interface-code isomorphism is reflexive, symmetric, and transitive. Thus a reachable-state computation may compare bounded receipts modulo coordinate renaming without changing their source meaning. This does not choose a canonical enumeration or identify unrelated source windows; those remain obligations of the combined letter and its initial-state construction.

The graph-free closure used by both recurrences is now isolated in `GoertzelV24InterfaceDeletionComponentFactorGraphFree`. It proves that closure in the three-row code is exactly ambient graph reachability whenever the finite coordinates are injective, and that this statement is natural under coordinate equivalence. Named successor coordinates are supplied by `GoertzelV24ClosedWebLocalLayerSerialCellUniformSuccessorCoordinates`: every successor port is located in the forty-nine-edge tracked carrier, every canonical face fragment and both dart occurrences over each port are located in the forty-eight-dart facial carrier, and all coordinates decode to the literal ambient edge or dart. Consequently the facial closure decides the actual occurrence-sensitive fragment–port incidence relation rather than an arbitrary Boolean over-approximation.

The former assembly junction is crossed by `GoertzelV24ClosedWebLocalLayerSerialCellUniformSuccessorL` Its single finite dependent record stores the two exact graph-free recurrence codes, the three named port colours, and the two residual finite face fields which connectivity alone cannot determine. Tracked connectivity is decoded from the endpoint colour guards and the tracked closure; fragment–port incidence is decoded from the two literal port darts and the facial closure. Lean proves field by field that the resulting five-coordinate profile is exactly the next terminal-aware source profile. This closes the exact *successor-output* letter, not the full transfer automaton: a realizable initial-state characterization and an executable relation tying predecessor, literal Cell, and successor letters remain open. The first certified reachable-closure count must wait for those two obligations; no earlier diagnostic count is authoritative.

One semantic correction is now explicit in `GoertzelV24ClosedWebLocalLayerSerialCellUniformSuccessorC` The preceding successor letter was exact only at the ambient closed-web colouring, whereas a positive literal `Count` witness may supply a different Cell colouring. The new theorem accepts an arbitrary colouring which is nonzero on the four named rebase roles, runs the same fixed forty-nine-edge and forty-eight-dart recurrences at that colouring, and proves that the decoded letter agrees in all five fields with the literal next terminal-aware profile at that colouring. Hence the successor calculation can follow the colouring carried by the eventual composed witness rather than silently reverting to the ambient one. This closes colour fidelity only: the compatible prefix–Cell composition, realizable initial states, and certified reachable closure remain open.

The tracked target of that successor calculation is now executable in the same stable state type as the next cumulative prefix. The generic module `GoertzelV24InterfaceExteriorSupportedPortProjection` recovers support from literal interface adjacency together with the diagonal exterior bit, projects an exact exterior code onto a smaller padded interface with named ports, and proves that exact padded states are invariant under finite coordinate equivalence. The source specialization `GoertzelV24ClosedWebLocalLayerSerialCellUniformTrackedTargetCode` contracts the fixed at-most-forty-nine-edge successor code onto the literal next rolling carrier. A target coordinate absent from the larger carrier is proved isolated using the established past-Cell overlap theorem; inactivity is not inferred from padding. Lean then proves that the contracted code is the canonical exact successor exterior code and that its stable twenty-one-slot projection is exactly the next `sourceLocalLayerSerialTrackedPrefixAttachmentStateForColorAt`. Consequently the tracked cumulative coordinate now has a literal recurrence, not merely a source receipt. The corresponding facial target-state projection, the combined five-field transition, realizable initial states, and the first certified reachable closure remain open.

Provenance

The August revision preserves the accumulated July audit and its negative results, and adds Section 7 as the controlling status for the current Goertzel/Fable reductive–compositional development. Future revisions should update that section when a closure obligation changes status and append failed constructions to Appendix A, rather than erasing the reason they were abandoned.

The manuscript discussed is Ben Goertzel, *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem*, version v23. The formalization is work of the MeT-

Tapedia formalization effort. The keystone and transparency are module `GoertzelLemma814`; the Pillar-A obstruction and its need-based repair are `FrontierGeometric`, `GoalAudit`, and `PillarASmallerSpace`; the Gate-2 architecture is `GoertzelLemma818SemanticBridge` and the frontier-automaton modules; the S2 bridge is `GoertzelLemma818ClosedCollarS2Bridge`; the S4 counterexample and the refuted two-sided-avoidance repair are `GoertzelLemma818TwoLayerEscape` and `GoertzelLemma818TwoSidedAvoidance`. The v24 route of §12 is the `GoertzelV24*` module family (roughly sixty modules; the primary mathematical source is the v24 working package *Deserts, Pivots, and the Necklace*, July 2026). Lean listings are ASCII transliterations of the source declarations (the sources are normative); every headline was checked with `#print axioms`. Per-declaration source links are omitted pending publication of the development tree.

A A graveyard of superseded sub-routes

Each entry is a route or reading that was tried and set aside; several are themselves machine-checked theorems that a stated step *fails*.

The spanning *equality* $W_0(H) = \mathrm{span}(\mathcal{G})$. Refuted on every closed-walk exact shell (§10); superseded by the need-based inclusion, which is what the synthesis actually consumes.

The four uniform geometric peel oracles. Each would have closed Theorem 4.9's synthesis uniformly; all four are machine-checked false. The route uses the manuscript's own per-embedding peeling witnesses instead.

The pointwise Step-1 preparation of Lemma 8.18. The manuscript's own pointwise lift (via Lemma 8.16) is refuted (`lemma818_pointwise_lift_preparation_obstruction`); replaced by the frontier-automaton / fibration composition.

The prefix-fibration shortcut for Gate-2. A tempting closed-form shortcut, machine-checked to fail (`not_concreteChainFiberAppendPrefixFibrationClosed`); kept dead.

The false "additivity" of Kempe switches. An $\mathbb{F}_2$-additive execution of the Pillar-A decomposition $\Delta = \sum v_{f,g}$. This is *not* the manuscript's route — Pillar C is titled "executability without false additivity," and v23 §8.1–8.3 show mixed color-pair switches on shared support do not commute. Recorded only to set aside; the faithful route is the operator/$\mathrm{LKR}_{\mathrm{in}}$ execution of §9.

An $\mathbb{F}_2$-additivity / evader reconstruction (disavowed). An early exploratory reconstruction outside the manuscript's route, which v23 explicitly disavows (additivity invalid for mixed color-pairs; Rmk. 8.10). Not part of this companion's formalized content.

Boundary color-word agreement as full profile agreement. Refuted by a two-state off-diagonal witness in `GoertzelV24ProfileDiagonalLiftFalsifier`: matching projected color support need not supply a diagonal entry for the full profile. The source splice therefore uses the finite full-profile diagonal audit, not its color projection.

Consecutive six-edge slab boundaries as radial splice cuts. Refuted by `sourceTwoTileAlignedBoundaryWalks_consecutive_not_disjoint`. Consecutive local boundary walks share their intervening dual interface. This rules them out as the two separated annular cuts while retaining them as useful local windows.

A single generic hole crosscut as the corridor splice. Generic dual connectedness produces one simple hole-to-hole slit. It supplies neither a second disjoint slit nor alignment with the selected Cell–3 corridor, so promoting it to the source splice would change the statement rather than prove it.

Monotonic reuse of the overlapping transition seam. After assigning shared transition edges exclusively to the accumulated prefix, the recomputed seam need not be a subgraph of the seam for the original overlapping factors. An old endpoint may also belong to the original right region, causing the original seam to suppress a turn that the exact exclusive factorization retains. The successor recurrence must encode those frontier turns explicitly rather than inherit the earlier support carrier by monotonicity.

One ambient-colouring path as the transfer word. A complete Boolean Cell receipt and its following rebase do give one source-supported transition between consecutive cumulative profiles. Listing those receipts over the corridor, however, records only the path selected by one ambient colouring. The source letter is the full support of the literal Cell's `Count` matrix over all endpoint profiles. The singleton-list interpretation was rejected before commit; the finite receipts remain validation witnesses for the later full relation.

An executable over-approximation as the exact closure. The finite one-Cell decoder and its compressed relation admit every proved source transition, but may admit spurious letters and omit the separate boundary-rebase composition. Their BFS closure is a diagnostic upper bound, not the source automaton, a numerical V_0, or a base certificate. Exact positive-`Count` adequacy and rooted cumulative-prefix factorization must precede an authoritative count.

Restricting the exterior row after interface forgetting. Pulling the predecessor's `exteriorConnected` table back along the successor coordinate map is not exact. A predecessor coordinate that is omitted by the successor becomes an interior vertex of the successor exterior and may join two retained coordinates even when the old table has no corresponding direct entry. The replacement is the proved finite contraction through all surviving forgotten coordinates, with alias survival read from the vertex-equality row; a plain submatrix must not be used again.

The Good-Word Closed-web Obstruction. Addenda XXII–XXVII assert that no totally closed web exists at a good five-stub inner word. Exact enumeration refutes this: the nonpolar C_{30} cap pair $(4, 12)$ has twelve totally closed webs, six at good words, while the same implementation exactly reproduces the manuscript's polar census. The closed-web reductive carrier with target `False` is therefore retired. Its intended use survives in stronger local form: total closure makes all strands radial, so each majority pair has four distinct cap-meeting components and hence menu B. The corridor factorization is retained parametrically for Cell 1 rather than extended toward the false GWCO base.

Unrestricted capped subtraction. The identity $\min(5, a + b) - d = \min(5, a + (b - d))$ fails without the premise $d \leq b$. The finite face-progress decoder carries this premise explicitly.

The unweighted twelve-pentagon shortcut. Addendum XXVII's statement that a minimal counterexample has exactly twelve pentagons does not follow from spherical cubicity and face length at least five. The curvature identity is weighted: one heptagon permits thirteen pentagons. The route must use the proved weighted budget (or a stronger source-specific hypothesis), never the unrestricted bound.

Small deletion orbit from a small alphabet. If every letter may be deleted, the orbit of a word with distinct letters contains every subset of its positions. Reachable-state compression must therefore come from the source transition semantics, not alphabet size alone.

Local rail validity as global simplicity. A closed three-edge walk has distinct consecutive endpoints while repeating its initial vertex. The long-rail construction consequently retains global support freshness, not merely a valid append certificate at each step.

Reusing a component receipt after a boundary switch. The cumulative tracked attachment state records support and connected components, but a rebase removes a bounded switch support before installing the new collar. Module `GoertzelV24TrackedComponentDeletionCounterexample` proves that this information is insufficient even for the path/cycle shapes of a properly two-colored tracked graph: a four-vertex path and even four-cycle have identical exact support-sensitive component receipts, while removing their common middle interface coordinate disconnects the path and leaves the cycle connected. Thus the already-constructed deletion-safe tracked core and facial residual receipts are load-bearing fields of the finite Cell–rebase factor; they cannot be reconstructed from the former cumulative component partition alone. This refutes an underspecified executable state, not the source `Count` factorization. The repair is now explicit: `GoertzelV24DeletionSensitivePortResidualFactorContraction` gives the exact finite one-mask core receipt, and `GoertzelV24ClosedWebLocalLayerSerialBoundaryRebaseDeletionSensitiveT` proves that selecting all four literal switch roles restricts the pre-rebase tracked graph to exactly the core already consumed by the successor recurrence. At the ambient coloring its residual code is definitionally the existing deletion-safe tracked core state. No exponential table of masks is materialized: one transition carries its one bounded removal mask and its verified core. The corrected one-mask receipt is therefore a correct source bridge, but it is not used as an exponential table in the cumulative state. The stronger generic module `GoertzelV24InterfaceDeletionComponentFactor` removes the whole bounded interface once, records literal interface adjacency and attachment through components strictly outside it, and proves that every deletion mask is recovered by closure over the surviving coordinates. Its size is polynomial in the interface width and is independent of the number of masks. Module `GoertzelV24ClosedWebLocalLayerSerialCellTrackedDeletionStablePrefixState` instantiates this factor on the literal at-most-twenty-one-edge Cell carrier for every tracked color pair and proves exact source-prefix reachability under every mask. That recurrence is now proved rather than assumed: `GoertzelV24InterfaceDeletionComponentFactorUpdate` shows that an interface-supported local graph updates only direct adjacency, and `GoertzelV24ClosedWebLocalLayerSerialCellTrackedDeletionStablePreRebaseState` applies the law to the existing finite Cell and seam codes. The resulting state reconstructs the literal pre-rebase graph after every deletion mask. The apparent shortcut "the four roles are four current coordinates" is now retired: consecutive local layer pairs are separately constructed, and the verified carrier transport gives role names precedence precisely because equality with predecessor coordinates is presentation-sensitive. The attempted induced mask on only the old carrier would therefore require an unjustified active-role coverage premise. Module `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedInterface` uses the exact replacement: the old coordinates and the four semantic role occurrences form the two summands of one finite interface. The mask selects the role summand, so fresh roles and aliases are handled uniformly. Lean proves unconditionally that the selected ambient edges are exactly the literal switch and that the restricted graph is the existing tracked core; its total finite closure computes the core relation on the two old output crossings. The generic component-graph presentation and the source decoder in `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedCoreState` now package that same exact relation as the pre-existing finite tracked core state, with exact adequacy on all four roles. This closes the representation seam but not, by itself, the rolling recurrence. The new alias-safe partial expansion does derive every old or aliased role coordinate from the cumulative state, and its exactness theorem exposes why the remaining "fresh role is inactive" premise cannot be inferred from clean-corridor overlap: its switch alternative is substantive. That smaller partial expansion is therefore retained only as a conditional lemma, not promoted to the executable letter. The exact repair is the bounded-lookahead receipt in `GoertzelV24ClosedWebLocalLayerSerialCellRebaseExpandedReceipt`. It carries the complete predecessor-plus-role exterior code in a closed finite type of at most twenty-five occurrences, uses the equality row to delete aliases, and proves that its graph-free role-masked relation and decoded tracked-core state are exactly the literal source core for every compatible colouring. Thus neither fresh-role coverage nor fresh-role inactivity is assumed. A reachable-closure number remains premature until this decoded core is composed with the local collar, the other profile fields receive the analogous exact update, and realizable initial states are executable.

Global simple-graph backing of a rotation carrier. The spherical theta rotation system is cubic, connected, and bridge-free but has three parallel edges on two vertices. Only the selected minimal-counterexample carrier may be graph-backed, through an explicit presentation theorem; no unrestricted carrier theorem is available.

A Generated status ledger for the controlling foundations

The inline badges in the controlling spine and foundation sections are the source of truth. This table is generated from them by `tools/check_status.py`; every VERIFIED entry is rejected if its named Lean declaration is absent or its module contains an admitted hole.

Result	Status	Machine-checkable claim
statement (compositional-proof-spine.tex:138)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SphericalMinim`
statement (compositional-proof-spine.tex:139)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SphericalMinim`
statement (compositional-proof-spine.tex:163)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FourEdgeCutGlu`
statement (compositional-proof-spine.tex:167)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24MinimalFaceSiz`
theorem (compositional-proof-spine.tex:178)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FiniteProfileP`
lemma (compositional-proof-spine.tex:225)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24PumpingEngine.`
statement (compositional-proof-spine.tex:268)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24CurvatureScope`
statement (compositional-proof-spine.tex:278)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24CurvatureScope`
statement (compositional-proof-spine.tex:362)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DodecahedralCh`
statement (compositional-proof-spine.tex:363)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DodecahedralCh`
statement (compositional-proof-spine.tex:364)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DodecahedralCh`
statement (compositional-proof-spine.tex:365)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DodecahedralCh`
statement (compositional-proof-spine.tex:366)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DodecahedralCh`
statement (compositional-proof-spine.tex:367)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DodecahedralCh`
statement (compositional-proof-spine.tex:403)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24WeightedOrbitF`
theorem (compositional-proof-spine.tex:416)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24OrbitFaceDualD`
corollary (compositional-proof-spine.tex:451)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24OrbitFaceDualD`
statement (compositional-proof-spine.tex:478)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24DeletionCompon`
lemma (compositional-proof-spine.tex:501)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SeparatedGeode`
statement (compositional-proof-spine.tex:534)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SeparatedGeode`
statement (compositional-proof-spine.tex:535)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SeparatedGeode`
definition (compositional-proof-spine.tex:579)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
lemma (compositional-proof-spine.tex:600)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
lemma (compositional-proof-spine.tex:637)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
lemma (compositional-proof-spine.tex:657)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
theorem (compositional-proof-spine.tex:698)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FourBlockEdgeL`
proposition (compositional-proof-spine.tex:785)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
proposition (compositional-proof-spine.tex:786)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
corollary (compositional-proof-spine.tex:810)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
corollary (compositional-proof-spine.tex:811)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24AnnularFrontie`
lemma (compositional-proof-spine.tex:1007)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1031)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1041)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedDualCycl`
statement (compositional-proof-spine.tex:1049)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedDualCycl`
statement (compositional-proof-spine.tex:1060)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedVerte`
corollary (compositional-proof-spine.tex:1065)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1088)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1090)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1094)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedDualCycl`
statement (compositional-proof-spine.tex:1096)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1116)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedVerte`
statement (compositional-proof-spine.tex:1117)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SourceTrailAli`
statement (compositional-proof-spine.tex:1119)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedVerte`
corollary (compositional-proof-spine.tex:1124)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1200)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1205)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1208)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
statement (compositional-proof-spine.tex:1232)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FramedCorridor`
theorem (compositional-proof-spine.tex:1266)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24FiniteTreeInte`
theorem (compositional-proof-spine.tex:1305)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ClosedCountRep`
theorem (compositional-proof-spine.tex:1327)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24PhysicalClosed`
corollary (compositional-proof-spine.tex:1368)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24PhysicalClosed`
theorem (compositional-proof-spine.tex:1397)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SphereCutDesce`
lemma (compositional-proof-spine.tex:1420)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24SphereCutMater`
statement (compositional-proof-spine.tex:1524)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedEdgeS`
statement (compositional-proof-spine.tex:1525)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedEdgeS`
statement (compositional-proof-spine.tex:1526)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedEdgeS`
statement (compositional-proof-spine.tex:1527)	VERIFIED	`Mettapedia.GraphTheory.FourColor.GoertzelV24ConnectedEdgeS`